# A MARKED-WIDTH REDUCTION OF MANDELBROT LOCAL CONNECTIVITY

RESEARCH MANUSCRIPT IN PROGRESS

Abstract. We develop a source-locked reduction of the Mandelbrot local connectivity conjecture to a marked backward transition theorem for mixed renormalization towers. The proposed state is a partially invariant virtual or pseudo-Siegel carrier with explicit postcritical, prime-end, obstacle, and curve-family records; an unmarked continuum is insufficient. We prove finite root-pullback and inverse-tree reductions, exact critical-value and attachment formulae, functorial finite routing, a common-domain theorem, finite Wave-horizon closure under stated hypotheses, and several source-locked KKL corrections and no-go results.

The stationary-Wave QA reconstructs the native proof line by line. It finds an unproved relative telescope before the separator argument, an undefined terminal index, a missing measurable final-exit lamination, and a false degree-one assertion; the canonical continuum degree is two. A cyclic-cover model also refutes the literal broad coefficient-one KKL Wave-pushforward statement under its printed definitions, while leaving the narrower Dudko–Lyubich one-common-interval annular theorem intact. The native repair is therefore split into four explicit gates: relative continuation, pointwise two-path breaking, measurable final fixed-component exit, and typed bounded-degree pushforward. The present manuscript proves the relative-continuation gate; the other three remain open.

We formulate the remaining construction, transition, compactness, and rigidity requirements as the Universal Protected-Tower Compactness and Rigidity Proposition and prove its implication to MLC only conditionally. That proposition, VAAPC-I, and MLC remain open; this manuscript does not claim an unconditional proof of Mandelbrot local connectivity.

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## 1. Status, scope, and principal reductions

1.1. **The target and the proof boundary.** For

$$f_c(z) = z^2 + c, \qquad \mathcal{M} = \{c \in \mathbb{C} : (f_c^n(0))_{n\geq 0} \text{ is bounded}\},$$

the Mandelbrot local connectivity conjecture is:

**Open Problem 1.1** (Mandelbrot local connectivity)**.** The set $\mathcal{M}$ is locally connected at every parameter $c \in \mathcal{M}$.

The machine-readable dependency ledger accompanying this manuscript has 465 obligations: 34 imported, 316 proved internally, and 115 open. Its target `MLC` is open. The source lock date is 2026-07-27.

*Warning* 1.2 (No theorem laundering). A result called "imported" below is used only with the hypotheses recorded in section 2. A conditional theorem proves an implication, not its open hypotheses. An open problem is never used as a lemma.

1.2. **The marked state.** At a renormalization checkpoint $n$, the proposed state is not a continuum alone but

$$\mathfrak{H}_n = (\Sigma_n, H_n, B_n, P_n^{\leq \nu_n}, \mathcal{I}_n, \Gamma_n). \tag{1.1}$$

Here $\Sigma_n$ is the ambient marked surface, $H_n$ is the regularized carrier, $B_n$ records satellite base-cycle or neutral access data, $P_n^{\leq \nu_n}$ is the finite critical orbit actually used by the proof, $\mathcal{I}_n$ is the ideal or sectorial boundary marking, and $\Gamma_n$ is the declared curve family. The corresponding marked degeneration is

$$D_n^{\#} = \mathrm{Ext}_{\Sigma_n \setminus (H_n \cup B_n \cup P_n^{\leq \nu_n})}(\Gamma_n). \tag{1.2}$$

Neither existence nor uniform control of (1.1) over all towers is assumed.

1.3. **Results proved in this manuscript.** The following results are unconditional within their stated finite topological or arithmetic hypotheses.

**Proved Theorem 1.3** (Root-pullback carrier equivalence; preview)**.** *One full root continuum, together with the routed components of all its finite iterated pullbacks containing the complete required packages, determines a unique finite saturated carrier forest. Selected sheets are distinct exactly when their packages lie in distinct same-depth pullback components.*

**Proved Theorem 1.4** (Critical-value corridor criterion; preview)**.** *For a quadratic polynomial, a selected pullback edge over a full target carrier has degree one exactly when the target avoids the critical value, and degree two exactly when it contains the critical value.*

**Proved Theorem 1.5** (Endpoint-birth accounting; preview)**.** *For successive boundary coverings of degrees $d_j$, terminal endpoint count $p_N$, and $b_j$ genuinely new endpoints born at stage $j$,*

$$p_0 = \Big(\prod_{j=0}^{N-1} d_j\Big)p_N + \sum_{j=0}^{N-1}\Big(\prod_{i=0}^{j-1} d_i\Big)b_j. \tag{1.3}$$

**Proved Theorem 1.6** (Attachment-support reduction; preview)**.** *Under the exact KKL-form interval pullback description, genuine endpoint births vanish away from components of the pulled-carrier excess attached to the marked interval. Births confined to a bounded terminal quadratic window have a period-independent degree-weighted cost.*

**Proved Theorem 1.7** (Canonical-carrier obstruction; preview)**.** *No least connected full carrier exists even for a two-point package. Moreover, an explicit critical-value-free branch of $z^2$ has a first pulled excess whose monotone absorption erases a protected upper prime-end access while producing no sheet, critical, obstacle, or localization collision.*

**Proved Theorem 1.8** (Protected free-component criterion; preview)**.** *After the complete finite protected-collar and obstacle data are frozen, a protected connector exists exactly when one component of their complement has simultaneous singleton ports to every required full atom. A protected separator between two required port classes is an exact obstruction inside the current localization.*

**Proved Theorem 1.9** (Augmented-atom common domain; preview)**.** *If the frozen forbidden inventory is covered by finitely many pairwise disjoint full augmented atom/obstacle packets compactly contained in one Jordan localization, and every original atom retains an explicit free singleton port, then all ports lie in one component of the original free domain. The result uses compacta Alexander duality and does not assume local connectivity.*

**Proved Theorem 1.10** (Fixed-packet closure audit; preview)**.** *For a fixed finite typed assignment, fullification is an order-independent closure. Post-fill merger, localization, and empty original-atom exposure defects persist under monotone same-label enlargement, while individual unprotected ports can be truncated in pairwise disjoint packet neighborhoods.*

**Proved Theorem 1.11** (Horizon-first inventory audit; preview)**.** *For a fixed finite typed Wave-horizon graph, dependency reachability from the complete proof-use seed set is the unique least inventory. Under a horizon extension, its old-stage slice is unchanged exactly when no extended seed reaches an omitted old record. Constructing the actual virtual stopping horizon remains open.*

**Proved Theorem 1.12** (Resolved Wave-horizon compiler; preview)**.** *A finite acyclic analytic Wave program with actual marked covering/immersion branches, exact iterate and degree words, a complete proof-use ledger, a finite dependency universe, and analytic stability proofs for every enlargement generator has a finite maximal iterate horizon and an order-independent analytically stable least simultaneous final-root closure. KKL's native recursion is a positive control with forbidden critical-visit chain length $216b^2$; this does not construct the marked virtual correspondence.*

**Proved Theorem 1.13** (First decoration-scale selector; preview)**.** *For a fixed non-principal decoration outside a directly attached satellite copy $M_{p/q}$, the least critical $q$-return escape $n$ selects the actual puzzle depth $\nu_{\mathrm{dec}}(1) = nq + 1$ and an actual first-return atlas. The terminal primitive little-Julia cycle does not fit in one critical return base: $K_1$ is the first excluded component, so the analytic replacement is a multi-base return chain. The complete raw separator fan has $2^n$ records; proof-complete bounded selection or faithful compression remains open.*

**Proved Theorem 1.14** (Terminal Hubbard-tree anchor; preview)**.** *In KKL's native periodically repelling prime primitive category, the complete terminal little-Julia cycle has its compact forward-invariant Hubbard tree as a connected carrier. Every finite selected vertex package has a unique convex subtree; an injective tree branch transports that hull exactly, and package enlargement adds a one-gate forest. The native ideal attachment side from $K_1$ toward $K_0$ is therefore defined. Selecting the actual intermediate virtual component package and its protected analytic prime ends remains open.*

**Proved Theorem 1.15** (Cantor-package finite-trace criterion; preview)**.** *An infinite family of distinct virtual small-Julia components cannot be identified injectively with a finite terminal-tree package. A finite quotient supports a declared joint observable exactly when that observable is constant on every quotient fiber. Copy-chain labels alone do not construct an embedded component family or analytic branch action, so the global Cantor-package realization and finite proof-use trace remain open.*

**Proved Theorem 1.16** (Direct-decoration pullback package; preview)**.** *In the directly visible decoration category, the two actual univalent $f^q$-pullbacks of the separator $P$ compactify the terminal root into a rooted family of actual preterminal Julia pullbacks and connected nested end fibers. Their planar union is compact, $f^q$ deletes the first address symbol, and geometric nesting deletes the last. Root enlargement is natural while it remains in the first-level gap. Transferring the second satellite portrait to actual bases on these fibers remains open.*

**Proved Theorem 1.17** (Normalized end tools, full-trace calculus, and collar obstruction; preview)**.** *For an independently realized root portrait, its actual finite-cylinder copies along one infinite direct-decoration address have a normalized direct limit canonically biholomorphic to the root. Finite shift-graded diagrams of already actual root data materialize at one common finite depth, and fully transported extremal-length and electrode-response data are conformally invariant. Under explicit no-bypass walls, serial resistances add. These facts do not prove scalar nodal response: on a length-$L$ cylinder the constant conductance is $2\pi/L$, while the $\cos(k\theta)$ trace has energy $\pi k \coth(kL)$. For a genuine rectangle, the exact full trace operator, trace-renormalized comparison, and compact half-cylinder limit are now proved. Their direct-decoration application remains open because inverse pullback divides the angular footprint by $2^q$, so the full parent and child equipotential arcs do not form one serial rectangle. The subsequent one-cell repair proves an active/inactive Jordan decomposition and exact one-junction trace refinement under explicit hypotheses. It does not close the iterated end: a side port can reactivate at a nonconsecutive shell, and whole-tail uniformization introduces an uncontrolled endpoint welding cocycle.*

The full statements and proofs appear in sections 5, 6, 10, 10.7 and 12. The finite inverse-tree Jordan-fan result is given in section 7.

### 1.4. **The conditional endpoint.**

**Open Problem 1.18** (Universal protected-tower package; preview)**.** Construct, with universal constants, the canonical protected carriers, their exact analytic realization, a finite Value-observable response state, the backward transition, coercive marked compactness, and tower rigidity with connected relative parameter neighborhoods. The full six-clause statement is open problem 9.1.

**Conditional Theorem 1.19** (Conditional MLC reduction; preview)**.** *Assume the marked construction, marked backward transition, dynamical coercivity, and parameter-rigidity hypotheses stated in section 8. Then $\mathcal{M}$ is locally connected.*

Equivalently, conditional theorem 9.4 proves that the open protected tower package implies MLC. These are logical endpoints of the project, not an unconditional proof of open problem 1.1.

## 2. Source lock and imported results

The bibliography is not used as a reservoir of unstated consequences. This section records the exact imports on which the reduction relies.

**Imported Theorem 2.1** (Yoccoz–Hubbard)**.** *If $c \in \mathcal{M}$ is not infinitely quadratic-like renormalizable, then $c$ has a basis of connected relative neighborhoods in $\mathcal{M}$.*

We use the written exposition in [Hub93]. This theorem handles the non-infinitely-renormalizable side only.

**Imported Theorem 2.2** (BLOT satellite root bound)**.** *For a quadratic polynomial with satellite relative period $s$, every quadratic-like root annulus $A$ satisfies*

$$\operatorname{mod}(A) < \frac{2\pi}{\log(4(s+1))}. \tag{2.1}$$

This is the quadratic specialization of the main estimate of [BLOT22]. It is an upper bound and therefore forbids a classical period-independent root-modulus lower bound.

**Imported Theorem 2.3** (KKL native terminal Hubbard tree)**.** *In KKL's periodically repelling, primitively renormalizable native $\psi$-ql category, the complete cycle of small filled Julia sets has a compact forward-invariant Hubbard tree whose vertices and edge paths form a finite abstract tree with a native ideal marking. If the filled carrier is not locally connected, KKL pass to the locally connected rational-lamination model and lift the corresponding objects.*

This imports only KKL's native terminal object. It does not select the partial Cantor component package, retained critical returns, or protected prime ends of an intermediate Dudko virtual scale.

**Imported Theorem 2.4** (Kiwi monotone model and finite-orbit fibers)**.** *Let $P$ be a complex polynomial with connected Julia set and no irrationally indifferent periodic cycle. Kiwi's real-lamination model admits a continuous monotone semiconjugacy from $J(P)$ onto its locally connected model. For every periodic or preperiodic $z \in J(P)$, the complete semiconjugacy fiber over its image is $\{z\}$* [Kiw04, Theorem 2, Proposition 4.2, and Theorem 5.12].

The full-fiber statement uses more than the restricted-injectivity sentence in Theorem 5.12: Theorem 2 makes the dynamical fiber of every periodic or preperiodic point trivial, Proposition 4.2 forms the quotient $q$ by those fibers, and the proof of Theorem 5.12 writes the semiconjugacy as $h = \widehat{w}^{-1} \circ q$. This import does not identify that quotient with the "natural projection" named at KKL lines 542–543, nor does it prove KKL's carrier full-preimage, selected-sheet, or geometric small-Julia-continuum correspondences.

**Imported Theorem 2.5** (Douady–Hubbard external prime-end dynamics)**.** *For a connected degree-$d$ polynomial-like map, straightening and the external construction identify the induced external prime-end map with a degree-$d$ expanding circle map topologically conjugate to $t \mapsto dt \pmod 1$.*

This is the polynomial-like external construction of [DH85]. We use its dynamical conjugacy and prove full impression covariance directly in theorem 12.79. We do not infer exactness from the degree of an arbitrary circle covering.

**Imported Theorem 2.6** (KKL native ideal-interval pullback)**.** *In the parabolically bounded primitive Hubbard-tree setting of* [KKL26]*, let $\mathcal{T}' = f^{-1}(\mathcal{T})$ and let $\iota$ be the immersion. For a marked multi-interval $I$,*

$$\iota^* I = \iota^{-1}(I) \cup \bigcup_{\substack{X \in \pi_0(\mathcal{T}' \setminus \mathcal{T}) \\ X \text{ attaches to } \mathcal{T} \text{ at } I}} \overline{\operatorname{Cara} X}. \tag{2.2}$$

*On KKL's central limb range $\mathcal{L}_{[B,q-B-2]}$, parabolic boundedness gives precisely*

$$\mathcal{T}' \cap \mathcal{L}_{[B,q-B-2]} = \mathcal{T} \cap \mathcal{L}_{[B,q-B-2]},$$

*and hence no pulled-carrier excess meets the proof-used intervals in that range. This restricted equality, not a global equality $\mathcal{T}' = \mathcal{T}$, yields the proper interval action there.*

Equation (2.2) is KKL Proposition 6.5; the restricted central properness statement is Proposition 6.7. We do not transfer that local no-excess conclusion to a virtual hull.

**Imported Theorem 2.7** (Kahn finite $\psi$-ql pullback structure)**.** *For a finite fiber-product tower of a $\psi$-polynomial-like map, Kahn's pullback construction makes every immersion leg a homeomorphism on the pulled filled set, preserves the embedded polynomial-like germ, and gives commuting covering/immersion squares at every finite level* [Kah06]*. No local connectivity of the filled set is assumed.*

The locked project source is `.tmp_kl06_source/psiql2.tex`. Its SHA-256 digest is recorded in two consecutive blocks:

`E4EFCCEC65D8C9C2BDFE2AF8203B3D4`
`D63ECF992B40D302F56FBA78259547228`.

The proof-used clauses are lines 20–41, 60–77, 155–178, 188–265, 350–416. This import supplies the finite graph coordinate proved later; it supplies no KKL Wave incidence statement.

**Imported Theorem 2.8** (Native static and tame positive-entropy branches)**.** *Within KKL's full category of parabolically bounded prime primitive $\psi$-ql maps, the static amplification/tame/wave trichotomy and the subsequent tame amplification theorem hold with the source's declared thresholds.*

**Open Problem 2.9** (Certification of KKL's source-stated Wave theorem)**.** KKL Proposition 3.9 states the controllable-Wave amplification conclusion, but this project does not certify its v1 proof. The bounded-visits paragraph requires a strict marked-interval growth lemma not implied merely by the Perron–Frobenius inequality. The stationary paragraph additionally requires a retained final proper-path/measurable-lamination extraction and the degree-two correction proved in this manuscript. Thus the ledger node

`KKL_PARABOLIC_WAVE_AMPLIFICATION`

remains open pending both

`KKL_BOUNDED_VISITS_INTERVAL_GROWTH_REPAIR`,
`KKL_PARABOLIC_WAVE_STATIONARY_PROOF_REPAIR`.

The source statements do not assert that a partially invariant virtual carrier has KKL's full cycle, Hubbard tree, canonical WAD, complete interval action, or long-iterate wave itinerary.

*Remark* 2.10 (Dudko's proposal is an open input)*.* Dudko's Interpolation Problem 4.4 proposes partially invariant connected hulls of Cantor small Julia sets containing the critical orbit through an appropriate finite number of first returns [Dud25]. The source does not define the Cantor address space, the embedded component family, its analytic branch action, that stopping number, or a stopping interface. A finite terminal-tree list is therefore not imported as the full virtual package; a finite quotient additionally requires a future-observability proof. The source also does not construct the root component routing, complete proof-use inventory, prime-end marking, KKL interval action, or uniform width comparison required here. KKL's native Hubbard-tree inventory is not transferred without a typed component/access bridge preserving every proof-used branch and prime-end side. In particular, the checked sources do not construct a virtual marked

hyperbolic complement or pairs $(g_r^{\mathrm{vir}}, \iota_r^{\mathrm{vir}})$ for which the pulled marking refines the immersed marking and the proof-used intervals are proper or end-proper.

The decoration source does supply two actual univalent pullbacks of its separator $P$. Theorem 10.37 uses them to construct a direct-decoration rooted-end package. This is an internal consequence of the published puzzle maps, not an import of Dudko's intended virtual package: the end fibers are not identified as small Julia sets and the second satellite marking transfer remains open.

Neutral pseudo-Siegel estimates are imported only in the settings proved in [DL22, DLL25]. Feigenbaum and high-type satellite results [DL23, CS15] are recovery tests, not substitutes for the missing mixed virtual transition.

## 3. Computational math and fractal atlas

This section gives a reproducible visual index to the equations, numerical experiments, exact counterexamples, and proof ledger used throughout the manuscript. The first four plates come from direct iteration of

$$f_c(z) = z^2 + c.$$

The remaining plates are generated from displayed formulas, exact certificates, checked JSON results, or the machine-audited dependency graph. The scripts are `figures/generate_fractal_plates.py` and `figures/generate_math_atlas.py`; exact parameters, array sizes, iteration cutoffs, source paths, and SHA-256 hashes are recorded in `figures/fractal_render_manifest.json` and `figures/math_atlas_manifest.json`.

*Warning* 3.1 (All plates are illustrations and audit views, not proof evidence). Dark pixels mean only that a sampled orbit did not escape before its printed cutoff. Smooth escape-time color, fitted exponents, finite spectra, and finite dependency layouts are visualization coordinates. No apparent self-similarity, landing, boundary regularity, numerical convergence, or plotted separation is used to prove local connectivity, a priori bounds, coercivity, rigidity, or any ledger obligation. Open and conditional statements remain open and conditional in every plate.

### 3.1. **Parameter and dynamical geometry.**

For the period-three panel below, the upper-half-plane rabbit center is the nonreal root with positive imaginary part of

$$c^3 + 2c^2 + c + 1 = 0, \qquad f_c^3(0) = c(c^3 + 2c^2 + c + 1).$$

For the neutral panel, put

$$\theta = \frac{\sqrt{5}-1}{2}, \qquad \lambda = e^{2\pi i\theta}, \qquad \alpha = \frac{\lambda}{2}, \qquad c_\theta = \alpha - \alpha^2. \tag{2.1}$$

Then $f_{c_\theta}(\alpha) = \alpha$ and $f'_{c_\theta}(\alpha) = \lambda$. The Misiurewicz example is exactly $c = i$, for which

$$0 \longmapsto i \longmapsto -1 + i \longmapsto -i \longmapsto -1 + i.$$

The Feigenbaum panel uses $c \approx -1.4011551890920506$.

### 3.2. **Quantitative dynamics and scaling laws.**

### 3.3. **Discrete topology, addresses, and attachment accounting.**

### 3.4. **Exact analytic test models.**

### 3.5. **Trace response and pullback geometry.**

### 3.6. **Machine-audited proof topology.**

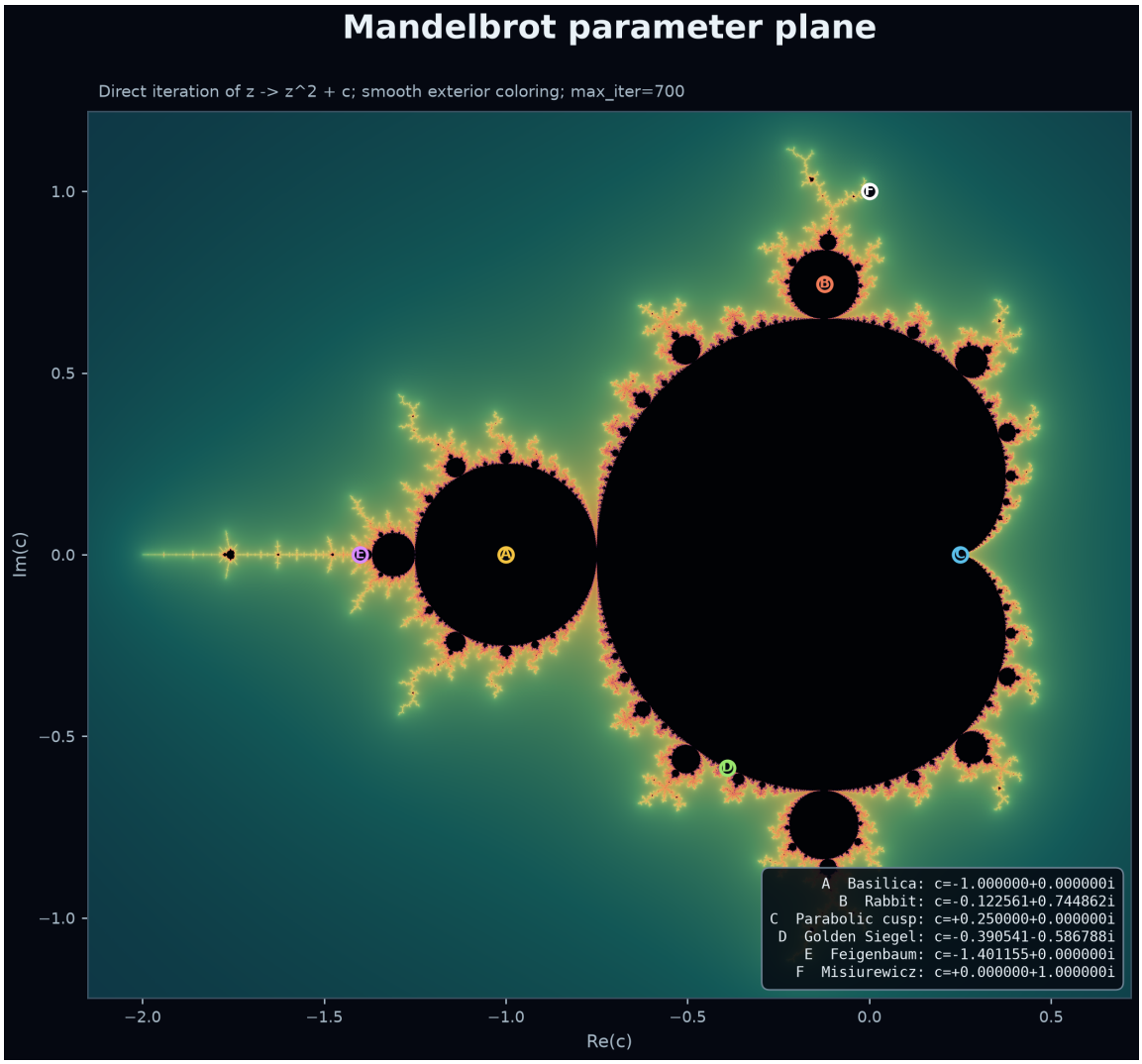



FIGURE 1. Finite escape-time rendering of the quadratic parameter plane. Markers A–F are the parameters used in figure 3: Basilica, Douady rabbit, parabolic cusp, golden-mean neutral fixed point, the numerical Feigenbaum parameter, and the Misiurewicz parameter $c = i$. This is an orientation map, not a certification of boundary membership or local connectivity.

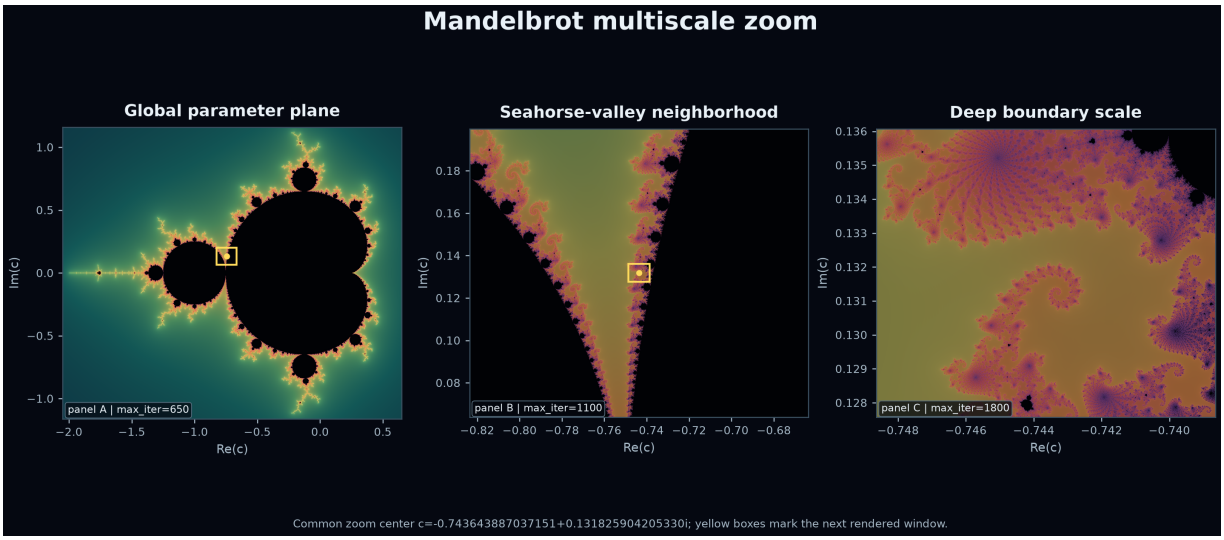



FIGURE 2. Three independently iterated parameter-plane windows centered at $c = -0.743643887037151 + 0.131825904205330\,i$. Each yellow rectangle is the exact next window. Visible recurrence motivates multiscale analysis but proves neither a renormalization theorem nor a transition estimate.

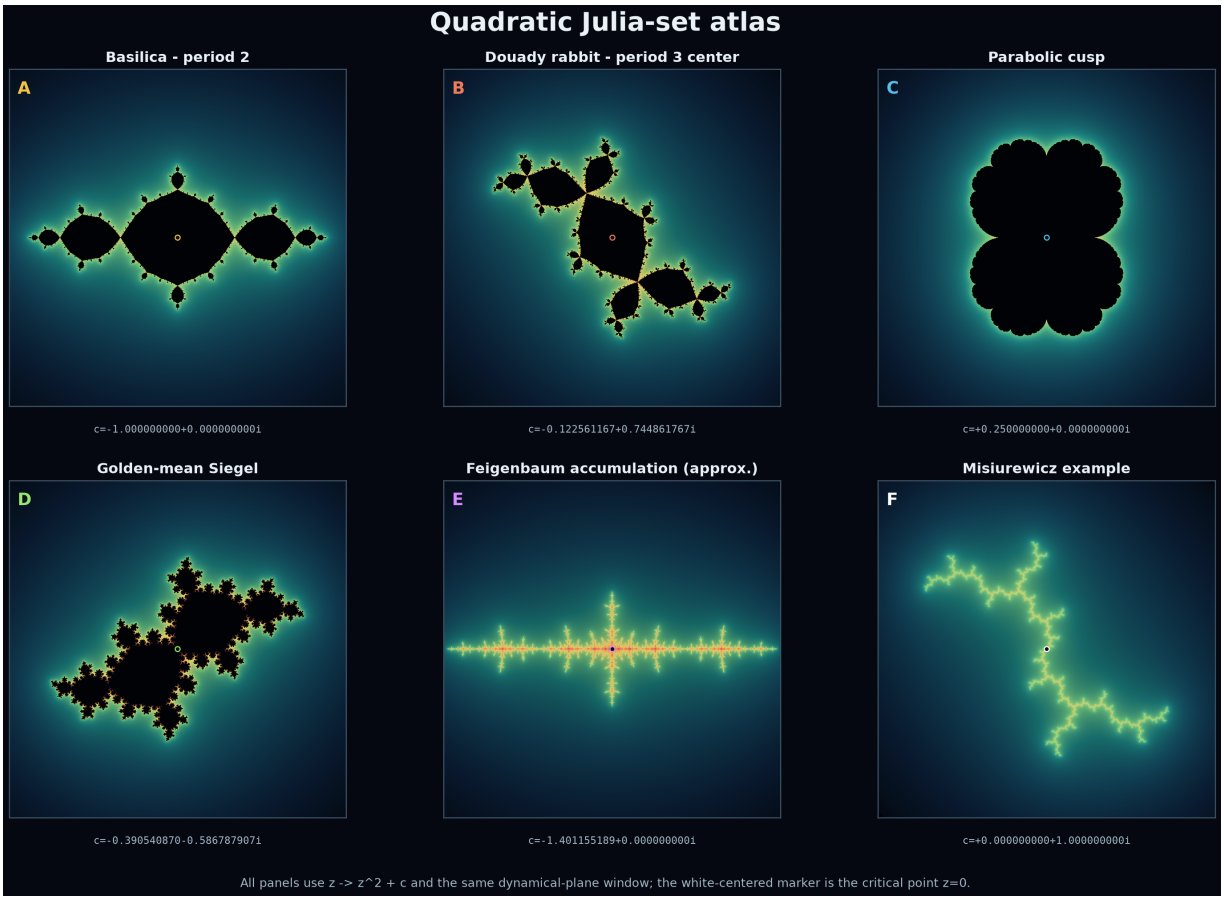



FIGURE 3. Finite escape-time dynamical-plane atlas in one common window. The dark set is the nonescaped finite-iteration approximation and the white-centered marker is the critical point 0. These motifs are not claimed to exhaust the KKL family classes.

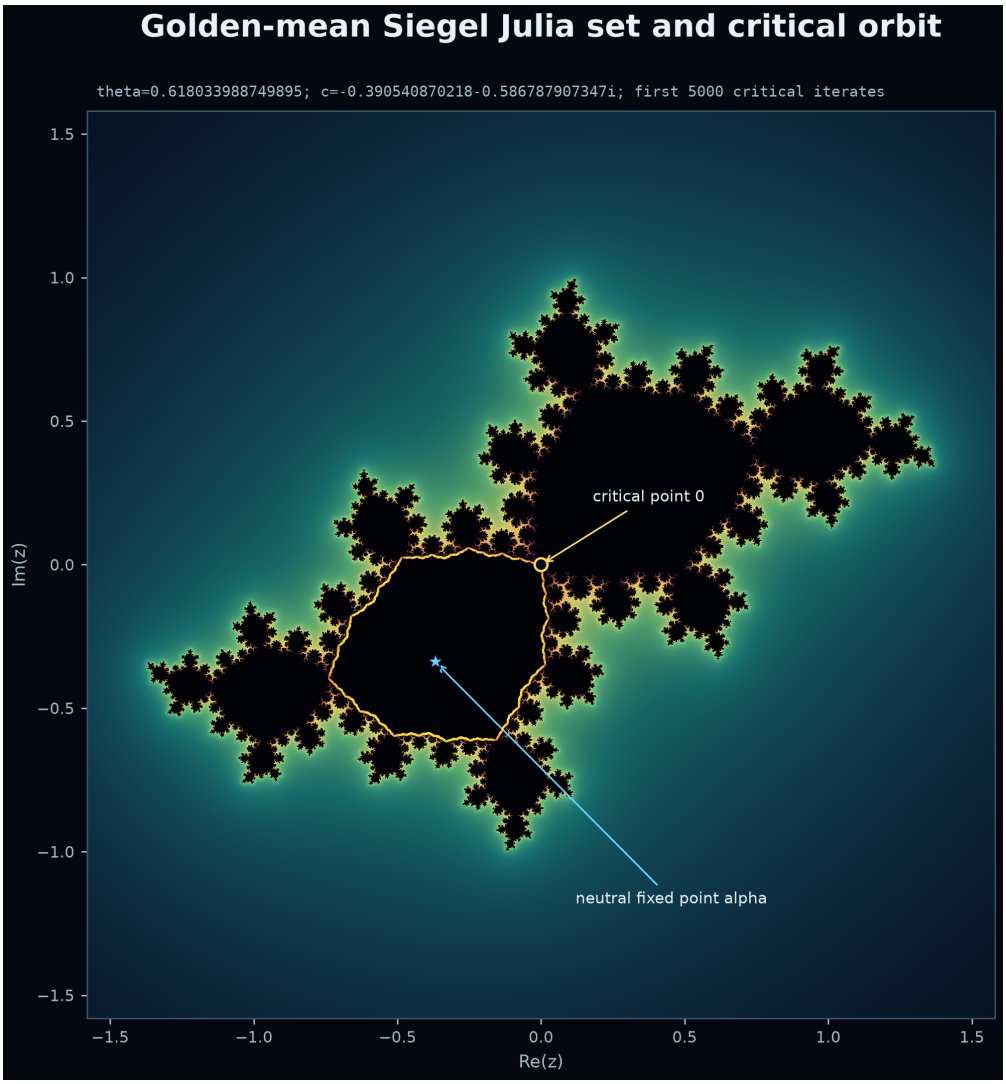



FIGURE 4. Golden-mean neutral example from (2.1). Colored points are the first 5000 directly computed critical iterates; the star is $\alpha$. The plot motivates recording critical-orbit and access data, but does not prove orbit density or boundary regularity.

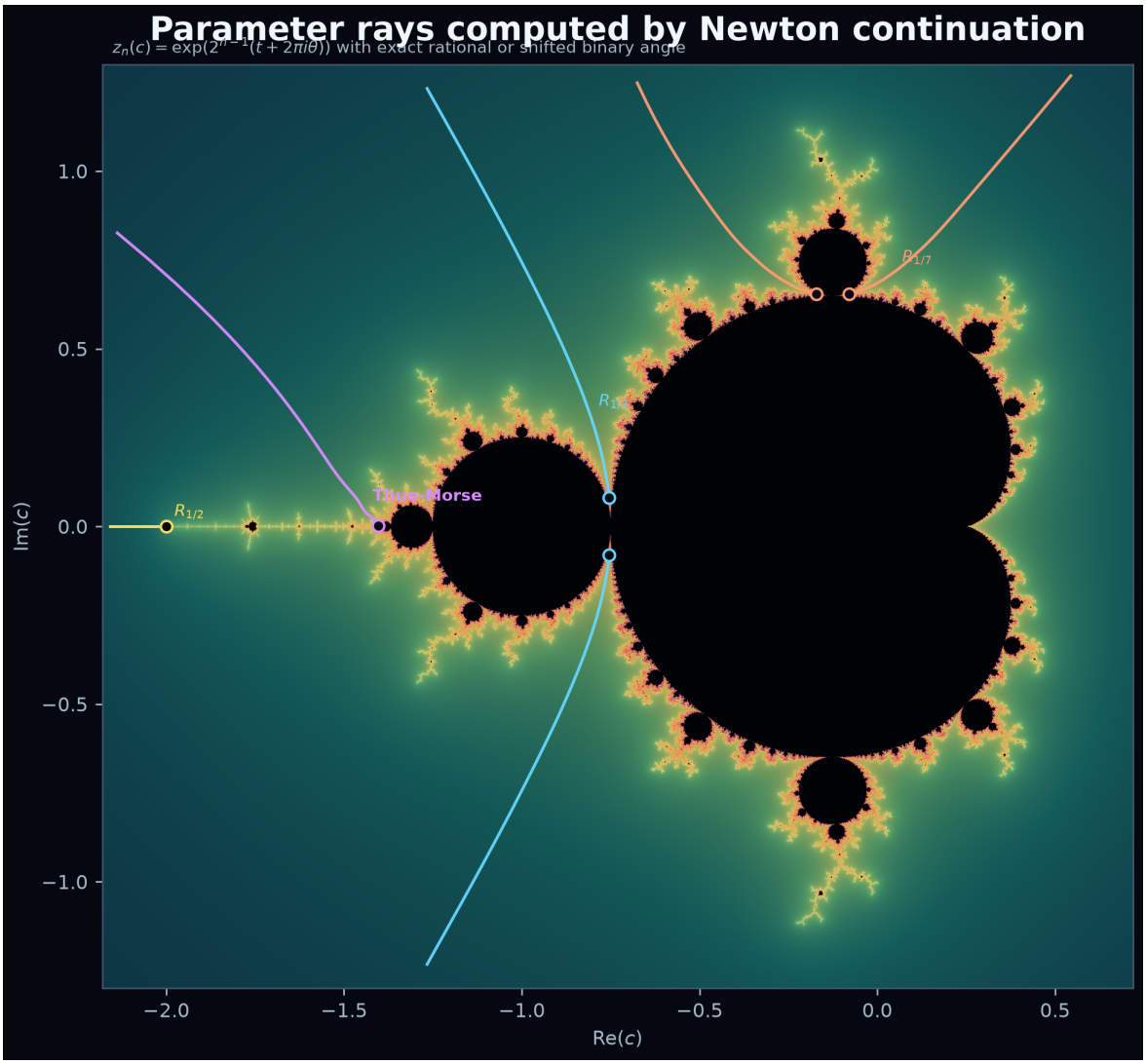



FIGURE 5. Parameter external rays computed by Newton continuation from $z_n(c) = \exp(2^{n-1}(t + 2\pi i\vartheta))$, using exact rational angles or the shifted Thue–Morse binary angle. The displayed endpoints and paths come from the checked ray-experiment data; finite traces are not landing proofs.

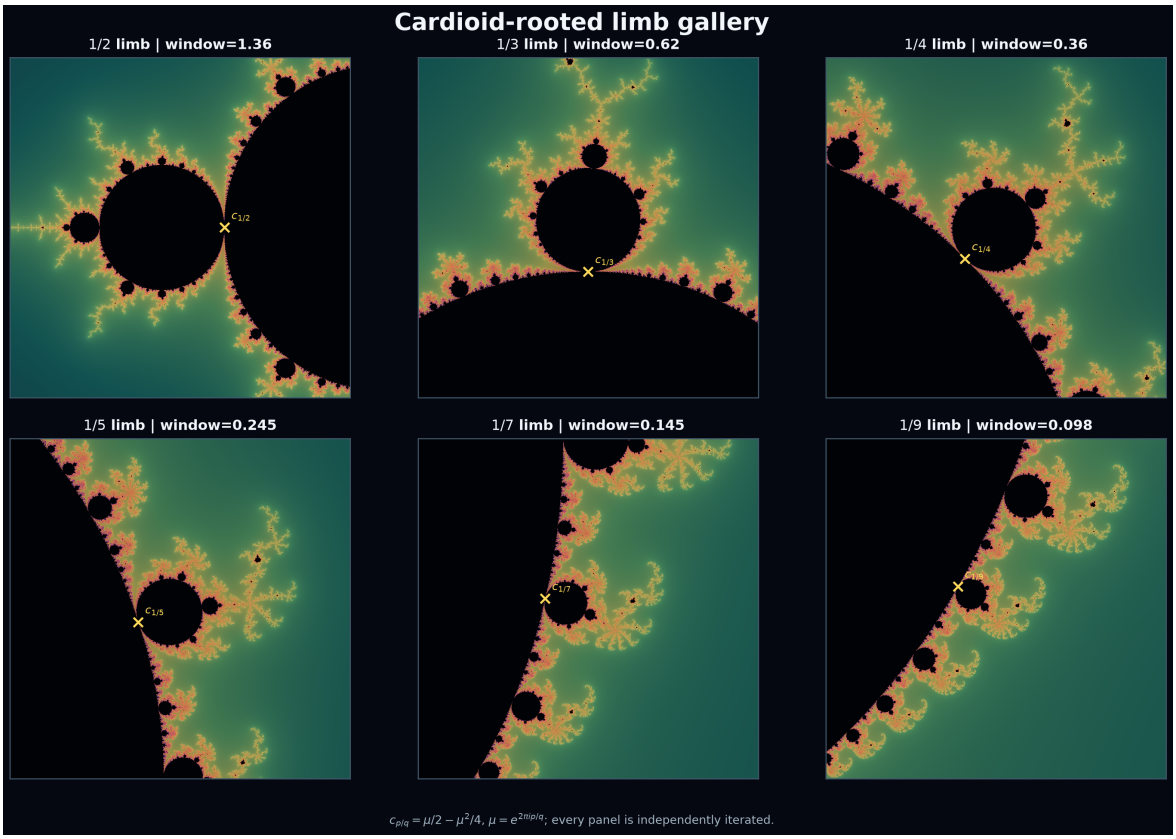



FIGURE 6. Independently iterated windows about cardioid roots $c_{p/q} = \mu/2 - \mu^2/4$, $\mu = e^{2\pi ip/q}$, for $q = 2, 3, 4, 5, 7, 9$. Each yellow cross is computed from the exact formula; the shrinking windows are printed in the plate.

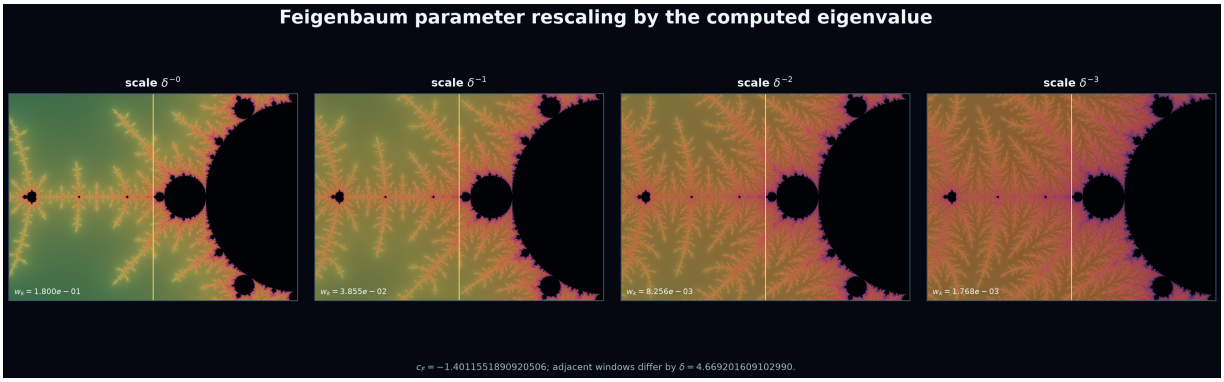



FIGURE 7. Four independently rendered windows centered at the computed Feigenbaum parameter. Adjacent widths differ by the computed unstable eigenvalue $\delta = 4.6692015955\ldots$, so the visible comparison uses the renormalization scale rather than an arbitrary zoom factor.

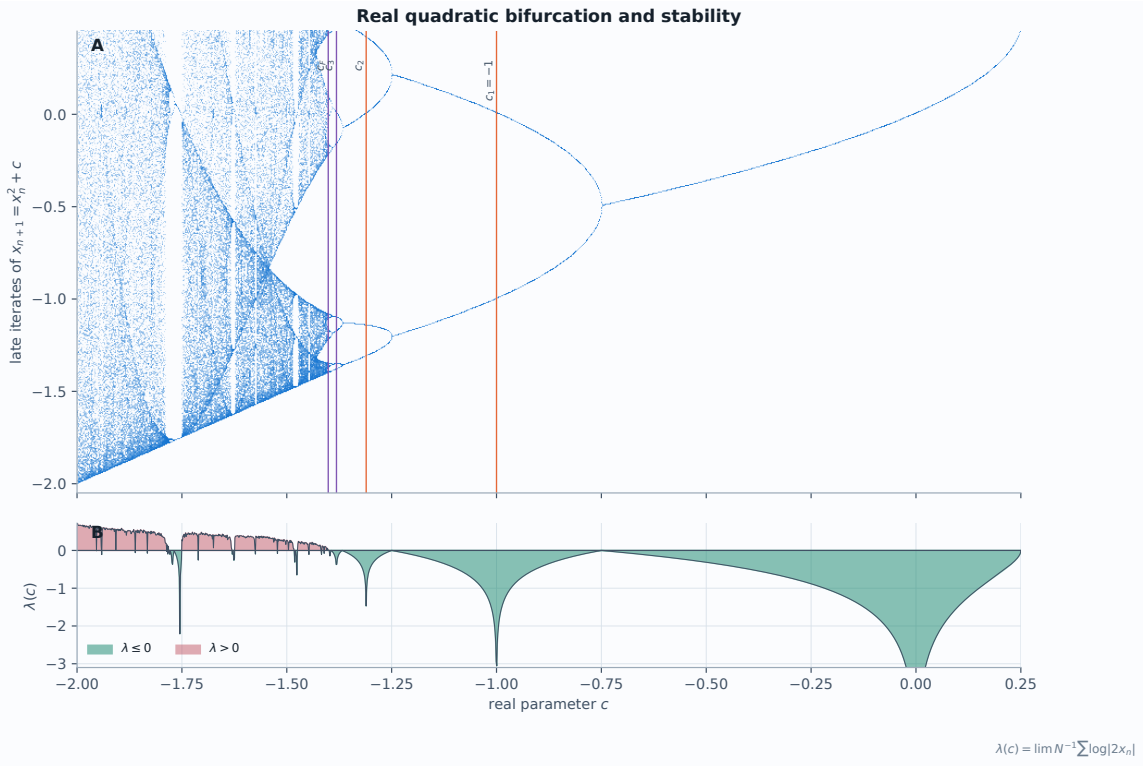



FIGURE 8. Direct iteration of $x_{n+1} = x_n^2 + c$ on the real slice. The upper panel retains late iterates and marks superstable parameters and $c_F$; the lower panel computes $\lambda(c) = N^{-1} \sum_n \log|2x_n|$. Both are finite numerical summaries.

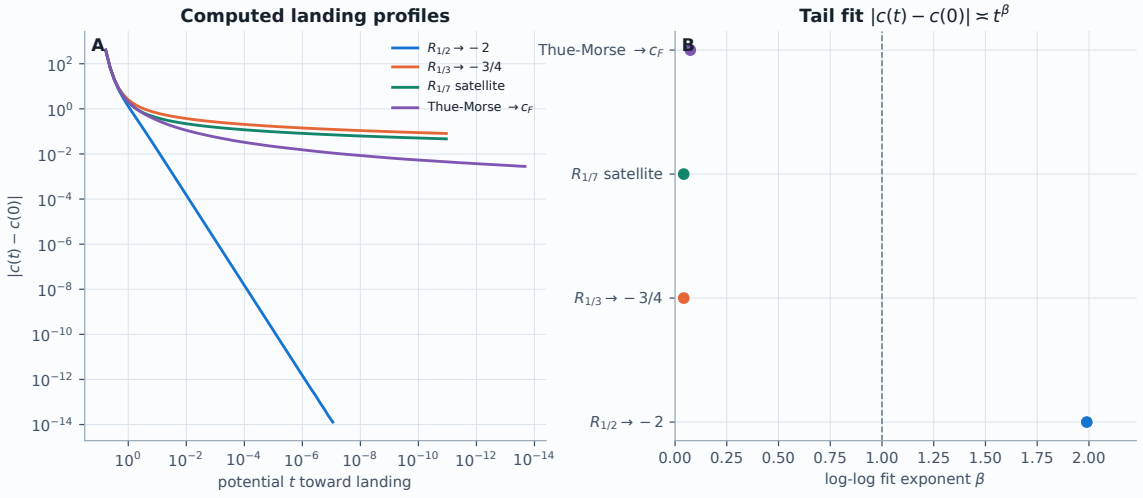



FIGURE 9. Landing profiles and fitted tail exponents from the checked ray-experiment data. The fits compare the rapid Misiurewicz direction with the slower parabolic, satellite, and Feigenbaum directions; they measure this computation only and assert no universal landing law.

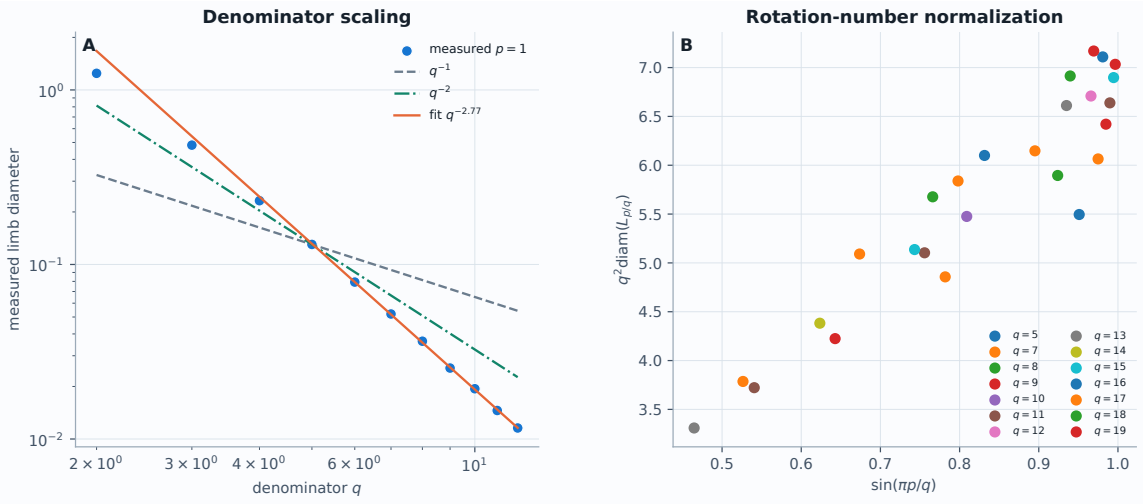



FIGURE 10. Measured limb diameters from the checked limb-experiment data, compared with $q^{-1}$, $q^{-2}$, and a log–log fit. The rotation-number normalization uses $q^2 \operatorname{diam}(L_{p/q})$ against $\sin(\pi p/q)$. These values inherit the numerical bracketing caveats of the source experiment.

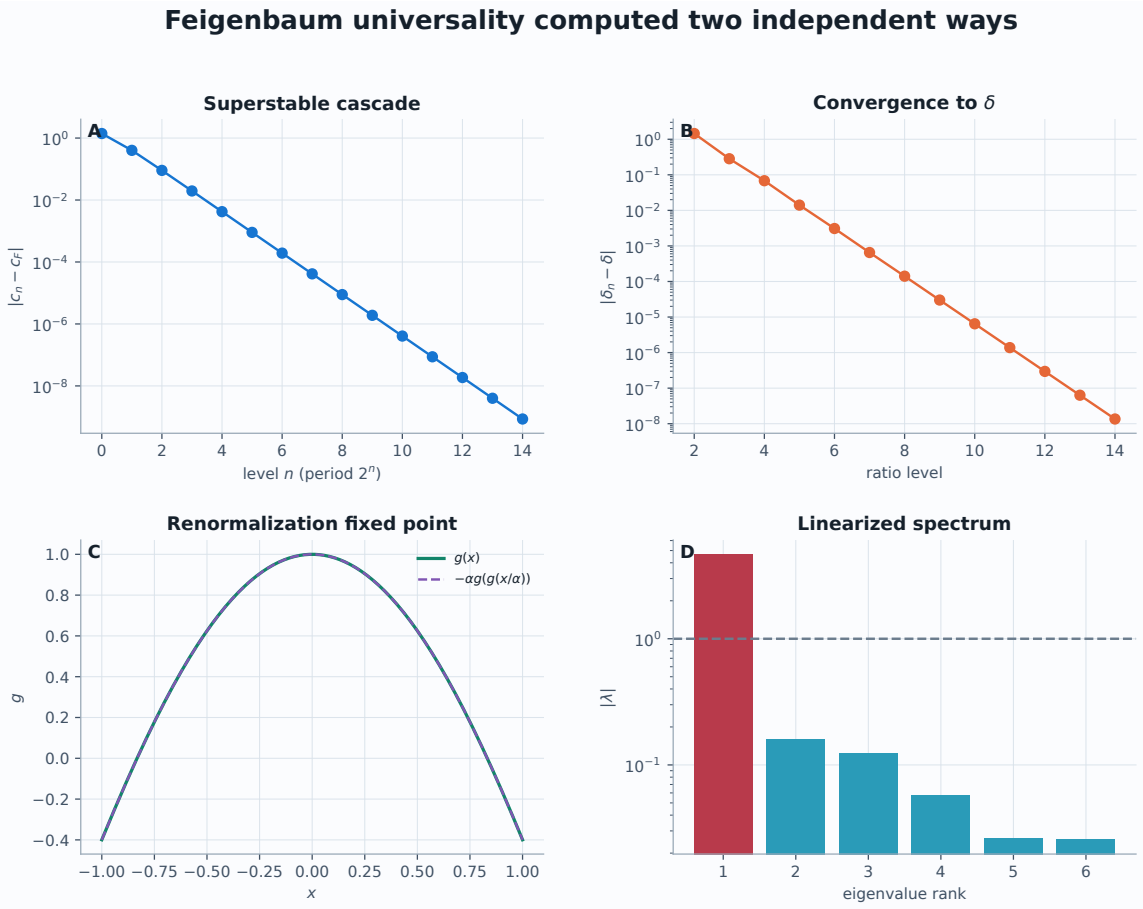



FIGURE 11. One source file, four independent numerical checks: superstable parameter convergence, convergence of $\delta_n$, the fixed-point identity $g(x) = -\alpha g(g(x/\alpha))$, and the spectrum of the linearized doubling operator. The residual and source data are recorded in the atlas manifest.

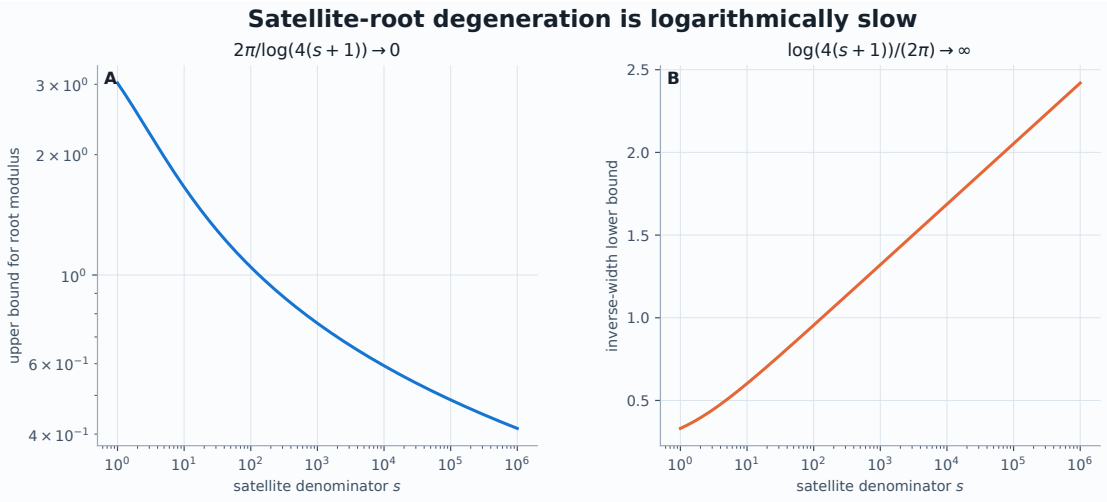



FIGURE 12. The source-locked satellite-root estimate $m_s \le 2\pi/\log(4(s+1))$ and its inverse-width consequence $m_s^{-1} \ge \log(4(s+1))/(2\pi)$. The logarithmic degeneration is exact for the displayed bound; the plot is not a theorem beyond that formula.

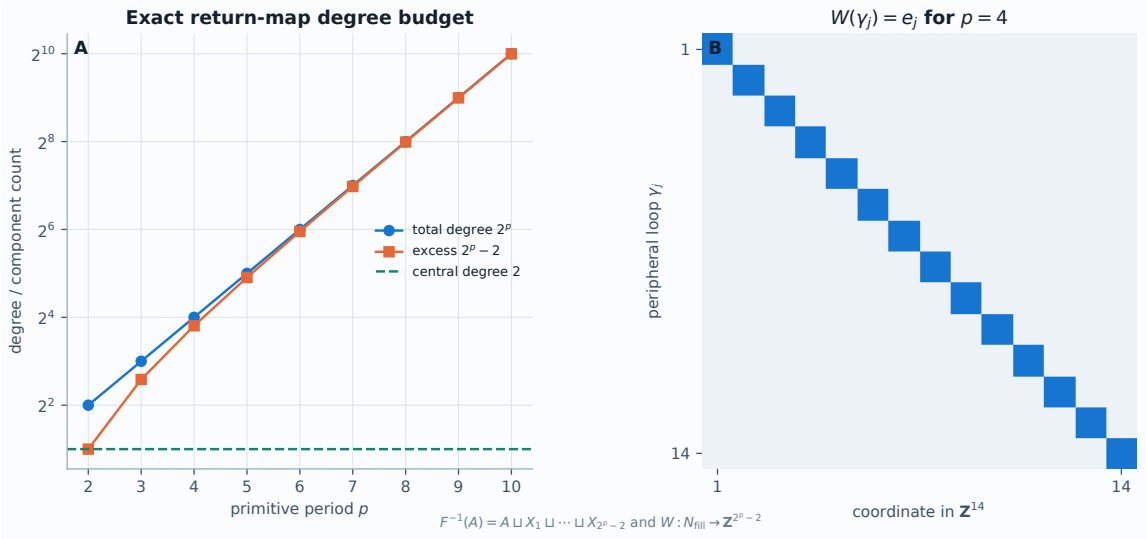



FIGURE 13. The exact return-map budget $F^{-1}(A) = A \sqcup X_1 \sqcup \cdots \sqcup X_{2^p-2}$: total degree $2^p$, central degree two, and $2^p - 2$ excess sheets. At $p = 4$, the winding coordinates display $W(\gamma_j) = e_j$ in $\mathbb{Z}^{14}$.

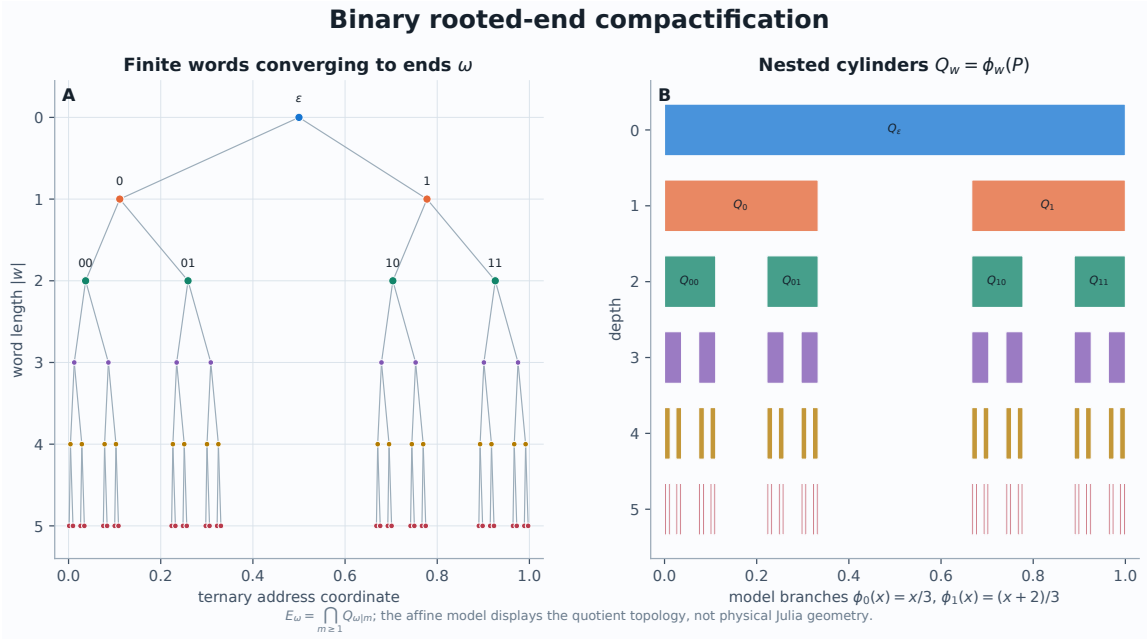



FIGURE 14. A binary rooted-end model. Finite words index nested cylinders $Q_w$; infinite words give $E_\omega = \bigcap_{m\geq 1} Q_{\omega|m}$. The affine Cantor construction shows the component quotient only: physical end fibers may be nondegenerate continua and are not identified with points or Julia sets.

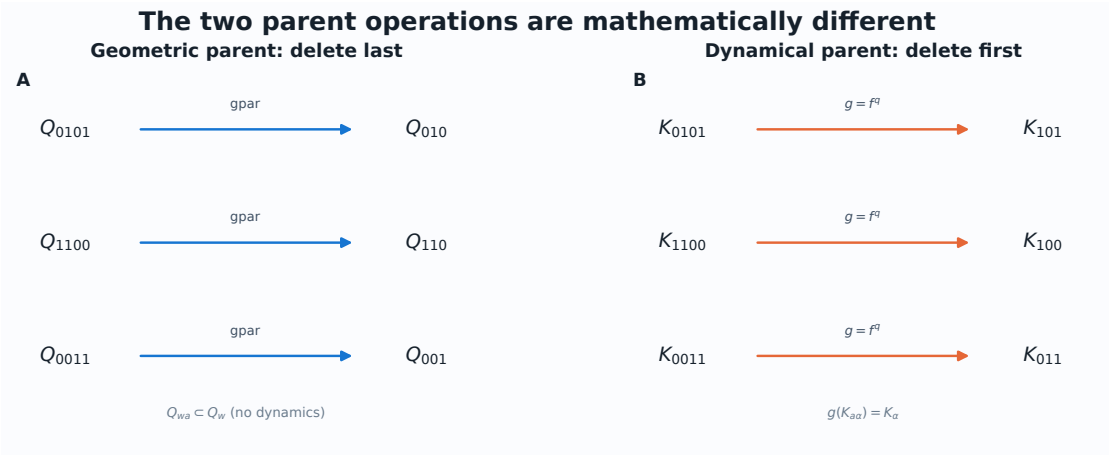



FIGURE 15. The two parent operations that must not be conflated. Geometric nesting deletes the last symbol, $\mathrm{gpar}(wa) = w$, while the dynamical shift deletes the first, $g(K_{a\alpha}) = K_\alpha$.

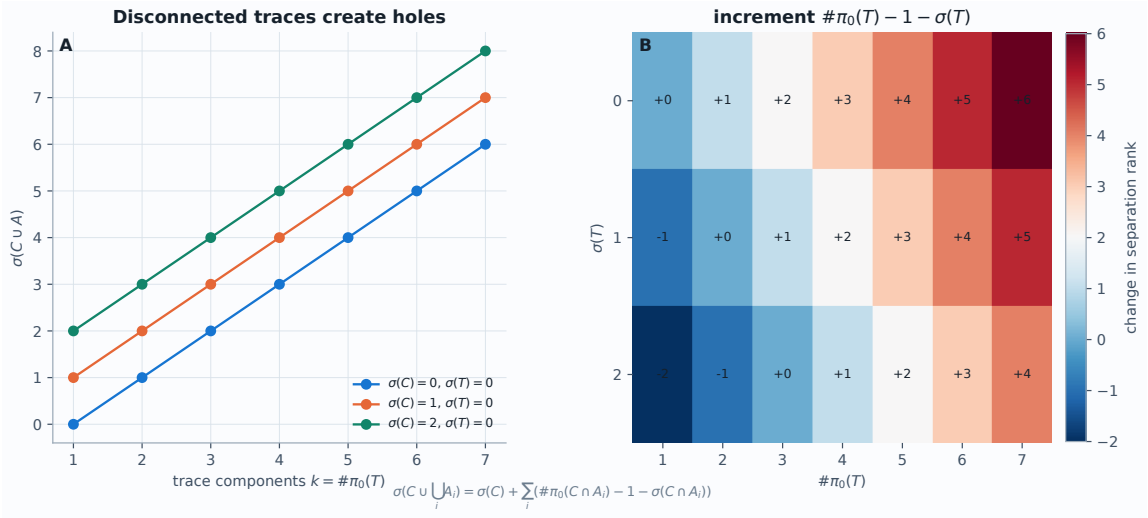



FIGURE 16. Exact attachment accounting for separation rank:

$$\sigma\left(C \cup \bigcup_i A_i\right) = \sigma(C) + \sum_i (\#\pi_0(C \cap A_i) - 1 - \sigma(C \cap A_i)).$$

The line plot and heat map expose when a disconnected trace creates a new complementary component and when trace separation consumes it.

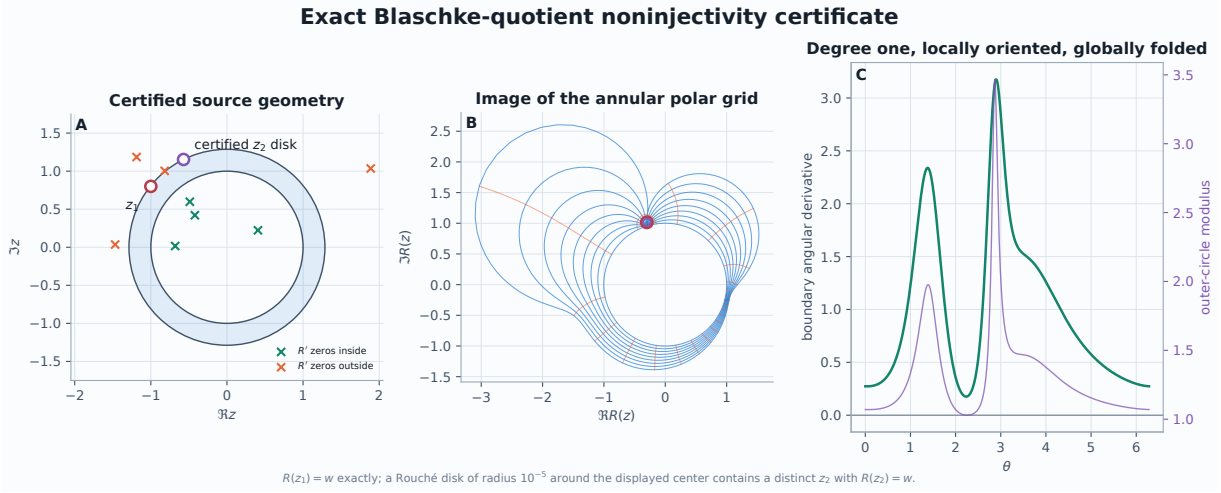



FIGURE 17. Exact Blaschke-quotient noninjectivity certificate from Paper 1. The source annulus, certified critical-point locations, annular polar-grid image, boundary angular derivative, and outer-circle modulus are all generated from the printed rational data. A Rouché disk of radius $10^{-5}$ certifies the second preimage of the displayed value.

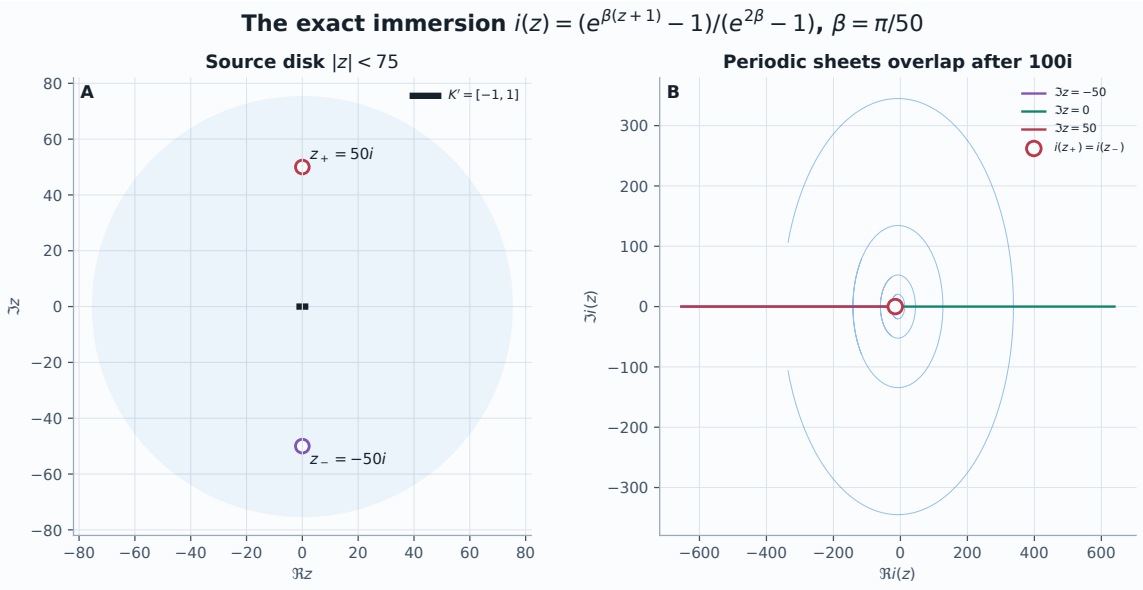



FIGURE 18. The exact immersion $i(z) = (e^{\beta(z+1)} - 1)/(e^{2\beta} - 1)$, $\beta = \pi/50$. Its identity $i(z + 100i) = i(z)$ produces the explicit collision $i(50i) = i(-50i)$, visible as overlapping source sheets in the target.

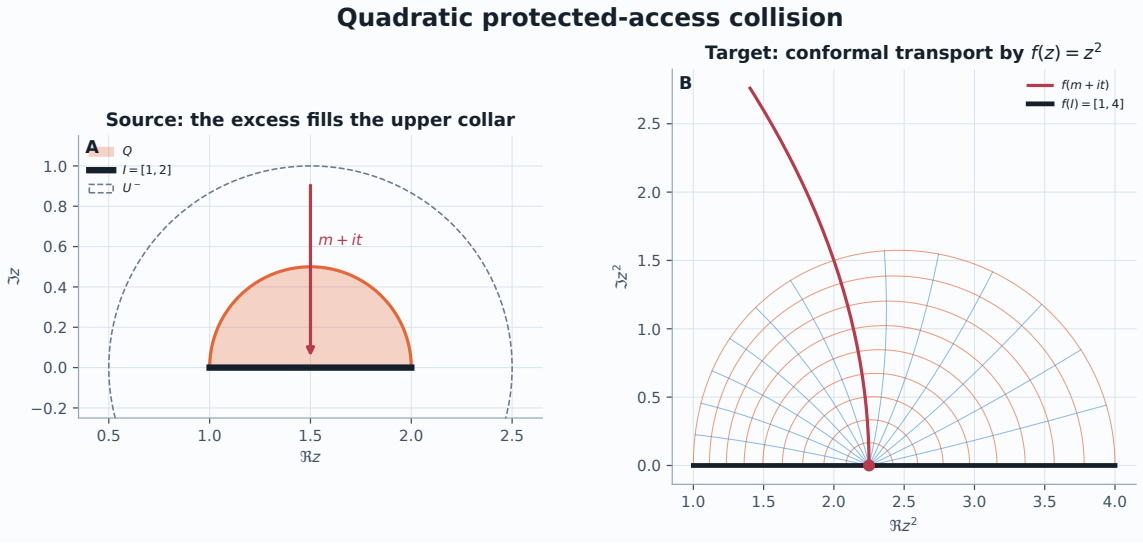



FIGURE 19. The exact quadratic access-collision model from Paper 3. The upper half-disk $Q = \{|z - 3/2| \le 1/2,\ \mathrm{Im}\, z \ge 0\}$ meets the protected interval $I = [1, 2]$, and conformal transport by $f(z) = z^2$ carries the full attachment trace and the access $3/2 + it$ into the target geometry.

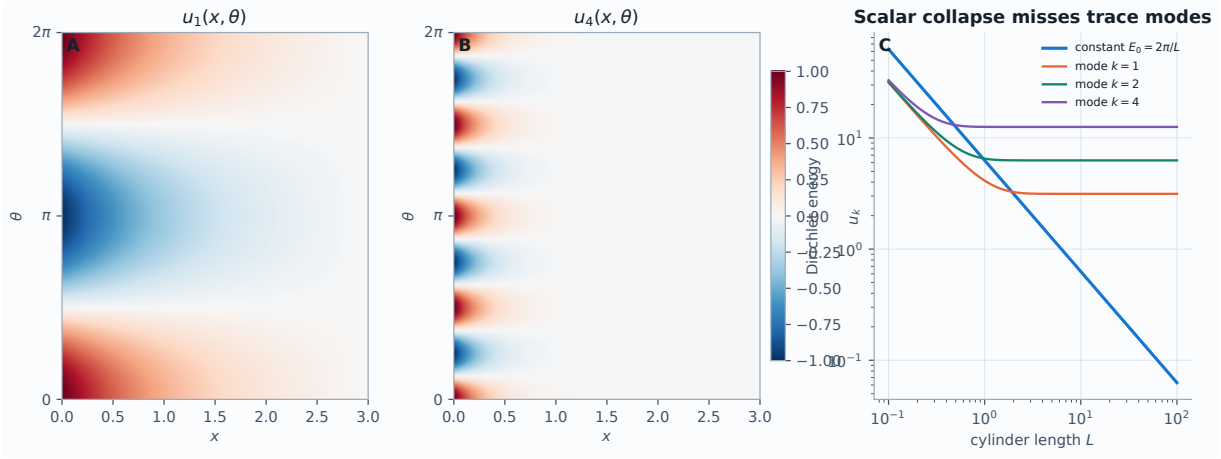



FIGURE 20. Cylinder trace modes $u_k(x,\theta) = \sinh(k(L-x))\cos(k\theta)/\sinh(kL)$. Their energies $E_k = \pi k \coth(kL)$ do not collapse with the constant-mode energy $E_0 = 2\pi/L$; hence a scalar width loses boundary-trace information.

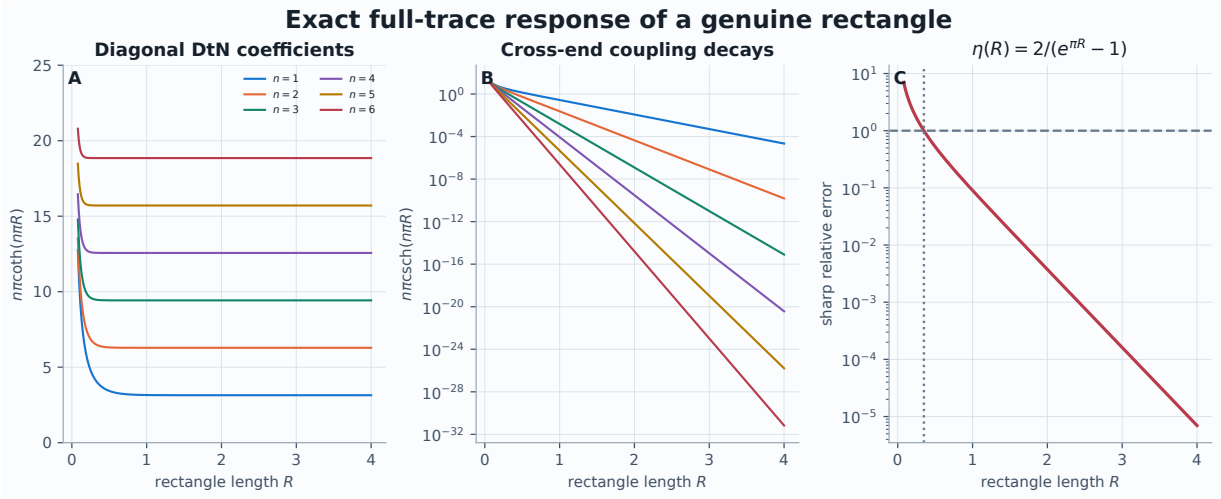



FIGURE 21. Exact Dirichlet-to-Neumann response of a rectangle. The diagonal coefficients are $n\pi\coth(n\pi R)$, cross-end coupling is $n\pi\operatorname{csch}(n\pi R)$, and the sharp relative error is $\eta(R) = 2/(e^{\pi R} - 1)$.

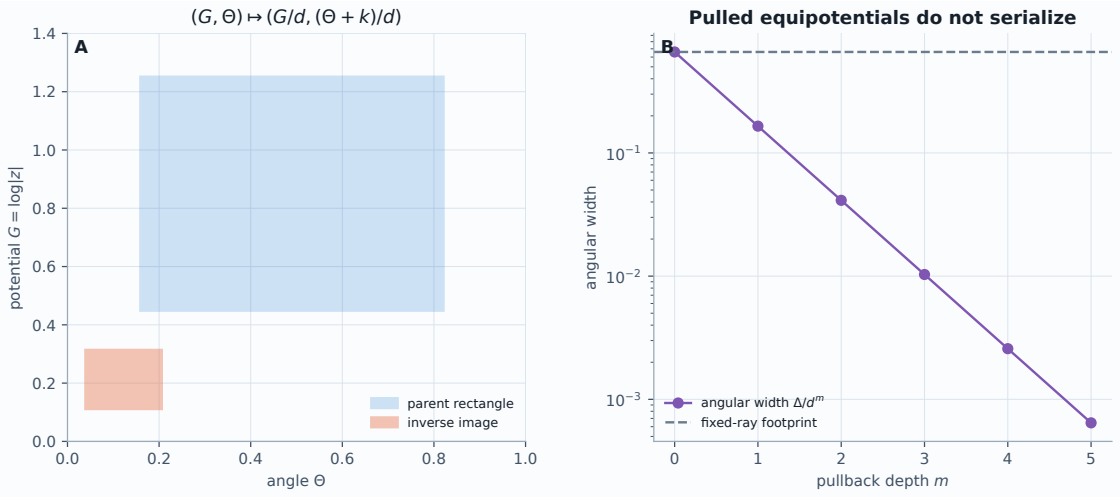



FIGURE 22. Böttcher log-polar pullback $(G,\Theta)\mapsto(G/d,(\Theta+k)/d)$. Angular width contracts as $\Delta/d^m$, while a fixed-ray footprint does not; the mismatch is the geometric obstruction to naive serialization.

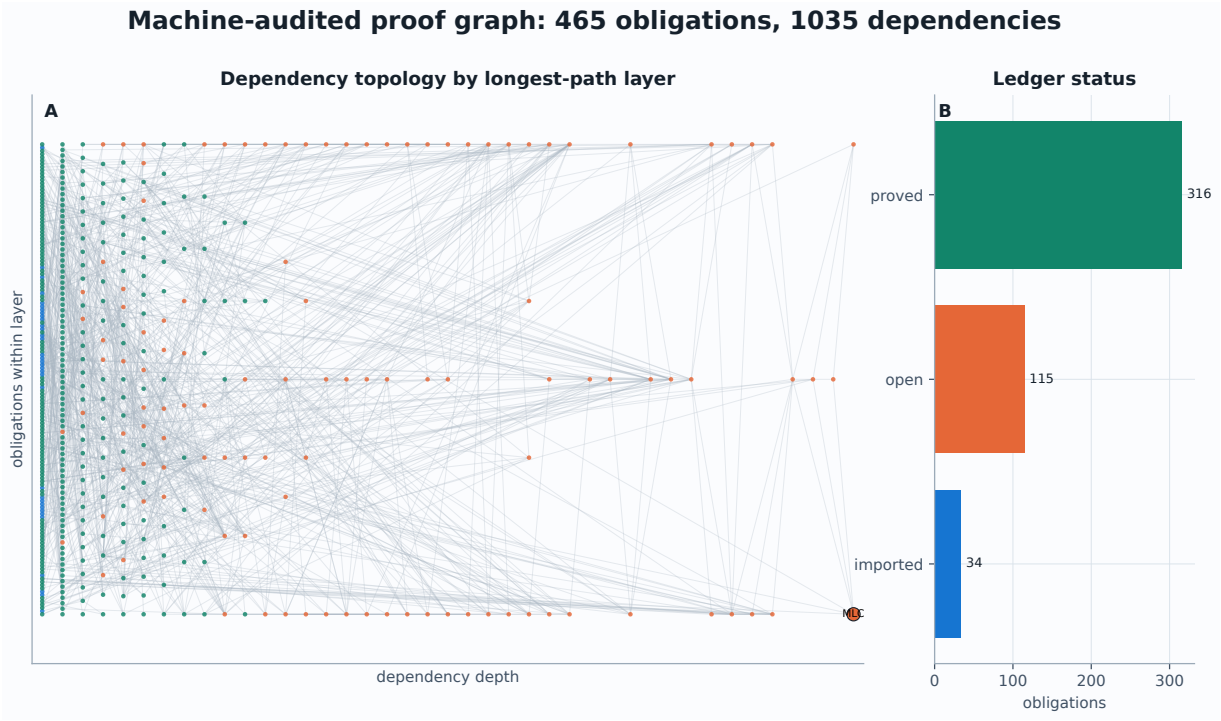



FIGURE 23. The complete machine-audited dependency DAG: 465 obligations and 1035 directed dependencies, layered by longest path and colored by ledger status. The status bars summarize 316 proved, 115 open, and 34 imported nodes at generation time. This image is an index to the ledger; the JSON and its audit script remain authoritative.

## 4. Marked state, the satellite obstruction, and backward records

### 4.1. Why ordinary root modulus cannot be the unified coordinate.

**Proved Theorem 4.1** (Unmarked satellite obstruction)**.** *There do not exist a bounded scalar functional $D$, a universal constant $C$, and an assignment of $D$ to every satellite quadratic-like root such that*

$$\left| D - \frac{1}{\operatorname{mod}(A)} \right| \le C \tag{3.1}$$

*for all satellite relative periods.*

*Proof.* By imported theorem 2.2,

$$\frac{1}{\operatorname{mod}(A)} > \frac{\log(4(s+1))}{2\pi}.$$

The right-hand side tends to $+\infty$ as $s \to \infty$. If $D$ were bounded and (3.1) held with universal $C$, then the inverse modulus would be bounded by $\sup D + C$, a contradiction. □

*Remark* 4.2. The conclusion does not prohibit a regularized marked functional. It says that the marked surface and curve family must genuinely change; renaming the ordinary root annulus cannot solve the satellite problem.

### 4.2. Forward growth is not a contradiction.

A lower recurrence toward deeper levels,

$$L_{n+k} \ge \lambda L_n - C,$$

is compatible with an unbounded sequence. The direction needed for a tower closure is the reverse one.

**Proved Lemma 4.3** (Backward-record lemma)**.** *Let $(L_n)_{n\ge 0}$ be a nonnegative sequence. Suppose there are constants*

$$D_* \ge 0, \qquad \lambda > 1, \qquad C \ge 0$$

*such that whenever $L_n > D_*$, there exists $p(n) < n$ satisfying*

$$L_{p(n)} \ge \lambda L_n - C. \tag{3.2}$$

*Then*

$$\sup_{n\ge 0} L_n \le \max\{D_*, L_0\} + \frac{C}{\lambda - 1}. \tag{3.3}$$

*Proof.* Fix $n$. If $L_n \le D_*$, there is nothing to prove. Otherwise iterate $p$ while the current value exceeds $D_*$. Since the index strictly decreases, the chain terminates at some $m$ with either $m = 0$ or $L_m \le D_*$. Rearranging (3.2) at every step gives

$$L_n \le \lambda^{-r} L_m + C \sum_{j=1}^{r} \lambda^{-j} \le \max\{D_*, L_0\} + \frac{C}{\lambda - 1},$$

where $r$ is the number of backward steps. □

### 4.3. The calibrated state.

For a complete entropy-clock block $B$, the provisional homogeneous geometric state is

$$\mathscr{R}_B^{\mathrm{aug}} = \left( \frac{Q_B}{r_B}, \frac{\Lambda_B}{r_B}, \frac{\mathbf{V}_B}{r_B}, \frac{W_B^{\mathrm{std}}}{r_B}, \frac{\boldsymbol{\Xi}_B}{r_B} \right). \tag{3.4}$$

The components respectively record retained active width, a bounded-terminal response form, future-required KKL Value moments, standard corridor width, and any nonregularized zero-entropy shear moments. Exact counters, stack words, endpoint frames, marking density, and Frobenius flags are side data, not positive homogeneous coordinates.

**Open Problem 4.4** (Marked backward transition theorem)**.** Construct a potential

$$L_B = L_{t_B}(\mathscr{R}_B^{\mathrm{aug}}; \sigma_B)$$

whose constants are uniform over all admissible side data and such that high potential at a deep checkpoint implies (3.2) at a preceding checkpoint.

Once open problem 4.4 is proved, lemma 4.3 supplies the global scalar closure. It does not supply the construction, calibration, or coercivity of the state.

## 5. Root-pullback virtual carriers

Fix

$$f(z) = z^2 + c_0$$

and a finite rooted combinatorial tree $\mathscr{T}$, with vertex depth $d(v)$.

**Definition 5.1** (Required package)**.** At a vertex $v$, a required package is a nonempty compact set

$$S_v \subset \mathbb{C}$$

containing every small Julia set, orbit point, dam, fjord boundary, or terminal-fan anchor that the selected carrier must retain at that stage. The package need not be connected.

**Definition 5.2** (Root component-routing certificate)**.** Let $H_r$ be a nondegenerate full continuum. A root component-routing certificate consists of components

$$H_v \in \pi_0\big(f^{-d(v)}(H_r)\big)$$

such that:

(RC1) $S_v \subset H_v$ for every $v$;
(RC2) if $u$ is a child of $v$, then $f(H_u) = H_v$;
(RC3) vertices intended to represent distinct same-depth sheets are assigned distinct components.

**Proved Theorem 5.3** (Root-pullback forest equivalence)**.** *Every certificate in definition 5.2 determines a unique finite saturated inverse carrier forest. Its carriers are nondegenerate full continua, it contains every required package, and its selected same-depth sheets are disjoint.*

*Conversely, every finite saturated carrier forest rooted at $H_r$ is the routed component family of the iterated pullbacks of $H_r$.*

*Proof.* A component of the polynomial pullback of a nondegenerate full continuum is again a nondegenerate full continuum. Thus every $H_v$ has the required topological type and contains $S_v$. If $u$ is a child of $v$, the routing condition makes $H_u$ a component of $f^{-1}(H_v)$; hence the family is pullback-component saturated. Distinct components of the same closed pullback set are disjoint.

Uniqueness follows because a nonempty package contained in one pullback component selects that component. Conversely, induction on depth in a saturated forest shows that each $H_v$ is a component of $f^{-d(v)}(H_r)$, and the edge relation supplies the routing condition. □

*Warning* 5.4 (Root-shadow inclusion is weaker)*.* The inclusions

$$f^{d(v)}(S_v) \subset H_r$$

do not imply $S_v \subset H_v$ for one component $H_v$. Different parts of a disconnected package may occupy different pullback components.

5.1. **Critical-value compatibility.**

**Proved Proposition 5.5** (Critical-value merger criterion)**.** *For a nondegenerate full continuum $K$,*

$$f^{-1}(K) \text{ has } \begin{cases} \text{one component of degree } 2, & c_0 \in K, \\ \text{two components of degree } 1, & c_0 \notin K. \end{cases} \tag{4.1}$$

*Consequently a selected carrier edge $H_u \to H_v$ is degree one exactly when $c_0 \notin H_v$.*

*Proof.* Away from the critical value, $f$ is an unbranched double cover, and the preimage of the full continuum splits into the two inverse-branch components. If the target contains $c_0$, the two sheets meet at the critical point 0, giving one connected degree-two component. □

**Criterion 5.6** (Degree-one fast-forward corridor)**.** A chain

$$H_N \longrightarrow H_{N-1} \longrightarrow \cdots \longrightarrow H_0$$

is degree one at every edge if and only if

$$c_0 \notin H_j \qquad (0 \le j < N). \tag{4.2}$$

A required package containing $c_0$ therefore cannot be assigned to the target of a claimed degree-one central step.

5.2. **Collision monotonicity.**

**Proved Proposition 5.7** (Root enlargement only merges classes)**.** *If $H_r \subset \widehat{H}_r$ are full continua, then every component of $f^{-n}(H_r)$ is contained in one component of $f^{-n}(\widehat{H}_r)$. Hence enlarging the root can only coarsen the component partition of required packages.*

*Proof.* The inclusion $f^{-n}(H_r) \subset f^{-n}(\widehat{H}_r)$ puts every connected component of the smaller closed set into one connected component of the larger set. □

**Open Problem 5.8** (Virtual root-carrier separation)**.** For every virtual-Molecule block through the complete Wave stopping horizon, construct one full root continuum for which:

(VRC1) every proof-used object belongs to a declared required package;
(VRC2) every whole package lies in one selected iterated-pullback component;
(VRC3) the selected components realize the prescribed parent routing;
(VRC4) all Wave and terminal-fan sheets remain distinct after every future-required root object has been included;
(VRC5) the critical value occurs exactly in the declared quadratic windows;
(VRC6) the components stay in the admissible localization; and
(VRC7) every remaining object passes the absorb-or-separate obstacle audit.

Theorem 5.3 eliminates free connector choices once open problem 5.8 is solved. It does not prove the local equality between a pulled carrier and the copy embedded by a $\psi$-quadratic-like immersion.

## 6. Endpoint births and attachment support

6.1. **Exact endpoint arithmetic.** Let

$$g_j : S_j \longrightarrow S_{j+1}, \qquad 0 \le j < N,$$

be proper maps between Jordan disks, of degrees $d_j$, with no boundary critical point. Let $P_N \subset \partial S_N$ be a terminal endpoint marking. Recursively set

$$P_j = g_j^{-1}(P_{j+1}) \sqcup B_j, \qquad b_j := |B_j|, \tag{5.1}$$

where $B_j$ contains precisely the endpoints first required at stage $j$. Put

$$D_{a,b} := \prod_{i=a}^{b} d_i, \qquad D_{a,a-1} := 1. \tag{5.2}$$

**Proved Theorem 6.1** (Degree-weighted endpoint births)**.** *Writing $p_j = |P_j|$, one has*

$$p_j = d_j p_{j+1} + b_j \tag{5.3}$$

*and*

$$p_0 = D_{0,N-1} p_N + \sum_{j=0}^{N-1} D_{0,j-1} b_j. \tag{5.4}$$

*Proof.* The boundary restriction of $g_j$ is a degree-$d_j$ circle covering, so every endpoint in $P_{j+1}$ has exactly $d_j$ boundary preimages. Equation (5.3) follows from the disjoint union (5.1). Iterating (5.3) gives (5.4). □

**Definition 6.2** (Birth charge)**.** The endpoint-birth charge seen at stage 0 is

$$\mathfrak{b}_0 := \sum_{j=0}^{N-1} D_{0,j-1} b_j. \tag{5.5}$$

**Proved Proposition 6.3** (Bounded terminal window)**.** *Suppose for some $M \le N$ and $b_* \ge 0$ that*

$$d_j = 1, \quad b_j = 0 \qquad (0 \le j < N - M),$$

*and*

$$d_j \le 2, \quad b_j \le b_* \qquad (N - M \le j < N).$$

*Then*

$$\mathfrak{b}_0 \le b_*(2^M - 1) \tag{5.6}$$

*and*

$$p_0 \le 2^M p_N + b_*(2^M - 1), \tag{5.7}$$

*independently of $N$.*

*Proof.* Only the last $M$ summands in (5.5) remain, and their prefix degrees are bounded by $1, 2, \dots, 2^{M-1}$. Summing the geometric series gives (5.6); (5.7) follows from theorem 6.1. □

*Warning* 6.4 (Bounded local births are not enough). In a degree-one corridor with $b_j = 1$ at every stage,

$$p_0 = p_N + N.$$

Thus a uniform per-stage incidence bound can still grow linearly with the satellite period.

6.2. **Attachment support.** Assume a marked carrier correspondence has the exact interval pullback description

$$\iota_j^* I = \iota_j^{-1}(I) \cup \bigcup_{X \in \mathcal{A}_j(I)} \overline{\operatorname{Cara} X}, \tag{5.8}$$

where $\mathcal{A}_j(I)$ is the family of components of the pulled-carrier excess attached at $I$. Define

$$\operatorname{Att}_j := \{I : \mathcal{A}_j(I) \neq \varnothing\}. \tag{5.9}$$

**Proved Proposition 6.5** (Local carrier equality implies no birth)**.** *Suppose $I \notin \operatorname{Att}_j$, $\iota_j^{-1}(I)$ is one marked interval, and the polynomial map is conformal on this interval and its endpoint prime ends. Then*

$$b_j(I) = 0.$$

*Proof.* The second union in (5.8) is empty, so the complete pulled interval is exactly $\iota_j^{-1}(I)$. Its two endpoints are the unique boundary lifts of the already declared endpoints of $I$. Hence no endpoint lies outside $g_j^{-1}(P_{j+1})$. □

**Proved Theorem 6.6** (Attachment-support reduction)**.** *Assume* (5.8)*, access completeness, and the hypotheses of proposition 6.5 outside* $\mathrm{Att}_j$*. Then endpoint births are supported on* $\mathrm{Att}_j$*.*

*If all attachment support and all genuine births are confined to a terminal window of depth* $M$*, with degree at most two and at most* $b_*$ *new endpoints per stage, then*

$$\mathfrak{b}_0 \le b_*(2^M - 1).$$

*Proof.* Proposition 6.5 gives $b_j(I) = 0$ off the attachment support. The remaining hypotheses are precisely those of proposition 6.3. □

*Remark* 6.7 (Native KKL realization)*.* In the category of imported theorem 2.6, KKL's actual equality $\mathcal{T}' = \mathcal{T}$ on the central range makes $\mathrm{Att}_j$ empty there. This is the source mechanism behind native central no-birth.

6.3. **Transport through a natural degree-one corridor.** For $0 \le j \le N$, let $\eta_j : T_j \hookrightarrow T'_j$ be a marked carrier inclusion and put

$$E_j := T'_j \setminus \eta_j(T_j). \tag{5.10}$$

Call the transition from $j+1$ to $j$ *marked-pair natural* if there are orientation-preserving homeomorphisms

$$\Phi_j : T'_{j+1} \longrightarrow T'_j, \qquad \phi_j : T_{j+1} \longrightarrow T_j \tag{5.11}$$

such that

$$\Phi_j \circ \eta_{j+1} = \eta_j \circ \phi_j, \tag{5.12}$$

and these maps extend to the proof-used prime-end compactifications, bijection the ideal markings, and preserve attachment incidence and cyclic access order.

**Proved Theorem 6.8** (Corridor attachment transport)**.** *For a marked-pair natural transition,* $\Phi_j$ *induces bijections*

$$\mathcal{A}_{j+1}(I) \longleftrightarrow \mathcal{A}_j(\phi_j(I)) \tag{5.13}$$

*for all proof-used intervals* $I$*. Consequently,*

$$\mathrm{Att}_j = \phi_j(\mathrm{Att}_{j+1}). \tag{5.14}$$

*Thus, along a marked-pair natural degree-one corridor, one attachment-free anchor stage forces empty attachment support at every central stage.*

*Proof.* The commuting square gives

$$\Phi_j(E_{j+1}) = T'_j \setminus \eta_j(T_j) = E_j.$$

A homeomorphism bijects the components of these excess sets. The assumed prime-end extension carries a component attached at $I$ to a component attached at $\phi_j(I)$, preserving every marked access and its cyclic order. This proves (5.13) and (5.14). Iteration proves the anchor claim. □

*Warning* 6.9 (Root routing does not give pair naturality)*.* The degree-one root-pullback certificate constructs a homeomorphism between the selected polynomial carrier components. It does not construct $\Phi_j$ on the pulled ambient carriers, prove the commuting square (5.12), or extend it to the marked prime ends. A hidden ambient branch can therefore move while the selected carrier labels remain unchanged.

**Proved Theorem 6.10** (Univalent construction of pair naturality)**.** *Let $0 \le j \le k$ be a central corridor with orientation-preserving conformal homeomorphisms*

$$U_j : \Omega_{j+1} \longrightarrow \Omega_j.$$

*Given one actual anchor pair $\eta_k : T_k \hookrightarrow T'_k$, define recursively*

$$T'_j = U_j(T'_{j+1}), \qquad T_j = U_j(T_{j+1}), \qquad \eta_j = U_j \eta_{j+1} U_j^{-1} \,\big|_{T_j}. \tag{5.15}$$

*Transport every package, obstacle, and ideal marking by the same maps. Then the inclusion squares commute. On every proof-used simply connected complement localization, the maps extend to orientation-preserving homeomorphisms of prime-end compactifications and preserve cyclic order, without assuming local connectivity of the Euclidean boundaries.*

*Proof.* Equation (5.15) gives exact transport of both carriers and the commuting square by conjugation. Each restricted $U_j$ is a conformal isomorphism between the corresponding marked complement localizations. Functoriality of prime-end compactification therefore supplies the claimed circle homeomorphism and cyclic-order preservation. Iteration handles the complete finite corridor. □

**Proved Proposition 6.11** (Anchor underdetermination)**.** *A required package, full connected carriers, critical-value-free degree-one selected-carrier dynamics, and identical coarse orbit labels do not imply an attachment-free anchor.*

*Proof.* In a Jordan disk, take the package $S = \{-1, 1\}$, the full interval

$$T = [-1, 1],$$

and the full triod

$$T^\dagger = [-1, 1] \cup [0, i].$$

The identity is univalent near $T^\dagger$ and gives the same dynamics on $T$. The pair $T \subset T$ is attachment-free, whereas $T \subset T^\dagger$ has the vertical excess branch attached at 0. The triod has a valence-three point and the interval does not, so no homeomorphism of carrier pairs removes this distinction. The listed data are identical but anchor equality holds in only one model. □

This is a counterexample to deriving anchor equality from Dudko's published connected-hull description; it is not a counterexample to the possible existence of a stronger canonical virtual carrier.

**Proved Theorem 6.12** (No least connected full carrier)**.** *There is no least compact connected full plane continuum containing the two-point package $S = \{-1, 1\}$, even with identity dynamics and no obstacles.*

*Proof.* Let $A_+$ and $A_-$ be the upper and lower semicircles joining $-1$ to 1. Each is an arc, hence a compact connected full continuum, and

$$A_+ \cap A_- = S.$$

If a least carrier $H$ existed, then $H \subset A_+ \cap A_- = S$. Since $S \subset H$, this would give $H = S$, contradicting connectedness. □

Thus "take the minimal connected full hull" is not a carrier rule. A regulated connector system, together with its access or isotopy class and a forward-coherence proof, is additional data.

**Definition 6.13** (Finite attachment-orbit closure)**.** For a finite resolved carrier with cell set $\mathscr{C}$, let

$$\mathsf{A} : 2^{\mathscr{C}} \longrightarrow 2^{\mathscr{C}} \tag{5.16}$$

be an inflationary monotone operator adding every cell forced by attachment, access completion, corridor transport, or fullification. Starting from the proposed carrier cells $C_0$, set

$$C_{n+1} := \mathsf{A}(C_n). \tag{5.17}$$

**Proved Theorem 6.14** (Least attachment-orbit closure)**.** *The sequence $(C_n)$ stabilizes after at most $|\mathscr{C} \setminus C_0|$ strict enlargements at the least fixed point $C_\infty$ containing $C_0$. If a corridor bijection $P$ conjugates two completion operators, then it transports their least fixed points.*

*Consequently, any selected-sheet merger, critical-value intrusion, obstacle collision, marked-access identification, or localization escape already present in $C_\infty$ persists in every monotone attachment-complete enlargement.*

*Proof.* Inflationarity makes the sequence increasing, and every strict step adds a new cell, proving finite stabilization. If $D \supset C_0$ is any fixed point, monotonicity gives $C_n \subset D$ inductively, hence $C_\infty \subset D$. If $P\mathsf{A} = \widetilde{\mathsf{A}}P$, induction gives $P(C_n) = \widetilde{C}_n$, and stabilization gives the conjugacy claim. Every monotone attachment-complete enlargement is such a fixed point and contains $C_\infty$, so it retains every listed collision. □

6.4. **An analytic protected-access collision.** Let

$$f(z) = z^2, \qquad I = [1,2], \qquad m = \frac{3}{2},$$

and let

$$Q = \left\{ z : \left| z - \frac{3}{2} \right| \le \frac{1}{2},\ \operatorname{Im} z \ge 0 \right\}. \tag{5.18}$$

Set $U^- = \{z : |z - 3/2| < 1\}$ and $U^+ = f(U^-)$. The disk $U^-$ lies in $\operatorname{Re} z > 1/2$, so

$$F = f|_{U^-} : U^- \longrightarrow U^+ \tag{5.19}$$

is a conformal homeomorphism. Take

$$T_0 = I, \quad T_0' = Q, \qquad T_1 = f(I), \quad T_1' = f(Q). \tag{5.20}$$

Within this branch,

$$T_0' = F^{-1}(T_1'), \qquad T_0 = F^{-1}(T_1). \tag{5.21}$$

**Proved Proposition 6.15** (Quadratic protected-access collision)**.** *The pulled excess has the single component*

$$X_0 = Q \setminus I$$

*with attachment trace*

$$\tau_{T_0}(X_0) := \overline{X_0}^{T_0'} \cap T_0 = I.$$

*The closure is taken in the actual pulled carrier $T_0'$. In particular, the trace is not $X_0 \cap T_0$, which is empty by the definition of the excess. Its complete one-stage itinerary is*

$$X_0 \longmapsto f(X_0),$$

*on the right sheet with critical word 0. If the upper access $p_0^+(m)$, represented by $m + it$, is protected, every monotone attachment-complete enlargement absorbing $X_0$ erases that access and its transported access $F_* p_0^+(m)$.*

*Proof.* The relative upper half-disk is connected, and its closure meets $I$ in all of $I$. Since $0 \notin U^-$, $F'(z) = 2z \neq 0$, and the selected branch avoids the critical value, the itinerary has one right-sheet, degree-one step and no critical encounter.

Absorbing $X_0$ while retaining $I$ forces the enlarged carrier to contain $Q$. But $Q$ contains an upper relative half-neighborhood of $m$. No curve in $U^- \setminus Q$ can therefore land at $m$ from the upper side. Further monotone enlargement cannot restore a complement access through a set already absorbed. Conformality transports the same conclusion to stage 1. □

In the order

$$(\chi_{\mathrm{sheet}}, \chi_{\mathrm{crit}}, \chi_{\mathrm{obs}}, \chi_{\mathrm{access}}, \chi_{\mathrm{loc}}),$$

the collision tuple is

$$\chi(C_\infty) = (\varnothing, \varnothing, \varnothing, \{p_0^+(m), F_* p_0^+(m)\}, \varnothing). \tag{5.22}$$

The replacement invariant must therefore retain, for every excess component $X$, both its full geometric attachment trace

$$\tau_T(X) = \overline{X}^{T'} \cap T \tag{5.23}$$

and the partial survival map on protected accesses. The closure is taken in the pulled carrier $T'$, or equivalently in the closed ambient localization. The associated prime-end attachment record is the family of protected crosscut chains accumulating on this geometric trace; it is not identified with the trace itself. A cell quotient that forgets either datum can report a false zero collision.

**Open Problem 6.16** (Virtual bounded attachment support)**.** For the virtual carrier through the complete Wave horizon, prove:

(1) the exact interval pullback formula (5.8);
(2) an actual anchor pair whose full transport by theorem 6.10 is Dudko's required virtual carrier;
(3) a regulated forward-coherent connector system and a component-exhaustion theorem whose attachment traces preserve every protected access; then prove either one attachment-free anchor or a collision-free analytic realization of theorem 6.14, making the central support empty by theorem 6.8;
(4) a $B, L$-bounded endpoint contribution in fixed entry, exit, and terminal windows;
(5) preservation of every proof-used prime-end access and cyclic order; and
(6) proper or end-proper relative-cover lifts on every interval used by the Wave itinerary.

Root component routing from section 5 does not imply open problem 6.16: pulled-carrier excess can attach inside the correct selected polynomial pullback component.

## 7. Finite inverse-tree fans and ribbon markings

**Definition 7.1** (Full inverse tree)**.** A finite rooted tree carries a full inverse tree for $f$ if every vertex $v$ carries a nondegenerate full continuum $H_v$, every child carrier is a component of the pullback of its parent, and distinct same-depth vertices are distinct components of the corresponding iterated root pullback.

For a vertex $v$, let $m(v)$ be the number of target carriers containing the critical value along its route to the root.

**Proved Lemma 7.2** (Simultaneous pullback shrinking)**.** *Let $P$ be a polynomial, $K$ a nondegenerate full continuum, and $H_1, \dots, H_t$ finitely many selected components of finite iterated pullbacks of $K$. Given neighborhoods $N_i \supset H_i$, there is a neighborhood $O \supset K$ such that, for every open connected neighborhood $B$ with $K \subset B \subset O$, the corresponding component containing $H_i$ exists and has closure in $N_i$. In particular this applies to every admissible Jordan root neighborhood used in the fan construction.*

*Proof.* For one selected component $H$ of $P^{-1}(K)$, choose a bounded neighborhood $N'$ with

$$H \Subset N' \Subset N, \qquad \partial N' \cap P^{-1}(K) = \varnothing,$$

containing no other pullback component. The compact sets $P(\partial N')$ and $K$ are disjoint. A sufficiently small neighborhood $O$ of $K$ misses $P(\partial N')$, so the component of $P^{-1}(O)$ containing $H$ cannot cross $\partial N'$.

Apply this argument to the appropriate iterate for every selected component and intersect the finitely many resulting root neighborhoods. For any $K \subset B$ in the intersection, the selected $H_i$ lies in the relevant iterated preimage of $B$, so its containing component exists. That component lies in the component over the larger root neighborhood, whose closure is contained in $N_i$. □

**Proved Theorem 7.3** (Simultaneous proper Jordan fan)**.** *Let $(H_v)$ be a finite full inverse tree for $f(z) = z^2 + c_0$. Suppose each $H_v$ has a neighborhood $N_v$ such that same-depth neighborhoods are disjoint and every prescribed compact obstacle is disjoint from $\overline{N_v}$. Then there is a Jordan disk $B \supset H_r$ such that the selected component*

$$A_v := \mathrm{Comp}_{H_v}(f^{-d(v)}(B)) \tag{6.1}$$

*is a Jordan disk compactly contained in $N_v$, and every edge map is proper. Moreover*

$$\deg(f^{d(v)} : A_v \to B) = 2^{m(v)}. \tag{6.2}$$

*Proof.* Use lemma 7.2 to choose one root neighborhood that controls all selected components. A full planar continuum has a basis of Jordan neighborhoods obtained from level curves of the exterior Riemann map; choose such a root disk $B$ inside that neighborhood and with boundary avoiding the finite set of critical values of the relevant iterates.

Each selected pullback component is then a simply connected proper preimage of a Jordan disk with no boundary branching, hence is a Jordan disk. Simultaneous shrinking puts it inside $N_v$, and same-depth disjointness follows. By proposition 5.5, an edge has degree two precisely at a critical target and degree one otherwise. Proper degrees multiply, giving (6.2). □

**Proved Proposition 7.4** (Dyadic terminal fan)**.** *If the selected branches have critical counts $0, 1, \ldots, M$, their total degree is*

$$\sum_{m=0}^{M} 2^m = 2^{M+1} - 1. \tag{6.3}$$

**Proved Theorem 7.5** (Exact ribbon pullback)**.** *Let $P_r \subset \partial B$ be a nonempty finite endpoint marking. On a selected branch $A_v \to B$ of degree $d_v$, define*

$$P_v := (f^{d(v)}|_{\partial A_v})^{-1}(P_r).$$

*Then*

$$|P_v| = d_v |P_r|, \tag{6.4}$$

*the same number of cyclic ideal intervals is obtained, and cyclic order is preserved locally on every sheet. For the fan in proposition 7.4, the total count is*

$$|P_r|(2^{M+1} - 1). \tag{6.5}$$

*Proof.* Every proper holomorphic map between the selected Jordan disks extends to an orientation-preserving boundary circle covering. A degree-$d_v$ circle cover has exactly $d_v$ preimages of each root endpoint. Removing $n$ points from an oriented circle gives $n$ cyclic intervals. Summing (6.4) over the dyadic degree profile gives (6.5). □

*Warning* 7.6 (Cardinality is not access completeness). Theorem 7.5 counts preimages of the declared root endpoints. It does not prove that every vertical, outgoing, lost, or amplifying Wave end projects to that root set. Hidden fjords or immersion sheets can expose additional ends governed by theorem 6.1.

## 8. Conditional reduction to Mandelbrot local connectivity

We now state the exact logical endpoint. Each hypothesis is an open uniform theorem, not a definition disguised as an assumption.

**Open Problem 8.1** (H1: marked construction)**.** Construct for every infinitely renormalizable quadratic polynomial a canonical entropy-clock checkpoint tower with:

(1) an access-complete virtual, primitive, or pseudo-Siegel marked state;
(2) selected finite-horizon $\psi$-quadratic-like correspondences whose declared return has one quadratic critical encounter;
(3) the root component-routing and finite obstacle certificates of sections 5 and 7; and
(4) complete orbit and relative-cover admissibility through every Wave, Wanderers, localization, and quasi-additivity stopping time.

**Open Problem 8.2** (H2: marked backward transition)**.** On the states from H1, prove period-independent comparison among the KKL WAD/Value, pseudo-Siegel sectorial, virtual response, and valid quadratic-like coordinates. Realize the tame, long-Wave, positive-entropy, and zero-entropy branches so that a high calibrated potential obeys

$$L_{p(n)} \geq \lambda L_n - C, \qquad p(n) < n, \quad \lambda > 1, \tag{7.1}$$

with universal constants. This includes open problem 6.16 and all error summability and chart-cycle tests.

**Proved Theorem 8.3** (Countable-follower Stone compactification)**.** *Let $X$ be countable, let $T_e : D_e \to X$ be countably many partial deterministic maps, and let $\mathcal{O}$ be countably many observable subsets of $X$. There are a compact metrizable zero-dimensional space $K$, an injective dense map $\iota : X \to K$, clopen extensions of all sets in $\mathcal{O}$, and, for every $e$, a continuous extension*

$$\widehat{T}_e : \widehat{D}_e \longrightarrow K$$

*on a clopen domain.*

*Proof.* Let $\mathfrak{B}$ be the smallest Boolean algebra containing all singletons, all members of $\mathcal{O}$, all $D_e$, and all $T_e^{-1}(A)$ for $A \in \mathfrak{B}$. Iterated closure under countably many inverse-image operations and finite Boolean operations keeps $\mathfrak{B}$ countable. Its Stone space $K = \mathrm{Ult}(\mathfrak{B})$ is compact, zero-dimensional, and second countable, hence metrizable. Principal ultrafilters embed $X$ injectively and densely.

For $u \in \widehat{D}_e$, put

$$\widehat{T}_e(u) = \{A \in \mathfrak{B} : T_e^{-1}(A) \in u\}.$$

Since inverse image preserves intersections and complements relative to $D_e$, this is an ultrafilter. The identity

$$\widehat{T}_e^{-1}(\widehat{A}) = \widehat{T_e^{-1}(A)} \cap \widehat{D}_e$$

proves continuity and the principal-ultrafilter formula proves extension. □

Theorem 8.3 removes bare symbolic discreteness once the actual countable follower and complete Boolean observable catalog are fixed. It does not construct that follower, compactify the virtual hull geometry, factor continuum- or real-valued observables, or produce compatible continuous geometric transitions.

**Proved Theorem 8.4** (Future-observable compactification)**.** *Let $X$ be an arbitrary exact state set, let $T_e : D_e \to X$, $e \in E$, be countably many partial deterministic transitions, and let*

*$O_j : X \to Y_j$, $j \in J$, be a countable catalog with compact metrizable targets. Add $O_* : X \to \{0\}$. For every finite transition word $w$, put*

$$\Phi(x)_{w,j} = \begin{cases} O_j(T_w x), & x \in D_w, \\ \bot, & x \notin D_w, \end{cases}$$

*where $Y_j^{\bot} = Y_j \sqcup \{\bot\}$, and let*

$$K_{\mathrm{fut}} = \overline{\Phi(X)} \subset \prod_{(w,j) \in E^{<\mathbb{N}} \times (J \cup \{*\})} Y_j^{\bot}. \tag{7.2}$$

*Then $K_{\mathrm{fut}}$ is compact metrizable,*

$$\widehat{D}_e = \{z : z_{e,*} = 0\}$$

*is clopen, and*

$$(\widehat{T}_e z)_{w,j} = z_{ew,j} \tag{7.3}$$

*continuously extends $T_e$. Every declared observable extends by its empty-word coordinate, and every finite exact transition identity persists on its compactified domain.*

*The quotient identifies exactly the states with equal declared observations after all finite continuations. Every compact Hausdorff realization extending the same data on clopen domains maps continuously onto $K_{\mathrm{fut}}$. If $X$ is countable and all point predicates are declared, then $\Phi$ is injective.*

*Proof.* The coordinate set in (7.2) is countable, so the ambient product and its closed subset $K_{\mathrm{fut}}$ are compact metrizable. The coordinate shift in (7.3) is continuous. If $\Phi(x_n) \to z \in \widehat{D}_e$, the discrete $(e,*)$-coordinate gives $x_n \in D_e$ eventually, and then

$$S_e \Phi(x_n) = \Phi(T_e x_n).$$

Thus $S_e z \in K_{\mathrm{fut}}$.

For every finite word $w$, its compactified domain $\{z : z_{w,*} = 0\}$ is clopen and the exact points from $D_w$ are dense in it. Continuous finite compositions agreeing on $D_w$ therefore agree on the compactified domain.

For a competing compact realization, take every extended future observation as one coordinate and use $\bot$ off the clopen word domain. The resulting continuous map agrees with $\Phi$ on the dense exact subset, so its compact image is exactly $K_{\mathrm{fut}}$. Point predicates separate exact states in the countable case. □

**Proved Theorem 8.5** (Symbolic side and future-trace compactness)**.** *For every already realized exact side or component transition system with countably many labels and a frozen countable proof-use catalog in specified compact metrizable encodings, theorem 8.4 gives a compact metrizable future-faithful end topology. All declared guards, hull encodings, branches, transports, outputs, and authorized enlargements extend continuously. Countably many predicates for a countable side projection keep distinct exact counter/stack configurations separated.*

*Proof.* Apply theorem 8.4; future-faithfulness is equality of the full coordinate table. Add the point predicates of the countable side projection when exact side separation is required. □

This proves the symbolic side-compactness node and the compact-end-topology branch of the virtual proof-use trace selector as conditional interfaces. It supplies neither their open virtual input systems nor a finite response state bound.

**Open Problem 8.6** (H3: coercivity and compactness)**.** Prove that bounded calibrated potential and the normalized incidence and congestion hypotheses prevent degeneration of the analytic carriers, branch graphs, relative covers, protected prime ends, response forms, and Value features,

so every sequence has a subsequence converging to an admissible nondegenerate marked renormalization. The compact topology from theorem 8.4 allows degenerate hyperspace limits and does not prove this analytic assertion.

**Open Problem 8.7** (H4: parameter rigidity)**.** Transfer the dynamical bounds to a connected relative-neighborhood basis in parameter space, or verify the hypotheses of a precise trivial-fiber theorem. In particular, supply the required puzzle Teichmüller-diameter or equivalent phase–parameter control and then exclude invariant line fields for the resulting controlled tower.

**Conditional Theorem 8.8** (Conditional MLC theorem)**.** *If open problems 8.1, 8.2, 8.6 and 8.7 are all proved with constants uniform over arbitrary mixtures of the four renormalization regimes, then the Mandelbrot set is locally connected.*

*Proof.* Let $c \in \mathcal{M}$. If $c$ is not infinitely quadratic-like renormalizable, imported theorem 2.1 gives a connected relative-neighborhood basis at $c$.

Suppose $c$ is infinitely renormalizable. H1 constructs the complete marked tower. H2 and lemma 4.3 uniformly bound its calibrated potentials. H3 converts this bound into compact controlled renormalization geometry. H4 converts the controlled dynamical tower into a connected relative-neighborhood basis at $c$. Thus $\mathcal{M}$ is locally connected at every parameter. □

*Warning* 8.9. The proof of conditional theorem 8.8 is a dependency argument. It does not prove H1–H4, and neither beau bounds alone nor shrinking of an arbitrary nested sequence supplies H4.

## 9. The universal protected-tower closure proposition

This section collects the four global gates into one falsifiable statement. Its purpose is logical compression, not promotion: the proposition below is OPEN. Only the implication from the proposition to MLC is proved. The complete typed statement and failure certificates are recorded in `mlc_closure_proposition.md`.

### 9.1. Source boundary and data.

Dudko's Interpolation Problem 4.4 and the following paragraph propose partially invariant connected hulls of Cantor small Julia sets [Dud25]. They do not supply the canonical carrier, protected-access maps, component exhaustion, analytic realization, transition, compactness, or rigidity asserted below.

KKL Theorem B states that prime primitive families with beau bounds and bounded puzzle Teichmüller diameter are rigid [KKL26]. We invoke that theorem only in this native scope. The mixed-tower rigidity clause below is not imported from KKL.

At an entropy-clock checkpoint $n$, write

$$\mathfrak{T}_n = (\Sigma_n, H_n, B_n, P_n, \mathcal{I}_n, \Gamma_n, \mathscr{Q}_n, \mathfrak{A}_{\mathrm{prot},n}), \tag{8.1}$$

where $\mathscr{Q}_n$ is the typed proof-use inventory and

$$\mathfrak{A}_{\mathrm{prot},n} = (C_{\infty,n}, \chi_n, \{\tau_n(X)\}_X, \{\varsigma_{n,X}\}_X), \quad \chi_n = (\chi_{\mathrm{sheet}}, \chi_{\mathrm{crit}}, \chi_{\mathrm{obs}}, \chi_{\mathrm{access}}, \chi_{\mathrm{loc}})_n. \tag{8.2}$$

Here $C_{\infty,n}$ is the attachment-orbit closure, $\tau_n(X)$ is the complete attachment trace and itinerary, and $\varsigma_{n,X}$ is the partial protected-access survival map. The collision coordinates are fatal for the current protected analytic core. A raw defect may be placed in a bounded window only after a declared retiming, localization, sheet-set, or carrier replacement gives a protected local model with zero collision invariant. The normalized response state is

$$\mathscr{R}_n = \frac{1}{r_{B,n}}(Q_n, \Lambda_n, \mathbf{V}_n, W_n^{\mathrm{std}}, \mathbf{\Xi}_n), \qquad r_{B,n} > 0. \tag{8.3}$$

Normalization in (8.3) does not quotient away exact counter, stack, frame, flag, chart, or return labels.

**Open Problem 9.1** (Universal Protected-Tower Compactness and Rigidity Proposition (UPTCR))**.** There exist

$$\begin{gathered} d_0, w_0, m_0 \in \mathbb{N}, \qquad \rho_0, \kappa_0, T_0, D_*, E_0 < \infty, \qquad \eta_0 > 0, \\ \lambda_{\mathrm{raw}} > 1, \qquad 0 \le \delta < \lambda_{\mathrm{raw}} - 1 \end{gathered} \tag{8.4}$$

chosen before the polynomial, such that every infinitely renormalizable quadratic polynomial admits canonical checkpoints $n$ and predecessors $p(n) < n$ satisfying all of the following.

UPTCR-1. **Canonical protected carrier.** The actual carrier (8.1) is constructed from a lamination-lifted regulated tree or rooted predecessor puzzle-cell tree satisfying theorem 10.4 and open problem 10.6. It contains every object used by Wave, Wanderers, quasi-additivity, covering, response, or width arguments through the complete horizon. Fullification is allowed only while the access-survival maps remain injective and cyclic-order preserving. Every realized protected core, including every central degree-one corridor, has $\chi_n = 0$. Every raw defect used by a replacement model lies in a declared entry, exit, or terminal window of depth at most $m_0$, and that replacement preserves all proof-used accesses. The planar carrier realizes $C_{\infty,n}$ and no unforced connector is inserted.

UPTCR-2. **Exact analytic realization.** At every stage $j+1 \to j$ of a complete Wave horizon,

$$g_j^{-1}(H_{j+1}) = H'_j = \iota_j^{-1}(H_j). \tag{8.5}$$

The selected quadratic-like return spine has exactly one declared degree-two stage; its other corridor maps are univalent and critical-value-free. Each auxiliary proof branch $\beta$ has its full critical word and degree

$$\deg(\beta) = 2^{m(\beta)},$$

so the bounded terminal Wave fan is not replaced by a lone degree-two component. The full packages, obstacles, attachment traces, and prime-end markings transport equivariantly and with preserved cyclic order. The relative-cover subgroups are compatible, the analytic core is proper, and every proof-used ideal interval is end-proper. None of these facts is inferred merely from a ribbon-tree label.

UPTCR-3. **Uniform finite response state.** The source-faithful core and WAD modules have dimensions at most $d_0$ and $w_0$, and

$$\rho_n \le \rho_0, \qquad \frac{\kappa_n}{r_{B,n}} \le \kappa_0. \tag{8.6}$$

The state (8.3) is future-observable for all required KKL Value functionals, transports exactly, and is universally comparable in the calibrated gauge to every applicable KKL, pseudo-Siegel, puzzle, and valid quadratic-like family. No ordinary satellite-root modulus is used uniformly in the relative period.

UPTCR-4. **Backward transition.** There is one nonnegative calibrated potential

$$L_n = L_{t_n}(\mathscr{R}_n; \sigma_n) \in [0, \infty).$$

If $L_n > D_*$, then

$$L_{p(n)} \ge (\lambda_{\mathrm{raw}} - \delta) L_n - E_0. \tag{8.7}$$

Every complete block has at least one certified route: it accumulates entropy at least $\eta_0$, is absorbed by controlled pseudo-Siegel regularization, or lies in a proved transient/shear alternative inside a larger strictly contracting block. The complete error charge satisfies

$$G_B(e_B) \le \delta L_n + E_0. \tag{8.8}$$

Neutral chart changes are grouped with the block supplying strict contraction. A deterministic bookkeeping priority may choose one route; disjointness of the three mathematical alternatives is not assumed.

UPTCR-5. **Coercive marked compactness.** Bounded $L_n$, together with (8.6) and the protected side constraints, makes normalized marked towers sequentially precompact in a specified topology that retains the carrier, access, relative-cover, response, and Value data. Unbounded counters and stack or nesting words are forgotten after proved block closure or embedded in a compact end topology on which all relevant maps and observables extend continuously.
UPTCR-6. **Tower rigidity and parameter fiber.** The associated puzzle families have Teichmüller diameter at most $T_0$. Equal full marked combinatorics produce uniformly quasiconformal finite-depth pullback conjugacies and a limiting global quasiconformal conjugacy of the sphere normalized at infinity. Its Beltrami form is invariant, and the controlled tower has no invariant line field; hence that form vanishes. The conjugacy is conformal, and comparison of coefficients in the monic centered normalization makes the quadratic parameters coincide.
Moreover, each parameter $c$ has nested nonempty compact connected relative neighborhoods $\Pi_N(c)$ with

$$\bigcap_N \Pi_N(c) = \{\text{parameters with the full marked combinatorics of } c\}. \tag{8.9}$$

Neither singleton intersection nor shrinking is assumed in constructing the conjugacies.

Every constant in (8.4) is independent of satellite period, rotation number, continued-fraction depth, Wave horizon, virtual-Molecule depth, raw marking count, counter or stack size, and the number or order of chart changes.

*Warning* 9.2. Open problem 9.1 is one six-clause open proposition. A proof of (8.7) alone does not prove compactness, rigidity, or MLC. The ledger node `UNIVERSAL_PROTECTED_TOWER_CLOSURE` remains open.

## 9.2. **The proved conditional implication.**

**Proved Lemma 9.3** (Nested compact singleton fibers shrink)**.** *If $K_0 \supseteq K_1 \supseteq \cdots$ are nonempty compact subsets of a metric space and $\bigcap_n K_n = \{x\}$, then* $\operatorname{diam} K_n \to 0$.

*Proof.* Otherwise there are $\epsilon > 0$ and $y_n \in K_n$ with $d(x, y_n) \geq \epsilon/2$. A subsequence converges in $K_0$ to $y$. For each fixed $N$, its tail lies in the closed set $K_N$, so $y \in \bigcap_N K_N = \{x\}$, contradicting the distance bound. □

**Conditional Theorem 9.4** (UPTCR implies MLC)**.** *If open problem 9.1 is proved, then the Mandelbrot set is locally connected.*

*Proof.* Let $c \in \mathcal{M}$. The non-infinitely-renormalizable case is imported theorem 2.1. In the infinitely renormalizable case, put $\lambda = \lambda_{\text{raw}} - \delta > 1$. From (8.7) and lemma 4.3,

$$\sup_n L_n \leq \max\left\{L_{\text{prefix}}(c), D_*, \frac{E_0}{\lambda - 1}\right\} < \infty. \tag{8.10}$$

Indeed, a later record strictly above this bound would force its predecessor to be still larger.

UPTCR-5 makes the marked tower precompact. UPTCR-6 supplies the limiting global normalized quasiconformal conjugacy. If its invariant Beltrami form were nonzero on a positive-measure invariant set, normalization by its modulus would give an invariant line field. Hence it vanishes. A conformal conjugacy is affine, say $h(z) = az + b$; comparing coefficients in $h \circ f_c = f_{c'} \circ h$ gives $a = 1$, $b = 0$, and $c = c'$. Thus (8.9) has intersection $\{c\}$.

Lemma 9.3 makes the diameters tend to zero. For a relatively open $U \subset \mathcal{M}$ containing $c$, choose $r > 0$ with $B(c, r) \cap \mathcal{M} \subset U$. For large $N$, $\operatorname{diam} \Pi_N(c) < r$, so $\Pi_N(c) \subset B(c, r) \cap \mathcal{M} \subset U$. Thus every point has arbitrarily small connected relative neighborhoods. Components of relatively open subsets are therefore open, which proves local connectivity at every point of $\mathcal{M}$. □

*Warning* 9.5. The status of conditional theorem 9.4 is conditional because its hypothesis open problem 9.1 is open. The theorem proves a dependency implication, not UPTCR or MLC.

9.3. **Failure certificates and promotion rule.** The first failure of UPTCR-1 must record the block, excess component, attachment access, full itinerary, and collision coordinate. For UPTCR-2 it must record the stage, components, subgroup image, and extra sheet or ideal end. For UPTCR-3 it must record the escaping dimension, missing Value functional, failed quotient square, or congestion ratio. For UPTCR-4 it must record the complete block word, matrix, chart cycle, entropy charge, error, and failed margin. For UPTCR-5 it must exhibit an escaping marked subsequence. For UPTCR-6 it must exhibit the finite-depth dilatation defect, nonzero limiting Beltrami form, or failed connected relative-neighborhood property.

No nonzero fatal collision is licensed by a bounded window; the replacement protected local model itself must have zero collision invariant. No UPTCR clause is promoted by a finite model. The upper-half-disk access collision, increasing satellite periods, alternating neutral/primitive blocks, vanishing entropy increments, late sheet mergers, subgroup mismatch, topology-preserving width rescaling, shear cycles, unbounded memory, prime-end monodromy, and possible invariant line fields remain mandatory attacks.

9.4. **Cross-validation verdict.**

**Proved Proposition 9.6** (Current data do not prove UPTCR-1)**.** *Dudko's published connected-hull description together with the finite monotone completion rules proved in this manuscript does not imply the canonical protected-carrier clause of open problem 9.1.*

*Proof.* Theorem 6.12 rules out selecting the carrier as the least connected full hull of the required package. If instead every first pulled excess is absorbed monotonically, proposition 6.15 gives a critical-value-free quadratic branch whose first absorption erases a protected prime-end access. Thus neither available selection rule proves UPTCR-1. This does not refute the existence of a more structured dynamically selected carrier. □

The topological realization from an explicit protected ribbon certificate is proved later in theorem 10.4. The exact free-component criterion theorem 10.8 then removes arbitrary tree choice. The attempted proof still stops at the first clause because open problems 10.6 and 10.58 are open. The first missing input is one future-complete free-domain component with simultaneous singleton ports to all root-shadow clusters; complete component and critical-word routing and collision-free replacement models remain afterward.

## 10. Protected regulated connectors

The cross-validation in section 9 stops at the connector part of UPTCR-1. This section proves the purely topological realization once an actual protected ribbon-routing certificate is supplied, and isolates the remaining virtual dynamical certificate. The complete definitions and failure record are in `protected_regulated_connector.md`.

10.1. **Source boundary.** Dudko describes the virtual Julia sets as connected hulls of Cantor small Julia sets carrying a finite part of the critical orbit [Dud25]; no connector or protected-access theorem is stated. KKL have a stronger native object: their Hubbard tree is constructed inside the full dynamical carrier, and in the non-locally-connected case it is lifted from the locally connected rational-lamination model [KKL26, §§2.5–2.6]. That construction is a positive control in the published primitive category, not a theorem for virtual-Molecule packages.

10.2. **Certificate and realization.** Let $U$ be a Jordan disk and let

$$A_1, \dots, A_s \Subset U$$

be pairwise disjoint full continua. Let $Z \Subset U \setminus \bigcup_i A_i$ be the forbidden obstacle and critical-value set. For every protected access to $A_i$, choose a prime-end crosscut chain

$$\Gamma_{i,a} = (\gamma_{i,a,n}, D_{i,a,n})_{n\geq 1}.$$

The chains are truncated so that the terminal collar tails $\overline{D_{i,a,1}} \setminus A_i$ are pairwise disjoint, lie in $U$, and miss $Z$ and the other atoms.

**Definition 10.1** (Protected ribbon-routing certificate)**.** A protected ribbon-routing certificate consists of the preceding data, pairwise disjoint free-port arcs $\alpha_i$ satisfying

$$\alpha_i \cap A_i = \{x_i\}, \qquad \alpha_i((0,1]) \cap \left( Z \cup \bigcup_j A_j \cup \bigcup_{j,a} \overline{D_{j,a,1}} \right) = \varnothing,$$

and an embedded finite tree $G$ joining the outer port endpoints while missing the same protected and forbidden sets. The planar cyclic order of half-edges at every vertex is retained. Every access read by a proof must be a protected chain or a declared side of this ribbon tree.

Define

$$H_{\mathfrak{R}} = G \cup \bigcup_i \alpha_i \cup \bigcup_i A_i. \tag{9.1}$$

**Proved Lemma 10.2** (One-point tree attachments are full)**.** *If a finite plane tree meets each of finitely many pairwise disjoint full continua in exactly one point and otherwise misses them, their union is a full continuum.*

*Proof.* A finite plane tree is nonseparating. Attach the full continua one at a time. At each step the two nonseparating continua meet in a connected singleton. The Janiszewski separation lemma implies that their union is nonseparating. Induction proves that the final compact connected set has connected complement.

Equivalently, the same step follows from Alexander duality and the Mayer–Vietoris sequence: connectedness of the intersection and vanishing first Čech cohomology of the two nonseparating continua force vanishing first Čech cohomology of their union. □

**Proved Lemma 10.3** (Protected crosscut survival and order)**.** *Let $H \supset A_i$ be a full continuum and let $(\gamma_n, D_n)$ be one of the protected chains above. If*

$$(H \setminus A_i) \cap (\overline{D_1} \setminus A_i) = \varnothing,$$

*then a tail of the same crosscuts is a prime-end chain in $\widehat{\mathbb{C}} \setminus H$. Its endpoints remain boundary points and its selected sides contain the old $D_n$. A finite family with pairwise disjoint terminal selected domains induces distinct prime ends with unchanged cyclic order.*

*Proof.* For large $n$, $D_n \subset D_1$. The interior of $\gamma_n$ misses $H$, while its two distinct endpoints lie in $A_i \subset H$. Every neighborhood of either endpoint meets the crosscut interior, so those endpoints remain on the boundary of the enlarged complement. Thus $\gamma_n$ remains a crosscut. A path joining $D_n$ to the opposite side without meeting $H \cup \gamma_n$ would give the same forbidden path in $\widehat{\mathbb{C}} \setminus A_i$. Hence the selected sides remain nested, and the unchanged crosscut diameters still tend to zero.

Equivalent chains eventually divide one another. Disjoint terminal domains prevent that, so the induced prime ends are distinct. For a finite family, choose one deep oriented crosscut from each chain. Prime-end cyclic order is the separation order of these crosscuts. Neither the crosscuts nor their selected sides changed, so every ordered triple retains its orientation. □

**Proved Theorem 10.4** (Protected regulated connector realization)**.** *Every protected ribbon-routing certificate has an actual carrier $H_{\mathfrak{R}}$ with the following properties:*

*(1) $H_{\mathfrak{R}} \Subset U$ is compact, connected, and full;*

*(2) it contains every $A_i$, misses $Z$, and has singleton attachment trace $\{x_i\}$ at each atom;*
*(3) every protected crosscut chain remains a prime-end chain in $\widehat{\mathbb{C}} \setminus H_{\mathfrak{R}}$;*
*(4) the survival map is injective and preserves cyclic order;*
*(5) the connector-created sides have the boundary order of the retained ribbon tree; and*
*(6) no local connectivity of an $A_i$ is required.*

*Proof.* Lemma 10.2 gives the first statement, and the certificate gives the second. For a protected chain, the connector misses $\overline{D_{i,a,1}} \setminus A_i$, hence misses every later closed collar tail. Lemma 10.3 proves crosscut survival, injectivity, and preservation of cyclic order. A small regular neighborhood of the finite connector tree traverses every edge on two sides in exactly the cyclic boundary word of its ribbon structure. The argument uses crosscut chains only, so no Euclidean boundary extension of an atom is assumed. □

**Proved Proposition 10.5** (Univalent naturality)**.** *If $\phi : U \to V$ is an orientation-preserving conformal homeomorphism of Jordan disks and every certificate entry is transported by $\phi$, then*

$$H_{\phi_*\mathfrak{R}} = \phi(H_{\mathfrak{R}}). \tag{9.2}$$

*The protected survival map and ribbon boundary order commute with the prime-end map induced by $\phi$.*

*Proof.* The Jordan boundary extensions can be glued to an orientation-preserving sphere homeomorphism. Such a map preserves fullness, crosscuts, selected sides, intersections, and cyclic order. Equation (9.1) then gives (9.2) term by term. □

Consequently one actual anchor certificate transports through the complete degree-one corridor. This does not prove equality with the carrier copy supplied by the $\psi$-ql immersion.

10.3. **The lamination-lift candidate and remaining gate.** Following KKL's non-locally-connected construction, suppose an ambient carrier has a monotone semiconjugacy

$$\pi : K \longrightarrow K_{\mathrm{com}}$$

to a locally connected combinatorial model and a forward-coherent regulated tree $T_{\mathrm{com}}$ containing every root shadow. The canonical candidate is

$$H_{\mathrm{lam}} = \mathrm{Fill}(\pi^{-1}(T_{\mathrm{com}})). \tag{9.3}$$

A monotone preimage of a connected compact set is connected, so (9.3) is a full continuum containing the required shadows. Semiconjugacy gives forward coherence before fullification, and fullification preserves genuine polynomial forward invariance.

The missing assertion is not connectedness. A projection fiber or a component added by fullification may enter a protected collar, and a disconnected stage package may still split across iterated pullback components.

**Open Problem 10.6** (Virtual Protected Routing Certificate)**.** For every virtual-Molecule block and complete Wave horizon, prove that the candidate (9.3), or an explicitly equivalent puzzle-cell carrier:

(1) is defined on the authorized localization and contains the complete proof-use inventory;
(2) has no projection fiber, regulated edge, or filled component in a protected collar except at a declared free singleton port;
(3) realizes every package component, parent route, selected sheet, and critical word after the final root enlargement;
(4) absorbs or separates every obstacle without leaving the localization;
(5) contains every future proof-used access in its protected-chain or ribbon-side marking; and
(6) has uniformly bounded incidence or a proved future-faithful compression.

The first failure must name the block, stage, projection fiber or component, full attachment trace, protected chain, complete critical word, nonzero collision coordinate, and localization. The upper-half-disk collision is rejected at Item 2 because its trace is an interval and its excess occupies the protected upper collar.

10.4. **The exact free-component criterion.** Freeze the complete finite future inventory before choosing a connector. Let $Q$ be the union of all root-shadow atoms, closed terminal protected collars, separated obstacles, the critical-value exclusion, and the localization-forbidden set, and put

$$\Omega_{\mathrm{free}} = U \setminus Q. \tag{9.4}$$

A component $\Omega_0$ has simultaneous singleton ports if there are pairwise disjoint arcs $\alpha_i$ meeting $A_i$ only at one free point and satisfying $\alpha_i((0,1]) \subset \Omega_0$.

**Proved Lemma 10.7** (One-ended slits do not disconnect a plane domain)**.** *Let $\Omega$ be a plane domain and let $\delta_1, \dots, \delta_s : [0,1] \to \overline{\Omega}$ be pairwise disjoint Jordan arcs with $\delta_i(0) \in \partial\Omega$ and $\delta_i((0,1]) \subset \Omega$. Then*

$$\Omega \setminus \bigcup_i \delta_i((0,1])$$

*is path connected.*

*Proof.* Treat one slit first. A polygonal path between two points of the complement is compactly contained in $\Omega$, so it misses an initial tail $\delta((0,\epsilon))$. Perturb it to meet the compact arc $\delta([\epsilon,1])$ transversely at finitely many points. In a thin topological rectangle around that arc, replace every crossing by travel along the two long sides; the sides join around the free endpoint $\delta(1)$. The replacement stays in $\Omega$ and avoids the whole slit. This proves path connectedness after one removal. The removed slit is relatively closed in $\Omega$, so its complement is again a domain. Induction over the finite disjoint family proves the claim. □

**Proved Theorem 10.8** (Protected Free-Component Criterion)**.** *A protected ribbon-routing certificate exists if and only if one component of $\Omega_{\mathrm{free}}$ has simultaneous singleton ports to every required atom. If a first protected collar or obstacle separates two required port classes, no further monotone connector enlargement inside the current localization can repair the failure.*

*Proof.* Any protected connector with its atoms deleted is connected and lies in $\Omega_{\mathrm{free}}$; hence all its port interiors lie in one component.

Conversely, choose the simultaneous port arcs in $\Omega_0$ and shorten them away from the atoms. Their retained interiors are finitely many disjoint one-ended slits; lemma 10.7 proves that their complement is path connected. Join the finitely many outer port endpoints by polygonal paths whose interiors avoid those slits, perturb the paths into general position, and subdivide at their finitely many crossings. An inclusion-minimal connected subgraph spanning the port endpoints is a finite tree. Together with the shortened port arcs it is a protected ribbon-routing certificate, so theorem 10.4 supplies the full carrier and prime-end marking.

For the final assertion, any connected set joining two different components of $U \setminus Q$ meets $Q$. If the separating part of $Q$ is protected or forbidden, that intersection violates the certificate. □

The construction is natural under univalent changes of Jordan localization, since such a map carries $Q$, the components of (9.4), the port arcs, and the spanning tree to their transported counterparts. It is qualitative: the number of ports and ribbon sides is not bounded uniformly.

10.5. **Augmented atoms force a common domain.** The exact free-domain test is still difficult to verify directly from a partial virtual package. We now give a source-checkable sufficient criterion. For a compact plane set $K$, let

$$\mathrm{Fill}(K) = K \cup \{\text{bounded components of } \mathbb{C} \setminus K\}. \tag{9.5}$$

**Definition 10.9** (Augmented-atom packet)**.** For the frozen set $Q$ in (9.4), an augmented-atom packet consists of full continua

$$\widehat{A}_1, \dots, \widehat{A}_s, \widehat{O}_1, \dots, \widehat{O}_t \Subset U$$

and pairwise disjoint port arcs $\alpha_i$ such that:

(1) every protected collar, obstacle, critical exclusion, and other component of $Q$ has one typed assignment to a connected atom seed $A_i \cup E_i$ or connected separated-obstacle seed $O_\ell$;
(2) $\widehat{A}_i = \mathrm{Fill}(A_i \cup E_i)$ and $\widehat{O}_\ell = \mathrm{Fill}(O_\ell)$;
(3) these completed packets are pairwise disjoint and cover $Q$;
(4) $\alpha_i(0) = x_i \in A_i$, $\alpha_i \cap \widehat{A}_i = \{x_i\}$, and

$$\alpha_i((0,1]) \subset U \setminus \left( \bigcup_k \widehat{A}_k \cup \bigcup_\ell \widehat{O}_\ell \right);$$

and $x_i$ is not the impression of a protected chain.

The assignment records the component name, proof-use stages, attachment trace, return word, selected sheet, and absorb-or-separate decision.

The proof uses the compacta form of Alexander duality [Hat02, pp. 256–257]:

$$\widetilde{H}_0(S^2 \setminus K; \mathbb{Z}) \cong \check{H}^1(K; \mathbb{Z}) \tag{9.6}$$

for an arbitrary nonempty proper compact $K \subset S^2$. The Čech form, rather than the locally contractible special case, is required here.

**Proved Lemma 10.10** (Finite disjoint full packets)**.** *If $K_0, \dots, K_N \subset S^2$ are pairwise disjoint nonempty full continua, then*

$$S^2 \setminus \bigcup_{r=0}^{N} K_r$$

*is connected.*

*Proof.* Fullness of $K_r$ makes $S^2 \setminus K_r$ connected, so (9.6) gives $\check{H}^1(K_r; \mathbb{Z}) = 0$. The finite pairwise disjoint compacta have pairwise disjoint open neighborhoods. Every sufficiently fine cover of their union therefore has nerve equal to the disjoint union of the individual nerves. First cohomology and then the direct limit defining Čech cohomology give

$$\check{H}^1\left( \bigcup_r K_r; \mathbb{Z} \right) \cong \bigoplus_r \check{H}^1(K_r; \mathbb{Z}) = 0.$$

A second application of (9.6) makes the complement connected. □

**Proved Theorem 10.11** (Augmented-Atom Common-Domain Theorem)**.** *Every augmented-atom packet determines one component of $\Omega_{\mathrm{free}}$ with simultaneous singleton ports to every original atom. Hence it supplies a protected regulated connector.*

*Proof.* Apply lemma 10.10 to the completed packets together with

$$K_0 = S^2 \setminus U.$$

The Jordan-disk hypothesis makes $K_0$ a full continuum, and compact containment makes it disjoint from all completed packets. Their common complement

$$\Omega_{\mathrm{aug}} = U \setminus \left( \bigcup_i \widehat{A}_i \cup \bigcup_\ell \widehat{O}_\ell \right)$$

is therefore connected. Coverage of $Q$ gives $\Omega_{\mathrm{aug}} \subset \Omega_{\mathrm{free}}$, so $\Omega_{\mathrm{aug}}$ lies in one component $\Omega_0$ of $\Omega_{\mathrm{free}}$. Every port interior lies in $\Omega_{\mathrm{aug}}$, and

$$\alpha_i \cap A_i \subset \alpha_i \cap \widehat{A}_i = \{x_i\}.$$

Thus the arcs are simultaneous singleton ports from $\Omega_0$. Theorem 10.8 gives the connector. □

No local connectivity or width estimate occurs in this argument. Under an orientation-preserving conformal homeomorphism of Jordan disks, the boundary extensions can be glued to a sphere homeomorphism. The filled packets, ports, common domain, and any selected connector therefore transport naturally. The spanning tree is not claimed canonical.

10.6. **The fixed-packet closure and port audit.** The preceding theorem leaves an apparent dependence on the order in which assigned pieces are filled. This dependence is illusory once the typed assignment is fixed.

**Proved Lemma 10.12** (Filled-hull absorption)**.** *For compact plane sets $K, L$,*

$$\mathrm{Fill}(\mathrm{Fill}(K) \cup L) = \mathrm{Fill}(K \cup L). \tag{9.7}$$

*Consequently, sequentially adjoining and filling finitely many pieces gives the filled hull of their union, independently of processing order.*

*Proof.* The operator Fill is extensive, monotone, and idempotent. Indeed, if $K \subset L$, then the unbounded component of $\mathbb{C} \setminus L$ lies in the unbounded component of $\mathbb{C} \setminus K$, which gives $\mathrm{Fill}(K) \subset \mathrm{Fill}(L)$.

Since $K \cup L \subset \mathrm{Fill}(K) \cup L$, monotonicity gives the first inclusion in

$$\mathrm{Fill}(K \cup L) \subset \mathrm{Fill}(\mathrm{Fill}(K) \cup L).$$

Conversely, $\mathrm{Fill}(K) \cup L \subset \mathrm{Fill}(K \cup L)$, and the set on the right is already full. Filling the left side therefore stays inside $\mathrm{Fill}(K \cup L)$. This proves equality. Finite order independence follows by induction. □

For a fixed packet label $\ell$, let $S_\ell$ be the union of its original atom, when present, and all compact pieces assigned to that label. Its unique completion is

$$K_\ell = \mathrm{Fill}(S_\ell). \tag{9.8}$$

For $x \in K_\ell$, record $\mathrm{ch}_\ell(x) = \texttt{seed}$ when $x \in S_\ell$; otherwise record the bounded component of $\mathbb{C} \setminus S_\ell$ containing $x$. Thus a completed-packet merger has the geometric witness

$$(\ell, \ell', x, \mathrm{ch}_\ell(x), \mathrm{ch}_{\ell'}(x)), \qquad x \in K_\ell \cap K_{\ell'}. \tag{9.9}$$

Monotonicity shows that this witness remains a merger after any same-label enlargement $S_\ell \subset S'_\ell$. The same argument makes a localization escape $K_\ell \cap (\mathbb{C} \setminus U) \neq \varnothing$ persistent.

Let $P_i \subset A_i$ be the union of impressions forbidden by the protected marking at the last proof use. Define the unprotected exposure locus

$$\begin{aligned} \mathrm{Exp}(A_i, K_i; P_i) = \{x \in A_i \setminus P_i : \text{there is a Jordan arc } \gamma : [0,1] \to U, \\ \gamma(0) = x, \quad \gamma \cap K_i = \{x\}\}. \end{aligned} \tag{9.10}$$

This is actual access data; boundary membership alone is not substituted for (9.10).

**Proved Lemma 10.13** (Individual ports localize simultaneously)**.** *Suppose the completed atom packets $K_1, \dots, K_s \Subset U$ and obstacle packets $L_1, \dots, L_t \Subset U$ form a pairwise disjoint family and $\mathrm{Exp}(A_i, K_i; P_i) \neq \varnothing$ for every original atom. Then the access arcs can be chosen pairwise disjoint with positive interiors in*

$$U \setminus \left( \bigcup_i K_i \cup \bigcup_\ell L_\ell \right).$$

*If $K_i \subset K_i'$, then*

$$\mathrm{Exp}(A_i, K_i'; P_i) \subset \mathrm{Exp}(A_i, K_i; P_i). \tag{9.11}$$

*Proof.* Finite compact separation gives pairwise disjoint neighborhoods $N_i \Subset U$ of the $K_i$, each missing every obstacle packet. Choose one access arc from (9.10). Continuity at its landing point allows restriction to an initial subarc inside $N_i$. The restricted arcs miss every other packet and are pairwise disjoint because the $N_i$ are. For (9.11), every arc avoiding the larger packet $K_i'$ also avoids $K_i$. □

**Proved Theorem 10.14** (Fixed-Packet Closure and Port Audit)**.** *For a fixed finite typed assignment, the geometric augmented-packet clauses hold if and only if:*

*(1) the frozen inventory is covered and every raw seed $S_\ell$ is compact and connected;*
*(2) the unique completions $K_\ell = \mathrm{Fill}(S_\ell)$ are pairwise disjoint and compactly contained in $U$; and*
*(3) every original atom has nonempty unprotected exposure locus $\mathrm{Exp}(A_i, K_i; P_i)$.*

*The completions are independent of processing order. Merger, localization, and empty-exposure defects persist under monotone same-label enlargement. The complete audit is natural under a conformal homeomorphism of Jordan localizations that transports the full packets and protected marking.*

*Proof.* Lemma 10.12 gives unique order-independent completions. Connected raw seeds have full continuum completions. Pairwise disjointness and compact containment are exactly the absence of the intersections in (9.9) and with $\mathbb{C} \setminus U$. Lemma 10.13 is equivalent to the simultaneous port clause and gives port-loss persistence; filled-hull monotonicity gives persistence of the other stated defects.

A conformal homeomorphism of Jordan disks extends over their Jordan boundaries and, by Schoenflies on the complementary disks, to a sphere homeomorphism fixing infinity. Such a map carries the unbounded complementary component of every compact set to the corresponding unbounded component, hence commutes with Fill, and it transports the access arcs and protected sets. This proves naturality. □

The theorem does not construct the typed assignment. It proves that a favorable completion order and a global untangling of already existing individual ports are not additional degrees of freedom.

**Proved Proposition 10.15** (Cellwise packet-and-port certificate)**.** *Let $U$ be a Jordan domain and let $\overline{D}_1, \dots, \overline{D}_N \Subset U$ be pairwise-disjoint closed Jordan disks. Suppose every packet has a compact connected raw seed $S_\ell \Subset D_\ell$, the seeds cover the complete frozen inventory, and every original atom $A_i \subset S_i$ has an unprotected point $x_i \in A_i \setminus P_i$ and a Jordan arc $\gamma_i : [0,1] \to \overline{D}_i$ satisfying*

$$\gamma_i(0) = x_i, \qquad \gamma_i((0,1)) \subset D_i \setminus S_i, \qquad \gamma_i(1) \in \partial D_i. \tag{9.11a}$$

*Then $K_\ell = \mathrm{Fill}(S_\ell) \Subset D_\ell$, the completed packets are pairwise disjoint, every $x_i \in \mathrm{Exp}(A_i, K_i; P_i)$, and the packet system has simultaneous ports, one common free component, and a protected regulated connector.*

*Proof.* The connected exterior of $\overline{D}_\ell$ belongs to the unbounded component of $\mathbb{C} \setminus S_\ell$. Thus every bounded complementary component of $S_\ell$ lies in $D_\ell$, proving $\mathrm{Fill}(S_\ell) \Subset D_\ell$; the disjoint cells give packet disjointness. Concatenating $\gamma_i(1)$ with an arc to infinity in $\mathbb{C} \setminus D_i$ shows that $\gamma_i((0,1])$ lies in the unbounded component of $\mathbb{C} \setminus S_i$. Hence $\gamma_i \cap K_i = \{x_i\}$. Now lemma 10.13 and theorems 10.11 and 10.14 give the remaining conclusions. □

This proposition is sufficient only. The actual virtual theorem must still construct the cells, complete seeds, unprotected arcs, and their transport through the final Wave horizon.

10.7. **Horizon-first inventory audit.** The phrase "through the complete horizon" has mathematical content before any packet can be formed. A declared horizon skeleton is

$$\mathcal{H} = \left(B, N, \sigma, \{U_j^-, U_j^+, F_j, d_j\}_{0 \le j < N}\right), \tag{9.12}$$

where $B$ is the actual block, $N < \infty$ is the final proof-use stage, $\sigma$ is its stopping interface, and the remaining entries are the authorized localizations, selected branches, degrees, and critical words.

Let $X_{\mathcal{H}}$ be a finite typed set of actual component, collar, critical-exclusion, ambient-component, localization, protected-impression, transport, and curve-family records. Each record carries its stage, attachment trace, return word, parent route, sheet, and proof-use interval. Let $S_{\mathcal{H}}$ be the union of the records read directly by every Wave, Wanderers, covering, quasi-additivity, and width invocation through $\sigma$. Finally, orient a dependency edge $x \to y$ when retaining $x$ forces retention of $y$.

Starting with $I_0 = S_{\mathcal{H}}$, set

$$I_{r+1} = I_r \cup \{y : \text{there is } x \in I_r \text{ with } x \to y\}, \qquad I(\mathcal{H}) = \bigcup_{r \ge 0} I_r. \tag{9.13}$$

**Proved Theorem 10.16** (Horizon-first inventory and extension law)**.** *For fixed finite data as above:*

*(1) the iteration in* (9.13) *stabilizes after at most* $|X_{\mathcal{H}}| - |S_{\mathcal{H}}|$ *strict enlargements;*
*(2)* $I(\mathcal{H})$ *is the unique least set containing all proof seeds and closed under every declared dependency;*
*(3) if* $\mathcal{H} \preceq \mathcal{H}'$ *retains all old seeds and dependency edges, then* $I(\mathcal{H}) \subset I(\mathcal{H}')$*; and*
*(4) the old slice is stable,*

$$I(\mathcal{H}') \cap X_{\mathcal{H}} = I(\mathcal{H}), \tag{9.14}$$

*if and only if no seed of the extended horizon has a directed dependency path to a record in* $X_{\mathcal{H}} \setminus I(\mathcal{H})$*.*

*Thus a provisional inventory is final exactly when every admissible later extension satisfies* (9.14)*.*

*Proof.* The sets $I_r$ form an increasing sequence in the finite set $X_{\mathcal{H}}$. Every strict enlargement adds at least one record, so the stated bound follows. The limit contains all seeds and is dependency-closed. If $J$ is any other such set, induction on $r$ gives $I_r \subset J$, hence $I(\mathcal{H}) \subset J$. This proves leastness and uniqueness.

Under extension, every old seed and every old dependency path remains, so monotonicity follows. If an extended seed reaches an omitted old record, that record belongs to the left side of (9.14) but not its right side. Conversely, any record in the set-theoretic difference of the two sides is, by (9.13), reached by a path from an extended seed. This proves the equivalence and the final assertion. □

10.7.1. *Resolved stopping and final-root compiler.* The closure theorem becomes an actual horizon theorem only after the analytic proof program is resolved. Concretely, let $D$ be a finite acyclic decision graph. Label every edge $e$ by an actual pair

$$(g_e, \iota_e) : (V'_e, I'_e) \rightrightarrows (V_e, I_e),$$

its finite iterate length $s(e)$, complete critical and degree words, and proofs of the marking-refinement and proper/end-proper hypotheses used on that edge. Require every maximal path to end at a terminal node. Let the ledger list every theorem hypothesis, component, access, obstacle, curve family, and last-use stage. Finally let $\Phi : 2^X \to 2^X$ be the inflationary monotone operator containing every root-predecessor, attachment, access-survival, obstacle, sheet, critical, localization, and intervening-transport implication read by that ledger. For every generating implication, additionally require a proof that the corresponding geometric enlargement stays in the authorized localization and preserves, on every already certified proof support, the covering/immersion maps, exact marked action, marking refinement, proper/end-proper assertions, degree and critical words, obstacle decisions, and injective cyclic order of protected accesses. A nonzero collision rejects the program.

**Proved Theorem 10.17** (Resolved finite Wave-horizon compiler)**.** *For the resolved data above:*

*(1) the dynamical horizon*

$$H = \max_{\pi : q_0 \rightsquigarrow q_{\mathrm{term}}} \sum_{e \in \pi} s(e)$$

*is finite;*

*(2) dependency reachability gives the unique least complete proof-use inventory $C_\infty$;*

*(3) the iteration $R_0 = C_\infty$, $R_{r+1} = \Phi(R_r)$ stabilizes at the unique least final-root fixed point $R_\infty$, independently of every fair processing order, and all certified analytic branch assertions remain valid there; and*

*(4) after the program itself is extended, the old inventory is stable exactly when no new seed reaches an old omitted record.*

*Proof.* A directed path in a finite acyclic graph visits no vertex twice. There are therefore finitely many root-to-terminal paths, each has finitely many edges, and every recorded $s(e)$ is finite; the displayed maximum exists.

Starting from the ledger seeds, repeatedly adjoin the targets of all dependency edges whose sources are present. This increasing sequence stabilizes in finite $X$. Its limit is dependency-closed, and induction on the number of closure steps puts it inside every other closed set containing the seeds, proving leastness and uniqueness.

Since $\Phi$ is inflationary, the $R_r$ increase; since $X$ is finite, they stabilize. If $Y$ is any $\Phi$-fixed point containing $C_\infty$, monotonicity gives $R_r \subset Y$ for every $r$, hence $R_\infty \subset Y$. A fair sequential processing of the individual implications produces a fixed point, never leaves $R_\infty$, and hence equals $R_\infty$. The extension statement is precisely the reachability argument of theorem 10.16.

For analytic stability, follow any fair finite ordering of the generating enlargements from $C_\infty$ to $R_\infty$. The required stability proof preserves every listed analytic assertion at the first generator and, inductively, at every later generator. Hence all assertions hold at the fixed point. Since the hypothesis covers every generator in every fair order, the result is order-independent. Without these analytic generator proofs, the preceding argument establishes record inclusion only. □

In KKL's native category, the initial mostly-short alternative is followed, unless terminal, by successive critical-visit alternatives. Their bounded-visits proof sets $N_b = 216b^2$, so the recursion stops before it forms $N_b$ waves linked by critical visits. Each selected visit has a finite witnessed iterate $r_\nu$, and the actual iterate horizon is $\sum_\nu r_\nu$. The native marked branch is

$$(f^r, \iota_r) : f^{-r}(U) \setminus f^{-r}(T) \rightrightarrows U \setminus T, \qquad (f^r)_* \iota_r^* I = F_*^r I.$$

All Wave constants in that section are allowed to depend on $b$. Moreover, the fast-forward branch has a degree-one central part and degree-$2^m$ auxiliary ends, so it is not one globally univalent corridor. This is a native positive control, not a period-independent virtual result.

**Open Problem 10.18** (Virtual Wave Marked Correspondence)**.** For every actual virtual-Molecule block $B$, first define, independently of a completed horizon, a prefix-closed admissible finite return-word language $\mathscr{R}(B)$. For every finite provisional root package $P$ and every compatible $r \in \mathscr{R}(B)$, construct a marked hyperbolic complement $(V_P^{\mathrm{vir}}, I_P^{\mathrm{vir}})$ and actual pairs

$$(g_r^{\mathrm{vir}}, \iota_r^{\mathrm{vir}}) : (V_{P,r}^{\mathrm{vir}}, I_{P,r}^{\mathrm{vir}}) \rightrightarrows (V_P^{\mathrm{vir}}, I_P^{\mathrm{vir}})$$

with exact marked action. Prove marking refinement, complete prime-end and cyclic-order transport, component exhaustion, obstacle separation, every required proper/end-proper interval assertion, the complete degree word, and containment of all proof-used intermediate stages in the authorized localization. Prove functorial preservation of all certified assertions under every certified collision-free enlargement $P \subset P'$.

10.7.2. *Return language and prime-end action criterion.* A typed return graph has a distinguished root and actual directed edges. Each edge records its source and target localizations, holomorphic branch, positive iterate, complete critical and degree words, selected components, parent route, sheet, obstacles, and protected accesses. Let $\mathscr{R}(\mathcal{G})$ be the finite directed paths from the root, including the empty path.

**Proved Lemma 10.19** (Return-path language)**.** *The language $\mathscr{R}(\mathcal{G})$ is prefix closed. If $r = (e_1, \dots, e_k)$, its actual iterate is the finite integer*

$$s(r) = \sum_{j=1}^{k} s(e_j),$$

*and its edge records retain every intermediate component and critical encounter.*

*Proof.* Every initial edge sequence of a composable rooted path is again a composable rooted path. The displayed sum has finitely many positive integer terms. Intermediate data are read from each retained edge, not inferred from the terminal label. □

The lemma is a compiler statement. Constructing the actual typed graph for a virtual-Molecule block is open.

**Proved Theorem 10.20** (Forward-coherent complement pair)**.** *Let $R$ be a degree-$d$ polynomial and let $H \subset \mathbb{C}$ be a nondegenerate full continuum containing every finite critical value of $R$. If $R(H) \subset H$, then, with*

$$H' = R^{-1}(H), \qquad V = \mathbb{C} \setminus H, \qquad V' = \mathbb{C} \setminus H',$$

*$H'$ is a full continuum, $H \subset H'$, and*

$$(R, \iota) : V' \rightrightarrows V$$

*is an actual pair in which $R : V' \to V$ is a connected unbranched degree-$d$ covering and $\iota$ is complement inclusion.*

*Proof.* Every finite critical point maps to $H$, hence belongs to $H'$. Put $\Omega = \widehat{\mathbb{C}} \setminus H$. Fullness and nondegeneracy of $H$ make $\Omega$ a simply connected proper domain. Let $\Omega_\infty$ be the component of $R^{-1}(\Omega)$ containing infinity. A regular value sufficiently near infinity has all $d$ preimages in $\Omega_\infty$; hence this component has degree $d$. Every other pullback component would map properly and surjectively to $\Omega$ with positive degree. Degree additivity therefore gives

$$R^{-1}(\Omega) = \Omega_\infty.$$

After removing infinity, the restriction

$$R : \Omega_\infty \setminus \{\infty\} \longrightarrow \Omega \setminus \{\infty\}$$

is a connected unbranched degree-$d$ cover. A Riemann map sending $\infty$ to 0 identifies the target with a punctured disk. Its connected finite covers correspond to the subgroups $d\mathbb{Z} < \mathbb{Z}$, so the source is also a punctured disk. Filling its unique puncture proves that $\Omega_\infty$ is simply connected. Therefore $H' = \widehat{\mathbb{C}} \setminus \Omega_\infty$ is connected, while $\mathbb{C} \setminus H' = \Omega_\infty \setminus \{\infty\}$ is connected. Thus $H'$ is a full continuum. Forward invariance gives $H \subset R^{-1}(H) = H'$, hence $V' \subset V$. The restriction of $R$ has no finite critical point. Polynomial properness makes it proper, and every regular value in $V$ has all its $d$ preimages in $V'$. Thus it is the stated covering. □

For $R = f_c^n$, it suffices for the critical-value clause that $H$ contain $f_c^j(0)$, $1 \le j \le n$. Entire-carrier forward invariance, cusp-free localization, selected-component routing, and protected access remain separate hypotheses.

Let $S$ be a prime-end circle and let $E \subset S$ be closed and nowhere dense. The components $\mathcal{I}(E) = \pi_0(S \setminus E)$ form an ideal marking. If $(g, \iota) : V' \rightrightarrows V$ has defined protected prime-end pullbacks, write

$$E_g = g_\partial^{-1}(E), \qquad E_\iota = \iota_\partial^{-1}(E).$$

**Proved Theorem 10.21** (Prime-end marking action criterion)**.** *The covering-pulled marking refines the immersion-pulled marking if and only if*

$$E_\iota \subset E_g.$$

*If $\iota_\partial$ is an orientation-preserving circle homeomorphism and $\beta = g_\partial \iota_\partial^{-1}$, this is equivalent to $\beta(E) \subset E$. For a finite protected endpoint set $A$, the least closed forward-invariant candidate is*

$$E_A = \overline{\bigcup_{n \ge 0} \beta^n(A)}.$$

*An invariant endpoint marking containing $A$ exists exactly when $E_A$ is nowhere dense. A finite one exists exactly when every point of $A$ has finite forward orbit.*

*Proof.* For two closed nowhere-dense endpoint sets $E_1, E_2$, the marking $\mathcal{I}(E_1)$ refines $\mathcal{I}(E_2)$ exactly when $E_2 \subset E_1$: adding endpoints subdivides intervals, and omitting an old endpoint makes a proposed finer interval cross a coarse endpoint. Apply this statement on the source circle to $E_g, E_\iota$.

After identifying the circles by $\iota_\partial$, the endpoint containment is $\beta(E) \subset E$. The closure in the definition makes $E_A$ closed, while continuity gives

$$\beta(E_A) \subset \overline{\bigcup_{n \ge 1} \beta^n(A)} \subset E_A.$$

Every closed forward-invariant set containing $A$ contains every $\beta^n(A)$, hence contains $E_A$. This proves leastness and the nowhere-dense equivalence. Finiteness is equivalent to finiteness of all forward orbits of the finite set $A$. □

**Proved Proposition 10.22** (Boundary-phase obstruction)**.** *Carrier, degree, and prime-end-circle data do not determine marked action.*

*Proof.* On $V = \mathbb{C} \setminus \overline{\mathbb{D}}$, set

$$g(z) = z^2, \qquad \iota_\alpha(z) = e^{i\alpha} z.$$

For the marking with endpoint $E = \{1\}$,

$$E_g = \{1, -1\}, \qquad E_{\iota_\alpha} = \{e^{-i\alpha}\}.$$

By theorem 10.21, the pair acts exactly when $\alpha = 0$ or $\pi \pmod{2\pi}$, although all carrier and degree data are unchanged. □

For an enlargement $P \preceq Q$, suppose embeddings $a, a'$ satisfy

$$ag_Q = g_P a', \qquad a\iota_Q = \iota_P a',$$

extend injectively and cyclic-order preservingly on every protected prime-end interval, and pull both endpoint sets back by exact inverse image. The same two identities show directly that $E_{\iota,Q} \subset E_{g,Q}$ follows from $E_{\iota,P} \subset E_{g,P}$. Thus exact marked action is natural under finite compositions of such certified enlargements. Carrier monotonicity alone does not supply these squares or endpoint equalities.

The corrected source-facing inputs to open problem 10.18 are:

(1) *Return branch language*: construct the actual typed graph before the Wave execution;
(2) *Provisional complement pair*: construct the forward-coherent carriers, selected components, localizations, and obstacle ledgers for every finite compatible path; and
(3) *Access action certificate*: fix the relative-cover phase by a base access, prove the endpoint containment and all proper/end-proper flags, and prove the enlargement squares.

All three remain open. They are quantified before the completed horizon, so this decomposition introduces no dependency cycle.

10.7.3. *Analytic return-atlas refinement.* The first item above requires more than a directed graph of actual edge labels. Label composability does not ensure that the corresponding holomorphic restrictions have a nonempty composite domain.

For a polynomial $f$ and an open return base $Y$, define the strict $q$-return region

$$\mathcal{R}_q(f, Y) = Y \cap f^{-q}(Y) \cap \bigcap_{j=1}^{q-1} f^{-j}(\mathbb{C} \setminus \overline{Y}). \tag{9.16}$$

It is open. Every connected component carries an actual restriction $f^q|_A : A \to Y$.

**Proved Proposition 10.23** (Return-base underdetermination)**.** *A quadratic polynomial, compact seed, and instruction to take first returns to a connected Jordan neighborhood do not determine the first return time.*

*Proof.* Take $f(z) = z^2 - 1$ and seed $K = \{-1\}$. Its orbit starts

$$-1 \longmapsto 0 \longmapsto -1.$$

For

$$Y_s = D(-1, 1/4), \qquad Y_\ell = D(-1/2, 3/4),$$

both disks contain $K$. The point 0 is outside $\overline{Y_s}$, while $-1 \in Y_s$, so $-1 \in \mathcal{R}_2(f, Y_s)$ and its first return time is two. Both $-1$ and 0 lie in $Y_\ell$, so $-1 \in \mathcal{R}_1(f, Y_\ell)$ and its first return time is one. □

Thus Dudko's compact small-Julia data do not determine the open return base. The checked KKL bibliography records the promised virtual-renormalization work as unpublished and in preparation; no theorem from it is imported.

Fix now actual open localizations $Y_\sigma$. An atomic branch is an actual restriction

$$F_b = f^{q_b}|_{A_b} : A_b \to Y_{\tau_b},$$

together with its complete critical and degree words, component and parent routes, obstacles, localization checks, selected sheets, protected accesses, and normalized access phase. An actual prefix is

$$\Phi_\nu = f^{S_\nu}|_{D_\nu} : D_\nu \to Y_{\sigma_\nu}.$$

Extending $\nu$ by $b$ requires selecting an actual nonempty component

$$C \in \pi_0\big(\Phi_\nu^{-1}(A_b)\big). \tag{9.17}$$

On that component,

$$F_b \circ \Phi_\nu|_C = f^{S_\nu+q_b}|_C. \tag{9.18}$$

A Wave demand relation $\mathscr{D}_B(\nu)$ is called frozen if it is defined from the block and the syntax of every possible theorem call before the stopping execution. It is proof-complete if every continuation that Wave, Wanderers, covering, quasi-additivity, or width arguments can request appears in the relation. These conditions may not be made tautological by defining admissibility from the already constructed atlas.

**Proved Theorem 10.24** (Demand-closed analytic return-atlas compiler)**.** *Suppose all atomic branches, components in* (9.17)*, metadata, and a frozen proof-complete demand relation are actual and verified. Starting from the identity root and repeatedly adjoining every demanded child gives a unique least rooted unfolding. Every node is an actual restriction of an iterate of $f$; a depth-$n$ word has iterate $\sum_{j=1}^{n} q_j$, its critical and degree words are the exact chronological concatenations, and its proper degree, when defined, is the product of the selected edge degrees. The language is prefix closed, horizon independent, and exhaustive for every finite declared continuation.*

*Proof.* Let $T_0$ contain only the identity prefix and let $T_{n+1}$ add every child prescribed by $\mathscr{D}_B$ to a prefix in $T_n$. Inductively, (9.18) makes each child the actual iterate with summed exponent. The two chronological iterate ranges before and after the extension are disjoint and ordered, so the critical and degree words concatenate in that order; proper degrees multiply under composition.

Retaining the full word and selected component in each node gives a unique parent by deleting the last extension, hence every prefix is retained. If $X$ contains the root and is closed under every demand, induction gives $T_n \subset X$ for all $n$. Therefore $T_\infty = \bigcup_n T_n$ is the unique least demand-closed unfolding. No stopping index occurs in this construction. Proof-completeness and induction on the finite continuation length give exhaustiveness. □

**Proved Theorem 10.25** (Return-atlas enlargement naturality)**.** *Suppose two atlases are related edgewise by injective analytic squares that preserve the iterate, exact pullback component, critical and degree words, components, obstacles, localization, sheets, protected-access cyclic order, and normalized phase. Require distinct demanded branch-component pairs to remain distinct. Then the old rooted unfolding injects uniquely and prefix-preservingly into the enlarged unfolding, with exact composite squares and unchanged iterate sums. It is an isomorphism exactly when the map is bijective on demanded branch-component pairs at every reached prefix.*

*Proof.* Map the root by the declared base embedding. Assuming a prefix square commutes, the atomic square and (9.17) select the declared child component; composing the two squares proves the child square. Fiberwise injectivity prevents two old children from merging. Induction gives the unique prefix map and preserves every recorded field. Surjectivity follows by the same depth induction precisely when every new demanded child has an old preimage. Without that bijection, a late demanded child may be born although all old squares still commute. □

**Open Problem 10.26** (Virtual Return Base Atlas and Wave Demand Schema)**.** For every actual virtual-Molecule block and provisional root package, construct the open return bases, nice-boundary rule, atomic branch components, integer iterates, and every metadata field required above. Before executing a stopping rule, construct the source-faithful, proof-complete Wave demand relation. Prove exact enlargement morphisms and fiberwise demand bijectivity wherever complete no-birth stability is claimed.

Theorems 10.24 and 10.25 are proved compiler theorems. Open problem 10.26 is the first source-facing construction and remains open. Consequently the return branch language, marked correspondence, horizon certificate, VAAPC-I, UPTCR, and MLC remain open.

10.8. **Native principal-nest return atlas and the virtual selector gap.** There is an actual source-backed positive control for the analytic-base part of open problem 10.26. Let

$$Y^0 \supset V^0 \supset V^1 \supset \cdots$$

be Lyubich's principal nest for a quadratic-like map $f$, and, for a returning level $n \geq 2$, set

$$B_n = \operatorname{int} V^{n-1}.$$

The source boundary avoidance gives

$$f^m(\partial V^{n-1}) \cap \operatorname{int} V^{n-2} = \varnothing \quad (m \geq 1), \qquad V^{n-1} \Subset \operatorname{int} V^{n-2}.$$

It follows immediately that

$$f^m(\partial V^{n-1}) \cap V^{n-1} = \varnothing,$$

so the base is nice. This uses the native theorem in its stated puzzle category; it does not identify a virtual scale with a principal-nest level [Lyu97].

**Proved Theorem 10.27** (Native principal-nest return atlas)**.** *For every returning native principal-nest level $n \geq 2$, the first return to $B_n$ decomposes into pairwise disjoint actual open puzzle components $A_{n,i} \Subset B_n$, with integer return times $q_{n,i}$, such that*

$$f^{q_{n,i}} : A_{n,i} \longrightarrow B_n$$

*is proper and onto. Exactly one domain is central. Its chronological degree word is $(2, 1, \ldots, 1)$; every other branch is univalent and has word $(1, \ldots, 1)$. The actual pullback chain records every component, parent, and sheet. The branch maps extend to the prime-end compactifications, preserve cyclic order, and admit the phase normalization induced by external angle, without assuming local connectivity of the Julia set.*

*Proof.* For a returning point $z \in B_n$, let $q$ be its first return time and pull $V^{n-1}$ back along

$$z, f(z), \ldots, f^q(z).$$

The Markov property makes the selected pullback a puzzle component. If an intermediate image of its interior met $B_n$, the nesting/disjointness alternative for puzzle interiors would force an earlier return. Hence its interior $A$ is an actual component of

$$B_n \cap f^{-q}(B_n) \cap \bigcap_{k=1}^{q-1} f^{-k}(\mathbb{C} \setminus \overline{B_n}).$$

Conversely every such puzzle component is a first-return domain. Markov disjointness gives pairwise disjointness.

If a closed return component met $\partial V^{n-1}$, its pullback chain would give a boundary point of $V^{n-1}$ returning to $V^{n-1}$, contrary to niceness. Thus $\overline{A} \Subset B_n$. The closed pullback maps onto $V^{n-1}$; restriction to the interiors is therefore proper and onto.

The unique critical point 0 belongs to $B_n$. If $0 \in f^k(A)$ for $0 < k < q$, then the intermediate image meets $B_n$, again contradicting first return. Thus a noncentral branch has no critical step. The unique central branch contains 0 at time zero, so its first step has degree two and every later step has degree one. This proves the two degree words and simultaneously fixes the chronological component and sheet records.

Finally, puzzle pieces are Jordan disks bounded by finitely many ray/equipotential arcs. Conjugating a branch by Riemann maps produces a disk automorphism in the univalent case and a degree-two Blaschke product in the central case. Both extend to their prime-end circles and

preserve cyclic order. The external ray angles mark the boundary vertices and fix the base phase. This argument uses the Jordan puzzle boundary, not local connectivity of all of $J(f)$. □

An isotopy class of such neighborhoods is not an analytic substitute for the embedded base.

**Proved Proposition 10.28** (Relative-isotopy return underdetermination)**.** *An ambient isotopy class of Jordan neighborhoods relative to a compact seed need not determine first-return time.*

*Proof.* For $f(z) = z^2 - 1$ and seed $\{-1\}$, consider

$$Y_s = D(-1, 1/4), \qquad Y_\ell = D(-1, 5/4).$$

The affine isotopy

$$h_t(z) = -1 + (1 + 4t)(z + 1), \qquad 0 \le t \le 1,$$

fixes $-1$ and takes $Y_s$ to $Y_\ell$. Nevertheless $-1 \mapsto 0 \mapsto -1$, with $0 \notin \overline{Y_s}$ and $0 \in Y_\ell$. The respective first-return times of the seed are two and one. □

The checked virtual sources provide a satellite-copy index $s$ and a terminal primitive Yoccoz level, but no map

$$s \longmapsto \nu(s)$$

selecting an embedded nice base at every intermediate virtual scale. They likewise do not prove that the virtual carrier, retained critical orbit, Wave accesses, obstacles, and demand relation all lie in the selected base. KKL's native neighborhood $U_I$, chosen only up to isotopy relative to the small Julia cycle, cannot supply this missing analytic choice in view of proposition 10.28 [KKL26, Dud25].

**Open Problem 10.29** (Virtual Scale-to-Puzzle Selector)**.** For every actual virtual-Molecule block and every intermediate copy index $s$, construct before the Wave stopping execution an index $\nu(s)$, an actual embedded nice puzzle or principal-nest base, the completed virtual carrier and retained critical orbit inside it, all proof-used prime-end accesses with normalized phase, and the complete obstacle and sheet ledger. Prove that every branch requested by the frozen Wave demand schema occurs in the selected atlas, and prove exact component and marking naturality under provisional-to-final root enlargement. All assertions must be independent of satellite period, rotation denominator, chain length, and every downstream width, compactness, or rigidity conclusion. If one base cannot carry the demands, construct an actual multi-base return groupoid with iterate restrictions as transition maps.

10.9. **The first direct-decoration scale.** The universal selector is open, but its first scale in the original connected dynamical plane has a source-defined analytic skeleton. Fix a non-principal decoration outside a directly attached satellite copy $M_{p/q}$, $q > 1$. In the Kahn–Lyubich decoration puzzle let $Y^1$ be the critical depth-one piece and let $Z^1_j$ be the off-critical depth-one pieces. Define

$$n = \min\{m \ge 1 : f^{mq}(0) \in \operatorname{int} Z^1_j \text{ for some } j\}. \tag{DS1}$$

The terminal component $Z^1_\kappa$ and all earlier pullback components give the escape route $\bar{\kappa}$. The selected escape piece is

$$V_{\mathrm{esc}} = f^{-nq}(Z^1_\kappa)|_0 = Y^{nq+1}, \qquad \nu_{\mathrm{dec}}(1) = nq + 1. \tag{DS2}$$

These objects and the associated modified principal nest are constructed in [KL08, KL09].

**Proved Theorem 10.30** (First decoration-scale escape atlas)**.** *At the direct-decoration scale, $V_{\mathrm{esc}}$ has actual first-return domains $W_i \Subset \operatorname{int} V_{\mathrm{esc}}$ and integer return times $r_i$ such that*

$$f^{r_i} : W_i \longrightarrow \operatorname{int} V_{\mathrm{esc}}$$

*is the corresponding return branch. Exactly one domain contains* 0*; its chronological degree word is* $(2,1,\dots,1)$*, while every other word is* $(1,\dots,1)$*. The ordered pullback components determine the parent and sheet records, and the ray/equipotential boundaries determine cyclic-order-preserving native prime-end markings.*

*Proof.* The least-event definition (DS1) is finite because the parameter is outside the candidate satellite copy. Disjointness of the $\operatorname{int} Z_j^1$ gives the unique terminal component, and Markov pullback gives the unique route and (DS2). Kahn–Lyubich define this piece as the first kingdom and prove that every domain of its full first-return map is compactly contained in it [KL08, KL09].

Each branch is selected as an actual connected pullback component before its word is recorded. If an intermediate image of a first-return domain contained 0, it would meet the return base because 0 belongs to that base, contradicting first return. Thus the unique central domain has its only critical encounter at time zero; every other one-step restriction is univalent. This proves the two degree words and fixes the ordered component and sheet records.

The puzzle domains are Jordan disks bounded by finitely many external-ray and equipotential arcs. Univalent branches extend homeomorphically to prime-end circles and preserve cyclic order; the central branch extends as an orientation-preserving degree-two boundary covering. External angles fix the native marking. □

The terminal primitive branch supplied by the modified nest has the form

$$f^P : A_0 \longrightarrow B, \qquad A_0 \Subset B. \tag{DS3}$$

Let $K_0$ be its filled Julia set and $K_i = f^i(K_0)$.

**Proved Proposition 10.31** (Critical-base cycle exclusion)**.** *For* $1 \le i < P$*,*

$$K_i \cap B = \varnothing.$$

*Thus the first component of the terminal little-Julia cycle excluded from the one critical base is* $K_1$*. Its complete cycle itinerary is*

$$K_1 \to K_2 \to \cdots \to K_{P-1} \to K_0,$$

*and the full chronological degree word, starting at* $K_0$*, is* $(2,1,\dots,1)$*.*

*Proof.* Equation (DS3) is a first-return branch, so

$$f^i(A_0) \cap B = \varnothing, \qquad 1 \le i < P.$$

Since $K_0 \subset A_0$, the same statement holds for $K_i = f^i(K_0)$. Primitivity makes the $K_i$ pairwise disjoint. The only critical point is $0 \in K_0$; an intermediate critical encounter would also contradict first return. Hence only the first factor has degree two. □

Consequently the actual replacement skeleton is the multi-base chain

$$A_0 \longrightarrow A_1 \longrightarrow \cdots \longrightarrow A_{P-1} \longrightarrow B, \qquad A_i = f^i(A_0), \tag{DS4}$$

together with the inclusion $A_0 \hookrightarrow B$. The checked source does not itself construct a continuum in $J(f)$ joining the pairwise-disjoint $K_i$. The native KKL terminal Hubbard tree constructed below supplies a connected terminal anchor and ideal attachment side, but not a protected analytic prime end for a partial virtual carrier.

**Proved Proposition 10.32** (Raw decoration-fan growth)**.** *The quadratic separator* $Y^{(n-1)q+1}$ *has exactly* $2^n$ *vertices. Kahn–Lyubich construct a distinct pairwise-disjoint univalent pullback* $Q^v$ *for every vertex* $v$*. Hence any raw marking retaining all separator pairs has at least* $2^n$ *entries and has no universal cardinality bound as* $n$ *varies.*

*Proof.* The vertex count and the one-per-vertex pairwise-disjoint pullbacks are the source statements in [KL08]. Distinct disjoint pullbacks require distinct marked records, so the cardinality is at least $2^n$, which is unbounded. □

Theorem 10.30 proves the first relative depth and native return atlas, but propositions 10.31 and 10.32 force a protected multi-base connector and require either a uniformly bounded proof-complete subset of the raw fan or a future-faithful access compression of the demanded subset. The decoration source alone does not prove that all $2^n$ vertices are Wave demands. At deeper virtual copies, repeating the construction would pass through a disconnected intermediate satellite object and would assume the virtual renormalization being constructed.

10.10. **Terminal Hubbard-tree anchor.** In KKL's native periodically repelling prime primitive $\psi$-ql category, the complete cycle $\mathcal{K}$ has a compact forward-invariant Hubbard tree $\mathcal{T}$. Its vertex continua and edge paths form a finite abstract tree $G_{\mathcal{T}}$. When the filled carrier is not locally connected, KKL use the locally connected rational-lamination model and lift the corresponding objects. We import this construction only in that native category.

For a nonempty finite vertex set $S \subset V(G_{\mathcal{T}})$, set

$$[S]_G = \bigcup_{u,v \in S} [u, v]_G, \tag{HT1}$$

the union of the unique tree paths, and let $H_{\mathcal{T}}(S)$ be its geometric realization in $\mathcal{T}$.

**Proved Theorem 10.33** (Native Hubbard-subtree functoriality)**.** *The set $[S]_G$ is the unique least connected subgraph containing $S$, and $H_{\mathcal{T}}(S)$ is compact and connected. For every declared tree map $\Phi$,*

$$[\Phi(S)]_G \subset \Phi([S]_G). \tag{HT2}$$

*Equality holds if $\Phi$ is injective on $[S]_G$. If $S \subset S^+$, every component of $[S^+]_G \setminus [S]_G$ has exactly one gate on the old subtree and contains a vertex of $S^+ \setminus S$.*

*Proof.* Every connected subgraph containing $S$ contains the unique path between each pair of its vertices, proving leastness. The geometric realization is the inverse image of a connected finite subtree under the monotone quotient that collapses each continuum vertex, hence is compact and connected.

The image $\Phi([S]_G)$ is connected and contains $\Phi(S)$, giving (HT2) by leastness. Under injectivity it is homeomorphic to $[S]_G$; all its leaves are images of leaves of $[S]_G$, hence lie in $\Phi(S)$. A finite tree is the hull of its leaves, proving equality.

For an enlargement, a component meeting the old subtree at two points would combine with the old path between those points to form a cycle. If it contained no new selected vertex, it could be deleted without losing $S^+$, contradicting leastness. □

**Proved Proposition 10.34** (Terminal-cycle Hubbard anchor)**.** *Under the native KKL hypotheses, for*

$$S_{\mathrm{cyc}} = \{K_0, \dots, K_{P-1}\}$$

*the geometric hull is $\mathcal{T}$. The first edge on the unique path from $K_1$ to $K_0$ selects a native ideal attachment side $I_{1\to 0}$. Its component itinerary is*

$$K_1 \to K_2 \to \cdots \to K_{P-1} \to K_0,$$

*and the cycle degree word beginning at $K_0$ is $(2, 1, \dots, 1)$.*

*Proof.* KKL define $\mathcal{T}$ as the least connected carrier of the complete cycle, so theorem 10.33 gives the first assertion. The abstract tree has one path between the two indicated vertices and one first edge at $K_1$; KKL's native ideal marking records its side. The itinerary and degree word are those of proposition 10.31. □

This refines the earlier access record to

$$\chi_{\text{ideal}} = 0, \qquad \chi_{\text{protected analytic}} = \bot. \tag{HT3}$$

Connectedness does not prove fullness, and the native ideal side does not automatically survive as a protected crosscut chain in a partial carrier.

**Proved Proposition 10.35** (Finite exact Cantor encoding obstruction)**.** *An infinite family of pairwise distinct virtual small-Julia components has no injective identification with a finite terminal-tree vertex package.*

*Proof.* An injection into a finite set would make the component family finite. □

**Proved Proposition 10.36** (Finite quotient observability criterion)**.** *Let $q : \mathscr{K} \to S$ be a map to a finite set and let $\mathcal{O} : \mathscr{K} \to \prod_{t\in\mathscr{H}} Y_t$ be the joint observable over a declared horizon. A map $\overline{\mathcal{O}}$ with $\mathcal{O} = \overline{\mathcal{O}} \circ q$ exists exactly when*

$$q(X) = q(Y) \Longrightarrow \mathcal{O}(X) = \mathcal{O}(Y). \tag{HT4}$$

*Proof.* Factorization implies (HT4). Conversely, (HT4) makes $\overline{\mathcal{O}}(a) := \mathcal{O}(X)$, $X \in q^{-1}(a)$, independent of the representative. □

Dudko's virtual object is described using Cantor small Julia sets, whereas KKL's terminal tree starts from one finite primitive cycle. The copy-chain data alone do not construct the scale-2 address space $\Omega_2$, embedded components $K_{2,\omega}$, or analytic branch action $\mathcal{R}_2$. Before the decoration pullbacks are inserted, the first failure record is therefore

$$\mathfrak{X}_2^{\text{type}} = (2, \mathcal{M}^{(0)}, \mathcal{M}^{(1)}, \mathcal{M}^{(2)}, \mathscr{K}_2, \Omega_2, \mathcal{R}_2, \rho_2, \chi_{\text{type}} = 1), \tag{HT5}$$

with itinerary, sheet, access, and localization coordinates undefined, not zero. Naming a first excess component before realizing $\omega \mapsto K_{2,\omega}$ would be circular.

10.11. **Direct-decoration pullback compactification.** The decoration source supplies more dynamical data than the copy chain alone. Let $P = Y^{(n-1)q+1}$, let $g = f^q$, and let

$$g : Q_a \longrightarrow P, \qquad a \in A := \{0, 1\}, \tag{HT6}$$

be the two actual univalent pullbacks inside $P$. Here $P, Q_0, Q_1$ are closed Jordan puzzle pieces, $Q_0 \cap Q_1 = \varnothing$, and each proper degree-one restriction extends to a homeomorphism of closed pieces. These closed-piece assertions, not merely disjointness of interiors, are used below. Write $\phi_a : P \to Q_a$ for their inverse branches. Let $K_0 \Subset \operatorname{int} V^0$ be the terminal primitive root and assume explicitly that $K(f) = J(f)$. The source construction makes $V^0, Q_0, Q_1$ distinct depth-$nq+1$ puzzle pieces. Their interiors are disjoint, and $K_0$ is compactly contained in the first one; hence

$$K_0 \cap (Q_0 \cup Q_1) = \varnothing. \tag{HT7}$$

For clarity, the separation used below is separation of the closed pieces. Suppose $Q_0, Q_1$ met at $z$. Injectivity of the restrictions alone would not make $z$ regular: a holomorphic map can be injective on a closed piece while critical on its boundary, and any meeting point lies on both boundaries. Instead, the interiors of distinct depth-$(nq+1)$ pieces are disjoint complementary components of the depth-$(nq+1)$ puzzle graph, so $z$ lies on that graph. The critical points of $g = f^q$ are the points $\zeta$ with $f^i(\zeta) = 0$ for some $0 \le i < q$, and $f$ maps the depth-$d$ puzzle graph into the depth-$(d-1)$ graph; a critical $z$ would therefore place 0 on a puzzle graph. But the critical pieces $Y^d$ are defined at every depth used—the critical orbit avoids the puzzle graphs [KL08]—so 0 lies in $\operatorname{int} Y^d$ and never on a graph. Hence $z$ is a regular point of $g$, and the local inverse of a small Jordan half-neighborhood of $g(z)$ in $P$ would belong to both pullback components, contradicting

their distinctness. Thus $Q_0 \cap Q_1 = \varnothing$. Proper degree one also makes each restriction $g : Q_a \to P$ a homeomorphism on the closed Jordan pieces.

For $w = a_1 \cdots a_m$, put

$$\phi_w = \phi_{a_1} \circ \cdots \circ \phi_{a_m}, \qquad Q_w = \phi_w(P), \qquad K_w = \phi_w(K_0). \tag{HT8}$$

For $\omega \in A^{\mathbb{N}}$, put

$$E_\omega = \bigcap_{m \geq 1} Q_{\omega|m}. \tag{HT9}$$

The address compactification is

$$\Omega_\partial = \{w\#^\infty : w \in A^{<\mathbb{N}}\} \cup A^{\mathbb{N}} \subset \{0, 1, \#\}^{\mathbb{N}}. \tag{HT10}$$

**Proved Theorem 10.37** (Direct-decoration pullback compactification)**.** *The space $\Omega_\partial$ is compact. The set*

$$\mathcal{H}_\partial = \bigcup_{w \in A^{<\mathbb{N}}} K_w \ \cup \ \bigcup_{\omega \in A^{\mathbb{N}}} E_\omega \tag{HT11}$$

*is a compact subset of $J(f)$, and its connected components are exactly the displayed finite pullbacks and end fibers. The realization*

$$w\#^\infty \mapsto K_w, \qquad \omega \mapsto E_\omega \tag{HT12}$$

*is upper semicontinuous. Moreover,*

$$g(K_{aw}) = K_w, \qquad g(E_{a\omega}) = E_\omega, \tag{HT13}$$

*by actual univalent branches of $f^q$. Geometric nesting and dynamical transport are the distinct maps*

$$\mathrm{gpar}(wa) = w, \qquad \sigma_{\mathrm{pre}}(aw) = w. \tag{HT14}$$

*The partial map $\sigma_{\mathrm{pre}}$ is defined on non-root finite addresses and on all infinite addresses. The deleted first symbol is the inverse sheet and every transition in (HT13) has time $q$. It is undefined at the empty word, whose separate root transition is*

$$g(K_0) = f^q(K_0) = K^{\mathrm{term}}_{q \bmod P_{\mathrm{term}}}, \tag{HT15}$$

*recorded as $\tau_{\mathrm{root}}$. It is not an extension of the deletion shift or an assertion of permanent escape from $\mathcal{H}_\partial$.*

*Proof.* The product $\{0, 1, \#\}^{\mathbb{N}}$ is compact. The subset (HT10) is closed: after the first $\#$, every later symbol must be $\#$. Finite words are isolated, and finite words of increasing length converge to an end exactly by eventual agreement on every fixed prefix.

Every $Q_w$ is a compact connected puzzle disk and $g^{|w|} : Q_w \to P$ is univalent. The cylinders are nested on appending a symbol. Two incomparable words lie in the images of the disjoint first-level components $Q_0, Q_1$ under their common inverse prefix and are therefore disjoint. Thus (HT9) is a nonempty compact connected set. Every $z \in E_\omega$ satisfies $g^m(z) \in P$ for all $m$. Its intermediate iterates lie in the finite bounded union $\bigcup_{r=0}^{q-1} f^r(P)$, so $z \in K(f) = J(f)$. Complete invariance of $J(f)$ puts every $K_w$ in $J(f)$.

If one finite word is a strict prefix of another, (HT7), pulled back by their common inverse branch, separates the ancestor $K_w$ from the descendant. The same argument separates a finite fiber from every end extending its word. First-differing cylinders separate every other pair.

To prove compactness of (HT11), let $x_j$ be a convergent sequence in it and pass to convergent address labels. At a finite label the labels are eventually constant. At an end $\omega$, for every $m$ the points eventually lie in the closed cylinder $Q_{\omega|m}$, hence the limit lies in $E_\omega$. This also proves closed graph, equivalently upper semicontinuity. The same cylinder separations, together with (HT7), show that no connected subset meets two displayed fibers, so they are the components.

Finally, $g\circ\phi_a = \mathrm{id}_P$ proves the first formula in (HT13). Since $g : Q_a \to P$ is a homeomorphism, it commutes with the nested intersection in (HT9), proving the second. The first symbol is therefore deleted dynamically, while cylinder inclusion deletes the last symbol. This defines only $\sigma_{\mathrm{pre}}$. Formula (HT15) follows from the terminal cycle and defines the distinct root arrow $\tau_{\mathrm{root}}$. □

**Proved Theorem 10.38** (Pullback-gap enlargement naturality)**.** *Let*

$$K_0 \subset S \subset S^+ \subset P \setminus (Q_0 \cup Q_1) \tag{HT16}$$

*be compact connected roots. Replacing $K_w$ by $\phi_w(S)$ or $\phi_w(S^+)$ preserves all component separations and (HT13)–(HT14). The inclusions $\phi_w(S) \subset \phi_w(S^+)$ commute with $g$ at every finite word, and the end fibers are unchanged.*

*Proof.* The proof of theorem 10.37 uses the root only through compactness, connectedness, and (HT7), all of which follow from (HT16). Injectivity gives $\phi_w(S) \subset \phi_w(S^+)$, and $g \circ \phi_a = \mathrm{id}$ gives the commuting squares. The end fibers depend only on $P, Q_0, Q_1$. □

This theorem concerns the topological finite-fiber package only. It does not assert that $S$ or $S^+$ is full, dynamically protected, access-complete, or admissible as an analytic virtual carrier. Those properties remain subject to the component, obstacle, and protected-access audits.

Outside the safe gap, define the actual collision ledger by all nonempty intersections between two distinct finite pullbacks or between a finite pullback and an end fiber. Leaving the gap invalidates the preceding sufficient proof but is not itself a collision.

**Proved Lemma 10.39** (Nested finite-full-packet nonseparation)**.** *Let $F_0 \supset F_1 \supset \cdots$ be compact subsets of the sphere, each a finite union of pairwise-disjoint full continua. Then $\bigcap_m F_m$ has connected complement. If $U$ is a Jordan domain, $F_m \Subset U$, and $F_m \cup (\widehat{\mathbb{C}} \setminus U)$ is such a packet, then $U \setminus \bigcap_m F_m$ is connected.*

*Proof.* Lemma 10.10 makes every $\widehat{\mathbb{C}} \setminus F_m$ connected. These complements increase, share infinity, and their union is the complement of the intersection. The localized statement adjoins the exterior full continuum. □

**Proved Theorem 10.40** (Direct-decoration full fibers and nonseparation)**.** *Every $K_w$ and $E_\omega$ in theorem 10.37 is full. Moreover, $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial$ is connected.*

*Proof.* The filled Julia set $K_0$ is full. Each homeomorphism $\phi_w : P \to Q_w$ of closed Jordan disks extends to a sphere homeomorphism, so $K_w = \phi_w(K_0)$ is full. Also,

$$\widehat{\mathbb{C}} \setminus E_\omega = \bigcup_m \left(\widehat{\mathbb{C}} \setminus Q_{\omega|m}\right)$$

is an increasing union of connected sets containing infinity.

For $m \geq 0$, define

$$\mathcal{H}_m = \bigcup_{|u|<m} K_u \;\cup\; \bigcup_{|w|=m} Q_w. \tag{DG1}$$

The terms are full and pairwise disjoint. Equal-depth cylinders are disjoint; finite fibers are disjoint; and if $u$ is a strict prefix of $w$, the root-gap relation (HT7) puts $K_u$ outside the child containing $Q_w$. The sets (DG1) decrease. Their intersection is exactly $\mathcal{H}_\partial$: outside the explicit finite fibers, a point in every approximation determines one unique compatible infinite cylinder address. Apply lemma 10.39. □

**Proved Lemma 10.41** (Full-continuum Jordan neighborhood)**.** *If a nondegenerate full continuum $K$ is compactly contained in an open set $G$, there is a closed Jordan disk $D$ such that*

$$K \subset \operatorname{int} D \subset D \Subset G. \tag{DG2}$$

*Proof.* Uniformize the simply connected exterior of $K$ by $\psi : \{|z| > 1\} \to \widehat{\mathbb{C}} \setminus K$. The filled interiors of $\psi(\{|z| = R\})$, $R > 1$, are closed Jordan disks and decrease to $K$ as $R \downarrow 1$. If none lay in $G$, compactness would put a point of the closed set $\widehat{\mathbb{C}} \setminus G$ in their intersection $K$. □

Let $G_0$ be the component of $\operatorname{int} P \setminus (Q_0 \cup Q_1)$ containing $K_0$. Equation (HT7) gives $K_0 \Subset G_0$. Choose $D$ by lemma 10.41 and put $D_w = \phi_w(D)$.

**Proved Theorem 10.42** (Direct-decoration gap-cell atlas)**.** *The closed Jordan cells $D_w$ are pairwise disjoint, including for ancestor–descendant words, and satisfy*

$$K_w \subset \operatorname{int} D_w, \qquad D_w \Subset \operatorname{int} Q_w \setminus (Q_{w0} \cup Q_{w1}), \tag{DG3}$$

$$D_w \cap \mathcal{H}_\partial = K_w, \qquad g(D_{aw}) = D_w. \tag{DG4}$$

*If $S_w \Subset \operatorname{int} D_w$ is compact and connected, then*

$$\operatorname{Fill}(S_w) \Subset \operatorname{int} D_w. \tag{DG5}$$

*Proof.* Incomparable words have disjoint cylinders. If $v = uat$, then $D_v$ lies in the child $Q_{ua}$, while $D_u$ lies outside both children of $Q_u$. This proves disjointness and, by the same dichotomy, the first identity in (DG4). The second follows from $g \circ \phi_a = \mathrm{id}$.

A collar of $\partial D_w$, together with the exterior of $D_w$, is a connected subset of the unbounded component of $\mathbb{C} \setminus S_w$. Every bounded complementary component is therefore contained away from that collar, proving (DG5). □

**Proved Theorem 10.43** (Finite direct-decoration packet replacement)**.** *Let $F \subset A^{<\mathbb{N}}$ be finite and let*

$$K_w \subset L_w \Subset \operatorname{int} D_w, \qquad w \in F, \tag{DG6}$$

*with every $L_w$ full. Replacing the selected $K_w$ by $L_w$ in $\mathcal{H}_\partial$ gives a compact set with connected sphere complement.*

*Proof.* Equation (DG4) makes each selected $K_w$ clopen in $\mathcal{H}_\partial$, so the replacement is compact. For $m > \max_{w \in F} |w|$, make the same replacements in (DG1). The modified approximants remain decreasing finite pairwise-disjoint full packets, and their intersection is the modified rooted-end compactum. Lemma 10.39 applies. □

**Proved Lemma 10.44** (Proper protected impressions leave a port)**.** *Let $K \Subset D^\circ$ be a non-degenerate full continuum in a closed Jordan disk and let $P_{\mathrm{prot}} \subsetneq \partial K$ be compact. There are $x \in \partial K \setminus P_{\mathrm{prot}}$ and a Jordan arc $\alpha$ from $x$ to $\partial D$, otherwise in $D^\circ \setminus K$. Finitely many protected crosscut-chain collars whose impressions have union in $P_{\mathrm{prot}}$ can be truncated so that, for their connected union $S$ with $K$,*

$$\alpha \cap \operatorname{Fill}(S) = \{x\}. \tag{DG7}$$

*Proof.* Choose $y \in \partial K \setminus P_{\mathrm{prot}}$, put $\delta = \operatorname{dist}(y, P_{\mathrm{prot}})$, choose $z \in D^\circ \setminus K$ with $|z - y| < \delta/4$, and let $x \in K$ minimize distance to $z$. The segment $(x, z]$ misses $K$, and

$$|x - y| \le 2|z - y| < \delta/2,$$

so $x \notin P_{\mathrm{prot}}$. The connected open complement of the full continuum is arcwise connected. Join $z$ to the exterior, take a simple subarc, and stop at its first hit on $\partial D$.

The compact arc misses $P_{\mathrm{prot}}$. Nested compact crosscut-chain closures have intersection equal to their impressions, so sufficiently deep tails lie in prescribed impression neighborhoods chosen away from $\alpha$. Extending $\alpha$ outside $D$ to infinity puts all of its points except $x$ in the unbounded component of $\mathbb{C} \setminus S$, proving (DG7). □

**Proved Theorem 10.45** (Finite direct-decoration protected carrier)**.** *Fix finitely many finite direct-decoration addresses and finitely many actual protected root prime-end chains whose total impression is a proper subset of $\partial K_0$. Transport sufficiently deep collar tails and the free port through the inverse branches. Their packet fullifications remain in the pairwise-disjoint cells $D_w$, avoid every unselected finite fiber and every end fiber, and admit a protected regulated connector preserving the transported prime ends and cyclic order.*

*Proof.* Apply lemma 10.44 at the root and transport by the orientation-preserving $\phi_w$. Theorem 10.42 gives localization, isolation, and transport. Theorem 10.43 gives one common complementary component after fullification. Apply theorems 10.4 and 10.8. □

This protected-carrier theorem is finite and direct-scale only. It does not construct a second satellite portrait, an end-fiber return map, a uniform incidence bound, or future fjord, dam, and attachment data.

**Proved Theorem 10.46** (Finite-cylinder commuting-iterate lift)**.** *Let*

$$h_w = f^N|_{Q_w} : Q_w \longrightarrow P_\sigma, \qquad h_v = f^N|_{Q_v} : Q_v \longrightarrow P_\tau \tag{FC1}$$

*be equal-depth homeomorphic cylinder maps. Let $F = f^r|_A : A \to B$ and put $A_w = h_w^{-1}(A)$. If*

$$f^r(A_w) \subset Q_v, \tag{FC2}$$

*then*

$$f^r|_{A_w} = h_v^{-1} \circ F \circ h_w. \tag{FC3}$$

*If $F(A) = B$, then*

$$f^r(A_w) = h_v^{-1}(B). \tag{FC4}$$

*Proper degree and local degree are preserved. Critical chronology is separate arrow metadata.*

*Proof.* For $x \in A_w$, condition (FC2) makes $h_v(f^r(x))$ defined. Since both cylinder maps use the same iterate $f^N$,

$$h_v(f^r(x)) = f^N(f^r(x)) = f^r(f^N(x)) = F(h_w(x)).$$

Apply $h_v^{-1}$ to obtain (FC3). Taking images gives $f^r(A_w) = h_v^{-1}(F(A))$, hence (FC4) under surjectivity. The homeomorphic conjugacy (FC3) preserves properness and (local and total) topological degree. It does not determine the ordered intermediate critical times. □

**Proved Proposition 10.47** (Fixed-base connected-image target uniqueness)**.** *Fix one target base $\tau$ and one depth $m$. Let $\{Q_{\tau,v} : v \in W_{\tau,m}\}$ be a finite pairwise-disjoint family of nonempty connected compact cylinders. Every nonempty connected set*

$$C \subset \bigcup_{v \in W_{\tau,m}} Q_{\tau,v} \tag{FTU1}$$

*lies in one unique $Q_{\tau,v}$.*

*Proof.* Finite pairwise-disjoint compacta have positive pairwise distance, so each cylinder is open and closed in the union in (FTU1). The cylinders are therefore its connected components, and connected $C$ lies in one component. □

The proposition applies to the direct-decoration cylinders after the target base and actual union containment are known. It does not select a target base or establish cross-base ownership.

**Proved Lemma 10.48** (Localized/global cylinder component)**.** *If $P, Q$ are closed Jordan disks and $f^N|_Q : Q \to P$ is a homeomorphism, then $Q^\circ$ is the component of $f^{-N}(P^\circ)$ containing it.*

*Proof.* Let $D$ be that component. The set $Q^\circ$ is open in $D$. If $x \in \overline{Q^\circ}^{\,D}$, then $x \in Q$ and $f^N(x) \in P^\circ$. Boundary preservation for the homeomorphism of closed Jordan disks excludes $x \in \partial Q$, so $x \in Q^\circ$. Thus $Q^\circ$ is also closed in connected $D$, hence equals $D$. □

**Proved Theorem 10.49** (Interior pullback component forcing)**.** *Retain the equal-depth source cylinder and branch data of theorem 10.46, assume $A$ is connected, and suppose*

$$F(A) \subset P_\tau^\circ. \tag{IC1}$$

*Then $f^r(A_w)$ lies in one component $C$ of $f^{-N}(P_\tau^\circ)$. If an authorized closed target cylinder $Q_v$ maps homeomorphically to $P_\tau$ under $f^N$ and $Q_v^\circ = C$, then $v$ is unique in every pairwise-disjoint fixed-base family and (FC3) holds. If the branch extends to the relevant closures and*

$$F(\overline{A}) \Subset P_\tau^\circ, \tag{IC2}$$

*then*

$$\overline{f^r(A_w)} \Subset C. \tag{IC3}$$

*Proof.* For $x \in A_w$,

$$f^N(f^r(x)) = f^r(f^N(x)) = F(h_w(x)) \in P_\tau^\circ.$$

The connected image therefore lies in one open pullback component $C$. Lemma 10.48 identifies an authorized cylinder interior with that global component, proposition 10.47 gives uniqueness, and theorem 10.46 gives (FC3).

Under (IC2), the lifted closure is a compact subset of

$$(f^N)^{-1}(F(\overline{A})) \cap \overline{C}.$$

Every point of $\overline{C}$ mapping into $P_\tau^\circ$ belongs to an open preimage component; because that component meets $C$, it equals $C$. The displayed compact set is therefore contained in the open set $C$, proving (IC3). □

**Open Problem 10.50** (Virtual cylinder target-address selector)**.** For every realized finite virtual copy-chain prefix, source base $\sigma$, equal-depth cylinder $w$, and source-defined branch $F = f^r|_A$, construct a unique target pair

$$\rho_{F,m}(\sigma, w) = (\tau, v)$$

satisfying (FC2). First type or uniquely select the target base $\tau$. Then use theorem 10.49 to obtain the open pullback component and prove component authorization: that component is one of the declared equal-depth target cylinders. A global-preimage formulation must invoke lemma 10.48. Once a fixed pairwise-disjoint target family is available, proposition 10.47 supplies word uniqueness. Finally prove compatibility with geometric parents, the partial pre-root shift, the separate root arrow, critical chronology, sheets, accesses, obstacles, localizations, presentation changes, and authorized root enlargement. A failure must name the first actual image component and every target cylinder it meets.

**Open Problem 10.51** (Disconnected marking transfer)**.** For every chain $\mathcal{M}^{(0)} \supsetneq \mathcal{M}^{(1)} \supsetneq \mathcal{M}^{(2)}$, transfer the second satellite portrait to actual puzzle or generalized polynomial-like bases and actual iterates on the finite cylinders and end fibers in theorem 10.37. Prove equivariance under (HT13), analytic realization at every end, the declared critical chronology, complete access and obstacle marking, terminal-return compatibility, and naturality under every authorized final-root enlargement.

The refined first bottom record is

$$\mathfrak{X}_2^{\mathrm{mark}} = (\Omega_\partial, \Xi_\partial, f^q, \mathrm{gpar}, \sigma_{\mathrm{pre}}, \tau_{\mathrm{root}}, \mathrm{sheet}, r;\ \mu_2 = \bot, \{B_{2,\xi}\} = \bot, \mathcal{R}_2 = \bot). \tag{HT17}$$

The end fibers have not been identified as small Julia sets. Hence open problem 10.51 is a prior subgate of the following global realization problem.

**Open Problem 10.52** (Virtual Cantor-Package Realization)**.** For each actual intermediate virtual scale $s$, construct an actual compact address space $\Omega_s$, nonempty compact connected components $K_{s,\omega}$, and exact polynomial branch, return, parent, sheet, critical-address, escape, localization, presentation, terminal-compatibility, and final-root-enlargement data. Construct them without using a downstream width, compactness, rigidity, or MLC conclusion.

**Proved Theorem 10.53** (Virtual Proof-Use Trace Selector; conditional interface)**.** *After open problem 10.52 and after the complete proof-use horizon is frozen, let the exact component/trace states, actual finite-path transitions, and complete countable proof-use catalog with compact metrizable target encodings be given. Equality of all declared observations after every finite continuation gives a future-faithful quotient satisfying (HT4) for every Wave, Wanderers, covering, quasi-additivity, width, transport, access, obstacle, and authorized enlargement observable. Its closure is a compact metrizable end topology.*

*Proof.* Apply theorem 8.4. Current and future component records are coordinates, transition domains are clopen, actual transitions extend by coordinate shift, and every declared observable factors through its coordinate. Equality of the complete coordinate table is exactly future-faithfulness. This produces no finite-cardinality or response- dimension bound and does not construct the open component package or horizon. □

**Open Problem 10.54** (Virtual Vertex-Package Compatibility)**.** Construct a partial factor from the trace in theorem 10.53 to the native terminal Hubbard tree. Prove exact component and return-word compatibility, protected-access survival, analytic degree-one realization of (HT2) as equality, fold-excess and one-gate enlargement audits, presentation invariance, and period-independent faithful compression.

The ambient terminal tree and a virtual copy label do not solve the remaining source-facing problems: the path $K_0 - K_1 - K_2$ has distinct hulls for $\{K_0, K_1\}$ and $\{K_0, K_1, K_2\}$, and a finite hull cannot be declared to enumerate an unconstructed Cantor family. Thus open problems 10.52 and 10.54 and the open hypotheses of the trace theorem are prior inputs to open problem 10.29. Only the formal compact-end-topology alternative is promoted.

Theorem 10.27 proves the native analytic positive control. Open problem 10.29 is open and is a necessary input to the virtual return-base atlas. Thus no source-facing virtual status is promoted.

The dependency closure does not select packet labels. The same unlabelled two-vertex graph can be realized with the second geometric component declared separated in one system and attached to the first atom in another. The respective source-faithful assignments differ. Actual attachment and routing data must therefore choose the labels before theorem 10.14 applies.

Dudko's public virtual-Molecule paragraph defines the nested copy chain and proposes connected hulls carrying the critical orbit for an "appropriate number" of first returns, but it supplies no stopping integer or rule. Consequently the first undefined inventory field is $N$ in (9.12). KKL's native Hubbard tree and ideal marking are a positive control only in their published primitive category. Copying that inventory requires a typed bridge preserving actual components, dependency paths, stage branches, critical words, sheets, and every proof-used prime end; the checked sources construct no such bridge.

**Open Problem 10.55** (Virtual Wave Horizon Certificate)**.** Assuming no conclusion sought later, solve open problem 10.18 and use its actual branches to construct (9.12), the complete finite proof-use seed and dependency data in (9.13), and actual metadata for every record. Prove either that the resolved universe is finite or that it has a finite future-faithful summary. After the final root enlargement, prove (9.14) for every admissible continuation. The stopping rule must be defined without using the a priori bound or compactness conclusion sought later.

For a proposed packet, record

$$\chi_{\mathrm{aug}} = (\chi_{\mathrm{cov}}, \chi_{\mathrm{conn}}, \chi_{\mathrm{merge}}, \chi_{\mathrm{loc}}, \chi_{\mathrm{port}}). \tag{9.19}$$

These coordinates detect incomplete coverage, disconnected seed assignment, post-fullification merger, escape from $U$, and loss of a required free port. A nonzero coordinate refutes this sufficient construction only. It refutes open problem 10.58 only when the original free domain also has the separator certified by theorem 10.8.

**Proved Proposition 10.56** (Connected-hull underdetermination)**.** *The atoms, one connected full carrier containing them, and identity package dynamics do not determine whether the common-free-component property holds.*

*Proof.* Take

$$U = \{|z| < 4\}, \qquad A_1 = \{|z| \le 1/4\}, \qquad A_2 = \{|z - 2| \le 1/4\},$$

and use in both systems the filled hull of

$$A_1 \cup [1/4, 7/4] \cup A_2$$

with identity package dynamics. With no additional forbidden set, the two atoms form an augmented packet and theorem 10.11 applies.

In the second system add

$$R = \{1/2 \le |z| \le 1\}$$

to the forbidden set. The component inside $R$ accesses $A_1$ but not $A_2$, while every component outside $R$ can access $A_2$ but not $A_1$. Thus no common component exists. Moreover, $\mathrm{Fill}(R) = \{|z| \le 1\}$: treating $R$ as separated merges its fill with $A_1$, while assigning it to $A_1$ makes $A_1$ interior to $\widehat{A}_1$ and erases every singleton port. The connected carrier and package data themselves are unchanged. □

This is an underdetermination model for the public connected-hull data, not an actual virtual-Molecule counterexample.

**Open Problem 10.57** (Virtual Augmented-Atom Port Certificate)**.** For every actual virtual-Molecule block and complete Wave horizon, after the final root enlargement, prove:

- I. a frozen actual component inventory and typed assignment, including every attachment trace, return word, sheet, protected impression, and proof-use interval; this requires open problem 10.55 before the packet labels are chosen;
- C. zero collision for the unique completed packets;
- P. one actual unprotected original-atom exposure per atom;
- H. full-packet equivariance through the complete horizon; and
- U. a period-independent incidence bound or a future-faithful compression.

**Open Problem 10.58** (Virtual Common Free-Component)**.** For every actual virtual-Molecule block, after all future root shadows, proof-used collars, obstacles, critical exclusions, and localization data through the complete Wave horizon are inserted, prove that one component of $\Omega_{\mathrm{free}}$ has simultaneous singleton ports to every required root-shadow cluster and remains valid after the final root enlargement.

Dudko's connected-hull paragraph does not define this completed free domain, and KKL's native Hubbard tree proves the assertion only in its published primitive category. A failure must record the first inserted separator, the free-component partition before and after insertion, the first two separated atoms, the protected chain and attachment trace when applicable, the complete return word, and the authorized localization.

*Warning* 10.59. Theorem 10.4 proves realization from a certificate. Theorem 10.8 characterizes when that certificate exists, theorem 10.11 proves one sufficient planar route, and theorem 10.14 removes completion order and global port untangling from that route. Theorem 10.16 proves only the closure after a horizon graph is given, while theorem 10.17 proves the stopping and final-root algorithms only after the analytic marked branches are supplied. Neither constructs open problem 10.18. None proves open problems 10.6, 10.55, 10.57 and 10.58. Therefore the virtual canonical interface carrier, UPTCR-1, and MLC remain open.

## 11. Critical chronology and the retimed-cylinder boundary

11.1. **Status boundary.** This section integrates the independently verified part of an external research push. The main new device is a retimed pair

$$(f^{N_w+r}, f^{N_w}),$$

but it is conditional on an independently constructed root portrait $F = f^r : A \to B$. No checked source constructs that portrait from the second virtual satellite copy, authorizes its complete marking, or proves its final-root stability. Accordingly `MAPPING_TELESCOPE`, `FINITE_HORIZON_ASSEMBLY`, `VIRTUAL_DISCONNECTED_MARKING_TRANSFER`, UPTCR, and MLC remain OPEN.

Throughout this section, $P$ and its cylinder pieces $Q_w$ are closed Jordan disks. The symbols $A, B, \dots$ in holomorphic restrictions denote bounded open Jordan domains. We use $h_w$ both for its conformal interior restriction and for its stated homeomorphic extension between the closed puzzle pieces. Thus $A \Subset B$ means $\overline{A}$ is a compact subset of the open domain $B$.

11.2. **Conservation of the critical cut.** For a nonconstant holomorphic map $H$, write $e_H(z)$ for its local degree. For the quadratic polynomial $f(z) = z^2 + c$, define the pointwise orbit word

$$\mathbf{d}_k(z) = (e_f(z), e_f(fz), \dots, e_f(f^{k-1}z)).$$

This is not yet a global branch word; the latter also requires specified proper intermediate domains.

**Proved Theorem 11.1** (Critical-cut conservation)**.** *For all $N, r \geq 1$,*

$$e_{f^r}(z) e_{f^N}(f^r z) = e_{f^N}(z) e_{f^r}(f^N z). \tag{11.1}$$

*For a quadratic polynomial,*

$$\mathbf{d}_r(z) \star \mathbf{d}_N(f^r z) = \mathbf{d}_N(z) \star \mathbf{d}_r(f^N z) = \mathbf{d}_{N+r}(z). \tag{11.2}$$

*Proof.* Both compositions $f^N \circ f^r$ and $f^r \circ f^N$ equal $f^{N+r}$. Multiplicativity of local degree gives (11.1). Both words in (11.2) list the one-step local degrees at the same absolute times $0, \dots, N + r - 1$, with the cut placed at different positions. □

If

$$m_k(z) = \#\{0 \leq j < k : f^j(z) = 0\},$$

then $e_{f^k}(z) = 2^{m_k(z)}$. Hence local univalence of $f^N$ at $z$ gives

$$m_N(f^r z) = m_r(f^N z) - m_r(z). \tag{11.3}$$

This conserves the number of critical encounters, not their order.

**Proved Theorem 11.2** (Proper-square degree law)**.** *Suppose finite-degree proper holomorphic maps between connected bordered surfaces form a commuting square*

$$\begin{array}{ccc} U & \xrightarrow{H} & V \\ {\scriptstyle h_0}\downarrow & & \downarrow{\scriptstyle h_1} \\ A & \xrightarrow{F} & B. \end{array}$$

*Then*

$$\deg(H)\deg(h_1) = \deg(h_0)\deg(F) \tag{11.4}$$

*and the analogous identity holds for local degrees. If $h_0$ is univalent and $\deg F = 2$, then*

$$(\deg H, \deg h_1) \in \{(2,1),(1,2)\}. \tag{11.5}$$

*Proof.* Degrees multiply under composition, so (11.4) follows from equality of the two compositions. The local statement is the local-degree chain rule. Under the final hypotheses, two positive integers have product two. □

The second alternative in (11.5) is a genuine criticality swap: the quadratic factor is stored in the transverse target projection.

**Proved Theorem 11.3** (Local off-critical square)**.** *Suppose*

$$h_0 = f^N|_U : U \longrightarrow A, \qquad F = f^r|_A : A \longrightarrow B$$

*are conformal isomorphisms of plane domains. Put*

$$H = f^r|_U, \qquad V = H(U), \qquad h_1 = f^N|_V.$$

*Then $H : U \to V$ and $h_1 : V \to B$ are conformal isomorphisms and*

$$h_1 \circ H = F \circ h_0.$$

*Every already certified chart-supported prime-end record transports through this square. The theorem neither authorizes $V$ as a virtual target nor makes $V$ Jordan without a closure embedding.*

*Proof.* If $H(x_1) = H(x_2)$, apply $f^N$ and use commutation:

$$F(h_0(x_1)) = f^N(H(x_1)) = f^N(H(x_2)) = F(h_0(x_2)).$$

Injectivity of $F$ and $h_0$ gives $x_1 = x_2$, so $H$ is conformal onto $V$. For $y = H(x)$, one has $h_1(y) = F(h_0(x))$; as $x$ runs through $U$, the right side runs bijectively through $B$. Thus $h_1$ is conformal and the square commutes. Conformal isomorphisms induce homeomorphisms of prime-end compactifications. □

*Warning* 11.4 (Interior univalence is not a closure embedding). The map $z \mapsto z^2$ is injective on the open right half-disk but identifies opposite points of its imaginary boundary. Its image is a slit disk, not a Jordan disk.

11.3. **The conditional retimed pair.** Retain a direct-decoration cylinder homeomorphism

$$h_w = f^{N_w}|_{Q_w} : Q_w \longrightarrow P$$

from section 10.11. The following result does not construct its root input.

**Proved Theorem 11.5** (Retimed finite-cylinder pair)**.** *Assume $A, B$ are bounded open Jordan domains satisfying*

$$\overline{A} \Subset B, \qquad \overline{B} \Subset \operatorname{int} P$$

*and $F = f^r|_A : A \to B$ is proper. Put*

$$U_w = h_w^{-1}(A), \qquad g_w = F \circ h_w = f^{N_w+r}|_{U_w}, \qquad \iota_w = \operatorname{incl}_{A\hookrightarrow B} \circ h_w.$$

*Then $g_w$ is proper with $\deg g_w = \deg F$, $\iota_w$ is a conformal embedding, and*

$$g_w \circ \iota_w^{-1} = F \tag{11.6}$$

*on $A$.*

*If a compact full $K \Subset A$ satisfies*

$$F^{-1}(K) = K = F(K),$$

*then the source carrier $K_w = h_w^{-1}(K)$ is compact and full and*

$$g_w^{-1}(K) = \iota_w^{-1}(K) = K_w, \qquad g_w(K_w) = \iota_w(K_w) = K. \tag{11.7}$$

*Every already certified chart-supported record transports by $h_w^{-1}$. No second-scale identification, arbitrary-marking admissibility, simultaneous ownership, or unrestricted enlargement is asserted.*

*Proof.* Commutation of polynomial iterates gives $f^{N_w+r} = F \circ h_w$. Composition with the conformal $h_w$ preserves proper degree, and cancellation gives (11.6). Equation (11.7) is the inverse-image calculation

$$(F \circ h_w)^{-1}(K) = h_w^{-1}(F^{-1}(K)) = h_w^{-1}(K).$$

The closed Jordan-disk homeomorphism $h_w$ extends to a sphere homeomorphism, so it preserves fullness. Transport of an already valid record is ordinary conjugacy. Conjugacy does not certify that an arbitrary record was dynamically admissible. □

In KKL's $\psi$-ql typing, $K$ is the target filled Julia set and $K_w$ is the pulled source carrier. The theorem does not construct a self-return with target $K_w$.

*Warning* 11.6 (Arbitrary markings and enlargements). For the disk pair $(z^2, \mathrm{id})$, the immersion pullback of the endpoint angle $1/3$ is $1/3$, while the covering pullbacks are $1/6$ and $2/3$. The required endpoint containment fails. Likewise, an enlargement can stay inside $P$ while meeting a child cylinder and merging sheets. The root marking must already satisfy the exact endpoint criterion, and enlargement must stay in the certified safe gap or carry a complete collision ledger.

11.4. **Chronology cannot be reconstructed at the endpoint.**

**Proved Proposition 11.7** (Criticality-location swap). *For $f(z) = z^2 - 1$, fix $0 < \rho < 1$, put*

$$A = \mathbb{D}(0, \rho), \qquad B = \mathbb{D}(-1, \rho^2),$$

*and let $U$ be the component of $f^{-1}(A)$ containing 1. Then $h_0 = f|_U : U \to A$ and $H = f|_U : U \to A$ are conformal, while $F = f|_A : A \to B$ and $h_1 = f|_A : A \to B$ are proper of degree two. The square $h_1 \circ H = F \circ h_0$ commutes, so its degree pair is $(\deg H, \deg h_1) = (1, 2)$.*

*Proof.* The disk $A$ omits the critical value $-1$, so each component of $f^{-1}(A)$ maps conformally onto $A$. The disk $A$ contains the critical point 0, and $f : A \to B$ is proper of degree two. Both compositions are $f^2|_U$. □

**Proved Proposition 11.8** (Critical-time shear). *For $f(z) = z^2 - 1$, there is a commuting square whose source and target projections are univalent and whose two horizontal returns both have degree two, while their chronological words are*

$$(2, 1) \qquad \textit{and} \qquad (1, 2). \tag{11.8}$$

*Proof.* Fix $0 < \rho < 1/2$, and let

$$A_0 = \mathbb{D}(0, \rho), \quad A_1 = f(A_0) = \mathbb{D}(-1, \rho^2), \quad A_2 = f(A_1),$$

and let $U_0$ be the component of $f^{-1}(A_0)$ containing 1. The maps

$$h_0 = f|_{U_0}, \qquad h_2 = f|_{A_1}$$

are conformal, while

$$F = f^2|_{A_0}, \qquad H = f^2|_{U_0}$$

are proper of degree two and satisfy $h_2 \circ H = F \circ h_0$. Moreover, $\overline{A}_1$ lies in the left half-plane and $-\overline{A}_1$ in the right half-plane. Since $f(z) = f(z')$ implies $z' = \pm z$, $f$ is injective on $\overline{A}_1$. Thus $A_2$ is Jordan and $h_2 : A_1 \to A_2$ is a proper conformal isomorphism. The first factor of $F$ is quadratic and the second is univalent, while the first factor of $H$ is univalent and the second is quadratic. □

**Proved Theorem 11.9** (Stagewise critical-word ladder)**.** *Assume every horizontal one-step map and every transverse map in a commuting ladder is proper and onto between connected Jordan disks. Put*

$$d_j = \deg(f : A_j \to A_{j+1}), \quad \widetilde{d}_j = \deg(f : U_j \to U_{j+1}), \quad e_j = \deg(f^N : U_j \to A_j).$$

*Then*

$$\widetilde{d}_j e_{j+1} = e_j d_j. \tag{11.9}$$

*If $e_0 = 1$, exact equality of the chronological words is equivalent to $e_j = 1$ for every $j$. The single endpoint condition $e_r = 1$ controls only total degree.*

*Proof.* Apply theorem 11.2 to every one-step square. If all $e_j = 1$, (11.9) gives equality of the one-step degrees. Conversely, equality of those degrees and $e_0 = 1$ imply $e_j = 1$ inductively. Multiplying (11.9) cancels the intermediate transverse degrees, explaining why endpoint data retain only the product. □

The Jordan-disk hypothesis matters. A critical-point-free proper map to a simply connected disk has degree one; the analogous claim is false on an annulus.

### 11.5. **The embedded-Jordan critical spine.**

**Proved Theorem 11.10** (Quadratic Jordan base change)**.** *Let $p : X \to Y$ be proper of degree two between Jordan disks, with critical value $v$. If $E \Subset Y$ is a Jordan disk and $v \notin \partial E$, then:*

*(1) if $v \in E$, $p^{-1}(E)$ is one Jordan disk mapped with degree two onto $E$;*
*(2) if $v \notin \overline{E}$, $p^{-1}(E)$ is the disjoint union of two Jordan disks, each mapped conformally onto $E$.*

*Proof.* When $v \in E$, let $C$ be the component containing the critical point. It has degree at least two and therefore exhausts the total degree. Its restriction over $E$ has one simple branch point, so Riemann–Hurwitz gives

$$\chi(C) = 2\chi(E) - 1 = 1.$$

Since $C$ is planar, it has one boundary component. When $v \notin \overline{E}$, every component is an unbranched finite cover of the simply connected disk $E$, hence has degree one; total degree gives two components. Because the critical value misses $\partial E$, an annular Jordan collar lifts without branching. In the branched case the Euler-characteristic calculation gives one lifted boundary component; in the unbranched case each inverse branch extends across the collar. Consequently all the stated components are Jordan disks. □

**Proved Theorem 11.11** (Planar critical-spine compiler)**.** *Consider a finite cyclic sequence of Jordan targets $Y_0, \dots, Y_m$ with $Y_m = Y_0$. At stage $j$, let $g_j : X_j \to Y_{j+1}$ be proper and let*

$$\iota_j : \overline{X}_j \longrightarrow Y_j$$

*be a closure embedding, conformal on the interior, whose image $A_j = \iota_j(X_j) \Subset Y_j$ is a Jordan disk. Assume exactly one $g_j$ has degree two, all others have degree one, and every post-quadratic partial critical value lies in the next $A_j$.*

*Then the recursive fiber-product base changes have unique connected Jordan components, and the endpoint pair has a conformal embedding and a proper degree-two covering into $Y_0$. If critical-value containment first fails with the critical value strictly outside the closure of the next $A_j$, the corresponding preimage splits into two separated degree-one Jordan components and no connected later route recovers degree two. A critical value on $\partial A_j$ is instead a boundary-critical collision; this theorem certifies no Jordan collar or separated-sheet continuation there.*

*Proof.* If $g_0$ is quadratic, it supplies the initial degree-two partial covering. Otherwise, before the quadratic stage the partial covering is conformal, so the preimage of $A_j$ is one Jordan disk; the unique later quadratic stage then creates the degree-two partial covering. Thereafter apply theorem 11.10: containment of the propagated critical value gives one connected degree-two preimage, while strict exterior omission gives exactly two separated degree-one components. Composition with each subsequent conformal $g_j$ preserves the retained degree. Restriction preserves the initial embedding, and $Y_m = Y_0$ closes the target route. The boundary-critical case has no unbranched collar and is stopped rather than passed through the base-change theorem. □

The closure-embedded Jordan hypothesis is essential. For $z^2 : \mathbb{D} \to \mathbb{D}$, a relatively compact slit domain containing 0 satisfies prime-end critical-value retention, but its preimage has pulled slits and is not a Jordan disk.

**Proved Proposition 11.12** (Finite planar telescope identity)**.** *On the selected route, put*

$$R_j = g_j \circ \iota_j^{-1},$$

*Then, as maps on the exact domain $I_m(T_m)$,*

$$G_m \circ I_m^{-1}\Big|_{I_m(T_m)} = R_{m-1} \circ \cdots \circ R_0|_{I_m(T_m)}\,. \tag{11.10}$$

*If $R_j = f^{s_j}$, the exponent is $\sum_j s_j$. This does not prove the relative-cover, protected-prime-end, or end-proper clauses of `FINITE_HORIZON_ASSEMBLY`.*

*Proof.* The recursion gives $G_{k+1} = R_k \circ G_k$ on $T_{k+1}$, while $I_{k+1} = I_k|_{T_{k+1}}$. Induction yields

$$G_k = R_{k-1} \circ \cdots \circ R_0 \circ I_k$$

on $T_k$, which is exactly (11.10). For actual polynomial iterates, exponents add on the stated chronological composition domain. □

*Warning* 11.13 (Why the marked telescope is still open)*.* The subgroup criterion proves existence of a based lift, not its embeddedness, properness, deck phase, full component orbit, or critical lift. Moreover, $\iota(z) = z/2 : \mathbb{D} \hookrightarrow \mathbb{D}$ sends the source prime-end circle to the interior circle $|z| = 1/2$, not to the target prime-end circle. Protected KKL markings live on carrier-complement ideal boundaries and require separately proved endpoint maps.

11.6. **What happens at infinite addresses.** Let

$$\Phi_m = \phi_{\omega|m} : \operatorname{int} P \to \operatorname{int} Q_{\omega|m}, \qquad E_\omega = \bigcap_m Q_{\omega|m} \subset J(f).$$

**Proved Theorem 11.14** (Planar end collapse)**.** *For every compact $L \Subset \operatorname{int} P$,*

$$\operatorname{diam} \Phi_m(L) \longrightarrow 0. \tag{11.11}$$

*Every locally uniform subsequential limit is constant with value in $E_\omega$.*

*Proof.* The family is bounded and hence normal. A locally uniform limit lies in every closed $Q_{\omega|n}$, hence in $E_\omega$. Since $E_\omega \subset J(f)$ has empty interior, the open mapping theorem forces the limit to be constant. Failure of (11.11) would give a subsequence with diameter bounded below and a locally uniformly convergent subsubsequence, contradicting its constant limit. □

**Proved Proposition 11.15** (Normalized retimed identity)**.** *For a root pair supplied to theorem 11.5,*

$$(g_w \circ h_w^{-1}, \iota_w \circ h_w^{-1}) = (F, \mathrm{incl}_{A \hookrightarrow B}) \tag{11.12}$$

*for every finite word $w$.*

*Proof.* This is cancellation in the definitions $g_w = F \circ h_w$ and $\iota_w = \mathrm{incl} \circ h_w$. □

Thus the specified planar charts collapse while the normalized representatives are identical. The identity constructs no end surface.

**Proved Theorem 11.16** (Projection-fiber no-go)**.** *Let $K$ be a compact connected set with at least two points. Suppose embeddings $j_m, j_\infty : K \to \mathcal{X}$ into a Hausdorff space satisfy $j_m(z) \to j_\infty(z)$ pointwise, and a continuous $\pi : \mathcal{X} \to \overline{P}$ satisfies*

$$\pi \circ j_m = \phi_{\omega|m}|_K.$$

*Then $\pi \circ j_\infty$ is constant. Hence $\pi$ is not locally injective at any point of $j_\infty(K)$.*

*Proof.* For $z, z' \in K$, theorem 11.14 gives

$$|\phi_{\omega|m}(z) - \phi_{\omega|m}(z')| \longrightarrow 0.$$

Pass to the pointwise marked limit and use continuity of $\pi$. The embedded compactum $j_\infty(K)$ is connected, nondegenerate, and has no isolated points. Every neighborhood of each of its points therefore contains a second carrier point with the same $\pi$-image. □

This theorem forces a nontrivial projection fiber under its hypotheses. It does not construct or uniquely force a conformal prime-end blow-up.

**Proved Theorem 11.17** (Fixed-iterate nested-fiber extension)**.** *Let $X, Y$ be Hausdorff spaces, $C_1 \supset C_2 \supset \cdots$ nonempty compact subsets of $X$, and $D_1 \supset D_2 \supset \cdots$ nonempty compact subsets of $Y$. Let $F$ be one fixed continuous map from a neighborhood of $C_1$ into $Y$. If $F(C_n) \subset D_n$ for all $n$, then*

$$F\left(\bigcap_n C_n\right) \subset \bigcap_n D_n.$$

*If $F(C_n) = D_n$ for every $n$, equality holds.*

*Proof.* The inclusion is immediate. For the reverse inclusion, fix $y \in \bigcap_n D_n$. The sets

$$C_n \cap F^{-1}(\{y\})$$

are nonempty, nested, and closed in the compact set $C_1$: compact subsets and singleton fibers are closed because the ambient spaces are Hausdorff. Their intersection is nonempty, giving a point of $\bigcap_n C_n$ mapped to $y$. □

*Warning* 11.18 (Why Hausdorffness is explicit). In $\mathbb{N}$ with the cofinite topology, the tails $C_n = \{n, n+1, \ldots\}$ are nested nonempty compact sets with empty intersection. A constant map satisfies $F(C_n) = \{*\}$ at every finite level, but equality fails after intersection.

This theorem concerns one fixed map. It does not apply directly to the growing exponents $f^{N_w+r}$, nor does it transport non-eventually constant access phases or protected markings.

11.7. **The three exposed realization gates.**

**Open Problem 11.19** (Virtual finite-word root authorization)**.** For every actual two-step virtual satellite-copy chain, construct and source-identify the required root portrait in a Jordan neighborhood compactly contained in

$$\operatorname{int} P \setminus (Q_0 \cup Q_1),$$

construct its proof-complete carrier and certified marking, authorize all simultaneous finite-word and forced off-critical components, and prove complete-Wave-demand and final-root naturality.

**Open Problem 11.20** (End prime-end cofinality)**.** Construct the actual multisector Böttcher footprint of every nested word, prove compatible inverse angular lifts, and prove that the exterior puzzle traces form a fundamental chain cofinal at one prime end, including endpoint addresses. Only then identify

$$E_\omega = \operatorname{Imp}(\theta_\omega).$$

Obstacle and collar protection remains separate.

**Open Problem 11.21** (Virtual end realization)**.** Attach the authorized normalized root pair over every proof-used infinite address in an analytic rescaled or nontrivial-projection-fiber space, and prove its protected accesses, obstacle and sheet records, critical chronology, shift and fixed-iterate actions, compatible relative covers, proper or end-proper intervals, and complete-horizon final-root naturality.

The proved results in this section sharpen the finite critical-spine and end constraints. They do not reduce the source-facing construction to "only an end limit": root authorization, relative-cover realization, protected marking, and final-root stability remain independent open requirements.

## 12. Normalized end rescaling, full-trace calculus, and the pulled-collar obstruction

This section records the portion of Research Push III that survives source, dependency, and analytic cross-examination. Throughout, the quadratic root portrait is independently supplied. Nothing here constructs the missing virtual scale-two portrait or identifies a normalized fiber with a planar infinite-address component.

12.1. **The normalized direct system.** Let $P$ be a closed Jordan puzzle piece and let $g = f^q$. Suppose

$$g : Q_a \longrightarrow P, \qquad a \in \mathcal{A},$$

are actual degree-one pullback branches with inverse maps $\phi_a : P \to Q_a$. For $w = a_1 \cdots a_m$, put

$$\phi_w = \phi_{a_1} \circ \cdots \circ \phi_{a_m}, \qquad h_w = g^m|_{Q_w} = \phi_w^{-1}.$$

Let $F : A \to B$, together with $\jmath : A \hookrightarrow B$, be an independently realized proper root pair, where $A, B$ are bounded open Jordan domains and $\overline{A} \subset P^\circ$. Put $A_w = h_w^{-1}(A)$.

Fix $\omega \in \mathcal{A}^{\mathbb{N}}$, write $w_m = \omega|m$, and define

$$t_m = \left(h_{w_{m+1}}|_{A_{w_{m+1}}}\right)^{-1} \circ h_{w_m}|_{A_{w_m}}. \tag{10.1}$$

Then

$$h_{w_{m+1}} \circ t_m = h_{w_m}. \tag{10.2}$$

**Proved Theorem 12.1** (Normalized end direct limit)**.** *The direct system $(A_{w_m}, t_m)$ has a Riemann-surface direct limit $\widehat{A}_\omega$ canonically biholomorphic to $A$. Under the biholomorphism $h_\omega : \widehat{A}_\omega \to A$, the descended retimed pair is*

$$\widehat{G}_\omega = F \circ h_\omega, \qquad \widehat{I}_\omega = \jmath \circ h_\omega. \tag{10.3}$$

*Every record explicitly pulled through the charts $h_{w_m}$ descends exactly. No planar projection to the nested impression $E_\omega$ is constructed.*

*Proof.* Take the quotient of $\bigsqcup_m A_{w_m}$ by the equivalence generated by $(z, m) \sim (t_m z, m + 1)$. Equation (10.2) makes

$$h_\omega([z, m]) = h_{w_m}(z) \tag{10.4}$$

well defined. Conversely, for $y \in A$, the points $(h_{w_m}|_{A_{w_m}})^{-1}(y)$ represent one equivalence class. Thus (10.4) is a bijection. Using it as a global chart gives the quotient its Hausdorff Riemann-surface structure and makes $h_\omega$ biholomorphic.

At level $m$, the retimed maps are $F \circ h_{w_m}$ and $\jmath \circ h_{w_m}$. Equation (10.2) proves compatibility and gives (10.3). The same calculation applies to each explicitly pulled record. It supplies no map into the original dynamical plane. □

For $w = a_1 \cdots a_{m+1}$, the inverse-word convention gives

$$h_{a_2 \cdots a_{m+1}} \circ g = h_w. \tag{10.5}$$

Therefore deletion of the first address symbol induces

$$\widehat{S}_\omega : \widehat{A}_\omega \longrightarrow \widehat{A}_{\sigma\omega}, \qquad h_{\sigma\omega} \circ \widehat{S}_\omega = h_\omega. \tag{10.6}$$

The normalized shift is an identity chart, but its ledger label remains the actual forward iterate $g = f^q$. It is not the geometric presentation change $t_m$.

12.2. **Finite shift-graded materialization.** A finite normalized diagram is *shift graded* if its vertices carry integers $s(v)$, root and presentation edges preserve $s$, and every forward shift $u \to v$ satisfies $s(u) = s(v) + 1$.

**Proved Theorem 12.2** (Depth-graded cylinder materialization)**.** *Let $\mathcal{D}$ be a finite shift-graded diagram made from actual composable root restrictions, actual root commuting squares, presentation maps, forward address shifts, and records already certified to be natural under the root charts. There is one $M$ for which vertex $v$ is represented at cylinder depth $M + s(v)$. Dynamical arrows are actual polynomial iterates, presentation arrows remain separately typed conformal maps, and every supplied root square still commutes.*

*The conclusion constructs no root portrait, relative cover, protected carrier, or marked finite-horizon assembly.*

*Proof.* Choose $M \geq -\min_v s(v)$. Pull the complete supplied root datum at $v$ back through the chart of depth $M + s(v)$. Equal grades make root and presentation formulas pull back at equal depth. Across a shift edge, the source depth is one more than the target depth, so (10.5) realizes it by $g$. Finiteness permits one common $M$ for all compactly contained supplied domains. Composition of the actual restrictions gives an actual iterate on its stated domain. The construction does not manufacture any absent root or marked structure. □

If two paths to the same vertex have different net shift counts, one depth would have to satisfy two incompatible equations. The diagram must then be unfolded by duplicating that vertex.

12.3. **Exact conformal observables.**

**Proved Theorem 12.3** (End-chart observable invariance)**.** *Every fully transported marked curve family has the same extremal length in the root, finite-cylinder, and normalized-end charts. A finite smooth electrode surface, with its ordered electrodes transported by the same charts, has the same Dirichlet-to-Neumann response matrix. Parallel sums and Schur-complement eliminations commute with these identities.*

*Proof.* Extremal length and real two-dimensional Dirichlet energy are conformally invariant. A marked conformal bijection identifies the electrode trace classes and their energy minimizers, hence their response quadratic forms. Parallel addition and Schur complementation are algebraic operations on the equal matrices. □

This theorem assumes the entire observable is transported. It gives no uniform port count, future-closed quotient, Value observability, or surface–WAD comparison.

12.4. **Conditional series law.**

**Proved Theorem 12.4** (Serial-exhaustion resistance bound)**.** *Let $C_1, \dots, C_m$ be pairwise interior-disjoint quadrilaterals. Suppose every $\gamma$ in a marked family $\Gamma$ contains, in chronological order, a crossing of each $C_j$ between its declared horizontal sides. If $R_j$ is the extremal length of the $j$-th crossing family, then*

$$\operatorname{Ext}(\Gamma) \geq \sum_{j=1}^{m} R_j. \tag{10.7}$$

*Where reciprocal conductance is defined, it is at most $(\sum_j R_j)^{-1}$.*

*Proof.* For each $j$, take an approximating admissible metric $\rho_j$, supported in $C_j$, for which every declared crossing has length at least one. For $a_j > 0$, put $\rho = \sum_j a_j \rho_j$. Every $\gamma \in \Gamma$ has $\rho$-length at least $\sum_j a_j$, while interior disjointness gives

$$\operatorname{Area}(\rho) = \sum_j a_j^2 \operatorname{Area}(\rho_j).$$

Optimizing the extremal-length quotient over the $a_j$, then taking the extremal-metric limit, gives (10.7). □

The intact-wall and no-bypass hypotheses are substantive. Boundary-ray labels alone do not prove them in every protected localization.

12.5. **The trace-mode obstruction.** The response approximation proposed in the external research packet replaces each cut cross-section by one scalar node. An ordinary analytic cut instead has an $H^{1/2}$ trace. Constant conductance does not control its nonconstant modes.

Let

$$C_L = [0, L] \times (\mathbb{R}/2\pi\mathbb{Z})$$

be the flat cylinder. Constant boundary values 1 and 0 at its ends have energy

$$E_0(L) = \frac{2\pi}{L}. \tag{10.8}$$

For $k \geq 1$, the harmonic function with boundary values $\cos(k\theta)$ and 0 is

$$u_k(x, \theta) = \frac{\sinh(k(L - x))}{\sinh(kL)} \cos(k\theta). \tag{10.9}$$

Green's identity gives

$$\int_{C_L} |\nabla u_k|^2 \, dA = \pi k \coth(kL). \tag{10.10}$$

**Proved Theorem 12.5** (Scalar-neck trace-mode no-go)**.** *No $L$-independent constant $C$ can bound the energy of every cross-section trace by*

$$C\,E_0(L)\,\mathrm{osc}(\varphi)^2. \tag{10.11}$$

*Consequently scalar two-terminal conductance alone cannot justify collapsing a cut analytic interface to one electrical node.*

*Proof.* Take $\varphi(\theta)=\cos\theta$. Its oscillation is 2. By (10.10), its energy is $\pi\coth L\ge\pi$; the right side of (10.11) is $8C\pi/L$, which tends to zero. □

12.6. **Exact full-trace response on a genuine rectangle.** The preceding no-go theorem is repaired completely for a *genuine* serial rectangle. It does not itself prove that a dynamically nested puzzle end is such a rectangle.

Let $C_R=(0,R)\times(0,1)$ with natural Neumann conditions at $y=0,1$. In the normalized Neumann basis

$$\varphi_0=1,\qquad \varphi_n=\sqrt{2}\cos(n\pi y),\quad n\ge1,$$

put $A\varphi_0=0$ and $A\varphi_n=n\pi\varphi_n$. The spectral trace space has norm

$$\|f\|_{\mathcal{H}}^2=|f_0|^2+\sum_{n\ge1}n\pi|f_n|^2.$$

**Proved Theorem 12.6** (Full rectangle trace response)**.** *For $f,g\in\mathcal{H}$, the minimizing energy is*

$$\begin{aligned}\mathcal{E}_R(f,g)=&\frac{|f_0-g_0|^2}{R}\\&+\sum_{n\ge1}n\pi\left[\coth(n\pi R)(|f_n|^2+|g_n|^2)-2\,\mathrm{csch}(n\pi R)\Re(f_n\overline{g_n})\right].\end{aligned} \tag{10.12}$$

*Equivalently,*

$$\mathcal{N}_R=\begin{pmatrix}A\coth(RA) & -A\,\mathrm{csch}(RA)\\ -A\,\mathrm{csch}(RA) & A\coth(RA)\end{pmatrix}, \tag{10.13}$$

*where both scalar functions take their continuous value $R^{-1}$ on $\ker A$. If two complete parameterized rectangle ends are glued, then*

$$\mathcal{N}_R\star\mathcal{N}_S=\mathcal{N}_{R+S}. \tag{10.14}$$

*Proof.* Mode zero has the affine minimizer and energy $|f_0-g_0|^2/R$. For $n\ge1$, the unique minimizing coefficient is

$$u_n(x)=\frac{\sinh(n\pi(R-x))}{\sinh(n\pi R)}f_n+\frac{\sinh(n\pi x)}{\sinh(n\pi R)}g_n.$$

Green's identity gives the $n$-th summand in (10.12); orthogonality gives the sum. For (10.14), adding the two rectangle energies and minimizing over their common complete trace is exactly the Dirichlet principle on $(0,R+S)\times(0,1)$. □

Let $P_0$ be projection onto constants and define

$$\mathcal{N}_R^{\#}=\begin{pmatrix}A+R^{-1}P_0 & -R^{-1}P_0\\ -R^{-1}P_0 & A+R^{-1}P_0\end{pmatrix}. \tag{10.15}$$

**Proved Theorem 12.7** (Sharp trace-renormalized comparison)**.** *Put*

$$\eta(R)=\frac{2}{e^{\pi R}-1}.$$

*Then*

$$\left|\mathcal{E}_R(f,g)-\mathcal{E}_R^{\#}(f,g)\right|\le\eta(R)(\langle Af,f\rangle+\langle Ag,g\rangle). \tag{10.16}$$

*If $\eta(R) < 1$, then*

$$(1-\eta(R))\mathcal{N}_R^{\#} \preceq \mathcal{N}_R \preceq (1+\eta(R))\mathcal{N}_R^{\#}. \tag{10.17}$$

*Moreover,*

$$\left\| \mathcal{N}_R - \begin{pmatrix} A & 0 \\ 0 & A \end{pmatrix} \right\|_{\mathcal{H}\oplus\mathcal{H}\to(\mathcal{H}\oplus\mathcal{H})^*} \le \max\left\{ \frac{2}{R}, \frac{2}{e^{\pi R}-1} \right\}. \tag{10.18}$$

*Proof.* On positive mode $n$, divide the difference by $n\pi$ and put $t = n\pi R$. The resulting matrix is

$$\begin{pmatrix} \coth t - 1 & -\operatorname{csch} t \\ -\operatorname{csch} t & \coth t - 1 \end{pmatrix},$$

whose eigenvalues are $2/(e^t-1)$ and $-2/(e^t+1)$. The largest absolute value is attained at $n=1$. The zero modes agree in $\mathcal{N}_R$ and $\mathcal{N}_R^{\#}$, proving (10.16) and (10.17). For (10.18), the constant block has normalized norm $2/R$, while the mean-zero block has norm at most $\eta(R)$. □

The comparison survives complete Hilbert-trace elimination. Indeed, if

$$q_N(x) = \inf_{\phi}\{Q(x,\phi) + N(\phi)\}, \qquad Q \ge 0,$$

and $(1-\delta)N_0 \preceq N_1 \preceq (1+\delta)N_0$, then

$$(1-\delta)(Q+N_0) \le Q + N_1 \le (1+\delta)(Q+N_0).$$

Taking the infimum over the same trace space gives

$$(1-\delta)q_{N_0} \le q_{N_1} \le (1+\delta)q_{N_0}. \tag{10.19}$$

Consequently, for a nonnegative homogeneous Loewner-monotone gauge,

$$L_{\#}^{\mathrm{out}} \le \mu L_{\#}^{\mathrm{in}} + E, \quad 0 < \mu < 1, \quad E \ge 0,$$

transfers under two-sided physical comparisons of size $\eta < 1$ to

$$L_{\mathrm{phys}}^{\mathrm{out}} \le \mu \frac{1+\eta}{1-\eta} L_{\mathrm{phys}}^{\mathrm{in}} + (1+\eta)E. \tag{10.20}$$

The factor remains strict precisely when $\eta < (1-\mu)/(1+\mu)$. This is a conditional transfer lemma: it does not supply the root contraction or the physical rectangle.

12.7. **Why pulled equipotential arcs do not serialize.** Research Push IV attempted to apply the preceding calculus by taking one equipotential cross-section and all of its inverse pullbacks. The exact Böttcher model refutes the required geometry.

**Proved Theorem 12.8** (Pulled-equipotential serial-rectangle no-go)**.** *Let $d = 2^q > 1$ and consider $z \mapsto z^d$ outside the unit disk. A nondegenerate equipotential arc and its image under one inverse branch cannot be the two full equipotential sides of a quadrilateral bounded by the same two external rays.*

*Proof.* Write $G = \log|z|$ and $\Theta = \arg z/(2\pi) \pmod 1$. If

$$\gamma(t, I) = \{G = t,\ \Theta \in I\}$$

and $I$ has angular width $\Delta > 0$, inverse branch $k$ sends

$$(G, \Theta) \longmapsto \left( \frac{G}{d}, \frac{\Theta + k}{d} \right).$$

Thus the pulled arc has angular width $\Delta/d$. The two equipotential sides of a quadrilateral bounded by two fixed rays span the same angular interval and therefore have equal width. Equality would require $\Delta = \Delta/d$, impossible for $d > 1$ and $\Delta > 0$. □

An inverse branch does map a rectangle conformally to a smaller rectangle. The region *between* the rectangle and its nested copy is not thereby a rectangle: the unmatched angular footprint produces additional boundary pieces and potentially additional trace ports. The exact semigroup (10.14) cannot be applied until those interfaces are constructed.

There is a separate quantitative requirement. A finite depth realizing a target $\eta$ exists only if the physical accumulated resistance tends beyond the required threshold. Positive cell resistances do not suffice: $\sum_j 2^{-j} < \infty$. Neither the pullback compactification nor the abstract serial-exhaustion theorem proves the required resistance divergence for the rejected cells.

**Open Problem 12.9** (End-plumbing trace-response control)**.** For every proof-used materialized neck, construct the actual complete parent and child trace spaces, their restriction or extension map, every unmatched side component, and the resulting possibly multi-port Dirichlet-to-Neumann form. Prove exact serial composition or compact multi-port closure, protected-carrier and future-port compatibility, and sufficient accumulated resistance for every claimed transition margin. Scalar collapse is allowed only under the proved constant-trace criterion.

12.8. **Ambient survivors, fixed-end WADs, and the first response gate.** The route-exit calculation uses two different finite horizons:

$$N = b + 3, \qquad m = N - 1 = b + 2. \tag{10.102}$$

Ambient KKL breakability is tested through $N$; named route pullbacks have depths $1 \le h \le m$.

**Proved Proposition 12.10** (Ambient route-exit survivor partition)**.** *Let $E_A$ be the family defined by a complete single-valued marked first-exit selector on a native KKL Wave. Then*

$$\begin{aligned} E_A^{\mathrm{br}} &= E_A \cap A^{N\text{-}br}, \\ E_A^{\mathrm{sh}} &= E_A \cap A^{N\text{-}nbr} \cap A^{\mathrm{sh}}, \\ E_A^{\mathrm{surv}} &= E_A \cap A^{N\text{-}nbr} \setminus A^{\mathrm{sh}} \end{aligned} \tag{10.103}$$

*are pairwise disjoint and exhaust $E_A$. The first two families belong to*

$$B_A = A^{N\text{-}br} \cup A^{\mathrm{sh}}; \tag{10.104}$$

*only $E_A^{\mathrm{surv}}$ is a new response family.*

*Proof.* First split $A$ into its $N$-breakable and $N$-unbreakable parts. Split the latter by membership in $A^{\mathrm{sh}}$. Intersecting both binary partitions with the single-valued family $E_A$ gives (10.103). The two inclusions in (10.104) are immediate. □

**Proved Theorem 12.11** (Pointwise survivor route-switch flag)**.** *For $\alpha \in E_A^{\mathrm{surv}}$, write*

$$\delta_0 = \alpha, \qquad F_{j*}\delta_j = \{\delta_{j+1}\}, \qquad 0 \le j < N. \tag{10.105}$$

*Define*

$$\begin{aligned} \mathrm{Ret}_j(\delta_j, \delta_{j+1}) = \{\, \gamma : \gamma \in \iota_j^* \delta_j, \quad & \gamma \text{ is full and proper in the fine marking,} \\ & \gamma \text{ survives the coarse subordinate filter,} \quad f_{j*}\gamma = \delta_{j+1} \,\}. \end{aligned} \tag{10.106}$$

*Then*

$$\mathrm{Ret}_j(\delta_j, \delta_{j+1}) \neq \varnothing \qquad (0 \le j < N). \tag{10.107}$$

*At the first named-route failure, either the named occurrence is not in this retained fiber and a different full retained path replaces it, or the underlying retained path exists but its complete marked successor state differs. For simple oriented Wave representatives, the stagewise choices and their unique occurrence embeddings form a linked segmented prefix flag. They need not form one full proper deepest fiber-product path.*

*Proof.* The source definition of the native stage action is

$$F_{j*} = f_{j*}\iota_j^*$$

after the proper-path, fine-marking, and coarse-subordination filters. Equation (10.105) is a nonempty singleton equality, so at least one filtered source restriction has covering image $\delta_{j+1}$. This proves (10.107). Finite pointwise choice gives one retained predecessor at every stage. At the first failure, the named object either fails membership in the retained fiber—including partiality, nonproperness, filter loss, or wrong covering image—or it is retained and one of the frozen marked-state predicates fails. These alternatives are exhaustive.

For simple representatives, $\iota_j(|\gamma_{j+1}|) \subset |f_j\gamma_j|$. The segmented-occurrence lemma extracts the unique orientation-preserving occurrence embedding at each stage. Its prefix reconstruction theorem gives only linked prefix flags; earlier coordinates can be strict nonproper truncations. No measurable choice or ideal-end survival is inferred. □

**Imported Theorem 12.12** (KKL fixed-end immersion decomposition)**.** *Let $i : V' \to V$ be a holomorphic immersion of finite-topology open Riemann surfaces. Let $E'$ be source ends on which $i$ is proper of degree one, put $E = i(E')$, and let $Y$ be a valid WAD on $V$ whose arcs each have an endpoint in $E$. Then the source construction gives a valid WAD $Y'$ and paired pieces*

$$Y = \sum_a v_a\beta_a, \qquad Y' \geq \sum_a v_a\beta'_a, \tag{10.108}$$

*where either both ends of $\beta'_a$ lie in $E'$ and $i_*\beta'_a = \beta_a$, or one end lies in $E'$ and the other does not.*

This is the source theorem 'thm:immer-decomp' in the locked KKL package [KKL26]. Its proof lifts one realizing target foliation from the prescribed source ends and subdivides the paired target and source foliations by their homotopy classes.

**Proved Proposition 12.13** (Fixed-end full/exit WAD split)**.** *Splitting the witness (10.108) by its two endpoint alternatives gives valid WADs*

$$Y_{\mathrm{full}}, \quad Y_{\mathrm{exit}}, \quad Z'_{\mathrm{full}}, \quad Z'_{\mathrm{exit}}$$

*such that*

$$\begin{aligned} Y &= Y_{\mathrm{full}} + Y_{\mathrm{exit}}, & Y' &\geq Z'_{\mathrm{full}} + Z'_{\mathrm{exit}}, \\ \|Y_{\mathrm{full}}\| &= \|Z'_{\mathrm{full}}\|, & \|Y_{\mathrm{exit}}\| &= \|Z'_{\mathrm{exit}}\|. \end{aligned} \tag{10.109}$$

*These are paired WAD coefficient-norm equalities for the source witness. They are not arbitrary marked-modulus equalities and do not imply $\|Y'\| = \|Y\|$.*

*Proof.* The source proof first partitions the realizing target foliation by target homotopy class and then the lifted source foliation by source homotopy class. Restriction to either endpoint alternative therefore gives proper Borel sublaminations and valid sub-WADs. The paired source and target pieces carry the same coefficient $v_a$; summing those coefficients over each alternative proves (10.109). □

**Proved Proposition 12.14** (Blockwise finite marked-end selection)**.** *Let $\Lambda$ be a standard Borel transversal for a realizing target lamination, with Borel saturation for every leaf subset used below. For $\varepsilon \in \{-,+\}$, let $E'_\varepsilon$ be the finite set of proof-used topological source ends and let*

$$\mathcal{C}^\varepsilon \subset \Lambda \times E'_\varepsilon \tag{10.110}$$

*be Borel with nonempty fibers. Assume every pair in $\mathcal{C}^\varepsilon$ satisfies the oriented target-end equality and every proper-degree-one fixed-end eligibility predicate. After ordering $E'_\varepsilon$, the selectors*

$$e^\varepsilon(\delta) = \min \mathcal{C}^\varepsilon_\delta \tag{10.111}$$

*are Borel, and their pair fibers form a finite Borel partition on which imported theorem 12.12 can be applied in both orientations.*

*If both maximal lifts are independently certified to exit at the same complete occurrence code, the partition couples the common target leaf, transverse parameter, and selected ends. It does not prove the two exits or marking-filter survival. The partition is finite only for the individual surface; no uniform bound over periods, marked accesses, ideal intervals, sheets, or occurrence codes follows.*

*Proof.* For a fixed ordered end $e_r^\varepsilon$,

$$\{\delta : e^\varepsilon(\delta) = e_r^\varepsilon\} = \{\delta : (\delta, e_r^\varepsilon) \in \mathcal{C}^\varepsilon\} \setminus \bigcup_{t<r} \{\delta : (\delta, e_t^\varepsilon) \in \mathcal{C}^\varepsilon\},$$

which is Borel. Intersect the two finite selector partitions. Borel saturation gives proper Borel target sublaminations, and the fixed-end eligibility assumptions permit the two source-theorem applications. The final coupling statement uses the same target leaf index and the independently supplied common occurrence code. □

**Proved Theorem 12.15** (Conditional vertical survivor-packet charge)**.** *Suppose a survivor sublamination $\mathscr{S}_h^{\mathrm{ver}}$ has an injective-modulo-zero Borel assignment to a genuine source lamination $\mathscr{B}_h^{\mathrm{ver}}$ of full proper paths joining the pulled-tree ideal boundary to the outer ideal boundary. Assume the native marking filters, covering extension on the ending locus, and endpoint degree*

$$D_h \le 2^h.$$

*Then*

$$\mathcal{W}(\mathscr{S}_h^{\mathrm{ver}}) \le D_h\big(V(F)+2\big) \le 2^h\big(V(F)+2\big). \tag{10.112}$$

*Consequently*

$$\sum_{h=1}^{b+2} \mathcal{W}(\mathscr{S}_h^{\mathrm{ver}}) \le (2^{b+4}-4)\big(V(F)+1\big). \tag{10.113}$$

*Proof.* Domination and the pushforward-lamination lemma give a target lamination $\mathscr{C}_h^{\mathrm{ver}}$ with

$$\mathcal{W}(\mathscr{S}_h^{\mathrm{ver}}) \le \mathcal{W}(\mathscr{B}_h^{\mathrm{ver}}) \le D_h \mathcal{W}(\mathscr{C}_h^{\mathrm{ver}}) + 2D_h.$$

The output joins the two boundary components of $\mathcal{U} \setminus \mathcal{T}$, so $\mathcal{W}(\mathscr{C}_h^{\mathrm{ver}}) \le V(F)$. This proves (10.112). Finally,

$$\sum_{h=1}^{b+2} 2^h = 2^{b+3} - 2, \qquad V + 2 \le 2(V+1),$$

which gives (10.113). □

**Open Problem 12.16** (Optional Freudenthal fixed-end survivor route)**.** For the optional fixed-end route, first construct an access-splitting localization and actual Freudenthal ends satisfying every proper-degree-one hypothesis of imported theorem 12.12. A marked $BL_0$ or $L_0^h$ access may not be relabeled as such an end. Then construct the Borel candidate relations and common-occurrence exits, retain all access, cyclic-order, sheet, obstacle, window, relative-cover, and filter data, and select the retained predecessor relation with uniformly bounded or summably charged multiplicity.

Equivalently, establish the six source-facing clauses

$$C_h^- \wedge X_h^- \wedge C_h^+ \wedge X_h^+ \wedge P_h \wedge M_h, \tag{10.114}$$

where $C_h^{\pm}$ are the proper-degree-one starting-end certificates, $X_h^{\pm}$ are exits at the same actual $L_0^h$-occurrence, $P_h$ is the complete common-leaf occurrence coupling, and $M_h$ is native filter and covering-image survival. Then classify every ambient survivor replacement into a proper vertical packet, an exact small-Julia-incidence packet, an independently terminal response, or a retained explicit residual.

This is one possible construction route, not a necessary one: actual paired source Waves may instead be supplied directly or by an access-local continuation theorem. The first uncontrolled output is a replacement projecting to a Wave ending at $BL_0$ with neither vertical nor prescribed small-Julia incidence. The KKL critical-limb proposition cannot be used to control this class because its proof contains the open line-1603 assertion. No part of this open problem is discharged by the finite schema or by pointwise choice.

*Warning* 12.17 (Promotion boundary). The results above prove local, conditional reductions. They do not construct the measurable survivor packets, prove QSDB, repair the stationary Wave proof, or promote VAAPC-I, UPTCR, or MLC.

12.9. **Typed access continuation and paired-Wave response.**

**Proved Proposition 12.18** (Freudenthal/access typing boundary)**.** *The source set $E'$ in imported theorem 12.12 consists of Freudenthal ends. KKL's stationary $BL_0$ and $L_0^h$ are marked ideal multi-intervals or accesses. The locked source constructs neither an access-splitting localization nor an eligible Freudenthal end representing the selected access.*

*Assume additionally that the actual source and target carriers satisfy* $\mathrm{TWO}_h$ *and* $\mathrm{TWO}_0$*: each is compact, connected, and full in its conformal disk. Assume also the boundary-compatible inclusion hypothesis* $\mathrm{INC}_h$*. Then each tree complement has two Freudenthal ends, the source inner end is nonproper for the immersion, and, whenever the fixed-end theorem is otherwise applicable to a target WAD with two inner endpoints, every paired exit term based at an eligible source outer end has vertical topology.*

*Proof.* KL06 defines the input of the fixed-end theorem as a subset of the Freudenthal end set of the source surface. KKL's ideal marking instead records cyclically ordered intervals and accesses on ideal boundary components. No implication from the latter datum to the former is present in the source, which proves the unconditional type boundary.

Under $\mathrm{TWO}_h$, the complement of the full connected carrier has one end at the carrier and one at the outer boundary. Under $\mathrm{INC}_h$, a sequence tending to the source inner end through the selected access has image converging to a point of the pulled carrier which lies in the interior of the target complement. Its image therefore remains in a compact target disk, contradicting properness at that end. The only other source end is the outer end. The exit alternative in the fixed-end theorem joins an eligible end to an end outside the eligible set, hence joins the outer and inner ends. This is precisely the asserted vertical topology. Each conclusion after the first paragraph uses the displayed conditional hypotheses. □

**Proved Theorem 12.19** (Access-seeded maximal continuation)**.** *Let $X, Y$ be second-countable locally compact metric spaces, let $I : X \to Y$ be a local homeomorphism, and let $\Lambda$ be a second-countable metric space. Suppose*

$$\Phi : \Lambda \times (0,1) \longrightarrow Y \tag{10.115}$$

*is continuous, each path $\Phi(\lambda, \cdot)$ is embedded, and distinct path images are disjoint. Fix $0 < t_0 < 1$, and suppose a continuous lift*

$$\widehat{\Phi}_0 : \Lambda \times (0, t_0] \longrightarrow X, \qquad I\widehat{\Phi}_0 = \Phi \tag{10.116}$$

*is supplied. If $(0, \tau(\lambda))$ is its maximal continuation interval, then*

$$\mathcal{D} = \{(\lambda, t) : 0 < t < \tau(\lambda)\} \tag{10.117}$$

*is open, the joint maximal lift $\widehat{\Phi} : \mathcal{D} \to X$ is continuous, and $\tau$ is lower semicontinuous and Borel.*

*If $X$ is open and dense in a compact metrizable space $\widehat{X}$, then the set on which the terminal lift has a limit in $\widehat{X}$ is Borel and the landing map is Borel. For a finite horizon of such systems on one common standard-Borel leaf domain, Borel-compatible continuous product charts, Borel boundary strata, and explicitly coded seams/no-route leaves give a Borel first-failure record.*

*Proof.* For one leaf, local inverse charts continue (10.116) uniquely. Fix $(\lambda_0, t_1) \in \mathcal{D}$ with $t_1 > t_0$. The lifted segment over $[t_0, t_1]$ is compact. Cover it by finitely many inverse charts, shrink around each transition so that its closure lies in the overlap of consecutive charts, and use continuity of $\Phi$ and the seed to preserve this finite chart chain for all nearby $\lambda$. A final inverse chart continues every nearby lift past $t_1$. Thus $\mathcal{D}$ is open and the joint lift is continuous. For $a \le t_0$, $\{\tau > a\} = \Lambda$; for $t_0 < a < 1$, the same finite-chart argument over $[t_0, a]$ makes $\{\tau > a\}$ open. This is lower semicontinuity.

Choose a bounded compatible metric $d$ on $\widehat{X}$, and let $\operatorname{osc}_n(\lambda)$ be the supremum of

$$d(\widehat{\Phi}(\lambda, q), \widehat{\Phi}(\lambda, r))$$

over rational $q, r$ in the last $1/n$ of the maximal interval. Each $\operatorname{osc}_n$ is a countable supremum of Borel functions. The terminal limit exists exactly when $\operatorname{osc}_n \to 0$. Hence the landing domain is Borel. Evaluating at the Borel times

$$q_j(\lambda) = t_0 + (\tau(\lambda) - t_0)(1 - j^{-1})$$

and taking the pointwise limit proves that the landing map is Borel. A finite minimum of Borel failure indices and gluing over a tagged Borel partition gives the last assertion. □

*Warning* 12.20 (Scope of access-seeded continuation). KL06 defines a proper lamination as a Borel union of disjoint proper paths; this does not supply the continuous product charts in (10.115). Nor does (10.116) force the lift to leave every compact set as $t \downarrow 0$. Thus theorem 12.19 proves no seed-side properness, native product exhaustion, valid source lamination, marking survival, transverse-width identity, or stationary response.

**Proved Theorem 12.21** (Exact occurrence Series conductance)**.** *Let $S$ be a full-width target path family. Suppose that for every target path one has selected $r$ pairwise-disjoint, untrimmed subpaths, and let $\Gamma^i$ be the family of $i$-th subpaths. Suppose $B^i$ is a paired source path family satisfying the exact parametrized identity $I_h(B^i) = \Gamma^i$. Then*

$$\mathcal{W}(S) \le \bigoplus_{i=1}^{r} \mathcal{W}(B^i), \qquad \bigoplus_{i=1}^{r} x_i = \left( \sum_{i=1}^{r} x_i^{-1} \right)^{-1}. \tag{10.119}$$

*Proof.* Every member of $S$ contains the disjoint union of one member of each $\Gamma^i$. The iterated KL06 Series Law gives

$$\mathcal{W}(S) \le \bigoplus_{i} \mathcal{W}(\Gamma^i).$$

Holomorphic-image monotonicity and the exact image identity give $\mathcal{W}(\Gamma^i) \le \mathcal{W}(B^i)$. The harmonic sum is monotone in every coordinate, proving (10.119). No immersion injectivity, deck noncollision, or Wave typing is used. □

**Proved Theorem 12.22** (Conditional paired-Wave response)**.** *Assume each $B^i$ in theorem 12.21 is a genuine source Wave ending on a whole marked multi-interval on which*

$$d_i \le 2^h, \qquad h \le b + 2.$$

*Assume one constant $c_{\mathrm{W}}$ is valid for every Wave in the packet in KKL's stated estimate $\mathcal{W}(B^i) \le \mathcal{W}(C^i) + c_{\mathrm{W}}(1+\log d_i)$. Put $C_b = c_{\mathrm{W}}(1+(b+2)\log 2)$. Then*

$$\max_i \mathcal{W}(C^i) \ge r\mathcal{W}(S) - C_b. \tag{10.120}$$

*For $r = 2$,*

$$\max\{\mathcal{W}(C^-), \mathcal{W}(C^+)\} \ge 2\mathcal{W}(S) - C_b. \tag{10.121}$$

*If all $C^i$ lie in one certified output Wave $T$, the same right-hand side is a lower bound for $\mathcal{W}(T)$.*

*Proof.* Let $M = \max_i \mathcal{W}(C^i)$. The assumed Wave estimate and $d_i \le 2^{b+2}$ give $\mathcal{W}(B^i) \le M + C_b$. Hence

$$\mathcal{W}(S) \le \bigoplus_i \mathcal{W}(B^i) \le \bigoplus_i (M + C_b) = \frac{M + C_b}{r}.$$

Rearrangement proves (10.120). If $C^i \subset T$, monotonicity gives $\mathcal{W}(C^i) \le \mathcal{W}(T)$, proving the collision statement. □

**Open Problem 12.23** (Typed paired-Wave realization and uniform response)**.** On the entire nonshort ambient-singleton $BL_0$-ending residual, construct a Borel first-horizon partition without duplicated transverse width and two full-width, leafwise-paired genuine source Waves with disjoint untrimmed target occurrence images. The start must be typed as `DIRECT_WAVE`, `ACCESS_LOCAL`, or a genuinely localized `FREUDENTHAL` construction; an access is never relabeled as an end. Prove exact image identities, source properness and lamination validity, marking/filter survival, whole-ending-set degrees, and a zero-width or valid $V(F)/R(F)$ charge for every omitted family.

Independently verify one packet-uniform $c_{\mathrm{W}}$ from a complete proof of the stated KKL Wave estimate. Finally prove common-output or maximizing-output controllability depending only on $b$, an ambient-width fraction, and an absolute threshold absorbing $C_b$. Only then may (10.121) be promoted from a packet response to KKL terminality.

*Warning* 12.24 (Stationary status after the response calculation). The exact Series inequality and response algebra are proved, but the native paired source Waves, the uniform hidden Wave constant, and the uniform survivor response remain open. Consequently the stationary Wave repair, VAAPC-I, UPTCR, and MLC remain open.

Thus the full-trace rectangle calculus is proved, but the proposed `BOTTCHER_TRACE_COLLAR_ATLAS` and `END_PLUMBING_TRACE_RESPONSE_CONTROL` promotions are not. The replacement geometric gate is `DIRECT_DECORATION_TRACE_COLLAR_REALIZATION`. Until it and open problem 12.9 are proved, direct-decoration finite-horizon inflation and strict physical transition transfer are not available. Independently, `FINITE_HORIZON_ASSEMBLY` and `MAPPING_TELESCOPE` remain open. Thus this section changes neither UPTCR nor MLC from open to proved.

12.10. **The physical one-cell repair.** The unequal-footprint obstruction does admit a precise local repair. Its hypothesis must be stated independently of the direct-decoration source.

**Proved Theorem 12.25** (Active/inactive attached-Jordan cell)**.** *Let $P, Q$ be closed Jordan disks with $Q \subset P$, and suppose that their complete boundary intersection is one nondegenerate closed arc $A$, with endpoints $x^-, x^+$. Put*

$$\Omega = \operatorname{int} P \setminus Q, \qquad C = \overline{\Omega}.$$

*Then*

$$P \setminus \operatorname{int} Q = C \cup A, \qquad C \cap A = \{x^-, x^+\}. \tag{10.21}$$

*Moreover, $C$ is a closed Jordan disk and the relative interior of $A$ is not approached from $\Omega$.*

*Proof.* At a relative-interior point of $A$, the two Jordan curves agree on a boundary neighborhood. Their interiors lie on the same local side because $Q \subset P$. Hence $\Omega$ does not approach $A^\circ$. Removing $A^\circ$ from the two Jordan curves leaves two simple arcs with endpoints $x^-, x^+$. Their relative interiors are disjoint because $\partial P \cap \partial Q = A$, so their union is a Jordan curve. Its bounded Jordan domain is $\Omega$, and the only points of $P \setminus \operatorname{int} Q$ outside its closure are the points of $A^\circ$. This proves (10.21). □

The theorem is conditional on the complete-intersection hypothesis. The locked decoration source defines $g = f^q|_{Q_L \cup Q_R}$ and identifies the $2^n$ boundary-attached pieces $Q^v$ with the components of $\operatorname{dom}(g^n)$. Thus the binary branches and length-$n$ macrobranches are related, but they are not interchangeable alphabets with the same contact record. Accordingly, theorem 12.25 does not, by itself, source-license the binary end package.

12.11. **Correct one-junction trace refinement.** Let $A^-, A^+$ be closed attachment arcs on a prime-end circle and put

$$V = \partial A^- \cup \partial A^+.$$

Raw set differences of the closed arcs need not be disjoint from $V$. The exact construction takes the components $J$ of $S^1 \setminus V$ and assigns the two activity bits

$$\big(\mathbf{1}_{A^-}(J), \mathbf{1}_{A^+}(J)\big).$$

The four possible values define

| | |
|---|---|
| $(0,0)$ | $I$ (common active), |
| $(0,1)$ | $E^{\mathrm{out}}$ (outer only), |
| $(1,0)$ | $E^{\mathrm{in}}$ (inner only), |
| $(1,1)$ | $Z$ (double inactive). |

Consequently

$$S^1 = I \sqcup E^{\mathrm{out}} \sqcup E^{\mathrm{in}} \sqcup Z \sqcup V \tag{10.22}$$

is a genuine disjoint partition. There are at most four genuine endpoints and four open atoms. This is only a pairwise local structural count: it does not bound the branch alphabet, proof markings, port lifetime, or analytic response dimension.

For two regular Dirichlet cells with one actual admissible common active interface $I$, the exact union response is the fiber-product infimum

$$q(x_1, x_2) = \inf_{\substack{r_1\tau_1 = r_2\tau_2 \\ e_i\tau_i = x_i}} \big(q_1(\tau_1) + q_2(\tau_2)\big). \tag{10.23}$$

This is the Sobolev gluing theorem and additivity of Dirichlet energy. One-sided atoms remain external variables. For arbitrary Jordan boundaries, the required Douglas-trace/capacity gluing statement must be proved separately; it is not a consequence of the word "Jordan" alone.

**Proved Theorem 12.26** (Finite tame contact-overlay response)**.** *Let a bounded planar domain $\Omega$ be partitioned up to area zero into finitely many bounded Lipschitz cells $\Omega_i$ with pairwise disjoint interiors. Suppose the positive-length interior interfaces are finitely many paired Lipschitz arcs with opposite normals and the remaining interior junction set has one-dimensional measure zero. Then restriction identifies $H^1(\Omega)$ with the finite fiber product of the $H^1(\Omega_i)$ over matching paired traces. The $H^1$ norm and Dirichlet energy are the sums of the cellwise quantities, and the global response is the corresponding finite fiber-product infimum.*

*Proof.* Restriction of a global $H^1$ function has matching traces on every common Lipschitz arc, and additivity holds because the interiors partition $\Omega$ almost everywhere. Conversely, define $u$ cellwise from a compatible tuple $u_i \in H^1(\Omega_i)$. For $\varphi \in C_c^\infty(\Omega)$, apply the weak Green formula in every cell. Exterior boundary terms vanish against $\varphi$. Every interior arc occurs twice with

opposite normals and equal traces, so its two boundary terms cancel; the junction set has zero $\mathcal{H}^1$-measure. Hence the piecewise weak derivatives define the distributional derivatives of $u$ and lie in $L^2(\Omega)$. Thus $u \in H^1(\Omega)$, with the asserted norm and energy identities. The response formula follows by minimizing over exactly these compatible tuples. □

This theorem assumes the tame finite overlay. It neither constructs that overlay from puzzle pullbacks nor supplies an arbitrary-Jordan prime-end-trace theorem, all-depth contact closure, or welding control.

12.12. **Why pairwise refinement does not serialize the end.** The first iterated inference in Research Push V is false without an all-depth contact audit.

**Proved Theorem 12.27** (Nonconsecutive attachment bypass)**.** *There are four nested closed Jordan disks whose three consecutive active/inactive decompositions are individually of the form in theorem 12.25, but for which the first and third active shells share an active boundary arc that is inactive for the middle shell.*

*Proof.* Take

$$\begin{aligned} D_0 &= [0,10] \times [0,10], \\ D_1 &= [1,9] \times [1,10], \\ D_2 &= [3,9] \times [3,8], \\ D_3 &= [3,8] \times [4,7], \end{aligned} \tag{10.24}$$

and set

$$C_j = \overline{\operatorname{int} D_{j-1} \setminus D_j}, \qquad j = 1,2,3.$$

The successive common arcs lie on the top, right, and left sides, respectively. On

$$J = \{9\} \times (3,8),$$

$C_1$ approaches from the right and $C_3$ approaches from the left. The middle shell $C_2$ does not approach $J$, since $D_1, D_2$ occupy the same local side of their common right arc. Thus contact-closing the physical decomposition glues $C_1$ directly to $C_3$ on $J$, while skipping $C_2$. □

At the first relevant shell-to-shell junction, $J$ is a one-sided port and becomes active again two levels later. Thus the response of the immediately preceding shell is not a sufficient state. There are two possible repairs: retain an all-pairs contact structure, or retain the exact response of the accumulated prefix on its current boundary. The next theorem proves that the latter exposes every raw past–future contact for one fixed nested chain. It does not yet include phase, orientation, protected-access, sibling-branch, trace-coordinate, or enlargement data.

In particular, the metric proving the conditional serial law need not charge the bypass in theorem 12.27. Hence the physical hypotheses required for

$$R_{\text{total}} \geq \sum_j R_j \tag{10.25}$$

are not established. The abstract `SERIAL_EXHAUSTION_RESISTANCE_BOUND` remains correct under its stated chronological-crossing, intact-wall, and no-bypass assumptions.

12.13. **The nested-separator repair and its exact boundary.**

**Proved Theorem 12.28** (Binary direct-decoration nested-separator calculus)**.** *Let*

$$D_0 \supset D_1 \supset \cdots \supset D_H$$

*be a finite nested chain of closed sets, and put*

$$\Omega_j = \operatorname{int} D_0 \setminus D_j, \qquad S_{j+1} = \operatorname{int} D_j \setminus D_{j+1},$$

$$J_j = (\operatorname{int} D_0 \cap \partial D_j) \setminus D_{j+1}.$$

*Then, for $k > j$,*

$$\overline{\Omega_j} \cap \overline{S_k} \subset \partial D_j, \tag{10.26}$$

*and*

$$\Omega_{j+1} = \Omega_j \sqcup J_j \sqcup S_{j+1}. \tag{10.27}$$

*If $i < k$ and $x \in \partial D_i \cap \partial D_k$, then $x \in \partial D_j$ for every $i \le j \le k$.*

*For the source-locked binary $f^q$ inverse branches, these sets are the actual nested puzzle disks. If the proof-used positive-length parts of $J_j$ have separately verified finite admissible Sobolev-interface decompositions, prefix-to-next-shell fiber-product minimization gives the exact actual-domain Dirichlet response.*

*Proof.* Nesting gives $S_k \subset D_{k-1} \subset D_j$, hence $\overline{S_k} \subset D_j$. Since $\Omega_j \subset \mathbb{C} \setminus D_j$,

$$\overline{\Omega_j} \subset \mathbb{C} \setminus \operatorname{int} D_j.$$

Intersecting the two inclusions proves (10.26).

For $x \in \Omega_{j+1}$, exactly one of

$$x \notin D_j, \qquad x \in \partial D_j, \qquad x \in \operatorname{int} D_j$$

holds. The corresponding alternatives place $x$ in $\Omega_j, J_j, S_{j+1}$, respectively. The converse inclusions are immediate, proving (10.27), including the finite vertices as a set identity.

If $x \in \partial D_i \cap \partial D_k$ and $i < j < k$, then $x \in D_k \subset D_j$. Were $x \in \operatorname{int} D_j$, the inclusion $D_j \subset D_i$ would imply $x \in \operatorname{int} D_i$, contradicting $x \in \partial D_i$. Thus $x \in \partial D_j$.

For the dynamical specialization, each inverse branch is a homeomorphism of closed puzzle disks and

$$D_{w_j a} = \phi_{w_j}(Q_a) \subset \phi_{w_j}(P) = D_{w_j}.$$

On an independently realized finite admissible interface, theorem 12.26 and (10.23) give the response assertion. □

The interval conclusion says that one actual physical open arc cannot leave the nested boundary and later reappear as that same arc. At the stage when $C_3$ is attached in theorem 12.27, the old $C_1$-to-$C_3$ contact lies on $\partial D_2$, so the exact prefix response sees it. This resolves the raw bypass bookkeeping for one address chain, not the physical resistance or compactness problem.

The exact root-shared boundary set has a set-valued recursion. With

$$B_j = h_{w_j}(\partial D_j \cap \partial D_0) \subset \partial P$$

and next symbol $a$, persistence and $h_{w_{j+1}} = g \circ h_{w_j}$ give

$$B_{j+1} = g(B_j \cap \partial Q_a). \tag{10.28}$$

This must not be replaced by the finite atom-to-atom map proposed in Research Push VI. On a shared ray, Böttcher height satisfies $t \mapsto dt$, $d = 2^q > 1$. If $v$ is the smallest positive vertex of a finite root partition, its first atom $(0, v)$ maps to $(0, dv)$, which crosses $v$ and is not one atom of the same partition. An exact finite replacement must use one-to-many incidence and trace-restriction data.

There is a blockwise relational repair under explicit additional hypotheses. Put

$$\Gamma = \partial P, \qquad C_a = \Gamma \cap \partial Q_a, \qquad C = C_L \cup C_R.$$

Assume that every $C_a$ is a finite union of closed root subarcs and vertices, positive Böttcher height is multiplied by $2^q$ while an endpoint orbit remains in $C$, every height-zero seed endpoint

is preperiodic, and zero-length access data are recorded separately. Starting with the finite root and contact endpoints $V_0$, take the least forward saturation

$$g(V \cap C) \subset V.$$

This $V$ is finite: positive heights leave the bounded contact graph after finitely many multiplications by $2^q > 1$, and the height-zero seed orbits are finite. For

$$\mathcal{E} = \pi_0(\Gamma \setminus V), \qquad \Theta_a(J) = \{K \in \mathcal{E} : K \subset g(J)\},$$

the endpoints of $g(J)$ lie in $V$, and the open-arc masks obey

$$R_{j+1} = \bigcup_{\substack{J \in R_j \\ J \subset C_a}} \Theta_a(J) \qquad (\mathrm{mod}\ V).$$

This proves a finite, block-dependent raw contact mask under the stated hypotheses. It does not source-realize the complete binary contact sets: the checked source does not list all $\partial P \cap \partial Q_a$ contacts or their endpoint itineraries. Nor does it give a finite-dimensional response state. When one input atom maps across several target atoms, the resulting traces are linked by one restriction/presentation operator and cannot be treated as independent ports. Protected or enlargement endpoints may also have infinite forward orbits.

There are two further analytic defects. For a frozen physical overlay $Z$,

$$h_{w_j}(S_{j+1} \setminus Z) = (\operatorname{int} P \setminus Q_a) \setminus h_{w_j}(Z \cap S_{j+1}), \tag{10.29}$$

and the last set generally depends on the word and stage. Hence two fixed protected root-cell forms do not generate the response unless overlay equivariance is proved. Also, the full $H^1$ norm is not conformally invariant, whereas Dirichlet energy is only a seminorm until the correct componentwise constants are quotiented or grounded. Therefore the submitted fixed contact-aware cocycle and its finite-state consequences are not promoted.

12.14. **Closedness and last-use elimination.** For an extended nonnegative functional $q$ and composable maps $p, r$, the identity

$$(r \circ p)_* q = r_*(p_* q), \qquad (p_* q)(y) = \inf_{p\tau = y} q(\tau), \tag{10.30}$$

is exact: both sides minimize over the same fiber. Closed response forms, however, are not preserved by arbitrary continuous forgetful maps.

Let $H = \ell^2$, let $Ae_n = ne_n$, and define on $D(A) \oplus H$

$$q(x, z) = \|Ax - z\|^2.$$

This is a closed nonnegative form. Projection $p(x, z) = x$ gives

$$(p_* q)(x) = \begin{cases} 0, & x \in D(A), \\ +\infty, & x \notin D(A). \end{cases} \tag{10.31}$$

Its form domain is dense and nonclosed in $H$, so the pushed form is not closed. Infinite-dimensional last-use elimination therefore needs finite-dimensionality, coercivity and closed range, or an explicitly controlled lower-semicontinuous relaxation.

12.15. **The omitted endpoint-welding cocycle.** Even if an independently contact-closed tail is a quadrilateral of modulus $R_w$, its standard response is expressed in the whole-tail uniformization coordinates. Let

$$J_w^{\pm} : \mathcal{H}^{\pm}_{\mathrm{phys}} \longrightarrow \mathcal{H}_{\mathrm{std}}$$

be the induced endpoint trace maps. In fixed physical coordinates the exact response is

$$\mathcal{N}_w^{\mathrm{phys}} = (J_w^- \oplus J_w^+)^* \mathcal{N}_{R_w} (J_w^- \oplus J_w^+). \tag{10.32}$$

Dropping $J_w^{\pm}$ while keeping the prefix and cap forms fixed is a type error. Absorbing them into the caps instead makes those cap forms depend on the complete word and depth.

Finite combinatorial type gives no norm-precompactness of the cumulative welding cocycle. No uniform quasisymmetry, bounded $H^{1/2}$-composition action, inverse bound, phase compatibility, or convergence theorem is currently available for these physical tails.

The fixed-coordinate analytic statement does survive. For fixed bounded positive cap forms on genuine Cartesian trace products and one exact standard neck, the scalar internal matrix

$$M_t = \begin{pmatrix} \alpha_L + t & -t \\ -t & \alpha_R + t \end{pmatrix}, \qquad t = R^{-1},$$

gives, when $\alpha_* = \min(\alpha_L, \alpha_R) > 0$,

$$\|q_R - q_\infty\| \le M \frac{2}{e^{\lambda_1 R} - 1} + \frac{2B^2}{\alpha_*^2 R}. \tag{10.33}$$

Indeed $M_t, M_0 \ge \alpha_* I$ and $\|M_t - M_0\| = 2/R$, so the resolvent identity bounds the scalar Schur correction by $2B^2/(\alpha_*^2 R)$; the first term is theorem 12.7. If one scalar coefficient vanishes, positivity kills the corresponding coupling and the scalar correction is zero. This proves only the fixed-standard-neck theorem, not (10.32) uniformly over dynamical words.

**Open Problem 12.29** (Marked separator, trace, and welding control)**.** For every proof-used source-authorized direct-decoration end, choose one exact branch granularity—binary one-step or length-$n$ macrobranch—and:

(1) realize the complete current-separator interface state, retaining sibling and multi-branch incidence, activity, phase, orientation, endpoints, prime ends, protected accesses, obstacles, and itinerary data;
(2) replace atom-to-single-atom transport by an exact incidence- or restriction-valued transport closed on a finite or future-faithfully compact state;
(3) construct the correctly grounded homogeneous multi-port trace category and the actual stage-normalized protected overlay;
(4) prove a genuine last-live frontier or a future-faithful compact multi-port quotient;
(5) prove no-bypass resistance divergence or an equivalent coercive full-response theorem;
(6) prove an algebra-compatible compactification of the endpoint welding maps in the fixed response and Value norm, preserving graphs, composition, inverse control, response gluing, phase, and orientation; and
(7) prove a commuting incidence, trace, response, and protected-access naturality square under authorized final-root enlargement.

Thus Research Push VI resolves raw past–future contact localization for one nested binary address chain but does not construct the marked analytic separator state or its compact response dynamics.

12.16. **Fixed-block structural contacts and the Push-VII boundary.** We now use the actual depth-$n$ attachment fan of [KL08]. In addition to the source statements, retain the already audited hypotheses that $P, Q_L, Q_R$ are closed Jordan puzzle pieces, that $Q_L \cap Q_R = \varnothing$, and that each proper degree-one branch extends to a homeomorphism of the closed pieces. Put

$$C_w = \partial P \cap \partial Q_w, \qquad \widehat{C}_w = h_w(C_w), \qquad \widehat{C}_\varnothing = \partial P. \tag{10.40}$$

For a word $s$, write $v_s = \phi_s(\alpha)$.

**Proved Theorem 12.30** (Fixed-block binary structural contact automaton)**.** *For one fixed source-locked direct-decoration block:*

*(1) the landing vertices of the open ray part of $C_a = \partial P \cap \partial Q_a$ are exactly*

$$\mathcal{V}_a = \{v_{at} : t \in \{L, R\}^{n-1}\}; \tag{10.41}$$

*(2) for every word $w$ and letter $a$,*

$$\boxed{\widehat{C}_{wa} = g_a(\widehat{C}_w \cap C_a)}; \tag{10.42}$$

*(3) incomparable cylinders are disjoint, while*

$$\partial Q_u \cap \partial Q_{ut} = \phi_u(C_t); \tag{10.43}$$

*(4) for $|w| = k \le n$, the contact landing vertices are exactly*

$$\{v_{wt} : |t| = n - k\}, \qquad h_w(v_{wt}) = v_t; \tag{10.44}$$

*(5) if $\mathscr{R}_\alpha$ is the complete pair of root ray atoms landing at $\alpha$, then for every $|s| = n$,*

$$\widehat{C}_s = \mathscr{R}_\alpha, \qquad T_L(\mathscr{R}_\alpha) = \mathscr{R}_\alpha, \qquad T_R(\mathscr{R}_\alpha) = \varnothing, \tag{10.45}$$

*and the empty state is absorbing; and*

*(6) after a connector-aware finite boundary subdivision, the maps $T_a(S) = g_a(S \cap C_a)$ form a finite exact structural contact automaton for this fixed block.*

*No uniform bound in $q, n$, protected or obstacle overlay, Sobolev trace, response, welding, Value transport, or analytic enlargement naturality is asserted.*

*Proof.* The source gives $2^n$ components of $\mathrm{dom}(g^n)$, one for each vertex of $P$; the component $Q_s$ contains $v_s$ and is attached to the two $P$-boundary rays landing there. If $s = at$, then $Q_s \subset Q_a$. Every common $Q_s/P$ ray segment lies in $\partial Q_a$: a point of $\partial P$ cannot be interior to the subset $Q_a \subset P$.

Conversely, suppose an open $Q_a/P$ ray contact lands at a $P$-vertex $v$. The source gives the unique depth-$n$ cylinder $Q_s$ attached there. If the first symbol of $s$ were the other letter $b$, then $v \in Q_a \cap Q_b$, contrary to closed branch disjointness. This proves (10.41).

Take $z \in C_{wa}$. Since

$$Q_{wa} \subset Q_w \subset P \quad \text{and} \quad z \in \partial P,$$

the point $z$ cannot lie in $\operatorname{int} Q_w$, and hence $z \in \partial Q_w$. Thus $x = h_w(z) \in \widehat{C}_w$. The identity $Q_{wa} = \phi_w(Q_a)$ also gives $x \in \partial Q_a$, and $h_{wa}(z) = g_a(x)$. Conversely, if $x \in \widehat{C}_w \cap C_a$, then $h_w^{-1}(x)$ lies on both $\partial P$ and $\partial Q_{wa}$. This proves (10.42).

At the first differing symbol, incomparable words lie in pullbacks of the disjoint pieces $Q_L, Q_R$. If the second word is $ut$, then

$$Q_{ut} = \phi_u(Q_t), \qquad Q_u = \phi_u(P).$$

Applying the closed-piece homeomorphism $\phi_u$ proves (10.43).

Every completion $t$ gives $Q_{wt} \subset Q_w$, attached at $v_{wt}$; its common ray remains on every intermediate boundary by the same boundary-versus-interior argument. Conversely, the unique

depth-$n$ cylinder attached at a contact vertex must begin with $w$, or two incomparable cylinders would share that vertex. Finally

$$h_w(v_{wt}) = g^k \phi_w \phi_t(\alpha) = \phi_t(\alpha) = v_t,$$

proving (10.44).

At $k = n$, the two attached ray segments map under $g^n$ to the complete root ray pair at $\alpha$; the deeper equipotential maps to the root equipotential. The $L$-branch fixes $\alpha$, and $f^q$ fixes each ray in its rotation orbit as a set. The first-symbol classification (10.41) shows that the $R$-contact has no ray atom landing at $\alpha = v_{L^n}$. This proves (10.45).

A faithful boundary graph has finitely many ray intervals, equipotential connectors, vertex joins, and flags. Subdivide it at the contact endpoints and their branch images. Landing endpoints move through the finite source preimage tree, and positive-height outer endpoints map to the root equipotential and leave the child contact interval. The forward-saturated subdivision is finite, and (10.42) acts on its finite set of subcomplexes. □

*Remark* 12.31 (Why the submitted compiler is not the theorem). The source boundary does not alternate a ray with an equipotential after every angle. Consecutive rays may meet at one landing vertex; only rays with different landing points are joined by an equipotential. A faithful record must therefore distinguish VertexJoin($v$) from EqJoin($m; \theta_i, \theta_{i+1}$) and retain landing equivalence, sides, flags, and cyclic order. The Push-VII $q = 1$ fixture is schematic, not an instantiated source block. Moreover its compiler still passes after deleting all boundary words, corrupting prefix shifts, adding multiple same-branch lifts, or erasing the flag atlas. Theorem 12.30 rests on the analytic/source proof, not on those tests.

**Proved Theorem 12.32** (Fixed-tree KKL pair-state moment factorization)**.** *Let $E$ be the edge set of one fixed finite terminal Hubbard tree. For a pulled bivertical arc $\alpha$, put*

$$s_m(\alpha) = (f_*^m \alpha, (\iota_m)_* \alpha) \in E \times E, \qquad a_{e,e'} = v(e)\sqrt{w(e')}.$$

*For a level-$m$ arc $\gamma$, define*

$$z_m(\gamma)_{e,e'} = \sum_{\alpha:\ s_m(\alpha)=(e,e')} \langle \alpha, \gamma \rangle,$$

*and for a finite nonnegative WAD $X$, set*

$$\mathsf{M}_m(X) = \sum_\gamma X(\gamma) z_m(\gamma) z_m(\gamma)^\top.$$

*Then*

$$\boxed{\mathrm{Val}(X) = \lambda^{-2m} \langle a a^\top, \mathsf{M}_m(X) \rangle} \tag{10.46}$$

*and the ambient symmetric-state dimension is*

$$\frac{|E|^2(|E|^2+1)}{2}. \tag{10.47}$$

*Proof.* KKL's pulled-arc formula gives

$$\ell_m(\gamma) = \lambda^{-m} \sum_\alpha \langle \alpha, \gamma \rangle v(f_*^m \alpha) \sqrt{w((\iota_m)_* \alpha)} = \lambda^{-m} a^\top z_m(\gamma).$$

Substitution into $\mathrm{Val}(X) = \sum_\gamma X(\gamma) \ell_m(\gamma)^2$ gives (10.46). The pair-state space has $|E|^2$ coordinates, proving (10.47). Positivity is immediate from the rank-one sum defining $\mathsf{M}_m(X)$. □

The theorem does not construct a direct-decoration branch incidence $P_a$, prove $\mathsf{M} \mapsto P_a \mathsf{M} P_a^\top$, bound $|E|$ universally, or give a future-observability gap. The generic Kronecker word-sum identity is exact supporting algebra but does not supply any of those geometric inputs.

Finally, the Kahn–Lyubich pseudo-functor supplies a dynamical covering and an inclusion-induced immersion, finite ideal-boundary arcs, and continuous boundary extension. It does not define the proposed $H^1_{\mathrm{fin}}$ ideal-arc trace quotient or its response. The inclusion immersion need not map source boundary into target boundary, so tracking only the dynamical cover omits the pulled-versus-immersed welding. The safe-gap condition $S^+ \subset P \setminus (Q_L \cup Q_R)$ also allows $S^+$ to reach an unused arc of $\partial P$. Therefore Push VII does not prove `DIRECT_DECORATION_TRACE_COLLAR_REALIZATION` or `END_PLUMBING_TRACE_RESPONSE_CONTROL`. The explicit open nodes remain `DIRECT_DECORATION_ALL_DEPTH_CONTACT_CLOSURE`, `FINITE_LIVE_COMMON_REFINEMENT_FRONTIER`, and `DIRECT_DECORATION_TRACE_WELDING_COMPACTNESS`. VAAPC-I, H2, MBTT, UPTCR, and MLC remain open.

12.17. **Finite label markings and the Push-VIII pull–push obstruction.** Push VIII makes one necessary distinction correctly: a response, access, or obstacle overlay may subdivide a KKL action interval without licensing the new subdivision points as KKL ideal endpoints. It does not follow that the submitted finite labels construct the immersion pullback. In the KKL definition, $\iota^* I$ is determined by unique proper-path lifts and subordination in the ideal compactifications. A label saying only that a source atom lands properly on a target atom, or instead enters the target interior, contains neither that lift relation nor the relative-cover and homotopy data needed to recover it.

**Proved Theorem 12.33** (Finite linear-label marking enumerator)**.** *Let $M_0, \dots, M_{N-1}$ and $A_0, \dots, A_{m-1}$ be finite linearly ordered target and source atom chains. Give each source atom a label*

$$r_j \in \{0, \dots, N-1\} \cup \{\bot\}.$$

*For a contiguous target block $B$, define the* formal label hull $H(B)$ *by*

*(1) deleting each $A_j$ with $r_j \notin B \cup \{\bot\}$;*
*(2) taking the components of the remaining finite source chain; and*
*(3) retaining precisely those components containing an $A_j$ with $r_j \in B$.*

*The interval-convex transitive closure of the pairs $p, q$ flanking every maximal two-sided $\bot$-run is the unique finest target partition whose distinct blocks have disjoint formal label hulls. Every hull-disjoint partition is its coarsening.*

*Suppose additionally that every source vertex $v$ has a target vertex label $c_\partial(v)$. For an endpoint set $E \subset \{0, \dots, N\}$ containing $0, N$, call the corresponding partition* formally admissible *when its hulls are pairwise disjoint and, for every connected component $K$ of every $H(B)$,*

$$c_\partial(\partial_- K), c_\partial(\partial_+ K) \in E. \tag{10.48}$$

*For every mandatory endpoint set $E_{\mathrm{req}}$, exhaustive enumeration of the coarsenings of the finest hull-disjoint partition returns a maximum-cardinality formally admissible partition containing $E_{\mathrm{req}}$, or the complete finite obstruction.*

*Proof.* Let $R$ be a maximal $\bot$-run flanked by proper labels $p, q$. If $p$ and $q$ lie in different target blocks, deleting the proper labels outside the $p$-block leaves $R$ attached to the $p$-seed; deleting those outside the $q$-block leaves the same run attached to the $q$-seed. Thus the two hulls overlap on the source atom indices of $R$.

If $p, q$ lie in one target block, only that block retains $R$. A one-sided $\bot$-run is retained by at most its sole proper neighbor's block, and a run with no proper neighbor is retained by none. Outside these runs, proper atoms assigned to different target blocks separate the remaining source-chain components. Therefore pairwise hull disjointness is equivalent to placing each such flanking pair $p, q$ in one block.

Target blocks are contiguous. Consequently every block containing $p$ and $q$ contains every target atom between them. The interval-convex transitive closure of all forced pairs is therefore necessary, sufficient, and the unique finest hull-disjoint partition; every other hull-disjoint partition is its coarsening.

There are finitely many such coarsenings. For each, its target endpoints, the finitely many components of every $H(B)$, the mandatory-endpoint condition, and (10.48) are decidable by finite enumeration. Testing all of them proves soundness and completeness. A maximum-cardinality valid coarsening exists whenever the valid list is nonempty; otherwise the finite list of failed tests is the obstruction. □

This theorem is deliberately formal. It does not prove $H(B) = \iota^* B$, existence of a partial ideal marking, or properness and end-properness. Hull intersection here means intersection of source atom indices; shared source vertices are governed separately by (10.48), so no geometric-contact claim is hidden in the finite-chain definition. The Push-VIII proof of its stronger pullback formula reverses the KKL quantifiers: uniqueness concerns a source lift of a target proper path, not the uniqueness of the immersed image of a source path.

The submitted computations do not bridge that gap. The successful fixture has twelve source atoms and eight target atoms, despite modelling a degree-one covering, and it declares an atom homeomorphic to $M_0$ while both of its boundary vertices map to the same target vertex. The random generator does not impose continuity, covering degree, cyclic order, or endpoint compatibility. Its exhaustive oracle assumes the same formal hull rule as the compiler. Finally, the implementation declares an immersion component end-proper when only its first and last micro-atoms are proper. KKL end-properness instead requires properness on the complete end-intervals of the $f^*\mathcal{I}$-multi-interval. The fixture therefore cannot serve as analytic evidence.

**Proved Theorem 12.34** (Pull–push word-composition no-go)**.** *Let*

$$F_1 = (f_1, \iota_1) : V_1 \rightrightarrows V_0, \qquad F_2 = (f_2, \iota_2) : V_2 \rightrightarrows V_1.$$

*If*

$$R_1 : \mathcal{T}(V_0) \to \mathcal{T}(V_1), \qquad P_1 : \mathcal{T}(V_1) \to \mathcal{T}(V_0)$$

*are the restriction and covering transport for $F_1$, and $R_2, P_2$ have the analogous $V_1, V_2$ types, then the expression $(P_2R_2)(P_1R_1)$ is not typed without a further identification of $\mathcal{T}(V_0)$ and $\mathcal{T}(V_1)$.*

*Even after making such an identification, ordinary associativity does not identify the componentwise pull–push composite*

$$\mathcal{W}_{\mathrm{comp}} = P_1P_2R_2R_1 \tag{10.49}$$

*with the product*

$$\mathcal{W}_2\mathcal{W}_1 = P_2R_2P_1R_1. \tag{10.50}$$

*Such an identity requires an additional Cartesian base-change, or Beck–Chevalley, interchange square.*

*Proof.* The displayed domains and codomains give the typing obstruction. For the algebraic obstruction, identify all four trace spaces with $\mathbb{R}^2$ and take

$$R_1 = P_2 = I, \qquad P_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad R_2 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Then

$$\mathcal{W}_{\mathrm{comp}} = P_1R_2 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \mathcal{W}_2\mathcal{W}_1 = R_2P_1 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}. \tag{10.51}$$

The matrices differ. A check that $W_3(W_2(W_1x)) = (W_3W_2W_1)x$ tests only associativity and cannot test the missing interchange relation. □

There is a valid local analytic lemma behind part of the proposal, but it is not yet realized on the dynamical interface. If a Lipschitz prime-end quadrilateral $C$ with designated side $J$, a bounded functional $\ell_C \in H^1(C)^*$ satisfying $\ell_C(1) = 1$, and a conformal rectangle chart are independently supplied, then

$$\mathcal{T}_C(J) = \operatorname{Tr}_J H^1(C)$$

with the minimum of $\int_C |\nabla u|^2 + |\ell_C(u)|^2$ is a Hilbert quotient trace space. Modulo constants, conformal collar transport is isometric for the Dirichlet norm. The full norm is natural only when $\ell_C$ is transported with the collar. For several arcs, compatibility must be defined as the quotient trace of one connected collar and core; an unnormed image in a product gives neither a Hilbert-space nor a closed-range theorem. KKL properness by itself does not construct any of these collars.

**Open Problem 12.35** (Pullback-complete collared interface certificate)**.** For every proof-used direct-decoration branch and finite horizon, construct:

(1) the actual source and target pseudo-pieces and both analytic maps;
(2) a finite prime-end atlas compatible with covering degree, endpoints, phase, and cyclic order;
(3) the complete unique-lift/subordination argument identifying every KKL immersion pullback;
(4) the actual $f^*\mathcal{I}$-intervals and proper/end-proper decisions on whole intervals;
(5) conformal collar maps, transported gauges, connected compatible trace quotients, and every proper trace arc;
(6) the exact collar-overlap multiplicity;
(7) selected analytic fiber products and Beck–Chevalley squares for word composition; and
(8) proper-path no-birth/no-loss, relative-cover compatibility, and collar naturality for every authorized final-root enlargement.

At the first failure, record the source component, proper path or missing lift, complete interval, collar, sheet, and failed square.

Consequently the Push-VIII headline obligation

`FIXED_HORIZON_MARKED_BINARY_INTERFACE_COMPLETION`

is not a proved analytic theorem and is not promoted. The action-versus-overlay distinction and Theorem 12.33 remain valid formal tools; Theorem 12.34 prevents the proposed matrix associativity shortcut. The actual marked interface, welding, normalized-word, and final-root naturality gates remain open, as do VAAPC-I, H2, MBTT, UPTCR, and MLC.

12.18. **The first source-realized pseudo-piece correspondence.** We now freeze one of the actual Kahn–Lyubich direct-decoration pullbacks

$$Q = Q_a \subset P = Y^{(n-1)q+1}, \qquad g = f^q|_Q : Q \longrightarrow P, \qquad a \in \{L, R\}.$$

Write $K_Q = Q \cap K(f)$, $K_P = P \cap K(f)$, and let

$$\mathbf{Q} \underset{\boldsymbol{\iota}}{\overset{\mathbf{g}}{\rightrightarrows}} \mathbf{P}$$

be the dynamical covering and inclusion immersion induced on the Kahn–Lyubich pseudo-pieces. KL prove that $\boldsymbol{\iota}$ is an immersion. Global injectivity at a general $\psi$-ql stage is a separate open condition.

**Proved Theorem 12.36** (Tile-injectivity glue criterion)**.** *For either fixed branch $Q_a$, the map* $\mathbf{g}$ *is a biholomorphism and the inclusion component map is bijective. The inclusion immersion*

*$\boldsymbol{\iota} : \mathbf{Q} \to \mathbf{P}$ is an interior holomorphic embedding if and only if every induced full universal-cover tile immersion $L_i$ is injective. In that case the collision-relative overlap identity is*

$$L_i^{-1}\Big(\widetilde{e}_{j(i)}(Q_i)\Big) = e_i(Q_i). \tag{10.34}$$

*Full inverse-image reflection over all of $P_{j(i)}$ is not required.*

*Proof.* The branch $g : Q \to P$ is univalent and onto. Complete invariance of the filled Julia set gives

$$g(K_Q) = K_P,$$

so $g$ bijects the finite component sets of $Q \setminus K_Q$ and $P \setminus K_P$. The component map $j$ induced by the inclusion is surjective by the pseudo-piece construction. Its finite domain and codomain have the same cardinality, hence $j$ is bijective.

Let $P_j$ be a target complementary component. Since

$$Q \setminus K_Q = Q \cap (P \setminus K_P),$$

partitioning both complements into components gives

$$Q \cap P_j = \bigcup_{\{k : j(k) = j\}} Q_k.$$

Bijectivity of $j$ therefore gives

$$Q \cap P_{j(i)} = Q_i.$$

Assume every full tile map $L_i$ is injective. In the quotient gluing, central–central and same-tile collisions are excluded. Distinct source tiles land in distinct target tiles because $j$ is bijective, and KL's target tiles are disjoint outside their declared central overlaps.

It remains to consider a tile–central collision. Let $x$ lie in source tile $i$, let $z \in Q$, and suppose their target quotient classes agree. The exact target tile–central intersection gives

$$z \in Q \cap P_{j(i)} = Q_i, \qquad L_i(x) = \widetilde{e}_{j(i)}(z).$$

Overlap commutation gives

$$L_i(e_i(z)) = \widetilde{e}_{j(i)}(z) = L_i(x).$$

Tile injectivity forces $x = e_i(z)$, so the two source representatives were already identified by the declared gluing. These four cases exhaust the quotient, proving injectivity of $\boldsymbol{\iota}$. Since KL already prove that it is a holomorphic immersion, it is an interior holomorphic embedding.

Conversely, every source tile embeds in $\mathbf{Q}$. If $L_i$ had two distinct points with one image, their source quotient classes would be distinct while their $\boldsymbol{\iota}$-images agreed. Thus glued injectivity implies tile injectivity. Finally, (10.34) follows directly: if $L_i(x) = \widetilde{e}_{j(i)}(q)$ with $q \in Q_i$, commutation gives the same value at $e_i(q)$, and tile injectivity gives $x = e_i(q)$. The dynamical map has covering degree one and is therefore a biholomorphism. □

*Warning* 12.37 (Full overlap reflection is a non-gating saturation question). Commutation on the declared source overlap proves only

$$L_i(e_i(Q_i)) \subset \widetilde{e}_{j(i)}(P_{j(i)}).$$

The checked KL source does not prove the stronger saturation equality

$$L_i^{-1}\Big(\widetilde{e}_{j(i)}(P_{j(i)})\Big) = e_i(Q_i)$$

for a general bold $\psi$-ql stage. It is not needed for interior embedding: a tile point over $P_{j(i)} \setminus Q$ has no source-central mate and cannot create a tile–central quotient collision. For example, glue $D_1$ into $D_2$ in the source and $D_{3/2}$ into $D_2$ in the target, using coordinate inclusions. The quotient map is the identity on $D_2$, while the full target-overlap preimage is $D_{3/2} \supsetneq D_1$.

The exact invariant is reflection of the gluing equivalence relation, or equivalently the mixed central–tile fiber product over the source-central image. The theorem proves that invariant for the full fixed KL pair once tile injectivity is known. The stronger open obligation

`DIRECT_DECORATION_TILE_CENTRAL_OVERLAP_REFLECTION`

is retained only where collar, footprint, or carrier saturation explicitly requires it. In particular, a collar restriction may omit the comparison point $e_i(q)$, so collar injectivity alone does not reproduce the full-tile proof.

**Proved Theorem 12.38** (Degree-one annulus immersion does not imply injectivity)**.** *Degree one, $\mathbb{Z}$-equivariance, and local univalence do not imply global injectivity of a universal-cover tile map.*

*Proof.* Fix $h > 0$, choose $a > e^{2\pi h}$, and set

$$F_a(z) = z + \frac{a}{2\pi i} e^{2\pi i z}.$$

Then $F_a(z+1) = F_a(z) + 1$, and

$$F_a'(z) = 1 + a e^{2\pi i z} \neq 0 \quad (0 < \Im z < h),$$

because $|a e^{2\pi i z}| > 1$. For $0 < d < 1$, let

$$r(d) = \frac{\pi d}{a \sin(\pi d)}.$$

It tends to $1/a < e^{-2\pi h}$ at 0 and to $+\infty$ at 1. Choose $d$ with $e^{-2\pi h} < r(d) < 1$, and choose $z$ by

$$e^{2\pi i z} = -\frac{2\pi i d}{a(e^{2\pi i d} - 1)}.$$

Then $z, z+d \in \Pi(h)$, $d \notin \mathbb{Z}$, and direct substitution gives $F_a(z+d) = F_a(z)$. The imaginary part is bounded on a closed fundamental strip, so a vertical translation and sufficiently tall target strip make this a degree-one annulus immersion between strips. The example refutes degree plus immersion alone; it is not claimed to realize all KL end data. □

**Proved Theorem 12.39** (Kahn's annulus hypotheses do not imply global injectivity)**.** *There are $d > 1$, $r > 1$, and a holomorphic immersion*

$$I : \mathbb{A}(1, r^{1/d}) \longrightarrow \mathbb{A}(1, r)$$

*that extends analytically to the inner circle as an orientation-preserving degree-one homeomorphism but is not globally injective.*

*Proof.* Put

$$r' = \frac{1287}{1000}, \qquad d = 16, \qquad r = (r')^{16},$$

and, with $B_a(z) = (z-a)/(1-\bar{a}z)$, set

$$R(z) = \frac{B_{(1+6i)/10}(z) B_{(-3-i)/10}(z) B_{(-7+2i)/10}(z)}{B_{(-6+2i)/10}(z) B_{4i/10}(z)}. \tag{10.35}$$

The zeros and poles are among $a, b, 1/\bar{a}, 1/\bar{b}$. Since

$$(r')^2 \frac{53}{100} < 1, \qquad (r')^2 \frac{40}{100} < 1,$$

none lies in $1 \le |z| \le r'$.

On $\mathbb{T}$, $|R| = 1$ and the degree is $3 - 2 = 1$. For $t = \tan(\theta/2)$, the angular derivative has the sign of

$$\begin{aligned} p(t) = {} & 619271t^{10} - 5100608t^9 + 35876739t^8 - 169721152t^7 \\ & + 491196326t^6 - 885663808t^5 + 1065558054t^4 \\ & - 957958592t^3 + 669681731t^2 - 236915328t + 60062535. \end{aligned}$$

The exact Sturm-chain sign strings at $-\infty, +\infty$ are

$$+--++++--+--, \qquad ++--+--++++-$$

and each has five variations. Since $p(0) > 0$ and its leading coefficient is positive, $p > 0$ on the extended real line. Thus the inner-circle map is an orientation-preserving homeomorphism.

After a common nonzero scaling write $R = N/D_0$, where the descending Gaussian-integer coefficient lists are

$$\begin{aligned} N : {} & 2000 + 4000i,\ 19600 + 2200i,\ 28460 - 23880i, \\ & 4072 - 29046i,\ -6630 - 10210i,\ -1950 - 1525i, \\ D_0 : {} & 1525 + 1950i,\ 10210 + 6630i,\ 29046 - 4072i, \\ & 23880 - 28460i,\ -2200 - 19600i,\ -4000 - 2000i. \end{aligned}$$

On $z = r'(1 - t^2 + 2it)/(1 + t^2)$, the sign of $|N|^2 - |D_0|^2$ is the sign of

$$\begin{aligned} q(t) = {} & 11751160313656400426951788919t^8 \\ & - 13757983963892527749297130080 0t^7 \\ & + 95890179923428844955183112847 6t^6 \\ & - 41069069242342877051673209424 00t^5 \\ & + 10779372189655006534093819771914t^4 \\ & - 16680328775739752952831727982400t^3 \\ & + 14042199262274463963774353314076t^2 \\ & - 55863540030421994211573783408 00t \\ & + 13949552417142237508054128817 19. \end{aligned}$$

Its exact Sturm sign strings are $+--++--++$ and $++--++--+$; both have four variations. Hence $q > 0$. The minimum principle for $\log|R|$ gives $|R| > 1$ in the source annulus. Moreover,

$$|R| < \prod_{s \in \{61/100, 32/100, 73/100\}} \frac{r' - s}{1 - r's} = \frac{5697579109375}{119480253283} < 48 < (r')^{16},$$

because, for $1 < \rho \le r'$, $|a| \le s$, and $r's < 1$,

$$\left| B_a(\rho e^{i\theta}) \right| \le \frac{\rho - |a|}{1 - \rho|a|} \le \frac{r' - s}{1 - r's}.$$

The first bound follows by maximizing the squared modulus in $\cos(\theta - \arg a)$; the second follows from monotonicity in both $\rho$ and $|a|$. Thus $R$ maps into $\mathbb{A}(1, r)$.

For completeness, global local-univalence and the collision are certified without floating-point root inference. Up to nonzero scale the numerator of $R'$ is the degree-eight polynomial

$$\begin{aligned}P(z) = {} & (-1268000 + 501000i)z^8 + (-1247000 + 7116800i)z^7 \\ & + (17024420 + 10563170i)z^6 + (33992992 - 21977616i)z^5 \\ & - 52469243iz^4 + (-33992992 - 21977616i)z^3 \\ & + (-17024420 + 10563170i)z^2 + (1247000 + 7116800i)z \\ & + 1268000 + 501000i.\end{aligned}$$

Let $L(w) = \max(|\Re w|, |\Im w|)$ and $U(w) = |\Re w| + |\Im w|$. With $\rho = 10^{-4}$, take centers

$$\begin{aligned} -1.469179 + .035737i, \quad & -1.187253 + 1.185352i, \\ -.818448 + 1.003716i, \quad & -.680250 + .016547i, \\ -.487955 + .598410i, \quad & -.421815 + .421140i, \\ .407227 + .223067i, \quad & 1.888874 + 1.034668i. \end{aligned}$$

Exact rational substitution gives

$$L(P'(c_j))\rho > m_j \left[ U(P(c_j)) + \sum_{k=2}^{8} U(P^{(k)}(c_j)/k!)\rho^k \right],$$

with

$$(m_1, \dots, m_8) = (126, 362, 118, 451, 122, 192, 182, 144).$$

Rouché gives one zero in each disjoint disk. Degree eight exhausts the zeros; exact squared-modulus comparisons put disks $4, \dots, 7$ in $|z| < 1$ and the others beyond $|z| = 1.2901 > r'$. Hence $R' \neq 0$ in the source annulus.

Finally set $z_1 = -1 + 4i/5$. Exact substitution gives

$$R(z_1) = -\frac{559602300}{1858025273} + \frac{1886709225}{1858025273}i.$$

After removing the exact factor $z - z_1$ from $N - R(z_1)D_0$, the remaining quartic is

$$\begin{aligned}Q(z) = {} & (1649705408450 + 1129218801775i)z^4 \\ & + (8374862949240 - 2102550942545i)z^3 \\ & + (5597363347032 - 11487498744901i)z^2 \\ & + (-2960676892680 - 7024430543585i)z \\ & - 1525645358450 - 501687535025i.\end{aligned}$$

At $c = -.570412 + 1.153470i$, $\rho_1 = 10^{-5}$, the analogous exact Rouché ratio is greater than 11. Thus $Q$ has a root $z_2$ in that disk. Since

$$|c|^2 = \frac{413965722661}{250000000000},$$

exact squared comparisons give $1 < |z_2| < r'$. The disk does not contain $z_1$, and $D_0 \neq 0$ there, so $z_2 \neq z_1$ and $R(z_2) = R(z_1)$. □

*Warning* 12.40 (Scope of the exact annulus counterexample). Theorem 12.39 satisfies the full abstract hypotheses of Kahn's annulus lemma and rules out a global injectivity proof from those hypotheses alone. It does not realize the extra dynamically generated path-space or fiber-product structure of the actual KL bold ambient immersion, so it does not solve open problem 12.58.

**Proved Theorem 12.41** (Full abstract $\psi$-ql axioms do not imply global injectivity)**.** *There is a degree-two $\psi$-quadratic-like pair*

$$(i, f) : (\mathbf{U}', K') \longrightarrow (\mathbf{U}, K)$$

*for which $f$ is a proper degree-two branched cover, $i$ is a holomorphic immersion,*

$$i^{-1}(K) = f^{-1}(K) = K'$$

*is connected, and $i|_{K'} : K' \to K$ is a homeomorphism, while the degree-one complementary-annulus immersion*

$$i : \mathbf{U}' \setminus K' \longrightarrow \mathbf{U} \setminus K$$

*has a self-collision.*

*Proof.* Put

$$R = 75, \qquad \beta = \frac{\pi}{50}, \qquad \Delta = e^{2\beta} - 1,$$

and define

$$\mathbf{U}' = \{z : |z| < 75\}, \quad \mathbf{U} = \{w : |w| < 5625\}, \quad f(z) = z^2,$$
$$K = [0, 1], \qquad K' = [-1, 1], \qquad i(z) = \frac{e^{\beta(z+1)} - 1}{\Delta}. \tag{10.36}$$

The intervals are compact, connected, full, and nondegenerate. The map $f : \mathbf{U}' \to \mathbf{U}$ is a proper degree-two branched cover and

$$f^{-1}([0, 1]) = [-1, 1]. \tag{10.37}$$

Moreover,

$$i'(z) = \frac{\beta e^{\beta(z+1)}}{\Delta} \neq 0.$$

For $|z| < 75$, the elementary inequalities

$$3 < \pi < \frac{22}{7}, \qquad e < 3, \qquad e^x > 1 + x \ (x > 0)$$

give

$$|i(z)| < \frac{e^{76\beta} + 1}{e^{2\beta} - 1} < \frac{3^5 + 1}{3/25} = \frac{6100}{3} < 5625. \tag{10.38}$$

Here $76\beta = 38\pi/25 < 836/175 < 5$; the strict bound $\pi < 22/7$ follows, for example, from

$$\frac{22}{7} - \pi = \int_0^1 \frac{x^4(1 - x)^4}{1 + x^2}\, dx > 0,$$

and $e < 3$ follows from its factorial series. Thus $i$ maps the source disk into the target disk.

For $t \in [0, 1]$,

$$i(z) = t \quad \Longleftrightarrow \quad z = -1 + \frac{\log(1 + \Delta t)}{\beta} + 100ik, \qquad k \in \mathbb{Z}. \tag{10.39}$$

The real part in (10.39) belongs to $[-1, 1]$, while every nonzero sheet has imaginary modulus at least $100 > 75$. Consequently

$$i^{-1}(K) = [-1, 1] = f^{-1}(K).$$

On this interval $i$ is strictly increasing and sends $-1 \mapsto 0$, $1 \mapsto 1$.

The circle $|z| = 2$ is a positive source-annulus generator. The equation $i(z) = 1/2$ has exactly one simple solution inside that circle: its solutions have the form

$$-1 + \beta^{-1} \log(1 + \Delta/2) + 100ik,$$

and the $k = 0$ real part lies strictly between $-1$ and 1. The argument principle therefore gives complementary-annulus degree $+1$.

Nevertheless, $z_+ = 50i$ and $z_- = -50i$ are distinct points of $\mathbf{U}' \setminus K'$, and

$$\beta(z_+ - z_-) = 2\pi i, \qquad i(z_+) = i(z_-) = \frac{-e^{\beta} - 1}{\Delta} < 0.$$

Their common image belongs to $\mathbf{U} \setminus K$, proving the collision. □

*Warning* 12.42 (Abstract $\psi$-ql scope). Theorem 12.41 satisfies the complete abstract $\psi$-quadratic-like axioms, including the proper companion branch and the exact common connected inverse. It is not claimed to be an actual KL canonical-localization or direct-decoration path-space stage. It proves that any global-injectivity theorem for those actual maps must use their additional path-space history, full properness, or an equivalent homotopy-reflection theorem.

**Proved Theorem 12.43** (KL tile/base injectivity reduction)**.** *Let $I : A_s \to A_t$ be a degree-one holomorphic immersion between connected annuli, and let*

$$\pi_s : \widetilde{A}_s \to A_s, \qquad \pi_t : \widetilde{A}_t \to A_t$$

*be universal covers. For compatible positive deck generators $T_s, T_t$, every lift $L$ of $I$ satisfies*

$$L \circ T_s = T_t \circ L,$$

*and*

$$L \text{ is injective} \quad \Longleftrightarrow \quad I \text{ is injective.}$$

*If $I$ is proper, then it is a biholomorphism.*

*Proof.* Suppose first that $I$ is injective and $L(x) = L(y)$. Projection gives $I(\pi_s x) = I(\pi_s y)$, hence $\pi_s x = \pi_s y$ and $y = T_s^k x$ for some $k \in \mathbb{Z}$. Equivariance gives

$$L(y) = T_t^k L(x) = L(x).$$

The target deck action is free, so $k = 0$ and $x = y$.

Conversely, suppose $I(u) = I(v)$ with $u \neq v$, and choose lifts $x, y$. There is a unique $k \in \mathbb{Z}$ with

$$L(y) = T_t^k L(x).$$

Then

$$L(T_s^{-k} y) = T_t^{-k} L(y) = L(x),$$

while $\pi_s(T_s^{-k} y) = v \neq u = \pi_s(x)$. Thus $L$ is not injective.

Finally, a proper local biholomorphism has open and closed image, hence is onto the connected target, and is a finite-sheeted covering. Its number of sheets equals its positive annular degree, namely one. It is therefore a biholomorphism. □

**Proved Proposition 12.44** (Localization subgroup-reflection criterion)**.** *Let $a : (X_s, x_s) \hookrightarrow (X_t, x_t)$ be a based embedding. Let connected based covers $p_s : \widetilde{X}_s \to X_s$ and $p_t : \widetilde{X}_t \to X_t$ correspond to subgroups*

$$H_s \leq \pi_1(X_s, x_s), \qquad H_t \leq \pi_1(X_t, x_t),$$

*and suppose $A : \widetilde{X}_s \to \widetilde{X}_t$ is a based lift:*

$$p_t A = a p_s.$$

*Then*

$$A \text{ is injective} \quad \Longleftrightarrow \quad a_*^{-1}(H_t) = H_s. \tag{10.40}$$

*Failure is certified by a class*

$$[\gamma] \notin H_s, \qquad a_*[\gamma] \in H_t, \tag{10.41}$$

*whose source lift has distinct endpoints identified by $A$.*

*Proof.* Lift existence gives $a_*(H_s) \subseteq H_t$. Suppose first that $a_*^{-1}(H_t) = H_s$ and $A(\widetilde{u}) = A(\widetilde{v})$. Projection and injectivity of $a$ give $p_s(\widetilde{u}) = p_s(\widetilde{v})$. Choose paths from the source base lift to $\widetilde{u}, \widetilde{v}$, and let $\gamma$ be the loop obtained by projecting the first and then the reverse of the second. Equality of the target endpoints says $a_*[\gamma] \in H_t$. Reflection gives $[\gamma] \in H_s$, so the source endpoints agree.

Conversely, take $[\gamma] \in a_*^{-1}(H_t)$ and lift $\gamma$ from the source base lift. Its image under $A$ is the target lift of $a\gamma$, which is closed because $a_*[\gamma] \in H_t$. If $A$ is injective, the source lift is also closed, so $[\gamma] \in H_s$. This proves (10.40). If reflection fails, the same lifted loop gives the two distinct endpoints and common target endpoint in (10.41). □

*Warning* 12.45 (Functoriality alone does not prove reflection). Kahn's categorical pullback iteration is Cartesian and preserves an embedding already supplied. Canonical localization is stated as a functorial commuting lift that preserves immersion. Its path formulation proves that source-equivalent paths become target-equivalent; it does not prove the converse

$$a\alpha \simeq a\beta \quad \Longrightarrow \quad \alpha \simeq \beta,$$

which is precisely $a_*^{-1}(H_t) = H_s$. Thus proposition 12.44 identifies a sufficient and necessary cover-level gate only after the relevant map has independently been realized as the lift of a physical embedding. The ordinary primitive canonical seed has exactly that realization, and theorem 12.48 below proves that reflection fails there. At a later general bold $\psi$-ql stage the underlying map may already be only an immersion; a base collision is then an independent earlier obstruction.

*Warning* 12.46 (The base annulus is bold, not physically nested). Before localization, an ordinary quadratic-like stage satisfies $U^{m+1} \subset U^m$, so that particular base-annulus map is a physical inclusion. Canonical localization changes the complementary surface: the source deletes a pulled packet and the target deletes only the original packet. Theorem 12.48 proves that the resulting localized inclusion is noninjective. At a general KL $\psi$-ql stage the tiles cover $\mathbf{U}^m \setminus K(f)$, and KL explicitly replace that inclusion by an immersion of bold ambient domains. They prove an embedded inner collar and the end-properness needed in their width argument, not global base univalence or full annulus properness. Physical nesting may therefore be used only for the unlocalized positive control, not for a full canonical tile.

**Proved Lemma 12.47** (Finite full pullbacks and peripheral isolation)**.** *Let $q : V' \to V$ be a proper holomorphic map of degree two between plane topological disks, and let $K \Subset V$ be a nonempty full continuum. If the critical value belongs to $K$, then $q^{-1}K$ is one full degree-two continuum. Otherwise it is the disjoint union of two full continua, each mapped homeomorphically onto $K$.*

*Consequently, under a chain of $p$ quadratic-like pullbacks, the inverse image of a finite pairwise-disjoint family of full continua is a finite pairwise-disjoint family of full continua. Every component maps onto its parent with positive integer degree. Each component admits a Jordan neighborhood disjoint from all the others, and the complement of the finite family in the source disk is path connected.*

*Proof.* A full continuum has a nested Jordan-domain basis

$$K \subset V_{r+1}, \qquad \overline{V}_{r+1} \subset V_r, \qquad \bigcap_r \overline{V}_r = K.$$

For a nondegenerate $K$, obtain it by uniformizing $\widehat{\mathbb{C}} \setminus K$ by the exterior disk and pulling back regular circles of radii decreasing to one; for a singleton use round disks.

If the critical value is outside $K$, choose all $V_r$ to avoid it. Then $q^{-1}(V_r) \to V_r$ is an unbranched two-sheeted cover of a simply connected disk and hence consists of two coherently

labeled biholomorphic disks $W_r^0, W_r^1$. Each

$$L_k = \bigcap_r \overline{W}_r^k$$

is connected, maps homeomorphically onto $K$, and is full because

$$\mathbb{C} \setminus L_k = \bigcup_r (\mathbb{C} \setminus \overline{W}_r^k)$$

is an increasing union of connected sets.

If the critical value lies in $K$, every $V_r$ contains it. The component $W_r$ containing the critical point already has degree two and therefore exhausts the total preimage. Riemann–Hurwitz gives

$$\chi(W_r) = 2\chi(V_r) - 1 = 1,$$

so $W_r$ is a disk; regularity of $\partial V_r$ makes its boundary Jordan. The same nested-intersection and increasing-complement argument makes $q^{-1}K = \bigcap_r \overline{W}_r$ connected and full.

Iteration gives at most $2^p$ children of one parent and proves the finite-family and surjectivity assertions. A finite disjoint family has positive pairwise separation, so nested Jordan neighborhoods isolate its members. To prove path connectivity, choose for each member $C_i$ nested isolating Jordan disks

$$C_i \subset D_i^-, \qquad \overline{D_i^-} \subset D_i^+,$$

with the $D_i^+$ pairwise disjoint. The disk with the $\overline{D_i^-}$ removed is path connected. Each $D_i^+ \setminus C_i$ is path connected by fullness and overlaps that core in $D_i^+ \setminus \overline{D_i^-}$. Their union is exactly the complement of the $C_i$. □

**Proved Theorem 12.48** (Forced collision in Kahn's canonical localization)**.** *Let $f : U^1 \to U^0$ be an ordinary quadratic-like restriction of a quadratic polynomial, primitively renormalizable of period $p > 1$. Put*

$$F = f^p : U^p \to U^0, \qquad A = K_p, \qquad B = \mathcal{K}_p = \bigcup_{j=0}^{p-1} K_p(j).$$

*Kahn's canonical localization* [Kah06] *uses*

$$T' = (U^p \setminus F^{-1}B) \cup A, \qquad T = (U^0 \setminus B) \cup A$$

*and localizes the physical inclusion $i^p : T' \to T$. The localized inclusion*

$$\widehat{i} : T'_A \longrightarrow T_A$$

*is not injective. More precisely, there is a based class*

$$[\gamma] \in (i^p)_*^{-1}(H_t) \setminus H_s$$

*for the source and target localization subgroups.*

*The collision propagates through every exact Kahn pullback inclusion $i_r$. Whenever the same canonical seed carries the KL direct-decoration data* [KL08]*, the full base-annulus map from depth $m = \nu q + 1$ to $n = (\nu - 1)q + 1$, every corresponding full universal-cover tile lift, and the full glued inclusion pseudo-piece map are noninjective.*

*Proof.* The proper map $F : U^p \to U^0$ has global degree $2^p$. On the central renormalization component its degree is two. Choose a sufficiently small Jordan neighborhood $V$ of $A$ in the range of the quadratic-like renormalization. The central component of $F^{-1}V$ has degree two, whereas the degrees of all components sum to $2^p$. Since $p > 1$, another component exists. By lemma 12.47, it contains a component

$$X \subset F^{-1}A, \qquad X \neq A,$$

and $X$ maps onto $A$.

Write $K_j = K_p(j) = f^j(A)$. Since $F = f^p$,

$$F(K_j) = f^j(F(A)) = f^j(A) = K_j.$$

If $X \cap K_j \neq \varnothing$, its image meets both $A$ and $K_j$. For $j > 0$, this contradicts primitive pairwise disjointness. For $j = 0$, it contradicts that $X$ and $A$ are distinct components of $F^{-1}A$. Thus

$$X \cap B = \varnothing.$$

The lemma makes $F^{-1}B$ a finite disjoint family of full continua. Choose a Jordan disk $D_X \Subset U^p$ satisfying

$$X \subset D_X, \qquad \overline{D}_X \cap ((F^{-1}B \setminus X) \cup B) = \varnothing.$$

Set

$$\Omega_s = U^p \setminus F^{-1}B, \qquad \Omega_t = U^0 \setminus B.$$

Because $F(B) = B$, one has $B \subset F^{-1}B$, and the physical inclusion maps $\Omega_s$ into $\Omega_t$.

Choose a small Jordan neighborhood $D_A \Subset T'$ containing $A$ and no other pulled component, a basepoint $z_* \in D_A \setminus A$, and a path in $\Omega_s$ from $z_*$ to $\partial D_X$; path existence is the last assertion of lemma 12.47. Conjugate the positive peripheral loop $\partial D_X$ by that path and call the based loop $\gamma$. For $x \in X$,

$$\mathrm{Ind}_x(\gamma) = 1.$$

Every loop in $D_A \setminus A$ has winding zero about $x$, and winding is invariant under homotopy in $\Omega_s$. Therefore

$$[\gamma] \notin H_s = \mathrm{im}[\pi_1(D_A \setminus A, z_*) \to \pi_1(\Omega_s, z_*)]\,.$$

On the other hand, $D_X \cap B = \varnothing$, so $\partial D_X$, and hence its based conjugate, contracts in $\Omega_t$. Thus

$$(i^p)_*[\gamma] = 1 \in H_t.$$

Lift $\gamma$ from a point in the source localization cover. Since its class is not in $H_s$, its two endpoints are distinct. Its target lift is closed because the image class lies in $H_t$. Commutativity of the localized lift therefore identifies the two source endpoints. Adding the marked continuum $A$ does not identify distinct points of the complement cover, so this is a collision of $\widehat{i}$.

For propagation, Kahn's exact equalizer is

$$\mathbf{U}^{r+2} = \{(x, y) : i_r(x) = f_r(y)\}, \qquad i_{r+1}(x, y) = y, \quad f_{r+1}(x, y) = x.$$

Suppose

$$x_r^+ \neq x_r^-, \qquad i_r(x_r^+) = i_r(x_r^-) = v_r$$

in the complementary annulus. The restriction of $f_r$ there is a surjective unbranched covering. Choose $y_r$ with $f_r(y_r) = v_r$. Then

$$(x_r^+, y_r) \neq (x_r^-, y_r)$$

are equalizer points with the same $i_{r+1}$-image $y_r$. Induction propagates the collision to every $i_r$ and hence to every composite $I_{m,n}$.

For the stated KL depths, the induced annular degree is $+1$. If $I_{m,n}(u) = I_{m,n}(v)$, $u \neq v$, choose source lifts $\widetilde{u}, \widetilde{v}$. Their target images differ by a deck translation $T_t^k$. Degree-one equivariance gives

$$L(T_s^{-k}\widetilde{v}) = L(\widetilde{u})$$

at distinct source points. Each full universal-cover copy embeds in the source pseudo-piece before the prescribed central gluing, so the same-tile collision survives in the glued map. □

*Warning* 12.49 (Scope and replacement invariant). The direct-decoration conclusion is conditional on the same canonical seed carrying the stated KL decoration data; no separate parameter realization is inferred here. The theorem closes neither a footprint nor a width gate. It instead forbids the universal full embedded-pseudo-piece carrier. Any replacement must retain the nontrivial coset state

$$(i^p)_*^{-1}(H_t)/H_s$$

taken as a pointed left-coset set, not as a quotient group unless normality is separately proved, together with the pulled components and their itineraries, synchronize the source and target obstacle packets, or restrict to a proved collar and charge every omitted sheet. VAAPC-I, UPTCR, and MLC remain open.

**Proved Theorem 12.50** (Canonical localization sheet resolution)**.** *Let $a : (X_s, x_s) \hookrightarrow (X_t, x_t)$ be a based embedding of Hausdorff, connected, locally path-connected, semilocally simply connected spaces. Let*

$$p_s : \widehat{X}_s \to X_s, \qquad p_t : \widehat{X}_t \to X_t$$

*be connected based covers with subgroups $H_s, H_t$, and let $A : \widehat{X}_s \to \widehat{X}_t$ be a based lift of $a$. Put*

$$K = a_*^{-1}(H_t)$$

*and let $P_K$ be the based component of the pullback*

$$X_s \times_{X_t} \widehat{X}_t = \{(x, \widehat{y}) : a(x) = p_t(\widehat{y})\}.$$

*Then $P_K \to X_s$ is the $K$-cover and there is a canonical factorization*

$$\widehat{X}_s \xrightarrow{q} P_K \xrightarrow{e} \widehat{X}_t, \qquad q(\widehat{u}) = (p_s(\widehat{u}), A(\widehat{u})), \quad e(x, \widehat{y}) = \widehat{y}, \tag{10.51a}$$

*where $q$ is a surjective covering, $e$ is a topological embedding, and $A = e \circ q$. Moreover,*

$$A(\widehat{u}) = A(\widehat{v}) \quad \Longleftrightarrow \quad q(\widehat{u}) = q(\widehat{v}), \tag{10.51b}$$

*the based fiber of $q$ is the pointed coset set $K/H_s$, and this is the unique connected intermediate-cover factorization in the following precise class: if $r_C : C \to X_s$ is a connected based cover and based maps*

$$\widehat{X}_s \xrightarrow{q_C} C \xrightarrow{e_C} \widehat{X}_t$$

*satisfy*

$$r_C q_C = p_s, \qquad p_t e_C = a r_C, \qquad A = e_C q_C,$$

*with $e_C$ injective, then there is a unique based cover isomorphism $C \cong P_K$ intertwining both legs.*

*Path lifting for $q$ defines a functor*

$$\Pi_1(P_K) \longrightarrow \mathbf{Set}_{\mathrm{bij}}, \qquad z \longmapsto q^{-1}(z), \tag{10.51c}$$

*which is the canonical sheet-transport functor; its category of elements is the canonical sheet-transport groupoid. If a compact target path $\gamma$ lies in $e(P_K)$, then all source paths mapped to $\gamma$ by $A$ are exactly the $q$-lifts of $e^{-1}\gamma$, one from each initial sheet.*

*Proof.* The pullback of a cover is a cover. A based loop $\alpha$ in $X_s$ lifts closed to the selected pullback component exactly when $a\alpha$ lifts closed to $\widehat{X}_t$, equivalently $a_*[\alpha] \in H_t$. Hence the pullback subgroup is $K$.

The second projection on the entire pullback is a homeomorphism onto $p_t^{-1}(a(X_s))$: its inverse is

$$\widehat{y} \longmapsto (a^{-1}(p_t(\widehat{y})), \widehat{y}).$$

Thus $e$ is embedded. Since lift existence gives $H_s \leq K$, the covering criterion gives the unique based map $q$. It is onto: lift to $\widehat{X}_s$ the projection of any path in $P_K$ from the basepoint; uniqueness

of lifting shows that its $q$-image is the original path. On evenly covered neighborhoods this map is sheetwise a homeomorphism, so it is a covering.

The formula for $q$ gives $A = eq$. Injectivity of $e$ gives (10.51b), hence equality of the two kernel pairs. The fiber of the map from the $H_s$-cover to the $K$-cover is $H_s\backslash K$, canonically identified by inversion with the project notation $K/H_s$; no normality is used.

For uniqueness, let an intermediate cover have subgroup $L$ and an injective second leg. Existence of its two maps gives $H_s \le L \le K$. The subgroup-reflection criterion applied to the injective second leg gives $L = a_*^{-1}(H_t) = K$, so based-cover classification identifies it uniquely with $P_K$.

Finally, uniqueness and homotopy invariance of covering path lifts give the transport functor. If $A\widehat{\delta} = \gamma = e\delta$, injectivity of $e$ gives $q\widehat{\delta} = \delta$, and the converse follows from $A = eq$. This proves the compact-path exhaustion. □

**Proved Theorem 12.51** (Infinite canonical sheet defect)**.** *At every ordinary primitive canonical seed,*

$$[K : H_s] = \infty.$$

*Hence $q$ is infinite-sheeted and every fiber*

$$A^{-1}(e(z)) = q^{-1}(z), \qquad z \in P_K,$$

*is infinite.*

*Proof.* Let $x$ lie in the excess component and let $\gamma$ be the peripheral class in the proof of theorem 12.48. Winding about $x$ is a homomorphism $w : \pi_1(\Omega_s) \to \mathbb{Z}$ with

$$w(H_s) = 0, \qquad w(\gamma) = 1, \qquad \gamma \in K.$$

If $\gamma^m H_s = \gamma^n H_s$, then $\gamma^{m-n} \in H_s$, so winding gives $m - n = 0$. Thus the cosets $\gamma^n H_s$, $n \in \mathbb{Z}$, are pairwise distinct. The fiber conclusion follows from theorem 12.50. □

**Proved Proposition 12.52** (Base-change propagation of sheet resolution)**.** *Let $X \xrightarrow{i} Z \xleftarrow{f} Y$, suppose*

$$i = e \circ q, \qquad X \xrightarrow{q} P \xrightarrow{e} Z,$$

*where $q$ is a covering and $e$ is an embedding, and form*

$$E = X \times_Z Y, \qquad R = P \times_Z Y.$$

*Then*

$$q' : E \to R, \quad q'(x, y) = (q(x), y)$$

*is a covering,*

$$e' : R \hookrightarrow Y, \quad e'(p, y) = y$$

*is an embedding, and the pullback projection $i' : E \to Y$ satisfies*

$$i' = e' \circ q'.$$

*The $i'$- and $q'$-kernel pairs agree, and*

$$(q')^{-1}(p, y) \cong q^{-1}(p).$$

*Thus the factorization and its infinite full fibers propagate through every full, unselected exact Kahn equalizer on the complementary-cover locus. This statement neither selects the proof-used connected component nor extends the factorization across a filled or prime-end completion.*

*Proof.* The square defining $q'$ is the pullback of the cover $q$, so $q'$ is a cover. If $e'(p_1, y_1) = e'(p_2, y_2)$, then $y_1 = y_2 = y$ and

$$e(p_1) = f(y) = e(p_2).$$

Injectivity of $e$ gives $p_1 = p_2$; the inverse on the image is $y \mapsto (e^{-1}f(y), y)$, so $e'$ is embedded. The remaining identities follow by substitution and injectivity of $e'$. □

**Proved Theorem 12.53** (Kahn iterate complementary-component alignment)**.** *Let*

$$(i_r, f_r) : (\mathbf{U}^{r+1}, K^{r+1}) \longrightarrow (\mathbf{U}^r, K^r)$$

*be any finite Kahn iterate of one fixed $\psi$-polynomial-like seed, and put*

$$X_r = \mathbf{U}^{r+1} \setminus K^{r+1}, \qquad Z_r = \mathbf{U}^r \setminus K^r.$$

*Then the full complementary equalizer is*

$$\mathbf{U}^{r+2} \setminus K^{r+2} = X_r \times_{Z_r} X_r = \{(x, y) \in X_r^2 : i_r(x) = f_r(y)\}, \tag{10.51c'}$$

*and is connected.*

*If*

$$i_r|_{X_r} = e_r q_r, \qquad X_r \xrightarrow{q_r} P_r \xrightarrow{e_r} Z_r,$$

*with $q_r$ a surjective covering and $e_r$ an embedding, then*

$$R_{r+1} = P_r \times_{Z_r} X_r$$

*is connected, and the base-changed factorization of $i_{r+1}$ is a surjective covering from the space in (10.51c') onto $R_{r+1}$, followed by an embedding.*

*For the same fixed seed satisfying the KL direct-decoration hypotheses, this alignment is tilewise as well. Fix a source tile index $a$ for one inclusion stage $\iota : A_s \to A_t$, let $j_\iota(a)$ be its target index, and assume the complementary inclusion has the factorization*

$$A_s \xrightarrow{q} P \xrightarrow{e} A_t, \qquad \iota = eq,$$

*where $q$ is a surjective covering and $e$ is an embedding. Let*

$$L_a : \widetilde{A}_{s,a} \longrightarrow \widetilde{A}_{t,j_\iota(a)}$$

*be KL's lifted tile immersion. Define the $a$-labelled copies*

$$\mathcal{E}_a = \{a\} \times (A_s \times_{A_t} \widetilde{A}_{t,j_\iota(a)}), \qquad \mathcal{R}_a = \{a\} \times (P \times_{A_t} \widetilde{A}_{t,j_\iota(a)}).$$

*Then $\mathcal{E}_a$ is connected and is the source universal cover. If $\pi_s : \widetilde{A}_{s,a} \to A_s$, then, after fixing the source and target lifted basepoints by KL's chosen component embeddings,*

$$\Phi_a(\widetilde{z}) = (a, \pi_s(\widetilde{z}), L_a(\widetilde{z})) : \widetilde{A}_{s,a} \xrightarrow{\cong} \mathcal{E}_a$$

*is the unique based cover isomorphism. Base change of $A_s \xrightarrow{q} P \xrightarrow{e} A_t$ gives*

$$\mathcal{E}_a \twoheadrightarrow \mathcal{R}_a \hookrightarrow \widetilde{A}_{t,j_\iota(a)}, \tag{10.51c"}$$

*where the first map is a surjective covering, the second is an embedding, the composite with $\Phi_a$ is $L_a$, and $\mathcal{R}_a$ is connected. The assertion is family-valued over source tile indices. If $a_1 \neq a_2$ but $j_\iota(a_1) = j_\iota(a_2)$, their labelled copies remain distinct even though the underlying fiber products are canonically isomorphic. The theorem neither collapses distinct tile copies nor equates the dynamical-covering and inclusion component maps. In particular, it does not identify the earlier canonical infinite sheet-resolution factorization with this whole actual KL tile factorization; theorem 12.67 shows that such a whole-tile identification would force the covering leg to have one sheet.*

*Proof.* Because $i_r$ is an immersion, Kahn's special pullback lemma identifies the underlying space of the complex-manifold pullback with the ordinary topological fiber product; the later general desingularization discussion is not used. On that pullback,

$$x \in K^{r+1} \iff i_r(x) \in K^r \iff f_r(y) \in K^r \iff y \in K^{r+1}.$$

Thus a point is outside $K^{r+2}$ exactly when both coordinates lie in $X_r$, which proves (10.51c'). Kahn proves that the entire pulled pair is again a disk/FJ pair. An FJ-set is full, so its complement in that disk is connected.

Proposition 12.52 supplies the covering and embedding. Surjectivity must also be recorded: if $(p, y) \in R_{r+1}$, choose $x \in X_r$ with $q_r(x) = p$. Then

$$i_r(x) = e_r q_r(x) = e_r(p) = f_r(y),$$

so $(x, y)$ belongs to the complementary equalizer and maps to $(p, y)$. The continuous surjective image of that connected equalizer is $R_{r+1}$, hence $R_{r+1}$ is connected.

For the tilewise statement, Kahn's annulus uniformization theorem gives oriented inner-boundary degree $+1$ for every $\psi$-ql inclusion stage. Hence

$$\iota_* : \pi_1(A_s) \cong \mathbb{Z} \longrightarrow \pi_1(A_t) \cong \mathbb{Z}$$

is the identity on oriented generators. The pullback of the target universal cover therefore has one component, because

$$[\, \pi_1(A_t) : \iota_* \pi_1(A_s) \,] = 1,$$

and its subgroup is $\iota_*^{-1}(0) = 0$, so it is the source universal cover. KL's lifted map satisfies $\pi_t L_a = \iota \pi_s$, so $\Phi_a$ maps into the pullback and preserves the basepoints fixed by the chosen source and target component embeddings. Based universal-cover uniqueness makes $\Phi_a$ an isomorphism and fixes the deck phase separately for each source index. Applying proposition 12.52 tilewise gives (10.51c"); the first map there is surjective, so $\mathcal{R}_a$ is connected. Substitution shows that the composite with $\Phi_a$ is $L_a$. KL construct their pseudo-pieces from the same supplied seed and its natural $\mathbf{U}^m$-restrictions. □

**Proved Proposition 12.54** (Exact based-coset merger and birth tests)**.** *Let $H \le K \le G$, $H^+ \le K^+ \le G^+$, and let $\varphi : G \to G^+$ satisfy $\varphi(H) \le H^+$ and $\varphi(K) \le K^+$. The induced pointed map*

$$\overline{\varphi} : K/H \to K^+/H^+, \qquad kH \longmapsto \varphi(k)H^+$$

*is injective exactly when*

$$K \cap \varphi^{-1}(H^+) = H, \tag{10.51d}$$

*and is surjective exactly when*

$$K^+ = \varphi(K)H^+. \tag{10.51e}$$

*Thus a class in $(K \cap \varphi^{-1}(H^+)) \setminus H$ is an exact based-coset-merger certificate, while a class in $K^+ \setminus \varphi(K)H^+$ is an exact based-coset-birth certificate.*

*Proof.* Two old cosets have the same image exactly when $\varphi(k_2^{-1}k_1) \in H^+$. Since $k_2^{-1}k_1 \in K$, reflection of equality is precisely (10.51d). A new coset $k^+H^+$ has an old preimage exactly when $k^+ \in \varphi(K)H^+$, which for every $k^+ \in K^+$ is precisely (10.51e). □

*Warning* 12.55 (What sheet resolution does not prove). At the ordinary primitive canonical seed, theorem 12.51 makes $q$ infinite-sheeted. The embedded resolved object is therefore the pair $(e(P_K), q)$, not the unresolved image with its fibers forgotten. Theorem 12.50 proves no conformal-disk fullification of $P_K$, prime-end extension, actual $O, \xi, \eta$ complement footprint, future-faithful compactification of the infinite sheet state, whole-end properness, or uniform width charge. Proposition 12.54 controls only the induced based-coset map. Even when both

(10.51d) and (10.51e) hold, a simply connected enlargement can add a new component or access without changing any fundamental group. Full final-root stability also requires a Cartesian cover square, component and path-fiber bijection, obstacle and nonproper-bridge audit, and the complete prime-end or proper-path endpoint check.

**Proved Theorem 12.56** (Finite-horizon coherent Kahn collars)**.** *Let*

$$\mathbf{Y}_N \xrightarrow{\iota_{N-1}} \mathbf{Y}_{N-1} \longrightarrow \cdots \longrightarrow \mathbf{Y}_1 \xrightarrow{\iota_0} \mathbf{Y}_0$$

*be a fixed finite chain of inclusion-induced pseudo-piece immersions. At every stage $j$, assume:*

*(1) the central map is an embedding;*
*(2) the finite source-to-target tile-component map is bijective;*
*(3) every tile map is a compatible lift of a base-annulus immersion*

$$I_j : \mathbb{A}(1, r_j^{1/d_j}) \longrightarrow \mathbb{A}(1, r_j), \qquad d_j \in \mathbb{Z}, \quad d_j \geq 2, \quad r_j > 1,$$

*which is holomorphically immersive, extends continuously to the inner circle, and has oriented degree one there, in compatible uniformizations of $A_{j+1}$ and $A_j$;*
*(4) central/tile gluing commutes, its overlap maps are embeddings, and the quotient has only the declared central/tile identifications.*
*(5) on the Kahn collar adjacent to the inner end, every selected tile lift reflects the central overlap: the inverse image of the target central/tile overlap is exactly the declared source central/tile overlap.*

*Then there are coherent open inner collars $C_j \subset A_j$ such that*

$$I_j : C_{j+1} \xrightarrow{\cong} C_j$$

*is biholomorphic for every $j$. After retaining only the central pieces, the selected lifted collars, and their declared overlaps, every restricted pseudo-piece map is a holomorphic embedding, as is their finite composite*

$$F_H : \mathbf{Y}_N^{\mathrm{col}} \hookrightarrow \mathbf{Y}_0^{\mathrm{col}}.$$

*Assume separately that the two endpoint pseudo-pieces are conformal disks, that $F_H$ extends continuously to their prime-end compactifications, and that every complete pointwise complement-access footprint $\Phi_H(p)$ is known. For a connected target ideal interval $I$, put*

$$\mathrm{Safe}_H(I) = \{p : \Phi_H(p) \subset I\}.$$

*A connected source ideal set is KKL-admissible for $I$ exactly when it is contained in $\mathrm{Safe}_H(I)$. Consequently the maximal connected admissible candidates are the connected components of this safe set; uniqueness requires an additional connectedness or distinguished-seed argument.*

*Proof.* Kahn's annulus argument gives, for every $j$, a sufficiently thin target collar $\Lambda_j$, the component $\Lambda'_{j+1} \subset I_j^{-1}(\Lambda_j)$ adjacent to the inner circle, and an inverse germ

$$\varphi_j : \Lambda_j \longrightarrow \Lambda'_{j+1} \quad \text{with} \quad I_j \varphi_j = \mathrm{id}\,.$$

For $1 \leq k \leq N$, the domain of $\varphi_{k-1} \circ \cdots \circ \varphi_0$ contains a neighborhood of the inner end of $A_0$. There are only finitely many such domains, so their intersection still contains an open inner collar $C_0$. Define

$$C_{j+1} = \varphi_j(C_j).$$

This proves the collar identities. It also exhibits the only use of the finite-horizon hypothesis; the collar choice is existential, not canonical.

Let $L_j$ be one selected universal-cover tile lift, with positive deck generators $T_{j+1}, T_j$. Degree one gives

$$L_j T_{j+1} = T_j L_j.$$

If $x, y$ in the selected source collar satisfy $L_j x = L_j y$, base injectivity gives $y = T_{j+1}^k x$. Equivariance then gives $T_j^k L_j x = L_j x$; freeness of the target deck action forces $k = 0$. Thus every restricted tile lift is injective.

It remains to audit the quotient. Central–central collisions are excluded by the central embedding, same-tile collisions by the preceding paragraph, and different-tile collisions by component bijectivity and disjoint target tile interiors. A tile–central collision lies in the declared target overlap. Overlap reflection puts its source tile representative in the declared source overlap; let $w$ be the glued source-central representative. If $z$ is the other source-central representative, commutation gives equal central images of $w$ and $z$, and central injectivity gives $w = z$. Thus the two original representatives were already glued in the source. These cases exhaust the quotient classes, so each restricted holomorphic immersion is injective and hence an embedding.

Under the additional marked hypotheses, apply theorem 12.63. Its first/last-traversal argument says that the target ideal endpoints attainable from a source prime end $p$ are exactly $\Phi_H(p)$. Thus admissibility is precisely pointwise footprint containment. Nothing in that argument makes the safe set connected, proving the final assertion. □

*Warning* 12.57 (Exact scope of the collar theorem). Theorem 12.56 embeds only the distinguished inner-collar truncations of one fixed finite chain. It does not prove embedding of a full KL pseudo-piece, exhaustion of omitted outer-return sheets, finiteness of access footprints, final-root no-birth, or either

$$|W_{\mathrm{KL}} - W_{\mathrm{col}}| \le D_{\mathrm{out}}, \qquad \sum_{\mathrm{block}} D_{\mathrm{out}} \le \delta L + E_0$$

with period-independent constants. Naturality under final-root enlargement requires, as additional hypotheses, a commuting analytic enlargement square, bijection of all proof-used retained and omitted components, preservation of subgroup, deck, itinerary, properness, and cyclic-order data, declared protected source and target prime-end sets containing all relevant footprints, boundary homeomorphisms on those protected sets, no new nonproper bridge, and the pointwise no-birth/no-loss identity

$$a_{0,\partial} \Phi_H^+(p) = \Phi_H(a_{N,\partial} p).$$

Kahn's existential collar lemma supplies neither the theorem's overlap-reflection hypothesis nor any of these enlargement hypotheses. If the exact relative fiber-product condition is omitted, a retained tile point can enter the part of a target central overlap also hit by the source central image while its source tile point is not in the declared source overlap. This produces a tile–central quotient collision despite injectivity on both retained pieces. The full overlap-reflection hypothesis stated in the theorem is a stronger sufficient form of that relative condition.

**Open Problem 12.58** (Restricted collision-free KL pseudo-piece classification)**.** Classify any restricted, noncanonical or separately truncated KL stages on which the bold ambient annulus immersion and every full retained tile lift are globally univalent. Theorem 12.36 then gives the corresponding restricted interior pseudo-piece embedding; no independent full-overlap reflection hypothesis is needed for that conclusion.

Ordinary primitive full canonical-localization seeds are excluded: theorem 12.48 proves their noninjectivity. Their universal footprint route passes instead through the embedded resolved component and sheet groupoid of theorem 12.50. This restricted classification problem is nongating and has no descendant in the MLC dependency chain.

If a later footprint or collar argument invokes the stronger full saturation equality, separately record its first failure $L_i(x) = \widetilde{e}_{j(i)}(p)$ with $x \notin e_i(Q_i)$, including component, sheet, access, itinerary, and whether $p \in Q$. The case $p \in Q$ is already a tile-injectivity failure; the case $p \in P_{j(i)} \setminus Q$ is a saturation or footprint defect rather than an interior quotient collision.

*Warning* 12.59 (Even interior embedding would not be ideal-boundary properness). Even a solution of open problem 12.58 would prove neither properness nor injectivity of the boundary extension. Kahn–Lyubich explicitly allow an inclusion immersion to send a source ideal-boundary arc into the target interior. Consequently interior embedding alone does not define $\boldsymbol{\iota}^*\mathcal{I}$, prove that the pair acts, or prove whole-end properness.

**Proved Theorem 12.60** (Tile-interior homogeneous trace)**.** *Let $J$ be a compact nondegenerate subarc in the relative interior of one universal-cover ideal-boundary tile arc of a Kahn–Lyubich pseudo-piece. Then $J$ has an intrinsic coordinate-Lipschitz quadrilateral collar. Its homogeneous minimum-Dirichlet-energy trace quotient is Hilbert, and a conformal isomorphism between selected transported collars induces an exact isometry of those homogeneous trace quotients.*

*Proof.* The tile is a simply connected universal cover of an annulus. In a disk uniformization, $J$ is compactly contained in a proper circular arc. The annular sector with outer side equal to that circular arc,

$$C = \{\rho e^{i\theta} : r < \rho < 1,\ \theta_- < \theta < \theta_+\}, \qquad \theta_+ - \theta_- < 2\pi,$$

is a bounded Lipschitz quadrilateral collar. Define

$$\mathcal{D}(C) = \left\{ u \in H^1_{\mathrm{loc}}(C) : \int_C |\nabla u|^2 < \infty \right\} / \mathbb{R}$$

and give the trace image on $J$ the quotient norm

$$\|[\tau]\|^2 = \inf_{\mathrm{Tr}_J\, u - \tau \in \mathbb{R}} \int_C |\nabla u|^2.$$

Poincaré's inequality makes $\mathcal{D}(C)$ a Hilbert space. The Lipschitz trace map is continuous with closed kernel, so its image with this quotient norm is Hilbert. A conformal isomorphism of the chosen collars bijects admissible extensions and preserves two-dimensional Dirichlet energy, proving the isometry. □

The collar in theorem 12.60 is intrinsic and chosen, not canonical or physically embedded in the dynamical plane. Independently chosen collars need have only equivalent norms, with no uniform equivalence constant. A full norm needs a transported constant-detecting gauge. The theorem does not identify arbitrary proof-used KKL intervals with these tile arcs.

**Proved Proposition 12.61** (Nonproper label-hull no-go)**.** *Proper target-atom labels together with a nonproper symbol do not determine the KKL immersion pullback, even for a holomorphic embedding of one Jordan disk in another.*

*Proof.* Let $V = \mathbb{D}$ and let

$$\Omega = \mathbb{D} \cap \{\mathrm{Im}\, z > 0\}.$$

Choose a Riemann map $\phi : \mathbb{D} \to \Omega$. Since $\Omega$ is Jordan, $\phi$ extends homeomorphically to the closures. The immersion

$$\iota : \mathbb{D} \xrightarrow{\phi} \Omega \hookrightarrow \mathbb{D}$$

is a holomorphic embedding. Mark the target ideal circle by the upper and lower open semicircles $I, J$. The source arc $E$ mapping to $I$ is proper, whereas the complementary source arc $N$ maps to the interior diameter. The deletion-and-seed rule for $I$ deletes no outside proper label and therefore retains both open atoms $E$ and $N$. Whether it also retains the two intervening vertices depends on the separate endpoint predicate and is not needed here.

Choose a nondegenerate $N_0 \Subset N$. A small crosscut of $\Omega$ subordinate to the corresponding diameter segment can be extended through the lower half-disk to a target proper crosscut with both ideal endpoints in $J$. Its intersection with $\Omega$ is one component. Hence its source inverse is the unique proper lift and is subordinate to $N_0$; the target crosscut is not subordinate to $I$.

The defining KKL implication therefore forbids any candidate for $\iota^* I$ from containing $N_0$. The formal label hull does contain $N_0$, so the two sets differ. □

**Proved Theorem 12.62** (KL pseudo-piece conformal ideal boundary)**.** *Each of the two Kahn–Lyubich pseudo-pieces $\mathbf{P}, \mathbf{Q}_a$ in the fixed direct-decoration pair is a hyperbolic conformal disk. Its named $\lambda, \xi, \eta$ boundary is its conformal ideal circle, and a puzzle inclusion immersion extends continuously between the corresponding closed ideal disks. This assertion uses no local connectivity of the physical Julia set.*

*Proof.* The physical puzzle piece $Y$ is a closed Jordan domain, so its closed copy is not used as an open member of the van Kampen cover. Instead put $U_0 = \operatorname{int} Y$ and let $T_i$ be the open universal-cover tile images. These cover $\mathbf{Y}$: $K_Y$ meets $\partial Y$ only at the vertices, every vertex belongs to $S_Y$ and is absent from the path surface, and every other central-boundary point lies in one tile. Distinct tiles are disjoint and $U_0 \cap T_i = \operatorname{int} Y \cap Y_i$ is path-connected. To see this, join two points by a path in the source-locked simply connected set $Y_i$. Since $Y_i$ is relatively open in $Y$, the compact path image has positive distance from $Y \setminus Y_i$. A sufficiently small Schoenflies-collar isotopy, fixed at the two interior endpoints, pushes every boundary contact into $\operatorname{int} Y$ while remaining in $Y_i$. The open cover $\{U_0, T_i\}$ therefore has finite star nerve, simply connected vertex spaces, and path-connected nonempty edge intersections. Repeated van Kampen gives $\pi_1(\mathbf{Y}) = 0$.

The surface $\mathbf{Y}$ is noncompact. Otherwise its nonconstant open projection would have compact, hence closed, image in the connected target and therefore would be surjective, contrary to omission of the nonempty set $S_Y$. The bold target $\mathbf{U}^m$ is a noncompact simply connected disk and therefore uniformizes as $\mathbb{D}$ or $\mathbb{C}$. The path-space projection $\pi_Y : \mathbf{Y} \to \mathbf{U}^m$ is a nonconstant holomorphic immersion whose image omits $S_Y$. If $\mathbf{U}^m \cong \mathbb{D}$, a hypothetical $\mathbf{Y} \cong \mathbb{C}$ contradicts Liouville. If $\mathbf{U}^m \cong \mathbb{C}$, it contradicts Little Picard: the proof-used piece $P$ has $2^n \geq 2$ distinct vertices in $S_Y$, and the univalent branch preserves the corresponding distinct vertices for $Q_a$. The sphere is excluded by noncompactness. Thus $\mathbf{Y} \cong \mathbb{D}$.

Kahn defines ideal closure from a disk universalization, its deck group, and its limit set. For a conformal disk the deck group is trivial and the limit set is empty, so that ideal closure is $\overline{\mathbb{D}}$. Kahn–Lyubich call the named arcs a tiling of this ideal boundary and state that the inclusion immersion extends continuously to it. No second symbolic compactification is introduced. □

**Proved Theorem 12.63** (Embedded-disk accessible-footprint formula)**.** *Let $U, V$ be conformal disks and let $\iota : U \hookrightarrow V$ be holomorphic with a continuous extension $\bar{\iota} : \widehat{U} \to \widehat{V}$ to their conformal ideal closures. Put $\Omega = \iota(U)$. For $p \in \partial_i U$, define*

$$\Phi_\iota(p) = \begin{cases} \{\bar{\iota}(p)\}, & \bar{\iota}(p) \in \partial_i V, \\ \left\{ \xi \in \partial_i V : \begin{array}{l} \exists a : [0,1] \to \widehat{V},\ a(0) = \xi,\ a(1) = \bar{\iota}(p), \\ a((0,1)) \subset V \setminus \Omega \end{array} \right\}, & \bar{\iota}(p) \in V. \end{cases}$$

*Then for every connected target ideal set $I$,*

$$\boxed{\iota^* I = \{p \in \partial_i U : \Phi_\iota(p) \subset I\}.} \tag{10.52}$$

*Proof.* We first prove that every connected boundary-fiber subset is a point. In disk coordinates write the embedding as $F : \mathbb{D} \to \mathbb{D}$, continuous on the closed disk. If the fiber of $w$ contained a nondegenerate connected subset of the circle, it would contain an arc $J$. Put $H = F - w$, precompose by a disk automorphism so that $H(0) \neq 0$, and set

$$M = \max(1, \|H\|_\infty), \qquad u_r(t) = \log M - \log |H(re^{it})| \geq 0.$$

Jensen's formula gives the uniform bound

$$\int_0^{2\pi} u_r(t)\,dt \le 2\pi\big(\log M - \log|H(0)|\big).$$

Continuity and $H|_J = 0$ give $u_r(t) \to +\infty$ on a positive-measure set as $r \uparrow 1$. Fatou contradicts the bound.

Now let $\gamma : (0,1) \to V$ be target proper and let $A$ be a component of $\gamma^{-1}(\Omega)$. The inverse restriction $\beta = \iota^{-1}\gamma|_A$ is topologically proper: for $K \Subset U$, the set

$$\beta^{-1}(K) = A \cap \gamma^{-1}(\iota(K))$$

is closed in the compact target preimage and cannot accumulate at a finite endpoint of $A$, since $\Omega$ is open. At either end of $A$, the nested closures of source tail images have a nonempty compact connected intersection in $\partial_i U$. Continuity places it in one boundary fiber of $\bar{\iota}$, so lightness makes it a singleton. Therefore every intersection component is a KKL proper source path.

Suppose the entire geometric preimage of a target proper trace contains a unique proper source path $\delta$. Every parameter intersection component is proper and must trace $\delta$; uniqueness also excludes a proper path obtained by switching branches at a target self-intersection or retraced segment. There are finitely many traversals: otherwise preimages of one interior point of $\delta$ accumulate at an interior target parameter, contradicting openness of $\Omega$. The first and last complement tails show that the two target endpoints belong to the footprints of the two source endpoints.

Conversely, for $p \in \partial_i U$ and $\xi \in \Phi_\iota(p)$, choose a source hairpin with both ends at $p$. If $\bar{\iota}(p)$ is ideal, its image is the required target hairpin. If it is interior, join the image hairpin to $\xi$ by one complement access traversed in both directions. KKL paths need not be simple, and the access interior is disjoint from $\Omega$, so the hairpin is the unique source restriction. Thus a source set $C$ is admissible for $I$ exactly when $\Phi_\iota(p) \subset I$ for every $p \in C$. The right side of (10.52) is itself admissible and contains every admissible set, which proves that it is the unique maximum in the KKL definition. □

**Proved Theorem 12.64** (A degree-one annulus immersion need not be a lower strip)**.** *For every $0 < h_s < h_t$, there is a $\mathbb{Z}$-equivariant degree-one holomorphic embedding $L : \Pi(h_s) \hookrightarrow \Pi(h_t)$ that carries the lower boundary to the lower boundary but whose upper boundary is nonhorizontal. Hence a general KL degree-one tile immersion cannot be replaced, from degree and boundary data alone, by $z \mapsto z + \theta$.*

*Proof.* For sufficiently small nonzero real $\varepsilon$, put

$$L_\varepsilon(z) = z + \varepsilon \sin(2\pi z).$$

Then $L_\varepsilon(z+1) = L_\varepsilon(z) + 1$. On the closed source strip,

$$|L_\varepsilon'(z) - 1| \le 2\pi|\varepsilon| \cosh(2\pi h_s).$$

Choose the right side $< 1$. Integrating along the straight segment between any two points gives

$$|L_\varepsilon(z) - L_\varepsilon(w)| \ge \big(1 - 2\pi|\varepsilon|\cosh(2\pi h_s)\big)|z - w|,$$

so the map is injective. Also $\mathbb{Z}$-equivariance makes the quotient map injective as well: if $[L_\varepsilon(z)] = [L_\varepsilon(w)]$, then $L_\varepsilon(z) = L_\varepsilon(w) + k = L_\varepsilon(w+k)$ for some $k \in \mathbb{Z}$, and lift injectivity gives $z = w + k$, hence $[z] = [w]$. Also

$$\Im L_\varepsilon(x + iy) = y + \varepsilon \cos(2\pi x)\sinh(2\pi y).$$

Taking

$$|\varepsilon| < \min\left\{ \frac{h_s}{\sinh(2\pi h_s)}, \frac{h_t - h_s}{\sinh(2\pi h_s)} \right\}$$

places the open source strip inside the target strip and maps $y = 0$ to the real line. At $y = h_s$, however, the displayed imaginary part is a nonconstant function of $x$. Thus the image is not a horizontal lower strip. □

**Proved Theorem 12.65** (Common-coordinate outer-cap model)**.** *Suppose, in addition to the fixed embedded pseudo-piece pair, that every source tile annulus is literally nested in its target tile annulus in one external coordinate and that the tile map is the lift of this inclusion. Then, for every connected target ideal set $I$,*

$$\operatorname{relint} \lambda_{s,i} \subset \boldsymbol{\iota}^* I \quad \Longleftrightarrow \quad \overline{\lambda}_{t,j(i)} \subset I. \tag{10.53}$$

*Consequently the open target $O$-atom excludes the complete open source $O$-side, while the native open three-atom tile-star contains it in its pullback.*

*Proof.* Under the stated common-coordinate hypothesis, the target and source tiles are $\Pi(h_t)$ and $\Pi(h_s)$, with $h_s = 2^{-q} h_t < h_t$, and a compatible lift is $z \mapsto z + \theta_i$. Tile-component bijectivity makes the source image fill every target lower strip.

The upper cap $C_i^O = \{h_s \le \Im z < h_t\}$ is then a full connected component of the complement. Indeed, the lifted Green height is continuous, vanishes on the central Julia seam, and the complement has no point with $0 < G < h_s$. A connected complement set meeting both cap and seam would have connected real image containing a forbidden intermediate height. A connected set avoiding the seam remains in one target tile, where the strip formula confines it to $C_i^O$.

For each fixed prime end $p$ on the open source $O$-side, its image is one finite point $x_p + i h_s$ on the cap's lower crosscut. From that fixed point, polygonal arcs in $h_s < \Im z < h_t$ reach every finite point of the top ideal side, while arcs with real part tending to $-\infty$ or $+\infty$ reach the two wrapping corners. Conversely, cap isolation prevents access to any other target ideal point. Thus the pointwise accessible footprint of every such $p$ is exactly the closed target $O$-port. Substitution in theorem 12.63 proves (10.53). The open $O$-atom omits the corners; the smallest native open ideal set containing the closed port consists of the $O$-atom, its two wrapping corners, and its two adjacent $J$-atoms, so it has combinatorial length three. □

**Conditional Theorem 12.66** (Sheet-isolated footprint criterion)**.** *Let $E$ be a resolved source ideal set, $J$ a target ideal set, and $\Phi : E \to 2^J$ a complete sheet-resolved footprint. For a target proper path $\Gamma$, let $\mathrm{PropRes}_{\mathscr{G}_q}(\Gamma)$ be its full set of KKL-proper source restrictions with coset and deck phase retained. Assume:*

*(1) if this set is the singleton $\{\delta\}$, with source endpoints $p_-, p_+$, then the two target endpoints lie in $\Phi(p_-)$ and $\Phi(p_+)$;*

*(2) for every $p \in E$ and $\xi \in \Phi(p)$, there is a target proper hairpin with endpoint $\xi$ whose full proper-restriction set consists of exactly one source hairpin with both source endpoints $p$.*

*(3) the authorized ideal compactifications satisfy the endpoint subordination rules: a source proper path subordinate to connected $C \subset E$ has both endpoints in $C$; a source hairpin with both endpoints $p \in C$ is subordinate to $C$; and a target proper path is subordinate to connected $I \subset J$ if and only if both target endpoints lie in $I$.*

*Then a connected source ideal set $C$ satisfies the KKL universal unique-proper-restriction implication for a connected target ideal set $I$ exactly when*

$$C \subset \{p \in E : \Phi(p) \subset I\}. \tag{10.54}$$

*The maximal connected candidates are the connected components of this safe set.*

*Proof.* If $p \in C$ and $\xi \in \Phi(p) \setminus I$, the hairpin in the second hypothesis has one proper source restriction with both source endpoints $p$. The third hypothesis makes that restriction subordinate to $C$ and, because one target endpoint lies outside $I$, makes the target hairpin non-subordinate

to $I$; hence $C$ is inadmissible. Conversely, if $C$ lies in the safe set and a target proper path has one proper restriction subordinate to $C$, its source endpoints $p_-, p_+$ lie in $C$ by the third hypothesis. The first hypothesis puts the target endpoints in $\Phi(p_-) \cup \Phi(p_+) \subset I$, so the target path is subordinate to $I$, again by the third hypothesis. This is exactly the KKL implication. These endpoint-subordination rules are explicit hypotheses; they are not inferred from local connectivity or from a ribbon-tree quotient. □

**Proved Theorem 12.67** (Resolved-cover boundary obstruction and KL compatibility)**.** *Let $A_s, A_t$ be finite-modulus annuli and suppose that a holomorphic annular immersion of oriented degree $+1$ has a factorization*

$$A_s \xrightarrow{q} P \xrightarrow{e} A_t, \qquad e \circ q = \iota,$$

*where $q$ is a connected surjective covering and $e$ is an embedding. After compatible basepoints identify $\iota_* : \pi_1(A_s) \to \pi_1(A_t)$ with $\mathrm{id}_{\mathbb{Z}}$, put*

$$G = \pi_1(P), \qquad H = q_*\pi_1(A_s), \qquad N = \ker e_*.$$

*Then:*

*(1) $G = N \rtimes H$. In one based universal-cover action,*

$$\mathcal{E}_a/H \cong A_s, \qquad \mathcal{E}_a/N \cong \mathcal{R}_a, \qquad \mathcal{E}_a/G \cong P, \tag{10.55}$$

*and $q_a : \mathcal{E}_a \to \mathcal{R}_a$ is the universal cover with deck group literally $N \le G$.*

*(2) If $N \neq \{1\}$, neither $q_a$, nor any nonconstant $N$-invariant composite, extends continuously through a point of $\Lambda(N)$ to a Hausdorff target. In particular, $q_a$ has no extension from the complete closed-disk prime-end compactification.*

*(3) If this factorization is asserted to be the whole actual Kahn–Lyubich tile map via a based source-cover biholomorphism $\Phi_a : \widehat{A}^{\mathrm{KL}}_{s,a} \to \mathcal{E}_a$, with the source-locked extension at the two wrapping vertices transported through $\overline{\Phi}_a$, then $N = \{1\}$. Thus a canonical infinite-sheet resolution cannot at the same time be that whole boundary-extended KL tile factorization.*

*(4) Let $\Gamma = N$, let $\Omega = \partial\mathbb{D} \setminus \Lambda(\Gamma)$, and let $J \subset \Omega$ be a $\Gamma$-invariant open set. The quotient*

$$R_J = (\mathbb{D} \cup J)/\Gamma \tag{10.56}$$

*is Hausdorff. Every continuous extension to $\mathbb{D} \cup J$ of the invariant interior composite descends uniquely to $R_J$.*

*Proof.* The identity $e_*q_* = \iota_* = \mathrm{id}_{\mathbb{Z}}$ makes $e_*$ onto and makes $e_*|_H : H \to \mathbb{Z}$ an isomorphism. For $g \in G$, set

$$h = q_*(e_*(g)), \qquad n = gh^{-1}.$$

Then $n \in N$, $g = nh$, and $N \cap H = \{1\}$. Since $N$ is the kernel of $e_*$, it is normal; hence $G = N \rtimes H$.

Pulling the universal cover of $A_t$ back along $\iota$ gives the cover classified by $\iota_*^{-1}(0) = 0$, so $\mathcal{E}_a$ is simply connected. Pulling it back along $e$ gives the connected cover of $P$ classified by $e_*^{-1}(0) = N$; connectedness follows from $e_*(G) = \mathbb{Z}$. The composite $\mathcal{E}_a \to \mathcal{R}_a \to P$ is therefore the based universal cover of $P$, which puts $G, H, N$ in one deck action and proves (10.55). The base-changed $q_a$ has simply connected source and deck subgroup exactly $N$.

For the boundary obstruction, let $\gamma_k z_0 \to \zeta \in \Lambda(N)$. For every $z \in \mathbb{D}$,

$$d_{\mathbb{D}}(\gamma_k z, \gamma_k z_0) = d_{\mathbb{D}}(z, z_0),$$

so bounded hyperbolic distance forces $\gamma_k z \to \zeta$. If an $N$-invariant nonconstant map $L$ had a continuous value at $\zeta$, then

$$\overline{L}(\zeta) = \lim_k L(\gamma_k z) = L(z)$$

for every $z$, a contradiction. Taking $L = q_a$ proves the second claim.

Let $h \in H$ be the source-annulus generator. Finite modulus makes $h$ hyperbolic, with the two standalone wrapping fixed points $\zeta_-, \zeta_+$. If $N \neq 1$, then $G$ cannot be elementary: an elementary torsion-free group containing $h$ is cyclic, and $e_*(h) = 1$ would force $H = G$ and $N = 1$. Hence a nontrivial normal subgroup $N \lhd G$ has infinite closed $G$-invariant limit set. Choosing $\alpha \in \Lambda(N) \setminus \{\zeta_-, \zeta_+\}$ and applying $h^k$ gives

$$h^k\alpha \longrightarrow \zeta_+ \quad (k \to +\infty), \qquad h^k\alpha \longrightarrow \zeta_- \quad (k \to -\infty),$$

so both resolved-source wrapping points lie in $\Lambda(N)$. The source-locked KL strip model has actual-tile wrapping points $\omega_-, \omega_+$ at its named infinitely wrapping vertices and extends the actual tile map $L_a$ there. Based covering uniqueness gives

$$\Phi_a t_a = h\Phi_a, \qquad \overline{\Phi}_a(\omega_\pm) = \zeta_\pm,$$

where $t_a$ is the positive actual-tile deck generator. Thus the correctly typed transported composite on $\mathcal{E}_a$ is

$$K_a := L_a\Phi_a^{-1} = e_a q_a.$$

It is nonconstant, $N$-invariant, and continuous at $\zeta_\pm$. The preceding invariant-map obstruction therefore contradicts $N \neq 1$, proving the third claim.

Finally, $\Gamma$ acts properly discontinuously on $\mathbb{D} \cup \Omega$, hence on $\mathbb{D} \cup J$, so the quotient in (10.56) is Hausdorff. If $\overline{L}$ continuously extends the invariant interior composite, then for $x \in J$, $\gamma \in \Gamma$, and $z_k \to x$,

$$\overline{L}(\gamma x) = \lim_k L(\gamma z_k) = \lim_k L(z_k) = \overline{L}(x).$$

It is therefore constant on quotient fibers and descends uniquely by the definition of the quotient topology. □

**Proved Proposition 12.68** (Canonical resolution cannot be the whole actual KL tile)**.** *Retain the nontrivial resolved tower in theorem 12.67, and write*

$$\mathcal{E} \xrightarrow{q} \mathcal{R} \xrightarrow{e} T, \qquad L = eq, \qquad N = \mathrm{Deck}(q) \neq 1.$$

*There do not exist maps*

$$\Phi : \mathcal{E} \longrightarrow \widehat{A}_s, \qquad B : T \longrightarrow \widehat{A}_t$$

*such that:*

1. *$\Phi$ is the based biholomorphic isomorphism of universal covers over the same oriented source annulus;*
2. *$\Phi h = t_s\Phi$ for the positive annular deck generators;*
3. *$B$ is continuous and injective on $L(\mathcal{E})$;*
4. *$L_{\mathrm{KL}}\Phi = BL$ on the complete source tile; and*
5. *the conformal ideal extension of $\Phi$ sends $\zeta_\pm$ to KL's named wrapping vertices $\omega_\pm$.*

*In particular, the ordinary primitive canonical infinite resolution is not a whole-tile covering/embedding factorization of the actual KL map.*

*Proof.* For $N \neq 1$, the proof of theorem 12.67 gives $\zeta_+ \in \Lambda(N)$. Choose $\gamma_k \in N$ with $\gamma_k z \to \zeta_+$ for every $z \in \mathcal{E}$. Deck invariance and the asserted square give

$$L_{\mathrm{KL}}(\Phi(\gamma_k z)) = BL(\gamma_k z) = BL(z).$$

End compatibility gives $\Phi(\gamma_k z) \to \omega_+$, and KL's source-locked continuity at $\omega_+$ therefore implies

$$\overline{L}_{\mathrm{KL}}(\omega_+) = BL(z) \quad \text{for every } z.$$

This is impossible because $L$ is nonconstant and $B$ separates its image. The canonical winding theorem and fiber-preserving tile pullback make $N$ infinite in the ordinary primitive canonical resolution, so the final assertion follows. □

*Warning* 12.69 (The no-go is not an injectivity theorem for the KL immersion). The map

$$F(z) = \exp(4(z-1)) : \mathbb{D} \longrightarrow \mathbb{D}$$

has $F'(z) \neq 0$, extends continuously to $\overline{\mathbb{D}}$, and satisfies

$$F(\pi i/4) = F(-\pi i/4) = -e^{-4}.$$

It is not a covering onto its image. If it were, universal-cover deck transitivity would give an automorphism $\delta$ carrying $\pi i/4$ to $-\pi i/4$ with $F\delta = F$. Connectedness would make $4(\delta(z) - z)/(2\pi i)$ one constant integer, namely $-1$; thus $\delta(z) = z - \pi i/2$, which is not a disk automorphism. Therefore noninjectivity can coexist with full boundary continuity when it is not induced by a global covering-deck relation. Proposition 12.68 forbids only the complete end-compatible comparison square.

*Warning* 12.70 (The replacement boundary state is relative). For each resolved tile the state must retain

$$\mathfrak{B}_a = (\Gamma_a, \Lambda_a, \Omega_a, \mathscr{I}_a^{\mathrm{KL}}, \mathrm{inc}_a, \mathrm{Stab}_a, \mathrm{phase}_a^{\mathrm{int}}, \mathscr{D}_a).$$

Here $\mathrm{inc}_a$, the incidence of the actual KL marking with the canonical resolved circle, is not yet constructed. Once it is typed, intersection of a marked set itself with $\Lambda_a$ forbids a point-valued factor extension on that set. Intersection of only its closure with $\Lambda_a$ is weaker: a component of $\Omega_a$ may still quotient to a complete relative ideal-boundary circle, while its standalone endpoints remain omitted. Interior deck phase does not supply boundary incidence or boundary phase.

**Proved Proposition 12.71** (Canonical/KL restricted-interface classification). *Fix a supplied finite labeled same-source canonical/KL tile datum, with chosen lifted basepoints. For each source label $i$, let*

$$q_i : \mathcal{E}_i \longrightarrow \mathcal{R}_i, \qquad N_i = \mathrm{Deck}(q_i),$$

*be the resolved universal-cover leg, let $L_i^{\mathrm{KL}} : \widehat{A}_{s,i}^{\mathrm{KL}} \to \widehat{A}_{t,j(i)}^{\mathrm{KL}}$ be the actual KL tile immersion, and let $\Phi_i : \mathcal{E}_i \to \widehat{A}_{s,i}^{\mathrm{KL}}$ be the based source-cover biholomorphism. Put $K_i = L_i^{\mathrm{KL}}\Phi_i$. Then:*

*(1) The based relation span*

$$\mathcal{R}_i \xleftarrow{q_i} \mathcal{E}_i \xrightarrow{K_i} \widehat{A}_{t,j(i)}^{\mathrm{KL}} \tag{10.57}$$

*is canonical for the supplied datum. If $N_i \neq 1$, it does not descend to a single-valued map on $\mathcal{R}_i$.*

*(2) For a path $\alpha : J \to \widehat{A}_{t,j}^{\mathrm{KL}}$, the tagged pullback*

$$\mathscr{P}^{\mathrm{tag}}(\alpha) = \bigsqcup_{\substack{i \\ j(i)=j}} \{i\} \times \{(t,z) \in J \times \mathcal{E}_i : \alpha(t) = K_i(z)\} \tag{10.58}$$

*has path components exactly the graphs of all maximal interior restrictions in the declared source tiles. Source labels, resolved traces, sheets, and deck phases remain distinct.*

*(3) Let $\zeta_{i,\pm}$ be the two resolved-source wrapping points transported from KL's named actual-tile wrapping vertices, and put $D_{\mathrm{ext},i} = \{\zeta_{i,-}, \zeta_{i,+}\}$. If $N_i \neq 1$, then for every $r \in \mathcal{R}_i$,*

$$\overline{K}_i(\Lambda(N_i) \cap D_{\mathrm{ext},i}) \subset \overline{\{K_i(\gamma z) : \gamma \in N_i\}}, \qquad q_i(z) = r, \tag{10.59}$$

*where the closure is taken in the Hausdorff target pseudo-piece compactification. No value on the rest of $\Lambda(N_i)$ is asserted.*

(4) *Kahn's degree-two annulus hypotheses alone do not imply a universal unscaled additive bound between complete source-collar and target crossing widths. Indeed, for*

$$A_s(\varepsilon) = \{1 < |z| < e^{\varepsilon/2}\} \hookrightarrow A_t(\varepsilon) = \{1 < |z| < e^{\varepsilon}\},$$

*every essential embedded source inner collar $C \subset A_s$ satisfies*

$$\mathcal{W}(\Gamma_C) - \mathcal{W}(\Gamma_t) \geq \frac{2\pi}{\varepsilon} \longrightarrow \infty. \tag{10.60}$$

(5) *At every exact Kahn base change $E = X \times_Z Y$, $R = P \times_Z Y$, the induced first factor $q'(x, y) = (q(x), y)$ satisfies*

$$(q')^{-1}(p, y) \cong q^{-1}(p). \tag{10.61}$$

*Hence no finite retiming inside the unchanged exact equalizer tower turns an infinite first factor into a one-sheet factor.*

(6) *For revised connected localization covers classified by $H_s \leq \pi_1(X_s)$ and $H_t \leq \pi_1(X_t)$, first assume the based lift condition $a_*(H_s) \leq H_t$. The first factor has pointed coset fiber $a_*^{-1}(H_t)/H_s$ and is one-sheeted exactly when*

$$a_*^{-1}(H_t) = H_s. \tag{10.62}$$

*Proof.* Both $\mathcal{E}_i$ and $\widehat{A}^{\mathrm{KL}}_{s,i}$ are based universal covers of the same oriented source annulus. Based lifting uniqueness gives the unique $\Phi_i$, hence (10.57). If $K_i = \kappa_i q_i$, then $K_i$ is $N_i$-invariant. The normal-kernel wrapping theorem puts $\zeta_{i,\pm}$ in $\Lambda(N_i)$, while KL's source-locked values at its named vertices transport through $\overline{\Phi_i}$ to continuous values of $K_i$ there. The deck-limit theorem forbids this for a nonconstant $N_i$-invariant map, proving non-descent.

Each $K_i$ is a local homeomorphism. Its pullback by $\alpha$ is therefore an étale, Hausdorff, second-countable space over $J$. If $J$ is a singleton, every component is a singleton graph and the claim is immediate. Assume $J$ is nondegenerate. Every path component is a connected one-manifold. It cannot be a circle over an interval, and the local orientation of its projection to $J$ is constant; hence that projection is injective. Its inverse is the graph of one maximal lift. Conversely, local lifting uniqueness prevents a maximal lift graph from being a proper path-connected subset of a component. Taking the disjoint union with the source label attached proves (10.58).

For $\zeta \in \Lambda(N_i) \cap D_{\mathrm{ext},i}$, choose distinct $\gamma_n \in N_i$ with $\gamma_n z_0 \to \zeta$. Hyperbolic isometry gives

$$d(\gamma_n z, \gamma_n z_0) = d(z, z_0),$$

so bounded hyperbolic distance gives $\gamma_n z \to \zeta$. Partial continuity at the two certified points yields $K_i(\gamma_n z) \to \overline{K}_i(\zeta)$, proving (10.59).

For the annulus family,

$$\mathcal{W}(\Gamma_t) = \frac{2\pi}{\varepsilon}.$$

Every essential $C \subset A_s(\varepsilon)$ has $\operatorname{mod} C \leq \varepsilon/(4\pi)$, hence $\mathcal{W}(\Gamma_C) \geq 4\pi/\varepsilon$. Subtraction proves (10.60). This calculation is unscaled; it does not exclude a separately proved calibrated or source-specific response.

Equation (10.61) follows directly: $(q')^{-1}(p, y) = \{(x, y) : q(x) = p\}$. Induction proves the finite-retiming claim. Finally, the lift condition is precisely $H_s \leq a_*^{-1}(H_t)$. Covering classification identifies the first fiber with the pointed coset set $a_*^{-1}(H_t)/H_s$, which is a singleton exactly under (10.62). □

*Warning* 12.72 (Exact scope of the restricted relation). Proposition 12.71 is conditional on a supplied fixed labeled same-source datum; it does not realize such a datum at every Wave stage. Its tagged family is an interior enumeration, not a geometric-trace KKL restriction, a whole-end-interval properness classification, or a defined finite groupoid. KL's glued pseudo-piece extension certifies the two wrapping points used in (10.59), not the full standalone tile circle.

The candidate uncompressed relation state is not proved minimal, finite, Value-observable, or future-faithful. No prime-end naturality, uniform omitted-family response, Cartesian final-root no-birth, compactness, rigidity, or MLC conclusion follows.

**Proved Proposition 12.73** (Full KL pseudo-piece maximal-lift properness)**.** *Let*

$$\iota : \mathbf{Y}_s \longrightarrow \mathbf{Y}_t$$

*be one supplied full KL pseudo-piece inclusion immersion. Assume that* $\mathbf{Y}_s, \mathbf{Y}_t$ *are conformal disks, their ideal closures are the corresponding closed disks, and KL's extension*

$$\bar{\iota} : \overline{\mathbf{Y}_s}^{\,i} \longrightarrow \overline{\mathbf{Y}_t}^{\,i} \tag{10.63}$$

*is continuous, but not necessarily boundary-valued. If* $\alpha : (0,1) \to \mathbf{Y}_t$ *is a KKL proper path, then every component of*

$$\mathscr{P}_\iota(\alpha) = \{(t,z) : \alpha(t) = \iota(z)\} \tag{10.64}$$

*is the graph of a maximal lift* $\widetilde{\alpha}_C : (a_C, b_C) \to \mathbf{Y}_s$ *having unique ideal limits*

$$p_C^- = \lim_{t \downarrow a_C} \widetilde{\alpha}_C(t), \qquad p_C^+ = \lim_{t \uparrow b_C} \widetilde{\alpha}_C(t). \tag{10.65}$$

*Thus every maximal parameter-compatible lift is a KKL proper source path, and*

$$\bar{\iota}(p_C^-) = \overline{\alpha}(a_C), \qquad \bar{\iota}(p_C^+) = \overline{\alpha}(b_C). \tag{10.66}$$

*An internal parameter endpoint maps to target interior and is a break end. The proposition does not assert properness of* $\iota$ *on a nondegenerate source ideal interval.*

*Proof.* The local-homeomorphism argument in proposition 12.71 makes each component of (10.64) one maximal lift graph. At the left end put

$$Z_C^- = \bigcap_{\epsilon > 0} \overline{\widetilde{\alpha}_C((a_C, a_C + \epsilon) \cap (a_C, b_C))}^{\,i}. \tag{10.67}$$

The closed source disk is compact. Each tail closure is compact and connected, and the family is nested; hence $Z_C^-$ is a nonempty compact continuum.

It cannot meet $\mathbf{Y}_s$. If $x \in Z_C^- \cap \mathbf{Y}_s$, choose a local inverse chart $\iota|_U : U \xrightarrow{\cong} V$ at $x$. A sufficiently short target tail lies in $V$, and one source tail point lies in $U$. The local inverse lift and $\widetilde{\alpha}_C$ then agree at that point and hence, by uniqueness of lifts, on their connected common domain. If $a_C \in (0,1)$, this continues the lift through $a_C$, contradicting maximality. If $a_C = 0$, it puts the target ideal limit $\overline{\alpha}(0)$ in target interior, again a contradiction. Therefore

$$Z_C^- \subset \partial_i \mathbf{Y}_s, \qquad \bar{\iota}(Z_C^-) = \{\overline{\alpha}(a_C)\}. \tag{10.68}$$

Every connected subset of a boundary fiber in (10.68) is a point. Indeed, in disk coordinates a nondegenerate connected subset would contain a circular arc on which $H = F - w$ has continuous boundary value zero. Precompose by a source-disk automorphism so that $H(0) \neq 0$, and put $M = \max(1, \|H\|_\infty)$. Jensen's formula uniformly bounds

$$\int_0^{2\pi} (\log M - \log |H(re^{it})|)\, dt, \qquad 0 < r < 1, \tag{10.69}$$

whereas Fatou's lemma makes the integral diverge because the integrand tends to $+\infty$ on the boundary arc. Thus $Z_C^-$ is a singleton. The right end is identical. Continuity gives (10.66). □

**Proved Proposition 12.74** (Parametrized-pullback KKL exhaustion no-go)**.** *Maximal components of* (10.64) *do not, in general, exhaust KKL's immersion restriction, which consists of every proper source path contained in the geometric inverse trace.*

*For the identity disk, choose a proper lollipop path that travels from* $-1$ *to an interior vertex* $p$*, crosses from* $p$ *to* $q$ *on an upper arc, returns from* $q$ *to* $p$ *on a lower arc, traverses that lower arc*

*again from $p$ to $q$, and exits to 1. Its parametrized fiber product is its single chronological graph. The path using only the entry segment, lower arc, and exit segment is proper and contained in the same geometric trace, but is not a reparametrization of that graph.*

*Moreover,*

$$C(z) = \frac{1+z}{1-z}, \qquad K(z) = e^{-C(z)} \tag{10.70}$$

*is a holomorphic disk immersion. For every $k \in \mathbb{Z}$ and $0 < t < 1$,*

$$u_k(t) = -\log t + 2\pi i k, \qquad z_k(t) = \frac{u_k(t)-1}{u_k(t)+1} \tag{10.71}$$

*satisfies $K(z_k(t)) = t$, with*

$$z_k(t) \longrightarrow 1 \quad (t \downarrow 0), \qquad z_k(t) \longrightarrow \frac{2\pi i k - 1}{2\pi i k + 1} \in \partial\mathbb{D} \quad (t \uparrow 1). \tag{10.72}$$

*Thus one proper target diameter has infinitely many distinct proper source restrictions.*

*Finally, for $0 < a < b$, $0 < \theta < \pi$, the disjoint logarithmic rectangles*

$$R_k = (a,b) \times (2\pi k - \theta, 2\pi k + \theta) \tag{10.73}$$

*map biholomorphically by $u \mapsto e^{-u}$ to one target sector. Their vertical-side crossing families satisfy*

$$\mathcal{W}\left(\bigcup_{|k|\le m} \Gamma_k\right) = (2m+1)\frac{2\theta}{b-a} \longrightarrow \infty. \tag{10.74}$$

*Equation (10.74) concerns compact interior branch collars; it is not a complete KKL-marked lamination or a WAD/Value-response counterexample. The proposition does not exclude a separately proved calibrated response or actual-stage sheet bound.*

*Proof.* The lollipop's lower route is a proper path in the identity inverse trace, but its image omits the upper arc, so it is absent from the parametrized graph. This gives strict non-exhaustion.

For (10.70),

$$K'(z) = -\frac{2e^{-C(z)}}{(1-z)^2} \neq 0,$$

and direct substitution proves $K(z_k(t)) = e^{-u_k(t)} = t$. The two limits in (10.72) are immediate; the second has modulus one because numerator and denominator have equal modulus, and it depends on $k$. Hence the proper restrictions are distinct.

The rectangles in (10.73) have disjoint supports and height $2\theta < 2\pi$, so the exponential is injective on each and periodicity gives the same image. The rectangle formula gives $\mathcal{W}(\Gamma_k) = 2\theta/(b-a)$; the disjoint-support parallel law gives (10.74). □

*Warning* 12.75 (Geometric-trace and properness type boundary). KKL distinguish a proper source path, properness of the immersion on an ideal subset, and end-properness on every complete end-interval of an already constructed pulled multi-interval. "An end-proper path component" is therefore ill-typed. The valid order is: construct every geometric restriction; construct every $\iota^* I$; prove pulled partial-marking disjointness and covering refinement; then test properness on whole intervals and complete end-intervals.

**Open Problem 12.76** (Sheet-sensitive actual KL outer-port footprint)**.** The full, unselected Kahn equalizers and their resolved intermediate components are connected by proposition 12.52 and theorem 12.53. That result is conditional at the whole-tile map-identification step: theorem 12.67 proves that the canonical infinite cover cannot be the whole actual boundary-extended KL tile factorization.

The whole-tile option is excluded by proposition 12.68, while proposition 12.71 supplies the relation span and tagged maximal parameter-compatible interior lifts for every supplied fixed same-source datum. By proposition 12.73, each maximal lift is a proper source path once the full KL pseudo-piece datum is supplied, but proposition 12.74 rules out identifying those lifts with the complete KKL restriction.

First freeze the complete target-family and universal-subordination ledger. Then construct the actual glued canonical/KL stage datum, the full tagged geometric inverse-trace path space, all pulled intervals, partial-marking disjointness, covering refinement, and map properness on every whole proof-used interval and complete end-interval. Next prove zero omission or a universal Value-observable response. Cartesian final-root no-birth is an independent geometric successor of the classification, not part of the response estimate.

Only after the response is established may one choose an authorized $\Gamma_a$-invariant domain-of-discontinuity set $J_a \subset \Omega_a$, form the relative quotient in (10.56), and construct an intermediate outer set $\lambda_a^R$ satisfying

$$\lambda_a^E = \overline{\Phi}_a(\lambda_a^Q), \qquad \overline{q}_a(\lambda_a^E) = \lambda_a^R, \qquad \overline{e}_a\overline{q}_a|_{\lambda_a^E} = \overline{L}_a\overline{\Phi}_a^{-1}|_{\lambda_a^E},$$

with exact deck-phase normalization and all boundary lifts retained. The raw based fiber is infinite by theorem 12.51, so retain the full monodromy and limit-set groupoid or prove a future-faithful compactification.

Only after constructing that boundary model may one enumerate, on every resolved $O$-side, all landing complement components and complete target access footprints with multiplicity. Every witness must isolate exactly one KKL-proper coset lift, or classify all competing proper lifts with deck phase. Prove the sheet-isolated $O$-port formula, or record the first extra, missing, merged, or competing access with its tile, coset, component, and itinerary. The unresolved immersion may not be inserted into the embedded-disk safe-set theorem, and the common-coordinate hypothesis in theorem 12.65 may not be assumed.

**Open Problem 12.77** (Direct-decoration path-lift exhaustion)**.** For every nontrivial proof-used target marking, use the propagated infinite-sheet groupoid or a proved future-faithful compactification and enumerate every landing component on a nonproper source $O$-, $\xi$-, or $\eta$-side. The $O$-side computation in open problem 12.76 comes first.

Prove the endpoint-subordination axioms on the authorized boundary model, endpoint necessity, and sheet-isolated hairpins before applying any safe-set formula: each target witness must have exactly one KKL-proper resolved restriction, or every competing proper lift must remain in the state. Then prove pairwise disjointness, covering-endpoint refinement, and immersion properness or end-properness on every complete covering end-interval. Retain subgroup, coset, deck phase, access order, multiplicity, and all proper intersection components. Final-root stability requires both the based-coset tests of proposition 12.54 and a Cartesian geometric component/path/endpoint no-birth theorem.

For nontrivial markings, open problem 12.77 occurs before trace welding. If two collared spans are later constructed, word composition still requires a Cartesian fiber product and the Beck–Chevalley identity

$$c^* b_* = q_* p^*;$$

a commuting non-Cartesian square can omit a fiber component. Likewise, commutation on an old final-root subdiagram does not prevent a new proper lift after enlargement; an actual no-birth component bijection is required.

Thus the fixed-branch analysis source-locks the actual KL immersion, proves its forced canonical-localization collision and exact pullback propagation, proves connected ambient and tilewise resolved-interior alignment for one fixed eligible seed, and retains the conditional

tile-injectivity gluing criterion, one intrinsic local trace space, the conformal ideal-boundary identification, and the general embedded-disk footprint formula in their valid scopes. It also proves that a general degree-one annulus immersion need not be a lower strip, while retaining the exact cap computation as a common-coordinate positive control. The sheet-sensitive restricted canonical-resolution/KL-tile interface, cover-relative ideal-boundary realization, $O$-footprint, and then the central $J$-seam footprints remain open. The analysis does not promote the all-depth contact, trace-collar, welding, H2, VAAPC-I, MBTT, UPTCR, or MLC obligations.

12.19. **Dense terminal obstacles and the ordered-multicurve replacement.**

**Proved Proposition 12.78** (Properness of the locally connected native carrier)**.** *For a locally connected, primitively renormalizable KKL map with primitive period $p > 1$, the native Hubbard carrier $T$ is a proper subcontinuum of the filled continuum $K$.*

*Proof.* KKL's nondividing fixed point $\beta_F$ satisfies $K \setminus \{\beta_F\}$ connected. If $\beta_F$ belonged to one small Julia continuum $K_j$, fixedness and $F(K_j) = K_{j+1}$ would place it in every member of the pairwise disjoint primitive cycle, a contradiction. Hence $K \setminus \{\beta_F\}$ is a connected set containing the complete cycle. Since KKL define $T$ as the least connected subset of $K$ containing that cycle,

$$T \subset K \setminus \{\beta_F\} \subsetneq K.$$

□

**Proved Theorem 12.79** (Dense terminal-obstacle no-go)**.** *Let $g : U' \to U$ be a connected degree-$d$ polynomial-like map, $d \geq 2$, whose filled continuum $K$ has empty interior. Let $T \subsetneq K$ be a nondegenerate compact connected continuum with $g(T) \subset T$. In a bounded Jordan localization put*

$$A = \Omega \setminus K, \qquad M = \Omega \setminus T,$$

*and let $p : \widetilde{M} \to M$ be the universal cover. Then*

$$\partial^{\mathrm{PE}}_{\mathrm{in,fin}} \widetilde{M} \subset \overline{p^{-1}(K \setminus T)}^{\,\mathrm{PE}}. \tag{10.75}$$

*Both deck-limit corners also belong to the right-hand closure. Therefore no nondegenerate inner ideal interval supports an obstacle-free native collar.*

*Proof.* By Douady–Hubbard straightening and external dynamics [DH85], the external prime-end map $\sigma_g$ is conjugate to $t \mapsto dt \pmod 1$; hence every nonempty open prime-end arc eventually maps onto the whole circle. Full prime-end impressions obey

$$g(\mathrm{Imp}_K \mathfrak{p}) = \mathrm{Imp}_K(\sigma_g \mathfrak{p}). \tag{10.76}$$

For the reverse inclusion in (10.76), choose an arbitrary defining exterior sequence at the image prime end and lift it with the local $d$-th-root branch converging to the prescribed source prime end. This proves covariance for full impressions, not merely for radial cluster sets.

If a nonempty open external arc had all impressions in $T$, forward invariance and exactness would give that property on the whole prime-end circle. Prime-end impressions cover $\partial K$, so $\partial K \subset T$. Empty interior gives $K = \partial K$, contrary to $T \subsetneq K$.

The access-survival equivalence proved earlier in the project says that a compact inner ideal interval missing the complete lifted obstacle closure would project, after shrinking to one evenly covered boundary chart, to an open external arc whose impressions lie in $T$. The preceding paragraph excludes that arc and proves (10.75). Finally, positive and negative deck translates of any lift of a point of $K \setminus T$ converge to the two deck-limit corners. □

**Proved Lemma 12.80** (Ordered-multicurve overflow)**.** *For a finite ordered multicurve $c = \bigsqcup_{r=1}^{s} c_r$, define $\ell_\rho(c) = \sum_r \ell_\rho(c_r)$. Let $f : X \to Z$ be a holomorphic immersion. Suppose every $h$ in a target curve family $\mathcal{H}$ contains pairwise parameter-disjoint ordered subarcs whose union is $f(c_h)$ for some $c_h$ in a source ordered-multicurve family $\mathcal{G}$, with image multiplicity retained. Then*

$$\mathcal{L}(\mathcal{H}) \geq \mathcal{L}(\mathcal{G}), \qquad \mathcal{W}(\mathcal{H}) \leq \mathcal{W}(\mathcal{G}). \tag{10.77}$$

*Proof.* For a finite-area conformal metric $\rho$ on $X$, put

$$\sigma(w)^2 = \sum_{z \in f^{-1}(w)} \frac{\rho(z)^2}{|f'(z)|^2}.$$

Tonelli and change of variables give $\mathrm{Area}_\sigma(Z) = \mathrm{Area}_\rho(X)$. Each selected lift summand gives $\ell_\sigma(f(c_r)) \geq \ell_\rho(c_r)$; summing over ordered components proves holomorphic image monotonicity for $\mathcal{G}$. For an arbitrary target metric, pairwise parameter-disjointness gives $\ell(h) \geq \ell(f(c_h))$. Taking the infimum over curves and then the supremum over metrics proves the extremal-length inequality; inversion gives the width inequality. □

**Proved Theorem 12.81** (KL ordered-submulticurve transfer)**.** *Let $Y$ be a KL geometric puzzle piece and let $v \neq w$ be vertices. If $\mathcal{H}_Y(v, w)$ is the physical family connecting $S_Y^v$ to $S_Y^w$, and $\Gamma_Y(v, w)$ is KL's horizontal ordered-multicurve family in $\Psi(Y)$, then*

$$\mathcal{W}(\mathcal{H}_Y(v, w)) \leq \mathcal{W}(\Gamma_Y(v, w)) = \mathrm{Bd}_Y(v, w)^{-1}. \tag{10.78}$$

*The complete relation $(\gamma, \gamma', \gamma^*)$, consisting of the physical curve, its ordered contained submulticurve, and its KL lift, is retained.*

*Proof.* KL Lemma 'lifts' [KL08] says that every physical $\gamma$ contains an ordered multicurve $\gamma'$ lifting to a horizontal $\gamma^*$ from $\partial^v \Psi(Y)$ to $\partial^w \Psi(Y)$. Apply lemma 12.80; KL's definition of $\mathrm{Bd}_Y(v, w)$ gives the equality. □

**Proved Proposition 12.82** (Raw untagged path-image collision)**.** *At the raw proper-path level, a degree-two pseudo-quadratic-like correspondence can have two distinct immersion-restriction components with one identical unparametrized image under the covering projection. The example does not produce two KKL Waves and does not assert a complete acting KKL outer marking.*

*Proof.* Put

$$\beta = \frac{\pi}{50}, \qquad \Delta = e^{2\beta} - 1, \qquad U' = \mathbb{D}_{75}, \qquad U = \mathbb{D}_{5625},$$

$$f(z) = z^2, \qquad K' = [-1, 1], \qquad K = [0, 1], \qquad i(z) = \frac{e^{\beta(z+1)} - 1}{\Delta}.$$

The derivative of $i$ never vanishes, $i(U') \subset U$, and direct solution of the two inverse equations gives

$$i^{-1}(K) = f^{-1}(K) = K'.$$

Thus $(i, f) : (U', K') \to (U, K)$ is a degree-two pseudo-quadratic-like pair. For

$$\alpha(t) = \frac{e^{2\beta t + 2\pi i t} - 1}{\Delta}, \qquad 0 < t < 1,$$

all immersion lifts are

$$z_k(t) = -1 + 2t + 100i(t + k), \qquad k \in \mathbb{Z}.$$

Only $k = 0, -1$ meet $U'$. If $t_* \in (1/2, 1)$ is the positive root of

$$10004t^2 - 4t - 5624 = 0,$$

the two proper components are

$$\gamma_+ = z_0((0,t_*)), \qquad \gamma_- = z_{-1}((1-t_*,1)).$$

They lie in opposite half-planes and hence are distinct. But

$$z_{-1}(1-t) = -z_0(t)$$

gives

$$f(\gamma_-) = f(\gamma_+).$$

The common image is the simple proper path

$$\delta(t) = (-1+2t+100it)^2, \qquad 0 < t < t_*,$$

from the inner to the outer ideal boundary of $U \setminus K$. Consequently it is not a KKL Wave, whose leaves are subordinate to one connected multi-interval. The calculation proves only that untagged set-valued pushforward forgets occurrence multiplicity. □

**Proved Theorem 12.83** (One retained chain, with possible collisions)**.** *Let $N = b+3$, let $b \le n \le N-2$, and let $\alpha \in A_{n+1}$. Put $\beta_r = F_*^r\alpha$. Then every $\beta_r$, $0 \le r \le n+1$, is a single retained marked path. At each stage the retained immersion restriction is nonempty, and every one of its components has the same covering image $\beta_{r+1}$. At least one component can therefore be selected at every stage.*

*This does not prove uniqueness, compatibility of the stagewise selections as one segmented prefix flag, or realization of either separator-side portion used at KKL line 1603.*

*Proof.* The subwaves are nested, so

$$\alpha \in A_{n+1} \subset A_r \subset A_0 \subset A^{N\text{-nbr}} \qquad (0 \le r \le n+1).$$

By KKL's definition of $N$-unbreakability, $F_*\beta_r$ is a single path for every $r \le n$. In particular it is nonempty. If $\mathscr{R}(\beta_r)$ denotes the retained proper components of the immersion restriction, the untagged action is

$$F_*\beta_r = \{f(\gamma) : \gamma \in \mathscr{R}(\beta_r)\}.$$

The left side is the singleton $\{\beta_{r+1}\}$. Hence $\mathscr{R}(\beta_r) \neq \varnothing$ and every retained component maps to $\beta_{r+1}$. Finite choice gives one stagewise chain. Finally, $\alpha \in A_r$ and the definition of $A_r$ put $\beta_r$ in the Wave $F_*^r A_r$ ending at $BL_0$. The untagged set equality records none of the excluded coherence data. □

**Open Problem 12.84** (Stationary two-sided Wave noncollision)**.** Fix a KKL stationary Wave with its complete ideal marking. Let $b \le n < N-1$, and suppose that the occurrence $\alpha \in A_{n+1}$ avoids the corrected continuum $Q_n$ (the role assigned to the undefined printed $K_0^n$). The following five assertions remain to be proved:

(1) prove that the incoming proper restriction has maximal target window $(0,u)$ and its far ideal end is the selected $BL_0$ prime-end access;
(2) bind the two topological portions to two distinct coherent components of the complete iterated immersion restriction; then prove that each separator-side component is a proper path through every immersion restriction, survives every pulled-marking filter, and has a terminal image that is a Wave;
(3) the root and deep parent routes determine disjoint fully marked prime-end access records through the complete horizon;
(4) both portions are confined to one selected collar occurrence, with no re-entry or ambient sheet change; and
(5) their selected collar crosscuts are not related by the nontrivial deck involution of the unique central degree-two collar.

The first clause is not supplied by KKL's interval itinerary, which permits adjacency and same-pulled-interval transitions instead of actual paths. New tree attachments can also be absorbed into one $\iota^*\mathcal{I}$-interval and filtered when subordinate. The source does not identify the complete attachment itinerary needed either to exclude this hazard or turn it into a counterexample. The endpoint-pair inequality is a sufficient certificate for the last clause, not a necessary one. KKL's component itinerary does not supply it: the root and deep parents route to the same component of $T \setminus K_1$. Until these four clauses hold, line 1603 and `KKL_STATIONARY_TWO_PULLBACK_PASS_THROUGH` remain open.

**Conditional Theorem 12.85** (Conditional off-by-one reduction)**.** *Assume open problem 12.84, so that KKL's line-1603 pass-through predicate is valid. Let the parabolic bound be $b$ and put $N = b + 3$. Then every occurrence in the actually defined $A_{N-1} = A_{b+2}$ passes through both $Q_b$ and $Q_{b+1}$ in KKL's successive-pullback sense. No final-level component identity or incidence order is asserted.*

*Proof.* KKL define $A_n \subset A_{n-1}$ only for $0 < n < N$. Hence

$$A_{b+2} \subset A_{b+1}.$$

Under the stated noncollision hypothesis, apply line 1603 first at $n = b$ to the occurrence viewed in $A_{b+1}$, and then at $n = b + 1$ to the same occurrence in $A_{b+2}$. This gives the two claimed predicates. Both indices are in range because $b + 2 = N - 1 < N$; the undefined $A_N$ printed in KKL's concluding sentence is not used. □

**Proved Lemma 12.86** (Finite $\psi$-ql filled graph)**.** *Let*

$$(i_n, f_n) : (U^{n+1}, K^{n+1}) \longrightarrow (U^n, K^n)$$

*be a finite Kahn fiber-product tower and put*

$$I_m = i_0 \circ \cdots \circ i_{m-1}, \qquad P_m = f_0 \circ \cdots \circ f_{m-1}, \qquad F = f_0 \circ (i_0|_{K^1})^{-1}.$$

*Then*

$$I_m|_{K^m} : K^m \longrightarrow K^0 \text{ is a homeomorphism}, \qquad P_m^{-1}(K^0) = K^m, \qquad P_m = F^m \circ I_m \quad \text{on } K^m. \tag{10.79}$$

*For every compact connected $A \subset K^0$, the set*

$$\widehat{A}^m = (I_m|_{K^m})^{-1}(A)$$

*is its unique connected filled graph and satisfies*

$$P_m(\widehat{A}^m) = F^m(A).$$

*If $A$ is a component of $F^{-m}(B)$, then $\widehat{A}^m$ is a component of $P_m^{-1}(B)$.*

*Moreover, an embedded quadratic-like chart supplies a neighborhood $V_m \supset K^0$ and a holomorphic graph $\Gamma_m : V_m \to U^m$ with*

$$I_m\Gamma_m = \mathrm{id}, \qquad P_m\Gamma_m = F^m. \tag{10.80}$$

*Thus local proper degrees and the corresponding ideal-boundary degrees transfer through the graph. No local connectivity is used.*

*Proof.* Kahn's pullback theorem makes every $i_n|_{K^{n+1}}$ a homeomorphism. Their composition gives the first claim in (10.79). Induction using $f_n^{-1}(K^n) = K^{n+1}$ gives the second. The commuting squares $i_n f_{n+1} = f_n i_{n+1}$, together with $f_0 = F i_0$ on $K^1$, telescope to the third. Consequently

$$P_m^{-1}(B) = (I_m|_{K^m})^{-1}(F^{-m}(B))$$

for $B \subset K^0$, proving graph uniqueness, the image formula, and component naturality.

For the analytic assertion, choose Kahn's embedded quadratic-like restriction $F = f_0 i_0^{-1} : D_1 \to D_0$. On the open set where $z, F(z), \dots, F^{m-1}(z)$ remain in $D_1$, the embedded inverse images of this orbit form a unique fiber-product chain. The universal property gives a holomorphic map $\Gamma_m$ satisfying (10.80). Since $I_m \Gamma_m = \mathrm{id}$, $\Gamma_m$ is a biholomorphism onto an open neighborhood of $K^m$. Conjugating isolating proper restrictions by this biholomorphism preserves their degree. Deleting the full continua leaves annular covering collars; uniformization extends them to prime-end circles with the same degree. □

**Proved Lemma 12.87** (Confluence, composition, and fullness)**.** *Let $h : X \to Y$ be a finite proper holomorphic branched cover of planar surfaces. If $A \subset Y$ is a continuum and $C$ is a component of $h^{-1}(A)$, then $h(C) = A$. If $A$ is a component of $g^{-1}(B)$, then $C$ is a component of $(g \circ h)^{-1}(B)$. If $A$ is full, then $C$ is full.*

*Proof.* Choose connected relatively compact neighborhoods $V_{k+1} \Subset V_k$ with $\bigcap \overline{V}_k = A$, and let $W_k$ be the component of $h^{-1}(V_k)$ containing $C$. The map $h|_{W_k}$ is proper and open, so its image is both open and closed in $V_k$, hence equals $V_k$. The compact connected sets $\overline{W}_k$ are nested and their intersection is $C$. For every $y \in A$, the nested compact sets $\overline{W}_k \cap h^{-1}(y)$ are nonempty, proving $h(C) = A$.

If a connected subset of $(g \circ h)^{-1}(B)$ meets $C$, its $h$-image is connected and lies in the component $A$; maximality then keeps the original subset in $C$. This proves component composition. Finally, a bounded complementary component $H$ of $C$ would satisfy $H \not\subset h^{-1}(A)$, since otherwise $C \cup H$ would be a larger connected subset of $h^{-1}(A)$. For $x \in H \setminus h^{-1}(A)$, openness and properness give $\partial h(H) \subset h(\partial H) \subset A$. The component of $\mathbb{C} \setminus A$ containing $h(x)$ would then lie in the bounded set $h(H)$, contradicting that fullness makes $\mathbb{C} \setminus A$ connected and unbounded. □

**Proved Proposition 12.88** (Stationary final components and degree correction)**.** *Put $m = N - 1 = b + 2$, let $Q_0$ be the critical member of the primitive small-Julia cycle, write $K_j = F^j(Q_0)$, let $Q_r = \phi_0^r(Q_0)$, where $\phi_0$ is the unique critical-limb inverse branch proved in lemma 12.106 below, and set*

$$\widehat{Q}_r = (I_m|_{K^m})^{-1}(Q_r), \qquad r \in \{b, b+1\}.$$

*Then $\widehat{Q}_b, \widehat{Q}_{b+1}$ are distinct whole components of $\mathcal{K}^m = P_m^{-1}(\mathcal{K})$, with*

$$P_m(\widehat{Q}_b) = K_2, \qquad P_m(\widehat{Q}_{b+1}) = K_1, \tag{10.81}$$

*and*

$$\deg(P_m|\widehat{Q}_b) = \deg(P_m|\widehat{Q}_{b+1}) = 2 \tag{10.82}$$

*for the isolating proper restrictions, equivalently on their ideal ends. Thus KKL's printed degree-one clause is false for the corrected canonical component; no identification of $Q_0$ with the undefined printed $K_0^0$ is made.*

*Proof.* Repeated component composition gives $Q_r \in \mathrm{Comp}\, F^{-r}(Q_0)$. The central restriction $F : Q_0 \to K_1$ contains the unique simple critical point and has degree two, exhausting the total quadratic degree; hence $F^{-1}(K_1) = Q_0$. The target $K_2$ avoids the critical value in the disjoint member $K_1$, so $K_1$ is a univalent component of $F^{-1}(K_2)$. Therefore

$$Q_{b+1} \in \mathrm{Comp}\, F^{-m}(K_1), \qquad Q_b \in \mathrm{Comp}\, F^{-m}(K_2).$$

Lemma 12.86 makes their graphs whole components. A connected component over one member of the finite pairwise-disjoint primitive cycle cannot meet a component over another, proving distinctness and (10.81).

For $s \geq 1$, the selected map $F : Q_s \to Q_{s-1}$ has degree one. For $s = 1$, its target $Q_0$ is disjoint from the critical value in $K_1$; for $s \geq 2$, its target lies in the critical limb while the critical value lies in the disjoint rotated limb. A Jordan target avoiding that value has only univalent

inverse components. Each displayed final itinerary consists of these univalent pulls followed by the degree-two central step, and, for $\widehat{Q}_b$, one further univalent cycle step. Multiplication gives (10.82); the analytic graph transfers this degree to the final ideal ends. □

**Conditional Theorem 12.89** (Correct stationary width charge)**.** *Suppose the final immersion restriction contains a measurable lamination $B \subset I_m^* A_m$ such that every source leaf contributes a retained proper leaf of $B$, every such leaf joins $\widehat{Q}_{b+1}$ to $\mathcal{K}^m \setminus \widehat{Q}_{b+1}$, and the declared prime-end landing survives KKL's marking filter. Then*

$$\mathcal{W}(A_m) \le \mathcal{W}(B) \le 2\mathcal{W}(C) + 4 \le 4\mathcal{W}(\mathcal{R}F) + 4. \tag{10.83}$$

*Proof.* The first inequality is KKL domination because every source leaf has a retained leaf in $B$. The relevant final end has degree two by proposition 12.88, so KKL's pushforward-lamination lemma gives the second inequality with $d = 2$. Their terminal comparison $\mathcal{W}(C) \le 2\mathcal{W}(\mathcal{R}F)$ gives the third. □

**Open Problem 12.90** (Stationary final path extraction and ordered restriction)**.** Prove first that KKL's final pass-through predicate yields the measurable lamination in conditional theorem 12.89, including one retained occurrence for every source leaf, prime-end landing, survival of the subordinate/non-marked restriction filter, and the non-locally-connected rational-lamination model. This is `KKL_STATIONARY_FINAL_PASS_THROUGH_PATH_EXTRACTION`.

Separately prove that the two source predicates refine occurrence by occurrence to $\widehat{Q}_b, \widehat{Q}_{b+1}$ in one connected restriction or one source-faithful ordered restriction multicurve. Prove their order, uniform retained width, and one authorized KL puzzle certificate. Until these clauses hold, neither the native stationary proof nor the stronger fixed-pair route is complete.

*Warning* 12.91 (A terminal crest is not a KL endpoint). KKL define a Wave over $I_{1\to 0}$ by placing that side in the crest, not at either end. At a KL vertex $v_i$, the two half-accesses are $\eta_i$ and $\xi_{i+1}$ in adjacent tiles, whereas KL's one-tile family $\mathcal{C}_{A_i}$ joins $\xi_i$ to $\eta_i$. These types cannot be identified. In a punctured sector, curves joining the two adjacent arcs can concentrate at their common vertex; logarithmic coordinates give extremal length 0, hence infinite width. A nondegenerate separating scale is indispensable.

**Conditional Theorem 12.92** (Crest-to-KL skipping width)**.** *Let $S_0$ be a KKL subwave and $\mathcal{H}$ a complete admissible immersion-restriction lamination retaining every occurrence and ordered component. Suppose fixed data $Y, Z, v, w$, with $v \ne w$, satisfy: $Z \subset Y$ has depth $m$ and separates $v$ from $w$; every leaf of $\mathcal{H}$ literally contains an ordered submulticurve lifting to KL's horizontal family $\mathcal{T}_{Y,Z}(v,w)$; and one lifted component skips $K_Z$. Then*

$$\mathcal{W}(S_0) \le \mathcal{W}(\mathcal{H}) \le \mathcal{W}(\widehat{\mathcal{H}}) \le \mathcal{W}(\mathcal{T}_{Y,Z}(v,w)) \le \frac{2^{2m+q+2}}{\mu}. \tag{10.84}$$

*Proof.* KKL domination gives the first inequality. The ordered-overflow lemma gives the second, and family monotonicity gives the third. KL Corollary 'separating pieces' gives $\mathcal{L}(\mathcal{T}_{Y,Z}(v,w)) \ge \mu/2^{2m+q+2}$; inversion gives the last inequality. The first three coefficients are exactly one. The last estimate is not uniform when $m$ or $q$ grows. □

**Open Problem 12.93** (Occurrence-level crest-to-skipping binder)**.** For every retained terminal Wave occurrence, construct a literal ordered restriction, never an appended connector, with fixed distinct endpoint continua $S_Y^v, S_Y^w$, a fixed separating piece $Z$, a lifted horizontal KL multicurve, and its componentwise skip witness. Retain the two adjacent terminal half-access occurrences, tile and deck indices, order, multiplicity, obstacle filters, and final-root behavior. This width-only binder does not establish equality of the original KKL and KL prime ends.

*Warning* 12.94 (A deepest relation path is not a sequential occurrence record). KKL's immersion restriction requires only

$$\iota(|\gamma_{r+1}|) \subset |f\gamma_r|. \tag{10.85}$$

The inclusion can be strict. Consequently the full proper paths at all stages need not satisfy one common-parameter equalizer equation, and a deepest fiber-product path retains only cumulative truncations of the earlier paths.

This occurs in an exact unbranched degree-two relation. For $0 < \rho < 1$, put

$$X_0 = A(\rho, 1), \qquad X_1 = A(\sqrt{\rho}, 1), \qquad \iota(z) = z, \qquad f(z) = z^2,$$

and

$$\beta(t) = \rho^{1-t}, \qquad \gamma(t) = \rho^{(1-t)/2}, \qquad a(t) = \frac{1+t}{2}.$$

Then $f\gamma = \beta$, $\iota\gamma = \beta a$, and both full paths are proper. But

$$\gamma a(t) = \rho^{(1-t)/4} \longrightarrow \rho^{1/4} \quad (t \downarrow 0),$$

an interior point of $X_1$. Thus the retained earlier coordinate is not proper and cannot certify the discarded endpoint, marking, or filter data. The exact common-parameter reconstruction theorem remains valid under equality of the full paths; it is only a strict special case.

**Proved Theorem 12.95** (Segmented occurrence reconstruction)**.** *Let*

$$X_m = \{(z_1, \dots, z_m) \in X_1^m : f(z_r) = \iota(z_{r+1}) \text{ for } 1 \le r < m\}$$

*be a finite Kahn relation tower. For $1 \le r \le m$, let*

$$\gamma_r : J_r \to X_1, \qquad \beta_r = f\gamma_r$$

*be full oriented stage paths. Suppose that for $2 \le r \le m$ there is an orientation-preserving occurrence map*

$$a_{r-1} : J_r \to J_{r-1}, \qquad \beta_{r-1}a_{r-1} = \iota\gamma_r. \tag{10.86}$$

*Write $a_{r|m} = a_r \circ \cdots \circ a_{m-1}$, with $a_{m|m} = \mathrm{id}$. Then*

$$\Theta_m(t) = (\gamma_1 a_{1|m}(t), \dots, \gamma_{m-1}a_{m-1}(t), \gamma_m(t)) \tag{10.87}$$

*is the unique path in $X_m$ having these stage coordinates. Its prefix and terminal projections satisfy*

$$q_{m,r}\Theta_m = \gamma_r a_{r|m}, \qquad i_{m-1}\Theta_m = \Theta_{m-1}a_{m-1}, \tag{10.88}$$

$$I_m\Theta_m = \beta_0 a_{0|m}, \qquad P_m\Theta_m = \beta_m. \tag{10.89}$$

*Conversely, a compatible flag of prefix paths with the declared full terminal projections uniquely recovers the stage paths and occurrence maps, modulo orientation-preserving reparameterization.*

*Proof.* For $r < m$, associativity of the cumulative windows and (10.86) give

$$f\gamma_r a_{r|m} = \beta_r a_{r|m} = \beta_r a_r a_{r+1|m} = \iota\gamma_{r+1}a_{r+1|m}.$$

Hence every adjacent pair in (10.87) satisfies the defining equation of $X_m$. The coordinate identities in (10.88) are immediate. Applying $\iota$ to the first coordinate and $f$ to the last gives (10.89).

For the converse, the full terminal projection of the $r$-th prefix is $\gamma_r$. Prefix compatibility reads $\Theta_r \circ a_r = i_r\Theta_{r+1}$; projecting to its last coordinate recovers $a_r$ and (10.86). The fiber-product equations then force every earlier coordinate by backward induction. Thus the two constructions are inverse. No assertion here supplies proper remote landing or survival in a KKL marking. □

**Conditional Theorem 12.96** (Strengthened quantitative defect suffices)**.** *Put*

$$V(F) = \mathcal{W}(\mathcal{U} \setminus \mathcal{T}), \qquad R(F) = \mathcal{W}(\mathcal{R}F), \qquad N = b + 3, \qquad m = N - 1.$$

*Suppose the last stationary subwave has a partition in KKL's native transverse width measure, modulo width zero,*

$$Y_m = G \sqcup D,$$

*such that the restricted transverse measure agrees with the proof-used lamination widths, $G$ has the segmented final incidence and proper measurable route realization required by conditional theorem 12.89, while*

$$\mathcal{W}(D) \le a_b V(F) + c_b R(F) + c_b. \tag{10.90}$$

*Then*

$$\mathcal{W}(Y^{N\text{-}nbr}) \le \mathcal{W}(Y^{\mathrm{sh}}) + a_b V(F) + (c_b + 4)R(F) + C_b. \tag{10.91}$$

*Assume additionally KKL's source-stated 'small breaking' proposition and either the native HTND scale or the absolute hypothesis*

$$\mathcal{W}(A) \ge C_{b,k,\lambda}(R(F) + 1). \tag{10.92}$$

*Then every sufficiently wide $k$-controllable Wave is either strictly $\lambda$-short or $(\delta, \eta)$-terminal, where $\delta > 0$ is independent of $k$ and $\eta$ is universal.*

*Proof.* Modulus subadditivity and the corrected degree-two estimate give

$$\mathcal{W}(Y_m) \le \mathcal{W}(G) + \mathcal{W}(D) \le a_b V(F) + (c_b + 4)R(F) + (c_b + 4).$$

Substitution in KKL's stationary telescoping estimate proves (10.91). The other Wave types in the source decomposition give the corresponding weak estimate.

For the source-level implication, put

$$B = A^{N\text{-br}} \cup A^{\mathrm{sh}}.$$

Partition $A$, in its native transverse measure, into $B$ and the nonshort $N$-unbreakable $X, Y, Z_j$ classes. The short term in the stationary $Y$-estimate then vanishes, so

$$\mathcal{W}(A) \le \mathcal{W}(B) + a_b V(F) + c_b R(F) + C_b(k + 1). \tag{10.93}$$

Choose $\lambda < \lambda_3 < 1$. Apply KKL 'small breaking' with $n = N$, $k_0 = 12b$, and parameter $\lambda_3$, and denote its constants by $\delta_3 > 0$ and universal $\eta_3$. Choose

$$\max\left\{\frac{\lambda}{\lambda_3}, \frac{1}{1 + \delta_3}\right\} < \theta < 1 \tag{10.94}$$

and set $\varepsilon_\theta = (1 - \theta)/2$. If $a_b > 0$, put $\delta_v = \min\{1/2, \varepsilon_\theta / a_b\}$; otherwise put $\delta_v = 1/2$.

If $V(F) > \delta_v \mathcal{W}(A)$, the vertical terminal alternative holds with any fixed universal second parameter. Otherwise $a_b V(F) \le \varepsilon_\theta \mathcal{W}(A)$. By (10.92), enlarge the scoped threshold until

$$c_b R(F) + C_b(k + 1) \le \varepsilon_\theta \mathcal{W}(A).$$

Equation (10.93) now gives

$$\mathcal{W}(B) \ge \theta \mathcal{W}(A). \tag{10.95}$$

KKL source lines 1375–1387 show that $B^{N\text{-nbr}}$ is short, hence $12b$-controllable. Apply 'small breaking' to $B$. If $B$ is $(\delta_3, \eta_3)$-terminal, its vertical alternative transfers with factor $\delta_3\theta$, while its wider-Wave alternative transfers with factor $(1 + \delta_3)\theta - 1$. Thus $A$ is $(\delta_A, \eta_3)$-terminal for

$$\delta_A = \min\{\delta_3\theta, (1 + \delta_3)\theta - 1\} > 0. \tag{10.96}$$

If instead $B$ is $(N, \lambda_3)$-unbreakable, then $B^{N\text{-nbr}} \subset A^{\mathrm{sh}}$, because no leaf in $A^{N\text{-br}}$ survives $N$ steps. Hence

$$\mathcal{W}(A^{\mathrm{sh}}) \ge \lambda_3 \mathcal{W}(B) \ge \lambda_3 \theta \mathcal{W}(A) > \lambda \mathcal{W}(A).$$

This is strict KKL shortness. The theorem is conditional on (10.90), its transverse-measure typing, and (10.92); those analytic hypotheses remain open. □

**Proved Proposition 12.97** (Block-dependent continuation-exit charge)**.** *Fix a finite Kahn horizon $h$. Inside a neighborhood on which the complete immersion $I_h$ is embedded, choose Jordan disks*

$$K^h \subset O_h^- \Subset O_h^+$$

*and put*

$$Q_h = I_h(O_h^+ \setminus \overline{O_h^-}), \qquad \mu_h = \operatorname{mod} Q_h > 0.$$

*If $E_h$ is a measurable family of selected continuation germs which start inside $I_h(O_h^-)$, first exit $I_h(O_h^+)$ before their declared remote ends, and admit a measurable first-exit selector, then*

$$\mathcal{W}(E_h) \le \frac{1}{\mu_h}. \tag{10.97}$$

*Proof.* The selected segment from its last inner-boundary crossing to its first outer exit contains a crosscut of $Q_h$. Ordered-subcurve overflow therefore bounds $\mathcal{W}(E_h)$ by the width of the annular crossing family, which is $\mu_h^{-1}$. □

**Proved Theorem 12.98** (Qualitative collar data do not imply a vertical charge)**.** *Fix $0 < a < 1$. For $1 < R < 2$, put*

$$U_R = \mathbb{D}_R, \qquad U_R' = \mathbb{D}_{\sqrt{R}}, \qquad K = \overline{\mathbb{D}}, \qquad T = [-a, a], \qquad f_R(z) = z^2, \qquad \iota_R(z) = z.$$

*This is a degree-two pseudo-polynomial-like map with globally embedded immersion and forward-invariant full tree $T$. Its vertical ring width is uniformly bounded as $R \downarrow 1$, but every horizon-$h$ embedded collar around $K$ satisfies*

$$\frac{1}{\operatorname{mod} Q_h} \ge \frac{2\pi\, 2^h}{\log R} \longrightarrow \infty. \tag{10.98}$$

*Thus no vertical-width-linear collar charge follows from the listed qualitative hypotheses. The model is not a native periodically repelling primitive KKL block and therefore does not refute the intended native theorem.*

*Proof.* The common filled preimage is $K$, and the finite relation domains are $U_R^h = \mathbb{D}_{R^{1/2^h}}$, with complete immersion equal to inclusion. Put $m_a = \operatorname{mod}(\mathbb{D} \setminus [-a, a]) > 0$. Ring monotonicity gives

$$\mathcal{W}(U_R \setminus T) = \operatorname{mod}(U_R \setminus T)^{-1} \le m_a^{-1}.$$

For every $K \subset O^- \Subset O^+ \Subset U_R^h$,

$$\operatorname{mod}(O^+ \setminus \overline{O^-}) \le \operatorname{mod}(U_R^h \setminus K) = \frac{\log R}{2\pi\, 2^h}.$$

Taking reciprocals proves (10.98). Here $z^2$ is superattracting and $T$ is not asserted to be a native primitive Hubbard tree, which gives the stated qualification. □

**Open Problem 12.99** (Uniform stationary collar calibration)**.** Define the optimized compatible collar conductance

$$\mathfrak{C}^{\mathrm{comp}}_{\le b+2}(F) = \inf_{\mathscr{Q}\ \text{compatible}} \max_{1 \le h \le b+2} \frac{1}{\operatorname{mod} Q_h}. \tag{10.99}$$

Either prove

$$\mathfrak{C}^{\mathrm{comp}}_{\le b+2}(F) \le C_b(\mathcal{W}(\mathcal{U} \setminus \mathcal{T}) + 1) \tag{10.100}$$

with measurable source first-exit selection, or prove the weaker route-localized estimate

$$\sum_{h=1}^{b+2} \mathfrak{E}_h(F) \le C_b(\mathcal{W}(\mathcal{U} \setminus \mathcal{T}) + 1), \tag{10.101}$$

where $\mathfrak{E}_h$ is the width of the fully marked measurable first-failure continuation family. The constants must be independent of satellite period, rotation denominator, and the individual block. Qualitative embedding and Kahn's response in the larger full width $\mathcal{W}(F)$ prove neither estimate. The full Quantitative Stationary Defect Bound must also charge nonlanding, marking/filter loss, incompatible windows, degree-two collisions, and measurable-incidence failures. None of these open clauses is promoted.

12.20. **Post-freeze QA correction of the stationary frontier.** An independent line-by-line and extremal-width audit changes the active stationary frontier. It proves no global closure theorem.

**Proved Theorem 12.100** (Cyclic-cover obstruction to the broad Wave claim)**.** *Under the broad printed definition of a KKL Wave, the coefficient-one estimate*

$$\mathcal{W}(A') \le \mathcal{W}(A) + O(1 + \log d), \qquad A \subset f_* A', \tag{10.122}$$

*is false without an additional coherence or nonrepetition hypothesis.*

*Proof.* Take a finite cyclic annulus cover and $N$ disjoint lifts of one proper marked target Wave of width $w$. If necessary, add one smooth outer leaf of width zero to force one connected source base. The finite Parallel Law gives

$$\mathcal{W}(A') = Nw,$$

while all positive-width lift components have the same set-theoretic pushforward, so every target Wave $A \subset f_* A'$ has $\mathcal{W}(A) \le w$. The covering degree is at most the fixed integer $M = N + 2$. Thus (10.122) would imply

$$(N-1)w \le C(1 + \log M),$$

which fails for $N = 2$ and $w \to \infty$. □

The theorem refutes the candidate full-base collision reduction and the literal broad wording of the KKL Wave lemma under the printed definitions. It does not refute the cited Dudko–Lyubich annular theorem, whose source family is one Wave above one fixed boundary interval. A usable coefficient-one theorem must prove single-crest/single-WAD-coordinate coherence for the actual source packet. Otherwise the state must retain the equivalence classes of source conductance coordinates that collapse downstairs and charge their multiplicity linearly.

12.20.1. *Closure of the stationary relative-continuation gate.*

**Proved Theorem 12.101** (Borel leafwise-directed stationary continuation)**.** *Fix $b \ge 1$, put $N = b + 3$, and let $F$ be a KKL parabolically $b$-bounded, periodically repelling, primitively-renormalizable pseudo-quadratic-like relation. Let $A$ be a proof-used proper KKL Wave ending at $BL_0$. There are nested proper Borel subwaves*

$$A_0 \supset A_1 \supset \cdots \supset A_{N-1}$$

*with*

$$A_0 = A^{N\text{-}nbr} \setminus A^{\mathrm{sh}}, \qquad S_n := F_*^n A_n,$$

*whose successive output Waves $S_n$ retain the required no-short, unbreakability, occurrence, prime-end marking, and leafwise direction data. There is $C_b < \infty$, independent of the rotation denominator, the input Wave width, the number of represented arc packets, and all selector histories, such that*

$$\mathcal{W}(A_n \setminus A_{n+1}) \le C_b \qquad (0 \le n < N - 1). \tag{10.123}$$

*Proof synopsis.* The complete line-by-line proof, occurrence ledger, source hashes, and independent backward audits are the source-locked Gate-N1 companion documents identified by the generated machine ledger. We record the dependency-complete argument here.

The native Dudko–Lyubich annular argument is first applied only to the rectifiable core. Indeed, the pre-pruning standard-Borel Wave atlas makes Euclidean total variation in a finite-area round-annulus model a Borel leaf function. Thus the input packet splits Borelly as

$$\mathscr{S} = \mathscr{S}_{\mathrm{rec}} \dot\cup \mathscr{N}_\infty, \qquad \mathcal{W}(\mathscr{N}_\infty) = 0.$$

The last equality follows because every constant density $\rho_\varepsilon = \varepsilon$ is admissible for every infinite-length leaf and has area tending to zero.

On $\mathscr{S}_{\mathrm{rec}}$, the proof of the native annular pruning lemma partitions the deleted family into $T = O(\log m)$ nonzero deck-span packets of $O(1)$ width and one final $O(1)$ outer buffer. The Parallel Law gives

$$\mathcal{W}(\mathscr{S} \setminus \mathscr{S}^{\mathrm{new}}) \le C(1 + \log m).$$

At the proof-used degree $m = 2$, this charge is absolute. This is a bound on the deleted family itself; it is not obtained by subtracting aggregate retained widths.

The deleted family need not have a Borel leaf predicate. Choose an admissible density for it and dominate that density pointwise by a strictly positive lower-semicontinuous density with arbitrarily small area overhead. Weighted path length is lower semicontinuous in the compact-open path topology. Its length-at-least-one leaf set is therefore a Borel super-discard $E$ satisfying

$$\mathcal{W}(E) \le C_{\mathrm{DL}} + 1.$$

The complementary retained family is Borel and remains inside the set-theoretically injective DL family.

The ordered-Wave realization supplies the filtered occurrence code, complete prime-end markings, and full source saturation. KKL Proposition A.14 is used singleton-by-singleton to obtain leafwise direction. The aggregate target is typed as a Wave because it is already a proper Borel lamination subordinate to one whole allowed target boundary component; no aggregate crest and no transported source base are asserted.

Write $S_n = F_*^n A_n$, and let $\Phi_n : A_n \to S_n$ be the Borel unique-untagged-output map supplied on the unbreakable packet. If $P_{C,n} \subset S_n$ is the retained central packet, define

$$A_{n+1} = A_n \cap \Phi_n^{-1}(P_{C,n}).$$

Full saturation gives

$$\Phi_n(A_n \setminus A_{n+1}) \subset S_n \setminus P_{C,n}.$$

Sequential use of the exact degree-two full-lift identity, retaining all colliding source occurrences, now yields

$$\begin{aligned} \mathcal{W}(A_n \setminus A_{n+1}) &\le 2^n \mathcal{W}(S_n \setminus P_{C,n}) \\ &\le 2^n c_b^{\mathrm{step}} \le 2^{b+1} c_b^{\mathrm{step}}. \end{aligned}$$

Together with the bounded initial loss this proves (10.123) with

$$C_b = \max\{c_b^{\mathrm{init}}, 2^{b+1} c_b^{\mathrm{step}}\}.$$

The last attempted transition is $n = N - 2$, so the source retains the two steps and the output the one step required by the final precritical estimate. No $A_N$, line-1603 alignment, or later gate is used. □

The exact frozen proof and backward audits are recorded in the source-locked project proof ledger and its Gate-N1 integration audit; the preceding manuscript proof is an explicit dependency synopsis, not a replacement for those line-by-line companions. The separate measurable-support/additivity rederivation is not a dependency: the source phrase "genuine subwave" does not explicitly source-lock measurable retained support.

12.20.2. *The first pointwise two-path boundary.*

**Proved Lemma 12.102** (Fixed host-component occurrence window)**.** *Let $\alpha : (0,1) \to Y$ be embedded, let $I : X \to Y$ be a local homeomorphism, and fix a connected component $C$ of $I^{-1}(|\alpha|)$. Then there are unique $0 \le a_C < c_C \le 1$ and a homeomorphism*

$$t_C : C \longrightarrow (a_C, c_C), \qquad I|_C = \alpha \circ t_C. \tag{10.125}$$

*Now orient a proof-used KKL leaf from its $BL_0$ endpoint, and fix one marked $L_0^h$ contact occurrence $\xi = (u, e_{\mathrm{deep}})$, $0 < u < 1$. Suppose independently that $C$ is an embedded component-maximal proper restriction, that the lifted incoming germ $\alpha((u-\epsilon, u))$ belongs to $C$, and that the $c_C$-end lands at $e_{\mathrm{deep}}$ with $c_C = u$. Then $C$ realizes the complete incoming portion $BL_0 \rightsquigarrow L_0^h$ exactly when $a_C = 0$ and its other end is the actual marked lift of the target $BL_0$ endpoint access.*

*Proof.* The map

$$t_C = \alpha^{-1} \circ I|_C : C \longrightarrow (0,1)$$

is a local homeomorphism. Thus $C$ is a connected second-countable one-manifold without boundary. It cannot be circular, since its image would then be a nonempty compact open subset of $(0,1)$. After orienting the line-type component, the local orientation sign of $t_C$ is constant. A finite ordered subdivision of every compact coordinate interval shows that $t_C$ is globally strictly monotone. It is therefore injective; local openness gives the open interval image $(a_C, c_C)$, and its inverse is continuous. This proves (10.125).

Under the additional KKL hypotheses, the incoming target window is $(0, u)$ and the host window is $(a_C, u)$. They are equal exactly when $a_C = 0$. Window equality does not identify a prime end, so the actual marked lift of the $BL_0$ endpoint is a separate necessary and sufficient clause. No assertion at source lines 1598 or 1600 is used. □

Properness of an arbitrary path contained in $C$ does not make it component-maximal or embedded. For example, with $X = Y = \mathbb{D}$, $I = \mathrm{id}$, $\alpha(t) = i(2t-1)$, and

$$\theta^*(t) = \alpha\Big(t + \tfrac{1}{10}\sin(4\pi t)\Big),$$

$\theta^*$ is proper but retraces because the inner derivative is negative at points where $\cos(4\pi t) = -1$. Also, a host ending at the deep contact may have $u = a_C$, the outgoing incidence, rather than $u = c_C$; reorientation of $C$ cannot change which side of the oriented target leaf it covers.

**Open Problem 12.103** (Incoming-window anchor)**.** Within a stationary block whose primitive renormalization is parabolically $b$-bounded, has rotation $1/q$, and satisfies the explicit high-type inequality $q > b$, take $n = b, b+1$, $h = n+1$, and every actual $\alpha \in A_h$ used at source line 1603, first select a complete contact occurrence $\xi = (u, e_{L_0^h})$ and its incoming one-sided germ. Construct its connected host $C$, exhaust or canonically distinguish every other host meeting the same coarse locus, and prove:

$$C \text{ is embedded, component-maximal, and proper at both ends,}$$

$$0 < u < 1, \qquad c_C = u, \qquad a_C = 0,$$

the $a_C$-end is the actual marked lift of the target $BL_0$ endpoint access.

The source currently supplies none of these clauses for one common restriction. A failure record must distinguish: no typed contact component; wrong one-sided incidence; retracing or nonmaximality; nonlanding or noninterior deep end; $a_C > 0$; and $a_C = 0$ with the wrong $BL_0$ lift. Direct domain exit, removed-tree incidence, and wrong ideal landing are correctly typed here; sequential sheet and filter words belong to the later two-occurrence binder.

In the locally connected case, the finite tree below is the combinatorial quotient of the physical Hubbard carrier. In the non-locally-connected case it is KKL's locally connected rational- lamination model tree. In either case the marked tree is the connected hull of the small-Julia vertices. We do not infer inclusion-minimality of the physical lift in the latter case; actual contact with a lifted small-Julia continuum remains a separate predicate.

**Proved Lemma 12.104** (Marked-tree terminal pruning)**.** *Let $G$ be a finite tree and let $M \subset V(G)$ satisfy $|G| = \mathrm{Hull}_{|G|}(M)$. For every $x \in |G|$, each component of $|G| \setminus \{x\}$ contains a vertex of $M$.*

*Proof.* If a component $C$ contained no marked vertex, then $|G| \setminus C$ would be a proper closed connected subtree containing $M$: the unique arc between two points outside $C$ cannot enter and leave $C$ without passing twice through $x$. This contradicts the hull hypothesis. □

The terminal-side hypothesis is essential. In the interval tree $[0, 3]$, marked only at $0, 3$, the connected middle subsegment $[1, 2]$ meets the tree but contains no mark. Thus a nonempty edge-only $L_0^b \cap T$ is not eliminated by minimality of the Hubbard carrier.

**Proved Lemma 12.105** (Critical-limb orbit distinctness)**.** *For KKL's primitive small-Julia cycle of period $P$ and rotation number $1/q$,*

$$P \geq q.$$

*Consequently, in the explicit high-type range $q > b$, the components $K, F(K), \ldots, F^b(K)$ are distinct for every $K \in \mathrm{Comp}\,\mathcal{K}$.*

*Proof.* Let $K_0$ contain the critical point. It is connected, avoids $\alpha_F$, and contains a point of $\mathfrak{L}_0$, so $K_0 \subset \mathfrak{L}_0$. Then $K_1 = F(K_0)$ contains the critical value, which lies in $\mathfrak{L}_1$. No $K_j$ contains $\alpha_F$, since its fixedness would put it in every pairwise-disjoint primitive component. Thus $K_1 \subset \mathfrak{L}_1$. The homeomorphisms $F : \mathfrak{L}_j \to \mathfrak{L}_{j+1}$, $j \neq 0$, give $K_j \subset \mathfrak{L}_j$ for $1 \leq j < q$. Hence the first $q$ cycle components are distinct and $P \geq q$. Every cycle component has period $P$, proving the consequence. □

Put $H = b + 2$. Write $\mathrm{PSC}_H$ for a preliminary complete-horizon chain

$$L_0^0, L_0^1, \ldots, L_0^H; \quad w_0, w_1, \ldots, w_H,$$

where $L_0^0 = T \cap \mathfrak{L}_0$, each $L_0^n$ is one recorded connected component of the full set $F^{-1}(L_0^{n-1})$, contained in $\mathfrak{L}_0$, with

$$F(L_0^n) \subset L_0^{n-1}.$$

Here $w_0 = ()$, $w_n = (L_0^1, \ldots, L_0^n)$, and $\mathrm{par}(w_n) = w_{n-1}$ for $n \geq 1$; equality of addresses means equality of the ordered actual connected subsets. Write separately

$$\mathrm{APC}_H = (\alpha^0 = \alpha_F, \alpha^1, \ldots, \alpha^H; v_0, v_1, \ldots, v_H), \qquad F(\alpha^n) = \alpha^{n-1},$$

where $v_0 = ()$, $v_n = (\alpha^1, \ldots, \alpha^n)$, and parent is truncation. Inverse sets are taken inside $K(F)$ under KKL's induced map $F = f \circ \iota^{-1}$.

**Proved Lemma 12.106** (Critical-limb inverse branch)**.** *Assume $q \geq 2$, as in the active high-type regime, and let*

$$D_0 = F^{-1}(\mathfrak{L}_0) \cap \mathfrak{L}_0.$$

*Then $F : D_0 \to \mathfrak{L}_0$ is a homeomorphism. Consequently* $\mathrm{PSC}_H$ *and* $\mathrm{APC}_H$ *exist through the complete horizon. If the first inverse of $L_0^0$ has several connected components, this preliminary construction does not identify the component met by the proof-used Wave.*

*Proof.* The noncritical limb map $F : \mathfrak{L}_{q-1} \to \mathfrak{L}_0$ is a homeomorphism. Every $y \in \mathfrak{L}_0$ is different from the critical value, which lies in $\mathfrak{L}_1$. Since $F : K(F) \to K(F)$ is degree two, $y$ has two distinct preimages. One is in $\mathfrak{L}_{q-1}$, while $F(\mathfrak{L}_0) = K(F)$ supplies one in $\mathfrak{L}_0$; these exhaust the two. The latter preimage is noncritical. Hence $F : D_0 \to \mathfrak{L}_0$ is a bijective local homeomorphism: the finitely many limbs make $\mathfrak{L}_0$ open in $K(F) \setminus \{\alpha_F\}$, and the restriction lies in the noncritical locus. A bijective local homeomorphism is open, hence a homeomorphism; denote its inverse by $\phi_0$.

Choose one connected component of $\phi_0(L_0^0)$ as $L_0^1$. Because $F^{-1}(L_0^0)$ omits the fixed point $\alpha_F$, that component cannot join a component in another limb. It is therefore a component of the full inverse, and $F(L_0^1) \subset L_0^0$. For $n \geq 2$, set

$$L_0^n = \phi_0(L_0^{n-1}).$$

The connected target then gives $F(L_0^n) = L_0^{n-1}$. This constructs the preliminary $\mathrm{PSC}_H$ and its actual-component address.

The equality $F(\mathfrak{L}_0) = K(F)$ supplies a preimage $\alpha^1 \in \mathfrak{L}_0$ of $\alpha_F$. It is unique because the fixed point $\alpha_F$ is the other, noncritical preimage. Define $\alpha^n = \phi_0(\alpha^{n-1})$ for $n \geq 2$, obtaining $\mathrm{APC}_H$. □

**Proved Lemma 12.107** (Canonical small-Julia predecessor selection)**.** *Let $Q_0$ be the critical member of the primitive small-Julia cycle and set $Q_n = \phi_0(Q_{n-1})$. Then $Q_n$ is a whole connected component of $F^{-1}(Q_{n-1})$, and $F : Q_n \to Q_{n-1}$ is a homeomorphism. Selecting $L_0^1$ as the unique component of $\phi_0(L_0^0)$ containing $Q_1$, and then using $L_0^n = \phi_0(L_0^{n-1})$, gives the unique $Q_0$-selected certificate* $\mathrm{SC}_H$ *with $Q_n \subset L_0^n$ through $H$. Henceforth* $\mathrm{SC}_H$ *denotes only this canonical chain;* $\mathrm{PSC}_H$ *remains the arbitrary preliminary chain.*

*Proof.* The continuum $Q_0$ contains the critical point, avoids $\alpha_F$, and lies in the cycle contained in $T$; hence $Q_0 \subset L_0^0$. The inverse-branch homeomorphism makes every $Q_n$ compact, connected, and univalently mapped to $Q_{n-1}$. The full inverse omits $\alpha_F$, so a connected component cannot cross limbs, and uniqueness of $\phi_0$ makes $Q_n$ the whole critical-limb component. Now $Q_1 \subset \phi_0(L_0^0)$, so its connected component uniquely selects $L_0^1$; induction gives $Q_n \subset L_0^n$ and uniqueness thereafter. □

This corrected canonical chain does not identify the undefined printed $K_0^0$ by fiat. The open full-carrier predicate $\mathrm{KSEP}_H$ must separately prove that $Q_0$ separates $\alpha^0$ from $L_0^1$, propagate separation by $Q_n$ without allowing routes outside the selected inverse domain, and establish the proof-used collar/access compatibility. Likewise $\mathrm{PUS}_H$ is only the conditional assertion that an independently supplied proper restriction following the declared $Q$-itinerary lands in the canonical $L_0^n$; it constructs no occurrence or host.

**Proved Lemma 12.108** (Selected predecessor avoids the primitive cycle)**.** *For the constructed* $\mathrm{APC}_H$*,*

$$\alpha^n \notin \mathcal{K}, \qquad 0 \leq n \leq H.$$

*Proof.* If $\alpha^n \in K_i$, then $F^n(\alpha^n) = \alpha_F \in K_{i+n}$. Since $F(\alpha_F) = \alpha_F$ and $F(K_j) = K_{j+1}$, the fixed point would belong successively to every member of the pairwise-disjoint primitive cycle, impossible for $P > 1$. □

This excludes the selected point from the cycle. It does not prove that this point is the attachment gate of $L_0^n$.

**Proved Lemma 12.109** (Selected-predecessor continuum saturation)**.** *For the constructed* $\mathrm{SC}_H$*, every cycle component* $K_i$ *and* $0 \le n \le H$ *satisfy*

$$K_i \cap L_0^n \neq \varnothing \quad \Longrightarrow \quad K_i \subset L_0^n.$$

*Proof.* No $K_i$ contains $\alpha_F$: fixedness would put $\alpha_F$ in both $K_i$ and $F(K_i) = K_{i+1}$, contradicting the pairwise disjointness of the primitive cycle. At $n = 0$, a connected $K_i \subset T \setminus \{\alpha_F\}$ that meets $\mathfrak{L}_0$ lies wholly in that component of $K(F) \setminus \{\alpha_F\}$, hence in $L_0^0 = T \cap \mathfrak{L}_0$.

Assume the assertion at $n-1$, and let $K_i$ meet $L_0^n$. Then $F(K_i)$ meets $L_0^{n-1}$, so the induction hypothesis gives

$$F(K_i) \subset L_0^{n-1}.$$

Thus $K_i \subset F^{-1}(L_0^{n-1})$. The continuum $K_i$ is connected and meets the selected connected component $L_0^n$ of this full inverse image, so it lies wholly in $L_0^n$. □

For a supplied $\mathrm{SC}_H$, define the minimal marked-core contact predicate

$$\mathrm{MCI}_b : \qquad L_0^b \cap T \neq \varnothing \Longrightarrow \exists K \in \operatorname{Comp} \mathcal{K}, \quad K \cap L_0^b \neq \varnothing.$$

The preceding lemma upgrades this contact to $K \subset L_0^b$.

Define $\mathrm{TMC}_b$, terminal marked-core incidence, to mean that whenever $L_0^b \cap T \neq \varnothing$, there are a recorded point $x_b$ and one component $C_b \in \pi_0(|G| \setminus \{x_b\})$ with $C_b \subset \pi_T(L_0^b \cap T)$. Pruning chooses $v_K \in C_b \cap M$, and this same vertex must be realized by $K \cap L_0^b \neq \varnothing$. Thus

$$\mathrm{TMC}_b \Longrightarrow \mathrm{MCI}_b.$$

The converse is not asserted: a different dynamical argument could produce actual contact without a complete terminal side. In the locally connected physical Hubbard-tree model, the marked quotient fiber is the corresponding continuum $K$. In a non-locally-connected prime-end or lamination lift, identification of the collapsed marked fiber with an actual $K$-contact is a separate analytic assertion.

**Proved Proposition 12.110** (Terminal critical-limb inverse sheets)**.** *Put*

$$\Omega_0 = \mathfrak{L}_0, \qquad \Omega_n = \phi_0^n(\mathfrak{L}_0).$$

*Without assuming local connectivity, for every* $n \ge 1$*,*

$$\overline{\Omega_n} = \Omega_n \cup \{\alpha^n\},$$

*and* $\Omega_n$ *is a connected component of* $K(F) \setminus \{\alpha^n\}$*.*

*Proof.* Set $D_0 = \phi_0(\mathfrak{L}_0)$. We first prove

$$\partial_{K(F)} D_0 = \{\alpha^1\}.$$

The fixed point $\alpha_F$ is noncritical. Suppose, for a contradiction, that $x_k \in D_0$ tends to $\alpha_F$, and put $y_k = F(x_k) \in \mathfrak{L}_0$. Let

$$z_k = (F|_{\mathfrak{L}_{q-1}})^{-1}(y_k).$$

Then $y_k \to \alpha_F$. Every accumulation point of $z_k$ belongs to $\overline{\mathfrak{L}_{q-1}} = \mathfrak{L}_{q-1} \cup \{\alpha_F\}$ and maps to $\alpha_F$. The other preimage $\alpha^1$ lies in $\mathfrak{L}_0$, so compactness gives $z_k \to \alpha_F$. For large $k$, the distinct points $x_k$ and $z_k$ lie in one injectivity neighborhood of $\alpha_F$, yet $F(x_k) = F(z_k)$, a contradiction. Hence $\alpha_F \notin \overline{D_0}$.

Let $x \in \partial D_0$, and choose $x_k \in D_0$ with $x_k \to x$. Continuity gives $F(x) \in \overline{\mathfrak{L}_0}$. If $F(x) \in \mathfrak{L}_0$, then continuity of the global inverse branch $\phi_0$ gives

$$x_k = \phi_0(F(x_k)) \longrightarrow \phi_0(F(x)),$$

and hence $x = \phi_0(F(x)) \in D_0$, contradicting $x \in \partial D_0$ because $D_0$ is open. Thus $F(x) \in \partial\mathfrak{L}_0 = \{\alpha_F\}$, and the only remaining preimage is $\alpha^1$. Moreover $F(\alpha^1) = \alpha_F \notin \mathfrak{L}_0$, so $\alpha^1 \notin D_0$. Conversely, if $y_k \in \mathfrak{L}_0$ tends to $\alpha_F$, every accumulation point of $\phi_0(y_k)$ is a preimage of $\alpha_F$, and the preceding exclusion forces it to be $\alpha^1$. This proves the boundary identity. Since $D_0$ is open and connected and is closed in $K(F) \setminus \{\alpha^1\}$, it is a component of that punctured continuum.

Induction now gives

$$\overline{\Omega_n} = \phi_0(\overline{\Omega_{n-1}}) = \Omega_n \cup \{\alpha^n\}.$$

For $n \geq 2$, the compact preceding closure lies wholly in $\mathfrak{L}_0$, where $\phi_0$ is a homeomorphism. Continuity and density give the inclusion from the image of the closure into the closure of the image; compactness makes that image closed and gives the reverse inclusion. Moreover,

$$F^n(\alpha^n) = \alpha^0 = \alpha_F \notin \mathfrak{L}_0, \qquad F^n(\Omega_n) = \mathfrak{L}_0,$$

so $\alpha^n \notin \Omega_n$. Each $\Omega_n$ is open and connected and is closed in $K(F) \setminus \{\alpha^n\}$, hence is the claimed punctured-set component. □

**Proved Theorem 12.111** (Locally connected terminal-sheet incidence)**.** *Assume $K(F)$ is locally connected and put $T_n = F^{-n}(T)$. For the $Q_0$-selected chain,*

$$L_0^n = T_n \cap \Omega_n.$$

*Consequently*

$$L_0^n \cap T \neq \varnothing \Longrightarrow \exists K_i \in \operatorname{Comp}\mathcal{K}, \quad K_i \subset L_0^n.$$

*Thus* $\mathrm{MCI}_b$ *is proved in the locally connected physical category.*

*Proof.* The periodically repelling filled continuum has empty interior by the same-filled-continuum quadratic-like restriction and the no-wandering-domain theorem. It is full; hence a Jordan curve in it would force the bounded Jordan domain into the filled set, a contradiction. Local connectivity therefore makes $K(F)$ a dendrite.

The carrier contains the critical cycle in at least $\mathfrak{L}_0$ and $\mathfrak{L}_1$, so it contains $\alpha_F$. Unique arcs in the dendrite show that $L_0^0 = T \cap \mathfrak{L}_0$ is connected.

Connectedness of $L_0^0$ and the selected recursion give $L_0^n = \phi_0^n(L_0^0)$. Since $F^n : \Omega_n \to \mathfrak{L}_0$ is a homeomorphism,

$$\begin{aligned} T_n \cap \Omega_n &= (F^n|_{\Omega_n})^{-1}(T \cap \mathfrak{L}_0) \\ &= \phi_0^n(L_0^0) = L_0^n. \end{aligned}$$

This $T_n$ is only the induced base-dynamical inverse inside $K(F)$; no ambient fiber-product equality is asserted here.

Finally $T \subset T_n$ by forward invariance. If $\alpha^n \notin T$, connectedness and a nonempty $T \cap \Omega_n$ give $T \subset \Omega_n$, hence $T \subset L_0^n$. If $\alpha^n \in T$, the component $C$ of $T \setminus \{\alpha^n\}$ meeting $L_0^n$ lies in $\Omega_n$, and therefore in $L_0^n$. Since $\alpha^n \notin K_i$ for every component $K_i$ of $\mathcal{K}$, any $K_i$ meeting $C$ lies wholly in $C$. If none met $C$, then $C$ is open in the dendrite $T$, so $T \setminus C$ is compact. It is connected because every $y \in T \setminus C$ joins to $\alpha^n$ by the unique arc in $T$, and that arc avoids $C$. Thus $T \setminus C$ would be a proper connected compact subtree containing all of $\mathcal{K}$, contrary to the source definition of $T$ as its least connected hull. Therefore some actual small-Julia continuum satisfies $K_i \subset C \subset L_0^n$. □

Let $\bar{\mathcal{K}} = \{\bar{K}_i\}$ denote the geometric model small-Julia continua. Only the subsequent finite-tree quotient collapses $\bar{K}_i$ to an abstract vertex $\bar{v}_i$. The identical argument in KKL's locally connected degree-two combinatorial model proves the correctly typed model incidence

$$L_{0,\mathrm{com}}^n \cap T_{\mathrm{com}} \neq \varnothing \Longrightarrow \exists i, \quad \bar{K}_i \subset L_{0,\mathrm{com}}^n.$$

The remaining physical non-locally-connected lift route is

$$\pi^{-1}(L_{0,\mathrm{com}}^b) = L_0^b, \qquad \pi(K_i) \subset \bar{K}_i. \tag{10.125a}$$

Together these inclusions force $K_i \subset L_0^b$. KKL's one-line instruction to lift the model objects does not state the selected full-preimage equality or the geometric continuum correspondence. This is a chosen sufficient package, not a minimality theorem. Given the selected full-preimage equality, a separately proved nonempty intersection $\pi(K_i) \cap \bar{K}_i \neq \varnothing$, together with continuum saturation, could also suffice.

**Proved Proposition 12.112** (Monotone full-lift criterion)**.** *Let barred symbols denote KKL's locally connected combinatorial model. Suppose, through $H = b + 2$, that:*

*(1) $K(F)_{\mathrm{com}}$ is a dendrite and $\pi : K(F) \to K(F)_{\mathrm{com}}$ is a continuous monotone surjective semiconjugacy;*
*(2) $\pi(\alpha^n) = \bar{\alpha}^n$ for $0 \le n \le H$, and $\pi^{-1}(\bar{\alpha}^0) = \{\alpha^0\}$;*
*(3) $T = \pi^{-1}(T_{\mathrm{com}})$;*
*(4) $\pi(c_F) = \bar{c}_F$, $\pi(Q_n) \subset \bar{\Omega}_n$ for $0 \le n \le H$, and $\pi(K_i) \subset \bar{K}_i$ for every marked small-Julia continuum and its corresponding geometric model continuum.*

*The last clause does not collapse $\bar{K}_i$ to the abstract finite-tree vertex $\bar{v}_i$; those are different quotient levels. Then*

$$\pi^{-1}(L^n_{0,\mathrm{com}}) = L^n_0, \qquad 0 \le n \le H.$$

*In particular,*

$$\bar{K}_i \subset L^n_{0,\mathrm{com}} \quad \Longrightarrow \quad K_i \subset L^n_0.$$

*Thus the proved model incidence implies physical marked-core incidence under this package.*

*Proof.* Put $E_n = \pi^{-1}(\bar{\alpha}^n)$. Monotonicity makes $E_n$ connected, while semiconjugacy gives

$$F^n(E_n) \subset \pi^{-1}(\bar{\alpha}^0) = \{\alpha^0\}.$$

The degree-two quadratic-like map is finite-to-one, so $F^{-n}(\alpha^0)$ is finite, of cardinality at most $2^n$. Thus $E_n$ is a connected subset of a finite set. Since it contains $\alpha^n$,

$$\pi^{-1}(\bar{\alpha}^n) = \{\alpha^n\}, \qquad 0 \le n \le H.$$

We first prove the component fact needed for the selected sheets. Let $\rho : X \to Y$ be a continuous monotone surjection from a compact continuum onto a dendrite, with $\rho^{-1}(y) = \{x\}$. For every compact subcontinuum $A \subset Y$, the inverse image $\rho^{-1}(A)$ is connected. Indeed, if it had compact separation $U \sqcup V$, connected point fibers would lie wholly in one side, and the disjoint compact sets $\rho(U), \rho(V)$ would separate $A$.

The same conclusion holds for every nonempty connected, possibly nonclosed, subset $A \subset Y$. Fix $a_0 \in A$. Connectedness in a dendrite forces every arc $[a_0, a]_Y$, $a \in A$, to lie in $A$; omitting an interior arc point would separate $A$. Therefore

$$A = \bigcup_{a \in A} [a_0, a]_Y.$$

The inverse images of these compact arcs are connected by the preceding case and share $\rho^{-1}(a_0)$, so their union $\rho^{-1}(A)$ is connected.

If $D$ is a component of $Y \setminus \{y\}$, fix $d_0 \in D$. The dendrite identity

$$D = \bigcup_{d \in D} [d_0, d]_Y$$

expresses $\rho^{-1}(D)$ as a union of connected inverse images of compact arcs, all containing $\rho^{-1}(d_0)$. Hence $\rho^{-1}(D)$ is connected and avoids $x$. The image of a component of $X \setminus \{x\}$ is connected and avoids $y$; maximality and surjectivity therefore give a bijection of punctured-set components, with inverse $D \mapsto \rho^{-1}(D)$.

Apply this fact at $\alpha^n \mapsto \bar{\alpha}^n$. The exact chain

$$Q_n \subset L_0^n \subset \Omega_n, \qquad \pi(Q_n) \subset \bar{\Omega}_n$$

shows that the component bijection sends the physical critical predecessor component to its model counterpart, while proposition 12.110 and its model counterpart show that those components are $\Omega_n$ and $\bar{\Omega}_n$. Thus

$$\pi^{-1}(\bar{\Omega}_n) = \Omega_n.$$

At depth zero, the full-lift and component identities give

$$L_0^0 = T \cap \mathfrak{L}_0 = \pi^{-1}(T_{\mathrm{com}} \cap \bar{\mathfrak{L}}_0) = \pi^{-1}(L_{0,\mathrm{com}}^0).$$

The model set on the right is connected but need not be compact, because it omits the root. The arbitrary-connected-set inverse argument above therefore makes $L_0^0$ connected. The set-theoretic calculation in theorem 12.111, which after this connectedness input does not use local connectivity, gives

$$L_0^n = F^{-n}(T) \cap \Omega_n$$

and the analogous model identity. Semiconjugacy now yields

$$\begin{aligned} \pi^{-1}(L_{0,\mathrm{com}}^n) &= \pi^{-1}\Big(F_{\mathrm{com}}^{-n}(T_{\mathrm{com}}) \cap \bar{\Omega}_n\Big) \\ &= F^{-n}\Big(\pi^{-1}(T_{\mathrm{com}})\Big) \cap \pi^{-1}(\bar{\Omega}_n) \\ &= F^{-n}(T) \cap \Omega_n = L_0^n. \end{aligned}$$

Finally,

$$K_i \subset \pi^{-1}(\bar{K}_i) \subset \pi^{-1}(L_{0,\mathrm{com}}^n) = L_0^n,$$

so geometric model-continuum incidence gives physical incidence. □

**Proved Proposition 12.113** (Small-Julia point-collapse obstruction)**.** *Let $\pi_{\mathrm{Kw}}$ be the monotone semiconjugacy obtained by transporting Kiwi's canonical model through the same-filled quadratic-like restriction and a straightening. For every member $K_i$ of the nondegenerate primitive small-Julia cycle,*

$$\#\pi_{\mathrm{Kw}}(K_i) \geq 2.$$

*Thus the geometric image $\pi_{\mathrm{Kw}}(K_i)$ is nondegenerate and cannot be identified with one abstract finite-tree vertex $\bar{v}_i$ at the level of $\pi_{\mathrm{Kw}}$. Only a later carrier-tree quotient may collapse that image to $\bar{v}_i$.*

*Proof.* Let $P$ be the primitive return period and let $G = F^P : V' \to V$ be the quadratic-like renormalization with filled Julia set $K_0$. Straighten $G$ to $P_c(z) = z^2 + c$. The two finite fixed points lie in $K(P_c)$. They are distinct: a repeated root of $z^2 - z + c = 0$ forces $c = 1/4$. But $K(P_{1/4})$ contains the open immediate parabolic basin. A global quasiconformal straightening takes $K_0$ onto $K(P_c)$ and preserves ambient interior. The active KKL map is periodically repelling, so the same-filled quadratic-like restriction and the no-wandering-domain theorem give $\operatorname{int} K(F) = \varnothing$. Since $K_0 \subset K(F)$, the parabolic case is impossible.

The straightening conjugacy gives distinct $G$-fixed points $x_0, y_0 \in K_0$. Put

$$x_i = F^i(x_0), \qquad y_i = F^i(y_0), \qquad 0 \leq i < P.$$

Both are finite-orbit points in $K_i$. Equality $x_i = y_i$ would, after applying $F^{P-i}$, give $x_0 = y_0$. Moreover, $\operatorname{int} K(F) = \varnothing$ gives $K(F) = J(F)$, so these periodic points belong to the finite-orbit Julia set on which Kiwi's Theorem 5.12 is injective by imported theorem 2.4, so $\pi_{\mathrm{Kw}}(x_i) \neq \pi_{\mathrm{Kw}}(y_i)$. □

**Open Problem 12.114** (Source identification for the monotone full lift)**.** Verify the hypotheses of proposition 12.112 for the exact projection and lift used at KKL lines 542–543. Those lines state the locally connected model, a natural projection, descent of $F$, and an unspecified lift of the corresponding objects. They do not identify their projection with the monotone model supplied by Kiwi's Theorem 5.12 after the same-filled restriction and straightening. The latter is injective on finite-orbit points, but Kiwi's Theorem 2, Proposition 4.2, and the factorization $h = \widehat{w}^{-1} \circ q$ in the proof of Theorem 5.12 do give singleton complete fibers at periodic and preperiodic points for the transported Kiwi map. A marked dynamical homeomorphism between the two quotient targets intertwining the projections, the selected predecessor labels, the full-lift identity $T = \pi^{-1}(T_{\mathrm{com}})$, and exact limb and geometric marked-continuum correspondence remain unproved. No non-locally-connected physical incidence is promoted until these identifications are source-locked or proved after straightening.

One selected sufficient architecture for the later access is $\mathrm{EGC}_H$, endpoint-gate coherence. For each $0 \le n \le H$, construct the selected pulled carrier $\widehat{T}_n$ and the full attachment set

$$\mathrm{Att}_n = \overline{L_0^n}^{\widehat{T}_n} \cap \overline{\widehat{T}_n \setminus L_0^n}^{\widehat{T}_n}.$$

For $1 \le n \le H$, let $\eta_n : \widehat{T}_n \to \widehat{T}_{n-1}$ be the actual branch-induced carrier homeomorphism extending to the declared collar and its prime-end boundary. The predicate requires

$$\alpha^n \in \overline{L_0^n}^{\widehat{T}_n}, \qquad \mathrm{Att}_n = \{x_n\}, \qquad x_n = \alpha^n,$$

and, for $1 \le n \le H$, transport by $\eta_n$ of the declared one-sided prime-end access at $x_n$ to that at $x_{n-1}$. The selected-predecessor avoidance lemma constrains such a gate once coherence is known; it does not establish coherence. This is a selected sufficient route, not a claim that every direct marked- access construction must factor through $\mathrm{EGC}_H$.

**Proved Proposition 12.115** (Conditional line-1598 reduction)**.** *Assume that $F$ has a parabolically $b$-bounded primitive renormalization with rotation number $1/q$. Assume $q > b$, an explicit $\mathrm{SC}_H$, and $\mathrm{MCI}_b$. Then*

$$L_0^b \cap T = \varnothing.$$

*For $h = b+1, b+2$, the complete-horizon recursion gives $F^{h-b}(L_0^h) \subset L_0^b$, and hence $L_0^h \cap T = \varnothing$.*

*Proof.* If $L_0^b \cap T \neq \varnothing$, $\mathrm{MCI}_b$ supplies a component $K_b$ contacting $L_0^b$. Lemma 12.109 gives $K_b \subset L_0^b$. The selected parent relations then give

$$F^j(K_b) \subset L_0^{b-j} \subset \mathfrak{L}_0, \qquad 0 \le j \le b.$$

These are $b+1$ distinct components by lemma 12.105, contradicting the parabolic $b$-bound. For the last assertion, a point in $L_0^h \cap T$ maps by forward invariance into $L_0^b \cap T$. □

**Proved Proposition 12.116** (Locally connected line-1598 repair)**.** *If $K(F)$ is locally connected, the primitive renormalization is parabolically $b$-bounded with rotation $1/q$, and $q > b$, then*

$$L_0^h \cap T = \varnothing, \qquad h = b, b+1, b+2.$$

*Proof.* Theorem 12.111 supplies $\mathrm{MCI}_b$; apply proposition 12.115. □

**Open Problem 12.117** (Non-LC incidence lift and selected access)**.** Prove the non-locally-connected selected-component and marked-fiber equalities (10.125a). On failure, record the actual-component address, model marked vertex, its full physical fiber, the intended $K_i$, the selected component's intersection with that fiber, and the first failed projection or saturation equality.

Separately prove the full-carrier separator predicate $\mathrm{KSEP}_H$, the conditional compatibility $\mathrm{PUS}_H$, and either $\mathrm{EGC}_H$ or a direct marked access theorem. None of these constructs may identify a fixed-point preimage with an attachment gate by label alone. The endpoint architecture is unnecessary for the set count and is only a sufficient route to the canonical marked endpoint access in open problem 12.103; a direct access theorem could replace it.

**Proved Theorem 12.118** (Wave-valued pairwise occurrence collision)**.** *There is an analytic degree-two pseudo-quadratic-like relation, a target Wave $\alpha$, and two distinct proper restriction Waves $\gamma_{\downarrow}, \gamma_{\uparrow}$ in the covering-pulled marking such that their common unparametrized covering image*

$$f(\gamma_{\downarrow}) = f(\gamma_{\uparrow}) = \delta \tag{10.126}$$

*is itself a Wave for the same target marking. Neither restriction meets the critical point.*

*Proof.* Use $U' = \mathbb{D}_{75}$, $U = \mathbb{D}_{5625}$, $K' = [-1, 1]$, $K = [0, 1]$,

$$f(z) = z^2, \qquad \iota(z) = \frac{e^{\pi(z+1)/50} - 1}{e^{\pi/25} - 1}.$$

Put $a = 25\sqrt{5}$ and

$$\gamma_{\downarrow}(x) = x - 50i, \qquad \gamma_{\uparrow}(x) = x + 50i, \qquad -a < x < a.$$

Both are outer-to-outer proper chords. The immersion has period $100i$, so their immersion images agree. Also $\gamma_{\uparrow}(-x) = -\gamma_{\downarrow}(x)$, which proves (10.126). The common image is

$$\delta(x) = x^2 - 2500 - 100xi.$$

Its imaginary part determines $x$, its only real point is $-2500$, and its endpoint moduli are 5625; thus it is a simple proper outer-to-outer path avoiding $K$. Extend the common immersion trace by a simple inner-to-inner polygonal arc, choosing its tangent outside both source-disk sheets at the two chord endpoints. This gives a proper target Wave $\alpha$ for a finite inner-boundary marking. Choose the outer base of $\delta$ as the square of the lower outer cap and subdivide it into at least three intervals. Its two covering lifts make both chords Waves in $f^*\mathcal{I}$.

This is a pairwise collision only. The full immersion-action refinement, line-2097 survival, uniqueness of raw restrictions, and unbreakability are not asserted. Hence it is not a stationary KKL counterexample. □

Theorem 12.118 proves that distinct components, sheets, properness, and even individual Wave image type do not replace the exact selected-collar predicate

$$[\lambda^+] \neq [\tau_n(\lambda^-)]. \tag{10.127}$$

The incoming anchor in open problem 12.103 occurs logically before this deck-path test.

The native stationary proof now has three distinct open gates:

(1) `KKL_STATIONARY_POINTWISE_TWO_PATH_BREAK`;
(2) `KKL_STATIONARY_FINAL_FIXED_COMPONENT_EXIT`;
(3) `KKL_STATIONARY_TYPED_BOUNDED_DEGREE_PUSHFORWARD`.

The first is pointwise: at $n = b, b+1$, the two line-1603 portions must be retained proper paths with distinct terminal images. No doubled-width paired-Wave response is consumed by that contradiction. The second assembles the final restrictions into a measurable fixed-component exit lamination. The third applies the covering lemma to the whole marked end.

The terminal index is $A_{N-1} = A_{b+2}$, not the undefined $A_N$. The canonical graph component has continuum degree 2, not 1, but whole-end degree remains part of the original fourth gate, now the third remaining gate. The self-contained conclusion has the form

$$O_b(1 + \mathcal{W}(\mathcal{R}F))\,; \tag{10.124}$$

the additive constant may be absorbed into $O_b(\mathcal{W}(\mathcal{R}F))$ only in an explicit large-width regime.

The paired-response construction developed above is retained as an optional stronger route. Its Series algebra remains a correct conditional implication, but no broad coefficient-one source-Wave theorem or full residual realization has been proved. The three gates listed above, the native stationary repair, VAAPC-I, UPTCR, and MLC remain open. Theorem 12.101 promotes Gate N1 only.

## 13. Validated side-library deductions

The side-research library is a cumulative proposal series, not an independent proof corpus. After deduplication and adversarial review, four narrow mathematical advances survive and two proposed analytic shortcuts fail. This section gives the proofs needed by the ledger. It does not construct a virtual carrier, close a response transition, or prove MLC.

### 13.1. **Primitive-root parabolic nondegeneracy.**

**Proved Theorem 13.1** (Quadratic primitive-root nondegeneracy)**.** *Let $\lambda$ be a primitive $q$-th root of unity, $q \geq 2$, and*

$$g(w) = \lambda w + w^2.$$

*Then*

$$g^q(w) = w + C_{\lambda,q} w^{q+1} + O(w^{q+2}), \qquad C_{\lambda,q} \neq 0. \tag{11.1}$$

*Proof.* Put $F = g^q$. Since $F'(0) = 1$, let

$$F(w) = w + a_m w^m + O(w^{m+1})$$

be its first nonlinear term. It exists because $F$ is a polynomial of degree $2^q$ and is not the identity. Commutation $F \circ g = g \circ F$ gives, at degree $m$,

$$a_m \lambda^m = a_m \lambda.$$

Thus $\lambda^{m-1} = 1$, and primitivity implies

$$m = 1 + kq, \qquad k \geq 1. \tag{11.2}$$

This proves cancellation in degrees $2, \ldots, q$.

The iterate $F$ has $kq$ attracting petals. The derivative of $g$ at the parabolic point rotates their tangent directions by the primitive root $\lambda$, so they form $k$ cycles of length $q$ under $g$. The classical Fatou petal/basin theorem associates a critical orbit to each such cycle; equivalently, each cycle of immediate parabolic basin components contains a critical point [Gey16, Corollaries 11.6 and 11.8]. The quadratic polynomial $g$ has only one finite critical point, $-\lambda/2$. Its critical point at infinity belongs to the separate superattracting basin of infinity. Therefore $k \leq 1$. Equation (11.2) has $k \geq 1$, so $k = 1$, proving (11.1). □

For an exact coefficient audit, let

$$\phi(z) = z + \sum_{n \geq 2} b_n z^n, \qquad h(z) = \lambda z + A z^{q+1} + O(z^{q+2}),$$

and solve $g \circ \phi = \phi \circ h$. If $b_1 = 1$ and

$$S_n = \sum_{i=1}^{n-1} b_i b_{n-i},$$

then coefficient comparison gives

$$b_n = \frac{S_n}{\lambda^n - \lambda} \quad (2 \leq n \leq q), \qquad A = S_{q+1}, \tag{11.3}$$

and

$$[z^{q+1}]\, h^q = q\lambda^{q-1}A. \tag{11.4}$$

The denominators in (11.3) are nonzero by primitivity. This recurrence checks the coefficient algebra; the basin-critical-point argument is what proves nonvanishing for every $q$. No uniform Ecalle-cylinder or parameter estimate follows.

13.2. **Finite modulus versus ambient end shrinking.**

**Proved Theorem 13.2** (Fixed-modulus ambient-end no-go)**.** *There are nested univalent images $D_m$ of $\mathbb{D}$ and conformally conjugate degree-two quadratic-like maps on $D_m$, all with the same quadratic-like modulus, for which*

$$\bigcap_{m\geq 0} \overline{D_m}$$

*contains a nondegenerate continuum.*

*Proof.* For $0 < r < 1$, let

$$E_r = \left\{ x + iy : x^2 + \frac{y^2}{r^2} < 1 \right\} \subset \mathbb{D}.$$

Let $\phi : \mathbb{D} \to E_r$ be normalized by $\phi(0) = 0$, $\phi'(0) > 0$. Uniqueness and the conjugation and central symmetries of $E_r$ imply

$$\phi(\bar{z}) = \overline{\phi(z)}, \qquad \phi(-z) = -\phi(z).$$

Carathéodory extension gives $\phi(\pm 1) = \pm 1$, hence

$$\phi([-1, 1]) = [-1, 1].$$

Since $E_r \subset \mathbb{D}$, the compact Jordan domains $\phi^m(\overline{\mathbb{D}})$ are nested, and all contain $[-1, 1]$.

Choose a quadratic-like $F : A \to B$ with $\overline{B} \subset \mathbb{D}$ and put

$$F_m = \phi^m \circ F \circ \phi^{-m} : \phi^m(A) \longrightarrow \phi^m(B).$$

Conformal invariance preserves degree and $\operatorname{mod}(B \setminus \overline{A})$ at every $m$, while the ambient intersection remains nondegenerate. □

This is an interface countermodel, not a quadratic-polynomial counterexample. The exact sufficient replacement is serial separation.

**Proved Theorem 13.3** (Serial-annulus singleton bridge)**.** *Suppose actual pairwise-disjoint annuli $A_n$ are nested around one common nonempty compact connected full end fiber $K$, separate $K$ from one fixed nondegenerate full outer continuum $L$, and admit no side bypass. If*

$$\sum_{n\geq 0} \operatorname{mod}(A_n) = \infty, \tag{11.5}$$

*then $K$ is a singleton.*

*Proof.* If $K$ were nondegenerate, the ring domain separating the two disjoint nondegenerate continua $K$ and $L$ would have finite modulus. Every $A_n$ is an essential pairwise-disjoint subannulus of that ring domain. Grötzsch superadditivity would give

$$\sum_{n=0}^{N} \operatorname{mod}(A_n) \leq \operatorname{mod}\big(\widehat{\mathbb{C}} \setminus (K \cup L)\big) < \infty$$

for every $N$, contradicting (11.5). Nested compactness gives $K \neq \varnothing$, hence $K$ is a singleton. □

The unresolved step is to realize these annuli in every proof-used physical end while auditing all side contacts, protected markings, and bypasses.

13.3. **The correct multi-port gauge.** The side proposal made three distinct response errors. They are useful because each identifies a datum that a future collar theorem must retain.

**Proved Theorem 13.4** (Multi-port gauge and strictness no-go)**.** *For one connected Dirichlet cell with ports $\Gamma_1, \dots, \Gamma_N$, the correct gauge is*

$$\left(\bigoplus_{i=1}^{N} H^{1/2}(\Gamma_i)\right) \Big/ \operatorname{span}\{(1, \dots, 1)\}. \tag{11.6}$$

*Independent quotienting of each port by constants is invalid. Moreover, addition of an arbitrary common nonnegative component energy does not preserve a strict response factor below one.*

*Proof.* The Dirichlet response is first the nonnegative quadratic form

$$q_D(u) = \min_{\operatorname{Tr} U = u} \int_D |\nabla U|^2, \tag{11.7}$$

and its bilinear form is obtained by polarization. Adding one common constant to all traces changes no gradient, which gives the quotient in (11.6). By contrast, on a width-one rectangle of length $R$, constant endpoint traces $a, b$ have energy

$$\frac{(a-b)^2}{R}.$$

Independent port quotients erase this positive relative-potential mode.

For strictness, let $q_{\text{out}} = \mu q_{\text{in}}$, $0 < \mu < 1$, and add the common form $c = t q_{\text{in}}$. Then

$$\frac{q_{\text{out}} + c}{q_{\text{in}} + c} = \frac{\mu + t}{1 + t} \longrightarrow 1.$$

Thus positivity alone cannot preserve the strict factor. □

The rejected side formula

$$\inf \int_D \nabla U \cdot \nabla V$$

is also unbounded below: with zero traces, $U = t\psi$ and $V = -t\psi$, $0 \neq \psi \in H_0^1(D)$, make it tend to $-\infty$. A valid transition after gluing must instead control the relative added energy, prove strictness for the complete state, or charge an explicit additive error.

13.4. **Weak operator topology does not preserve welding.**

**Proved Theorem 13.5** (Weak-operator welding-algebra no-go)**.** *For bounded $J : H \to H$, the orthogonal projection onto its graph is*

$$\Pi_J = \begin{pmatrix} A & AJ^* \\ JA & JAJ^* \end{pmatrix}, \qquad A = (I + J^*J)^{-1}. \tag{11.8}$$

*Weak operator subsequence convergence of uniformly bounded weldings does not preserve graph constraints, composition, inverses, or response gluing.*

*Proof.* For $Rx = (x, Jx)$, one has $R^*R = I + J^*J$, and $\Pi_J = R(R^*R)^{-1}R^*$, proving (11.8). The side proposal instead used

$$\frac{1}{2}\begin{pmatrix} I & J^* \\ J & I \end{pmatrix}.$$

For scalar $J = 2$, this matrix has eigenvalues $3/2, -1/2$, so it is not positive.

For the compactness failure, let $H = \ell^2(\mathbb{Z})$, let $Ue_k = e_{k+1}$, and put $J_n = U^n$. Then

$$J_n \rightharpoonup 0, \qquad J_n^* J_n = I.$$

If $u_n = e_{-n}$, then $u_n \rightharpoonup 0$, whereas $J_n u_n = e_0$. Thus the varying graph relations are not weakly closed. Since every $J_n$ is unitary,

$$\Pi_{J_n} = \frac{1}{2}\begin{pmatrix} I & J_n^* \\ J_n & I \end{pmatrix} \rightharpoonup \frac{1}{2} I_{H\oplus H},$$

which is not idempotent and hence is not a graph projection. □

Uniform boundedness still gives weakly convergent subsequences of matrix coefficients, and positivity of a separately retained response passes to a weak limit. The theorem proves that this elementary fact is not a gluing-compatible compactification of physical weldings.

13.5. **One external trace component is uniform.**

**Proved Theorem 13.6** (Sectorwise dyadic Böttcher trace isometry)**.** *Let $T : I \to J$ be an affine bijection of intervals, with either orientation. Then*

$$[u \circ T]_{\dot H^{1/2}(I)} = [u]_{\dot H^{1/2}(J)}. \tag{11.9}$$

*Consequently, on one declared external sector, the inverse angular lift*

$$T_{k,q}(\theta) = \frac{\theta + k}{2^q}$$

*has exact normalized homogeneous trace norm one, independently of $q$.*

*Proof.* Write $T(\xi) = a + s\xi$, $s \neq 0$. In the Gagliardo seminorm, the two-variable Jacobian contributes $|s|^2$, while

$$|T(\xi) - T(\eta)|^2 = |s|^2 |\xi - \eta|^2.$$

The factors cancel exactly. □

This result is sectorwise and homogeneous. It does not provide a multisector address system, a physical Böttcher-to-core welding, all-depth contact closure, coercivity, or a strict transition. In particular, theorem 13.5 prevents replacing those missing fields by an unstructured weak limit.

13.6. **Status consequence.** The source-side subclass audit also confirms that Cheraghi–Shishikura's $\mathcal{Q}G_N$ theorem and Levin's Theorems 1, 2, and 7 close their specified high-growth satellite families [CS15, Lev08]. Kahn–Lyubich prove a-priori bounds for the exact Elephant-Eyes E1–E3 class, while the Ecalle-cylinder little-copy input in their parameter discussion is cited as in preparation [KL26]. BLOT's non-parabolic limit theorem concerns the exact annulus $U_n \setminus K_n$, not the outer annulus $V_n \setminus U_n$. These sufficient subclasses are not exhaustive. A neutral-attractor theorem reported through an unavailable 2026 manuscript is not imported from a proof summary.

Accordingly, the results in this section refine the response and local parabolic interfaces but discharge none of `DIRECT_DECORATION_TRACE_WELDING_COMPACTNESS`, `DIRECT_DECORATION_TRACE_COLLAR_REALIZATION`, `END_PLUMBING_TRACE_RESPONSE_CONTROL`, UPTCR, or MLC.

## 14. Exact open gates and falsification requirements

14.1. **The global closure gate.** The exact global proposition is open problem 9.1. Its six clauses package, without erasing, the marked construction, analytic realization, finite response state, backward transition, coercive compactness, and rigidity gates. The implication UPTCR ⇒ MLC is proved in conditional theorem 9.4; UPTCR and MLC both remain open.

14.2. **The immediate geometric gate.** The connector-realization and selection criteria are now theorems 10.4, 10.8 and 10.11; the immediate dynamical target is open problem 10.57, which suffices for open problem 10.58, followed by open problem 10.6 together with open problems 5.8 and 6.16. A successful construction must use Dudko's actual Cantor small Julia sets and prescribed first returns, not an abstract replacement with better topology.

The first current analytic subgate occurs earlier than the Wave correspondence. At the proved direct-decoration scale, theorem 10.49 produces one open pullback component only after a target base is typed; it neither authorizes that component as one of the declared cylinders nor supplies the complete marking. Thus the machine-ledger gate `VIRTUAL_REACHED_CYLINDER_COMPONENT_EXHAUSTION` and then open problems 10.50 and 10.51 precede the global Cantor-package and return-atlas construction. The later first Wave subgate is open problem 10.18. The finite stopping and final-root algorithms are supplied by theorem 10.17 only after that marked correspondence and its complete ledger exist.

The critical-chronology audit sharpens this boundary without closing it. Theorem 11.5 compiles a finite-cylinder pair only after the root portrait has independently been constructed. Thus open problem 11.19 remains upstream of the disconnected marking transfer. At infinite addresses, open problems 11.20 and 11.21 remain open; the planar collapse theorem supplies a necessary no-go constraint, not the missing analytic end object. Likewise theorem 11.11 is only a planar compiler: relative-cover realization and protected ideal-boundary action remain in `MAPPING_TELESCOPE` and `FINITE_HORIZON_ASSEMBLY`. The normalized end direct limit in theorem 12.1 is a coordinate theorem, not a planar component theorem. Moreover, theorem 12.5 shows that scalar serial conductance cannot control the nonconstant traces on an analytic cut. The response route therefore also requires open problem 12.9 before any prescribed plumbing-error or transition-transfer conclusion is available.

**Open Problem 14.1** (Virtual Seed and Attachment Theorem)**.** For each virtual-Molecule block and the full finite Wave horizon:

(1) construct one root carrier realizing all packages, component routing, sheet separation, and critical windows;
(2) prove that saturation stays in the admissible virtual localization;
(3) absorb or separate every ambient Cantor component, orbit point, fjord, and dam;
(4) construct a future-complete bounded root endpoint marking;
(5) construct a regulated connector system with full attachment traces and protected-access survival maps, construct an actual anchor pair, transport the complete pair univalently, and prove either anchor equality or a collision-free attachment-orbit closure, then bound the defect-window endpoint charge; and
(6) verify the relative-cover subgroup lift, core properness, and end-proper Wave interval action.

14.3. **Adversarial tests that every completion must pass.**

**A1. Root shadow split.** Two atoms of one package may have forward images in the root while occupying different inverse components.
**A2. Late merger.** Adding future-required root data may enlarge the root across the critical value and merge formerly distinct sheets.
**A3. False translation.** A target containing the critical value is a degree-two step by proposition 5.5, regardless of its label.
**A4. Hidden attachment.** Correct polynomial component routing does not prevent pulled-carrier excess from attaching to the copy embedded by the immersion.
**A5. Linear endpoint leakage.** One new endpoint at every degree-one stage gives $p_0 = p_N + N$.
**A6. Access explosion.** One attachment component can expose arbitrarily many proof-used prime-end accesses; component count does not bound endpoint count.

**A7. Relative-cover failure.** Set-theoretic forward closure does not imply the subgroup criterion needed to lift the immersion.
**A8. Horizon leakage.** Every Wave pullback must remain inside the declared finite horizon, including repeated critical visits.
**A9. Chart-cycle neutrality.** Strict estimates on individual chart edges do not imply strict gain around a directed cycle.
**A10. Rigidity circularity.** Parameter-piece shrinkage cannot be assumed in order to prove the phase–parameter theorem responsible for that shrinkage.
**A11. No least connector.** The upper and lower semicircles through a two-point package are full connected carriers whose intersection is disconnected; inclusion-minimality cannot define the virtual carrier.
**A12. Protected-access loss.** In proposition 6.15, monotone absorption is critical-free and sheet-preserving but erases a proof-used upper prime-end access.
**A13. Lamination-fiber collision.** A monotone preimage of a regulated model tree is connected, but one projection fiber may enter a protected collar. Connectedness alone does not prove open problem 10.6.
**A14. Free-component split.** Two atoms may each have a free singleton access while a protected collar or separated obstacle places those accesses in different components of $\Omega_{\mathrm{free}}$. Individual accessibility does not prove open problem 10.58.
**A15. Late separator.** A connector selected before the final horizon may lose its only free bridge when a future-used collar, obstacle, or root shadow is inserted. The free-domain test must be run after future completion.
**A16. Fullification merger.** A seed disjoint from an annular obstacle can be swallowed when the obstacle is filled. Packet disjointness must be checked after fullification.
**A17. Port erasure by assignment.** Assigning that annulus to the enclosed atom can make the original atom interior to its augmentation. An explicit singleton port is still required.
**A18. Sufficient-certificate overreach.** Failure of one augmented packet does not refute open problem 10.58; the original free-domain separator must be exhibited separately.
**A19. Trace-mode loss.** Replacing a cut cross-section by one scalar electrical node deletes its nonconstant $H^{1/2}$ trace modes. Long-neck constant conductance alone does not bound their energy by theorem 12.5.

14.4. **Promotion rule.** An open obligation may be promoted only after recording:

(1) the exact quantified statement;
(2) a complete proof or a theorem source with matching hypotheses;
(3) every constant dependence;
(4) an attack aimed at every hypothesis;
(5) its position in the dependency graph; and
(6) a successful machine-ledger and manuscript build audit.

Computational experiments may falsify a proposed formula or normalization, but no numerical experiment discharges an analytic proof obligation.

## Appendix A. Machine-readable proof ledger

The following appendix is generated from `../proof_obligations.json`. It is included to make the manuscript's proof boundary mechanically checkable. The generator rejects unknown statuses, missing dependencies, duplicate identifiers, and a non-open `MLC` target.

### Imported obligations (34)

**YOCCOZ** IMPORTED

MLC at parameters that are not infinitely quadratic-like renormalizable.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* source_ledger.md#primary-sources

---

**BLOT_ROOT** IMPORTED

Satellite root modulus is at most 2*pi/log(4(s+1)) for quadratic polynomials.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* source_ledger.md#primary-sources

---

**BLOT_NONPARABOLIC_LIMIT_MODULUS_COLLAPSE** IMPORTED

For a convergent sequence of fixed-degree rational maps carrying polynomial-like renormalizations f_n^(q_n):U_n->V_n with connected K_n converging in Hausdorff distance to a nondegenerate compact K, if q_n tends to infinity and no parabolic periodic domain of the limiting map contains K, then mod(U_n minus K_n) tends to zero. This is BLOT Theorem B and concerns a different annulus from the outer extension annulus in its Lavaurs example.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* unbounded_satellite_modulus_source_reaudit_20260726.md#conditional-branch-and-its-exact-boundary

---

**FATOU_PARABOLIC_BASIN_CRITICAL_CYCLE** IMPORTED

Every cycle of immediate attracting or parabolic basin components of a rational map contains a critical point or, in the transcendental setting, an asymptotic value; for a polynomial parabolic cycle the required singular value is a finite critical point.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* source_ledger.md#side-research-source-addendum

---

**CHERAGHI_SHISHIKURA_QG_MLC** IMPORTED

There exist N,C and 0<lambda<1 such that for every Cheraghi-Shishikura QG_N satellite sequence, with b_(1,1)>=N, b_(i,j+1)>=b_(i,j)^2, and b_(i+1,1)>=q_i^2, the corresponding depth-n parameter prefix set has diameter at most C*lambda^n; every infinitely satellite-renormalizable quadratic parameter in this class is combinatorially rigid and MLC holds at it.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* source_ledger.md#satellite-subclass-theorems

---

**LEVIN_EXPONENTIAL_RENORMALIZATION_MLC** IMPORTED

For an infinitely renormalizable quadratic parameter with simply renormalizable selected periods n_m and separating-fixed-point rotation numbers p_m/q_m, limsup log(abs(p_m))/n_m>0 implies MLC at the parameter; if every selected renormalization is satellite, limsup log(q_m)/n_m>0 suffices.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* source_ledger.md#satellite-subclass-theorems

---

**LEVIN_FAST_SMALL_ROTATION_MLC** IMPORTED

For Levin's all-satellite cascade with nonzero internal arguments t_m=p_m/q_m, if q_m tends to infinity and limsup abs(t_(m+1))^(1/q_m)<1, equivalently limsup abs(t_m)^(1/q_(m-1))<1, then the cascade converges to an infinitely renormalizable parameter, MLC holds there, and its Julia set is not locally connected.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* source_ledger.md#satellite-subclass-theorems

---

**LEVIN_THEOREM7_SATELLITE_MLC** IMPORTED

For Levin's satellite cascade, conditions (Y0a) sup |t_m|q_0...q_(m-1)<infinity and (Y0b) inf log(abs(p_(m-1)q_(m-1))/abs(t_m))/(q_0...q_(m-2))>0 give convergence; if additionally (Y1) limsup q_m/max((q_0...q_(m-1))^2,exp(beta*q_0...q_(m-2)))>0 for some beta>0, then MLC holds at the limit.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* source_ledger.md#satellite-subclass-theorems

---

**ELEPHANT_EYES_APRIORI_BOUNDS** IMPORTED

Kahn-Lyubich prove a priori bounds for their exact Elephant-Eyes E1-E3 class: even quadratic-like maps with connected Julia set and repelling fixed points, rotation 1/q, the stated central little-Julia uniqueness, and renormalization period p=q+b with bounded b. The separate Ecalle-cylinder little-copy input used in their parameter-space discussion is not imported.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* source_ledger.md#satellite-subclass-theorems

---

**KKL_TRANSLATION** IMPORTED

For an (n,p)-translation marking with N=2(n+1)p, long canonical WAD weight is at most 2*W_std+6*N.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* source_ledger.md#exact-source-locations

---

**KKL_THICK_THIN** IMPORTED

Total marked width differs from twice canonical WAD norm by at most a universal constant times Euler and marking complexity.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* source_ledger.md#localization-and-wad-source-lock-addendum

---

**KKL_WAD_SUPPORT** IMPORTED

The support of a WAD on a marked surface has cardinality at most 3*abs(chi(V))+abs(I).

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* source_ledger.md#localization-and-wad-source-lock-addendum

---

**KKL_VALUE_CONTRACTION** IMPORTED

For a primitively renormalizable periodically repelling psi-ql map with core eigenvalue lambda_F, KKL constructs a nonnegative WAD valuation that is monotone under WAD domination and contracts under pullback by lambda_F^{-1}.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* source_ledger.md#bc1-finite-port-and-value-observability-addendum

---

**KKL_OCCUPANCY_TRICHOTOMY** IMPORTED

For every B and M there is A(B,M) such that a periodically repelling prime primitively renormalizable psi-ql map already carrying KKL's full cycle, Hubbard tree, canonical WAD, and canonical renormalization, with at most B small filled Julia sets in each limb and W(RF)>=A(B,M), is M-amplifying, M-tame, or has M-wide controllable waves. This imports Proposition 3.7 plus Remark 4.14 only, not the tame or wave branch closures.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* kkl_virtual_occupancy_source_audit.md#41-exact-source-theorem

---

**KKL_PARABOLIC_TAME_AMPLIFICATION** IMPORTED

In KKL's full parabolically B-bounded, prime primitively renormalizable psi-ql category, for every requested factor P, sufficiently high type and sufficiently large successor width turn the P-tame branch into P-amplification, with thresholds depending on B and P but not on the satellite period.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_tame_full_return.md#3-exact-source-theorem-and-notation`

---

**KKL_IDEAL_PULLBACK_NATIVE** IMPORTED

In KKL's native parabolically B-bounded Hubbard-tree category, Proposition 6.5 lists the immersion pullback of an ideal interval as its inverse image plus the ideal boundaries of attached components of T' minus T, while Proposition 6.7 proves T' intersect L_[B,q-B-2] = T intersect L_[B,q-B-2], so no pulled excess meets the relevant central intervals and the interval action is proper there; no global equality T'=T is imported.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `kkl_translation_no_birth_source_audit.md#2-checked-source`

---

**ALEXANDER_DUALITY_COMPACTA** IMPORTED

For every nonempty proper compact subset K of the 2-sphere, reduced singular H_0 of its complement is naturally isomorphic to first reduced Cech cohomology of K; this is the arbitrary-compacta, not merely locally contractible, form of Alexander duality.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_common_free_component.md#23-topological-input`

---

**DL_TRANSITIONAL_PERIPHERAL** IMPORTED

For an eventually-golden-mean psi-bullet-ql Siegel map satisfying DL25's transition-level grounded-interval hypotheses, transitional peripheral width is O(sqrt(ell_m*K_F)).

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `source_ledger.md#exact-source-locations`

---

**DL_TILE_SCALING** IMPORTED

For an eventually-golden-mean psi-bullet-ql Siegel map satisfying DL25's transition-level grounded-interval and equidistribution hypotheses, there are r_m=q_{m+1} tiles, per-tile vertical width is comparable to A_m=ell_m*K_F, and total parallel vertical width is comparable to r_m*A_m and hence K_F.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `source_ledger.md#neutralcorridor-normalization-correction`

---

**LYUBICH_PRINCIPAL_RETURN_MAP** IMPORTED

For a principal-nest puzzle piece in the native Yoccoz-puzzle setting, Lyubich's Markov pullback and boundary-avoidance lemmas decompose the first return into pairwise disjoint puzzle domains compactly contained in the base; the central branch is a degree-two branched covering and every noncentral branch is univalent.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_return_base_atlas.md#21-lyubichs-yoccoz-puzzle-source`

---

**KL_DECORATION_ESCAPE_PUZZLE** IMPORTED

For a quadratic parameter in a fixed non-principal decoration outside a directly attached satellite copy M_{p/q}, Kahn–Lyubich construct the marked ray puzzle, the least q-return escape number n and route kappa, the actual critical escape piece Y^{nq+1}, its compactly contained first-return domains, the modified principal nest, the terminal primitive little-Julia cycle, and a pairwise-disjoint univalent pullback Q^v for each of the 2^n vertices of Y^{(n-1)q+1}.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_scale_to_puzzle_selector.md#2-source-lock`

---

**KKL_NATIVE_TERMINAL_HUBBARD_TREE** IMPORTED

In KKL's native periodically repelling, primitively renormalizable psi-ql category, the complete cycle of small filled Julia sets has a compact forward-invariant Hubbard tree with a finite vertex-edge tree and native ideal marking; in the non-locally-connected case KKL construct the corresponding objects through the rational-lamination model and its lift.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `terminal_hubbard_anchor.md#2-source-lock-and-scope`

---

**PERIODICALLY_REPELLING_FILLED_JULIA_EMPTY_INTERIOR** IMPORTED

For a periodically repelling quadratic or quadratic-like map, the filled Julia continuum has empty interior and hence equals its Julia set. This is imported only in the periodically repelling category.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `terminal_relative_cover_alignment.md#2-source-lock`

---

**PQL_SAME_FILLED_CONTINUUM_QUADRATIC_LIKE_RESTRICTION** IMPORTED

Kahn's induced-quadratic-like theorem, as recalled by KKL, restricts a finite-width psi-quadratic-like map near the same physical filled continuum K to a quadratic-like germ. The restriction has the identical dynamics and periodic multipliers on K.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `terminal_relative_cover_alignment.md#2-source-lock`

---

**DOUADY_HUBBARD_CONNECTED_QUADRATIC_LIKE_STRAIGHTENING** IMPORTED

Every degree-two quadratic-like map with connected filled Julia set is hybrid conjugate to a quadratic polynomial, uniquely up to affine conjugacy in the connected case; the hybrid conjugacy restricts to a homeomorphism conjugating the dynamics on the filled Julia sets.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `kkl_stationary_n2_kiwi_model_identification_audit_20260727.md#22-dudko-lyubich`

---

**KIWI_MONOTONE_MODEL_FINITE_ORBIT_INJECTIVITY** IMPORTED

For a polynomial with connected Julia set and no irrationally indifferent cycle, Kiwi's real-lamination theorem supplies a monotone surjective semiconjugacy to a locally connected model. Theorem 2 makes every periodic or preperiodic dynamical fiber trivial; Proposition 4.2 forms the quotient q by those fibers; and the proof of Theorem 5.12 factors the semiconjugacy as h=w^(-1) composed with q. Hence every periodic or preperiodic point has a singleton complete semiconjugacy fiber.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `kkl_stationary_n2_kiwi_model_identification_audit_20260727.md#23-kiwi`

---

**DOUADY_HUBBARD_QL_EXTERNAL_PRIME_END_EXACTNESS** IMPORTED

For a connected degree-d polynomial-like map, d>=2, the inner prime-end dynamics is topologically conjugate to angle multiplication t->dt mod 1 and is topologically exact. Prime-end impressions are covariant with equality under the map. This is the topological external-dynamics content of Douady-Hubbard straightening and the strongly expanding external circle construction; degree alone would not imply exactness.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `terminal_dense_obstacle_and_submulticurve_replacement.md#2-source-lock-and-exact-hypotheses`

---

**KKL_FIXED_END_IMMERSION_DECOMPOSITION** IMPORTED

For a holomorphic immersion i:V'->V of finite-topology open Riemann surfaces and a source-end set E' at every end of which i is proper of degree one, let E=i(E'). Every valid target WAD Y whose arcs each have at least one endpoint in E admits a valid source WAD Y' and paired positive-coefficient decompositions Y=sum_a v_a beta_a and Y'>=sum_a v_a beta'_a such that each beta'_a either has both endpoints in E' and i_*beta'_a=beta_a, or has one endpoint in E' and the other outside E'. No canonical fixed-end choice, marked occurrence coupling, or uniform end-count bound is imported.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `kkl_fixed_end_immersion_switch_wad.md#3-source-theorem-in-coefficient-form`

---

**KL06_EXTREMAL_WIDTH_SERIES_HOLOMORPHIC_RULES** IMPORTED

For path families on Riemann surfaces, if every path in Gamma contains pairwise disjoint members of Gamma_1,...,Gamma_r, then W(Gamma)<=oplus_i W(Gamma_i); and for a holomorphic map f, L(f(Gamma))>=L(Gamma), equivalently W(f(Gamma))<=W(Gamma). These are the KL06 Series Law and holomorphic transformation rule in the SHA-locked appendix source.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `kkl_stationary_bl0_survivor_analysis.md#22-kl06-source-locks`

---

**DL23_ANNULAR_WAVE_BUFFER_EXTRACTION** IMPORTED

For a degree-m covering between closed annuli, one Wave above one fixed boundary interval J contains a genuine subwave on which the covering is injective and whose width is at least the source width minus an absolute O(log m) buffer. This is the native scope proved in Dudko–Lyubich; it does not cover an arbitrary KKL Wave assembled from several independent positive-width source coordinates.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `qa_extremal_wave_2026-07-26.md#2-immutable-evidence-used`

---

**KKL_LAMINATION_PUSHFORWARD_DEGREE_BOUND** IMPORTED

KKL master lines 2075–2078 state the general finite-degree pushforward bound: if a source lamination ends at a set on which the covering has degree at most d, its pushforward contains a lamination A with W(A')<=d*W(A)+2d. Unlike the coefficient-one Wave claim, this linear-degree estimate is consistent with repeated source sheets collapsing to one target coordinate.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `qa_extremal_wave_2026-07-26.md#2-immutable-evidence-used`

---

**KKL_WAVE_PUSHFORWARD_STATED** IMPORTED

Documentary source record only: KKL master lines 2149–2152 print the assertion that a Wave ending at a multi-interval of degree at most d has a pushed Wave C with W(B)<=W(C)+O(1+log d). This node records what the source says; it does not license the inequality mathematically. The literal broad scope is refuted by KKL_BROAD_WAVE_PUSHFORWARD_COUNTEREXAMPLE.

*Ledger role:* documentary source record

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `qa_extremal_wave_2026-07-26.md#11-required-corrections-and-safe-status`

---

**DL_NATIVE_RECTIFIABLE_RELATIVE_DISCARD_BOUND** IMPORTED

For a degree-m covering of closed annuli and a rectifiable Wave above one connected source boundary interval, the Dudko–Lyubich proof-internal deletion has width O(1+log m): it consists of T=O(log m) nonzero deck-span packets of O(1) width and one final O(1) outer buffer. This is the native source scope and uses no subtraction or measurable-support additivity.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `kkl_stationary_n1_integration_audit_20260726.md#32-proof-internal-derivation`

`KKL_PRIMITIVE_ROTATION_LIMB_CYCLE_DATA` IMPORTED

In KKL's native periodically repelling primitive psi-quadratic-like category with rotation number 1/q and q>=2, the induced map F=f composed with iota^(-1) is defined near K(F), maps K(F) to itself as a degree-two branched cover, fixes alpha_F, satisfies F(limb 0)=K(F), places the unique critical value in limb 1, and maps every noncritical limb j homeomorphically to limb j+1. The small-Julia cycle is a period-P cyclic family of pairwise-disjoint connected continua whose critical member contains the critical point. This imports only source definitions and limb dynamics, not a selected predecessor or stationary-Wave conclusion.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `kkl_stationary_n2_selected_predecessor_endpoint.md#2-source-lock`

---

## Proved obligations (316)

`KKL_TYPED_VALUE_INCIDENCE` PROVED

For the raw source-to-target occurrence incidence N in KKL Value, the project source-column count map is B=N^T; KKL v is the project left eigenweight, KKL w is the project right eigenweight, and the Value coefficient is v^2*w. The checked source symbol T_F is orientation-ambiguous and is not used for this identification.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `source_ledger.md#kkl-typed-incidence-and-clone-source-audit`

---

`KKL_PRINTED_CLONE_EQUATION_NO_GO` PROVED

Under the literal checked KKL clone definitions, the displayed +lambda^k finite-n reduced equation fails on an explicit primitive three-vertex cross-section graph; this invalidates that printed proof bridge, while the asymptotic limit is recovered only by the separate internal negative-weight theorem.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `kkl_transfer_orientation_audit.md#61-literal-finite-graph-counterexample-to-the-displayed-clone-equation`

---

`KKL_CLONE_EIGEN` PROVED

For a fixed strongly connected literal cross-section clone with strict return entropy, the exact typed reduction uses H(t)=sum_k t^(-k)R_k and rho(H(lambda_n))=lambda_n^n; consequently lambda_n -> 1, lambda_n^n -> rho(H(1))>1, and both gate-normalized typed Perron vectors converge. This is an internal correction, not an import of the false printed +lambda^k bridge.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `corrected_clone_theorem_derivation.md#1-status-and-result`

---

`REGIME_PARTITION` PROVED

The KKL precedence gives an exhaustive disjoint four-way partition of families.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `regime_and_hulls.md#2-the-family-level-four-regime-partition`

---

`UNMARKED_NO_GO` PROVED

No bounded scalar is uniformly additively comparable to inverse root modulus over unbounded satellite periods.

*Formal hypotheses:* *none*

*Proof dependencies:* `BLOT_ROOT`

*Evidence:* `master_transition.md#2-exact-obstruction-from-unbounded-satellites`

---

`BACKWARD_RECORD` PROVED

A uniform backward amplification recurrence bounds every scalar tower.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `master_transition.md#7-record-lemma`

---

`MARKING_QUOTIENT_NO_GO` PROVED

A quotient preserving N distinct labeled return accesses has at least N target marks.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_block_quotient.md#2-marking-cardinality-obstruction`

---

`TRANSLATION_ORBIT_TYPES` PROVED

For fixed interaction radius R and translation shift s, oriented short interactions away from defect ends have at most s*(2R+1) translation-orbit types, independently of corridor length.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_block_quotient.md#4-translation-orbit-compression`

---

`RELATIVE_TAIL_CONTROL` PROVED

KKL long-tail width is a small relative error when standard width and marking complexity are small relative to degeneration.

*Formal hypotheses: none*

*Proof dependencies:* `KKL_TRANSLATION`

*Evidence:* `virtual_block_quotient.md#5-width-of-interactions-that-cannot-be-compressed`

---

`CONE_REDUCTION` PROVED

Positive expanding chart covectors convert componentwise transition inequalities with controlled error into one backward scalar recurrence.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `master_transition.md#vector-state-and-cone-calibration`

---

`HOMOGENEOUS_LYAPUNOV_REDUCTION` PROVED

Monotone subadditive homogeneous chart gauges convert expanding homogeneous transition inequalities with controlled error into the backward scalar recurrence.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `master_transition.md#vector-state-and-cone-calibration`

---

`ERROR_STABLE_LYAPUNOV_REDUCTION` PROVED

Monotone homogeneous gauges satisfying an explicit error-stability inequality, including bottleneck minima, convert expanding homogeneous transitions into the backward scalar recurrence.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `master_transition.md#vector-state-and-cone-calibration`

---

`FORWARD_CONTRACTION_REDUCTION` PROVED

A forward componentwise upper transition with an error-stable gauge and strict forward scalar contraction implies backward scalar amplification with an explicit relative-error margin.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `master_transition.md#vector-state-and-cone-calibration`

---

`GROUPED_BLOCK_CONTRACTION_REDUCTION` PROVED

A complete grouped entropy-clock block with strict forward contraction and output-relative error below the contraction margin yields the MBTT backward amplification factor (1-delta)/mu>1.

*Formal hypotheses: none*

*Proof dependencies:* `FORWARD_CONTRACTION_REDUCTION`

*Evidence:* `block_contraction_program.md#4-closed-abstract-reduction`

---

**PRIME_END_RETURN** PROVED

For nondegenerate full planar continua whose complements are hyperbolic domains containing infinity, a proper holomorphic return between the complements induces a finite-degree covering of their prime-end circles.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* virtual_block_quotient.md#7-fullification-and-prime-end-realization

---

**CRITICAL_VALUE_PULLBACK** PROVED

If a full continuum contains every finite critical value of a polynomial return, its full preimage is connected and the complement return is proper.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* virtual_block_quotient.md#7-fullification-and-prime-end-realization

---

**FULLIFICATION** PROVED

Filling a connected compact set produces a full continuum and preserves genuine forward invariance under a polynomial.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* virtual_block_quotient.md#7-fullification-and-prime-end-realization

---

**GLOBAL_DEGREE_NO_GO** PROVED

When a full hull contains all critical values of f^q, the global complement return is a proper map of degree 2^q and cannot be the uniformly quadratic virtual return; for nondegenerate hulls with hyperbolic complements the same degree occurs on prime-end circles.

*Formal hypotheses: none*

*Proof dependencies:* PRIME_END_RETURN, CRITICAL_VALUE_PULLBACK

*Evidence:* virtual_block_quotient.md#7-fullification-and-prime-end-realization

---

**DEGREE_TWO_FACTORIZATION** PROVED

Any proper factorization of a degree-two return has one quadratic factor and all remaining factors conformal.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* virtual_block_quotient.md#7-fullification-and-prime-end-realization

---

**QUADRATIC_PULLBACK_DICHOTOMY** PROVED

A full continuum pulls back to two conformal full components when it avoids the quadratic critical value, and to one quadratic full component when it contains that value.

*Formal hypotheses: none*

*Proof dependencies:* CRITICAL_VALUE_PULLBACK

*Evidence:* virtual_block_quotient.md#7-fullification-and-prime-end-realization

---

**STAGEWISE_FACTORIZATION** PROVED

A finite inverse itinerary of nondegenerate full quadratic hulls with exactly one critical-value target admits a simultaneous compatible Jordan-neighborhood chain whose proper stage degrees multiply to two.

*Formal hypotheses: none*

*Proof dependencies:* QUADRATIC_PULLBACK_DICHOTOMY, DEGREE_TWO_FACTORIZATION

*Evidence:* virtual_block_quotient.md#7-fullification-and-prime-end-realization

---

**QL_RETURN_FACTORIZATION** PROVED

Every genuine quadratic-like endpoint return factors canonically through successive proper preimage components with one quadratic step.

*Formal hypotheses: none*

*Proof dependencies:* DEGREE_TWO_FACTORIZATION

*Evidence:* virtual_block_quotient.md#8-the-pseudo-quadratic-like-inflation-blueprint

---

**FINITE_HORIZON_TRUNCATION** PROVED

Every genuine all-horizon psi-ql correspondence canonically restricts to a depth-q datum for each finite q.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* finite_horizon_inflation.md#3-finite-horizon-category

---

**STAGE_FORWARD_CLOSURE** PROVED

A stage complement inclusion exists exactly when the excluded marking is forward closed into the next stage.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* finite_horizon_inflation.md#4-stage-dependent-partial-invariance

---

**RELATIVE_COVER_LIFT** PROVED

A based map lifts between selected connected covers exactly under the fundamental-group subgroup inclusion criterion.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* finite_horizon_inflation.md#5-relative-cover-lifting

---

**TOPOLOGY_WIDTH_NO_GO** PROVED

Topological finite-horizon psi-ql data alone control neither fundamental-annulus nor core-to-boundary inverse modulus.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* finite_horizon_inflation.md#7-quantitative-no-go-theorem

---

**INFLATION_MODULUS_MONOTONICITY** PROVED

Once a controlled local target collar embeds conformally in its inflation, target inverse annular modulus cannot increase.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* finite_horizon_inflation.md#8-one-sided-inflation-monotonicity

---

**FINITE_LOCAL_SATURATION** PROVED

A finite sequence of nondegenerate full continua in which each H_j is a selected component of f^{-1}(H_{j+1}), with separated compact obstacles, exactly one target containing the critical value, and every other target collar avoiding it, admits a qualitative locally saturated Jordan telescope with exactly one quadratic stage.

*Formal hypotheses: none*

*Proof dependencies:* QUADRATIC_PULLBACK_DICHOTOMY, DEGREE_TWO_FACTORIZATION

*Evidence:* finite_horizon_inflation.md#10-conditional-local-saturation-construction

---

**CONNECTED_EXTERIOR_CALIBRATION** PROVED

For disjoint nondegenerate continua H,E with sphere minus (H union E) connected, treating E as the connected exterior marking gives a doubly connected complement whose marked annular width is recovered by Jordan target collars with arbitrarily small error.

*Formal hypotheses: none*

*Proof dependencies:* INFLATION_MODULUS_MONOTONICITY

*Evidence:* marked_collar_calibration.md#2-exact-two-continuum-calibration

---

**WAD_CORRIDOR_SANDWICH** PROVED

A translation-realized corridor decomposition satisfies Q <= W <= Q + 4*S + C*kappa.

*Formal hypotheses: none*

*Proof dependencies:* KKL_TRANSLATION, KKL_THICK_THIN

*Evidence:* corridor_network_calibration.md#4-corridor-sandwich

---

**COMPLEXITY_COORDINATE_NO_GO** PROVED

Raw unbounded marking complexity cannot have a uniformly positive coefficient in a uniformly bounded coercive potential.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* corridor_network_calibration.md#5-raw-complexity-cannot-be-a-coercive-coordinate

---

**HARMONIC_LINEAR_NO_GO** PROVED

The harmonic sum of two unrestricted positive widths is not uniformly comparable to any fixed positive linear form.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* corridor_network_calibration.md#6-harmonic-transport-is-not-uniformly-linear

---

**BOUNDED_FANOUT_BOTTLENECK** PROVED

An m-fold harmonic sum lies between min(x_i)/m and min(x_i), so bounded fanout is controlled by a bottleneck coordinate.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* corridor_network_calibration.md#6-harmonic-transport-is-not-uniformly-linear

---

**ORBIT_PACKING** PROVED

If each good translation class injects at least q-b arcs into distinct global WAD coordinates, the coordinate sets of different classes are disjoint, and their representatives form an arc diagram, then the number of classes is bounded by marked arc-support complexity divided by q-b.

*Formal hypotheses: none*

*Proof dependencies:* KKL_WAD_SUPPORT

*Evidence:* corridor_network_calibration.md#9-orbit-packing-and-normalized-complexity

---

**TRANSLATION_DENSITY_IDENTITY** PROVED

An (n,p)-translation marking has exactly 2p ideal labels per translation step, so increasing n alone cannot absorb unbounded p.

*Formal hypotheses: none*

*Proof dependencies:* KKL_TRANSLATION

*Evidence:* block_contraction_program.md#3-open-analytic-theorem

---

**SQRT_ERROR_ABSORPTION** PROVED

Every C*sqrt(x) interface loss is bounded by delta*x plus a delta-dependent universal additive constant.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* neutral_transition_interface.md#4-aggregated-square-root-absorption

---

**CONDITIONAL_NEUTRAL_CORRIDOR_ABSORPTION** PROVED

If total orbit width is comparable to r*A and aggregate interface loss is O(r*sqrt(A)+r), then the relative loss is O(A^(-1/2)+A^(-1)), independently of tile count r.

*Formal hypotheses: none*

*Proof dependencies:* DL_TILE_SCALING, SQRT_ERROR_ABSORPTION

*Evidence:* neutral_transition_interface.md#4-aggregated-square-root-absorption

---

**COMMON_WIDTH_BRIDGE** PROVED

Two chart coordinates are quantitatively comparable when both are calibrated, with explicit additive and relative errors, to the same marked curve-family width.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* neutral_transition_interface.md#5-a-common-width-bridge

---

**REPLICATION_AMORTIZATION** PROVED

If active width grows like r*D while standard width and marked complexity grow at most linearly in r, their entire corridor charge is O(1/D) relative to active width.

*Formal hypotheses:* *none*

*Proof dependencies:* `WAD_CORRIDOR_SANDWICH`

*Evidence:* `corridor_network_calibration.md#11-replication-absorbs-linear-complexity`

---

**CONGESTION_AMORTIZATION** PROVED

If marking charge grows with r_mark while active width grows with r_parallel*D, the relative corridor error is O((r_mark/r_parallel)/D).

*Formal hypotheses:* *none*

*Proof dependencies:* `WAD_CORRIDOR_SANDWICH`

*Evidence:* `corridor_network_calibration.md#12-density-and-congestion-form`

---

**TREE_NERVE_FULLNESS** PROVED

A finite recursive union of full plane branch continua attached one at a time along nonempty connected intersections is full.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `virtual_hull_realization.md#2-recursive-full-carriers`

---

**TREE_HULL_COMPLEXITY** PROVED

An O(R)-occupancy recursive full carrier has corridor complexity O(R) after explicit bounded-fiber assignments for ideal/access/translation markings and an O(R) finite-type Euler bound are proved.

*Formal hypotheses:* *none*

*Proof dependencies:* `TREE_NERVE_FULLNESS`

*Evidence:* `virtual_hull_realization.md#3-access-complete-incidence`

---

**VIRTUAL_LIMB_TREE_BOUND** PROVED

A reduced anchored limb tree spanning m non-anchor access-complete marked atoms has at most 4m-2 oriented tree sides; charging at most two local ideal decorations at every prescribed vertex, including the anchor, gives a complete limb multi-interval of size at most 6m. Selected-component occupancy alone gives no such bound when extra proof-used access atoms are unbounded.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `virtual_limb_marking_bound.md#3-exact-tree-count`

---

**VIRTUAL_BRANCH_QUANTIFIER_ASSEMBLY** PROVED

A static split invoked at the wave-selector parameter P=P_w(B,L), a same-parameter tame closure Tame(P)->Amp(P), the wave closure Wave(P_w)->Amp(L), monotonicity of amplification, and the maximum of all displayed thresholds imply Amp(L). Separate tame and wave input parameters do not compose without a common-parameter or monotonicity theorem.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `virtual_branch_assembly.md#4-assembly-theorem`

---

**ABSTRACT_SECTION4_STATIC_TRICHOTOMY** PROVED

Every static virtual Section-4 datum satisfying S4-AC through S4-END and D>=max(1,5E_X(B)) is M-amplifying, M-tame in the declared normalization, or carries an M-wide controllable wave. This is a formal interface theorem and does not realize those axioms for a virtual hull.

*Formal hypotheses:* *none*

*Proof dependencies:* `VIRTUAL_LIMB_TREE_BOUND`

*Evidence:* `virtual_section4_interface.md#7-abstract-static-trichotomy`

---

`LOCALIZED_CRITICAL_ENCOUNTER_DEGREE` PROVED

A selected proper telescope with exactly m_I quadratic stages and all other stages conformal has total selected degree 2^m_I; bounding selected degree therefore requires a dynamical bound on critical encounters, not merely a bound on retained components.

*Formal hypotheses: none*

*Proof dependencies:* `DEGREE_TWO_FACTORIZATION`

*Evidence:* `virtual_tame_full_return.md#lemma-71-critical-encounter-degree`

---

`LINEAR_MARKING_DENSITY_ABSORPTION` PROVED

If complete marking cardinality is at most C_mark*s while active WAD norm is at least c_0*s*W_R-C_0*s, then above W_R>2C_0/c_0 the topology error ratio is at most 12C_mark/(c_0 W_R), independently of the period s.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_tame_full_return.md#lemma-72-linear-marking-density-absorbs-the-topology-term`

---

`TAME_VALUE_OBSTRUCTION` PROVED

For WADs 0<=Y<=Z, domination by g^*Z, Value contraction V(g^*Z)<=mu^(-1)V(Z), and loss V(Z-Y)<=C_V*delta*V(Z) are incompatible whenever C_V*delta<1-mu^(-1).

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_tame_full_return.md#lemma-73-abstract-value-obstruction`

---

`CONDITIONAL_VIRTUAL_TAME_TRANSFER` PROVED

A virtual block satisfying VR0-VR5 and VR7 admits the same-parameter implication Deg_{Q_t(B,P),A_t(B,P)} and Tame_P imply Amp_P in the realized KKL coordinate. Realization of those clauses and the unified-chart closure VR6 remain open.

*Formal hypotheses: none*

*Proof dependencies:* `KKL_PARABOLIC_TAME_AMPLIFICATION`, `KKL_VALUE_CONTRACTION`, `LOCALIZED_CRITICAL_ENCOUNTER_DEGREE`, `LINEAR_MARKING_DENSITY_ABSORPTION`, `TAME_VALUE_OBSTRUCTION`

*Evidence:* `virtual_tame_full_return.md#theorem-74-conditional-virtual-tame-full-return-transfer`

---

`ROTATION_RELABEL_NO_GO` PROVED

For a general reduced rotation a/q, multiplication by a^{-1} modulo q conjugates the permutation to translation by one but does not preserve the cyclic order unless a is congruent to 1 modulo q; a virtual proof therefore needs geometric acceleration, not a formal relabeling.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_tame_full_return.md#92-rotation-relabeling-obstruction`

---

`FINITE_HORIZON_WAVE_BOUND` PROVED

Assuming the separately open strict interval-growth implication used by KKL's bounded-visits lemma, all pullbacks used in one KKL wave-terminal argument fit inside the conservative horizon H_B(q)=(216B^2+1)(q+B+4), including the complete chain of critical visits.

*Formal hypotheses:* `KKL_BOUNDED_VISITS_INTERVAL_GROWTH_REPAIR`

*Proof dependencies: none*

*Evidence:* `virtual_long_iterate_wave.md#proposition-61-finite-horizon-sufficiency`

---

`DEGREE_TWO_FAST_FORWARD_NO_GO` PROVED

A single proper long-return component of total degree two cannot realize the KKL terminal fast-forward fan containing a proof-used region of degree 2^L for L>=2; auxiliary selected sheets or a replacement argument are necessary.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_long_iterate_wave.md#proposition-51-a-degree-two-endpoint-return-is-insufficient`

---

**FINITE_HORIZON_KKL_WAVE_REDUCTION** PROVED

Assuming the source-stated KKL Wave theorem and a wave-admissible virtual realization satisfying corrected WA1-WA12 through H_B(q), including the actual stationary degree-two graph component, a retained proper-path/measurable lamination, and the bounded marked auxiliary fan, the source proof schema admits synchronized selectors and period-independent thresholds for Wave_{P_w(B,L)} to imply Amp_L. This is a conditional reduction, not an independent certification of the KKL theorem.

*Formal hypotheses:* KKL_PARABOLIC_WAVE_AMPLIFICATION

*Proof dependencies:* FINITE_HORIZON_TRUNCATION, STAGE_FORWARD_CLOSURE, RELATIVE_COVER_LIFT, FINITE_HORIZON_WAVE_BOUND, DEGREE_TWO_FAST_FORWARD_NO_GO

*Evidence:* virtual_long_iterate_wave.md#8-the-minimal-conditional-theorem

---

**FINITE_INVERSE_TREE_FAN_SATURATION** PROVED

For a finite full inverse-component tree of a quadratic polynomial with separated compact obstacles, one root Jordan neighborhood can be chosen so all selected iterated pullback components are simultaneous disjoint proper Jordan sheets, each branch has degree 2 to the number of its critical encounters, and selected branches with encounter counts 0 through M form a proper dyadic fan of total degree 2^(M+1)-1.

*Formal hypotheses:* *none*

*Proof dependencies:* QUADRATIC_PULLBACK_DICHOTOMY, LOCALIZED_CRITICAL_ENCOUNTER_DEGREE

*Evidence:* selected_marked_fast_forward_fan.md#5-finite-inverse-tree-fan-saturation

---

**FORWARD_HULL_COMPONENT_NO_GO** PROVED

Fullness, inclusion of the critical point/value, and even surjective forward invariance f(H_-)=H_+ do not imply that H_- is a connected component of f^(-1)(H_+); pullback-component saturation is an independent requirement.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* selected_marked_fast_forward_fan.md#proposition-33-forward-full-hull-component-no-go

---

**SATURATED_SEED_FOREST** PROVED

A finite forward-coherent forest of connected compact seeds admits a unique pullback-component saturation relative to its filled root; with sibling separation, it is a disjoint full inverse forest whose branch degree is 2 to the number of critical targets.

*Formal hypotheses:* *none*

*Proof dependencies:* FULLIFICATION, QUADRATIC_PULLBACK_DICHOTOMY

*Evidence:* saturated_virtual_seed_forest.md#3-canonical-saturation-theorem

---

**ROOT_PULLBACK_FOREST_EQUIVALENCE** PROVED

A finite saturated carrier forest rooted at a full continuum is equivalently the routed component family of its iterated root pullbacks: whole required packages must lie in single components, child components must map to the prescribed parents, and distinct selected sheets must be distinct same-depth components.

*Formal hypotheses:* *none*

*Proof dependencies:* SATURATED_SEED_FOREST, QUADRATIC_PULLBACK_DICHOTOMY

*Evidence:* root_pullback_virtual_carrier.md#3-root-pullback-equivalence

---

**CRITICAL_VALUE_CORRIDOR_CRITERION** PROVED

For a quadratic polynomial and full target carrier, the selected pullback edge has degree one exactly when the target avoids the critical value and degree two exactly when it contains the critical value; hence a central corridor is degree one exactly when every one of its target carriers avoids the critical value.

*Formal hypotheses:* *none*

*Proof dependencies:* ROOT_PULLBACK_FOREST_EQUIVALENCE, QUADRATIC_PULLBACK_DICHOTOMY

*Evidence:* root_pullback_virtual_carrier.md#4-exact-degree-one-corridor-criterion

---

`FINITE_OBSTACLE_FORWARD_CLOSURE` PROVED

For a finite saturated seed forest, compact obstacles disjoint from the selected carriers admit simultaneous separated neighborhoods; if their images lie in the next obstacle-or-carrier set, the marked excluded sets are stagewise forward closed and give the canonical complement inclusions.

*Formal hypotheses:* *none*

*Proof dependencies:* `SATURATED_SEED_FOREST`, `STAGE_FORWARD_CLOSURE`

*Evidence:* `saturated_virtual_seed_forest.md#5-finite-obstacle-closure`

---

`FINITE_FAN_RIBBON_PULLBACK` PROVED

On a simultaneous proper Jordan fan, a finite root endpoint marking pulls back functorially with cyclic order and exact cardinality degree times the root cardinality; a dyadic fan with critical counts 0 through M has total pulled marking |P_r|(2^(M+1)-1), conditional on root future completeness.

*Formal hypotheses:* *none*

*Proof dependencies:* `SATURATED_SEED_FOREST`, `FINITE_INVERSE_TREE_FAN_SATURATION`

*Evidence:* `saturated_virtual_seed_forest.md#6-pullback-ribbon-marking`

---

`ENDPOINT_BIRTH_ACCOUNTING` PROVED

For boundary coverings of degrees d_j with b_j genuinely new endpoints at stage j, the initial complete marking count is D_(0,N-1)p_N plus sum_j D_(0,j-1)b_j. One birth per degree-one corridor stage grows linearly, while births confined to a bounded terminal quadratic window have a period-independent geometric-sum bound.

*Formal hypotheses:* *none*

*Proof dependencies:* `FINITE_FAN_RIBBON_PULLBACK`

*Evidence:* `marked_endpoint_birth_ledger.md#3-exact-birth-formula`

---

`LOCAL_CARRIER_EQUALITY_NO_BIRTH` PROVED

On a degree-one marked corridor interval, if the pulled carrier has no new component attached there, the immersion pullback is one interval, and the polynomial map is conformal on its endpoint prime ends, then the complete pulled marking has zero genuine endpoint births.

*Formal hypotheses:* *none*

*Proof dependencies:* `FINITE_FAN_RIBBON_PULLBACK`, `ENDPOINT_BIRTH_ACCOUNTING`

*Evidence:* `marked_endpoint_birth_ledger.md#proposition-71-local-carrier-equality-implies-no-birth`

---

`ATTACHMENT_SUPPORT_REDUCTION` PROVED

For any access-complete marked carrier satisfying the KKL-form interval pullback description, endpoint births vanish off the support of attached pulled-carrier excess; if that support and all genuine births are confined to a terminal depth-M degree-at-most-two window with at most b_* births per stage, the initial weighted birth charge is at most b_*(2^M-1).

*Formal hypotheses:* *none*

*Proof dependencies:* `KKL_IDEAL_PULLBACK_NATIVE`, `ENDPOINT_BIRTH_ACCOUNTING`, `LOCAL_CARRIER_EQUALITY_NO_BIRTH`

*Evidence:* `kkl_translation_no_birth_source_audit.md#4-abstract-attachment-support-reduction`

---

`CORRIDOR_ATTACHMENT_TRANSPORT` PROVED

A commuting orientation-preserving homeomorphism of access-complete marked carrier pairs bijects the components of pulled-carrier excess and their attachment intervals, preserving proof-used prime-end sides and cyclic order. Hence attachment support transports exactly along a marked-pair natural degree-one corridor, and one attachment-free anchor makes the whole central support empty.

*Formal hypotheses:* *none*

*Proof dependencies:* `CRITICAL_VALUE_CORRIDOR_CRITERION`, `ATTACHMENT_SUPPORT_REDUCTION`

*Evidence:* `corridor_attachment_transport.md#4-attachment-transport-theorem`

---

`UNIVALENT_CARRIER_PAIR_TRANSPORT` PROVED

One actual marked anchor pair transported in its entirety through critical-value-free conformal corridor branches gives commuting pulled/immersed carrier pairs, equivariant package and obstacle ledgers, and orientation-preserving prime-end marking maps through the full finite horizon; no local connectivity of the Euclidean carrier boundaries is required.

*Formal hypotheses:* *none*

*Proof dependencies:* `CRITICAL_VALUE_CORRIDOR_CRITERION`, `CORRIDOR_ATTACHMENT_TRANSPORT`

*Evidence:* virtual_corridor_pair_naturality.md#4-univalent-transport-construction

---

**ATTACHMENT_FREE_ANCHOR_UNDERDETERMINATION** PROVED

A required package, full connected carriers, critical-value-free degree-one selected-carrier dynamics, and identical coarse orbit or limb labels do not force an attachment-free anchor: an interval pair and an interval embedded in a triod have the same package and selected dynamics, while only the triod pair has attached excess.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* virtual_corridor_pair_naturality.md#5-exact-anchor-underdetermination

---

**FINITE_ATTACHMENT_ORBIT_CLOSURE** PROVED

Every inflationary monotone attachment/access/transport completion rule on a finite resolved carrier stabilizes at the least fixed carrier after at most the number of initially omitted cells; the least closure is functorial under conjugate corridor bijections, and any sheet, critical-word, obstacle, access, or localization collision in it persists in every monotone attachment-complete enlargement.

*Formal hypotheses: none*

*Proof dependencies:* ATTACHMENT_SUPPORT_REDUCTION, CORRIDOR_ATTACHMENT_TRANSPORT

*Evidence:* virtual_corridor_pair_naturality.md#6-finite-attachment-orbit-closure

---

**INTERFACE_PROOF_USE_INVENTORY** PROVED

A finite Wave proof-use ledger has a unique minimal source-complete inventory obtained by taking the typed union of every component, orbit index, access, obstacle, return word, curve family, and intervening transport stage read by the proof.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* canonical_virtual_interface_carrier.md#proposition-31-minimal-proof-use-inventory

---

**HORIZON_FIRST_INVENTORY_AUDIT** PROVED

For a fixed finite typed Wave-horizon dependency graph, reachability from the complete proof-use seed set gives the unique least source-complete inventory and stabilizes in finitely many steps. Under a horizon extension the inventory is monotone, and its old-stage slice is unchanged exactly when no extended proof seed has a dependency path to an old omitted record.

*Formal hypotheses: none*

*Proof dependencies:* INTERFACE_PROOF_USE_INVENTORY

*Evidence:* vaapc_inventory_source_audit.md#theorem-41-horizon-first-inventory-and-extension-law

---

**RESOLVED_FINITE_WAVE_HORIZON_COMPILER** PROVED

A finite acyclic analytic Wave decision program with actual marked covering/immersion branches, exact iterate and degree words, complete theorem-hypothesis and proof-use ledgers, a finite resolved dependency universe, and generator-by-generator proofs preserving marked action, properness, degree, obstacles, localization, and protected access under root enlargement has a finite maximal iterate horizon, a unique least proof-use inventory, and an order-independent analytically stable least simultaneous final-root enlargement; under a program extension the old inventory remains complete exactly when no new seed reaches an old omitted record.

*Formal hypotheses: none*

*Proof dependencies:* HORIZON_FIRST_INVENTORY_AUDIT

*Evidence:* virtual_wave_horizon_certificate.md#theorem-71-resolved-finite-wave-horizon-compiler

---

**KKL_NATIVE_WAVE_HORIZON_CONTROL** PROVED

Conditional on the separately open strict interval-growth implication in KKL's bounded-visits lemma, the native Wave theorem-call program stops by the mostly-short/critical-visit alternatives before forming a chain of 216b^2 waves; each selected visit carries a finite witnessed iterate and uses the marked pair (f^r,iota_r) on the pulled Hubbard-tree complement. This is a b-dependent conditional source-program certificate, not a proof of Wave amplification.

*Formal hypotheses:* KKL_BOUNDED_VISITS_INTERVAL_GROWTH_REPAIR

*Proof dependencies:* RESOLVED_FINITE_WAVE_HORIZON_COMPILER

*Evidence:* virtual_wave_horizon_certificate.md#4-explicit-theorem-call-stopping-rule

---

**RETURN_PATH_LANGUAGE** PROVED

The finite directed paths from the distinguished root of an actual typed branch graph form a horizon-independent prefix-closed return-word language; each word has finite iterate length equal to the sum of its edge lengths and retains every intermediate component, critical, obstacle, localization, sheet, and access record.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_wave_marked_correspondence.md#theorem-31-return-path-language`

---

**FORWARD_COHERENT_COMPLEMENT_PAIR** PROVED

If a nondegenerate full continuum contains every finite critical value of a polynomial return and is forward invariant under that return, its full preimage contains it, and the polynomial return together with complement inclusion gives a connected unbranched finite-degree covering/embedding pair on the plane complements.

*Formal hypotheses: none*

*Proof dependencies:* `CRITICAL_VALUE_PULLBACK`, `STAGE_FORWARD_CLOSURE`

*Evidence:* `virtual_wave_marked_correspondence.md#theorem-41-forward-coherent-complement-pair`

---

**PRIME_END_MARKING_ACTION_CRITERION** PROVED

For a covering/immersion pair with defined complete protected prime-end pullbacks, the covering-pulled marking refines the immersion-pulled marking exactly when every immersion-pulled endpoint is a covering-pulled endpoint; when the immersion boundary map is a homeomorphism this is forward invariance under the induced boundary transition, and the least endpoint closure exists as an ideal marking exactly when its orbit closure is nowhere dense.

*Formal hypotheses: none*

*Proof dependencies:* `PRIME_END_RETURN`

*Evidence:* `virtual_wave_marked_correspondence.md#5-exact-prime-end-marking-criterion`

---

**MARKED_PAIR_ENLARGEMENT_NATURALITY** PROVED

A carrier-package enlargement preserves exact marked action when the covering and immersion squares commute analytically, their protected prime-end extensions are injective and cyclic-order preserving, and both endpoint sets are exact inverse images under the enlargement maps; the conclusion is stable under finite composition.

*Formal hypotheses: none*

*Proof dependencies:* `PRIME_END_MARKING_ACTION_CRITERION`

*Evidence:* `virtual_wave_marked_correspondence.md#theorem-61-marked-pair-enlargement-naturality`

---

**RETURN_BASE_UNDERDETERMINATION** PROVED

A quadratic polynomial, a compact seed, and the instruction to take first returns to a connected Jordan neighborhood do not determine a return time or analytic branch: for f(z)=z^2-1 and seed {-1}, two explicit Jordan disks give return times two and one respectively.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_return_branch_language.md#theorem-31-return-base-underdetermination`

---

**NATIVE_PRINCIPAL_NEST_RETURN_ATLAS** PROVED

At every returning native principal-nest level n>=2, the open nice base int(V^{n-1}) has an actual first-return atlas with connected open puzzle components, exact integer iterates, chronological degree words, component/parent/sheet itineraries, native obstacle data, and cyclic-order-preserving prime-end markings normalized by external angle.

*Formal hypotheses: none*

*Proof dependencies:* `LYUBICH_PRINCIPAL_RETURN_MAP`, `PRIME_END_RETURN`

*Evidence:* `virtual_return_base_atlas.md#4-native-principal-nest-return-atlas-theorem`

---

**ISOTOPY_CLASS_RETURN_UNDERDETERMINATION** PROVED

Even the ambient isotopy class of a Jordan return neighborhood relative to a compact seed does not determine first-return time: for f(z)=z^2-1, the concentric disks D(-1,1/4) and D(-1,5/4) are explicitly isotopic relative to {-1} but give return times two and one.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* virtual_return_base_atlas.md#6-isotopy-does-not-select-a-return-atlas

---

**FIRST_DECORATION_SCALE_ESCAPE_ATLAS** PROVED

At the first satellite scale visible in the original connected dynamical plane, a fixed non-principal decoration canonically selects relative puzzle depth nq+1 by the least critical q-return escape, and its actual first-return components have exact integer iterates, chronological critical words, component/parent/sheet records, and native prime-end markings.

*Formal hypotheses: none*

*Proof dependencies:* KL_DECORATION_ESCAPE_PUZZLE, PRIME_END_RETURN

*Evidence:* virtual_scale_to_puzzle_selector.md#4-first-decoration-scale-escape-atlas

---

**CRITICAL_BASE_CYCLE_EXCLUSION** PROVED

For the terminal primitive first-return branch f^P:A_0->B selected by the decoration nest, every intermediate little Julia set K_i=f^i(K_0), 1<=i<P, lies outside the open critical return base B; the first excluded component is K_1 and the complete return degree word is (2,1,...,1).

*Formal hypotheses: none*

*Proof dependencies:* KL_DECORATION_ESCAPE_PUZZLE, FIRST_DECORATION_SCALE_ESCAPE_ATLAS

*Evidence:* virtual_scale_to_puzzle_selector.md#6-a-single-critical-base-cannot-carry-the-cycle

---

**DECORATION_FAN_GROWTH** PROVED

The raw quadratic decoration separator Y^{(n-1)q+1} has exactly 2^n vertices and pairwise-disjoint marked pullbacks Q^v, so any access inventory retaining every such separator has cardinality at least 2^n and is not uniformly bounded over decoration depth.

*Formal hypotheses: none*

*Proof dependencies:* KL_DECORATION_ESCAPE_PUZZLE

*Evidence:* virtual_scale_to_puzzle_selector.md#7-raw-access-growth

---

**PERIODICALLY_REPELLING_PQL_FILLED_JULIA_EMPTY_INTERIOR** PROVED

A periodically repelling psi-quadratic-like map has empty-interior filled continuum: its same-filled-continuum induced quadratic-like restriction has no wandering Fatou components, and any nonempty interior would eventually yield an attracting, parabolic, or Siegel periodic component and hence a nonrepelling periodic cycle.

*Formal hypotheses: none*

*Proof dependencies:* PERIODICALLY_REPELLING_FILLED_JULIA_EMPTY_INTERIOR, PQL_SAME_FILLED_CONTINUUM_QUADRATIC_LIKE_RESTRICTION

*Evidence:* terminal_relative_cover_alignment.md#theorem-31

---

**NATIVE_HUBBARD_SUBTREE_FUNCTORIALITY** PROVED

A finite marked vertex package in KKL's native combinatorial Hubbard tree has a unique least convex subtree and a compact connected geometric realization; every tree map sends the hull over the hull of the image package, with equality when it is injective on the selected subtree, and package enlargement adds a forest whose components each have one old-tree gate.

*Formal hypotheses: none*

*Proof dependencies:* KKL_NATIVE_TERMINAL_HUBBARD_TREE

*Evidence:* terminal_hubbard_anchor.md#5-functoriality-under-a-tree-map

---

**TERMINAL_CYCLE_HUBBARD_ANCHOR** PROVED

Under KKL's native periodically repelling hypotheses, the complete terminal primitive little-Julia cycle selected by the direct-decoration nest has the native Hubbard tree as a compact connected forward-invariant carrier; the unique tree path from K_1 to K_0 selects its native ideal attachment side and retains the exact cycle itinerary and degree word.

*Formal hypotheses: none*

*Proof dependencies:* KKL_NATIVE_TERMINAL_HUBBARD_TREE, NATIVE_HUBBARD_SUBTREE_FUNCTORIALITY, CRITICAL_BASE_CYCLE_EXCLUSION

*Evidence:* terminal_hubbard_anchor.md#7-the-terminal-primitive-anchor

---

**TERMINAL_NATIVE_HUBBARD_CARRIER_FULLNESS** PROVED

A compact connected subcontinuum T of a full continuum K with empty interior is full: any bounded complementary component of T would be an open subset of K. Therefore the native periodically-repelling terminal Hubbard carrier is a nondegenerate full continuum, without assuming local connectivity.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_CYCLE_HUBBARD_ANCHOR, PERIODICALLY_REPELLING_PQL_FILLED_JULIA_EMPTY_INTERIOR

*Evidence:* terminal_relative_cover_alignment.md#theorem-31

---

**TERMINAL_LC_NATIVE_HUBBARD_CARRIER_PROPER** PROVED

In KKL's locally connected periodically-repelling primitive category, the native Hubbard carrier T is a proper subcontinuum of K. The nondividing fixed point beta_F lies in no member of the pairwise-disjoint primitive little-Julia cycle, while K minus beta_F is connected and contains that cycle; inclusion-minimality of T therefore gives beta_F not in T.

*Formal hypotheses: none*

*Proof dependencies:* KKL_NATIVE_TERMINAL_HUBBARD_TREE, TERMINAL_CYCLE_HUBBARD_ANCHOR

*Evidence:* terminal_dense_obstacle_and_submulticurve_replacement.md#proposition-21

---

**TERMINAL_RELATIVE_COVER_SUBGROUP_ALIGNMENT** PROVED

For any common simply connected plane localization with terminal full continua T subset K compactly contained in Omega, A=Omega minus K and M=Omega minus T are doubly connected and the inclusion A->M induces the positive identity isomorphism on pi_1 under the common outer-collar generator. The inverse image of A in the universal cover of M is therefore connected and universal. Identification with a KL tile additionally requires an explicit orientation-preserving conformal base-domain isomorphism beta and paired lifted basepoints; in the terminal specialization Omega=U^m and beta=id. This aligns only the subgroup and cover component, not proper paths or ideal ends.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_NATIVE_HUBBARD_CARRIER_FULLNESS, KL_DECORATION_ESCAPE_PUZZLE

*Evidence:* terminal_relative_cover_alignment.md#theorem-51

---

**TERMINAL_PROPER_ACCESS_CHANNEL_ATLAS** PROVED

Inside the one connected terminal pulled-back universal cover, define a certified source ideal arc by an actual one-sided quadrilateral whose inclusion has a topologically embedded ambient prime-end extension, preserves two flanks and cyclic order, and whose ambient compactified image avoids the complete lifted obstacle ledger. The union of such arcs over a fixed native ideal interval is open; its components are at most countably many intervals, or the whole circle, and every compact subarc has a finite collar atlas. This formal atlas proves neither nonemptiness nor whole-channel injectivity and does not count arbitrary KKL proper paths.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_RELATIVE_COVER_SUBGROUP_ALIGNMENT, TERMINAL_ANCHOR_COMPACT_HALF_COLLARS

*Evidence:* terminal_relative_cover_alignment.md#theorem-62

---

**TERMINAL_LOCAL_CHANNEL_TO_KL_WELDING** PROVED

Conditional on one actual certified proper-access arc J in a fixed conformally based terminal alignment and on its intrinsic KL prime-end image lying in one declared relative-interior eta/xi tile side with compatible orientations, the ambient native arc and KL arc are joined by an explicit orientation-preserving prime-end homeomorphism. Every compact subarc has a conformal welding of marked compactified quadrilateral collars with distinct ordered flanks and exactly equal marked modulus. The maps agree on source-compatible overlaps. No channel existence, coverage, cross-seam continuation, endpoint modulus, or global access theorem is asserted.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_RELATIVE_COVER_SUBGROUP_ALIGNMENT, TERMINAL_PROPER_ACCESS_CHANNEL_ATLAS, FINITE_GRID_AND_TERMINAL_KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY

*Evidence:* terminal_relative_cover_alignment.md#theorem-63

---

**TERMINAL_KL_FINITE_SIDE_REFINEMENT** PROVED

For any certified terminal proper-access arc in a fixed proof-used KL tile, cut at the preimages of the two finite central endpoints and two deck-limit endpoints in the exact strip model. Each complementary source subarc remains certified and maps into exactly one of the outer side, the two exposed J-sides, or the intrinsic central/buried interval. Only a correctly tile-tagged eta/xi component in the v_* access receives the local welding; outer/wrong-side, central, finite-incidence, and wrapping cases remain explicit obstructions.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_LOCAL_CHANNEL_TO_KL_WELDING, FINITE_GRID_AND_TERMINAL_KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY

*Evidence:* terminal_relative_cover_alignment.md#theorem-64

---

**TERMINAL_FINAL_ROOT_COVER_OBSTACLE_DECOMPOSITION** PROVED

For every authorized connected terminal enlargement T subset T+ subset K, all carriers are full; in the universal cover of Omega minus T, the inverse image of Omega minus T+ is connected and universal, and the lifted obstacle set K minus T decomposes exactly and disjointly as the new-carrier part T+ minus T plus the residual part K minus T+. Thus no hidden interior cover obstacle is born. Ideal-channel survival and end-proper factorization remain open.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_NATIVE_HUBBARD_CARRIER_FULLNESS, TERMINAL_RELATIVE_COVER_SUBGROUP_ALIGNMENT

*Evidence:* terminal_relative_cover_alignment.md#theorem-71

---

**TERMINAL_CERTIFIED_CHANNEL_FINAL_ROOT_STABILITY** PROVED

Every already certified terminal proper-access collar whose compactified ambient image avoids the complete lifted obstacle ledger factors through every authorized connected final-root enlargement T subset T+ subset K. The same physical Jordan quadrilateral defines injective local prime-end maps through the intermediate universal cover, preserves the two flanks and cyclic order, and gives the identical local KL welding and marked modulus. This is local stability of certified channels, not existence or whole-side coverage.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_PROPER_ACCESS_CHANNEL_ATLAS, TERMINAL_LOCAL_CHANNEL_TO_KL_WELDING, TERMINAL_FINAL_ROOT_COVER_OBSTACLE_DECOMPOSITION

*Evidence:* terminal_relative_cover_alignment.md#theorem-72

---

**TERMINAL_OUTER_SEAM_DECK_NORMALIZATION** PROVED

An oriented spanning seam in the prime-end closed annulus canonically constructs the cut-and-stack universal cover, a zero sheet, and the positive deck generator. Finite transverse paths have deck phase equal to signed seam crossings. In the terminal KL annulus the marked external ray of angle zero supplies the source seam; a separately retained target KL lift fixes the unique based comparison map. The seam does not itself select the target lift or an inner protected access.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_RELATIVE_COVER_SUBGROUP_ALIGNMENT, KL_DECORATION_ESCAPE_PUZZLE

*Evidence:* terminal_relative_cover_alignment.md#theorem-112

---

**TERMINAL_SEAM_LIFT_END_PROPERNESS** PROVED

For the exact annular terminal base, KKL's ideal closure is canonically and presentation-independently identified with the prime-end closed annulus. Every KKL-proper physical path has a unique seam-based lift extending to finite points on the boundary lines of the cut-and-stack cover, so it is a proper lifted path and cannot land at either deck-limit corner. The conformal based KL comparison transports those finite endpoints. This proves path-level properness only; access-label membership, boundary-subset map properness, and whole-end end-properness remain separate.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_OUTER_SEAM_DECK_NORMALIZATION, TERMINAL_RELATIVE_COVER_SUBGROUP_ALIGNMENT

*Evidence:* terminal_relative_cover_alignment.md#theorem-113

---

`TERMINAL_KL_FINITE_ENDPOINT_FIVE_WAY_CLASSIFICATION` PROVED

In the exact strip model of a tagged terminal KL tile, the intrinsic ideal circle is the disjoint union of the finite-coordinate outer line, two exposed J-side components, the interior of the compact central interval representing K_Y, its two finite endpoints, and the two deck-limit points. Seam-lift end properness excludes the deck limits, leaving exactly five cases: outer, either exposed J-side, central/buried, or finite incidence. The desired v_* access is a tile-tagged eta_i or xi_{i+1} predicate; the classifier does not prove that the actual endpoint avoids the central interval or has the desired tag.

*Formal hypotheses: none*

*Proof dependencies:* `TERMINAL_SEAM_LIFT_END_PROPERNESS`, `FINITE_GRID_AND_TERMINAL_KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY`

*Evidence:* `terminal_relative_cover_alignment.md#theorem-114`

---

`DENSE_ONE_GATE_HAIR_ACCESS_NO_GO` PROVED

There is an explicit compact path-connected full empty-interior planar continuum obtained by attaching shrinking vertical hairs at a dense set of one-gate points of a base interval. After the enlargement, no nonempty upper subinterval has an obstacle-free strong ambient physical-side collar. Thus fullness, empty interior, and one-gate attachment alone do not imply protected side survival. The model does not rule out weaker prime-end or snaking channels and is not asserted to be a polynomial Julia carrier.

*Formal hypotheses: none*

*Proof dependencies:* `ALEXANDER_DUALITY_COMPACTA`

*Evidence:* `terminal_relative_cover_alignment.md#theorem-121`

---

`TERMINAL_ACCESS_SURVIVAL_EQUIVALENCE` PROVED

For a nondegenerate compact native terminal interval I_0, the following are equivalent: the complete lifted obstacle closure misses I_0 in the native prime-end disk; I_0 has an ambient-ledger-free native Jordan collar; and I_0 is contained in the native image of a certified source proper-access arc. The nontrivial transfer is proved in a disk coordinate where the source and target domains literally agree near the obstacle-free boundary arc, so crosscut chains construct the source prime-end arc without local connectivity. Correctly tagged refined components then have local KL welding and certified final-root stability.

*Formal hypotheses: none*

*Proof dependencies:* `TERMINAL_RELATIVE_COVER_SUBGROUP_ALIGNMENT`, `TERMINAL_ANCHOR_COMPACT_HALF_COLLARS`, `TERMINAL_PROPER_ACCESS_CHANNEL_ATLAS`, `TERMINAL_KL_FINITE_SIDE_REFINEMENT`, `TERMINAL_CERTIFIED_CHANNEL_FINAL_ROOT_STABILITY`

*Evidence:* `terminal_protected_access_reduction.md#theorem-41`

---

`TERMINAL_FORWARD_INVARIANT_CARRIER_DENSE_OBSTACLE_NO_GO` PROVED

Let g be a connected polynomial-like map with empty-interior filled continuum K, and let T be a nondegenerate compact connected forward-invariant proper subcontinuum. In every finite-modulus annular localization, the PE closure of K minus T contains the entire inner prime-end boundary of the carrier complement; the lifted relative obstacle likewise contains every finite inner ideal point and both deck-limit corners. Otherwise an obstacle-free arc would transfer to an open external-prime-end arc whose impressions lie in T, and external exactness would force all of K into T. KKL's locally connected native primitive carrier is proper, so the proposed universal terminal nonaccumulation collar is false already in that native subcategory.

*Formal hypotheses: none*

*Proof dependencies:* `DOUADY_HUBBARD_QL_EXTERNAL_PRIME_END_EXACTNESS`, `PQL_SAME_FILLED_CONTINUUM_QUADRATIC_LIKE_RESTRICTION`, `PERIODICALLY_REPELLING_PQL_FILLED_JULIA_EMPTY_INTERIOR`, `TERMINAL_ACCESS_SURVIVAL_EQUIVALENCE`, `TERMINAL_LC_NATIVE_HUBBARD_CARRIER_PROPER`

*Evidence:* `terminal_dense_obstacle_and_submulticurve_replacement.md#theorem-42`

---

`TERMINAL_KL_ORDERED_MULTICURVE_OVERFLOW` PROVED

For finite ordered multicurves, define metric length as the sum of component lengths. Holomorphic image monotonicity and ordered-subarc overflow then give L(H)>=L(G), equivalently W(H)<=W(G), whenever every target curve contains pairwise parameter-disjoint ordered subarcs forming the image of a source multicurve. The proof uses the pushforward metric sigma(w)^2=sum rho(z)^2/|f'(z)|^2 and retains image multiplicity.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* terminal_dense_obstacle_and_submulticurve_replacement.md#lemma-50-ordered-multicurve-overflow

---

**TERMINAL_KL_SUBMULTICURVE_OVERFLOW_TRANSFER** PROVED

For a KL geometric puzzle piece Y and distinct vertices v,w, every physical curve connecting S_Y^v to S_Y^w contains an ordered submulticurve lifting to a horizontal multicurve in Psi(Y) between the actual tile-tagged accesses. The resulting relation has nonempty provenance fibers and gives the exact sharp inequality W(H_Y(v,w))<=W(Gamma_Y(v,w))=Bd_Y(v,w)^(-1), with no degree, additive, or collar charge.

*Formal hypotheses: none*

*Proof dependencies:* KL_DECORATION_ESCAPE_PUZZLE, TERMINAL_KL_ORDERED_MULTICURVE_OVERFLOW

*Evidence:* terminal_dense_obstacle_and_submulticurve_replacement.md#theorem-51

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**KKL_STATIONARY_ONE_RETAINED_CHAIN** PROVED

For every proof-used n and every alpha in KKL's A_(n+1), N-unbreakability supplies at least one retained proper restriction component at every one-step stage; all retained components at that stage have the same untagged proper image, namely the leaf selected by the next A_r-Wave. This proves neither uniqueness, compatibility as one segmented prefix flag, nor realization of either line-1603 separator-side portion.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* kkl_stationary_retained_realization_source_audit.md#theorem-51-one-retained-chain-with-possible-collisions

---

**KKL_FIXED_END_ACCESS_TYPE_OBSTRUCTION** PROVED

The imported fixed-end immersion theorem is typed by Freudenthal ends, while KKL's stationary BL_0 and L_0^h data are marked ideal multi-intervals/accesses; the locked source does not construct an eligible topological source end representing the selected access. Conditional on the actual physical-carrier hypotheses TWO_0 and TWO_h, boundary-compatible inclusion INC_h, and applicability of the imported theorem, the source and target tree complements have two Freudenthal ends, the source inner end is nonproper, and every paired exit term based at an eligible outer end has vertical topology. This proves no proper packet, filter survival, transverse accounting, or width charge.

*Formal hypotheses: none*

*Proof dependencies:* KKL_FIXED_END_IMMERSION_DECOMPOSITION

*Evidence:* kkl_stationary_fixed_end_topological_obstruction.md#6-effect-on-the-six-clause-certificate

---

**KKL_ACCESS_SEEDED_MAXIMAL_LIFT_MEASURABILITY** PROVED

For a local homeomorphism I:X->Y, a supplied continuous product family of pairwise-disjoint embedded target paths, and a supplied continuous lifted access seed on each Borel-compatible product piece, the joint maximal-lift domain is open, the maximal lift is continuous, and the exit time is lower semicontinuous and Borel. In any supplied compact metrizable ideal model, terminal oscillation, the landing domain, landing map, boundary code, and finite-horizon first-failure code are Borel. A merely Borel KL06 lamination does not supply the required product pieces, and the theorem proves neither seed-side access properness, native KKL product exhaustion, valid-lamination assembly, marking/filter survival, transverse-width accounting, nor a stationary response.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* kkl_access_seeded_maximal_lift_measurability.md#1-purpose-and-status

---

**KKL_BROAD_WAVE_PUSHFORWARD_COUNTEREXAMPLE** PROVED

Under KKL's printed broad Wave definitions, a finite cyclic annulus cover admits a proper marked source Wave of width M*w ending at a boundary component of degree M whose set-theoretic pushforward is one target Wave of width w. With fixed M=2 and w tending to infinity, this refutes the literal coefficient-one estimate W(A')<=W(A)+O(1+log M). The model does not refute Dudko–Lyubich's native one-common-interval annular theorem.

*Formal hypotheses: none*

*Proof dependencies:* KL06_EXTREMAL_WIDTH_SERIES_HOLOMORPHIC_RULES

*Evidence:* qa_extremal_wave_2026-07-26.md#7-positive-width-translated-rectangle-counterexample

---

`KKL_STATIONARY_EXACT_OCCURRENCE_SERIES_CONDUCTANCE` PROVED

For an exact full-width r-branch occurrence realization, if every target leaf contains r pairwise-disjoint untrimmed occurrence segments and each segment family is the exact holomorphic image of its paired source path family B_i, then W(S)<=oplus_i W(B_i). No immersion injectivity, deck noncollision, Wave typing, endpoint degree, controllability, or fixed-end construction is used.

*Formal hypotheses:* *none*

*Proof dependencies:* `KLO6_EXTREMAL_WIDTH_SERIES_HOLOMORPHIC_RULES`

*Evidence:* `kkl_stationary_bl0_survivor_analysis.md#5-the-series-inequality-before-pushforward`

---

`KKL_STATIONARY_CONDITIONAL_PAIRED_WAVE_RESPONSE_ALGEBRA` PROVED

Given the exact occurrence series inequality, actual marked source Waves ending on whole multi-intervals of degree d_i<=2^h, and a supplied common Wave-pushforward constant c_W, the pushed Waves satisfy max_i W(C_i)>=r*W(S)-C_b; for r=2, max(W(C_-),W(C_+))>=2W(S)-C_b. Distinct pushed images are unnecessary: if all outputs lie in one certified output Wave T, the same bound holds for W(T). This is packet-local and gives ambient terminality only after an ambient-width fraction, absolute threshold, and output controllability are independently proved.

*Ledger role:* optional alternative

*Formal hypotheses:* `KKL_WAVE_PUSHFORWARD_UNIFORM_PACKET_CONSTANT`

*Proof dependencies:* `KKL_STATIONARY_EXACT_OCCURRENCE_SERIES_CONDUCTANCE`

*Evidence:* `kkl_stationary_bl0_survivor_analysis.md#6-quantitative-wave-response`

---

`N1_PREPRUNING_BOREL_WAVE_ATLAS` PROVED

For every proof-used stationary KKL Wave, the native finite arc-packet inventory, strict planar nesting, injective area coordinate, standard-Borel leaf space, jointly Borel proper-path evaluation, complete prime-end marking, and filtered occurrence relation B_I(S) are constructed before any DL pruning. The construction uses BOWR Sections 2–5 and 7.2, assumes no transverse width measure, and is stable under the finite stationary restrictions used by Gate N1.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `kkl_stationary_n1_integration_audit_20260726.md#5-ordered-wave-realization-and-target-typing`

---

`N1_BOREL_RECTIFIABLE_CORE_REDUCTION` PROVED

Every actual proof-used one-common-interval packet S=B_I(S), equivalently for Gate N1 every proof-used proper Borel Wave already leafwise above one fixed connected source-boundary interval, splits Borelly into a rectifiable core, still leafwise above that interval, and a nonrectifiable family of extremal width zero. The total-variation predicate is Borel in the pre-pruning Wave atlas after round-annulus uniformization, and constant epsilon densities on the finite-area round annulus prove the zero-width claim.

*Formal hypotheses:* *none*

*Proof dependencies:* `N1_PREPRUNING_BOREL_WAVE_ATLAS`

*Evidence:* `kkl_stationary_n1_dl_relative_discard_bound_20260726.md#lemma-24-borel-rectifiable-core-reduction`

---

`DL_RELATIVE_DISCARD_FAMILY_BOUND` PROVED

For every actual proof-used proper Borel KKL Wave above one connected boundary interval, first remove its Borel zero-width nonrectifiable subfamily, then apply the native Dudko–Lyubich proof-internal deletion to the rectifiable core. The union of the zero-width family, T=O(log m) bounded-width deck-span packets, and one bounded outer buffer has width O(1+log m); for Gate N1 at degree two this is absolute.

*Formal hypotheses:* *none*

*Proof dependencies:* `N1_BOREL_RECTIFIABLE_CORE_REDUCTION`, `DL_NATIVE_RECTIFIABLE_RELATIVE_DISCARD_BOUND`

*Evidence:* `kkl_stationary_n1_integration_audit_20260726.md#3-native-relative-discard-bound`

**BOREL_WIDTH_ENVELOPE_PRUNING** PROVED

If an arbitrary, possibly non-Borel, subfamily D of a proper Borel lamination has extremal width at most C, then for every epsilon>0 it is contained in a Borel leaf subfamily E of width at most C+epsilon, and both E and its complement have Borel geometric support. The construction uses a pointwise lower-semicontinuous admissible-density majorant and a Borel weighted-length threshold; it assumes no transverse measure.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* kkl_stationary_n1_borel_width_envelope_20260726.md#theorem-11-borel-width-envelope

---

**STANDARD_BOREL_ORDERED_WAVE_REALIZATION** PROVED

Every proof-used stationary KKL Wave and every retained one-common-interval packet in Gate N1 admits a standard-Borel ordered leaf realization with jointly Borel proper-path evaluation, complete prime-end marking, Borel filtered occurrence code, Borel width-envelope pruning, exact singleton target direction, common whole-component target-Wave typing, and full source saturation through the complete N-step horizon.

*Formal hypotheses: none*

*Proof dependencies:* N1_PREPRUNING_BOREL_WAVE_ATLAS, DL_RELATIVE_DISCARD_FAMILY_BOUND, BOREL_WIDTH_ENVELOPE_PRUNING, KKL_STATIONARY_ONE_RETAINED_CHAIN

*Evidence:* kkl_stationary_n1_borel_ordered_wave_realization_20260726.md#theorem-bowr-stationary-borel-wave-realization-and-pruning

---

**KKL_STATIONARY_RELATIVE_WAVE_CONTINUATION_ANALYTIC** PROVED

For fixed b and N=b+3, the leafwise-directed critical, precritical, adjacent-limb, away, and one-packet estimates, combined with full source saturation and sequential degree-two relative transfer, construct nested stationary source families A_0 through A_(N-1) and prove W(A_n minus A_(n+1))<=C_b for 0<=n<N-1, with C_b independent of the rotation denominator, Wave width, arc-packet count, and selector history.

*Formal hypotheses: none*

*Proof dependencies:* DL_RELATIVE_DISCARD_FAMILY_BOUND, BOREL_WIDTH_ENVELOPE_PRUNING, STANDARD_BOREL_ORDERED_WAVE_REALIZATION, KKL_STATIONARY_ONE_RETAINED_CHAIN

*Evidence:* kkl_stationary_relative_wave_continuation.md#9-full-stationary-induction

---

**KKL_STATIONARY_RELATIVE_WAVE_CONTINUATION** PROVED

For KKL's native stationary families with N=b+3, there are explicitly source-saturated nested proper Borel subwaves A_0 through A_(N-1), whose output Waves retain the required no-short, unbreakability, leafwise direction, occurrence, endpoint, and crest data, and one C_b independent of the rotation denominator and all analytic widths such that W(A_n minus A_(n+1))<=C_b for every 0<=n<N-1. The proof uses the native one-common-interval DL discard bound, a Borel width envelope, exact singleton target typing, and multiplicity-aware degree-two relative transfer; it does not use the false broad coefficient-one KKL Wave claim.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_RELATIVE_WAVE_CONTINUATION_ANALYTIC

*Evidence:* kkl_stationary_n1_integration_audit_20260726.md#9-final-classification

---

**KKL_FIXED_END_WAD_SPLIT_FINITE_SELECTOR** PROVED

For one KKL immersion-decomposition witness, subdividing its paired realizing foliations into full-lift and other-end pieces gives valid target and source sub-WADs with equal paired coefficient norms in each piece. If finite eligible proper-degree-one source-end candidate relations with nonempty Borel fibers, oriented target-end equality, and Borel-saturated target transversals are separately supplied, least-index choice gives Borel selectors and a finite blockwise end-pair partition on which the source theorem applies in both orientations. This constructs neither the candidate relations nor a uniform end bound, same-exit coupling, marking-filter survival, or a two-sided proper source path.

*Formal hypotheses: none*

*Proof dependencies:* KKL_FIXED_END_IMMERSION_DECOMPOSITION

*Evidence:* kkl_fixed_end_immersion_switch_wad.md#6-finite-end-borel-selection-and-two-orientation-refinement

---

`KKL_STATIONARY_SURVIVOR_PARTITION_AND_POINTWISE_SWITCH` PROVED

On a supplied single-valued selected-route occurrence space, route exits split disjointly into ambient N-breaks, ambient-unbroken members of the native maximal short subwave, and nonshort ambient-singleton survivors; only the survivor class lies outside B_A=A^(N-br) union A^sh. For each survivor, every native one-step retained-predecessor fiber is nonempty, finite pointwise choice gives a full stagewise retained flag through N=b+3, and the first named-route failure is either strict path replacement or retained-path marked-state mismatch. With separately verified simple oriented Wave representatives, the segmented-occurrence lemma supplies the occurrence windows. No Borel replacement selector, uniform multiplicity, ideal-end survival, width bound, or response is proved.

*Formal hypotheses: none*

*Proof dependencies:* `KKL_STATIONARY_ONE_RETAINED_CHAIN`, `PQL_SEGMENTED_OCCURRENCE_RECONSTRUCTION`

*Evidence:* `kkl_stationary_survivor_route_switch_reduction.md#1-result-and-status`

---

`KKL_STATIONARY_ABSTRACT_FINITE_ROUTE_EXIT_SCHEMA` PROVED

For a supplied blockwise finite route skeleton with ambient horizon N and route horizon at most N-1, supplied finite Borel quotient maps, and a marked Borel occurrence presentation carrying an injective physical-leaf selector, Borel maximal continuation time, complete first-cause set, raw retained switch witness, and literal nonconstant exit subarc, the finite signature is Borel with finite image, its fibers and exit-path fields are Borel, and ordinary overflow gives W(full selected route family)<=W(selected exit-subarc family)<=the sum of the finitely many signature-fiber widths. Raw paths, real windows, and switch witnesses remain uncompressed. The result is blockwise and conditional on the supplied measurable and width-faithful presentation; it proves no native KKL presentation, response-sound uniform quotient, packet construction, or analytic conductance bound.

*Formal hypotheses: none*

*Proof dependencies:* `KKL_STATIONARY_SURVIVOR_PARTITION_AND_POINTWISE_SWITCH`, `TERMINAL_KL_ORDERED_MULTICURVE_OVERFLOW`

*Evidence:* `kkl_stationary_route_exit_invariant.md#1-result-and-status`

---

`KKL_STATIONARY_CONDITIONAL_VERTICAL_PACKET_CHARGE` PROVED

Let 1<=h<=b+2 and let a measurable stationary survivor sublamination have a proper vertical exit packet: a single-valued target-to-source restriction selection, full proper source paths joining the pulled-tree ideal boundary to the outer ideal boundary, native fine/coarse filter survival, a genuine source lamination without duplicate transverse mass, and a covering extension whose degree on the relevant ideal-end locus is D_h<=2^h. Then W(S_h^ver)<=D_h(V(F)+2)<=2^h(V(F)+2), and summing the first route horizon gives sum_h W(S_h^ver)<=(2^(b+4)-4)(V(F)+1). The theorem does not construct or measurably select such packets on the native survivor family.

*Formal hypotheses: none*

*Proof dependencies:* `KKL_STATIONARY_SURVIVOR_PARTITION_AND_POINTWISE_SWITCH`

*Evidence:* `kkl_stationary_route_exit_conductance_reduction.md#theorem-62-explicit-vertical-charge`

---

`PQL_STRICT_SYNCHRONIZED_CHAIN_RECONSTRUCTION` PROVED

For a finite Kahn relation tower, full one-step paths satisfying the exact pointwise identities f gamma_r = i gamma_(r+1) under one common parameter reconstruct a unique continuous interior path in the complete fiber product, and every such complete path gives those identities. This is the strict full-image special case only; it does not characterize KKL's sequential set-valued restriction by containment and proves no properness, marking, or retention.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `kkl_stationary_two_occurrence_binder_design.md#theorem-61-synchronized-chain-reconstruction`

---

`PQL_SEGMENTED_OCCURRENCE_RECONSTRUCTION` PROVED

A finite sequential restriction occurrence is recorded by all full proper stage paths together with occurrence maps beta_(r-1) a_(r-1) = i gamma_r. These data reconstruct a unique compatible path at every Kahn-tower prefix, with cumulative nested-window coordinates and full terminal projection; conversely the complete prefix flag reconstructs the segmented record. A proper degree-two annulus model proves that strict windows occur and that earlier truncated fiber-product coordinates can be nonproper, so one deepest common trace is not the minimal sequential invariant.

*Formal hypotheses: none*

*Proof dependencies:* PQL_STRICT_SYNCHRONIZED_CHAIN_RECONSTRUCTION

*Evidence:* kkl_stationary_segmented_occurrence_binder.md#6-finite-segmented-reconstruction

---

KKL_STATIONARY_FINITE_EMBEDDED_COLLAR_NATURALITY PROVED

For each fixed finite Kahn horizon, the complete immersion is an embedding on a neighborhood of the pulled filled set. On the complement of the pulled tree inside such a neighborhood it is a conformal isomorphism and preserves the induced ideal accesses and their cyclic order. An actual whole route portion confined to this collar, with access survival and obstacle avoidance, therefore lifts to a proper path. Wave subordination alone does not prove the required literal collar containment.

*Formal hypotheses: none*

*Proof dependencies:* PQL_ITERATE_FILLED_GRAPH_NATURALITY

*Evidence:* kkl_stationary_two_occurrence_source_attack.md#lemma-55-the-strongest-embedded-collar-route-theorem

---

KKL_STATIONARY_COLLAR_EXIT_OVERFLOW PROVED

For a fixed finite embedded Kahn collar Q_h of modulus mu_h, any measurable family of selected route germs that begins inside the inner collar and has a measurably selected source-lift segment crossing to the outer collar overflows the annular crossing family and therefore has extremal width at most 1/mu_h. This estimate covers only that explicitly selected crossing class; it is block-dependent and supplies no uniform comparison of 1/mu_h with the native vertical width.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_FINITE_EMBEDDED_COLLAR_NATURALITY, TERMINAL_KL_ORDERED_MULTICURVE_OVERFLOW

*Evidence:* kkl_stationary_quantitative_defect_repair.md#proposition-102-exact-continuation-exit-overflow

---

KAHN_QUALITATIVE_COLLAR_VERTICAL_CHARGE_NO_GO PROVED

The qualitative pseudo-polynomial-like, embedded-neighborhood, finite-horizon, degree-two, connected-filled-set, and bounded-vertical-width hypotheses do not imply a vertical-width-linear compatible collar charge. In the exact family U_R=D_R, U'_R=D_sqrt(R), K=closed D, T=[-a,a], f=z^2, i=inclusion, the vertical width stays bounded as R decreases to 1 while every horizon-h filled-set collar has reciprocal modulus at least 2*pi*2^h/log R. The model is not a native periodically repelling primitive KKL block, so it refutes source derivability rather than the intended full theorem.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* kkl_stationary_uniform_collar_charge_attack.md#theorem-41

---

KKL_STATIONARY_QSDB_SUFFICIENCY PROVED

Assuming the strengthened Quantitative Stationary Defect Bound with its transverse-measure clause and an HTND or R(F)+1 absolute-scale hypothesis, the corrected degree-two good family and charged defect family yield a disjoint estimate W(A)<=W(B)+a_b W(U minus T)+c_b W(RF)+C_b(k+1), where B=A^(N-br) union A^sh. After the vertical split, choose theta>max(lambda/lambda_3,1/(1+delta_3)) and apply KKL small-breaking to B. Terminality then transfers quantitatively from B to A, while the unbreakable branch lies in A^sh and is strictly lambda-short. Thus strengthened QSDB is sufficient for the stationary contribution without pointwise line-1603 pass-through.

*Formal hypotheses:* KKL_STATIONARY_QUANTITATIVE_DEFECT_BOUND

*Proof dependencies:* KKL_STATIONARY_DEGREE_TWO_CORRECTION

*Evidence:* kkl_stationary_quantitative_defect_repair.md#theorem-62-strengthened-qsdb-suffices-for-the-stationary-contribution

---

FINITE_MARKED_TREE_TERMINAL_PRUNING PROVED

If a finite tree G is the convex hull of a finite marked vertex set M, then every connected component of |G| minus one point contains a marked vertex. Consequently any subset containing a complete terminal side contains a marked vertex. Nonempty compact connected intersection alone is insufficient: a middle subsegment of an interval tree with only its two endpoints marked contains no mark.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#3-finite-marked-tree-pruning

---

`KKL_STATIONARY_CRITICAL_LIMB_ORBIT_DISTINCTNESS` PROVED

For KKL's primitive small-Julia cycle with rotation number 1/q, the critical component K_0 lies in limb 0, its image K_1 contains the critical value in limb 1, and the noncritical limb homeomorphisms place K_j in limb j for 1<=j<q. Hence the component period P is at least q; in the explicit high-type range q>b, every K,F(K),...,F^b(K) consists of b+1 distinct cycle components.

*Formal hypotheses: none*

*Proof dependencies:* `KKL_NATIVE_TERMINAL_HUBBARD_TREE`, `KKL_PRIMITIVE_ROTATION_LIMB_CYCLE_DATA`

*Evidence:* `kkl_stationary_n2_selected_predecessor_endpoint.md#5-endpoint-avoidance-and-distinctness-of-the-predecessor-cycle`

---

`KKL_STATIONARY_CRITICAL_LIMB_INVERSE_BRANCH` PROVED

For rotation 1/q, D_0=F^(-1)(limb 0) intersect limb 0 maps homeomorphically onto limb 0. The noncritical limb q-1 already supplies one preimage of each y in limb 0; y is not the critical value in limb 1; degree two and F(limb 0)=K(F) therefore supply exactly one noncritical preimage in limb 0. The resulting bijective local homeomorphism is a homeomorphism.

*Formal hypotheses: none*

*Proof dependencies:* `KKL_PRIMITIVE_ROTATION_LIMB_CYCLE_DATA`

*Evidence:* `kkl_stationary_n2_selected_predecessor_endpoint.md#lemma-21-critical-limb-inverse-branch-and-horizon-selection`

---

`KKL_STATIONARY_SELECTED_SMALL_JULIA_PREDECESSOR_CHAIN` PROVED

Let Q_0 be the actual critical member of the primitive small-Julia cycle, rather than silently identifying it with the undefined printed K_0^0 at KKL line 1597. Through H=b+2 define Q_n=phi_0(Q_(n-1)). Each Q_n is a whole connected component of F^(-1)(Q_(n-1)), F maps Q_n homeomorphically to Q_(n-1), and Q_n canonically selects the unique component L_0^n containing it. Thus Q_n subset L_0^n for every 0<=n<=H. No separator, collar, prime-end access, or proof-used Wave occurrence is asserted.

*Formal hypotheses: none*

*Proof dependencies:* `KKL_NATIVE_TERMINAL_HUBBARD_TREE`, `KKL_PRIMITIVE_ROTATION_LIMB_CYCLE_DATA`, `KKL_STATIONARY_CRITICAL_LIMB_INVERSE_BRANCH`

*Evidence:* `kkl_stationary_n2_selected_predecessor_endpoint.md#lemma-22-canonical-small-julia-predecessor-selection`

---

`KKL_STATIONARY_SELECTED_PREDECESSOR_CHAIN` PROVED

For H=b+2, construct the canonical Q_0-selected predecessor chain L_0^0,...,L_0^H in the critical limb, with L_0^0=T intersect limb 0 and each L_0^n a recorded connected component of the full set F^(-1)(L_0^(n-1)); record containment, Q_n subset L_0^n, and the actual-component address w_0=empty and w_n=(L_0^1,...,L_0^n), whose parent is truncation for n>=1, with equality meaning equality of the ordered actual subsets. Prove F(L_0^n) subset L_0^(n-1). No proof-used Wave binding is asserted.

*Formal hypotheses: none*

*Proof dependencies:* `KKL_NATIVE_TERMINAL_HUBBARD_TREE`, `KKL_STATIONARY_CRITICAL_LIMB_INVERSE_BRANCH`, `KKL_STATIONARY_SELECTED_SMALL_JULIA_PREDECESSOR_CHAIN`

*Evidence:* `kkl_stationary_n2_selected_predecessor_endpoint.md#lemma-22-canonical-small-julia-predecessor-selection`

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`KKL_STATIONARY_SELECTED_ALPHA_PREDECESSOR_CHAIN` PROVED

For H=b+2, construct the fixed-point predecessor chain alpha^0=alpha_F,alpha^1,...,alpha^H with F(alpha^n)=alpha^(n-1), actual-point address v_0=empty and v_n=(alpha^1,...,alpha^n), and parent truncation for n>=1. Degree two, the fixed preimage alpha_F, and the unique critical-limb inverse branch give existence and uniqueness. This certificate neither asserts alpha^n lies in closure(L_0^n) nor identifies alpha^n with an attachment gate.

*Formal hypotheses: none*

*Proof dependencies:* `KKL_STATIONARY_CRITICAL_LIMB_INVERSE_BRANCH`

*Evidence:* `kkl_stationary_n2_selected_predecessor_endpoint.md#lemma-21-critical-limb-inverse-branch-and-horizon-selection`

---

KKL_STATIONARY_TERMINAL_CRITICAL_LIMB_INVERSE_SHEETS PROVED

Without assuming local connectivity, the selected critical-limb inverse domain D_0=phi_0(limb 0) has boundary exactly {alpha^1} and is a connected component of K(F) minus {alpha^1}. More generally Omega_n=phi_0^n(limb 0) has closure Omega_n union {alpha^n} and is a connected component of K(F) minus {alpha^n}. The proof excludes accumulation at alpha_F by pairing the two distinct preimages in limbs 0 and q-1 and using local injectivity at alpha_F; it does not assume a locally finite arm-germ structure.

*Formal hypotheses: none*

*Proof dependencies:* KKL_PRIMITIVE_ROTATION_LIMB_CYCLE_DATA, KKL_STATIONARY_CRITICAL_LIMB_INVERSE_BRANCH, KKL_STATIONARY_SELECTED_ALPHA_PREDECESSOR_CHAIN

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#theorem-43-terminal-inverse-sheets-and-locally-connected-incidence

---

KKL_STATIONARY_SELECTED_PREDECESSOR_CONTINUUM_SATURATION PROVED

For the constructed chain through every 0<=n<=H=b+2, in which L_0^n is a connected component of the full inverse set F^(-1)(L_0^(n-1)) for n>=1, actual contact K_i intersect L_0^n nonempty forces the whole connected small-Julia component K_i to lie in L_0^n. The base case uses that no primitive cycle component contains alpha_F; the induction uses F(K_i) contact, the preceding depth, and connected-component maximality. Thus whole-continuum containment is not an additional open gate after actual contact.

*Formal hypotheses: none*

*Proof dependencies:* KKL_NATIVE_TERMINAL_HUBBARD_TREE, KKL_PRIMITIVE_ROTATION_LIMB_CYCLE_DATA, KKL_STATIONARY_SELECTED_PREDECESSOR_CHAIN

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#4-marked-core-contact-and-dynamic-continuum-saturation

---

KKL_STATIONARY_SELECTED_ENDPOINT_CYCLE_AVOIDANCE PROVED

For the constructed fixed-point predecessor chain through H=b+2, alpha^n is outside the primitive small-Julia cycle for every 0<=n<=H: otherwise F^n(alpha^n)=alpha_F and fixedness would place alpha_F successively in pairwise-disjoint cycle components. This does not identify alpha^n as the attachment gate of L_0^n.

*Formal hypotheses: none*

*Proof dependencies:* KKL_PRIMITIVE_ROTATION_LIMB_CYCLE_DATA, KKL_STATIONARY_SELECTED_ALPHA_PREDECESSOR_CHAIN

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#5-endpoint-avoidance-and-distinctness-of-the-predecessor-cycle

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KKL_STATIONARY_LC_TERMINAL_SHEET_MCI PROVED

If the physical KKL filled continuum K(F) is locally connected, then it is a dendrite, L_0^0=T intersect limb 0 is connected, and for T_n=F^(-n)(T) one has the actual set equality L_0^n=T_n intersect Omega_n. The terminal inverse-sheet theorem and finite-tree pruning then prove L_0^n intersect T nonempty implies some actual small-Julia continuum K_i is contained in L_0^n. Hence MCI_b and the stronger containment form hold in the locally connected physical category.

*Formal hypotheses: none*

*Proof dependencies:* FINITE_MARKED_TREE_TERMINAL_PRUNING, KKL_NATIVE_TERMINAL_HUBBARD_TREE, KKL_STATIONARY_SELECTED_PREDECESSOR_CHAIN, KKL_STATIONARY_SELECTED_ENDPOINT_CYCLE_AVOIDANCE, KKL_STATIONARY_TERMINAL_CRITICAL_LIMB_INVERSE_SHEETS, PERIODICALLY_REPELLING_PQL_FILLED_JULIA_EMPTY_INTERIOR

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#theorem-43-terminal-inverse-sheets-and-locally-connected-incidence

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KKL_STATIONARY_MODEL_TERMINAL_SHEET_INCIDENCE PROVED

In KKL's locally connected degree-two rational-lamination model, the terminal inverse-sheet and finite-tree pruning argument proves the correctly typed geometric statement: a selected model predecessor meeting the model Hubbard tree contains an entire marked model small-Julia continuum bar K_i. Only the subsequent finite-tree quotient collapses bar K_i to an abstract marked vertex. This model statement does not itself supply physical contact in a non-locally-connected lift.

*Formal hypotheses: none*

*Proof dependencies:* FINITE_MARKED_TREE_TERMINAL_PRUNING, KKL_NATIVE_TERMINAL_HUBBARD_TREE, KKL_STATIONARY_TERMINAL_CRITICAL_LIMB_INVERSE_SHEETS

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#corollary-44-model-incidence-and-exact-non-lc-residue

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KKL_STATIONARY_MONOTONE_FULL_LIFT_IMPLIES_MCI PROVED

Assume the corrected geometric monotone full-lift identification package through H=b+2. The singleton fixed fiber first forces all correctly labelled selected predecessor fibers to be singleton connected subsets of finite preimage sets. A monotone map onto a dendrite with such a singleton root fiber bijects the punctured-set components. Hence pi^(-1)(Omega_(n,com))=Omega_n; inverse images of arbitrary connected model subsets are connected by dendrite arc exhaustion; semiconjugacy and T=pi^(-1)(T_com) give pi^(-1)(L_(0,com)^n)=L_0^n. Model incidence supplies a geometric small-Julia continuum bar K_i subset L_(0,com)^n, and pi(K_i) subset bar K_i then forces K_i subset L_0^n. Thus the corrected package is sufficient for physical non-LC MCI; the package itself remains open.

*Ledger role:* optional alternative

*Formal hypotheses:* KKL_STATIONARY_MONOTONE_FULL_LIFT_IDENTIFICATION

*Proof dependencies:* KKL_STATIONARY_MODEL_TERMINAL_SHEET_INCIDENCE, KKL_STATIONARY_TERMINAL_CRITICAL_LIMB_INVERSE_SHEETS

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#proposition-45-monotone-full-lift-criterion

---

KKL_STATIONARY_SMALL_JULIA_POINT_COLLAPSE_OBSTRUCTION PROVED

For the monotone semiconjugacy obtained by transporting Kiwi's canonical model through KKL's same-filled quadratic-like restriction and straightening, no nondegenerate primitive small-Julia continuum K_i can map to one point. Periodic repulsion gives K(F)=J(F). The return quadratic-like map has two distinct fixed points: a repeated straightened fixed point would produce an open parabolic basin and, under the global quasiconformal straightening, nonempty interior in K_i inside K(F). Their images give two distinct F-periodic Julia points in every K_i, and Kiwi's map is injective on the finite-orbit Julia set. Thus the nondegenerate Kiwi geometric image pi_Kw(K_i) and any abstract vertex obtained by a later carrier-tree quotient belong to two different quotient levels.

*Formal hypotheses: none*

*Proof dependencies:* KKL_NATIVE_TERMINAL_HUBBARD_TREE, PQL_SAME_FILLED_CONTINUUM_QUADRATIC_LIKE_RESTRICTION, PERIODICALLY_REPELLING_PQL_FILLED_JULIA_EMPTY_INTERIOR, DOUADY_HUBBARD_CONNECTED_QUADRATIC_LIKE_STRAIGHTENING, KIWI_MONOTONE_MODEL_FINITE_ORBIT_INJECTIVITY

*Evidence:* kkl_stationary_n2_kiwi_model_identification_audit_20260727.md#4-point-collapse-obstruction

---

KKL_STATIONARY_LINE1598_TREE_DISJOINTNESS_REDUCTION PROVED

Assume F has a parabolically b-bounded primitive renormalization with rotation 1/q, q>b, the complete-horizon selected predecessor chain through H=b+2, and marked-core incidence MCI_b for its depth-b member. Continuum saturation upgrades the contacted K_b to K_b subset L_0^b; iteration and orbit distinctness then place b+1 distinct cycle components in the critical limb, contradicting its b-component bound and proving L_0^b intersect T is empty. The certified horizon recursion and forward invariance give L_0^h intersect T empty for h=b+1,b+2. This proves only set disjointness, not proof-used Wave binding or a proper interior path contact.

*Formal hypotheses:* KKL_STATIONARY_SELECTED_PREDECESSOR_MARKED_COMPONENT_INCIDENCE

*Proof dependencies:* KKL_STATIONARY_CRITICAL_LIMB_ORBIT_DISTINCTNESS, KKL_STATIONARY_SELECTED_PREDECESSOR_CONTINUUM_SATURATION, KKL_STATIONARY_SELECTED_PREDECESSOR_CHAIN

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#6-conditional-completion-of-line-1598

---

KKL_STATIONARY_LC_LINE1598_REPAIR PROVED

For a locally connected physical KKL filled continuum with parabolically b-bounded primitive renormalization of rotation 1/q and explicit q>b, the proved locally connected marked incidence supplies the hypothesis of the selected-predecessor count. Consequently L_0^h intersect T is empty for h=b,b+1,b+2. This repairs KKL line 1598 only in the locally connected physical category and proves no proper path, Wave occurrence, incoming host, or non-locally-connected lift.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_LC_TERMINAL_SHEET_MCI, KKL_STATIONARY_LINE1598_TREE_DISJOINTNESS_REDUCTION

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#corollary-63-unconditional-lc-line-1598-repair

---

**KKL_STATIONARY_MAXIMAL_ONE_SIDED_WINDOW_REDUCTION** PROVED

Let alpha be an embedded proper target path and I_h a holomorphic immersion. Every fixed connected component C of I_h^(-1)(|alpha|) has a unique canonical target-parameter window (a,c), because alpha^(-1) composed with I_h|C is a homeomorphism C->(a,c). If C is independently proved simple, component-maximal, proper at both marked ends, its specified deep access is the upper endpoint c<1, and its other end is the actual lifted BL_0 endpoint access of alpha, then C realizes the marked incoming BL_0-to-L_0^h route exactly when a=0. Properness of an arbitrary path contained in C does not imply that it is the component-maximal representative, and a deep-ending component can instead have the deep access at the lower endpoint.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `kkl_stationary_n2_incoming_window_anchor.md#3-host-component-occurrence-window-theorem`

---

**KKL_STATIONARY_OFF_BY_ONE_PASS_THROUGH_REDUCTION** PROVED

Assuming the pointwise two-path break certificate at n=b and n=b+1, nestedness of the actually defined A_(N-1)=A_(b+2) formally gives both corrected Q_b and Q_(b+1) pass-through predicates, without invoking the undefined A_N.

*Formal hypotheses:* `KKL_STATIONARY_POINTWISE_TWO_PATH_BREAK`

*Proof dependencies: none*

*Evidence:* `qa_native_kkl_stationary_2026-07-26.md#56-the-corrected-terminal-index`

---

**PQL_RAW_UNTAGGED_PATH_IMAGE_COLLISION** PROVED

There exists an explicit degree-two pseudo-quadratic-like map with two distinct proper immersion-restriction path components having one identical unparametrized proper image under the covering projection. The model proves that raw untagged pushforward forgets occurrence multiplicity; its common image is not a KKL Wave and no complete acting KKL outer marking is asserted.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `stationary_occurrence_collision_obstruction.md#7-the-two-distinct-restriction-components-have-one-image`

---

**PQL_WAVE_TO_WAVE_UNTAGGED_PATH_IMAGE_COLLISION** PROVED

There exists an explicit analytic degree-two pseudo-quadratic-like relation, one proper target Wave alpha, and two distinct proper immersion-restriction Waves in the covering-pulled marking f^*I whose identical unparametrized covering image delta is also a Wave for the same finite target marking I. Neither source path meets the critical point. The construction proves that distinct components, sheets, properness, and individual Wave image type do not imply untagged terminal noncollision. It does not assert that I satisfies the full KKL immersion-action/refinement condition, that either component survives line 2097, that these are the only raw restrictions, or that alpha is unbreakable; it is not an actual stationary KKL counterexample.

*Formal hypotheses: none*

*Proof dependencies:* `PQL_RAW_UNTAGGED_PATH_IMAGE_COLLISION`

*Evidence:* `kkl_stationary_n2_wave_to_wave_collision.md#5-the-two-proper-covering-images-are-one-wave`

---

**PQL_ITERATE_FILLED_GRAPH_NATURALITY** PROVED

For every finite Kahn fiber-product iterate of a psi-polynomial-like map, the iterated immersion I_m restricts to a homeomorphism K^m -> K^0, P_m^(-1)(K^0)=K^m, and P_m=F^m o I_m on K^m. Hence every compact connected base-filled-set package has a unique compact connected graph lift, with exact P_m image and component naturality for inverse images.

*Formal hypotheses: none*

*Proof dependencies:* `PQL_SAME_FILLED_CONTINUUM_QUADRATIC_LIKE_RESTRICTION`

*Evidence:* `pql_iterate_filled_graph_naturality.md#theorem-32-canonical-filled-graph`

---

**KKL_STATIONARY_FINAL_COMPONENT_REALIZATION** PROVED

At KKL's stationary final level m=N-1=b+2, apply filled-set graph naturality to the corrected canonical Q-chain: the graph lifts of Q_b and Q_(b+1) exist in U^m, are whole distinct components of Kcal^m=(f^m)^(-1)(Kcal), and satisfy f^m(Qhat_r)=F^(m-r)(Q_0). This retyping does not identify Q_0 with the undefined printed K_0^0 and asserts no final path-incidence persistence, separator role, or order.

*Formal hypotheses: none*

*Proof dependencies:* PQL_ITERATE_FILLED_GRAPH_NATURALITY, QL_RETURN_FACTORIZATION, KKL_STATIONARY_SELECTED_SMALL_JULIA_PREDECESSOR_CHAIN

*Evidence:* pql_iterate_filled_graph_naturality.md#theorem-41-stationary-final-component-realization

---

KKL_STATIONARY_DEGREE_TWO_CORRECTION PROVED

At m=b+2, the canonical graph restrictions over the corrected Q_b and Q_(b+1) lifts both have continuum degree exactly two under f^m, not degree one: the selected Q inverse chain is univalent, while the remaining itinerary crosses the unique critical point in Q_0 exactly once. This does not identify Q_0 with the undefined printed K_0^0, certify degree on an entire marked ending multi-interval, construct the final lamination B, or license a pushforward-width estimate.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_FINAL_COMPONENT_REALIZATION

*Evidence:* qa_native_kkl_stationary_2026-07-26.md#7-final-degree

---

TERMINAL_WAVE_CREST_TO_KL_WIDTH_CONDITIONAL PROVED

For any supplied admissible terminal subwave restriction satisfying the explicit literal ordered-subcurve, holomorphic KL comparison-map, fixed endpoint, separating-piece, and componentwise skip hypotheses, KKL domination, ordered-multicurve overflow, family monotonicity, and KL's separating-pieces estimate give W(S_0)<=W(H)<=W(Hhat)<=W(T_(Y,Z)(v,w))<=2^(2m+q+2)/mu. The first three comparison coefficients are exactly one; the final KL bound is not uniform when m or q grows.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_KL_ORDERED_MULTICURVE_OVERFLOW, KL_DECORATION_ESCAPE_PUZZLE

*Evidence:* terminal_wave_crest_to_kl_skipping_bridge.md#theorem-81

---

FINITE_EXACT_CANTOR_ENCODING_OBSTRUCTION PROVED

An infinite family of pairwise distinct virtual small-Julia components has no injective identification with a finite terminal-tree vertex package; therefore a finite exact enumeration cannot represent an infinite Cantor component family.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* virtual_cantor_package_interface.md#4-finite-exact-encoding-cannot-be-assumed

---

FINITE_QUOTIENT_OBSERVABILITY_CRITERION PROVED

A finite quotient of a virtual component family supports a declared collection of present or future observables exactly when every quotient fiber is contained in one joint observational equivalence class; equality of coarse component labels alone is insufficient.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* virtual_cantor_package_interface.md#4-finite-exact-encoding-cannot-be-assumed

---

DIRECT_DECORATION_PULLBACK_COMPACTIFICATION PROVED

Under the source-locked direct-decoration hypotheses that P,Q_0,Q_1 are closed Jordan puzzle pieces, Q_0 and Q_1 are disjoint, and each proper degree-one f^q restriction is a homeomorphism of closed pieces, together with K(f)=J(f), the pullbacks and terminal root K_0 construct a compact rooted-end address space whose finite fibers are actual univalent preterminal Julia pullbacks, whose infinite fibers are connected nested puzzle continua, and whose planar union has exactly those connected components. Exact dynamics is the partial pre-root shift deleting the first address symbol; the empty root has a separately typed terminal-cycle transition.

*Formal hypotheses: none*

*Proof dependencies:* KL_DECORATION_ESCAPE_PUZZLE, FIRST_DECORATION_SCALE_ESCAPE_ATLAS, CRITICAL_BASE_CYCLE_EXCLUSION

*Evidence:* virtual_cantor_pullback_compactification.md#7-compact-planar-component-package

---

NESTED_FULL_PACKET_NONSEPARATION PROVED

A decreasing intersection of compacta, each a finite union of pairwise-disjoint full continua, has connected sphere complement. The Jordan-localized version follows after adjoining the exterior full continuum. Every application must separately construct nested finite approximants and prove their fullness, disjointness, and localization.

*Formal hypotheses: none*

*Proof dependencies:* ALEXANDER_DUALITY_COMPACTA

*Evidence:* research_push_1_assessment_and_integration.md#3-nested-full-packets

---

DIRECT_DECORATION_FULL_FIBERS PROVED

Under the proved direct-decoration pullback hypotheses, every finite pullback K_w and every infinite end fiber E_omega is a full continuum.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_PULLBACK_COMPACTIFICATION

*Evidence:* virtual_cantor_pullback_compactification.md#theorem-131-direct_decoration_full_fibers

---

DIRECT_DECORATION_GLOBAL_NONSEPARATION PROVED

The complete direct-decoration rooted-end compactum is the decreasing intersection of finite pairwise-disjoint full-packet approximants and therefore has connected sphere complement.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_FULL_FIBERS, NESTED_FULL_PACKET_NONSEPARATION

*Evidence:* virtual_cantor_pullback_compactification.md#theorem-132-direct_decoration_global_nonseparation

---

DIRECT_DECORATION_GAP_CELL_ATLAS PROVED

The terminal root of the direct-decoration compactification has a closed Jordan cell compactly inside the root gap whose inverse images form pairwise-disjoint equivariant cells around all finite pullback fibers, including ancestor-descendant pairs. Each cell meets the complete rooted-end compactum only in its designated fiber, and fullification of every compact connected seed stays compactly in its cell.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_PULLBACK_COMPACTIFICATION

*Evidence:* virtual_cantor_pullback_compactification.md#theorem-134-direct_decoration_gap_cell_atlas

---

DIRECT_DECORATION_FINITE_PACKET_REPLACEMENT PROVED

Replacing finitely many direct-decoration finite fibers by full packets compactly contained in their pairwise-disjoint gap cells preserves compactness and connectedness of the sphere complement of the complete rooted-end package.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_GLOBAL_NONSEPARATION, DIRECT_DECORATION_GAP_CELL_ATLAS, NESTED_FULL_PACKET_NONSEPARATION

*Evidence:* virtual_cantor_pullback_compactification.md#theorem-135-direct_decoration_finite_packet_replacement

---

PROPER_PROTECTED_IMPRESSION_EXPOSURE PROVED

For a nondegenerate full continuum compactly contained in a Jordan disk, every proper compact protected subset of its boundary leaves an accessible boundary point. Finitely many protected crosscut chains with impressions in that subset can be truncated away from the access arc, and fullification of their connected union with the continuum preserves the exposed port.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* virtual_cantor_pullback_compactification.md#lemma-136-proper_protected_impression_exposure

---

DIRECT_DECORATION_FINITE_PROTECTED_CARRIER PROVED

For finitely many already realized direct-decoration finite fibers and finitely many actual transported protected prime-end chains whose total root impression is a proper subset of the root boundary, localized packet fullifications preserve free ports against the complete rooted-end background and admit a protected regulated connector preserving the transported prime ends and cyclic order.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_FINITE_PACKET_REPLACEMENT, PROPER_PROTECTED_IMPRESSION_EXPOSURE, PROTECTED_FREE_COMPONENT_CRITERION, PROTECTED_REGULATED_CONNECTOR_REALIZATION

*Evidence:* virtual_cantor_pullback_compactification.md#theorem-137-direct_decoration_finite_protected_carrier

---

**PULLBACK_GAP_ENLARGEMENT_NATURALITY** PROVED

For the direct-decoration pullback compactification, every connected root enlargement that remains disjoint from the two first-level pullback cylinders transports topologically through every finite inverse word by commuting inclusion squares, with unchanged end fibers, sheets, branch times, partial pre-root shift, and separate root transition. This does not prove fullness, protected-access stability, or analytic-carrier admissibility; outside the safe gap an actual component-intersection ledger is required.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_PULLBACK_COMPACTIFICATION

*Evidence:* virtual_cantor_pullback_compactification.md#9-root-enlargement-naturality

---

**CRITICAL_CUT_POINTWISE_CONSERVATION** PROVED

For commuting iterates f^N and f^r, the two local-degree factorizations of f^(N+r) agree; for a quadratic polynomial the two cuts concatenate to the same pointwise chronological orbit word. This does not by itself define a global proper-branch word.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* critical_chronology_and_retimed_telescope.md#3-local-degree-and-chronology

---

**PROPER_COMMUTING_SQUARE_DEGREE_LAW** PROVED

In a commuting square of finite-degree proper holomorphic maps between connected bordered surfaces, the products of the horizontal and vertical degrees agree, pointwise local degrees obey the analogous identity, and a quadratic degree with univalent source projection lies either in the lifted branch or in the target projection.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* critical_chronology_and_retimed_telescope.md#4-proper-commuting-squares

---

**LOCAL_OFF_CRITICAL_CYLINDER_SQUARE** PROVED

Given conformal polynomial-iterate maps h_0=f^N:U->A and F=f^r:A->B, the forced image V=f^r(U) makes f^r:U->V and f^N:V->B conformal and gives an exact commuting square. Certified chart-supported prime-end data transport, but V is not thereby an authorized virtual target or a Jordan domain without an additional closure embedding.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* critical_chronology_and_retimed_telescope.md#5-the-local-off-critical-square

---

**RETIMED_FINITE_CYLINDER_PAIR** PROVED

Given an independently realized proper root branch F=f^r:A->B between bounded open Jordan domains compactly nested inside the closed direct-decoration root, and a proved cylinder map whose interior restriction is conformal and whose closed-piece extension is a homeomorphism, the pair (F composed h_w, inclusion composed h_w) is a proper covering/conformal-embedding correspondence whose relative map is exactly F. Full invariant target carriers and already certified chart-supported records pull back exactly; no scale-two identification, simultaneous authorization, arbitrary marking, or unrestricted enlargement is asserted.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_PULLBACK_COMPACTIFICATION

*Evidence:* critical_chronology_and_retimed_telescope.md#6-retiming-an-independently-realized-root-pair

---

**CRITICALITY_LOCATION_SWAP_NO_GO** PROVED

For f(z)=z^2-1 there is an explicit proper commuting square with univalent source projection and quadratic root branch in which the lifted branch is univalent and the target projection is quadratic, so critical degree need not stay in the branch.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* critical_chronology_and_retimed_telescope.md#7-exact-chronology-counterexamples

---

`CRITICAL_TIME_SHEAR_NO_GO` PROVED

For f(z)=z^2-1 there is an explicit exact commuting square whose endpoint projections are univalent and whose root and lifted returns both have degree two, while their chronological words are respectively (2,1) and (1,2).

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `critical_chronology_and_retimed_telescope.md#7-exact-chronology-counterexamples`

---

`STAGEWISE_CRITICAL_WORD_LADDER` PROVED

For a ladder whose one-step branch maps and transverse maps are proper and onto between connected Jordan disks, the recurrence lift_degree_j*projection_degree_(j+1)=projection_degree_j*root_degree_j holds. With univalent initial projection, exact chronological words are equivalent to univalence of every transverse projection; endpoint degree alone is insufficient.

*Formal hypotheses: none*

*Proof dependencies:* `PROPER_COMMUTING_SQUARE_DEGREE_LAW`

*Evidence:* `critical_chronology_and_retimed_telescope.md#8-the-stagewise-degree-ladder`

---

`QUADRATIC_JORDAN_BASE_CHANGE_DICHOTOMY` PROVED

For a proper degree-two holomorphic map between Jordan disks and a relatively compact Jordan target subdisk whose boundary avoids the critical value, the full preimage is one quadratic Jordan disk when the subdisk contains the critical value and two conformal Jordan disks when it omits the critical value.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `critical_chronology_and_retimed_telescope.md#9-quadratic-jordan-base-change`

---

`PLANAR_CRITICAL_SPINE_TELESCOPE_COMPILER` PROVED

A finite planar stage cycle with closure embeddings onto relatively compact Jordan subdisks, exactly one proper quadratic stage, all other stages conformal, and postquadratic containment of every propagated critical value has a unique connected Jordan base-change route and a proper degree-two endpoint covering. The theorem constructs no relative-cover lift or protected marking.

*Formal hypotheses: none*

*Proof dependencies:* `QUADRATIC_JORDAN_BASE_CHANGE_DICHOTOMY`

*Evidence:* `critical_chronology_and_retimed_telescope.md#10-embedded-jordan-critical-spine-telescope`

---

`FINITE_PLANAR_TELESCOPE_IDENTITY` PROVED

For the selected embedded-Jordan planar telescope, the endpoint relative map on its exact domain I_m(T_m) is the chronological composition of the relative stage maps; actual polynomial iterate exponents add, pointwise critical words concatenate, and the nested selected domains retain the intermediate route. No relative-cover, protected-marking, end-properness, or enlargement conclusion is included.

*Formal hypotheses: none*

*Proof dependencies:* `PLANAR_CRITICAL_SPINE_TELESCOPE_COMPILER`, `CRITICAL_CUT_POINTWISE_CONSERVATION`

*Evidence:* `critical_chronology_and_retimed_telescope.md#11-the-exact-finite-planar-telescope-identity`

---

`PLANAR_END_COLLAPSE_NO_GO` PROVED

Along every direct-decoration infinite address, the specified unrescaled inverse-word branches converge only to constants on locally uniform subsequences and shrink the diameter of every compact subset of the root interior to zero. This does not exclude rescaled or unrelated charts.

*Formal hypotheses: none*

*Proof dependencies:* `DIRECT_DECORATION_PULLBACK_COMPACTIFICATION`

*Evidence:* `critical_chronology_and_retimed_telescope.md#12-planar-end-collapse`

---

`NORMALIZED_RETIMED_GERM_CONSTANCY` PROVED

For a given independently realized root pair and its retimed finite-cylinder pairs, conjugating each common source by its univalent cylinder coordinate gives exactly the same root covering/embedding pair at every word. This identity constructs no analytic end attachment.

*Formal hypotheses: none*

*Proof dependencies:* `RETIMED_FINITE_CYLINDER_PAIR`

*Evidence:* critical_chronology_and_retimed_telescope.md#13-normalized-retimed-germ-and-projection-collapse

---

NORMALIZED_END_DIRECT_LIMIT PROVED

For an independently realized root portrait and its actual direct-decoration finite-cylinder copies along one infinite address, the geometric-prefix coordinate changes form a direct system whose Riemann-surface limit is canonically biholomorphic to the root portrait; the normalized shift is the identity in root coordinates while retaining its separate dynamical label. Only records explicitly pulled through the cylinder charts descend, and no planar end component or projection is constructed.

*Formal hypotheses: none*

*Proof dependencies:* NORMALIZED_RETIMED_GERM_CONSTANCY

*Evidence:* research_push_3_assessment_and_integration.md#theorem-41-normalized_end_direct_limit

---

DEPTH_GRADED_CYLINDER_MATERIALIZATION PROVED

A finite integer-shift-graded diagram of already actual composable root restrictions, actual root squares, presentation maps, forward address shifts, and chart-supported records has one exact finite-cylinder materialization after a common depth offset; dynamical arrows are actual polynomial iterates and presentation arrows remain separately typed. The theorem constructs no root portrait, relative cover, protected carrier, or marked finite-horizon assembly.

*Formal hypotheses: none*

*Proof dependencies:* RETIMED_FINITE_CYLINDER_PAIR

*Evidence:* research_push_3_assessment_and_integration.md#theorem-51-depth_graded_cylinder_materialization

---

END_CHART_OBSERVABLE_INVARIANCE PROVED

Marked conformal cylinder and normalized-end charts preserve every fully transported extremal-length family and finite smooth-electrode Dirichlet-to-Neumann response exactly; algebraic parallel sums and Schur complements commute with those identities. No bounded port count, future closure, Value observability, or uniform comparison follows.

*Formal hypotheses: none*

*Proof dependencies:* NORMALIZED_END_DIRECT_LIMIT, FINITE_ELECTRODE_RESPONSE

*Evidence:* research_push_3_assessment_and_integration.md#theorem-61-end_chart_observable_invariance

---

SERIAL_EXHAUSTION_RESISTANCE_BOUND PROVED

For pairwise interior-disjoint marked quadrilaterals that every authorized connecting curve crosses in chronological series with intact side walls and no bypass, the extremal resistance of the complete family is at least the sum of the individual crossing resistances; reciprocal conductance is therefore at most the reciprocal sum. Source realization of the serial-exhaustion hypotheses is separate.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* research_push_3_assessment_and_integration.md#theorem-62-serial_exhaustion_resistance_bound

---

SCALAR_NECK_TRACE_MODE_NO_GO PROVED

On a flat cylinder of length L, constant-mode two-terminal conductance is 2pi/L, while the harmonic energy of boundary trace cos(k theta) against zero is pi k coth(kL); hence scalar conductance does not control arbitrary analytic cross-section traces and cannot alone justify collapsing a cut interface to one electrical node.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* research_push_3_assessment_and_integration.md#theorem-71-scalar_neck_trace_mode_no_go

---

FULL_TRACE_RECTANGLE_DTN PROVED

For a width-one Euclidean rectangle of length R with natural Neumann side walls, the complete end-trace energy on the spectral H^{1/2} trace space is the functional-calculus form with diagonal A coth(RA), off-diagonal -A csch(RA), and continuous zero-mode value 1/R.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* research_push_4_assessment_and_integration.md#4-independent-audit-of-the-rectangle-calculus

---

**TRACE_RESPONSE_SEMIGROUP** PROVED

Genuine trace-compatible rectangles glued along one complete parameterized cross-section satisfy the exact variational response law N_R star N_S = N_{R+S}. This theorem does not construct such a serial decomposition in a dynamically nested puzzle end.

*Formal hypotheses: none*

*Proof dependencies:* FULL_TRACE_RECTANGLE_DTN

*Evidence:* research_push_4_assessment_and_integration.md#41-exact-concatenation

---

**EQUIPOTENTIAL_COLLAPSE_CRITERION** PROVED

For a linear admissible trace space V, the rectangle energy factors through the two trace means exactly when the mean map is injective on V. If constants are admissible or V is invariant under mean subtraction, this is equivalent to V consisting only of constants; a geometric dynamical equipotential is not thereby an electrical electrode.

*Formal hypotheses: none*

*Proof dependencies:* FULL_TRACE_RECTANGLE_DTN

*Evidence:* research_push_4_assessment_and_integration.md#42-exact-scalar-collapse-criterion

---

**SHORTED_EQUIPOTENTIAL_NODAL_RESPONSE** PROVED

If every plumbing cross-section is explicitly restricted to constant traces, its complete rectangle response is exactly the scalar conductance 1/R and serial conductances combine by the harmonic law. The constant-trace hypothesis must be independently verified in every application.

*Formal hypotheses: none*

*Proof dependencies:* EQUIPOTENTIAL_COLLAPSE_CRITERION, TRACE_RESPONSE_SEMIGROUP

*Evidence:* research_push_4_assessment_and_integration.md#42-exact-scalar-collapse-criterion

---

**TRACE_RENORMALIZED_NECK_ESTIMATE** PROVED

For a genuine width-one rectangle, retaining the two half-cylinder mean-zero responses and the exact zero-mode resistor gives the sharp relative full-trace bound eta(R)=2/(exp(pi R)-1), with (1-eta)N_R# <= N_R <= (1+eta)N_R# whenever eta<1.

*Formal hypotheses: none*

*Proof dependencies:* FULL_TRACE_RECTANGLE_DTN

*Evidence:* research_push_4_assessment_and_integration.md#43-sharp-trace-renormalized-estimate

---

**END_TRACE_RESPONSE_COMPACTIFICATION** PROVED

The genuine rectangle response converges in the normalized spectral H^{1/2}-to-dual norm to two independent half-cylinder responses, with error at most max(2/R,2/(exp(pi R)-1)); the one-parameter response family extends continuously to R=infinity.

*Formal hypotheses: none*

*Proof dependencies:* TRACE_RENORMALIZED_NECK_ESTIMATE

*Evidence:* research_push_4_assessment_and_integration.md#43-sharp-trace-renormalized-estimate

---

**HILBERT_TRACE_GLUE_CALCULUS** PROVED

Relative Loewner bounds between nonnegative closed interface trace forms survive addition of the same nonnegative component energy and minimization over the same complete Hilbert trace space.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* research_push_4_assessment_and_integration.md#44-elimination-and-strict-transition

---

**TRACE_STABLE_STRICT_TRANSITION_TRANSFER** PROVED

For a nonnegative positively homogeneous Loewner-monotone response gauge, a trace-renormalized contraction L_out <= mu L_in + E with 0<mu<1 and E>=0 transfers through two-sided endpoint response comparisons of size eta with factor mu(1+eta)/(1-eta), which remains strict when eta<(1-mu)/(1+mu). This does not supply the root contraction, a physical collar, or resistance growth.

*Formal hypotheses: none*

*Proof dependencies:* TRACE_RENORMALIZED_NECK_ESTIMATE, HILBERT_TRACE_GLUE_CALCULUS

*Evidence:* research_push_4_assessment_and_integration.md#44-elimination-and-strict-transition

---

**PULLED_EQUIPOTENTIAL_SERIAL_RECTANGLE_NO_GO** PROVED

In the exact Bottcher model z maps to z^(2^q), a nondegenerate equipotential arc of angular width Delta pulls back to width Delta/2^q. The full parent and pulled-child arcs therefore cannot be the two equipotential sides of one quadrilateral bounded by the same two rays. Pullback conformality does not make the physical region between nested puzzle cylinders a serial rectangle.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_PULLBACK_COMPACTIFICATION

*Evidence:* research_push_4_assessment_and_integration.md#5-exact-obstruction-to-the-proposed-pulled-collar

---

**ACTIVE_INACTIVE_PHYSICAL_CELL** PROVED

If Q is a closed Jordan disk contained in a closed Jordan disk P and their complete boundary intersection is exactly one nondegenerate closed arc A, then P minus int(Q) is the union of A and one active closed Jordan disk C=closure(int(P) minus Q); C meets A only at the two endpoints, and the relative interior of A is not approached from the active domain. This is a conditional one-cell topology theorem, not a source realization for the binary direct-decoration package.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* research_push_5_assessment_and_integration.md#4-the-valid-one-cell-theorem

---

**ATTACHMENT_ARC_OPEN_ATOM_REFINEMENT** PROVED

For two specified closed attachment arcs on a prime-end circle, removing their at most four genuine endpoints and classifying each resulting open component by its two activity bits gives a disjoint common-active, outgoing-only, incoming-only, double-inactive, and endpoint partition. There are at most four open atoms and four endpoint atoms per ordered pair; this does not bound alphabet size, proof markings, port lifetime, or analytic response dimension.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* research_push_5_assessment_and_integration.md#5-correct-one-junction-refinement

---

**REGULAR_INTERFACE_FIBER_PRODUCT_RESPONSE** PROVED

For two regular Dirichlet domains with disjoint interiors and one actual admissible common interface, the response of their union is the infimum of the sum of their response forms over the fiber product of traces agreeing on that interface, while all one-sided trace variables remain external. This requires the stated Sobolev gluing hypothesis and does not construct a puzzle interface.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* research_push_5_assessment_and_integration.md#61-fiber-product-gluing

---

**FINITE_TAME_CONTACT_OVERLAY_RESPONSE** PROVED

If a bounded planar domain is partitioned up to measure zero into finitely many bounded Lipschitz cells whose positive-length interior interfaces are finitely many paired Lipschitz arcs with opposite normals and whose remaining junction set has one-dimensional measure zero, then H1 of the union is exactly the finite fiber product of the cellwise H1 spaces with matching paired traces; norm and Dirichlet energy are the cellwise sums, and the global response is the corresponding finite fiber-product infimum. This does not construct a source overlay, handle arbitrary Jordan or prime-end traces, close all-depth contacts, or control weldings.

*Formal hypotheses: none*

*Proof dependencies:* REGULAR_INTERFACE_FIBER_PRODUCT_RESPONSE

*Evidence:* push5_independent_reverification_and_extrapolation.md#5-a-finite-tame-overlay-theorem

---

**INFIMUM_PUSHFORWARD_FUNCTORIALITY** PROVED

For any extended nonnegative functional q and composable maps p and r, defining p_*q(y)=inf over p(x)=y of q(x) gives (r composed with p)_*q = r_*(p_*q) as an equality of extended-valued infima. No preservation of closed quadratic forms is asserted.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* research_push_5_assessment_and_integration.md#62-functoriality-of-infima

---

**CLOSED_FORM_PUSHFORWARD_NO_GO** PROVED

A continuous linear forgetful map need not push a closed nonnegative quadratic form to a closed form: on H=l2 with Ae_n=n e_n, q(x,z)=norm(Ax-z)^2 on D(A) direct-sum H is closed, but projection to x has zero pushforward on the dense nonclosed domain D(A). Closed response elimination requires additional coercivity, closed-range, finite-dimensional, or lower-semicontinuous-relaxation hypotheses.

*Formal hypotheses: none*

*Proof dependencies:* INFIMUM_PUSHFORWARD_FUNCTORIALITY

*Evidence:* research_push_5_assessment_and_integration.md#62-functoriality-of-infima

---

**FIXED_STANDARD_NECK_MULTIPORT_COMPACTIFICATION** PROVED

For fixed bounded positive cap forms on genuine Cartesian trace products and one exact standard strip or cylinder neck in fixed endpoint coordinates, the composed response extends continuously to infinite resistance. With positive scalar cap coefficients its norm error is at most 2M/(exp(lambda_1 R)-1)+2B^2/(alpha_*^2 R); if one scalar coefficient vanishes, positivity kills its coupling and the scalar correction vanishes. This theorem supplies neither physical tail topology nor welding control.

*Formal hypotheses: none*

*Proof dependencies:* TRACE_RENORMALIZED_NECK_ESTIMATE, HILBERT_TRACE_GLUE_CALCULUS

*Evidence:* research_push_5_assessment_and_integration.md#9-what-remains-valid-in-the-compact-tail-calculation

---

**QUADRATIC_PRIMITIVE_ROOT_PARABOLIC_NONDEGENERACY** PROVED

If lambda is a primitive q-th root of unity and g(w)=lambda*w+w^2, then g^q(w)=w+C_(lambda,q)w^(q+1)+O(w^(q+2)) with C_(lambda,q) nonzero. Commutation gives cancellation through degree q, and the parabolic basin critical-cycle theorem forces exactly one q-cycle of petals for the quadratic polynomial.

*Formal hypotheses: none*

*Proof dependencies:* FATOU_PARABOLIC_BASIN_CRITICAL_CYCLE

*Evidence:* side_research_retained_lemmas.md#1-primitive-root-parabolic-nondegeneracy

---

**FIXED_MODULUS_AMBIENT_END_NO_GO** PROVED

A nested sequence of univalent images may carry conformally conjugate degree-two quadratic-like maps with one fixed modulus while its ambient intersection contains a nondegenerate continuum. The symmetric-ellipse construction refutes fixed-modulus ambient end shrinking but is not a quadratic-polynomial or MLC counterexample.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* side_research_retained_lemmas.md#2-fixed-modulus-does-not-force-ambient-end-shrinking

---

**SERIAL_ANNULUS_SINGLETON_BRIDGE** PROVED

If actual pairwise-disjoint annuli are serially nested around one common compact end fiber, separate it from one fixed nondegenerate outer continuum with no side bypass, and have divergent total conformal modulus, then the end fiber is a singleton.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* side_research_retained_lemmas.md#3-serial-annuli-force-a-singleton-end

---

**MULTIPORT_TRACE_GAUGE_AND_STRICTNESS_NO_GO** PROVED

For one connected Dirichlet cell, the complete port trace product is quotiented only by the single diagonal constant, or equivalently retains all relative port potentials; independent port-constant quotients erase positive rectangle energy. Moreover a common arbitrary nonnegative component energy can drive any strict response factor toward one, so strictness after gluing requires a relative-error, coercive full-state, or additive-budget estimate.

*Formal hypotheses: none*

*Proof dependencies:* FULL_TRACE_RECTANGLE_DTN

*Evidence:* side_research_retained_lemmas.md#5-correct-multi-port-gauge-and-strictness-obstruction

---

`WOT_WELDING_ALGEBRA_NO_GO` PROVED

The orthogonal graph projection of a bounded J is built with (I+J*J)^(-1), not one half of [[I,J*],[J,I]] in general. Bilateral shifts converge weakly to zero while their products and graph constraints do not; their graph projections converge weakly to one half of the identity, which is not a graph projection. Weak-operator subsequence extraction alone does not compactify welding or response-gluing algebra.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `side_research_retained_lemmas.md#6-weak-operator-compactness-does-not-compactify-welding-algebra`

---

`DYADIC_BOTTCHER_EXTERNAL_TRACE_ISOMETRY` PROVED

On one declared external angular sector, pullback by an affine inverse Bottcher lift is an exact isometry of the homogeneous H^(1/2) seminorm because its two-dimensional Jacobian cancels the squared distance scaling. This supplies no multisector or physical-to-core welding control.

*Formal hypotheses: none*

*Proof dependencies:* `DIRECT_DECORATION_PULLBACK_COMPACTIFICATION`

*Evidence:* `side_research_retained_lemmas.md#4-sectorwise-dyadic-bottcher-trace-isometry`

---

`BINARY_DIRECT_DECORATION_NESTED_SEPARATOR_CALCULUS` PROVED

For every fixed finite address in the source-locked binary f^q direct-decoration inverse branches, every contact between the accumulated prefix and the next or any future shell is confined to the current separator, and membership of each actual physical open boundary arc in the nested boundaries is interval-convex. Under separately verified finite admissible Sobolev-interface hypotheses, prefix-to-next-shell fiber-product minimization gives the exact actual-domain Dirichlet response. This asserts no finite invariant atomization, protected-overlay naturality, final-root-enlargement stability, fixed trace catalog, compactness, Value observability, or uniform constants.

*Formal hypotheses: none*

*Proof dependencies:* `DIRECT_DECORATION_PULLBACK_COMPACTIFICATION`, `REGULAR_INTERFACE_FIBER_PRODUCT_RESPONSE`, `FINITE_TAME_CONTACT_OVERLAY_RESPONSE`

*Evidence:* `research_push_6_assessment_and_integration.md#5-the-theorem-that-survives`

---

`FIXED_BLOCK_BINARY_CONTACT_AUTOMATON` PROVED

Under the source-locked Kahn–Lyubich direct-decoration package and the audited closed-disjoint Jordan branch hypotheses, one fixed binary f^q block has exact first-symbol contact vertices, the normalized recursion Chat_{wa}=g_a(Chat_w cap C_a), exact pairwise cylinder contacts, prefix endpoint deletion through source depth n, and a two-state structural contact skeleton {the complete alpha-ray port, empty} thereafter. A connector-aware boundary incidence graph gives a finite exact automaton for this fixed structural puzzle-contact system. No uniform bound in q or n, Sobolev trace atlas, protected/obstacle/Value overlay, response, welding, or final-root analytic naturality is asserted.

*Formal hypotheses: none*

*Proof dependencies:* `KL_DECORATION_ESCAPE_PUZZLE`, `DIRECT_DECORATION_PULLBACK_COMPACTIFICATION`, `BINARY_DIRECT_DECORATION_NESTED_SEPARATOR_CALCULUS`

*Evidence:* `research_push_7_assessment_and_integration.md#5-proved-structural-contact-theorem`

---

`FIXED_KKL_PAIR_STATE_MOMENT_FACTORIZATION` PROVED

For one fixed finite KKL terminal Hubbard tree with edge set E, grouping every pulled bivertical intersection by the pair state (f_*^m alpha,(iota_m)_* alpha) rewrites Value exactly as lambda^(-2m) times the Frobenius contraction of aa^T with a positive-semidefinite second-moment matrix. The ambient symmetric-state dimension is |E|^2(|E|^2+1)/2. This does not construct the binary branch incidence factor, prove its completely positive transport, bound |E| universally, or yield future observability or a strict transition.

*Formal hypotheses: none*

*Proof dependencies:* `KKL_VALUE_CONTRACTION`, `KKL_TYPED_VALUE_INCIDENCE`

*Evidence:* `research_push_7_assessment_and_integration.md#10-fixed-tree-kkl-pair-state-moment-factorization`

**FINITE_LINEAR_LABEL_MARKING_ENUMERATOR** PROVED

For an abstract finite linearly ordered target micro-atom set and source atom string labelled by target atoms or a nonproper symbol, define each target-block hull by deleting proper labels outside the block and retaining the remaining components containing an in-block proper seed. The interval-convex closure of the labels flanking every maximal nonproper run is the unique finest partition with pairwise disjoint label hulls, and exhaustive coarsening enumeration with a supplied boundary-vertex endpoint predicate returns a maximum-cardinality admissible partition or a finite obstruction. This does not identify the label hull with a KKL immersion pullback, validate analytic realizability, or certify properness, traces, welding, composition, or enlargement.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `research_push_8_assessment_and_integration.md#5-the-finite-theorem-that-survives`

---

**PULL_PUSH_WELDING_WORD_COMPOSITION_NO_GO** PROVED

For composable correspondences, a one-step pull-push trace map W_j=P_j R_j is an endomorphism of the target trace space at its own stage, so consecutive W_j are not even typed to compose without additional identifications. After artificially identifying all spaces, the componentwise composite P_1 P_2 R_2 R_1 need not equal the chronological product P_2 R_2 P_1 R_1; explicit two-by-two matrices separate them. Word welding therefore requires a fiber-product correspondence and a Beck-Chevalley interchange square, not matrix associativity alone.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `research_push_8_assessment_and_integration.md#12-exact-failure-of-word-functoriality`

---

**DEGREE_ONE_ANNULUS_IMMERSION_INJECTIVITY_NO_GO** PROVED

Degree one, Z-equivariance, and holomorphic immersion between annular universal-cover strips do not imply global injectivity. For a>exp(2 pi h), F_a(z)=z+a exp(2 pi i z)/(2 pi i) has nonvanishing derivative on Pi(h), is Z-equivariant, and has distinct points z,z+d with the same image; after vertical translation it maps into a sufficiently tall strip and descends to a degree-one annulus immersion. This refutes same-tile injectivity from degree and immersion alone, without asserting that the example satisfies every KL end condition.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `direct_decoration_path_lifted_collar_certificate.md#proposition-42-degree-one-and-immersion-do-not-imply-ti`

---

**KAHN_ANNULUS_GLOBAL_INJECTIVITY_NO_GO** PROVED

Kahn's stated annulus-lemma hypotheses do not imply global injectivity. An explicit finite Blaschke quotient maps A(1,1287/1000) holomorphically and immersively into A(1,(1287/1000)^16), extends analytically to the inner circle as an orientation-preserving degree-one homeomorphism, and has an exact self-collision. Exact Gaussian-integer expansions, Sturm certificates, rational Rouche disks, the harmonic minimum principle, and rational modulus bounds prove every clause. This is an abstract annulus no-go, not an actual counterexample to the additional dynamically generated KL path-space structure.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `kahn_annulus_global_injectivity_no_go.md#1-theorem-and-scope`

---

**PSI_QL_GLOBAL_INJECTIVITY_NO_GO** PROVED

The complete abstract degree-two psi-quadratic-like axioms do not force global injectivity of the immersion, even on the complementary annulus and even when its induced fundamental-group degree is +1. Explicitly, on D_75 take f(z)=z^2 into D_5625, K=[0,1], K'=[-1,1], and i(z)=(exp(pi(z+1)/50)-1)/(exp(pi/25)-1). Then f is a proper degree-two branched cover, i is an immersion into D_5625, i^(-1)(K)=f^(-1)(K)=K' is connected, and i|K' is a homeomorphism, while i(50i)=i(-50i) outside K. This is an abstract psi-ql counterexample, not an actual collision in a KL canonical-localization or direct-decoration path-space stage.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `psi_ql_global_injectivity_no_go.md#1-theorem-and-exact-scope`

**KL_LOCALIZATION_SUBGROUP_REFLECTION_CRITERION** PROVED

Let a:X_s->X_t be a based embedding and let connected covers of X_s,X_t correspond to subgroups H_s,H_t. If the based lift A exists, then a_*(H_s) is contained in H_t, and A is injective exactly when a_*^(-1)(H_t)=H_s. Failure is certified by a loop class gamma outside H_s whose image lies in H_t; its source lift has distinct endpoints that A identifies. KL pullback iteration is Cartesian, but canonical localization is stated only as a functorial commuting lift preserving immersion, and the checked source proves forward path equivalence rather than homotopy reflection. KL_CANONICAL_LOCALIZATION_FORCED_COLLISION applies this criterion to the physical inclusion underlying every ordinary primitive canonical seed and proves that equality fails there. At a later genuinely immersed bold stage, a base-point collision remains a separately typed obstruction.

*Formal hypotheses: none*

*Proof dependencies:* KL_DECORATION_ESCAPE_PUZZLE

*Evidence:* kl_localization_subgroup_reflection_gate.md#1-result-and-scope

---

**KL_CANONICAL_LOCALIZATION_FORCED_COLLISION** PROVED

For every ordinary quadratic-like stage of a quadratic polynomial that is primitively p-renormalizable with p>1, Kahn's canonical localization at A=K_p with removal B=union_j f^j(A) has a noninjective localized inclusion. Global degree 2^p versus central local degree 2 supplies an excess full component X of (f^p)^(-1)A; primitivity and f^p-invariance give X disjoint B. A Jordan peripheral loop around X lies outside the source local subgroup but is null after the physical inclusion into the target complement, so its distinct source-cover endpoints have one target endpoint. The collision propagates through every exact KL pullback equalizer; whenever the same seed carries direct-decoration data, it reaches m=nu q+1 to n=(nu-1)q+1 and forces same-tile and full glued-pseudo-piece noninjectivity. The parameter/stage alignment quantifier is recorded separately and is not used to promote any global gate.

*Formal hypotheses: none*

*Proof dependencies:* KL_LOCALIZATION_SUBGROUP_REFLECTION_CRITERION, KL_TILE_INJECTIVITY_BASE_ANNULUS_REDUCTION

*Evidence:* kl_canonical_localization_forced_collision.md#1-result-scope-and-status

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**KL_LOCALIZATION_SHEET_RESOLUTION_FACTORIZATION** PROVED

For a based physical embedding a:X_s->X_t, connected localization covers with subgroups H_s,H_t, and a based lift A, put K=a_*^(-1)(H_t). The selected component P_K of the pullback X_s times_{X_t} Xhat_t is the K-cover of X_s and embeds in Xhat_t. The lift factors canonically as A=e o q, where q is the covering from the H_s-cover to the K-cover; A and q have exactly the same kernel pair, the based q-fiber is the pointed coset set K/H_s, and covering path transport constructs the complete sheet groupoid and exhausts every compact resolved path lift whose target path is wholly contained in e(P_K). This is the unique connected intermediate-cover factorization with injective second leg in the based commuting comparison class stated in the proof. Under an enlargement homomorphism phi, the induced based-coset fiber map is injective exactly when K intersect phi^(-1)(H_s^+)=H_s and surjective exactly when K^+=phi(K)H_s^+; failure gives an explicit based-coset merger or birth witness. For the ordinary primitive canonical seed, winding of all powers of the forced peripheral class proves [K:H_s]=infinity, so every resolved collision fiber is infinite and raw finite sheet enumeration is impossible. No prime-end realization, actual O/xi/eta footprint, future-faithful compactification, whole-end properness, width control, or global gate follows.

*Formal hypotheses: none*

*Proof dependencies:* KL_LOCALIZATION_SUBGROUP_REFLECTION_CRITERION, KL_CANONICAL_LOCALIZATION_FORCED_COLLISION

*Evidence:* kl_localization_sheet_resolution_factorization.md#1-result-scope-and-status

---

**KAHN_ITERATE_COMPLEMENTARY_COMPONENT_ALIGNMENT** PROVED

For every finite pullback iterate of one fixed Kahn psi-polynomial-like seed, the full complementary equalizer U^(r+2) minus K^(r+2) is exactly (U^(r+1) minus K^(r+1)) times over (U^r minus K^r) with itself and is connected. If an inclusion leg has a q/e covering-then-embedding factorization, exact base change gives a surjective covering from this connected equalizer onto one connected resolved intermediate component, so that ambient complementary factorization has no second component. The Kahn-Lyubich direct-decoration pseudo-pieces use the same U^m iterate tower for the same fixed seed. Separately and conditionally, given a whole annular q/e factorization of an actual KL base immersion, each source tile pulls its declared target universal-cover tile back to the connected source universal-cover tile after based lifts fix the interior deck phase, and q/e base change yields a connected resolved intermediate. Source labels remain distinct when several source

indices have the same target index. This does not prove that the canonical infinite sheet-resolution factorization is that whole actual KL tile factorization; the later resolved-boundary incompatibility theorem shows such an identification conflicts with KL's wrapping-end extension. It does not equate the covering and inclusion component maps, construct a boundary or footprint, prove parameter existence, or close a global gate.

*Formal hypotheses: none*

*Proof dependencies:* `KL_CANONICAL_LOCALIZATION_FORCED_COLLISION`, `KL_LOCALIZATION_SHEET_RESOLUTION_FACTORIZATION`

*Evidence:* `kahn_iterate_complementary_component_alignment.md#1-result-and-exact-scope`

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**`RESOLVED_INTERMEDIATE_RELATIVE_IDEAL_BOUNDARY`** PROVED

For a source-indexed holomorphic annular q/e factorization iota=e o q with oriented degree +1, put G=pi_1(P), H=q_*pi_1(A_s), and N=ker(e_*). Then G=N semidirect H; in the common universal-cover action E_a/H=A_s, E_a/N=R_a, and E_a/G=P. The resolved pullback R_a is connected with pi_1(R_a)=N, and q_a:E_a->R_a is its universal cover with fibers canonically equal to those of q. A nontrivial universal cover, or any nonconstant deck-invariant composite, cannot extend continuously through a deck-limit point to a Hausdorff target. Thus q_a has a full prime-end factor extension exactly in the one-sheet branch N=1. If N is nontrivial, the two resolved-source finite-modulus wrapping points zeta_+ and zeta_- lie in Lambda(N). KL's actual-tile points omega_+ and omega_- are its named wrapping vertices and L_a extends there. The based source-cover biholomorphism satisfies bar(Phi_a)(omega_+/-)=zeta_+/-; hence the correctly typed invariant composite is K_a=L_a o Phi_a^(-1)=e_a o q_a. The deck-limit obstruction forces every whole-tile covering/embedding factorization of that actual KL map to be one-sheeted. Hence the former identification of the canonical infinite resolution with the whole actual KL tile is quarantined; the later whole-tile correspondence no-go closes that option negatively. On a Gamma-invariant subset J of the domain of discontinuity, the quotient (D union J)/Gamma is Hausdorff and every continuous composite extension descends uniquely. Canonical-to-KL restricted boundary correspondence, target clusters beyond the two certified wrapping points, access fidelity, end-properness, and enlargement remain open.

*Formal hypotheses: none*

*Proof dependencies:* `KL_LOCALIZATION_SHEET_RESOLUTION_FACTORIZATION`, `KAHN_ITERATE_COMPLEMENTARY_COMPONENT_ALIGNMENT`

*Evidence:* `resolved_intermediate_relative_ideal_boundary.md#1-verdict-and-exact-scope`

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**`CANONICAL_KL_WHOLE_TILE_CORRESPONDENCE_NO_GO`** PROVED

Let E->R->T be a canonical resolved holomorphic factorization whose first leg is a universal cover with nontrivial deck group N, and let G=N semidirect H be the common annular tower with wrapping fixed points zeta_+ and zeta_- in Lambda(N). There is no whole-tile end-compatible comparison to an actual KL tile map consisting of a based source universal-cover biholomorphism conjugating the annular deck generators, an injective target comparison on the canonical composite image, equality of the two interior composites, and transport of zeta_+ and zeta_- to KL's named wrapping vertices. Such a square would make the nonconstant actual KL composite constant by deck-limit convergence, contradicting KL's source-locked continuous values at those vertices. Since the ordinary primitive canonical resolved fiber is infinite, its covering/embedding factorization cannot be the whole actual KL tile factorization. This no-go does not prove actual KL injectivity and leaves proper collar/interior, non-covering relational, retimed one-sheet, and revised-localization comparisons open.

*Formal hypotheses: none*

*Proof dependencies:* `KL_LOCALIZATION_SHEET_RESOLUTION_FACTORIZATION`, `KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY`, `RESOLVED_INTERMEDIATE_RELATIVE_IDEAL_BOUNDARY`

*Evidence:* `canonical_resolution_kl_tile_correspondence_no_go.md#1-result-scope-and-status`

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**`FINITE_HORIZON_KAHN_COLLAR_EMBEDDING`** PROVED

For a fixed finite chain of inclusion-induced pseudo-piece immersions with embedded central maps, bijective tile-component maps, the full Kahn round-annulus degree-one data, exact declared gluing, and the additional overlap-reflection equality that the inverse image of each target central/tile overlap on the retained Kahn collar is exactly the declared source overlap, one can choose coherent distinguished inner collars by finite inverse-germ shrinking. Every restricted lifted tile map and every glued collar-truncated pseudo-piece map is then a holomorphic embedding. Under separately supplied conformal-disk prime-end extensions and complete pointwise footprints on the asserted domain, connected KKL-admissible sets are exactly connected subsets of the safe set {p:Phi(p) subset I}; maximal connected candidates are its connected components, not automatically unique. Kahn and KL do not themselves supply overlap reflection. The theorem gives neither full pseudo-piece

embedding, proof-use exhaustion of omitted sheets, width comparison, uniform defect charge, nor final-root no-birth.

*Formal hypotheses: none*

*Proof dependencies:* KL_DECORATION_ESCAPE_PUZZLE, KL_TILE_INJECTIVITY_BASE_ANNULUS_REDUCTION, TILE_INJECTIVITY_GLUE_EMBEDDING_CRITERION, EMBEDDED_DISK_ACCESSIBLE_FOOTPRINT_THEOREM

*Evidence:* finite_horizon_kahn_collar_embedding.md#1-the-conditional-theorem

---

**CANONICAL_KL_RESTRICTED_INTERFACE_CLASSIFICATION** PROVED

For every supplied fixed labeled same-source canonical/KL tile datum with chosen lifted basepoints, the canonical resolved source disk and the actual KL source tile have a unique based biholomorphism and hence a canonical relation span R<-q-E-K->Ahat_t^KL. Pulling a target path back by each local homeomorphism K_i and taking the tagged disjoint union over all declared source labels gives an exact etale family whose path components are precisely all maximal interior restrictions, with source label, resolved trace, sheet, and deck phase retained. If the canonical kernel N is nontrivial, the span cannot descend through q: the correctly typed K is nonconstant and N-invariant under a descent, while its source-locked values at the two transported wrapping points zeta_+ and zeta_- in Lambda(N) contradict the deck-limit theorem. Those two values, but not an unproved value on the whole limit set, are common cluster values of every resolved fiber orbit. Separately, the degree-two identity inclusions A(1,e^(epsilon/2))->A(1,e^epsilon) prove that Kahn's annulus hypotheses alone cannot bound the unscaled additive difference between complete collar and target crossing widths; this does not rule out a calibrated or source-specific response. Exact finite Kahn base change preserves every q-fiber, so retiming inside the unchanged equalizer tower cannot make an infinite factor one-sheeted. For revised connected localization covers, after the based lift condition a_*(H_s)<=H_t is checked, the first factor is one-sheeted exactly when a_*^(-1)(H_t)=H_s. The result does not construct the same-source alignment at every Wave stage, an actual retained KL collar, proper or end-proper endpoint classification, a uniform omitted-family response, prime-end naturality, final-root no-birth, or parameter realization.

*Formal hypotheses: none*

*Proof dependencies:* KL_LOCALIZATION_SHEET_RESOLUTION_FACTORIZATION, KAHN_ITERATE_COMPLEMENTARY_COMPONENT_ALIGNMENT, RESOLVED_INTERMEDIATE_RELATIVE_IDEAL_BOUNDARY, CANONICAL_KL_WHOLE_TILE_CORRESPONDENCE_NO_GO, FINITE_HORIZON_KAHN_COLLAR_EMBEDDING

*Evidence:* canonical_kl_restricted_interface_classification.md#1-verdict-scope-and-status

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**FULL_KL_PSEUDOPIECE_MAXIMAL_LIFT_PROPERNESS** PROVED

For one supplied full KL pseudo-piece inclusion immersion iota:Y_s->Y_t, assume the source and target ideal closures are closed conformal disks, iota is a nonconstant holomorphic immersion, and the source-locked KL extension bar(iota) is continuous on the full ideal closure but not necessarily boundary-valued. For every KKL proper target path alpha, every component of the parameterized fiber product {(t,z):alpha(t)=iota(z)} is the graph of a maximal lift on an interval (a,b), and that lift has unique source ideal endpoints p_- and p_+. Exact endpoint naturality gives bar(iota)(p_-)=bar(alpha)(a) and bar(iota)(p_+)=bar(alpha)(b). The proof uses maximality to exclude source-interior cluster points and boundary uniqueness to show every connected source-boundary fiber of bar(iota) is light. Therefore every maximal parameter-compatible lift is a KKL proper source path. An internal parameter endpoint maps to target interior and is a break end, so this theorem gives neither map properness, end-properness, geometric-trace exhaustion, uniqueness, nor a Wave-uniform statement.

*Formal hypotheses: none*

*Proof dependencies:* CANONICAL_KL_RESTRICTED_INTERFACE_CLASSIFICATION, KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY, KKL_IDEAL_PULLBACK_NATIVE

*Evidence:* canonical_kl_geometric_trace_proper_end_response.md#3-fixed-full-pseudo-piece-maximal-lift-properness

---

**PARAMETRIZED_PULLBACK_KKL_EXHAUSTION_NO_GO** PROVED

KKL's immersion restriction of a target proper path is the set of every proper source path contained in the geometric inverse trace, whereas a parameterized fiber product enumerates only maximal parameter-compatible lifts. For the identity disk, a proper lollipop trace contains a loop-omitting proper path absent from the parameterized graph, so the latter does not exhaust the former. Separately, the holomorphic immersion K(z)=exp(-(1+z)/(1-z)) gives one proper target diameter infinitely many distinct proper source restrictions indexed by logarithmic deck phase; pairwise disjoint logarithmic rectangles mapping biholomorphically to one target sector have total raw interior crossing width (2m+1)w_0. Thus local univalence, exact sheet labels, and endpoint data alone imply neither unique proper restriction nor a uniform raw branchwise crossing-width charge.

The rectangles are not asserted to be complete KKL-marked laminations. This no-go does not exclude a proved calibrated Value response and is not an actual quadratic counterexample.

*Formal hypotheses: none*

*Proof dependencies:* CANONICAL_KL_RESTRICTED_INTERFACE_CLASSIFICATION, FULL_KL_PSEUDOPIECE_MAXIMAL_LIFT_PROPERNESS, KKL_IDEAL_PULLBACK_NATIVE

*Evidence:* canonical_kl_geometric_trace_proper_end_response.md#4-parametrized-pullback-does-not-exhaust-kkl-restriction

---

**FINITE_BINDER_PROOF_USE_LEDGER_COMPILER** PROVED

For every supplied certified finite binder-aware source package, the atomic dependency graph oriented from each proof use to its prerequisites has a unique least complete-seed-containing closed ledger, equal to the reachable set with length-zero paths allowed. Its SCC condensation is a finite DAG, every fair work-list terminates at the same ledger, and a missing seed or first missing edge on a shortest seed path gives an omission certificate. Typed seeded graph isomorphisms preserve the compiled ledger. Under an injective typed seeded graph extension, exact fixed-skeleton syntactic stability is equivalent to every enlarged seed remaining in the old image and no enlarged dependency edge leaving it, equivalently every reachable SCC being contained in the old image. This theorem compiles a supplied certificate; it constructs neither the actual direct-decoration execution nor geometric final-root no-birth.

*Formal hypotheses: none*

*Proof dependencies:* INTERFACE_PROOF_USE_INVENTORY, PARAMETRIZED_PULLBACK_KKL_EXHAUSTION_NO_GO

*Evidence:* direct_decoration_wave_proof_use_ledger.md#theorem-82-finite-binder-ledger-compiler

---

**KKL_WAVE_ATOMIC_SOURCE_OCCURRENCE_MANIFEST** PROVED

For the three SHA-locked active KKL Wave source slices, the checked generator records every active ref, eqref, and citation occurrence with stable source coordinates and hashes, resolves all 161 distinct cross-reference labels, retains exactly eight outside-slice label boundaries and six external citation-key boundaries, and emits a deterministic dependency/SCC artifact. The manifest has 300 ref, 78 eqref, and 18 citation occurrences, 571 vertices, 424 edges, 571 singleton SCCs, and eight explicit unresolved formal-to-actual substitutions. It is a lexical occurrence theorem, not a total binder table, theorem-hypothesis instantiation, or analytic execution.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* direct_decoration_wave_proof_use_ledger.md#theorem-85-locked-atomic-source-occurrence-manifest

---

**KKL_BOUNDED_VISITS_ARITHMETIC_AFTER_GROWTH** PROVED

Assuming the interval-growth implication ell(I_{n+18b})>ell(I_n), the indices 1,1+18b,...,1+(12b-1)18b give exactly 12b-1 strict increases, so the last crest has length at least 12b and contradicts shortness. Thus the numerical cap 216b^2 is sufficient conditional on the separately open interval-growth repair.

*Formal hypotheses:* KKL_BOUNDED_VISITS_INTERVAL_GROWTH_REPAIR

*Proof dependencies: none*

*Evidence:* direct_decoration_wave_proof_use_ledger.md#32-conditional-arithmetic-after-a-growth-lemma

---

**KKL_LOCALIZED_WANDERERS_APPLICATION_REDUCTION** PROVED

K06 Property D gives the bounded domination residual. Conditional on a complete localized route-realization certificate and the fully repaired Wanderers theorem, apply Wanderers only to the endpoint-safe tail with first non-good index at least m+1, charge the excluded k=m stratum by 4(6b)^(m-1)||X_I^bv||+C_gen(b,m-1,L1), and retain the Wanderers, recurrence, and residual additive constants. This proves the corrected conditional inequality, not the route certificate, Wanderers theorem, or complete KKL application.

*Formal hypotheses:* KKL_WANDERERS_THEOREM_COMPLETE_REPAIR, KKL_LOCALIZED_WANDERERS_ROUTE_REALIZATION_CERTIFICATE

*Proof dependencies:* KKL_WAVE_ATOMIC_SOURCE_OCCURRENCE_MANIFEST

*Evidence:* direct_decoration_wave_proof_use_ledger.md#proposition-74-endpoint-safe-conditional-tail-reduction

---

`DIRECT_DECORATION_BIGRADED_KL_STAGE_GRID` PROVED

For the actual direct-decoration separator puzzle piece and every finite binary pullback cylinder, geometric parenthood deleting the last symbol and the univalent f^q shift deleting the first symbol form strict commuting squares. Every shift resolves into q univalent unit stages. Intrinsic KL path-space functoriality lifts the grid and the actual first-escape return atlas to pseudo-pieces, with separately typed covering and inclusion arrows, exact critical degree words, central charts, tile-component maps, and central/tile seams commuting. This theorem assigns no KKL Wave trace or carrier-relative access to a grid cell.

*Formal hypotheses: none*

*Proof dependencies:* `FIRST_DECORATION_SCALE_ESCAPE_ATLAS`, `DIRECT_DECORATION_PULLBACK_COMPACTIFICATION`, `KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY`

*Evidence:* `direct_decoration_bigraded_kl_stage_grid.md#theorem-52-bigraded-kl-pseudo-piece-grid`

---

`FINITE_GRID_AND_TERMINAL_KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY` PROVED

Every finite cylinder phase Q_{w,r} and the terminal first-return target Y_B have conformal-disk KL pseudo-pieces. The proof constructs the finite open-star cover, obtains at least two omitted values from the transported separator vertices or the nondegenerate excluded continuum K_1, excludes plane uniformization by Liouville or Little Picard, identifies Kahn's trivial-deck ideal boundary with the conformal circle, and scopes continuous extension to actual KL puzzle-inclusion immersions. No physical local connectivity, properness, or end-properness is asserted.

*Formal hypotheses: none*

*Proof dependencies:* `KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY`, `DIRECT_DECORATION_BIGRADED_KL_STAGE_GRID`, `CRITICAL_BASE_CYCLE_EXCLUSION`

*Evidence:* `direct_decoration_bigraded_kl_stage_grid.md#theorem-54-finite-grid-and-terminal-target-conformal-ideal-circles`

---

`ONE_DIMENSIONAL_CYLINDER_STAGE_COLLAPSE_NO_GO` PROVED

The direct-decoration parent/shift grid cannot be collapsed to one physical-inclusion stage chain: equality of the root inclusions after the prefix shift would imply f^q(z)=z on a nonempty open pullback cylinder and hence f^q=id, contradicting degree 2^q>1. Any carrier-pair naturality must retain both grid directions or an equivalent transported root chart.

*Formal hypotheses: none*

*Proof dependencies:* `DIRECT_DECORATION_BIGRADED_KL_STAGE_GRID`

*Evidence:* `direct_decoration_bigraded_kl_stage_grid.md#proposition-61-physical-inclusionshift-no-go`

---

`DIRECT_DECORATION_COMPONENT_ADDRESS_FINITE_PHASE_SKELETON` PROVED

The connected components of the actual direct-decoration pullback package have canonical finite or infinite binary addresses. For every independently identified finite component K_w, the chronological phase pairs (Q_{w,r},f^r(K_w)), 0<=r<=q, are actual puzzle-disk/Julia-continuum pairs; every unit map is univalent and the phase-q pair is the prefix-deleted pair. Cell suffix-parent inclusions commute with unit dynamics, but no false inclusion between the pairwise disjoint finite Julia components is asserted. Infinite-address continua retain their address and planar support, with no open pseudo-piece or numerical deck phase invented.

*Formal hypotheses: none*

*Proof dependencies:* `DIRECT_DECORATION_PULLBACK_COMPACTIFICATION`, `DIRECT_DECORATION_BIGRADED_KL_STAGE_GRID`

*Evidence:* `wave_to_kl_address_access_selector.md#theorem-32`

---

`CARRIER_ACCESS_TO_KL_MULTICURVE_BRIDGE` PROVED

Given an actual carrier-relative Wave access, two authorized puzzle vertices, endpoint subordination, injective cyclic-order-preserving prime-end/crosscut collars identifying the carrier sides with those vertex accesses, and obstacle avoidance, the KL lifting lemma produces an ordered horizontal pseudo-piece submulticurve with the transported carrier-side tags. The result is conditional on the protected bridge certificate and gives neither whole-Wave equality, width preservation, lift exhaustion, proper/end-proper behavior, nor final-root stability.

*Formal hypotheses: none*

*Proof dependencies:* `DIRECT_DECORATION_BIGRADED_KL_STAGE_GRID`, `FINITE_GRID_AND_TERMINAL_KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY`

*Evidence:* `wave_to_kl_address_access_selector.md#lemma-51`

---

**TERMINAL_K1_EXTERIOR_PUZZLE_VERTEX_CANDIDATE_SET** PROVED

For the terminal primitive first-return target puzzle piece Y_B=cl(B), the component C_1 of K(f) minus B containing K_1 meets the finite puzzle-vertex set. The canonical candidate set V_1(Y_B)=C_1 intersect Vert(Y_B) is therefore finite and nonempty, and for every candidate v one has C_1=S_{Y_B}^v and K_1 contained in that exterior continuum. This is set-level component incidence; it selects no side prime end and constructs no protected collar.

*Formal hypotheses: none*

*Proof dependencies:* CRITICAL_BASE_CYCLE_EXCLUSION, KL_DECORATION_ESCAPE_PUZZLE

*Evidence:* wave_to_kl_address_access_selector.md#theorem-61

---

**TERMINAL_K1_TO_K0_PATH_PUZZLE_VERTEX_CANDIDATES** PROVED

Intersecting the initial outside-return-base component of the actual terminal K_1-to-K_0 Hubbard carrier path with the terminal target puzzle vertices gives a finite nonempty subset of the exterior-component candidate set. Every retained vertex lies on that actual path component and its exterior Julia continuum contains K_1. This narrows the access audit to a canonical finite relation but does not identify the side prime end or construct a crosscut collar.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_K1_EXTERIOR_PUZZLE_VERTEX_CANDIDATE_SET, TERMINAL_CYCLE_HUBBARD_ANCHOR

*Evidence:* wave_to_kl_address_access_selector.md#corollary-63-path-refined-finite-candidate-relation

---

**PUZZLE_VERTEX_EXTERIOR_COMPONENT_INJECTIVITY** PROVED

For every closed geometric piece in the source-locked KL Yoccoz puzzle, the exterior Julia component S_Y^v cut off by the two boundary rays landing at v contains no other puzzle vertex. Hence distinct puzzle vertices determine distinct components of K(f) minus int(Y). The proof uses the ray-pair Jordan separator and no local connectivity of K(f).

*Formal hypotheses: none*

*Proof dependencies:* KL_DECORATION_ESCAPE_PUZZLE

*Evidence:* wave_to_kl_address_access_selector.md#lemma-64-puzzle-vertex-exterior-component-injectivity

---

**TERMINAL_K1_UNIQUE_PUZZLE_VERTEX_ANCHOR** PROVED

The terminal exterior component C_1 containing K_1 attaches to exactly one target puzzle vertex v_*. Both the component candidate set and the candidate set cut out by the actual K_1-to-K_0 Hubbard carrier path equal {v_*}. This selects the KL puzzle vertex canonically but does not select a lateral prime end, a KL boundary subarc, a based deck class, or a protected collar.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_K1_EXTERIOR_PUZZLE_VERTEX_CANDIDATE_SET, TERMINAL_K1_TO_K0_PATH_PUZZLE_VERTEX_CANDIDATES, PUZZLE_VERTEX_EXTERIOR_COMPONENT_INJECTIVITY

*Evidence:* wave_to_kl_address_access_selector.md#corollary-65-unique-terminal-puzzle-vertex-anchor

---

**TERMINAL_ANCHOR_NOT_SMALL_JULIA_VERTEX** PROVED

The unique terminal target puzzle vertex v_* is a finite preimage of the repelling fixed point alpha and cannot lie in any continuum K_i of the primitive small-Julia cycle: otherwise forward invariance and f(alpha)=alpha would put alpha in every pairwise-disjoint K_i. Hence in KKL's locally connected native Hubbard-tree category v_* has singleton quotient fiber. The non-locally-connected lifted-fiber case remains separate.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_K1_UNIQUE_PUZZLE_VERTEX_ANCHOR, TERMINAL_CYCLE_HUBBARD_ANCHOR, KL_DECORATION_ESCAPE_PUZZLE

*Evidence:* wave_to_kl_address_access_selector.md#lemma-66-the-anchor-is-not-a-small-julia-continuum-vertex

---

**TERMINAL_ANCHOR_COMPACT_HALF_COLLARS** PROVED

Every proper compact subinterval of the native terminal side I_{1->0} has an embedded conformal quadrilateral collar in the carrier complement, avoiding any finite closed obstacle family whose ideal closures miss that interval. Every proper compact subarc in the relative interior of either KL tile arc attached at v_* has its own intrinsic analytic collar. These are independent half-collars; no common physical access, welding map, uniform modulus, or full-end theorem is asserted.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_K1_UNIQUE_PUZZLE_VERTEX_ANCHOR, KL_TILE_INTERIOR_HOMOGENEOUS_TRACE

*Evidence:* wave_to_kl_address_access_selector.md#lemma-67-the-two-compact-half-collars

---

**TERMINAL_LOCAL_KL_KKL_ALIGNMENT_RECORD** PROVED

Conditional on an actual frozen terminal proof-use binder path already lying in the exact KL disk and a complete closed or locally-finite-closed obstacle ledger, the common disk, physical complement inclusion, KL central components, selected tile seams, paired pulled-cover/tile basepoints, identity subgroup comparison, and based conformal universal-cover identifications assemble a fixed marked terminal KL/KKL alignment record. The record retains its choices and asserts no presentation invariance, proper restriction, access welding, or whole-horizon result.

*Formal hypotheses:* DIRECT_DECORATION_WAVE_PROOF_USE_LEDGER

*Proof dependencies:* TERMINAL_K1_UNIQUE_PUZZLE_VERTEX_ANCHOR, TERMINAL_RELATIVE_COVER_SUBGROUP_ALIGNMENT, TERMINAL_OUTER_SEAM_DECK_NORMALIZATION

*Evidence:* terminal_relative_cover_alignment.md#theorem-91

---

**TERMINAL_SEAM_NORMALIZED_PATH_LIFT** PROVED

For each physical KKL proper restriction in one fixed marked terminal alignment, the frozen binder determines a raw authorization relation of actual seam occurrences or based connector classes. Evaluation sends authorization records to based KL/pulled-cover lift phases by unique path lifting. A literal unique lift phase is equivalent to the evaluation image having cardinal one; canonical reconstructibility additionally requires its provenance fiber to have cardinal one. All raw records are quotiented only by simultaneous oriented reparametrization, never by deck translation inside the fixed certificate.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_LOCAL_KL_KKL_ALIGNMENT_RECORD, TERMINAL_RELATIVE_COVER_SUBGROUP_ALIGNMENT, TERMINAL_OUTER_SEAM_DECK_NORMALIZATION

*Evidence:* wave_to_kl_address_access_selector.md#lemma-69-seam-normalized-path-lift

---

**TERMINAL_ANCHOR_COLLAR_OR_RESTRICTION_OBSTRUCTION** PROVED

Conditional on one fixed complete terminal local alignment record, count the full physical KKL proper-path restriction universe for the actual bound Wave trace, never merely connected inverse-trace components or deck translates. The ordered filter distinguishes a non-singleton lifted vertex fiber; zero, finite-multiple, or infinite physical restrictions; missing or competing seam lift phases; competing occurrence/connector provenance; failure of proper local embedding; collision with the declared closed forbidden set; finite endpoint access/subordination failure; and failure of two-flank ambient prime-end survival. Path-level lift properness and exclusion of deck-limit endpoints are already proved by the seam-lift theorem. Only when exactly one physical restriction and one reconstructible fixed-marked lift pass every remaining test does a variable-width ambient-ledger-free Jordan strip have an injective KL-tile lift with distinct cyclic-order-preserving flanks in the declared vertex access. The actual binder universe and positive branch remain open.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_LOCAL_KL_KKL_ALIGNMENT_RECORD, TERMINAL_K1_UNIQUE_PUZZLE_VERTEX_ANCHOR, TERMINAL_ANCHOR_NOT_SMALL_JULIA_VERTEX, TERMINAL_ANCHOR_COMPACT_HALF_COLLARS, TERMINAL_RELATIVE_COVER_SUBGROUP_ALIGNMENT, TERMINAL_SEAM_NORMALIZED_PATH_LIFT, TERMINAL_SEAM_LIFT_END_PROPERNESS, TERMINAL_KL_FINITE_ENDPOINT_FIVE_WAY_CLASSIFICATION, TERMINAL_PROPER_ACCESS_CHANNEL_ATLAS, TERMINAL_FINAL_ROOT_COVER_OBSTACLE_DECOMPOSITION, FINITE_GRID_AND_TERMINAL_KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY, PARAMETRIZED_PULLBACK_KKL_EXHAUSTION_NO_GO

*Evidence:* wave_to_kl_address_access_selector.md#proposition-610-filtered-collar-or-obstruction-theorem

---

**KL_TILE_INJECTIVITY_BASE_ANNULUS_REDUCTION** PROVED

For a degree-one holomorphic immersion I between connected annuli and any compatible universal-cover lift L, deck equivariance gives L∘T_s=T_t∘L and L is injective if and only if I is injective. A base collision u!=v, I(u)=I(v), lifts to an exact same-tile collision after one uniquely determined deck shift. A proper degree-one annulus immersion is a one-sheeted covering and hence a biholomorphism. In the general KL psi-ql stage the tile maps lift a bold ambient annulus immersion, not the physical nested-annulus inclusion of the ordinary ql model; KL prove only an embedded inner collar and the limited end-properness used in their width argument. Thus this is an exact reduction and sufficient criterion, not a proof of actual KL base univalence.

*Formal hypotheses: none*

*Proof dependencies:* KL_DECORATION_ESCAPE_PUZZLE

*Evidence:* actual_kl_tile_injectivity_gate.md#4-tilebase-injectivity-equivalence

---

**TILE_INJECTIVITY_GLUE_EMBEDDING_CRITERION** PROVED

For the fixed direct-decoration pair, the dynamical branch bijects complementary components and KL surjectivity therefore makes the inclusion component map bijective. The set identity Q intersect P_j=Q_i for the unique i with j(i)=j shows that only the target overlap lying in the source-central image can participate in a tile–central collision. If every full source-to-target universal-cover tile immersion L_i is injective, overlap commutation gives L_i^(-1)(tilde e_j(Q_i))=e_i(Q_i), and a complete central/central, same-tile, different-tile, and tile/central quotient audit proves that the glued pseudo-piece immersion is an interior holomorphic embedding. Conversely, glued injectivity implies every tile map is injective. Full reflection over all of P_j is stronger than necessary and is not an independent interior-embedding gate.

*Formal hypotheses: none*

*Proof dependencies:* KL_DECORATION_ESCAPE_PUZZLE

*Evidence:* kl_relative_overlap_cartesianity.md#6-corrected-full-pseudo-piece-embedding-theorem

---

**KL_TILE_INTERIOR_HOMOGENEOUS_TRACE** PROVED

A compact nondegenerate subarc contained in the relative interior of one universal-cover ideal-boundary tile arc of a Kahn–Lyubich pseudo-piece has an intrinsic coordinate-Lipschitz quadrilateral collar and a complete homogeneous minimum-Dirichlet-energy Hilbert trace quotient. A conformal isomorphism between selected transported collars acts isometrically. This gives no canonical collar, global glued-boundary identification, physical Julia-boundary collar, full gauge, uniform constants, immersion transport, or word naturality.

*Formal hypotheses: none*

*Proof dependencies:* KL_DECORATION_ESCAPE_PUZZLE

*Evidence:* direct_decoration_path_lifted_collar_certificate.md#6-tile-interior-homogeneous-trace-theorem

---

**NONPROPER_LABEL_HULL_PULLBACK_NO_GO** PROVED

There is a holomorphic embedding of one Jordan disk into another and a two-interval target ideal marking for which the deletion-and-seed finite label hull of one target interval retains every open source atom, including a nonproper atom N, while a nondegenerate subarc of N cannot belong to the KKL immersion pullback: an explicit target proper crosscut has that subarc as its unique subordinate source proper lift but is not subordinate to the selected target interval. This conclusion does not require the separate endpoint predicate to retain the two intervening vertices. Proper/nonproper labels and flanking labels therefore do not determine the KKL pullback.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* direct_decoration_path_lifted_collar_certificate.md#8-analytic-counterexample-to-the-finite-label-hull

---

**KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY** PROVED

Each of the two audited Kahn–Lyubich direct-decoration pseudo-pieces Pbold and Qbold_a is a hyperbolic conformal disk. Kahn's universal-uniformization ideal closure is therefore the closed disk, the named KL lambda/xi/eta arcs tile that conformal ideal circle, and their puzzle-inclusion immersion has the stated continuous extension to it. The proof uses KL's convention that a disk is an open simply connected domain, the finite open star gluing and van Kampen, a Liouville/Little-Picard dichotomy excluding plane uniformization for these pieces, Kahn's ideal-boundary definition, and KL's boundary paragraph; it assumes no local connectivity of the physical Julia set and asserts no unaudited broader pseudo-piece class.

*Formal hypotheses: none*

*Proof dependencies:* KL_DECORATION_ESCAPE_PUZZLE

*Evidence:* direct_decoration_outer_port_footprint.md#theorem-91-conformal-ideal-boundary-identification

---

**EMBEDDED_DISK_ACCESSIBLE_FOOTPRINT_THEOREM** PROVED

For a holomorphic embedding of conformal disks with continuous extension to their conformal ideal closures, every parameter component of a target-path intersection has two source ideal limits, and the unique maximum KKL pullback of a connected target ideal set I is exactly the source safe set of prime ends whose complete complement-access footprint is contained in I. The proof establishes light boundary fibers by Jensen and Fatou, treats the entire geometric restriction rather than claiming a parameter-component bijection, excludes

branch-switching paths under the uniqueness hypothesis, handles repeated traversals, and realizes the converse with a repeated-access hairpin.

*Formal hypotheses: none*

*Proof dependencies:* `KKL_IDEAL_PULLBACK_NATIVE`

*Evidence:* `embedded_disk_accessible_footprint_theorem.md#6-full-access-footprint`

---

`GENERAL_ANNULUS_IMMERSION_LOWER_STRIP_NO_GO` PROVED

Degree one, Z-equivariance, holomorphic immersion or even global embedding, preservation of the lower boundary, and mapping of a source strip into a taller target strip do not force a translation or an exact horizontal lower-strip image. For sufficiently small nonzero real epsilon, L_epsilon(z)=z+epsilon sin(2 pi z) is a Z-equivariant holomorphic embedding of Pi(h_s) into Pi(h_t), descends to degree one on annular quotients, and has a nonhorizontal upper boundary. Thus the actual complement-access footprint, not nominal height, is the required invariant for a general KL tile immersion.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `direct_decoration_outer_port_footprint.md#proposition-42-degree-one-immersion-does-not-imply-a-lower-strip`

---

`COMMON_COORDINATE_NESTED_ANNULUS_OUTER_CAP` PROVED

For a pseudo-piece pair whose source and target tile annuli are literal nested annuli in one external coordinate, whose tile maps lift those inclusions, and whose lifted inclusions additionally reflect the declared central overlaps, tile-component bijectivity gives an embedded pseudo-piece map, exact lower-strip images, and an isolated connected upper cap bounded by the source O-side. For every prime end p on that open source O-side, the union of accesses through all complement components landing at p is exactly the closed target O-port. The embedded-disk footprint formula then excludes the open target O-atom and absorbs the source O-side into the native three-atom open tile-star. This is a conditional common-coordinate positive control; neither the common coordinate nor overlap reflection is proved for a general KL psi-ql immersion.

*Formal hypotheses: none*

*Proof dependencies:* `TILE_INJECTIVITY_GLUE_EMBEDDING_CRITERION`, `KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY`, `EMBEDDED_DISK_ACCESSIBLE_FOOTPRINT_THEOREM`

*Evidence:* `direct_decoration_outer_port_footprint.md#theorem-93-conditional-exact-outer-port-criterion`

---

`END_BLOWUP_PROJECTION_NO_GO` PROVED

If pointwise coherent embeddings of a nondegenerate connected compact root carrier converge in a Hausdorff marked space and a continuous projection agrees with the collapsing direct-decoration planar copies, then the limiting carrier lies in one nontrivial projection fiber and the projection is not locally injective at any point of that carrier. This does not construct or force a unique conformal blow-up category.

*Formal hypotheses: none*

*Proof dependencies:* `PLANAR_END_COLLAPSE_NO_GO`

*Evidence:* `critical_chronology_and_retimed_telescope.md#13-normalized-retimed-germ-and-projection-collapse`

---

`FIXED_ITERATE_END_FIBER_EXTENSION` PROVED

For one fixed continuous map between Hausdorff spaces and nested nonempty compact source and target sets, levelwise inclusion passes to the intersections; levelwise surjectivity gives equality by nested closed compact preimage fibers. Only explicitly compatible or eventually constant observables pass formally.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `critical_chronology_and_retimed_telescope.md#14-fixed-iterate-action-on-nested-end-fibers`

---

`FINITE_CYLINDER_COMMUTING_ITERATE_LIFT` PROVED

For equal-depth univalent cylinders h_w=f^N|Q_w and h_v=f^N|Q_v, an actual branch F=f^r|A satisfying f^r(h_w^{-1}(A)) subset Q_v obeys f^r=h_v^{-1} composed with F composed with h_w. If F maps A onto B, the lifted image is exactly h_v^{-1}(B); proper degree and local degree are preserved, while critical chronology remains separate metadata.

*Formal hypotheses: none*

*Proof dependencies:* *none*

*Evidence:* `finite_cylinder_commuting_lift.md#theorem-31-finite_cylinder_commuting_iterate_lift`

---

`FIXED_BASE_CONNECTED_IMAGE_TARGET_UNIQUENESS` PROVED

For one typed target base and one finite equal-depth family of pairwise-disjoint nonempty connected compact cylinders, every nonempty connected image contained in their union lies in one unique cylinder. This does not select the target base or prove cross-base component ownership.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `finite_cylinder_commuting_lift.md#proposition-41-fixed_base_connected_image_target_uniqueness`

---

`INTERIOR_PULLBACK_COMPONENT_FORCING` PROVED

After a target base is typed, commutation forces every connected equal-depth lifted branch image whose root image lies in the target interior into one open pullback component. If that component is an authorized closed Jordan cylinder, fixed-base uniqueness and the commuting lift force the target word and branch formula; compact containment of the root closure forces the lifted closure away from every pullback boundary.

*Formal hypotheses:* *none*

*Proof dependencies:* `FINITE_CYLINDER_COMMUTING_ITERATE_LIFT`, `FIXED_BASE_CONNECTED_IMAGE_TARGET_UNIQUENESS`

*Evidence:* `finite_cylinder_commuting_lift.md#theorem-43-interior_pullback_component_forcing`

---

`VIRTUAL_PROOF_USE_TRACE_SELECTOR` PROVED

After the complete proof-use horizon and an actual virtual component transition system with a countable compact-target proof-use catalog are frozen, equality of all finite-future observations gives a future-faithful quotient whose compact metrizable end completion continuously extends every declared Wave, Wanderers, covering, quasi-additivity, width, transport, access, obstacle, and authorized enlargement observable. This does not give a universal finite quotient or response-dimension bound.

*Formal hypotheses:* *none*

*Proof dependencies:* `FUTURE_FAITHFUL_END_TOPOLOGY`

*Evidence:* `virtual_cantor_package_interface.md#theorem-72-virtual_proof_use_trace_selector-proved-conditional-interface`

---

`DEMAND_CLOSED_RETURN_ATLAS_COMPILER` PROVED

A sound analytic return atlas with actual open bases, atomic iterate restrictions, selected pullback components, complete metadata, and a frozen proof-complete Wave demand relation has a unique least horizon-independent rooted unfolding; every node is an actual polynomial iterate on an actual domain, with exact chronological metadata and a prefix-closed language exhaustive for the declared demand schema.

*Formal hypotheses:* *none*

*Proof dependencies:* `RETURN_PATH_LANGUAGE`

*Evidence:* `virtual_return_branch_language.md#theorem-51-demand-closed-analytic-atlas-compiler`

---

`RETURN_ATLAS_ENLARGEMENT_NATURALITY` PROVED

An injective edgewise analytic morphism of demand-closed return atlases that commutes on actual branch domains and preserves selected components and all metadata induces a unique injective prefix morphism of their rooted unfoldings; complete no-birth/no-loss language stability requires bijection on demanded children at every prefix.

*Formal hypotheses:* *none*

*Proof dependencies:* `DEMAND_CLOSED_RETURN_ATLAS_COMPILER`

*Evidence:* `virtual_return_branch_language.md#theorem-61-return-atlas-enlargement-naturality`

---

`FINITE_GEOMETRIC_SATURATION` PROVED

On a fixed finite resolved cell universe, any finite family of inflationary monotone package, transport, fullification, access, attachment, and collision rules has one order-independent least common fixed point; a cell quotient agrees with geometric saturation exactly when it is implication-complete and implication-sound.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `canonical_virtual_interface_carrier.md#theorem-41-finite-geometric-saturation`

---

`CONNECTED_CARRIER_NO_MINIMUM` PROVED

Even two prescribed points have no least compact connected full planar carrier under inclusion; consequently a connected virtual hull requires additional regulated connector data and cannot be defined as the minimal connected full hull.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `canonical_virtual_interface_carrier.md#theorem-51-nonexistence-of-a-least-full-connected-hull`

---

`QUADRATIC_PROTECTED_ACCESS_COLLISION` PROVED

For an explicit critical-value-free branch of z^2 on a right-half-plane Jordan disk, the first pulled-carrier excess is an upper half-disk over the immersed interval, and every monotone absorption of that excess erases a protected upper prime-end access; the complete itinerary has no critical, sheet, obstacle, or localization collision.

*Formal hypotheses: none*

*Proof dependencies:* `CRITICAL_VALUE_CORRIDOR_CRITERION`, `FINITE_GEOMETRIC_SATURATION`

*Evidence:* `canonical_virtual_interface_carrier.md#theorem-62-protected-access-collision`

---

`PROTECTED_REGULATED_CONNECTOR_REALIZATION` PROVED

Every finite protected ribbon-routing certificate on pairwise disjoint full planar atoms has an actual compact connected full carrier with singleton connector traces, injective cyclic-order-preserving survival of all declared prime-end crosscut chains, a finite ribbon-side marking, and exact functoriality under univalent changes of Jordan localization; no local connectivity of the atoms is required.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `protected_regulated_connector.md#theorem-43-protected-regulated-connector-realization`

---

`PROTECTED_FREE_COMPONENT_CRITERION` PROVED

After the finite proof-use collars and obstacles are frozen, a protected ribbon-routing carrier exists exactly when one component of their planar complement has simultaneous singleton access ports to every required full atom; a first collar or obstacle separating two required port classes is a monotone-unrepairable obstruction within the current localization.

*Formal hypotheses: none*

*Proof dependencies:* `PROTECTED_REGULATED_CONNECTOR_REALIZATION`

*Evidence:* `virtual_protected_routing_domain.md#theorem-41-protected-free-component-criterion`

---

`AUGMENTED_ATOM_COMMON_DOMAIN` PROVED

If a finite frozen forbidden inventory is covered by pairwise disjoint full augmented root-shadow atoms and full separated obstacles compactly contained in one Jordan localization, and every original atom retains a pairwise disjoint explicit free singleton port from their complement, then that complement is connected, all ports lie in one component of the original free domain, and a protected regulated connector exists.

*Formal hypotheses: none*

*Proof dependencies:* `ALEXANDER_DUALITY_COMPACTA`, `PROTECTED_FREE_COMPONENT_CRITERION`

*Evidence:* `virtual_common_free_component.md#theorem-43-augmented-atom-common-domain-theorem`

---

`AUGMENTED_PACKET_CLOSURE_AUDIT` PROVED

For a fixed finite typed packet assignment, planar fullification is an order-independent closure; its coverage, connectedness, post-fill merger, localization, and original-atom exposure conditions exactly decide the geometric augmented-packet clauses. Individual unprotected ports can be truncated to pairwise disjoint simultaneous ports, post-fill mergers, localization escapes, and empty exposure loci persist under monotone enlargement with fixed labels, and the complete audit is natural under sphere-extending univalent transport.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `augmented_packet_closure_audit.md#theorem-71-fixed-packet-closure-and-port-audit`

**CELLWISE_PACKET_PORT_CERTIFICATE** PROVED

Pairwise-disjoint closed Jordan cell envelopes compactly contained in one localization, each carrying one connected complete packet seed and, for every original atom, an unprotected arc from that atom through its cell complement to the cell boundary, force pairwise-disjoint localized fullifications, nonempty unprotected exposure loci, simultaneous singleton ports, one common free component, and a protected regulated connector.

*Formal hypotheses: none*

*Proof dependencies:* AUGMENTED_PACKET_CLOSURE_AUDIT, AUGMENTED_ATOM_COMMON_DOMAIN

*Evidence:* augmented_packet_closure_audit.md#proposition-54a-cellwise-packet-and-port-certificate

---

**PARALLEL_REPLICATION_CRITERION** PROVED

r_parallel retained WAD summands injecting into disjoint global canonical-WAD support coordinates and having individual weight at least a*D give total active weight at least a*r_parallel*D.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* virtual_hull_realization.md#4-active-replication-is-analytic

---

**TOPOLOGY_REPLICATION_NO_GO** PROVED

Finite tree topology and bounded incidence do not imply active replication: serial equal-width cells and repeated isotopic returns have vanishing replication efficiency.

*Formal hypotheses: none*

*Proof dependencies:* HARMONIC_LINEAR_NO_GO

*Evidence:* virtual_hull_realization.md#4-active-replication-is-analytic

---

**CONDITIONAL_TREE_REPLICATION_CALIBRATION** PROVED

An access-complete translation-realized carrier with O(R) standard width and complexity and positive replication efficiency has WAD width between Q and (1+O(1/D))*Q; with separate counts the error is O(rho_B/D).

*Formal hypotheses: none*

*Proof dependencies:* TREE_HULL_COMPLEXITY, WAD_CORRIDOR_SANDWICH, PARALLEL_REPLICATION_CRITERION, CONGESTION_AMORTIZATION

*Evidence:* virtual_hull_realization.md#5-conditional-wad-calibration

---

**DISCOUNTED_DUHAMEL** PROVED

Iterating v_{j+1} <= exp(-epsilon_j)*v_j+eta_j yields the exact terminal contraction plus the future-entropy-discounted error sum.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* entropy_and_cycle_bookkeeping.md#2-exact-iteration-formula

---

**DISCOUNTED_ENTROPY_ABSORPTION** PROVED

A discounted error at most theta*(1-exp(-E))*v_0+C with theta<1 and E bounded below yields a block-independent affine contraction coefficient mu<1, hence strict decrease above C/(1-mu).

*Formal hypotheses: none*

*Proof dependencies:* DISCOUNTED_DUHAMEL

*Evidence:* entropy_and_cycle_bookkeeping.md#3-sufficient-entropy-charge

---

**ENTROPY_THRESHOLD_NO_GO** PROVED

A positive lower bound on total entropy alone does not control block error without a discounted or telescoping local charge.

*Formal hypotheses: none*

*Proof dependencies:* DISCOUNTED_DUHAMEL

*Evidence:* entropy_and_cycle_bookkeeping.md#4-no-go-for-qualitative-entropy-payment

---

`CHART_CYCLE_OBSTRUCTION` PROVED

A common backward expansion factor lambda>1 forces lambda^k expansion around every k-cycle, while a common forward contraction mu<1 forces mu^k contraction; a positive-gauge identity cycle obstructs either strict calibration.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `entropy_and_cycle_bookkeeping.md#6-directed-chart-cycles`

---

`RESPONSE_GLUE_CALCULUS` PROVED

For finite conductance networks, the terminal response matrix represents parallel gluing by addition and intermediate-terminal elimination by Schur complement (with the quotient-safe pseudoinverse when needed), recovering the scalar harmonic law exactly.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `network_response_state.md#3-exact-gluing-laws`

---

`FINITE_ELECTRODE_RESPONSE` PROVED

Every fixed finite smooth electrode surface has an exact positive-semidefinite Dirichlet-to-Neumann response matrix whose kernel is componentwise constants and which is a finite conductance-graph Laplacian.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `bc1_finite_port_response.md#2-finite-electrode-surfaces`

---

`PORT_FUTURE_CLOSURE_NO_GO` PROVED

A port quotient that merges electrodes which can separate later loses response information and cannot be a sufficient state through that horizon.

*Formal hypotheses: none*

*Proof dependencies:* `FINITE_ELECTRODE_RESPONSE`

*Evidence:* `bc1_finite_port_response.md#3-port-compression`

---

`WAD_DOMINATION_RESPONSE` PROVED

Under an itinerary-faithful port assignment, KKL coefficientwise harmonic-sum WAD domination implies Loewner domination of the terminal response after eliminating intermediate itinerary ports.

*Formal hypotheses: none*

*Proof dependencies:* `RESPONSE_GLUE_CALCULUS`

*Evidence:* `bc1_finite_port_response.md#5-kkl-domination-is-electrical-domination`

---

`FIXED_BLOCK_SPECTRAL_COMPARISON` PROVED

On a fixed finite terminal space, exact surface and WAD response forms with the same kernel are two-sided comparable by their generalized Rayleigh eigenvalues; this gives no uniform bound over blocks.

*Formal hypotheses: none*

*Proof dependencies:* `FINITE_ELECTRODE_RESPONSE`, `RESPONSE_GLUE_CALCULUS`

*Evidence:* `bc1_finite_port_response.md#6-fixed-block-comparison-and-the-uniformity-gap`

---

`VALUE_RESPONSE_NO_GO` PROVED

Total WAD weight, terminal response, and an unweighted port pattern do not determine KKL Value when compression merges support classes with different Value coefficients; adjoining any aggregate coordinate fixed by the redistribution does not repair the loss.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `bc1_finite_port_response.md#7-value-is-not-observable-from-response-alone`

---

**VALUE_FEATURE_ROWSPACE** PROVED

For a finite lifted linear WAD state z=T*w and valuation family C*w, exact observability on the nonnegative cone is equivalent to row(C) being contained in row(T); the minimal number of added linear moments is rank([T;C])-rank(T).

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* bc2_value_feature.md#3-exact-observability-theorem

---

**PRIMITIVE_VALUE_AFFINE_CONTRACTION** PROVED

On one source-licensed KKL primitive block, one retained scalar Value coordinate obeys V_out <= lambda_F^{-n}*V_in + V(error), so an entropy-clock lower bound gives strict homogeneous contraction.

*Formal hypotheses: none*

*Proof dependencies:* KKL_VALUE_CONTRACTION, KKL_TYPED_VALUE_INCIDENCE

*Evidence:* bc2_value_feature.md#5-source-licensed-primitive-value-block

---

**CLONED_VALUE_BLOCK_CONTRACTION** PROVED

For every fixed source-licensed literal cloning sequence satisfying the corrected fixed-cross-section theorem, every sufficiently long complete primitive block has a uniform strict total Value contraction even though its elementary contraction tends to one.

*Formal hypotheses: none*

*Proof dependencies:* KKL_VALUE_CONTRACTION, KKL_TYPED_VALUE_INCIDENCE, KKL_CLONE_EIGEN

*Evidence:* bc2_value_feature.md#6-corrected-fixed-cross-section-cloning-special-case

---

**PRIMITIVE_INTEGER_ENTROPY_GAP** PROVED

Every irreducible non-permutation nonnegative integer matrix of dimension at most d has Perron root at least 2^(1/d).

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* virtual_clone_entropy.md#3-explicit-bounded-dimension-entropy-gap

---

**VARIABLE_CLONE_BLOCK_CONTRACTION** PROVED

Any source-licensed KKL primitive block satisfying the corrected negative-weight clone equation, with irreducible non-permutation gate dimension at most d, maximum first-return length K<=N, common delay n>N, and coefficientwise WAD domination Y<=(f^r)^*X+E for an actual pullback exponent r>=n, has homogeneous complete-block Value factor at most 2^(-1/(2d)), independently of period and graph template.

*Formal hypotheses: none*

*Proof dependencies:* KKL_VALUE_CONTRACTION, KKL_TYPED_VALUE_INCIDENCE, KKL_CLONE_EIGEN, PRIMITIVE_INTEGER_ENTROPY_GAP

*Evidence:* virtual_clone_entropy.md#4-uniform-total-contraction-for-corrected-clone-templates

---

**PERTURBED_CLONE_CONTRACTION** PROVED

If an actual weighted reduced transition dominates a (1-epsilon) fraction of a corrected bounded-dimensional clone family at its own root and satisfies its own negative-weight clone equation, then its multiplier is at least ((1-epsilon)*2^(1/d))^(n/(n+K)); it remains uniformly expanding for epsilon < 1-2^(-1/d).

*Formal hypotheses: none*

*Proof dependencies:* KKL_CLONE_EIGEN, PRIMITIVE_INTEGER_ENTROPY_GAP

*Evidence:* virtual_clone_entropy.md#4-uniform-total-contraction-for-corrected-clone-templates

---

**BOUNDED_ENTROPY_EIGENCONE** PROVED

For exact corrected clone solutions with bounded gate dimension and return depth, reduced return Perron root at most H, and n>=N, the total multiplier is at most H and both reduced Perron rays have coordinate ratio at most H^(2(d-1)); one-sided perturbation domination alone does not imply this compactness.

*Formal hypotheses: none*

*Proof dependencies:* KKL_CLONE_EIGEN, PRIMITIVE_INTEGER_ENTROPY_GAP

*Evidence:* virtual_clone_entropy.md#5-bounded-entropy-eigencone-compactness

---

`UNBOUNDED_DIMENSION_ENTROPY_NO_GO` PROVED

Primitive integer transition graphs can have Perron roots tending to one when their dimension is unbounded.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_clone_entropy.md#8-necessity-of-reduced-complexity-control`

---

`EXACT_CORRIDOR_CLONE_CRITERION` PROVED

A finite directed multigraph is a KKL cross-section clone exactly when its corridor has only the declared label-preserving chain edges, every other incidence occurs at the two endpoints, and deleting the distinguished chain edges before label identification gives the stated cross-sectional reduced quotient with the same multiplicities.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `corridor_clone_criterion.md#3-exact-translation-corridor`

---

`ENDPOINT_DEFECT_ABSORPTION` PROVED

Corridor defects confined to fixed entry and exit windows can be absorbed into the active graph without increasing reduced cross-section cardinality, provided the central reduced quotient retains the cross-section property.

*Formal hypotheses: none*

*Proof dependencies:* `EXACT_CORRIDOR_CLONE_CRITERION`

*Evidence:* `corridor_clone_criterion.md#4-endpoint-defect-absorption`

---

`FINITE_CORRIDOR_CLONE_UNION` PROVED

Equal-length disjoint exact corridors combine into one clone when their union is a cross section of the combined reduced quotient, with reduced dimension equal to the sum of component cross-section cardinalities.

*Formal hypotheses: none*

*Proof dependencies:* `EXACT_CORRIDOR_CLONE_CRITERION`

*Evidence:* `corridor_clone_criterion.md#5-several-parallel-corridors`

---

`CROSS_SECTION_DAG_EQUIVALENCE` PROVED

For a finite directed graph, every sufficiently long directed walk meets E if and only if the induced graph on its complement is acyclic; every E-avoiding walk then has at most |V minus E|-1 edges.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `dynamical_corridor_certificate.md#2-cross-sections-are-exactly-feedback-vertex-sets`

---

`CLONE_STRONG_CONNECTIVITY` PROVED

Cloning a nonempty cross section of a strongly connected finite directed graph preserves strong connectivity for every clone delay when every gate vertex lies on a positive-length closed walk, as is automatic for a nontrivial strongly connected base.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `dynamical_corridor_certificate.md#3-primitivity-under-arbitrary-clone-delay`

---

`ROBUST_CLONE_PRIMITIVITY` PROVED

If a strongly connected cross-section graph has two closed walks with the same number of gate passages and lengths differing by one, every delayed clone is primitive.

*Formal hypotheses: none*

*Proof dependencies:* `CLONE_STRONG_CONNECTIVITY`

*Evidence:* `dynamical_corridor_certificate.md#3-primitivity-under-arbitrary-clone-delay`

---

**PRIMITIVE_BASE_CLONE_NO_GO** PROVED

Primitivity of the uncloned base does not imply primitivity of all clones: the bouquet of directed 3- and 5-cycles has a primitive base and a period-two length-one clone.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* dynamical_corridor_certificate.md#3-primitivity-under-arbitrary-clone-delay

---

**FINITE_STATE_CORRIDOR_CERTIFICATE** PROVED

A complete finite core-transition model with pure equal-length corridor interiors, bounded reduced active and gate sets, acyclic active complement, and delay-robust mixing yields an exact primitive clone graph with uniform reduced dimension and label depth.

*Formal hypotheses: none*

*Proof dependencies:* EXACT_CORRIDOR_CLONE_CRITERION, ENDPOINT_DEFECT_ABSORPTION, FINITE_CORRIDOR_CLONE_UNION, CROSS_SECTION_DAG_EQUIVALENCE, ROBUST_CLONE_PRIMITIVITY

*Evidence:* dynamical_corridor_certificate.md#4-the-finite-state-corridor-certificate

---

**FINITE_STATE_CORRIDOR_VALUE_COROLLARY** PROVED

Under the full periodically repelling prime-primitive psi-ql hypotheses, typed incidence B=N^T with KKL left/right Value roles, coefficientwise WAD domination Y<=(f^r)^*X+E for an actual pullback exponent r at least the literal clone delay n, non-permutation return, and certified n>K, the finite-state graph certificate activates the corrected clone theorem and gives homogeneous complete-block Value factor at most 2^(-1/(2d0)); its error remains separate.

*Formal hypotheses: none*

*Proof dependencies:* FINITE_STATE_CORRIDOR_CERTIFICATE, KKL_VALUE_CONTRACTION, KKL_TYPED_VALUE_INCIDENCE, VARIABLE_CLONE_BLOCK_CONTRACTION

*Evidence:* dynamical_corridor_certificate.md#4-the-finite-state-corridor-certificate

---

**FRAMED_RETURN_SPECTRAL_INVARIANCE** PROVED

A closed source-column return changes by similarity under a base-frame relabeling, so its spectral data are invariant, whereas an open transfer's spectral radius can change under an independent endpoint relabeling.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* transfer_convention_and_frame.md#4-framed-returns

---

**PERMUTATION_COCYCLE_GAUGE** PROVED

A source-column layered interior whose every core transfer is a permutation has an identity label gauge after transporting layer labels; this alone does not prove corridor purity, a cross section, or WAD compatibility.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* corridor_entropy_shear_dichotomy.md#2-layered-core-transfers

---

**IRREDUCIBLE_TRANSFER_ENTROPY_DICHOTOMY** PROVED

A nonzero irreducible nonnegative integer matrix of dimension d is either one directed-cycle permutation with spectral radius one or has spectral radius at least 2^(1/d).

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* corridor_entropy_shear_dichotomy.md#3-irreducible-layers-translation-or-uniform-entropy

---

**ZERO_ENTROPY_SHEAR_NORMAL_FORM** PROVED

Every nonnegative integer matrix of spectral radius at most one has cycle-or-zero Frobenius diagonal blocks and polynomially bounded feed-forward transport; if it has no recurrent cycle block it is nilpotent.

*Formal hypotheses: none*

*Proof dependencies:* IRREDUCIBLE_TRANSFER_ENTROPY_DICHOTOMY

*Evidence:* corridor_entropy_shear_dichotomy.md#4-reducible-nonexpansion-is-extinction-or-polynomial-shear

---

`DEFECT_RUN_DECOMPOSITION` PROVED

A layered corridor with q non-permutation transfer defects splits outside those defects into at most q+1 identity-gauged permutation interior runs, while bounded local type count does not bound q; clone purity remains independent.

*Formal hypotheses:* *none*

*Proof dependencies:* `PERMUTATION_COCYCLE_GAUGE`

*Evidence:* `corridor_entropy_shear_dichotomy.md#5-defect-decomposition`

---

`ZERO_ENTROPY_SHEAR_NO_GO` PROVED

The constant two-state unipotent shear has one local type and zero exponential entropy but unbounded linear transport, so exact-clone or positive-entropy is not exhaustive without a regularized shear branch.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `corridor_entropy_shear_dichotomy.md#4-reducible-nonexpansion-is-extinction-or-polynomial-shear`

---

`BOOLEAN_SUPPORT_FUNCTOR` PROVED

For nonnegative matrices, the support of a product is exactly the Boolean product of the supports.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `corridor_support_automaton.md#2-labeled-transfer-automaton`

---

`SUPPORT_AUTOMATON_PUMPING` PROVED

A path in an s-state automaton labeled by d-dimensional nonnegative transfers has a support skeleton of length at most s*2^(d^2)-1, with every deleted segment retained as a loop preserving both automaton state and prefix support.

*Formal hypotheses:* *none*

*Proof dependencies:* `BOOLEAN_SUPPORT_FUNCTOR`

*Evidence:* `corridor_support_automaton.md#3-support-loop-pumping`

---

`BOUNDED_BRANCH_SUPPORT_WITNESS` PROVED

If a finite transfer automaton admits a product whose support has a cyclic strongly connected component that is not a simple directed cycle, it admits one of length at most s*2^(d^2)-1, and every such displayed square product has spectral radius at least 2^(1/d); dynamical entropy additionally requires a source-licensed framed repeatable return.

*Formal hypotheses:* *none*

*Proof dependencies:* `SUPPORT_AUTOMATON_PUMPING`, `IRREDUCIBLE_TRANSFER_ENTROPY_DICHOTOMY`, `FRAMED_RETURN_SPECTRAL_INVARIANCE`

*Evidence:* `corridor_support_automaton.md#4-bounded-witnesses-for-support-entropy`

---

`FUTURE_EQUIVALENCE_AUTOMATON` PROVED

The minimum state count of an exact deterministic decorated block-language automaton equals the number of distinct right future-equivalence classes.

*Formal hypotheses:* *none*

*Proof dependencies:* *none*

*Evidence:* `virtual_language_memory.md#2-source-faithful-future-equivalence`

---

`FINITE_HORIZON_LANGUAGE_NO_GO` PROVED

A fixed local alphabet and finite horizon do not bound exact automaton size: copy languages require exponentially many future classes, and one prescribed exit time requires linearly many unless represented by a counter.

*Formal hypotheses:* *none*

*Proof dependencies:* `FUTURE_EQUIVALENCE_AUTOMATON`

*Evidence:* `virtual_language_memory.md#3-finite-horizon-and-alphabet-do-not-bound-the-index`

---

`BOUNDED_MEMORY_AUTOMATON` PROVED

If decorated transition legality, output, mode update, and endpoint acceptance depend on a finite mode and at most R previous atoms, an exact automaton has at most |M|*sum_{j=0}^R |A|^j states.

*Formal hypotheses: none*

*Proof dependencies:* `FUTURE_EQUIVALENCE_AUTOMATON`

*Evidence:* `virtual_language_memory.md#4-a-sufficient-bounded-memory-theorem`

---

`COUNTER_SAFE_PUMPING` PROVED

For a loop increment Delta and continuation increment eta, deleting the loop preserves an endpoint predicate Phi exactly on the reachable counter fiber where Phi(nu+eta)=Phi(nu+Delta+eta); zero increment or reachable-fiber invariance is sufficient but not pathwise necessary.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_language_memory.md#5-counter-aware-finite-control`

---

`MEMORY_PROJECTION_PUMPING_NO_GO` PROVED

Projecting a counter or stack follower to finite control does not license accepted-word pumping: repeated projected control/support pairs can all carry nonidentity memory action, as in the one-state exact-exit language Pref{a^n b}.

*Formal hypotheses: none*

*Proof dependencies:* `COUNTER_SAFE_PUMPING`, `FINITE_HORIZON_LANGUAGE_NO_GO`

*Evidence:* `virtual_language_memory.md#5-counter-aware-finite-control`

---

`ADDITIVE_COUNTER_MEMORY_NO_GO` PROVED

Fixed finite control with k counters and uniformly bounded additive increments reaches only polynomially many configurations after m symbols, so it cannot hide arbitrary exponential future memory; opaque integer coding is not a geometric counter.

*Formal hypotheses: none*

*Proof dependencies:* `FINITE_HORIZON_LANGUAGE_NO_GO`

*Evidence:* `virtual_language_memory.md#5-counter-aware-finite-control`

---

`REVERSE_WORD_STACK_MODEL` PROVED

Reverse-copy prefix languages have exponential finite-automaton future index at length m, while their union is recognized by finite control with one finite-alphabet LIFO stack; this is only an abstract memory model until reverse-order antirenormalization is proved.

*Formal hypotheses: none*

*Proof dependencies:* `FUTURE_EQUIVALENCE_AUTOMATON`

*Evidence:* `virtual_language_memory.md#3-finite-horizon-and-alphabet-do-not-bound-the-index`

---

`DISCRETE_SIDE_DATA_COMPACTNESS_NO_GO` PROVED

Unbounded integer counters or unbounded-length finite stack words with their discrete topology are not precompact, and adjoining a compact geometric coordinate cannot make the product precompact.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_language_memory.md#6-corrected-virtual-language-theorem`

---

`COUNTABLE_FOLLOWER_STONE_COMPACTIFICATION` PROVED

Every fixed countable exact symbolic configuration set with countably many partial deterministic transitions and a countable Boolean catalog of proof-used predicates embeds densely and injectively in a compact metrizable zero-dimensional Stone space on which every declared predicate is clopen and every transition extends continuously over its clopen domain. Coupling to geometry additionally requires an independently compact geometric state and compatible continuous geometric transitions.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_language_memory.md#theorem-62-countable-follower-stone-compactification`

---

`FUTURE_OBSERVABLE_COMPACTIFICATION` PROVED

Every partial deterministic transition system with countably many labels and a countable compact-metrizable observable catalog has a canonical compact metrizable future-observable quotient and completion. Exact word domains are clopen, transitions and observables extend continuously by coordinate shifts, exact finite transition identities persist, and every other compact realization of the same catalog maps continuously onto this completion.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_language_memory.md#theorem-64-future-observable-compactification`

---

`FUTURE_FAITHFUL_END_TOPOLOGY` PROVED

For every already realized exact component or side-state transition system with countably many labels and a frozen countable compact-target proof-use catalog, equality of all finite-future observations gives a future-faithful quotient with compact metrizable end completion on which every declared guard, branch, transport, output, and authorized enlargement extends continuously.

*Formal hypotheses: none*

*Proof dependencies:* `FUTURE_OBSERVABLE_COMPACTIFICATION`

*Evidence:* `virtual_language_memory.md#corollary-66-future-faithful-trace-topology`

---

`SOURCE_FUTURE_QUOTIENT` PROVED

A finite resolved source-locked presentation has a canonical minimal source-faithful future-equivalence quotient preserving decorated trace language, terminal colors, and edge-labeled core/WAD cocycles.

*Formal hypotheses: none*

*Proof dependencies:* `FUTURE_EQUIVALENCE_AUTOMATON`

*Evidence:* `virtual_support_automaton_construction.md#3-the-canonical-future-equivalence-automaton`

---

`MARKOV_SIGNATURE_AUTOMATON` PROVED

After exact memory compilation, a universally bounded Markov-complete marked-cell signature produces a uniformly finite source-faithful core-transfer automaton with exact decorated compiled language and cocycle; this state bound does not imply uniform core or WAD dimension.

*Formal hypotheses: none*

*Proof dependencies:* `SOURCE_FUTURE_QUOTIENT`

*Evidence:* `virtual_support_automaton_construction.md#5-the-abstract-cell-tree-construction-theorem`

---

`WAD_TRANSPORT_QUOTIENT_CRITERION` PROVED

A proposed compressed WAD transport is compositional exactly when every enabled raw branch satisfies P_out*R_gamma=Rbar_gamma*P_in, equivalently ker(P_in) is contained in ker(P_out*R_gamma); terminal output observability alone is insufficient.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_support_automaton_construction.md#6-wad-to-core-incidence-and-future-observability`

---

`WAD_FUTURE_OBSERVABILITY` PROVED

A linear WAD compression is exact for a declared complete family of future observables exactly when its kernel lies in every observable kernel; for partition sums, every observable row must be constant on each merged class.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_support_automaton_construction.md#6-wad-to-core-incidence-and-future-observability`

---

`PURE_CORRIDOR_MACRO` PROVED

A checkpoint-pure deterministic run may be replaced by a parameterized macro edge carrying its exact length and transfer power, preserving checkpoint paths and products but not the layerwise prefix language.

*Formal hypotheses: none*

*Proof dependencies: none*

*Evidence:* `virtual_support_automaton_construction.md#52-pure-checkpoint-macros`

---

`COUNTER_STACK_SAFE_LOOP` PROVED

A specified loop preserves every augmented counter/stack configuration when its total counter increment is zero and its induced partial stack map is identity on every admissible entering stack; this neither proves that a pigeonhole-selected safe loop exists nor deletes its emitted transfer, critical/WAD, Value, error, or shear record.

*Formal hypotheses:* *none*

*Proof dependencies:* `COUNTER_SAFE_PUMPING`

*Evidence:* `virtual_support_automaton_construction.md#53-four-memory-mechanisms-that-must-not-be-conflated`

---

`H3_SYMBOLIC_SIDE_COMPACTNESS` PROVED

For every realized exact side or joint marked-state transition system with countably many transition labels and a countable proof-use catalog in specified compact metrizable target encodings, the future-observable completion gives a compact metrizable end topology on which all declared counter/stack predicates, virtual-hull encodings, endpoint guards, and transition maps extend continuously; countably many point predicates retain distinct exact side configurations. Analytic admissibility of limiting hulls remains in H3_COERCIVITY.

*Formal hypotheses:* *none*

*Proof dependencies:* `FUTURE_OBSERVABLE_COMPACTIFICATION`

*Evidence:* `virtual_language_memory.md#corollary-65-symbolic-side-compactness`

---

`CONDITIONAL_REDUCTION` PROVED

YOCCOZ plus H1, MBTT, coercivity, and rigidity imply MLC.

*Formal hypotheses:* `H1_MARKED_CONSTRUCTION`, `MBTT`, `H3_COERCIVITY`, `H4_RIGIDITY`

*Proof dependencies:* `YOCCOZ`, `BACKWARD_RECORD`

*Evidence:* `manuscript.md#7-proof-of-the-conditional-reduction`

---

`UPTCR_IMPLIES_MLC` PROVED

The Universal Protected-Tower Compactness and Rigidity Proposition, together with Yoccoz's theorem for the complementary parameters, implies that the Mandelbrot set is locally connected.

*Formal hypotheses:* `UNIVERSAL_PROTECTED_TOWER_CLOSURE`

*Proof dependencies:* `YOCCOZ`, `BACKWARD_RECORD`

*Evidence:* `uptcr_cross_validation.md#4-revalidated-proof-that-uptcr-implies-mlc`

---

`UPTCR_CROSS_VALIDATION` PROVED

The UPTCR specification requires one degree-two stage only on the selected quadratic return spine, an explicit degree-2^m auxiliary Wave fan, zero fatal collision on every realized protected core, and inclusive certified block routes; with these corrections the conditional implication UPTCR=>MLC is valid, while the checked source package still does not imply UPTCR-1.

*Formal hypotheses:* `UNIVERSAL_PROTECTED_TOWER_CLOSURE`

*Proof dependencies:* `BACKWARD_RECORD`, `DEGREE_TWO_FAST_FORWARD_NO_GO`, `CONNECTED_CARRIER_NO_MINIMUM`, `QUADRATIC_PROTECTED_ACCESS_COLLISION`

*Evidence:* `uptcr_cross_validation.md#1-verdict`

---

## Open obligations (115)

`KKL_PARABOLIC_WAVE_AMPLIFICATION` OPEN

Open source-stated KKL theorem: in the full parabolically B-bounded, prime primitively renormalizable psi-ql category, sufficiently wide controllable waves and sufficiently high type are claimed to imply any requested amplification factor, with thresholds depending on B and the requested factor but not on the satellite period. The project does not certify the v1 proof until the bounded-visits interval-growth step and the stationary proper-path/measurable-lamination step are repaired, with the proved degree-two correction propagated downstream.

*Formal hypotheses:* *none*

*Proof dependencies:* `KKL_BOUNDED_VISITS_INTERVAL_GROWTH_REPAIR`, `KKL_PARABOLIC_WAVE_STATIONARY_PROOF_REPAIR`

*Evidence:* `virtual_long_iterate_wave.md#3-exact-source-statement-and-certification-status`

---

**MAPPING_TELESCOPE** OPEN

Every actual finite virtual stage cycle admits compatible selected relative-cover lifts that are embedded on relatively compact Jordan cores, preserve the full component and deck-phase route, and retain the propagated critical value after the unique quadratic stage, so that the planar critical-spine compiler produces a closed degree-two selected return.

*Formal hypotheses:* *none*

*Proof dependencies:* PLANAR_CRITICAL_SPINE_TELESCOPE_COMPILER, RELATIVE_COVER_LIFT, STAGE_FORWARD_CLOSURE

*Evidence:* critical_chronology_and_retimed_telescope.md#153-relative-cover-and-marked-telescope

---

**FINITE_HORIZON_ASSEMBLY** OPEN

Every actual finite virtual horizon supplies the open mapping telescope together with exact depth-q core identities, iterated covering embeddings, protected carrier-complement prime-end action, compatible relative covers, proper or end-proper proof-used intervals, and exact enlargement squares, so that the planar telescope identity promotes to a closed depth-q psi-ql correspondence.

*Formal hypotheses:* *none*

*Proof dependencies:* MAPPING_TELESCOPE, FINITE_PLANAR_TELESCOPE_IDENTITY, PRIME_END_MARKING_ACTION_CRITERION, MARKED_PAIR_ENLARGEMENT_NATURALITY

*Evidence:* critical_chronology_and_retimed_telescope.md#153-relative-cover-and-marked-telescope

---

**VIRTUAL_ROOT_CARRIER_SEPARATION** OPEN

Every virtual-Molecule block through the complete Wave horizon has one full root carrier containing all required forward shadows whose iterated pullback component partition contains each whole required package, realizes the prescribed parent routing and selected-sheet separation after all future data are included, matches the declared critical word, and stays in the admissible localization with an absorb-or-separate remainder.

*Formal hypotheses:* *none*

*Proof dependencies:* H1_CELL_TREE_EXTRACTION, ROOT_PULLBACK_FOREST_EQUIVALENCE, CRITICAL_VALUE_CORRIDOR_CRITERION, FINITE_OBSTACLE_FORWARD_CLOSURE

*Evidence:* root_pullback_virtual_carrier.md#7-refined-virtual-root-carrier-target

---

**VIRTUAL_WAVE_HORIZON_CERTIFICATE** OPEN

Every actual virtual-Molecule block admits an explicitly stopped finite Wave horizon, with authorized stage localizations and branches, a complete finite proof-use ledger, a finite resolved component/dependency universe or proved future-faithful summary, actual component metadata, and a final-root-enlargement certificate excluding every later dependency path to an omitted earlier record.

*Formal hypotheses:* *none*

*Proof dependencies:* HORIZON_FIRST_INVENTORY_AUDIT, RESOLVED_FINITE_WAVE_HORIZON_COMPILER, VIRTUAL_WAVE_MARKED_CORRESPONDENCE

*Evidence:* virtual_wave_horizon_certificate.md#9-virtual-instantiation-audit

---

**TERMINAL_NONLC_HUBBARD_CARRIER_PROPER** OPEN

For KKL's non-locally-connected rational-lamination lift, identify the physical lifted carrier and the nondividing fixed-point fiber precisely enough to prove that the physical carrier omits that fiber and is a proper subcontinuum of K. The locked one-line instruction to lift the combinatorial objects does not by itself establish the needed inverse-image equality or fiber exclusion.

*Formal hypotheses:* *none*

*Proof dependencies:* KKL_NATIVE_TERMINAL_HUBBARD_TREE, TERMINAL_LC_NATIVE_HUBBARD_CARRIER_PROPER

*Evidence:* terminal_dense_obstacle_and_submulticurve_replacement.md#proposition-21

---

**KKL_WAVE_PUSHFORWARD_UNIFORM_PACKET_CONSTANT** OPEN

Construct a valid coefficient-one Wave pushforward theorem for the actual proof-used stationary source class. Either prove that every such source is one coherent single-crest or single-WAD-coordinate packet to which a complete DL-style logarithmic pruning proof applies, or replace coefficient one by a multiplicity-aware estimate with all repeated source conductance coordinates charged. The theorem must define boundary degree and document constants independent of period, horizon, access count, selector history, and leaf.

*Formal hypotheses:* *none*

*Proof dependencies:* DL23_ANNULAR_WAVE_BUFFER_EXTRACTION, KKL_BROAD_WAVE_PUSHFORWARD_COUNTEREXAMPLE

*Evidence:* qa_extremal_wave_2026-07-26.md#12-minimal-next-analytic-gate

---

**KKL_STATIONARY_POINTWISE_TWO_PATH_BREAK** OPEN

Within KKL's parabolically b-bounded stationary high-type context with rotation 1/q and q>b, for n=b and n=b+1 and every actual occurrence alpha in A_(n+1) avoiding the corrected Q_n continuum, construct the two line-1603 portions as retained proper depth-(n+1) components tied to the same leaf and stopping horizon; prove complete fine-marking and coarse-subordination survival and prove that their f^(n+1)-images are distinct unparametrized terminal marked paths. Record the stage, sheet, parent component, entry and exit accesses, fine and coarse ends, and terminal images. This is a pointwise break certificate and asserts no doubled-width paired-Wave response.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_TWO_SIDED_WAVE_NONCOLLISION

*Evidence:* qa_native_kkl_stationary_2026-07-26.md#gate-n2-pointwise-two-path-break

---

**KKL_STATIONARY_FINAL_FIXED_COMPONENT_EXIT** OPEN

For m=b+2, choose one actual canonical graph component Q of Kcal^m and construct a measurable retained lamination B such that every alpha in A_m has a selected proper component beta_alpha in iota_m^*(alpha) joining Q to Kcal^m minus Q, with proper prime-end landing, fine/coarse filter survival, a repeated-incidence convention, and no duplicated transverse mass. Prove that every iota_m^*(alpha) contains its selected beta_alpha and hence W(A_m)<=W(B).

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_FINAL_COMPONENT_REALIZATION, KKL_STATIONARY_MARKED_ACCESS_ITINERARY, KKL_STATIONARY_POINTWISE_TWO_PATH_BREAK

*Evidence:* qa_native_kkl_stationary_2026-07-26.md#gate-n3-final-retained-fixed-component-exit

---

**KKL_STATIONARY_TYPED_BOUNDED_DEGREE_PUSHFORWARD** OPEN

For the actual lamination B and whole marked ending multi-interval supplied by the final fixed-component exit theorem, apply the covering pushforward lemma to f^m, not F^m. Prove the degree on that whole end, using d=2 only when the ideal-boundary degree is certified and otherwise the safe bound d<=2^(b+2). Deduce W(B)<=d*W(C)+2d, combine it with the correctly typed native comparison W(C)<=2W(RF), and conclude W(A_m)=O_b(1+W(RF)); replace this by O_b(W(RF)) only under an explicit lower-width hypothesis.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_FINAL_FIXED_COMPONENT_EXIT, KKL_STATIONARY_DEGREE_TWO_CORRECTION, KKL_LAMINATION_PUSHFORWARD_DEGREE_BOUND

*Evidence:* qa_native_kkl_stationary_2026-07-26.md#gate-n4-correctly-typed-bounded-degree-pushforward

---

**KKL_STATIONARY_MARKED_FIXED_END_PAIR_CERTIFICATE** OPEN

For the optional Freudenthal fixed-end route and every proof-used stationary horizon h<=b+2, first construct an access-splitting localization and actual eligible Freudenthal source ends; a marked BL_0 or L_0^h access may not be relabeled as such an end. Then prove the six route-specific certificates C_h^-,X_h^-,C_h^+,X_h^+,P_h,M_h: proper degree-one oriented starting ends, same-occurrence exits, full common-leaf/window/sheet/access coupling, and native filter/Wave-image survival. Under TWO_h and INC_h the unsplit physical inner end is nonproper, but those hypotheses are not promoted unconditionally. This fixed-end route is sufficient neither for family-level full-width pairing nor for transverse accounting, and it is not logically necessary if actual paired source Waves are constructed by an access-local or direct method.

*Formal hypotheses: none*

*Proof dependencies:* KKL_FIXED_END_WAD_SPLIT_FINITE_SELECTOR, KKL_STATIONARY_SURVIVOR_PARTITION_AND_POINTWISE_SWITCH, KKL_FIXED_END_ACCESS_TYPE_OBSTRUCTION

*Evidence:* kkl_stationary_fixed_end_topological_obstruction.md#9-exact-next-task

---

**KKL_STATIONARY_TYPED_PAIRED_WAVE_REALIZATION** OPEN

On the entire nonshort ambient-singleton BL_0-ending stationary survivor, make a Borel first-horizon partition with no duplicated transverse mass and construct two full-width leafwise-paired genuine source Waves with disjoint untrimmed target occurrence images at the same actual L_0^h occurrence. The typed start must be DIRECT_WAVE, an actually eligible FREUDENTHAL end on an access-splitting localization, or an ACCESS_LOCAL RLS-A+ product seed; an ideal access is never relabeled as a topological end. Prove exact complete image identities, seed-side properness when needed, source-lamination validity, fine/coarse filter survival, whole-ending-multi-interval degrees, omitted-family zero-width or explicit V(F)/R(F) charge, and

first-horizon/no-duplication accounting. The first failed typed-start, exit, pairing, marking, degree, or accounting predicate is the obstruction certificate.

*Ledger role:* optional alternative

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_SURVIVOR_PARTITION_AND_POINTWISE_SWITCH, PQL_SEGMENTED_OCCURRENCE_RECONSTRUCTION, KKL_ACCESS_SEEDED_MAXIMAL_LIFT_MEASURABILITY, KKL_FIXED_END_ACCESS_TYPE_OBSTRUCTION

*Evidence:* kkl_stationary_bl0_survivor_analysis.md#12-historical-pre-qa-next-theorem-superseded

---

**KKL_STATIONARY_BL0_UNIFORM_SURVIVOR_RESPONSE** OPEN

Cover the entire nonshort ambient-singleton BL_0-ending residual, without duplicate extremal width, by typed paired-Wave packets or explicit valid V(F)/R(F)/zero-width charges. On every response packet prove the uniform Wave constant, common-output containment or a certified maximizing output, output controllability depending only on b, the necessary ambient-width fraction, and the absolute threshold needed to turn the packet-local factor-two estimate into KKL terminality. No use of the line-1603-dependent critical-limb proposition is allowed.

*Ledger role:* optional alternative

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_CONDITIONAL_PAIRED_WAVE_RESPONSE_ALGEBRA, KKL_STATIONARY_TYPED_PAIRED_WAVE_REALIZATION, KKL_WAVE_PUSHFORWARD_UNIFORM_PACKET_CONSTANT

*Evidence:* kkl_stationary_bl0_survivor_analysis.md#12-historical-pre-qa-next-theorem-superseded

---

**KKL_STATIONARY_SURVIVOR_EXCLUSION_OR_CHARGE** OPEN

On the actual native stationary critical-end Wave, construct a width-faithful survivor partition and a Borel retained-predecessor selector with multiplicity bounded in terms of b or charged summably. Prove that every strict-path or marked-state survivor switch is excluded by marked witness completeness or belongs measurably to a proper vertical packet, an exact small-Julia-incidence packet with its R(F) charge retained, or a full-width paired-Wave response packet. The paired packet may use a direct, eligible Freudenthal, or access-local typed start, but must satisfy exact untrimmed image, no-duplication, marking, degree, uniform Wave-error, output-controllability, and ambient-mass hypotheses. Critical-end, stage-output-short, marking, nonlanding, and omitted residuals must have zero width or enter a charged/response class without citing the line-1603-dependent critical-limb proposition. All packet counts and constants must depend only on b.

*Ledger role:* optional alternative

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_ABSTRACT_FINITE_ROUTE_EXIT_SCHEMA, KKL_STATIONARY_CONDITIONAL_VERTICAL_PACKET_CHARGE, KKL_STATIONARY_BL0_UNIFORM_SURVIVOR_RESPONSE

*Evidence:* kkl_stationary_bl0_survivor_analysis.md#12-historical-pre-qa-next-theorem-superseded

---

**KKL_STATIONARY_UNIFORM_COLLAR_EXIT_CHARGE** OPEN

For every proof-used horizon h<=b+2, prove either that the optimized prefix-compatible collar conductance is at most C_b(W(U minus T)+1), together with measurable source first-exit selection, or directly that the sum of marked route-exit conductances is at most that quantity. C_b must be independent of satellite period, rotation denominator, route depth within the fixed horizon, and the individual block. Qualitative collar existence and Kahn's response in the larger full width W(F) are insufficient.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_COLLAR_EXIT_OVERFLOW, KAHN_QUALITATIVE_COLLAR_VERTICAL_CHARGE_NO_GO

*Evidence:* kkl_stationary_uniform_collar_charge_attack.md#10-final-conclusion-and-exact-next-hypothesis

---

**KKL_STATIONARY_QUANTITATIVE_DEFECT_BOUND** OPEN

At the final stationary subwave Y_(N-1), construct a partition Y_(N-1)=G disjoint-union D modulo zero width in KKL's native transverse width measure, with the restrictions of that measure agreeing with the proof-used lamination widths. Every leaf of G must have the actual segmented final incidence and measurable route realization needed for the corrected degree-two estimate, while all untyped encounters, continuation exits, nonlanding, marking/filter loss, window incompatibility, terminal-family failures, deck collisions, and measurable-incidence failures in D satisfy W(D)<=a_b W(U minus T)+c_b W(RF)+c_b. Constants depend

only on b, and use in the source-wide induction requires the native HTND absolute scale or an explicit W(A)>=C(R(F)+1) hypothesis.

*Formal hypotheses: none*

*Proof dependencies:* PQL_SEGMENTED_OCCURRENCE_RECONSTRUCTION, KKL_STATIONARY_COLLAR_EXIT_OVERFLOW, KKL_STATIONARY_DEGREE_TWO_CORRECTION

*Evidence:* kkl_stationary_quantitative_defect_repair.md#3-quantitative-stationary-defect-bound

---

**KKL_STATIONARY_SELECTED_SEPARATOR_REALIZATION** OPEN

For the canonical Q_0-selected chain through H=b+2, prove in the full physical carrier that Q_0 separates alpha^0 from L_0^1 and, for every 1<=n<H, Q_n separates alpha^n from L_0^(n+1). Construct the proof-used collars and marked accesses and prove that competing carrier routes outside the selected inverse sheet do not bypass Q_n. Applying phi_0 inside one sheet is insufficient because a separating path in the full carrier may leave that sheet.

*Formal hypotheses: none*

*Proof dependencies:* KKL_NATIVE_TERMINAL_HUBBARD_TREE, KKL_STATIONARY_SELECTED_SMALL_JULIA_PREDECESSOR_CHAIN, KKL_STATIONARY_SELECTED_ALPHA_PREDECESSOR_CHAIN, KKL_STATIONARY_TERMINAL_CRITICAL_LIMB_INVERSE_SHEETS

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#lemma-22-canonical-small-julia-predecessor-selection

---

**KKL_STATIONARY_PROOF_USED_PREDECESSOR_SELECTION** OPEN

Prove conditional proof-used compatibility through H=b+2: every independently supplied depth-n proper restriction whose deep endpoint follows the declared central small-Julia parent itinerary lands in the canonical Q_0-selected component L_0^n and carries its actual-component address w_n. This is a universal compatibility assertion only; it does not construct the restriction, its endpoint occurrence, or its host. Those existential clauses remain in the incoming-window anchor, so no occurrence is assumed circularly.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_SELECTED_PREDECESSOR_CHAIN, KKL_STATIONARY_SELECTED_SMALL_JULIA_PREDECESSOR_CHAIN

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#7-minimal-gate-sufficient-certificate-and-endpoint-coherence

---

**KKL_STATIONARY_NONLC_SELECTED_GEOMETRIC_CONTINUUM_LIFT** OPEN

For KKL's exact non-locally-connected natural projection, prove the geometric two-level lift through the required horizon: pi^(-1)(L_(0,com)^n)=L_0^n for the selected predecessors, and pi(K_i) is contained in the corresponding geometric model small-Julia continuum bar K_i. Since model incidence gives bar K_i subset L_(0,com)^n, these clauses force K_i subset L_0^n. Record separately failure of the selected full-preimage equality and failure of physical/model continuum correspondence; an abstract finite-tree vertex is not a point of the geometric model.

*Formal hypotheses: none*

*Proof dependencies:* KKL_NATIVE_TERMINAL_HUBBARD_TREE, KKL_STATIONARY_MODEL_TERMINAL_SHEET_INCIDENCE, KKL_STATIONARY_SELECTED_PREDECESSOR_CHAIN

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#corollary-44-model-incidence-and-exact-non-lc-residue

---

**KKL_STATIONARY_MONOTONE_FULL_LIFT_IDENTIFICATION** OPEN

For the exact projection and lift meant by KKL lines 542-543, prove through H=b+2 that the model is a dendrite, pi is a continuous monotone surjective semiconjugacy, pi(alpha^n)=alpha_(n,com) for every selected predecessor, the fixed alpha fiber is a singleton, T=pi^(-1)(T_com), pi maps the physical critical point to the model critical point, pi(Q_n) lies in the selected model inverse sheet Omega_(n,com), and each marked physical small-Julia continuum K_i maps into its corresponding geometric model continuum bar K_i. Monotonicity, semiconjugacy, and the singleton fixed fiber then force every selected predecessor fiber to be a singleton. The geometric continua bar K_i are not the abstract vertices obtained by a later finite-tree quotient; collapsing K_i to such a point would conflict with finite-orbit injectivity of the canonical monotone model. The locked KKL and Dudko-Lyubich passages state a natural semiconjugating projection and an unspecified lift, but do not state all these identifications.

*Ledger role:* optional alternative

*Formal hypotheses: none*

*Proof dependencies:* KKL_NATIVE_TERMINAL_HUBBARD_TREE, KKL_STATIONARY_SELECTED_ALPHA_PREDECESSOR_CHAIN, KKL_STATIONARY_SELECTED_SMALL_JULIA_PREDECESSOR_CHAIN

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#proposition-45-monotone-full-lift-criterion

---

KKL_STATIONARY_SELECTED_PREDECESSOR_MARKED_COMPONENT_INCIDENCE OPEN

Global physical line-1598 marked-core incidence MCI_b says that if the canonical selected L_0^b intersects the original physical Hubbard carrier T, then it contacts an actual component K of the primitive small-Julia cycle. The theorem is proved for locally connected physical K(F), and whole-continuum incidence is proved in the locally connected combinatorial model. The exact remaining model-lift route is the non-locally-connected selected full-preimage and geometric small-Julia-continuum correspondence theorem; an abstract vertex of the later finite-tree quotient is not a point of the geometric model.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_SELECTED_PREDECESSOR_CHAIN, KKL_NATIVE_TERMINAL_HUBBARD_TREE, KKL_STATIONARY_LC_TERMINAL_SHEET_MCI, KKL_STATIONARY_MODEL_TERMINAL_SHEET_INCIDENCE, KKL_STATIONARY_NONLC_SELECTED_GEOMETRIC_CONTINUUM_LIFT

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#corollary-44-model-incidence-and-exact-non-lc-residue

---

KKL_STATIONARY_SELECTED_PREDECESSOR_TERMINAL_VERTEX_SATURATION OPEN

For the canonical selected predecessor chain, prove the stronger optional certificate TMC_b: whenever L_0^b intersects T, choose one component C_b of |G| minus the recorded point x_b with C_b contained in pi_T(L_0^b intersect T); pruning supplies v_K in C_b intersect M, and the geometric model small-Julia continuum represented by v_K must lift to actual contact K intersect L_0^b. The locally connected physical theorem realizes this certificate. In a non-locally-connected lift the remaining issue is the exact selected full-preimage and geometric continuum correspondence, not collapse to the abstract vertex. TMC_b implies MCI_b but is not necessary. Record the first failed depth, inverse component, complete attachment set, actual-component address, missing terminal side, and model continuum not realized by K-contact.

*Ledger role:* optional alternative

*Formal hypotheses: none*

*Proof dependencies:* FINITE_MARKED_TREE_TERMINAL_PRUNING, KKL_NATIVE_TERMINAL_HUBBARD_TREE, KKL_STATIONARY_SELECTED_PREDECESSOR_CHAIN, KKL_STATIONARY_LC_TERMINAL_SHEET_MCI, KKL_STATIONARY_NONLC_SELECTED_GEOMETRIC_CONTINUUM_LIFT

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#7-minimal-gate-sufficient-certificate-and-endpoint-coherence

---

KKL_STATIONARY_PREDECESSOR_ENDPOINT_GATE_COHERENCE OPEN

As one sufficient access architecture, for every 0<=n<=H=b+2 construct the selected pulled carrier T_hat_n and full attachment set Att_n=closure(L_0^n) intersect closure(T_hat_n minus L_0^n), all in T_hat_n. For 1<=n<=H construct eta_n as the actual branch-induced carrier homeomorphism extending to a declared collar and its prime-end boundary. Prove alpha^n lies in closure(L_0^n), Att_n={x_n}, x_n=alpha^n, and eta_n transports the declared one-sided access at x_n to that at x_(n-1). Endpoint cycle avoidance constrains a coherent gate but does not establish this identity. A direct marked-access theorem could replace this sufficient architecture.

*Ledger role:* optional alternative

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_SELECTED_PREDECESSOR_CHAIN, KKL_STATIONARY_SELECTED_ALPHA_PREDECESSOR_CHAIN, KKL_STATIONARY_SELECTED_ENDPOINT_CYCLE_AVOIDANCE, KKL_STATIONARY_PROOF_USED_PREDECESSOR_SELECTION

*Evidence:* kkl_stationary_n2_selected_predecessor_endpoint.md#7-minimal-gate-sufficient-certificate-and-endpoint-coherence

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KKL_STATIONARY_INCOMING_WINDOW_ANCHOR OPEN

Within a stationary block with parabolically b-bounded primitive renormalization, rotation 1/q, and explicit high-type inequality q>b, bind the complete-horizon selected chain to the proof-used Wave components and, for every n in {b,b+1}, h=n+1, and actual alpha in A_h to which KKL line 1603 is applied, construct a specified incoming-side germ at one marked L_0^h contact and its connected host C. Prove that C is an embedded component-maximal proper restriction, the deep access is the upper endpoint c<1 of its canonical window (a,c),

the other end is alpha's actual lifted BL_0 endpoint access, and a=0. The marked access may be obtained through EGC_H or by a separate direct theorem; EGC_H is not a logically necessary factorization. Multiple components at the same coarse locus must be exhausted or canonically selected. On failure distinguish: no typed incoming germ/component; wrong one-sided incidence; retracing or nonmaximal representative; nonlanding or noninterior deep end; a>0; or a=0 with the wrong BL_0 lift.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_MAXIMAL_ONE_SIDED_WINDOW_REDUCTION, KKL_STATIONARY_SELECTED_PREDECESSOR_CHAIN, KKL_STATIONARY_PROOF_USED_PREDECESSOR_SELECTION, KKL_STATIONARY_SELECTED_PREDECESSOR_MARKED_COMPONENT_INCIDENCE, KKL_STATIONARY_LINE1598_TREE_DISJOINTNESS_REDUCTION

*Evidence:* kkl_stationary_n2_incoming_window_anchor.md#7-exact-anchor-and-failure-records

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**KKL_STATIONARY_TWO_OCCURRENCE_BINDER** OPEN

Within the same parabolically b-bounded stationary high-type context with rotation 1/q and q>b, for every proof-used b<=n<N-1, if an actual alpha in A_(n+1) fails the source pass-through predicate at the corrected Q_n continuum, construct two segmented retained extension towers realizing exactly the two topological portions named at KKL line 1603. The incoming tower must be anchored at alpha's BL_0 end and terminate at one specified L_0^(n+1) contact; the outgoing tower must start at that same contact occurrence and terminate at alpha's non-BL_0 end. At every stage record the full proper restriction path, its actual occurrence window in the preceding full output, endpoint intervals and accesses, sheet, parent and attachment provenance, filter witness, and covering image; prove complete prefix compatibility and terminal Wave membership.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_INCOMING_WINDOW_ANCHOR, KKL_STATIONARY_ONE_RETAINED_CHAIN, PQL_SEGMENTED_OCCURRENCE_RECONSTRUCTION, KKL_STATIONARY_FINITE_EMBEDDED_COLLAR_NATURALITY

*Evidence:* kkl_stationary_segmented_occurrence_binder.md#9-what-remains-for-the-actual-two-occurrence-binder

---

**KKL_STATIONARY_RETAINED_PORTION_REALIZATION** OPEN

Within the same parabolically b-bounded stationary high-type context with rotation 1/q and q>b, for every proof-used b<=n<N-1 and every actual occurrence alpha in KKL's A_(n+1) that avoids the corrected Q_n continuum, prove that each line-1603 route has a complete segmented extension tower: every full one-step restriction is proper, belongs to the applicable fine marking, survives the coarse single-interval filter, and is connected to the next full output by its actual nested occurrence window; the final full output must be a Wave in the declared marking. Record the first failed continuation, remote ideal end, window, filter, or terminal membership.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_TWO_OCCURRENCE_BINDER

*Evidence:* kkl_stationary_two_occurrence_source_attack.md#121-the-segmented-retained-extension-tower

---

**KKL_STATIONARY_MARKED_ACCESS_ITINERARY** OPEN

For every proof-used selected Q_s pull in KKL's stationary horizon, first realize the full-carrier Q-separator predicate, then prove that the root and deep parent routes determine nonempty disjoint marked prime-end sets, retain their complete side, cyclic-order, sheet, parent, and endpoint-interval data through every selected conformal pull and immersion filter, and yield the ordered access records required at the central quadratic collar. Continuum separation in one inverse sheet or a coarse component label alone is insufficient.

*Formal hypotheses: none*

*Proof dependencies:* PQL_ITERATE_FILLED_GRAPH_NATURALITY, KKL_STATIONARY_SELECTED_SMALL_JULIA_PREDECESSOR_CHAIN, KKL_STATIONARY_SELECTED_SEPARATOR_REALIZATION

*Evidence:* kkl_stationary_tsnc_source_reduction.md#3-notation-and-the-two-separator-sides

---

**KKL_STATIONARY_SELECTED_CORRIDOR_CONFINEMENT** OPEN

Within the same parabolically b-bounded stationary high-type context with rotation 1/q and q>b, for each retained line-1603 portion construct one connected crosscut in the selected corrected Q_n collar and prove that its complete target-collar occurrence under the full fiber-product projection equals the image under the selected degree-two map G_n, with no additional entry, exit, re-entry, ambient sheet change, or late enlargement. Local germ containment is insufficient.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_RETAINED_PORTION_REALIZATION, KKL_STATIONARY_MARKED_ACCESS_ITINERARY, KKL_STATIONARY_DEGREE_TWO_CORRECTION

*Evidence:* kkl_stationary_tsnc_source_reduction.md#7-the-two-actual-portions-and-the-earlier-realization-gap

---

KKL_STATIONARY_SELECTED_COLLAR_DECK_NONCOLLISION OPEN

For the two actual retained and corridor-confined portions at every proof-used stationary index, prove the exact geometric predicate that their selected collar crosscuts are not related by the nontrivial deck involution of the unique central degree-two collar. It is sufficient, but not necessary, to prove that their complete unordered prime-end access pairs are not deck-related; with independently proved endpoint roles and orientations one may use the sharper ordered non-transposition certificate. Equality of the deck-related crosscut paths is the exact obstruction and requires an occurrence/deck-phase replacement state.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_MARKED_ACCESS_ITINERARY, KKL_STATIONARY_SELECTED_CORRIDOR_CONFINEMENT, KKL_STATIONARY_DEGREE_TWO_CORRECTION

*Evidence:* kkl_stationary_tsnc_source_reduction.md#8-the-exact-deck-obstruction-and-an-endpoint-certificate

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KKL_STATIONARY_TWO_SIDED_WAVE_NONCOLLISION OPEN

Under KKL's complete stationary-Wave and ideal-marking hypotheses in the parabolically b-bounded high-type context with rotation 1/q and q>b, for every proof-used b<=n<N-1, if a leaf occurrence in A_(n+1) avoids the corrected Q_n continuum, combine retained portion realization, complete marked-access itineraries, selected-corridor confinement, and the exact selected-collar path predicate [lambda^+] != [tau_n(lambda^-)] to prove that the two terminal Wave occurrences have distinct proper images under the full f^(n+1) projection. Inequality of unordered endpoint pairs is only a sufficient certificate; equality of the untagged geometric images is the exact failure certificate and forces an occurrence/deck-phase replacement state.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_RETAINED_PORTION_REALIZATION, KKL_STATIONARY_MARKED_ACCESS_ITINERARY, KKL_STATIONARY_SELECTED_CORRIDOR_CONFINEMENT, KKL_STATIONARY_SELECTED_COLLAR_DECK_NONCOLLISION, KKL_STATIONARY_DEGREE_TWO_CORRECTION

*Evidence:* kkl_stationary_tsnc_source_reduction.md#12-exact-next-theorem

---

KKL_STATIONARY_TWO_PULLBACK_PASS_THROUGH OPEN

In KKL's actual stationary Wave branch, with parabolic bound b and N=b+3, prove that every occurrence in the defined A_(N-1)=A_(b+2) passes through both corrected source continua Q_b and Q_(b+1) in the successive-pullback sense. This requires the full-carrier separator/access theorem and the two-sided Wave noncollision theorem validating line 1603; it asserts no final component incidence, order, proper restriction, or width.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_TWO_SIDED_WAVE_NONCOLLISION, KKL_STATIONARY_OFF_BY_ONE_PASS_THROUGH_REDUCTION, KKL_STATIONARY_SELECTED_SEPARATOR_REALIZATION

*Evidence:* kkl_stationary_wave_component_incidence.md#theorem-31-conditional-off-by-one-reduction

---

KKL_STATIONARY_FINAL_PASS_THROUGH_PATH_EXTRACTION OPEN

Legacy composite formulation: for m=b+2 and the actual final component over K_0^(m-1), prove that KKL's depth-m pass-through predicate produces a retained proper path in U^m minus Kcal^m from that component to Kcal^m minus it, including connected occurrence transport, restriction-filter survival, prime-end landing, the non-locally-connected rational-lamination model, measurable lamination assembly, and W(A_m)<=W(B). This does not certify degree on the whole marked end or a pushforward-width estimate; the active native route separates those clauses.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_TWO_SIDED_WAVE_NONCOLLISION, KKL_STATIONARY_TWO_PULLBACK_PASS_THROUGH, KKL_STATIONARY_FINAL_COMPONENT_REALIZATION

*Evidence:* qa_native_kkl_stationary_2026-07-26.md#6-from-pass-through-to-a-component-to-component-lamination

**KKL_PARABOLIC_WAVE_STATIONARY_PROOF_REPAIR** OPEN

Repair KKL's native stationary proposition by using relative Wave continuation for the preliminary telescope, the pointwise retained two-path contradiction at exactly n=b and n=b+1, a measurable final fixed-component exit lamination, and a correctly typed bounded-degree pushforward. Correct A_N to A_(N-1), Kcal^n to Kcal^(N-1), F^(N-1) to f^(N-1), and the false degree-one assertion. Prove the resulting O_b(1+W(RF)) stationary estimate and invoke O_b(W(RF)) only in an explicitly stated large-width regime; then recheck the downstream mostly-short, terminal-Wave, and amplification deductions.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_RELATIVE_WAVE_CONTINUATION, KKL_STATIONARY_POINTWISE_TWO_PATH_BREAK, KKL_STATIONARY_FINAL_FIXED_COMPONENT_EXIT, KKL_STATIONARY_TYPED_BOUNDED_DEGREE_PUSHFORWARD

*Evidence:* qa_native_kkl_stationary_2026-07-26.md#13-final-qa-determination

---

**KKL_STATIONARY_FINAL_INCIDENCE_REFINEMENT** OPEN

For every retained occurrence in KKL's stationary A_(N-1), prove that the two separately established shallower pass-through predicates refine to incidences with C_b and C_(b+1) in one actual component, or one source-faithful finite ordered component chain, of the final restriction in U^(N-1) minus Kcal^(N-1). Retain every intervening component and record the first split, missing end lift, or occurrence mismatch.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_TWO_SIDED_WAVE_NONCOLLISION, KKL_STATIONARY_TWO_PULLBACK_PASS_THROUGH, KKL_STATIONARY_FINAL_COMPONENT_REALIZATION, KKL_STATIONARY_DEGREE_TWO_CORRECTION, KKL_STATIONARY_FINAL_PASS_THROUGH_PATH_EXTRACTION, DIRECT_DECORATION_WAVE_PROOF_USE_LEDGER

*Evidence:* pql_iterate_filled_graph_naturality.md#6-exact-residual-incidence-gate

---

**KKL_STATIONARY_NESTED_INCIDENCE_ORDER** OPEN

After final graph-lift realization, prove a nested-domain and unique-or-controlled-crossing theorem in U^(N-1) that orders the two final component incidences for every retained oriented restriction, including repeated incidences. Separate-level separation statements do not by themselves determine the final order.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_FINAL_INCIDENCE_REFINEMENT

*Evidence:* kkl_stationary_wave_component_incidence.md#obligation-52

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**KKL_STATIONARY_FIXED_PAIR_PROPER_RESTRICTION** OPEN

For every bound occurrence in KKL's stationary A_(N-1), construct either one connected proper restriction with endpoints on the realized fixed pair C_b,C_(b+1), or a source-faithful finite ordered restriction multicurve carrying every intervening component. Retain every sheet, deck phase, obstacle, order, and occurrence multiplicity, prove the literal-subarc overflow relation, and prove W(H_stat)>=theta W(A_(N-1))-E uniformly.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_FINAL_INCIDENCE_REFINEMENT, KKL_STATIONARY_NESTED_INCIDENCE_ORDER, DIRECT_DECORATION_WAVE_PROOF_USE_LEDGER, PARAMETRIZED_PULLBACK_KKL_EXHAUSTION_NO_GO

*Evidence:* kkl_stationary_wave_component_incidence.md#obligation-71

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**TERMINAL_STATIONARY_RESTRICTION_KL_PUZZLE_SELECTOR** OPEN

For the complete retained stationary proper-restriction family, construct one authorized common KL puzzle certificate (Y,v,w), v!=w, or a uniformly controlled tagged atlas, whose physical endpoints lie in S_Y^v and S_Y^w. Preserve localization, obstacle, tile, deck, cyclic-order, occurrence, and final-root data; component incidence labels alone are insufficient.

*Formal hypotheses: none*

*Proof dependencies:* KKL_STATIONARY_FIXED_PAIR_PROPER_RESTRICTION, TERMINAL_K1_UNIQUE_PUZZLE_VERTEX_ANCHOR, TERMINAL_RELATIVE_COVER_SUBGROUP_ALIGNMENT, TERMINAL_SEAM_NORMALIZED_PATH_LIFT

*Evidence:* kkl_stationary_wave_component_incidence.md#obligation-72

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**TERMINAL_WAVE_KL_CONNECTING_SUBFAMILY** OPEN

For every actual terminal proof-use Wave binder, retain a sublamination S_0 with W(S_0)>=theta W(S)-E and construct a complete admissible proper-restriction lamination whose leaves are physical curves connecting one fixed pair S_Y^(v_*),S_Y^(w_*), with all occurrence multiplicities, ordered subintervals, obstacle/localization filters, and endpoint-component records. Prove that I_(1->0) is a proof-used end or supply contained physical subcurves; a crest designation alone is insufficient and appended connectors are forbidden.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_WAVE_PROOF_USE_LEDGER, TERMINAL_K1_UNIQUE_PUZZLE_VERTEX_ANCHOR, TERMINAL_KL_SUBMULTICURVE_OVERFLOW_TRANSFER, KKL_STATIONARY_FIXED_PAIR_PROPER_RESTRICTION, TERMINAL_STATIONARY_RESTRICTION_KL_PUZZLE_SELECTOR

*Evidence:* terminal_dense_obstacle_and_submulticurve_replacement.md#6-exact-replacement-obligations

---

**TERMINAL_WAVE_CREST_TO_KL_SKIPPING_BINDER** OPEN

For every retained terminal Wave occurrence whose terminal side lies in the crest, construct a literal ordered proper restriction, never an appended connector, with fixed distinct endpoint continua S_Y^v,S_Y^w, one fixed separating piece Z, a lifted horizontal KL multicurve, and a named componentwise skip witness. Retain the adjacent tagged terminal half-access occurrences, tile, deck, order, multiplicity, localization, obstacle, and final-root data, and prove a uniform response-class/depth charge or record the first B1–B10 failure.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_WAVE_KL_CONNECTING_SUBFAMILY, TERMINAL_K1_UNIQUE_PUZZLE_VERTEX_ANCHOR, TERMINAL_RELATIVE_COVER_SUBGROUP_ALIGNMENT, TERMINAL_SEAM_NORMALIZED_PATH_LIFT, TERMINAL_CERTIFIED_CHANNEL_FINAL_ROOT_STABILITY, TERMINAL_WAVE_CREST_TO_KL_WIDTH_CONDITIONAL

*Evidence:* terminal_wave_crest_to_kl_skipping_bridge.md#10-exact-open-binder-certificate

---

**TERMINAL_KL_SUBMULTICURVE_BINDER_SELECTOR** OPEN

Using the physical connecting subfamily and the crest-to-skipping binder, instantiate the provenance relation for every retained binder occurrence and propagate its ordered KL horizontal submulticurves, full pregluing eta/xi incidence tags, skip witness, width charge, and final-root restriction behavior through the complete terminal stage. This is a width selector, not an equality between original KKL and KL prime ends.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_WAVE_KL_CONNECTING_SUBFAMILY, TERMINAL_WAVE_CREST_TO_KL_SKIPPING_BINDER, TERMINAL_KL_SUBMULTICURVE_OVERFLOW_TRANSFER, DIRECT_DECORATION_WAVE_PROOF_USE_LEDGER

*Evidence:* terminal_dense_obstacle_and_submulticurve_replacement.md#6-exact-replacement-obligations

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**TERMINAL_TAGGED_ENDPOINT_ACCESS_MATCH** OPEN

For every proof-used terminal restriction retained by the full physical response, compute the exact five-way endpoint records and prove the protected endpoint occurrences avoid the outer and central cases and lie on the correctly tile-tagged eta_{i_*} or xi_{i_*+1} occurrence attached to v_*. Preserve multiple ideal incidences and record the first wrong tag, central endpoint, or competing occurrence; physical vertex incidence alone is insufficient.

*Formal hypotheses: none*

*Proof dependencies:* TERMINAL_K1_UNIQUE_PUZZLE_VERTEX_ANCHOR, TERMINAL_SEAM_NORMALIZED_PATH_LIFT, TERMINAL_KL_FINITE_ENDPOINT_FIVE_WAY_CLASSIFICATION, DIRECT_DECORATION_WAVE_PROOF_USE_LEDGER

*Evidence:* terminal_protected_access_reduction.md#obligation-52

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**DIRECT_DECORATION_TILE_CENTRAL_OVERLAP_REFLECTION** OPEN

For every actual fixed Kahn–Lyubich direct-decoration inclusion tile L_i, determine the stronger central-image saturation equality L_i^(-1)(tilde e_j(P_j))=e_i(Q_i), or the corresponding equality on any retained proof-used collar whose theorem explicitly needs it. This full equality is not required for injectivity of the full glued pseudo-piece: the collision-relevant equality over tilde e_j(Q_i) follows automatically from full tile injectivity and component bijectivity. A failure over p in P_j minus Q is a saturation, footprint, or collar-exhaustion defect rather than an interior quotient collision; record its component, sheet, access, itinerary, and target-central location. This is a non-gating open question for the full interior embedding but may remain a hypothesis of stronger carrier or footprint statements.

*Formal hypotheses: none*

*Proof dependencies:* KL_DECORATION_ESCAPE_PUZZLE

*Evidence:* kl_relative_overlap_cartesianity.md#10-ledger-consequence

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**DIRECT_DECORATION_DEGREE_ONE_PSEUDO_EMBEDDING** OPEN

Classify the restricted psi-ql direct-decoration stages, if any, on which the full bold ambient annulus immersion and full glued pseudo-piece inclusion are globally univalent. The former universal route is unavailable: KL_CANONICAL_LOCALIZATION_FORCED_COLLISION proves a nontrivial subgroup-kernel collision at every ordinary primitive canonical-localization seed, and propagates it to the exact direct-decoration depths whenever the same seed carries the KL decoration data. This is now a non-gating restricted-class question; it may not be used by the universal footprint, contact-closure, VAAPC-I, UPTCR, or MLC chains.

*Formal hypotheses:* *none*

*Proof dependencies:* KL_DECORATION_ESCAPE_PUZZLE, KL_TILE_INJECTIVITY_BASE_ANNULUS_REDUCTION, TILE_INJECTIVITY_GLUE_EMBEDDING_CRITERION, KL_CANONICAL_LOCALIZATION_FORCED_COLLISION

*Evidence:* kl_canonical_localization_forced_collision.md#8-alignment-and-realization-quantifiers

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**KKL_NATIVE_WAVE_WANDERERS_QUERY_SCHEMA_BOUNDARY_MAP** OPEN

Complete the target-first source boundary map for the active KKL Wave, Value, WAD, translation-marking, and Wanderers program by binding every symbol and universal geometric restriction, assigning every atomic source occurrence and prose invocation to its exact typed demand, and filling the total theorem-hypothesis/formal-to-actual substitution table. Parameterized lifts remain separately typed from the entire geometric restriction. The generated lexical occurrence manifest and overlapping D/W/T display buckets are inputs, not a completed source package.

*Formal hypotheses:* *none*

*Proof dependencies:* FINITE_BINDER_PROOF_USE_LEDGER_COMPILER, KKL_WAVE_ATOMIC_SOURCE_OCCURRENCE_MANIFEST, KKL_IDEAL_PULLBACK_NATIVE, PARAMETRIZED_PULLBACK_KKL_EXHAUSTION_NO_GO

*Evidence:* direct_decoration_wave_proof_use_ledger.md#11-promotion-decision-and-next-theorem

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**KKL_BOUNDED_VISITS_INTERVAL_GROWTH_REPAIR** OPEN

Prove the marked-interval combinatorial-growth implication used by KKL's bounded-visits lemma: establish the exact at-most 6b+k-1 count for length-k multi-intervals meeting the critical limb, obtain correctly indexed repetition only from a strict pigeonhole excess, and prove an interval-level no-periodicity or strict-growth statement. Global Perron-Frobenius expansion lambda_F>1 alone may not be cited as excluding a periodic marked interval.

*Formal hypotheses:* *none*

*Proof dependencies:* KKL_WAVE_ATOMIC_SOURCE_OCCURRENCE_MANIFEST

*Evidence:* direct_decoration_wave_proof_use_ledger.md#32-conditional-arithmetic-after-a-growth-lemma

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**KKL_WANDERERS_THEOREM_COMPLETE_REPAIR** OPEN

Supply a complete source-checkable proof of KKL's stated Wanderers theorem, including the work-backwards lemma left to the reader, every imported good-path, route, Q_k, and delta_k definition and hypothesis, the length-area step, the non-Galois reduction, and the exact additive topology/degree constant. The first missing route implication, buffer estimate, multiplicity bound, or cover-reduction identity is the failure certificate.

*Formal hypotheses:* *none*

*Proof dependencies:* KKL_NATIVE_WAVE_WANDERERS_QUERY_SCHEMA_BOUNDARY_MAP

*Evidence:* direct_decoration_wave_proof_use_ledger.md#7-conditional-wanderers-repair-interface

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**KKL_LOCALIZED_WANDERERS_ROUTE_REALIZATION_CERTIFICATE** OPEN

For the localized typical-interval stage, construct the proper filled map with no fullification-created critical values outside the selected compact package; prove one selected source component over each represented target and degree at most two on the whole selected union; realize the residual-subtracted domination by one simultaneous complete-itinerary foliation; and prove route exhaustion, internal-joint, terminal-component, no-return, and stratum-recurrence certificates for the same realized foliation.

*Formal hypotheses:* *none*

*Proof dependencies:* KKL_NATIVE_WAVE_WANDERERS_QUERY_SCHEMA_BOUNDARY_MAP

*Evidence:* direct_decoration_wave_proof_use_ledger.md#definition-73-localized-route-realization-certificate

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**DIRECT_DECORATION_WAVE_PROOF_USE_LEDGER** OPEN

Freeze an actual finite, target-first ledger of every theorem call, target curve-family schema, ideal interval, and universal KKL subordination query used by Wave, Wanderers, domination, quasi-additivity, covering, or width comparison on the complete direct-decoration horizon. The ledger must distinguish parameterized lifts from all geometric inverse-trace paths and may not define proof use by the desired properness or response conclusion. The first omitted theorem call, family, interval, or query is the failure certificate.

*Formal hypotheses:* *none*

*Proof dependencies:* HORIZON_FIRST_INVENTORY_AUDIT, INTERFACE_PROOF_USE_INVENTORY, PARAMETRIZED_PULLBACK_KKL_EXHAUSTION_NO_GO, FINITE_BINDER_PROOF_USE_LEDGER_COMPILER, KKL_NATIVE_WAVE_WANDERERS_QUERY_SCHEMA_BOUNDARY_MAP

*Evidence:* direct_decoration_wave_proof_use_ledger.md#11-promotion-decision-and-next-theorem

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**ESCAPE_RETURN_KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY** OPEN

Extend the conformal-disk/ideal-circle theorem to V_esc and every proof-used first-return phase P_{i,k}. For each piece verify the finite open-star cover and prove two omitted values or ambient hyperbolicity; then identify Kahn's trivial-deck ideal boundary with the conformal circle and scope continuous extension to actual puzzle-inclusion immersions. A closed gluing plus an assumed ambient Riemann map is insufficient.

*Formal hypotheses: none*

*Proof dependencies:* FINITE_GRID_AND_TERMINAL_KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY, FIRST_DECORATION_SCALE_ESCAPE_ATLAS

*Evidence:* direct_decoration_bigraded_kl_stage_grid.md#proposition-53-audited-root-pseudo-piece-ideal-circles

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**WAVE_TO_KL_ADDRESS_ACCESS_SELECTOR** OPEN

For every bound proof-use geometric trace, construct a compatible atlas of actual finite puzzle pseudo-pieces and carrier-relative complements. Cut the trace at every central/tile seam, chart only central segments centrally, retain tile segments as actual path classes with multiplicity, and preserve covering-versus-inclusion typing, localization subgroups, cyclic order, both grid directions, the separate root transition, and final-root behavior. The strong obstacle-free native collar is ruled out for proper forward-invariant carriers. The width branch must instead instantiate the provenance-preserving KL submulticurve selector on a binder-complete connecting subfamily; exact carrier-pair prime-end naturality separately requires the original endpoint's full tagged strip address.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_COMPONENT_ADDRESS_FINITE_PHASE_SKELETON, CARRIER_ACCESS_TO_KL_MULTICURVE_BRIDGE, FINITE_GRID_AND_TERMINAL_KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY, TERMINAL_LOCAL_KL_KKL_ALIGNMENT_RECORD, TERMINAL_LOCAL_CHANNEL_TO_KL_WELDING, TERMINAL_CERTIFIED_CHANNEL_FINAL_ROOT_STABILITY, TERMINAL_FORWARD_INVARIANT_CARRIER_DENSE_OBSTACLE_NO_GO, TERMINAL_WAVE_KL_CONNECTING_SUBFAMILY, TERMINAL_KL_SUBMULTICURVE_BINDER_SELECTOR, TERMINAL_TAGGED_ENDPOINT_ACCESS_MATCH, TERMINAL_ANCHOR_COLLAR_OR_RESTRICTION_OBSTRUCTION, DIRECT_DECORATION_WAVE_PROOF_USE_LEDGER

*Evidence:* wave_to_kl_address_access_selector.md#6-terminal-anchor-attempt-and-exact-first-obstruction

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**DIRECT_DECORATION_CANONICAL_KL_STAGE_ALIGNMENT** OPEN

For every stage named by DIRECT_DECORATION_WAVE_PROOF_USE_LEDGER, construct rather than assume the actual same-source canonical/KL seed, source and target labels, lifted basepoints, localization subgroups, separately typed dynamical covering and inclusion immersion, central chart, tile charts, and exact central/tile seam equalizers, and instantiate the proved relation spans on that glued datum. The first map, seed, basepoint, label, subgroup, or seam mismatch is the obstruction certificate.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_WAVE_PROOF_USE_LEDGER, DIRECT_DECORATION_BIGRADED_KL_STAGE_GRID, ONE_DIMENSIONAL_CYLINDER_STAGE_COLLAPSE_NO_GO, WAVE_TO_KL_ADDRESS_ACCESS_SELECTOR, CANONICAL_KL_RESTRICTED_INTERFACE_CLASSIFICATION, DIRECT_DECORATION_PULLBACK_COMPACTIFICATION, KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY

*Evidence:* canonical_kl_geometric_trace_proper_end_response.md#pe-2-canonicalkl-stage-alignment

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**DIRECT_DECORATION_RELATION_PROPER_PATH_END_INTERVAL_CLASSIFICATION** OPEN

On every aligned stage, construct the tagged end-compactified geometric inverse-trace path space for every proof-use family, classify proper source paths and their actual KL, canonical relative-or-limit-defect, and target ideal-or-interior endpoint frames, retain all 0/1/>=2/infinite restriction multiplicities and subordination predicates, construct every iota^*I, prove pulled partial-marking disjointness and refinement by g^*I, and only then prove immersion properness on every whole proof-used interval and every complete end-interval required for KKL end-properness. A path is never labelled end-proper. The first competing or missing proper path, endpoint defect, access mismatch, overlapping pullback, refinement failure, or nonproper complete end-interval is the obstruction certificate.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_CANONICAL_KL_STAGE_ALIGNMENT, FULL_KL_PSEUDOPIECE_MAXIMAL_LIFT_PROPERNESS, PARAMETRIZED_PULLBACK_KKL_EXHAUSTION_NO_GO, KKL_IDEAL_PULLBACK_NATIVE, RESOLVED_

INTERMEDIATE_RELATIVE_IDEAL_BOUNDARY, KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY, NONPROPER_LABEL_HULL_PULLBACK_NO_GO, PRIME_END_MARKING_ACTION_CRITERION

*Evidence:* canonical_kl_geometric_trace_proper_end_response.md#pe-3-proper-path-and-end-interval-classification

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**DIRECT_DECORATION_CANONICAL_KL_RELATION_PROPER_END_RESPONSE** OPEN

For every classified proof-use geometric restriction family, retain every component and limit contact with zero omission or construct an explicit nonnegative omitted-family WAD/Value response, prove row-space and Value observability, and bound the complete transition charge by universal constants independent of satellite period, rotation number, and horizon length. PARAMETRIZED_PULLBACK_KKL_EXHAUSTION_NO_GO forbids a bounded-sheet or raw-width shortcut. This node contains no final-root enlargement claim.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_RELATION_PROPER_PATH_END_INTERVAL_CLASSIFICATION, KKL_TYPED_VALUE_INCIDENCE, VALUE_RESPONSE_NO_GO, VALUE_FEATURE_ROWSPACE, WAD_DOMINATION_RESPONSE, PORT_FUTURE_CLOSURE_NO_GO, HILBERT_TRACE_GLUE_CALCULUS

*Evidence:* canonical_kl_geometric_trace_proper_end_response.md#pe-4-relation-response

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**DIRECT_DECORATION_FINAL_ROOT_RELATION_NO_BIRTH** OPEN

Under every certified final-root enlargement, prove the required subgroup reflection and exhaustion equalities and a Cartesian bijection of compactified geometric trace components, proper-path multiplicities, source and target endpoint frames, accesses, obstacles, nonproper bridges, and cyclic order. The first born, lost, or merged sheet, component, proper path, endpoint, access, obstacle, or bridge is the obstruction certificate. This geometric theorem is independent of the relation width-response estimate.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_RELATION_PROPER_PATH_END_INTERVAL_CLASSIFICATION, KL_LOCALIZATION_SHEET_RESOLUTION_FACTORIZATION, PULLBACK_GAP_ENLARGEMENT_NATURALITY, MARKED_PAIR_ENLARGEMENT_NATURALITY, RETURN_ATLAS_ENLARGEMENT_NATURALITY

*Evidence:* canonical_kl_geometric_trace_proper_end_response.md#pe-5-final-root-relation-no-birth

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**DIRECT_DECORATION_OUTER_CAP_FOOTPRINT** OPEN

After the complete proof-use ledger, actual canonical/KL stage alignment, tagged geometric inverse-trace proper-path and whole-end-interval classification, and uniform relation response are constructed, determine the physical complement-access footprint of every retained lambda, xi, and eta source end on every source label, sheet, coset, and deck phase. Prove endpoint subordination, preserve distinct accesses and cyclic order, merge exactly the target ideal intervals met by one physical footprint, and give an explicit exclusion crosscut for every omitted source arc. The canonical infinite q/e cover is not the whole actual KL tile, and the two certified wrapping clusters do not supply the complete physical footprint. The first uncertified landing component, target-interior access, missing end, access merger, omitted window, or competing lift is the obstruction certificate. Final-root no-birth remains a separate obligation.

*Formal hypotheses: none*

*Proof dependencies:* KL_CANONICAL_LOCALIZATION_FORCED_COLLISION, KL_LOCALIZATION_SHEET_RESOLUTION_FACTORIZATION, KAHN_ITERATE_COMPLEMENTARY_COMPONENT_ALIGNMENT, RESOLVED_INTERMEDIATE_RELATIVE_IDEAL_BOUNDARY, CANONICAL_KL_WHOLE_TILE_CORRESPONDENCE_NO_GO, CANONICAL_KL_RESTRICTED_INTERFACE_CLASSIFICATION, KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY, GENERAL_ANNULUS_IMMERSION_LOWER_STRIP_NO_GO, DIRECT_DECORATION_CANONICAL_KL_RELATION_PROPER_END_RESPONSE

*Evidence:* direct_decoration_outer_port_footprint.md#10-remaining-actual-immersion-footprint-record

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**DIRECT_DECORATION_PSEUDO_PIECE_PATH_LIFT_EXHAUSTION** OPEN

For every nontrivial proof-used direct-decoration marking, combine the proved geometric inverse-trace proper-path and whole-end-interval classification, the uniform relation response, the complete physical lambda/xi/eta footprint, and the independent Cartesian final-root no-birth theorem. Then prove the KKL universal pullback predicates, maximal safe-set components, pairwise disjoint pulled partial marking, refinement by the dynamical covering marking, and properness or end-properness on every complete proof-used interval. Every competing proper path, nonproper bridge, target-interior endpoint, source label, sheet, coset, deck phase, and response contribution must be retained or explicitly excluded. Parameterized maximal lifts alone do not exhaust the KKL geometric restriction.

*Formal hypotheses: none*

*Proof dependencies:* KL_CANONICAL_LOCALIZATION_FORCED_COLLISION, KL_LOCALIZATION_SHEET_RESOLUTION_FACTORIZATION, KAHN_ITERATE_COMPLEMENTARY_COMPONENT_ALIGNMENT, NONPROPER_LABEL_HULL_PULLBACK_NO_GO, KL_PSEUDOPIECE_CONFORMAL_IDEAL_BOUNDARY, CANONICAL_KL_WHOLE_TILE_CORRESPONDENCE_NO_GO, CANONICAL_KL_RESTRICTED_INTERFACE_CLASSIFICATION, DIRECT_DECORATION_OUTER_CAP_FOOTPRINT, DIRECT_DECORATION_FINAL_ROOT_RELATION_NO_BIRTH

*Evidence:* direct_decoration_outer_port_footprint.md#10-remaining-actual-immersion-footprint-record

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**DIRECT_DECORATION_ALL_DEPTH_CONTACT_CLOSURE** OPEN

For every proof-used direct-decoration end and complete horizon, choose one exact source branch granularity and realize the complete current-separator interface state on the restricted canonical-to-KL comparison certified by DIRECT_DECORATION_PSEUDO_PIECE_PATH_LIFT_EXHAUSTION, including sibling or multi-branch incidence, active sides, phase, orientation, endpoint and prime-end markings, protected accesses, obstacles, branch itineraries, localization subgroups, and every live coset in a_*^(-1)(H_t)/H_s. Construct an exact incidence- or restriction-valued transport closed on a finite or future-faithfully compact sheet-sensitive state, and prove completeness on that asserted domain, Sobolev trace admissibility, pulled-versus-immersed compatibility, protected-data compatibility, and stability under authorized root enlargement. Neither KL_CANONICAL_LOCALIZATION_FORCED_COLLISION nor the whole-tile correspondence no-go permits silently replacing this restricted state by an embedded or factorwise-completed whole pseudo-piece. The first untracked omitted window, coset merger, missing footprint, exclusion crosscut, complete end-interval, marked interface atom, trace map, or enlargement transport is the failure certificate.

*Formal hypotheses:* *none*

*Proof dependencies:* DIRECT_DECORATION_PULLBACK_COMPACTIFICATION, ACTIVE_INACTIVE_PHYSICAL_CELL, ATTACHMENT_ARC_OPEN_ATOM_REFINEMENT, BINARY_DIRECT_DECORATION_NESTED_SEPARATOR_CALCULUS, FIXED_BLOCK_BINARY_CONTACT_AUTOMATON, FINITE_LINEAR_LABEL_MARKING_ENUMERATOR, KL_CANONICAL_LOCALIZATION_FORCED_COLLISION, KL_LOCALIZATION_SHEET_RESOLUTION_FACTORIZATION, DIRECT_DECORATION_PSEUDO_PIECE_PATH_LIFT_EXHAUSTION

*Evidence:* direct_decoration_path_lifted_collar_certificate.md#10-the-conditional-path-lift-exhaustion-gate

---

**FINITE_LIVE_COMMON_REFINEMENT_FRONTIER** OPEN

After the complete source-faithful horizon and marked current-separator incidence state are frozen, every proof-used unequal-footprint, access, obstacle, response, ideal, enlargement, and transported-contact trace atom has a proved finite last-live stage, or admits a future-faithful compact quotient preserving every response and Value observable. Raw contact localization for one nested chain does not supply this marked analytic frontier.

*Formal hypotheses:* *none*

*Proof dependencies:* DIRECT_DECORATION_ALL_DEPTH_CONTACT_CLOSURE, CLOSED_FORM_PUSHFORWARD_NO_GO

*Evidence:* research_push_5_assessment_and_integration.md#13-replacement-invariant-and-next-theorem

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**DIRECT_DECORATION_TRACE_WELDING_COMPACTNESS** OPEN

For every contact-closed proof-used physical end, the entrance and exit prime-end trace identifications induced by whole-tail uniformization act boundedly and precompactly in the fixed response and Value trace norm, with uniform inverse, phase, orientation, and protected-marking control; alternatively retain the full welding cocycle in a proved compact state. The state must retain the separately typed immersion-restriction and covering-transport factors, together with the fiber-product or Beck-Chevalley interchange data needed for word composition; multiplying one-step endomorphisms is not a substitute.

*Formal hypotheses:* *none*

*Proof dependencies:* DIRECT_DECORATION_PULLBACK_COMPACTIFICATION, DIRECT_DECORATION_ALL_DEPTH_CONTACT_CLOSURE, REGULAR_INTERFACE_FIBER_PRODUCT_RESPONSE, FIXED_STANDARD_NECK_MULTIPORT_COMPACTIFICATION, WOT_WELDING_ALGEBRA_NO_GO, DYADIC_BOTTCHER_EXTERNAL_TRACE_ISOMETRY, PULL_PUSH_WELDING_WORD_COMPOSITION_NO_GO

*Evidence:* research_push_8_assessment_and_integration.md#12-exact-failure-of-word-functoriality

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**DIRECT_DECORATION_TRACE_COLLAR_REALIZATION** OPEN

For every proof-used direct-decoration end, choose and source-lock the actual branch package and the restricted canonical-to-KL interface certified upstream; construct its physical cells and complete marked current-separator trace interfaces on that asserted domain; retain the least compatible multi-port response state, incidence-valued boundary transport, stage-normalized protected overlay, phase, orientation, and separately typed pulled-versus-immersed trace factors; prove safe closed elimination or compact retention, no-bypass resistance or

full-response coercivity, fiber-product word composition, final-root no-new-sheet naturality, and sufficient uniform control for every claimed transition margin. A homogeneous Hilbert trace quotient is now proved only on a compact subarc in the relative interior of one intrinsic KL universal-cover tile collar. The proof-used physical interval identification, complete-interval immersion transport within the restricted interface, connected compatible collar core, uniform norm, Beck–Chevalley square, omitted-window charge, and enlargement component bijection remain open; no full factorwise or whole-tile completion is an admissible substitute.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_PULLBACK_COMPACTIFICATION, KL_TILE_INTERIOR_HOMOGENEOUS_TRACE, PULLBACK_GAP_ENLARGEMENT_NATURALITY, SERIAL_EXHAUSTION_RESISTANCE_BOUND, FULL_TRACE_RECTANGLE_DTN, PULLED_EQUIPOTENTIAL_SERIAL_RECTANGLE_NO_GO, ACTIVE_INACTIVE_PHYSICAL_CELL, ATTACHMENT_ARC_OPEN_ATOM_REFINEMENT, REGULAR_INTERFACE_FIBER_PRODUCT_RESPONSE, FIXED_MODULUS_AMBIENT_END_NO_GO, SERIAL_ANNULUS_SINGLETON_BRIDGE, MULTIPORT_TRACE_GAUGE_AND_STRICTNESS_NO_GO, DIRECT_DECORATION_ALL_DEPTH_CONTACT_CLOSURE, FINITE_LIVE_COMMON_REFINEMENT_FRONTIER, DIRECT_DECORATION_TRACE_WELDING_COMPACTNESS, FINITE_LINEAR_LABEL_MARKING_ENUMERATOR, PULL_PUSH_WELDING_WORD_COMPOSITION_NO_GO

*Evidence:* direct_decoration_path_lifted_collar_certificate.md#15-dependency-and-promotion-audit

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**END_PLUMBING_TRACE_RESPONSE_CONTROL** OPEN

For every proof-used materialized end neck, realize the all-depth contact-closed physical trace interfaces and either prove exact constant-trace shorting or control the full multi-port Dirichlet-to-Neumann response and its endpoint welding cocycle by a compact summable state, compatibly with protected carriers, genuinely last-live ports, Value features, and the transition norm; prove uniform no-bypass resistance or an equivalent coercive full-response estimate before invoking a finite-depth strictness margin.

*Formal hypotheses: none*

*Proof dependencies:* FINITE_ELECTRODE_RESPONSE, SERIAL_EXHAUSTION_RESISTANCE_BOUND, SCALAR_NECK_TRACE_MODE_NO_GO, FULL_TRACE_RECTANGLE_DTN, TRACE_RESPONSE_SEMIGROUP, EQUIPOTENTIAL_COLLAPSE_CRITERION, TRACE_RENORMALIZED_NECK_ESTIMATE, END_TRACE_RESPONSE_COMPACTIFICATION, HILBERT_TRACE_GLUE_CALCULUS, TRACE_STABLE_STRICT_TRANSITION_TRANSFER, FIXED_STANDARD_NECK_MULTIPORT_COMPACTIFICATION, SERIAL_ANNULUS_SINGLETON_BRIDGE, MULTIPORT_TRACE_GAUGE_AND_STRICTNESS_NO_GO, WOT_WELDING_ALGEBRA_NO_GO, DIRECT_DECORATION_TRACE_COLLAR_REALIZATION, DIRECT_DECORATION_ALL_DEPTH_CONTACT_CLOSURE, FINITE_LIVE_COMMON_REFINEMENT_FRONTIER, DIRECT_DECORATION_TRACE_WELDING_COMPACTNESS

*Evidence:* research_push_5_assessment_and_integration.md#7-exact-all-depth-bypass

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**VIRTUAL_REACHED_CYLINDER_COMPONENT_EXHAUSTION** OPEN

After an actual target base is typed or uniquely selected, every demanded finite-cylinder branch image has the open pullback component forced by INTERIOR_PULLBACK_COMPONENT_FORCING, and that reached localized component, or the globally identical component justified by the Jordan-disk component lemma, is one of the declared equal-depth target cylinders at every proof-used intermediate iterate with complete metadata and enlargement stability.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_PULLBACK_COMPACTIFICATION, FIXED_BASE_CONNECTED_IMAGE_TARGET_UNIQUENESS, INTERIOR_PULLBACK_COMPONENT_FORCING

*Evidence:* finite_cylinder_commuting_lift.md#5-the-exact-remaining-address-selector

---

**VIRTUAL_CYLINDER_TARGET_ADDRESS_SELECTOR** OPEN

For every realized finite virtual copy-chain prefix, equal-depth source cylinder, and source-defined return branch, there is a unique actual target base and cylinder containing the image component, compatibly with geometric parents, the partial pre-root shift, the separate root transition, complete marked metadata, presentation, and authorized root enlargement.

*Formal hypotheses: none*

*Proof dependencies:* DIRECT_DECORATION_PULLBACK_COMPACTIFICATION, PULLBACK_GAP_ENLARGEMENT_NATURALITY, FIXED_BASE_CONNECTED_IMAGE_TARGET_UNIQUENESS, VIRTUAL_REACHED_CYLINDER_COMPONENT_EXHAUSTION

*Evidence:* finite_cylinder_commuting_lift.md#open-problem-51-virtual_cylinder_target_address_selector

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**VIRTUAL_FINITE_WORD_ROOT_PORTRAIT_AUTHORIZATION** OPEN

For every actual two-step virtual satellite-copy chain, construct and source-identify the required second-scale root portrait in a Jordan neighborhood compactly contained in the direct-decoration safe gap, realize its proof-complete carrier and certified covering-versus-immersion marking, authorize all simultaneous finite-word and forced off-critical component ownership, and prove exact final-root and complete-Wave-demand naturality.

*Formal hypotheses:* *none*

*Proof dependencies:* RETIMED_FINITE_CYLINDER_PAIR, DIRECT_DECORATION_FINITE_PROTECTED_CARRIER, PRIME_END_MARKING_ACTION_CRITERION, PULLBACK_GAP_ENLARGEMENT_NATURALITY, VIRTUAL_CYLINDER_TARGET_ADDRESS_SELECTOR, VIRTUAL_WAVE_DEMAND_SCHEMA

*Evidence:* critical_chronology_and_retimed_telescope.md#16-exact-source-facing-residual

---

**DIRECT_DECORATION_END_PRIME_END_COFINALITY** OPEN

For every direct-decoration infinite address, construct the actual multisector Bottcher angular footprint with compatible inverse lifts and prove that the exterior traces of the nested puzzle cylinders form a fundamental chain cofinal at one external prime end, including endpoint addresses, so that its impression is exactly the nested end fiber. Protection against collars, obstacles, and enlargement is separate.

*Formal hypotheses:* *none*

*Proof dependencies:* DIRECT_DECORATION_PULLBACK_COMPACTIFICATION

*Evidence:* critical_chronology_and_retimed_telescope.md#152-prime-end-impression-equality

---

**VIRTUAL_END_BLOWUP_REALIZATION** OPEN

For every proof-used infinite address, construct an analytic rescaled or nontrivial-projection-fiber end space carrying the authorized normalized root pair over the identified prime-end impression, with protected accesses, obstacles, sheets, critical chronology, shift and fixed-iterate actions, compatible relative covers, proper or end-proper intervals, and exact complete-horizon final-root naturality.

*Formal hypotheses:* *none*

*Proof dependencies:* VIRTUAL_FINITE_WORD_ROOT_PORTRAIT_AUTHORIZATION, DIRECT_DECORATION_END_PRIME_END_COFINALITY, END_BLOWUP_PROJECTION_NO_GO, NORMALIZED_RETIMED_GERM_CONSTANCY, FIXED_ITERATE_END_FIBER_EXTENSION

*Evidence:* critical_chronology_and_retimed_telescope.md#16-exact-source-facing-residual

---

**VIRTUAL_DISCONNECTED_MARKING_TRANSFER** OPEN

For every two-step virtual satellite-copy chain, the next satellite portrait canonically realizes actual second-scale puzzle or generalized polynomial-like bases and polynomial iterates on the direct-decoration pullback cylinders and end fibers, equivariantly under the univalent shift and naturally under authorized root enlargement, with complete critical, sheet, access, obstacle, terminal-return, and presentation data.

*Formal hypotheses:* *none*

*Proof dependencies:* DIRECT_DECORATION_PULLBACK_COMPACTIFICATION, PULLBACK_GAP_ENLARGEMENT_NATURALITY, FINITE_CYLINDER_COMMUTING_ITERATE_LIFT, VIRTUAL_CYLINDER_TARGET_ADDRESS_SELECTOR, VIRTUAL_FINITE_WORD_ROOT_PORTRAIT_AUTHORIZATION, DIRECT_DECORATION_END_PRIME_END_COFINALITY, VIRTUAL_END_BLOWUP_REALIZATION

*Evidence:* virtual_cantor_pullback_compactification.md#10-attempted-scale-two-marking

---

**VIRTUAL_CANTOR_PACKAGE_REALIZATION** OPEN

Every actual intermediate virtual-Molecule scale has an analytically realized compact address space of actual small-Julia components with exact polynomial branch, return, parent, sheet, critical-address, escape, localization, presentation, terminal-compatibility, and final-root-enlargement data, constructed without downstream width, compactness, or rigidity conclusions.

*Formal hypotheses:* *none*

*Proof dependencies:* DIRECT_DECORATION_PULLBACK_COMPACTIFICATION, PULLBACK_GAP_ENLARGEMENT_NATURALITY, VIRTUAL_DISCONNECTED_MARKING_TRANSFER

*Evidence:* virtual_cantor_pullback_compactification.md#10-attempted-scale-two-marking

**VIRTUAL_VERTEX_PACKAGE_SELECTOR** OPEN

Every actual intermediate virtual-Molecule scale has a realized virtual Cantor component package and, after its complete proof-use horizon is frozen, a finite actual-component trace or future-faithful quotient together with a compatible partial factor to the native terminal Hubbard tree, preserving exact analytic corridor transport, protected accesses, obstacles, final-root enlargement traces, presentation invariance, and every claimed observable.

*Formal hypotheses: none*

*Proof dependencies:* NATIVE_HUBBARD_SUBTREE_FUNCTORIALITY, TERMINAL_CYCLE_HUBBARD_ANCHOR, DECORATION_FAN_GROWTH, VIRTUAL_CANTOR_PACKAGE_REALIZATION, VIRTUAL_WAVE_HORIZON_CERTIFICATE, VIRTUAL_PROOF_USE_TRACE_SELECTOR

*Evidence:* virtual_cantor_package_interface.md#7-revised-proof-obligations

---

**VIRTUAL_PREFIX_LOCAL_SCALE_SELECTOR** OPEN

For every finite actual virtual-Molecule copy-chain prefix, the realized component package canonically selects an embedded nice puzzle or principal-nest base, actual component branches, sheets, critical chronology, and presentation/enlargement maps using only prefix-local dynamics, compatibly under prefix extension and without reading a completed Wave horizon, final proof-use trace, downstream bound, compactness, rigidity, or MLC conclusion.

*Formal hypotheses: none*

*Proof dependencies:* NATIVE_PRINCIPAL_NEST_RETURN_ATLAS, ISOTOPY_CLASS_RETURN_UNDERDETERMINATION, RETURN_ATLAS_ENLARGEMENT_NATURALITY, FIRST_DECORATION_SCALE_ESCAPE_ATLAS, CRITICAL_BASE_CYCLE_EXCLUSION, DECORATION_FAN_GROWTH, TERMINAL_CYCLE_HUBBARD_ANCHOR, VIRTUAL_CANTOR_PACKAGE_REALIZATION

*Evidence:* virtual_scale_to_puzzle_selector.md#13-prefix-local-selector-and-downstream-trace-split

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**VIRTUAL_SCALE_TO_PUZZLE_SELECTOR** OPEN

After the prefix-local selector, marked correspondence, complete Wave horizon, and proof-use trace have been constructed, every actual intermediate virtual-Molecule scale has a final proof-use-complete embedded base, virtual carrier placement, return and access inventory, obstacle and sheet data, and exact presentation/enlargement maps, uniformly in satellite period and chain length and without using any downstream bound or rigidity conclusion.

*Formal hypotheses: none*

*Proof dependencies:* VIRTUAL_PREFIX_LOCAL_SCALE_SELECTOR, VIRTUAL_WAVE_HORIZON_CERTIFICATE, VIRTUAL_PROOF_USE_TRACE_SELECTOR, VIRTUAL_VERTEX_PACKAGE_SELECTOR

*Evidence:* virtual_cantor_package_interface.md#7-revised-proof-obligations

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**VIRTUAL_RETURN_BASE_ATLAS** OPEN

Every actual virtual-Molecule block and finite provisional root package has source-defined open nice return localizations and actual atomic branch components with integer iterates, complete critical and degree words, exact component, parent, obstacle, localization, sheet, and protected-access phase data, together with exact enlargement morphisms and demand-bijective no-birth maps wherever complete language stability is claimed.

*Formal hypotheses: none*

*Proof dependencies:* RETURN_BASE_UNDERDETERMINATION, NATIVE_PRINCIPAL_NEST_RETURN_ATLAS, ISOTOPY_CLASS_RETURN_UNDERDETERMINATION, VIRTUAL_PREFIX_LOCAL_SCALE_SELECTOR, CONNECTED_CARRIER_NO_MINIMUM, PROTECTED_REGULATED_CONNECTOR_REALIZATION, CRITICAL_VALUE_CORRIDOR_CRITERION

*Evidence:* virtual_return_base_atlas.md#11-promotion-decision

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**VIRTUAL_WAVE_DEMAND_SCHEMA** OPEN

Before any stopping execution, every actual virtual-Molecule block has a source-faithful syntactic continuation relation that is sound for actual branch-component extensions, complete for every Wave, Wanderers, covering, quasi-additivity, and width theorem call, stable under certified root enlargement, and independent of the desired bounds, compactness, rigidity, and MLC conclusions.

*Formal hypotheses: none*

*Proof dependencies:* HORIZON_FIRST_INVENTORY_AUDIT, RESOLVED_FINITE_WAVE_HORIZON_COMPILER

*Evidence:* virtual_return_branch_language.md#virtual_wave_demand_schema-open

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**VIRTUAL_RETURN_BRANCH_LANGUAGE** OPEN

Every actual virtual-Molecule block has, before any Wave stopping execution, an actual typed analytic branch graph whose prefix-closed language contains every branch-component extension that an admissible Wave continuation can request, whose words compose on nonempty actual domains with complete iterate, component, critical, degree, obstacle, localization, sheet, and protected-access data, and whose certified words persist naturally under final-root enlargement with no new demand when complete stability is asserted.

*Formal hypotheses: none*

*Proof dependencies:* RETURN_PATH_LANGUAGE, DEMAND_CLOSED_RETURN_ATLAS_COMPILER, RETURN_ATLAS_ENLARGEMENT_NATURALITY, VIRTUAL_RETURN_BASE_ATLAS, VIRTUAL_WAVE_DEMAND_SCHEMA

*Evidence:* virtual_return_branch_language.md#10-promotion-decision-and-next-analytic-task

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**VIRTUAL_PROVISIONAL_COMPLEMENT_PAIR** OPEN

For every actual virtual-Molecule block, finite provisional root package, and compatible finite return path, there is a nondegenerate full forward-coherent carrier containing all critical values of the return, with an actual covering/inclusion complement pair, compatible cusp-free localizations, exact selected-component routing, and a complete absorb-or-separate obstacle ledger.

*Formal hypotheses: none*

*Proof dependencies:* VIRTUAL_RETURN_BRANCH_LANGUAGE, FORWARD_COHERENT_COMPLEMENT_PAIR, ROOT_PULLBACK_FOREST_EQUIVALENCE, FINITE_LOCAL_SATURATION, PROTECTED_REGULATED_CONNECTOR_REALIZATION, CRITICAL_VALUE_CORRIDOR_CRITERION

*Evidence:* virtual_wave_marked_correspondence.md#virtual_provisional_complement_pair-open

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**VIRTUAL_ACCESS_ACTION_CERTIFICATE** OPEN

Every virtual provisional complement pair has a normalized relative-cover phase, complete prime-end endpoint and interval data, exact covering-versus-immersion endpoint containment, all required proper/end-proper flags, and commuting analytic and prime-end squares under every certified collision-free package enlargement.

*Formal hypotheses: none*

*Proof dependencies:* VIRTUAL_PROVISIONAL_COMPLEMENT_PAIR, PRIME_END_MARKING_ACTION_CRITERION, RELATIVE_COVER_LIFT, PRIME_END_RETURN, PROTECTED_REGULATED_CONNECTOR_REALIZATION

*Evidence:* virtual_wave_marked_correspondence.md#virtual_access_action_certificate-open

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**VIRTUAL_WAVE_MARKED_CORRESPONDENCE** OPEN

Every actual virtual-Molecule block has a horizon-independent prefix-closed language of admissible finite return words and, for every finite provisional root package and compatible word, an actual marked hyperbolic carrier complement and covering/immersion pair whose pulled marking refines the immersed marking, whose marked action is exact, whose used intervals are proper or end-proper, and whose component, access, obstacle, degree, and localization data commute with every certified collision-free package enlargement.

*Formal hypotheses: none*

*Proof dependencies:* VIRTUAL_RETURN_BRANCH_LANGUAGE, VIRTUAL_PROVISIONAL_COMPLEMENT_PAIR, VIRTUAL_ACCESS_ACTION_CERTIFICATE, MARKED_PAIR_ENLARGEMENT_NATURALITY

*Evidence:* virtual_wave_marked_correspondence.md#9-dependency-correction

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**VIRTUAL_UNPROTECTED_ATOM_EXPOSURE** OPEN

For every actual virtual-Molecule Wave block after its complete horizon inventory and final root enlargement, each original root-shadow atom has an actual unprotected singleton access in the complement of its completed packet, witnessed by an access arc or the proved metric-collar criterion, with no port landing only on an added attachment and no protected impression relabeled free.

*Formal hypotheses: none*

*Proof dependencies:* INTERFACE_PROOF_USE_INVENTORY, VIRTUAL_WAVE_HORIZON_CERTIFICATE, AUGMENTED_PACKET_CLOSURE_AUDIT

*Evidence:* augmented_packet_closure_audit.md#vaapc-p-unprotected-exposure

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**VIRTUAL_AUGMENTED_ATOM_PORT_CERTIFICATE** OPEN

Every actual virtual-Molecule Wave block, after the complete finite future inventory and final root enlargement, admits a source-faithful assignment whose connected fullifications cover all forbidden data, remain pairwise disjoint inside the authorized localization, retain pairwise disjoint explicit unprotected singleton ports to every original root-shadow atom through the complete horizon, transport equivariantly, and have uniformly bounded or future-faithfully compressed incidence.

*Clauses:*

- **VAAPC-I**: Frozen actual input. *Maps to:* INTERFACE_PROOF_USE_INVENTORY, VIRTUAL_WAVE_HORIZON_CERTIFICATE.
- **VAAPC-C**: Zero completed-packet collision. *Maps to:* VIRTUAL_ROOT_CARRIER_SEPARATION.
- **VAAPC-P**: Unprotected original-atom exposure. *Maps to:* VIRTUAL_UNPROTECTED_ATOM_EXPOSURE.
- **VAAPC-H**: Complete-horizon transport. *Maps to:* VIRTUAL_CORRIDOR_PAIR_NATURALITY.
- **VAAPC-U**: Uniform incidence or faithful compression. *Maps to:* H2_UNIFORM_CORE_WAD_DIMENSION.

*Formal hypotheses: none*

*Proof dependencies:* INTERFACE_PROOF_USE_INVENTORY, VIRTUAL_WAVE_HORIZON_CERTIFICATE, VIRTUAL_UNPROTECTED_ATOM_EXPOSURE, AUGMENTED_PACKET_CLOSURE_AUDIT, AUGMENTED_ATOM_COMMON_DOMAIN

*Evidence:* augmented_packet_closure_audit.md#8-reduction-of-the-live-dynamical-gate

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**VIRTUAL_COMMON_FREE_COMPONENT** OPEN

For every actual virtual-Molecule Wave block, after the complete root-shadow, protected-collar, obstacle, critical-value, and localization inventory is frozen through the full horizon, one free-domain component has simultaneous singleton ports to all required root-shadow clusters and remains valid after the final root enlargement.

*Formal hypotheses: none*

*Proof dependencies:* INTERFACE_PROOF_USE_INVENTORY, PROTECTED_FREE_COMPONENT_CRITERION, AUGMENTED_ATOM_COMMON_DOMAIN, VIRTUAL_AUGMENTED_ATOM_PORT_CERTIFICATE

*Evidence:* virtual_common_free_component.md#8-exact-remaining-dynamical-proposition

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**VIRTUAL_PROTECTED_ROUTING_CERTIFICATE** OPEN

Every actual virtual-Molecule Wave block admits a source-faithful lamination-lifted or puzzle-cell regulated root tree whose fibers, edges, and fullification avoid all protected collar tails, whose iterated pullback components realize the complete routing and critical words, and whose obstacle, localization, future-access, and uniform-complexity clauses hold through the complete Wave horizon.

*Formal hypotheses: none*

*Proof dependencies:* INTERFACE_PROOF_USE_INVENTORY, VIRTUAL_ROOT_CARRIER_SEPARATION, PROTECTED_REGULATED_CONNECTOR_REALIZATION, PROTECTED_FREE_COMPONENT_CRITERION, VIRTUAL_COMMON_FREE_COMPONENT

*Evidence:* protected_regulated_connector.md#8-exact-next-virtual-theorem

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**VIRTUAL_CANONICAL_INTERFACE_CARRIER** OPEN

Every actual virtual-Molecule Wave block admits a regulated forward-coherent carrier selected from the simultaneous root pullback, with complete proof-use inventory, full attachment traces, injective cyclic-order-preserving protected-access survival through the complete horizon, and KKL-style component exhaustion on the degree-one central corridor.

*Formal hypotheses: none*

*Proof dependencies:* H1_CELL_TREE_EXTRACTION, VIRTUAL_ROOT_CARRIER_SEPARATION, INTERFACE_PROOF_USE_INVENTORY, FINITE_GEOMETRIC_SATURATION, CONNECTED_CARRIER_NO_MINIMUM, QUADRATIC_PROTECTED_ACCESS_COLLISION, PROTECTED_REGULATED_CONNECTOR_REALIZATION, VIRTUAL_PROTECTED_ROUTING_CERTIFICATE

*Evidence:* canonical_virtual_interface_carrier.md#11-promotion-decision-and-next-theorem

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**VIRTUAL_CORRIDOR_PAIR_NATURALITY** OPEN

Every degree-one central virtual Wave corridor admits commuting orientation-preserving homeomorphisms of the actual pulled and immersed carrier pairs, extending to access-complete prime-end compactifications and preserving every proof-used ideal interval, attachment incidence, and cyclic access order.

*Formal hypotheses: none*

*Proof dependencies:* VIRTUAL_CANONICAL_INTERFACE_CARRIER, VIRTUAL_ROOT_CARRIER_SEPARATION, H1_ANALYTIC_ACCESS_REALIZATION, CORRIDOR_ATTACHMENT_TRANSPORT, UNIVALENT_CARRIER_PAIR_TRANSPORT

*Evidence:* virtual_corridor_pair_naturality.md#7-application-to-the-planned-virtual-construction

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**VIRTUAL_ATTACHMENT_FREE_ANCHOR** OPEN

Each central virtual Wave corridor contains one stage whose actual pulled carrier has no excess component attached to any proof-used central ideal interval, after all required Cantor components, first-return orbit data, dams, fjords, and future endpoint accesses are included.

*Formal hypotheses: none*

*Proof dependencies:* VIRTUAL_CANONICAL_INTERFACE_CARRIER, H1_CELL_TREE_EXTRACTION, H1_ANALYTIC_ACCESS_REALIZATION, VIRTUAL_ROOT_CARRIER_SEPARATION, ATTACHMENT_FREE_ANCHOR_UNDERDETERMINATION, FINITE_ATTACHMENT_ORBIT_CLOSURE

*Evidence:* virtual_corridor_pair_naturality.md#7-application-to-the-planned-virtual-construction

---

**VIRTUAL_ATTACHMENT_RESOLUTION** OPEN

For every central virtual Wave corridor, either an actual attachment-free anchor exists, or the least attachment-orbit closure has zero sheet, critical, obstacle, access, and localization collision, admits an access-complete analytic realization, and has empty attachment support on every proof-used central interval. The two alternatives are sufficient routes and are not asserted to be logically equivalent.

*Formal hypotheses: none*

*Proof dependencies:* VIRTUAL_CANONICAL_INTERFACE_CARRIER, H1_CELL_TREE_EXTRACTION, H1_ANALYTIC_ACCESS_REALIZATION, VIRTUAL_ROOT_CARRIER_SEPARATION, ATTACHMENT_FREE_ANCHOR_UNDERDETERMINATION, FINITE_ATTACHMENT_ORBIT_CLOSURE, CORRIDOR_ATTACHMENT_TRANSPORT, UNIVALENT_CARRIER_PAIR_TRANSPORT

*Evidence:* virtual_corridor_pair_naturality.md#virtual-attachment-resolution-gate

---

**VIRTUAL_BOUNDED_ATTACHMENT_SUPPORT** OPEN

Every virtual block routed to the Wave branch has an exact ideal-interval pullback description whose pulled-carrier excess has empty attachment support throughout the degree-one central corridor and uniformly bounded endpoint contribution in fixed entry, exit, and terminal windows, with access and cyclic order preserved and all proof-used intervals carrying proper or end-proper relative-cover lifts.

*Formal hypotheses: none*

*Proof dependencies:* H1_ANALYTIC_ACCESS_REALIZATION, RELATIVE_COVER_LIFT, SATURATED_SEED_FOREST, FINITE_OBSTACLE_FORWARD_CLOSURE, KKL_IDEAL_PULLBACK_NATIVE, ATTACHMENT_SUPPORT_REDUCTION, CORRIDOR_ATTACHMENT_TRANSPORT, UNIVALENT_CARRIER_PAIR_TRANSPORT, FINITE_ATTACHMENT_ORBIT_CLOSURE, VIRTUAL_CORRIDOR_PAIR_NATURALITY, VIRTUAL_ATTACHMENT_RESOLUTION

*Evidence:* kkl_translation_no_birth_source_audit.md#5-the-corrected-virtual-target

---

**H2_VIRTUAL_CORE_COCYCLE_REALIZATION** OPEN

Every virtual-Molecule block has an access-complete selected finite-horizon correspondence whose resolved tokens carry the exact typed source-column incidence B=N^T, KKL left/right Value weights, endpoint frames, raw WAD transport, and WAD-to-core incidence through every proof-used stopping time.

*Formal hypotheses: none*

*Proof dependencies:* H1_CELL_TREE_EXTRACTION, H1_ANALYTIC_ACCESS_REALIZATION, H1_FINITE_HORIZON_INFLATION, FRAMED_RETURN_SPECTRAL_INVARIANCE, KKL_TYPED_VALUE_INCIDENCE

*Evidence:* virtual_core_cocycle_program.md#3-the-raw-realization-theorem

---

**H2_UNIFORM_COUNTER_FOLLOWER** OPEN

Every virtual-Molecule block has a universal restricted counter/stack follower: finite decorated control and suffix memory, universally many bounded-increment geometric counters with a fixed guard catalog, a finite top-only stack alphabet, no hidden epsilon computation, fixed trace resolution, read-only macro parameters, and all emitted observables retained.

*Formal hypotheses: none*

*Proof dependencies:* H1_CELL_TREE_EXTRACTION, H1_ANALYTIC_ACCESS_REALIZATION, H1_FINITE_HORIZON_INFLATION, H2_VIRTUAL_CORE_COCYCLE_REALIZATION, FUTURE_EQUIVALENCE_AUTOMATON, FINITE_HORIZON_LANGUAGE_NO_GO, BOUNDED_MEMORY_AUTOMATON, COUNTER_SAFE_PUMPING, MEMORY_PROJECTION_PUMPING_NO_GO, ADDITIVE_COUNTER_MEMORY_NO_GO, REVERSE_WORD_STACK_MODEL, COUNTER_STACK_SAFE_LOOP

*Evidence:* virtual_core_cocycle_program.md#5-the-uniform-compression-package

---

**H2_UNIFORM_CORE_WAD_DIMENSION** OPEN

Every virtual-Molecule block admits source-faithful core and compositional WAD modules of universal dimensions d<=d0 and w<=w0, with every enabled WAD branch descending through its quotient and the complete fixed family of future observations remaining exact.

*Formal hypotheses: none*

*Proof dependencies:* H2_VIRTUAL_CORE_COCYCLE_REALIZATION, WAD_TRANSPORT_QUOTIENT_CRITERION, WAD_FUTURE_OBSERVABILITY, UNBOUNDED_DIMENSION_ENTROPY_NO_GO

*Evidence:* virtual_core_cocycle_program.md#5-the-uniform-compression-package

---

**H2_MEMORY_SAFE_PUMPING_COMPILATION** OPEN

Before support pumping, every memory-changing run is compiled into an exact typed checkpoint macro or the pumping summary is enlarged by a finite memory congruence whose repeated classes act identically on every relevant reachable counter/stack fiber, with endpoint guards, frames, terminal colors, and emitted observables preserved.

*Formal hypotheses:* *none*

*Proof dependencies:* H2_UNIFORM_COUNTER_FOLLOWER, MEMORY_PROJECTION_PUMPING_NO_GO, PURE_CORRIDOR_MACRO, COUNTER_STACK_SAFE_LOOP

*Evidence:* virtual_core_cocycle_program.md#5-the-uniform-compression-package

---

**H2_VIRTUAL_SUPPORT_AUTOMATON** OPEN

After exact memory-safe compilation, every virtual-Molecule corridor has a universally finite source-faithful core-transfer automaton at a fixed trace resolution, with universal core/WAD dimensions, exact decorated language, typed source-column endpoint frames, compositional future-observable WAD incidence, retained loop emissions, and contextual returns routed to closed source-licensed entropy, transient extinction, or a calibrated common Frobenius/shear flag.

*Formal hypotheses:* *none*

*Proof dependencies:* H1_CELL_TREE_EXTRACTION, H1_ANALYTIC_ACCESS_REALIZATION, H1_FINITE_HORIZON_INFLATION, H2_UNIFORM_COUNTER_FOLLOWER, H2_UNIFORM_CORE_WAD_DIMENSION, H2_MEMORY_SAFE_PUMPING_COMPILATION, SOURCE_FUTURE_QUOTIENT, MARKOV_SIGNATURE_AUTOMATON, WAD_TRANSPORT_QUOTIENT_CRITERION, WAD_FUTURE_OBSERVABILITY, PURE_CORRIDOR_MACRO, COUNTER_STACK_SAFE_LOOP, FRAMED_RETURN_SPECTRAL_INVARIANCE, BOOLEAN_SUPPORT_FUNCTOR, SUPPORT_AUTOMATON_PUMPING, BOUNDED_BRANCH_SUPPORT_WITNESS, ZERO_ENTROPY_SHEAR_NORMAL_FORM, ZERO_ENTROPY_SHEAR_NO_GO, KKL_TYPED_VALUE_INCIDENCE, H2_VIRTUAL_CORE_COCYCLE_REALIZATION

*Evidence:* virtual_support_automaton_construction.md#8-virtual-analytic-realization-theorem

---

**H2_VIRTUAL_CORRIDOR_GRAPH_REALIZATION** OPEN

Every virtual block assigned to the translation-clone branch has an actual finite-horizon core/Hubbard transition multigraph satisfying the exact corridor-to-clone criterion with universal reduced cross-section dimension, endpoint-window size, parallel-corridor count, label depth, and primitivity, together with complete critical and WAD-to-core itinerary incidence.

*Formal hypotheses:* *none*

*Proof dependencies:* H1_ANALYTIC_ACCESS_REALIZATION, H2_VIRTUAL_CORE_COCYCLE_REALIZATION, H2_UNIFORM_CORE_WAD_DIMENSION, KKL_TYPED_VALUE_INCIDENCE, KKL_TRANSLATION, EXACT_CORRIDOR_CLONE_CRITERION, ENDPOINT_DEFECT_ABSORPTION, FINITE_CORRIDOR_CLONE_UNION, CROSS_SECTION_DAG_EQUIVALENCE, ROBUST_CLONE_PRIMITIVITY, PRIMITIVE_BASE_CLONE_NO_GO, FINITE_STATE_CORRIDOR_CERTIFICATE, UNBOUNDED_DIMENSION_ENTROPY_NO_GO

*Evidence:* dynamical_corridor_certificate.md#6-refined-realization-theorem

---

**H1_CELL_TREE_EXTRACTION** OPEN

Every virtual-molecule block admits a monotone finite cell-tree quotient of its dynamically relevant Cantor carrier with universal atomic and gate incidence and compatible partial cellular dynamics.

*Formal hypotheses:* *none*

*Proof dependencies:* TREE_NERVE_FULLNESS, TOPOLOGY_REPLICATION_NO_GO

*Evidence:* virtual_hull_realization.md#61-finite-cell-tree-extraction-open

---

**H1_ANALYTIC_ACCESS_REALIZATION** OPEN

Every extracted cell tree lifts to an access-complete prime-end marked analytic correspondence satisfying finite-horizon psi-ql axioms, exact KKL translation rules, bounded incidence, standard-width charge, and controlled parallel replication.

*Formal hypotheses:* *none*

*Proof dependencies:* H1_CELL_TREE_EXTRACTION, H1_FINITE_HORIZON_INFLATION, KKL_TRANSLATION, TREE_HULL_COMPLEXITY, PARALLEL_REPLICATION_CRITERION, CONGESTION_AMORTIZATION

*Evidence:* virtual_hull_realization.md#62-analytic-access-realization-open

**H2_VIRTUAL_SECTION4_INTERFACE** OPEN

Every sufficiently degenerate virtual-Molecule block has access-complete small-set and tree marked complements, a complete per-limb ideal multi-interval bound, virtual WAD/return-width comparison, restriction multiplicity, and faithful flux/wave decomposition sufficient to reproduce KKL Proposition 3.7's static amplification/tame/controllable-wave split with constants independent of period and virtual depth.

*Formal hypotheses: none*

*Proof dependencies:* H1_ANALYTIC_ACCESS_REALIZATION, KKL_OCCUPANCY_TRICHOTOMY, TREE_HULL_COMPLEXITY, VIRTUAL_LIMB_TREE_BOUND, KKL_THICK_THIN, ABSTRACT_SECTION4_STATIC_TRICHOTOMY

*Evidence:* virtual_section4_interface.md#10-exact-unresolved-realization-obligations

---

**H2_VIRTUAL_TAME_FULL_RETURN** OPEN

Every virtual block routed to the tame branch realizes the complete full-cycle carrier and local complexity vector (m_I,|chi(V_I)|,|B_I|,d_I), a geometric rotation corridor or accelerated replacement, localized full-return cover with critical-encounter control, ordered canonical-WAD domination, Wanderers estimates, typed Value contraction, and unified-chart closure, with the actual pullback exponent at least the graph delay.

*Formal hypotheses: none*

*Proof dependencies:* H2_VIRTUAL_SECTION4_INTERFACE, H1_FINITE_HORIZON_INFLATION, H2_VIRTUAL_CLONE_REALIZATION, KKL_PARABOLIC_TAME_AMPLIFICATION, KKL_VALUE_CONTRACTION, WAD_DOMINATION_RESPONSE, LOCALIZED_CRITICAL_ENCOUNTER_DEGREE, LINEAR_MARKING_DENSITY_ABSORPTION, TAME_VALUE_OBSTRUCTION, CONDITIONAL_VIRTUAL_TAME_TRANSFER, ROTATION_RELABEL_NO_GO

*Evidence:* virtual_tame_full_return.md#8-exact-unresolved-lemmas

---

**H2_VIRTUAL_LONG_ITERATE_WAVE** OPEN

Every virtual block routed to the wave branch realizes the marked interval action, critical-limb alignment, and complete stagewise carrier through H_B(q)=(216B^2+1)(q+B+4), with proper/end-proper dynamics, exhaustive leaf itineraries, a degree-one central fast-forward corridor, and a bounded marked auxiliary fan with the required degree profile, yielding the virtual Wave amplification alternative in the unified chart.

*Formal hypotheses: none*

*Proof dependencies:* H2_VIRTUAL_SECTION4_INTERFACE, H1_ANALYTIC_ACCESS_REALIZATION, H1_FINITE_HORIZON_INFLATION, KKL_PARABOLIC_WAVE_AMPLIFICATION, KKL_PARABOLIC_WAVE_STATIONARY_PROOF_REPAIR, FINITE_HORIZON_TRUNCATION, RELATIVE_COVER_LIFT, FINITE_HORIZON_WAVE_BOUND, DEGREE_TWO_FAST_FORWARD_NO_GO, FINITE_INVERSE_TREE_FAN_SATURATION, FORWARD_HULL_COMPONENT_NO_GO, SATURATED_SEED_FOREST, ROOT_PULLBACK_FOREST_EQUIVALENCE, CRITICAL_VALUE_CORRIDOR_CRITERION, VIRTUAL_ROOT_CARRIER_SEPARATION, FINITE_OBSTACLE_FORWARD_CLOSURE, FINITE_FAN_RIBBON_PULLBACK, ENDPOINT_BIRTH_ACCOUNTING, LOCAL_CARRIER_EQUALITY_NO_BIRTH, ATTACHMENT_SUPPORT_REDUCTION, VIRTUAL_BOUNDED_ATTACHMENT_SUPPORT, FINITE_HORIZON_KKL_WAVE_REDUCTION

*Evidence:* virtual_long_iterate_wave.md#9-three-irreducible-construction-targets

---

**H2_RESPONSE_REALIZATION** OPEN

The KKL, DL25, and virtual marked surfaces admit a common bounded-terminal response form with universal extremal-energy comparison, distortion, lift multiplicity, congestion, and electrical-orientation control.

*Formal hypotheses: none*

*Proof dependencies:* H1_ANALYTIC_ACCESS_REALIZATION, RESPONSE_GLUE_CALCULUS, FINITE_ELECTRODE_RESPONSE, PORT_FUTURE_CLOSURE_NO_GO, WAD_DOMINATION_RESPONSE, FIXED_BLOCK_SPECTRAL_COMPARISON, COMMON_WIDTH_BRIDGE, DL_TILE_SCALING, WAD_CORRIDOR_SANDWICH, END_PLUMBING_TRACE_RESPONSE_CONTROL

*Evidence:* network_response_state.md#6-analytic-response-realization-theorem

---

**H2_VIRTUAL_CLONE_REALIZATION** OPEN

Every virtual corridor routed to the translation-clone branch admits a source-licensed core whose irreducible non-permutation gate dimension and actual first-return depth K are universally bounded, whose common delay satisfies n>K, and whose complete typed incidence obeys the corrected negative-weight clone equation or a weighted perturbation below the explicit entropy gap at its own root. Its outgoing retained WAD satisfies Y<=(f^r)^*X+E for an actual pullback exponent r>=n, and every required finite-horizon WAD itinerary is realized.

*Formal hypotheses: none*

*Proof dependencies:* H2_VIRTUAL_CORRIDOR_GRAPH_REALIZATION, KKL_TYPED_VALUE_INCIDENCE, KKL_PRINTED_CLONE_EQUATION_NO_GO, KKL_CLONE_EIGEN, PRIMITIVE_INTEGER_ENTROPY_GAP, VARIABLE_CLONE_BLOCK_

CONTRACTION, PERTURBED_CLONE_CONTRACTION, BOUNDED_ENTROPY_EIGENCONE, UNBOUNDED_DIMENSION_ENTROPY_NO_GO, WAD_DOMINATION_RESPONSE

*Evidence:* virtual_clone_entropy.md#7-virtual-clone-realization-theorem

---

**H2_VIRTUAL_TRANSFER_TRICHOTOMY** OPEN

Every virtual-Molecule corridor has a universally bounded-dimensional source-faithful counter/stack-follower layered core model whose fiber-safe stack-neutral complete framed returns are exact clone translations, uniformly positive-entropy primitive windows, or nonexpanding transfers routed to transient extinction, pseudo-Siegel regularization, or a calibrated finite shear state, with nonstationary frame and flag changes controlled.

*Formal hypotheses: none*

*Proof dependencies:* H1_ANALYTIC_ACCESS_REALIZATION, H2_VIRTUAL_SUPPORT_AUTOMATON, H2_VIRTUAL_CLONE_REALIZATION, PERMUTATION_COCYCLE_GAUGE, IRREDUCIBLE_TRANSFER_ENTROPY_DICHOTOMY, ZERO_ENTROPY_SHEAR_NORMAL_FORM, DEFECT_RUN_DECOMPOSITION, ZERO_ENTROPY_SHEAR_NO_GO, PRIMITIVE_VALUE_AFFINE_CONTRACTION, CONDITIONAL_NEUTRAL_CORRIDOR_ABSORPTION

*Evidence:* corridor_entropy_shear_dichotomy.md#6-refined-virtual-transfer-trichotomy

---

**H2_VALUE_FEATURE_REALIZATION** OPEN

Every positive-entropy block uses Value-faithful future-closed ports, proves uniform coefficient comparability inside each merged class, or retains a uniformly finite homogeneous vector of all outgoing KKL Value moments and transports it across the complete block.

*Formal hypotheses: none*

*Proof dependencies:* KKL_VALUE_CONTRACTION, WAD_DOMINATION_RESPONSE, VALUE_RESPONSE_NO_GO, VALUE_FEATURE_ROWSPACE, PRIMITIVE_VALUE_AFFINE_CONTRACTION, CLONED_VALUE_BLOCK_CONTRACTION, VARIABLE_CLONE_BLOCK_CONTRACTION, H2_VIRTUAL_CLONE_REALIZATION, H2_VIRTUAL_TRANSFER_TRICHOTOMY, H2_RESPONSE_REALIZATION

*Evidence:* bc1_finite_port_response.md#8-exact-remaining-uniform-theorem

---

**H2_DISCOUNTED_ERROR_CHARGE** OPEN

Every entropy-clock block converts all localization and chart errors into calibrated Value units and satisfies the universal discounted or telescoping error budget.

*Formal hypotheses: none*

*Proof dependencies:* H2_RESPONSE_REALIZATION, H2_VALUE_FEATURE_REALIZATION, DISCOUNTED_ENTROPY_ABSORPTION, ENTROPY_THRESHOLD_NO_GO, CONGESTION_AMORTIZATION

*Evidence:* entropy_and_cycle_bookkeeping.md#7-open-analytic-theorem

---

**H2_CHART_CYCLE_EXPANSION** OPEN

Every admissible directed cycle of primitive, neutral, and virtual chart transitions passes the applicable backward-expansion or forward-contraction gauge test, or the block decomposition removes that residual cycle.

*Formal hypotheses: none*

*Proof dependencies:* H2_RESPONSE_REALIZATION, CHART_CYCLE_OBSTRUCTION

*Evidence:* entropy_and_cycle_bookkeeping.md#7-open-analytic-theorem

---

**H1_FINITE_HORIZON_INFLATION** OPEN

Finite virtual postcritical data inflate to locally saturated depth-q psi-ql stages whose selected return is degree two, whose required pullbacks are admissible through the stopping time, and whose localization loss is uniformly relative or amortized.

*Formal hypotheses: none*

*Proof dependencies:* PRIME_END_RETURN, CRITICAL_VALUE_PULLBACK, FULLIFICATION, GLOBAL_DEGREE_NO_GO, DEGREE_TWO_FACTORIZATION, STAGEWISE_FACTORIZATION, MAPPING_TELESCOPE, QL_RETURN_FACTORIZATION, FINITE_HORIZON_TRUNCATION, STAGE_FORWARD_CLOSURE, RELATIVE_COVER_LIFT, FINITE_HORIZON_ASSEMBLY, TOPOLOGY_WIDTH_NO_GO, INFLATION_MODULUS_MONOTONICITY, FINITE_LOCAL_SATURATION

*Evidence:* finite_horizon_inflation.md#9-revised-open-theorem

---

**H1_MARKED_CONSTRUCTION** OPEN

Every tower has canonical entropy-clock checkpoints, access-faithful finite virtual skeletons, selected pullback factorizations with one quadratic step, and marked complements split into bounded active cores and resolved raw corridor/transfer carriers containing every proof-used cell, access, critical return, and stopping time. Uniform finite-control, counter, and stack realization belongs to H2.

*Formal hypotheses: none*

*Proof dependencies:* REGIME_PARTITION, MARKING_QUOTIENT_NO_GO, TRANSLATION_ORBIT_TYPES, RELATIVE_TAIL_CONTROL, H1_FINITE_HORIZON_INFLATION, ORBIT_PACKING, TREE_HULL_COMPLEXITY, PARALLEL_REPLICATION_CRITERION, TOPOLOGY_REPLICATION_NO_GO, CONDITIONAL_TREE_REPLICATION_CALIBRATION, H1_CELL_TREE_EXTRACTION, H1_ANALYTIC_ACCESS_REALIZATION

*Evidence:* virtual_hull_realization.md#6-finite-virtual-wad-realization-theorem

---

**H2_CALIBRATION** OPEN

QL, WAD, sectorial, and virtual marked coordinates are calibrated on declared interfaces with universal parameters and explicit additive, relative, density-normalized, or discounted error budgets.

*Formal hypotheses: none*

*Proof dependencies:* H1_MARKED_CONSTRUCTION, CONNECTED_EXTERIOR_CALIBRATION, WAD_CORRIDOR_SANDWICH, DL_TILE_SCALING, COMMON_WIDTH_BRIDGE, H2_RESPONSE_REALIZATION, H2_VALUE_FEATURE_REALIZATION, COMPLEXITY_COORDINATE_NO_GO, HARMONIC_LINEAR_NO_GO, BOUNDED_FANOUT_BOTTLENECK

*Evidence:* corridor_network_calibration.md#8-remaining-virtual-realization-theorem

---

**H2_FORWARD_CONTRACTION_CALIBRATION** OPEN

After identity localization/chart steps are grouped with the dynamical step, every complete forward upper entropy-clock block has endpoint gauges homogeneous in geometric state, uniformly strict and error-stable over admissible side records, or is replaced by an independently proved backward lower transition.

*Formal hypotheses: none*

*Proof dependencies:* FORWARD_CONTRACTION_REDUCTION, GROUPED_BLOCK_CONTRACTION_REDUCTION, H2_CALIBRATION, CHART_CYCLE_OBSTRUCTION, H2_CHART_CYCLE_EXPANSION

*Evidence:* block_contraction_program.md#3-open-analytic-theorem

---

**H2_CONE_CALIBRATION** OPEN

The mixed-chart transitions admit either a common backward-expanding Lyapunov calibration or a common forward-contracting calibration that inverts to backward amplification.

*Formal hypotheses: none*

*Proof dependencies:* CONE_REDUCTION, HOMOGENEOUS_LYAPUNOV_REDUCTION, ERROR_STABLE_LYAPUNOV_REDUCTION, FORWARD_CONTRACTION_REDUCTION, H2_CALIBRATION, H2_FORWARD_CONTRACTION_CALIBRATION, HARMONIC_LINEAR_NO_GO, BOUNDED_FANOUT_BOTTLENECK, CHART_CYCLE_OBSTRUCTION, RESPONSE_GLUE_CALCULUS, H2_CHART_CYCLE_EXPANSION

*Evidence:* corridor_network_calibration.md#7-forward-calibration-and-the-missing-contraction

---

**H2_SUMMABILITY** OPEN

The aggregated antirenormalization and chart-change error of every complete block is absolutely bounded, below the required relative contraction margin, or controlled by the exact future-entropy-discounted/telescoping budget in the calibrated gauge.

*Formal hypotheses: none*

*Proof dependencies:* H1_MARKED_CONSTRUCTION, H2_CALIBRATION, DL_TRANSITIONAL_PERIPHERAL, CONDITIONAL_NEUTRAL_CORRIDOR_ABSORPTION, CONGESTION_AMORTIZATION, DISCOUNTED_ENTROPY_ABSORPTION, ENTROPY_THRESHOLD_NO_GO, H2_DISCOUNTED_ERROR_CHARGE

*Evidence:* entropy_and_cycle_bookkeeping.md#7-open-analytic-theorem

---

**H2_ENTROPY_DICHOTOMY** OPEN

Every block is routed so that any entropy-driven contraction occurs on a closed source-framed return, while every nonexpanding residual is routed to transient extinction, sectorial regularization, or a calibrated finite shear state that itself lies in a proved strictly contracting augmented complete block.

*Formal hypotheses: none*

*Proof dependencies:* H1_MARKED_CONSTRUCTION, H2_SUMMABILITY, H2_VIRTUAL_SECTION4_INTERFACE, H2_VIRTUAL_TAME_FULL_RETURN, H2_VIRTUAL_LONG_ITERATE_WAVE, VIRTUAL_BRANCH_QUANTIFIER_ASSEMBLY, H2_VIRTUAL_CLONE_REALIZATION, H2_VIRTUAL_TRANSFER_TRICHOTOMY, TRANSLATION_DENSITY_IDENTITY

*Evidence:* corridor_network_calibration.md#10-arithmetic-density-gate

---

**MBTT** OPEN

A high calibrated potential of the augmented geometric response-density/Value/shear state amplifies to a preceding entropy-clock checkpoint, uniformly over admissible side data including exact counter values, finite-alphabet stack words, endpoint frames, and the Frobenius flag.

*Formal hypotheses:* *none*

*Proof dependencies:* H1_MARKED_CONSTRUCTION, H2_CALIBRATION, H2_FORWARD_CONTRACTION_CALIBRATION, H2_CONE_CALIBRATION, H2_SUMMABILITY, H2_ENTROPY_DICHOTOMY

*Evidence:* master_transition.md#6-corrected-master-transition-theorem

---

**H3_COERCIVITY** OPEN

Bounded calibrated potentials and the uniform normalized incidence and congestion hypotheses prevent degeneration of the analytic carriers, branch graphs, relative-cover structures, protected prime ends, response forms, and Value features, so every sequence has a subsequence converging to an admissible nondegenerate marked renormalization. Exact side memory is handled topologically by H3_SYMBOLIC_SIDE_COMPACTNESS, but compact encodings alone do not establish analytic admissibility.

*Formal hypotheses:* *none*

*Proof dependencies:* H1_MARKED_CONSTRUCTION, MBTT, BACKWARD_RECORD, H3_SYMBOLIC_SIDE_COMPACTNESS

*Evidence:* master_transition.md#m6-coercivity

---

**H4_RIGIDITY** OPEN

Controlled marked towers yield connected relative parameter neighborhoods forming a basis at the parameter with diameters tending to zero, either directly or through a precisely applicable trivial-fiber theorem.

*Formal hypotheses:* *none*

*Proof dependencies:* H3_COERCIVITY

*Evidence:* master_transition.md#m7-parameter-rigidity

---

**UNIVERSAL_PROTECTED_TOWER_CLOSURE** OPEN

Every infinitely renormalizable quadratic polynomial admits the six-clause Universal Protected-Tower Compactness and Rigidity package: canonical protected carriers, exact analytic realization, a uniformly finite Value-observable response state, a universal backward transition, coercive marked compactness, and tower rigidity with singleton parameter fiber and connected relative neighborhoods.

*Clauses:*

- **UPTCR-1**: Canonical protected carrier. *Maps to:* VIRTUAL_AUGMENTED_ATOM_PORT_CERTIFICATE, VIRTUAL_COMMON_FREE_COMPONENT, VIRTUAL_PROTECTED_ROUTING_CERTIFICATE, VIRTUAL_CANONICAL_INTERFACE_CARRIER, H1_CELL_TREE_EXTRACTION, H1_MARKED_CONSTRUCTION.
- **UPTCR-2**: Exact analytic realization. *Maps to:* H1_ANALYTIC_ACCESS_REALIZATION, VIRTUAL_CORRIDOR_PAIR_NATURALITY, VIRTUAL_ATTACHMENT_RESOLUTION, VIRTUAL_BOUNDED_ATTACHMENT_SUPPORT, H1_MARKED_CONSTRUCTION.
- **UPTCR-3**: Uniform finite response state. *Maps to:* H2_RESPONSE_REALIZATION, H2_UNIFORM_CORE_WAD_DIMENSION, H2_VALUE_FEATURE_REALIZATION.
- **UPTCR-4**: Backward transition. *Maps to:* H2_DISCOUNTED_ERROR_CHARGE, H2_CHART_CYCLE_EXPANSION, H2_ENTROPY_DICHOTOMY, MBTT.
- **UPTCR-5**: Coercive marked compactness. *Maps to:* H3_SYMBOLIC_SIDE_COMPACTNESS, H3_COERCIVITY.
- **UPTCR-6**: Tower rigidity and parameter fiber. *Maps to:* H4_RIGIDITY.

*Formal hypotheses:* *none*

*Proof dependencies:* H1_MARKED_CONSTRUCTION, MBTT, H3_COERCIVITY, H4_RIGIDITY

*Evidence:* mlc_closure_proposition.md#5-universal-protected-tower-compactness-and-rigidity-proposition

---

**MLC** OPEN

The Mandelbrot set is locally connected.

*Formal hypotheses:* *none*

*Proof dependencies:* YOCCOZ, H1_MARKED_CONSTRUCTION, MBTT, H3_COERCIVITY, H4_RIGIDITY, CONDITIONAL_REDUCTION

*Evidence:* dependency_audit.md#1-final-statement

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## Appendix B. Source and evidence policy

The research directory retains separate source, dependency, and adversarial ledgers:

- `../source_ledger.md` records exact source scope and checked locations;
- `../dependency_audit.md` records logical dependencies and forbidden shortcuts;
- `../adversarial_audit.md` records counterexamples and unresolved attacks;
- `../research_log.md` records chronological decisions; and
- `../proof_obligations.json` is the authoritative status graph.

The generated appendix is a status index, not a substitute for the cited proof text.

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